

Yen-Ting Wu

Mathematical Methods for Taylor's CM and Griffith's E&M

Lecture Notes for First-year Physics students

Department of Physics, National Central University

Cross-Campus Teaching Assistant from NTU PHY/MATH

Preface

This book explains and introduces the mathematical methods used in John R. Taylor's *Classical Mechanics* and David J. Griffiths' *Introduction to Electrodynamics*. These methods are usually not taught in full during high school. Unlike students at many other universities, we did not begin with "General Physics." In fact, even our calculus course was taught by the Department of Physics itself. Instead, we started directly with classical mechanics, one of the four fundamental branches of mechanics, and in the following semester we moved straight into electromagnetism. This book is written for students who are just beginning their new academic journey in college, especially students at NCU. :)

Classical mechanics and high school physics do not belong to the same world at all, not to mention classical electrodynamics. Of course, they still talk about motion, force, and the properties of electricity, but trust me, this is an entirely different universe. In mechanics, high school physics uses the Newtonian framework to describe motion, but only with very simple tools, such as triangles, basic vector addition and subtraction, and maybe the occasional dot product or cross product. As for electrodynamics, high school physics basically gives up pretending and just asks you to memorise formulas directly. You probably still remember that Biot-Savart law that you could never quite memorise. Back then, the goal was mostly just to get "high scores" and enter a "good university," right? But in this course, in next semester's course, and in this book, things are no longer only about scores, nor are they just about grinding through endless problem sets. Now, we are aiming for something deeper. We are trying to get closer to the **essence of the world**. Naturally, we cannot stay forever inside the comfort zone of simple equations and still expect to explain a world this complicated. We need a deeper mathematical language. This language speaks through calculus, vectors, and differential equations. Only with it do we have a chance to move a little closer to how the world truly works.

You will first encounter this book in the first semester of your freshman year, during the classical mechanics course. To prevent everyone from silently thinking, "What the hell is this?" during class, I put together these notes to introduce almost all the mathematical tools you will need when reading Taylor's *Classical Mechanics*. Roughly speaking, they cover about ninety percent of the necessary background. When I first encountered these topics, I was genuinely shocked by how different they were from high

school mathematics. I struggled a lot and felt deeply frustrated. I often felt completely lost because I did not have enough background knowledge, and I had not yet built the theoretical intuition needed to understand them. So, to fill in that gap, I spent some time organising the mathematics that I believe is essential for this course, and I arranged the material specifically to match the structure of Taylor's textbook. In this way, you will have a reference that walks alongside the physics instead of always falling behind it. After all, you probably do not want to be solving differential equations and taking gradients in mechanics class while your calculus course is still teaching you how to take limits.

Unlike a typical textbook, this booklet aims to explain the mathematics and the analysis of classical systems in a more rigorous and complete way. Most textbooks skip many elegant details, mainly because of page limits. Since Professor Kuo has only one semester to cover so much material, the course inevitably moves quickly, and some topics are naturally rushed or skipped. That is why I decided to create something that can complement both the textbook and the lectures. It is a companion guide designed to fill in the mathematical gaps. Pair it with our professor's highly recognisable voice, and you are probably in for a top-tier classroom experience. Also, unless something unexpected happens, you will probably need to write a lot of reports this semester. This booklet can serve as your mathematical background reference, so that you will not get lost in the mathematical jungle, wondering where to even begin. Anyway, enough rambling. I hope this semester gives you a smooth and meaningful journey into mechanics, and I hope you will experience for yourself just how beautiful the structure of physics and mathematics can be.

Even after this course ends, you can still carry this book with you into next semester's classical electrodynamics course. In the original V.1 version, this booklet only covered the mechanics part. Later, I sincerely hoped that it could help you throughout the entire year, so I added many, many more chapters. (In fact, I think my first version was done terribly.) These additions also prepare the mathematical background needed for electrodynamics, and they may even extend naturally into sophomore-level thermal and statistical physics, quantum physics, and your laboratory courses. To be honest, though, the content of this book is not easy. Honestly, even I often look back at what I wrote and think, "How did I even understand this back then?" Then I go back and reread my own booklet. (Only the first unit is relatively "cute.") Still, if you study while doing experiments, treat this booklet as a tool, and slowly chew through the parts you do not yet understand, you will make enormous progress. I hope all of you can make good use of this book. Of course, this is not mandatory. Maybe your mathematical ability is already far beyond mine. At the same time, I also hope that when you hold this book in your hands, you will remember that there was once a garrulous, annoying but deeply caring TA, together with a truly serious and dedicated Professor Kuo, quietly staying by your side throughout this journey. Trust me, you will not regret choosing this course.

Acknowledgements

First, I would like to thank the Department of Physics, the Department of Mathematics, and the Department of Atmospheric Sciences at National Taiwan University for the training and course materials they have provided. These experiences gave me broader directions for revising the content of this book. I would also like to express my special gratitude to Dr. Pei-Ming Ho from NTU Phy, Dr. Kwok-Wing Tsoi from NTU Math, and Dr. Kai-Chih Tseng from NTU AS. Their theoretical perspectives and guidance gave me new ideas and opportunities for intellectual exchange, allowing me to correct many of the errors and omissions in the original version.

I am also grateful to the students of the 114-1 Mechanics class for their feedback. In particular, the fried-fish bro Yu-Xi Lai often discussed the material with me and offered valuable theoretical directions and questions. These conversations made me realize that there was still much room for improvement in this book. To those who told me that this book had helped them, your words have been an enormous encouragement to me.

Most of all, I would like to thank Professor Chia-Ming Kuo from the Department of Physics at National Central University. He never dismissed me because of my formal status or academic path, and he even made an exceptional decision beyond ordinary administrative procedures by allowing me to serve as a teaching assistant across institutions. His support not only affirmed my ability but also gave me both the responsibility and the opportunity to improve my theoretical understanding, to do what I love most, and to meet more students from the Department of Physics at NCU. This has also given me more people with whom I may engage in academic discussion in the future. Most importantly, Professor Chia-Ming Kuo chose to believe that this book could improve the quality of teaching in the class, which gave the book the chance to be read by others. The opportunity he created has become my strongest academic support.

I also acknowledges the use of OpenAI GPT-5.5 as an AI-assisted tool during the preparation of this work. The tool was used for mathematical discussion, proof organization, explanatory restructuring, $\LaTeX$ editing, indexing assistance, and consistency checking of notation. All final mathematical statements, derivations, examples, citations, and editorial decisions were reviewed, verified, and approved by the author, who remains solely responsible for the content of this manuscript.

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Chapter 1

Basic Calculus for Physics

Abstract This chapter serves as the foundational prerequisite for entering theoretical physics. Whether in classical mechanics, electrodynamics, or any other discipline in the physics curriculum, calculus is regarded as a basic tool, akin to addition, subtraction, multiplication, and division in the eyes of high school students. Therefore, before formally beginning Taylor's *Classical Mechanics*, this book will provide a concise summary to help readers review the key concepts of differentiation, integration, and partial differentiation. The chapter will also include some common formula derivations used in classical mechanics. Unlike other chapters, this section will not focus on rigorous mathematical proofs or formal derivations. Instead, it will serve as a transitional bridge to more advanced content. Readers who are interested in more detailed mathematical justifications are encouraged to consult their own calculus textbooks.

1.1 Norm, Functions, and Continuity

1.1.1 p Norm in Metric Space

In high school, the size of a vector $\mathbf{x}$ is denoted by $|\mathbf{x}|$, which is equal to

$$|\mathbf{x}| = \sqrt{x^2 + y^2 + z^2}$$

under three-dimensional space. However, $|\cdot|$ is only valid under the normal Euclidean space. While we change the space, we need to change the way of representing the size of a vector. In general, we call the space as **Metric Space**:

Definition 1.1 (Metric Space) A *metric* on a set M is a function $d : M \times M \rightarrow \mathbb{R}$ such that for all $x, y, z \in M$,

1. **Positive definiteness:** $d(x, y) \geq 0$; when $d(x, y) = 0$, if and only if $x = y$.
2. **Symmetry:** $d(x, y) = d(y, x)$.
3. **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (M, d) is called a *metric space*.

In Euclidean space, we call it ℓ_2 metric so that the size representation $|\cdot|$ holds. One may replace the exponent 2 with an arbitrary real number $p \geq 1$. We call it ℓ_p metric, which is defined by settings $\mathbf{x} = (x_1, x_2, \dots)$ and $\mathbf{y} = (y_1, y_2, \dots)$:

$$d_p(\mathbf{x}, \mathbf{y}) := \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}} \quad (1.1)$$

To simplify the notation, fix $p \geq 1$ and for $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, we set

$$d_p(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_p$$

The quantity $\|\mathbf{x} - \mathbf{y}\|_p$ measures the size of the vector $\mathbf{x} - \mathbf{y}$. This is called *p-norm*. When $p = 2$, this becomes the ordinary Euclidean length

$$\|\mathbf{x}\|_2 = \left(\sum_{i=1}^n x_i^2 \right)^{\frac{1}{2}}.$$

This is the length used in ordinary geometry and classical mechanics.¹

In many places, especially in limits, continuity, and differentiability, we do not only care about the value of a vector itself. We care about a small change in a vector. For this reason, we often write

$$\Delta \mathbf{x} = (\Delta x_1, \Delta x_2, \dots, \Delta x_n)$$

for a small displacement or perturbation. Its p norm is

$$\|\Delta \mathbf{x}\|_p = \left(\sum_{i=1}^n |\Delta x_i|^p \right)^{\frac{1}{p}}.$$

Therefore, the notation

$$\|\Delta \mathbf{x}\|_p \rightarrow 0$$

means that the whole displacement vector tends to the zero vector. It means that all components become small at the same time, without choosing a specific direction or path.

For example, in $\mathbb{R}^3$, if

$$\Delta \mathbf{r} = (\Delta x, \Delta y, \Delta z),$$

then

$$\|\Delta \mathbf{r}\|_2 = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}.$$

Thus

$$\|\Delta \mathbf{r}\|_2 \rightarrow 0$$

¹ You can try to establish the ℓ_∞ metric and take the "limit" of d_∞ , which has been taught in high school. Verify that $d_\infty(x, y) = \max_i |x_i - y_i|$.

means that the spatial displacement becomes infinitesimally small in three-dimensional space. This is stronger and more geometric than only saying $\Delta x \rightarrow 0$, because the particle may approach the point from any direction.

In finite-dimensional Euclidean space, the exact choice of p does not change the meaning of approaching zero. For example,

$$\|\Delta \mathbf{x}\|_p \leq \|\Delta \mathbf{x}\|_\infty \leq n^{\frac{1}{p}} \|\Delta \mathbf{x}\|_\infty.$$

Therefore, in $\mathbb{R}^n$, writing $\|\Delta \mathbf{x}\|_p \rightarrow 0$ is equivalent to saying that every component Δx_i tends to zero. This is why many textbooks simply write $\|\Delta \mathbf{x}\| \rightarrow 0$ without specifying the norm. The notation emphasizes that the limit is taken with respect to the size of the whole perturbation vector. In many textbooks, the notation $\|\Delta\| \rightarrow 0$ appears in the definition of limits, continuity, and differentiability. Here Δ is not meant to be a scalar. It represents a small increment vector in $\mathbb{R}^n$. More explicitly, one may write

$$\Delta = (\Delta_1, \Delta_2, \dots, \Delta_n) \in \mathbb{R}^n.$$

Therefore, for $p \geq 1$, its p norm is

$$\|\Delta\|_p = \left(\sum_{i=1}^n |\Delta_i|^p \right)^{\frac{1}{p}}.$$

The notation

$$\|\Delta\|_p \rightarrow 0$$

means that the whole increment vector tends to the zero vector. In other words, every component Δ_i becomes small at the same time. It does not specify a particular direction or path of approach.

For example, in $\mathbb{R}^2$, if

$$\Delta = (\Delta x, \Delta y),$$

then

$$\|\Delta\|_2 = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

Thus $\|\Delta\|_2 \rightarrow 0$ means that the point approaches the reference point from any possible direction in the plane, not only along the x axis or the y axis. In physics, this is the same idea as taking an infinitesimal displacement. If a particle is displaced by

$$\Delta = (\Delta x, \Delta y, \Delta z),$$

then

$$\|\Delta\|_2 = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

is the size of the displacement. When we write $\|\Delta\| \rightarrow 0$, we are saying that the displacement becomes infinitesimally small in space, without restricting the direction of the displacement.

1.1.2 Continuity and Differentiability

Definition 1.2 *Continuity* Let $f : [a, b] \rightarrow \mathbb{R}$. We say that f is continuous on $[a, b]$ if for every $x_0 \in [a, b]$ and every $\varepsilon > 0$, there exists $\delta > 0$ such that for every $x \in [a, b]$,

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Proposition. If $f : [a, b] \rightarrow \mathbb{R}$ is differentiable at $x_0 \in (a, b)$, then f is continuous at x_0 .

Proof. Assume that f is differentiable at x_0 . Then

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

exists as a finite real number. By the definition of limit, there exists $\delta_1 > 0$ such that whenever

$$0 < |x - x_0| < \delta_1,$$

we have

$$\left| \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right| < 1.$$

Therefore,

$$\left| \frac{f(x) - f(x_0)}{x - x_0} \right| \leq \left| \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right| + |f'(x_0)| < |f'(x_0)| + 1.$$

For $x \neq x_0$, we may write

$$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0} (x - x_0).$$

Taking absolute values gives

$$|f(x) - f(x_0)| = \left| \frac{f(x) - f(x_0)}{x - x_0} \right| |x - x_0| < (|f'(x_0)| + 1) |x - x_0|.$$

Now let $\varepsilon > 0$ be given. Choose

$$\delta = \min \left\{ \delta_1, \frac{\varepsilon}{|f'(x_0)| + 1} \right\}.$$

If $0 < |x - x_0| < \delta$, then

$$|f(x) - f(x_0)| < (|f'(x_0)| + 1) \delta \leq \varepsilon.$$

If $x = x_0$, then $|f(x) - f(x_0)| = 0 < \varepsilon$ automatically. Hence, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Thus f is continuous at x_0 . $\square$

Corollary. If f is differentiable on $[a, b]$, with one sided differentiability at the endpoints, then f is continuous on $[a, b]$.

Proof. For every interior point $x_0 \in (a, b)$, the previous proposition shows that differentiability at x_0 implies continuity at x_0 .

At a , the same argument applies by replacing the ordinary limit $x \rightarrow a$ with the right hand limit $x \rightarrow a^+$. At b , the same argument applies by replacing the ordinary limit $x \rightarrow b$ with the left hand limit $x \rightarrow b^-$. Therefore f is continuous at every point of $[a, b]$. $\square$

Remark. The converse is not true. A function can be continuous but not differentiable. For example,

$$f(x) = |x|$$

is continuous at $x = 0$, but it is not differentiable at $x = 0$, since

$$\lim_{x \rightarrow 0^+} \frac{|x| - |0|}{x - 0} = 1, \quad \lim_{x \rightarrow 0^-} \frac{|x| - |0|}{x - 0} = -1.$$

The two one sided limits are different, so $f'(0)$ does not exist. In mechanics, let $x(t)$ be the position of a particle along a line. If q is differentiable at time t_0 , then the velocity

$$v(t_0) = \frac{dx}{dt}(t_0)$$

exists as a finite number. By the proposition above, q must be continuous at t_0 .

Physically, this means that a particle with a well defined finite velocity cannot suddenly jump from one position to another at the same instant. A discontinuous trajectory would make

$$\frac{x(t) - x(t_0)}{t - t_0}$$

fail to approach a finite limit, so the velocity at t_0 would not be well defined. This is why ordinary Newtonian mechanics usually describes physical trajectories by continuous, and often differentiable, functions of time.

For example, if

$$x(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0, \end{cases}$$

then $x(t)$ jumps at $t = 0$. The position is not continuous at 0, so it cannot be differentiable there. Such a path represents an instantaneous teleportation rather than an ordinary mechanical motion.

In electrostatics, the electric field is related to the electric potential ϕ by

$$\mathbf{E} = -\nabla\phi.$$

If ϕ is differentiable in a region, then each partial derivative of ϕ exists there. In particular, differentiability implies continuity, so the potential cannot jump abruptly inside that region.

This has a direct physical meaning. A finite electric field describes a finite local rate of change of potential. If the potential had a sudden jump across a surface, then its derivative would not be an ordinary finite function at that surface. Instead, one would need an idealized singular description, such as a distribution or a boundary condition.

Thus, in the ordinary smooth region of space, differentiability of the electromagnetic potential gives

$$\boxed{\mathbf{E} = -\nabla\phi \text{ well defined} \implies \phi \text{ locally continuous}}$$

Discontinuities in electromagnetic problems usually signal idealized sources, material interfaces, or the need to replace pointwise differential equations by integral laws and boundary conditions.

Definition 1.3 (Continuous function sets) Let $a < b$ and $[a, b]$ be a closed interval of $\mathbb{R}$. We define

$$C[a, b] \stackrel{\text{def}}{=} \{f : [a, b] \rightarrow \mathbb{R} : f \text{ is continuous on } [a, b]\}$$

Sometimes we may write $C[I]$, where I means an interval.

Definition 1.4 ($C^n(I)$ function sets) Let I be an interval of $\mathbb{R}$ and let $n \in \mathbb{N}$. We define

$$C^n(I) \stackrel{\text{def}}{=} \{f : I \rightarrow \mathbb{R} : f \text{ has continuous derivatives up to order } n \text{ on } I\}.$$

That is, $f \in C^n(I)$ if

$$f, f', f'', \dots, f^{(n)}$$

exist and are continuous on I .

Definition 1.5 ($C^\infty(I)$ function sets) Let I be an interval of $\mathbb{R}$. We define

$$C^\infty(I) \stackrel{\text{def}}{=} \{f : I \rightarrow \mathbb{R} : f \text{ has continuous derivatives of all orders on } I\}.$$

A function in $C^\infty(I)$ is also called a smooth function on I . Physicists love it!

1.1.3 Types of Functions

Functions describe how one object is assigned to another object. In mathematics, this assignment must be well defined. This means that each input has exactly one output. In physics, this simple idea is extremely important, because a physical law often plays the role of a map from one state to another state. For example, time evolution maps an initial state to a later state. If the same initial state could give two different later states

under the same physical law, then the evolution would not be a well defined function, and deterministic prediction would fail.

Definition 1.6 (Injective Function) Let A and B be sets, and let $f : A \rightarrow B$ be a function. We say that f is *injective* if for all $x_1, x_2 \in A$,

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

Equivalently, distinct elements in A have distinct images in B .

The intuitive meaning of injectivity is that the function does not collapse two different inputs into the same output. If f is injective, then the output still remembers which input it came from. This is why injectivity is closely related to information preservation.

A useful physical example appears in quantum mechanics. Let the state of a closed quantum system evolve by a unitary operator $U(t)$,

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle.$$

Since $U(t)$ is unitary, it preserves inner products,

$$\langle \phi(t) | \psi(t) \rangle = \langle \phi(0) | \psi(0) \rangle.$$

In particular, if two initial states are different and physically distinguishable, unitary evolution cannot send them to the same final state. Thus the time evolution of a closed quantum system is injective. If injectivity were lost, two different initial states could become the same final state. Then the final state would no longer contain enough information to reconstruct the initial state, and quantum information would be lost.

Definition 1.7 (Surjective Function) Let A and B be sets, and let $f : A \rightarrow B$ be a function. We say that f is *surjective* if for every $y \in B$, there exists $x \in A$ such that

$$f(x) = y.$$

Equivalently, every element of B is the image of at least one element of A .

The intuitive meaning of surjectivity is that the function reaches the whole target set. No element in B is missed. If $f : A \rightarrow B$ is surjective, then every possible output in B can be produced from at least one input in A .

In physics, surjectivity often depends on how we choose the target space. For example, consider Gauss's law in electrostatics,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.$$

This equation may be viewed as a map from an electric field $\mathbf{E}$ to a charge density ρ ,

$$\mathbf{E} \mapsto \epsilon_0 \nabla \cdot \mathbf{E}.$$

This map is not injective, because many different electric fields can have the same divergence. For example, if $\mathbf{E}_1$ and $\mathbf{E}_2$ satisfy

$$\nabla \cdot \mathbf{E}_1 = \nabla \cdot \mathbf{E}_2,$$

then their difference satisfies

$$\nabla \cdot (\mathbf{E}_1 - \mathbf{E}_2) = 0.$$

Therefore, the same charge density does not determine the full electric field unless additional physical information is given, such as boundary conditions or the curl condition in electrostatics. Without these extra conditions, the electromagnetic problem is not unique, and the physical prediction is incomplete.

Definition 1.8 (Bijective Function) Let A and B be sets, and let $f : A \rightarrow B$ be a function. We say that f is *bijective* if f is both injective and surjective. Equivalently, for every $y \in B$, there exists a unique $x \in A$ such that

$$f(x) = y.$$

The intuitive meaning of bijectivity is perfect matching. Every input gives one output, every output is reached, and no two different inputs share the same output. Therefore, a bijective function has an inverse function

$$f^{-1} : B \rightarrow A.$$

This means that the process described by f can be reversed.

This idea is central in quantum mechanics. For a closed system, the time evolution operator $U(t)$ is unitary, so

$$U(t)^\dagger U(t) = U(t)U(t)^\dagger = I.$$

Hence the inverse evolution exists and is given by

$$U(t)^{-1} = U(t)^\dagger.$$

Therefore, the map

$$|\psi(0)\rangle \mapsto |\psi(t)\rangle = U(t) |\psi(0)\rangle$$

is bijective on the Hilbert space of states. Every initial state determines one final state, and every final state comes from a unique initial state. This is the mathematical reason why the time evolution of a closed quantum system is reversible.

In contrast, if a physical law gave

$$x(0) \mapsto x(t)$$

but the same initial condition allowed two different solutions, then this rule would not be a function. If two different initial conditions evolved into the same final condition, then the map would fail to be injective. The first failure destroys deterministic prediction, while the second failure destroys reversibility and information preservation.

1.2 Differentiation

1.2.1 The Chain Rule

If $y = f(u)$ and $u = g(x)$ are both differentiable functions, then the composite function $y = (f \circ g)(x) = f(g(x))$ is also differentiable, and its derivative at the point $x = a$ is

$$(f \circ g)'(a) = f'[g(a)] \cdot g'(a) \quad (1.2)$$

Which can also be denoted by

$$\left. \frac{df[g(x)]}{dx} \right|_{x=a} = \left. \frac{df(y)}{dy} \right|_{y=g(a)} \cdot \left. \frac{dg(x)}{dx} \right|_{x=a} \quad (1.3)$$

This can also be intuitively understood using Leibniz notation as:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad (1.4)$$

Although this notation is not a rigorous mathematical proof, it is very commonly seen and extremely useful in physics and engineering.

Example 1.1 Find the derivative of the following function

1. $f(x) = (2x^2 + 3x)^5$

2. $f(x) = \sqrt{\frac{1}{x^2 - 1}}$

Solution 1.1

1. Let $f(x) = g(u)$, where $g(x) = x^5$ and $u = 2x^2 + 3x$, by chain rule

$$\frac{df}{dx} = \frac{dg}{du} \cdot \frac{du}{dx} = g'(u) \cdot u'$$

So the final answer is

$$f'(x) = g'(u) \cdot u' = 5(2x^2 + 3x)^4 \cdot (4x + 3)$$

2. Let $f(x) = h(u)$, where $h(x) = \sqrt{x} = x^{1/2}$ and $u = 1/(x^2 - 1)$, by chain rule

$$\frac{df}{dx} = \frac{dh}{du} \cdot \frac{du}{dx} = h'(u) \cdot u'$$

Note that the derivative of u is

$$u' = \frac{-2x}{(x^2 - 1)^2}$$

So the final answer is

$$f'(x) = h'(u) \cdot u' = \frac{1}{2}x^{-\frac{1}{2}} \cdot \frac{-2x}{(x^2 - 1)^2} = \frac{-x}{\sqrt{x}(x^2 - 1)^2}$$

1.2.2 Trigonometric Differentiation

Let us first review the definitions of the basic trigonometric ratios and the fundamental trigonometric identities.

$$\left\{ \begin{array}{l} \tan x = \frac{\sin x}{\cos x} \\ \cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x} \\ \sec x = \frac{1}{\cos x} \\ \csc x = \frac{1}{\sin x} \end{array} \right. \quad \left\{ \begin{array}{l} \sin^2 x + \cos^2 x = 1 \\ \tan^2 x + 1 = \sec^2 x \\ \cot^2 x + 1 = \csc^2 x \end{array} \right. \quad (1.5)$$

The basic trigonometric differentiations are sin and cos. Which is

$$\left\{ \begin{array}{l} \frac{d}{dx} \sin x = \cos x \\ \frac{d}{dx} \cos x = -\sin x \end{array} \right. \quad (1.6)$$

Note that when the x term become $u = g(x)$, which is $\sin(u)$, we should apply the chain rule on it. For example,

$$\frac{d}{dx} \sin(g(x)) = \cos(g(x)) \cdot g'(x)$$

So is the cosine term. After discussing the basic trigonometric differentiations, I will prove the other four differentiations. Starting with $\tan x$, because $\tan x = \sin x / \cos x$, by the differentiation rule, we get

$$\frac{d}{dx} \tan x = \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) = \frac{\frac{d}{dx}(\sin x) \cos x - \sin x \frac{d}{dx}(\cos x)}{\cos^2 x} = \frac{1}{\cos^2 x}$$

So, the derivative of $\tan x$ is

$$\boxed{\frac{d}{dx} \tan x = \sec^2 x} \quad (1.7)$$

Apply the same logic, we can calculate the other three derivatives

$$\boxed{\begin{cases} \frac{d}{dx} \cot x = -\csc^2 x \\ \frac{d}{dx} \sec x = \sec x \tan x \\ \frac{d}{dx} \csc x = -\csc x \cot x \end{cases}} \quad (1.8)$$

Example 1.2 Find the derivative of the following function

1. $f(t) = A \cos(\omega t - \delta) + B \sin(\omega_0 t - \delta)$
2. $f(x) = \cos(3x^2 + 4x - 1)$
3. $f(t) = \sin(u)$, where $u = \theta(t^2 - 3t)$

Solution 1.2

1. The differentiation rule

$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

So the derivative of $f(t)$ is

$$f'(t) = \frac{d}{dt}A \cos(\omega t - \delta) + \frac{d}{dt}B \sin(\omega_0 t - \delta)$$

Remember the chain rule,

$$\frac{d}{dt}A \cos(\omega t - \delta) + \frac{d}{dt}B \sin(\omega_0 t - \delta) = -A\omega \sin(\omega t - \delta) + B\omega_0 \cos(\omega_0 t - \delta)$$

So the final answer is

$$f'(t) = -A\omega \sin(\omega t - \delta) + B\omega_0 \cos(\omega_0 t - \delta)$$

2. Let $f(x) = g(u)$, where $g(x) = \cos x$ and $u = 3x^2 + 4x - 1$, then

$$\frac{d}{dx}f(x) = \frac{d}{du}g(u) \cdot \frac{d}{dx}u$$

So the final answer is

$$f'(x) = -\sin(3x^2 + 4x - 1) \cdot (6x + 4)$$

3. In this problem, there are three layers of composition, $\sin t$, $\theta(t)$ and $u = t^2 - 3t$, so we need to apply the chain rule repeatedly.

$$\frac{d}{dx}(f(g(u))) = f'(g(u)) \cdot g'(u) \cdot u'$$

So

$$\frac{d}{dt}f(t) = \dot{f}(t) = \frac{d}{d\theta} \sin(\theta(t^2 - 3t)) \cdot \frac{d}{du}\theta(u) \times \frac{d}{dt}u$$

The final answer is

$$\dot{f}(t) = (2t - 3)\dot{\theta} \cos(\theta)$$

1.2.3 Derivative of Exponential Function

An exponential function is defined as

$$f(x) = b^x \quad (1.9)$$

Where $b > 0$, and x can be a rational or irrational number. If $b > 0, b \neq 1$, the exponential function is continuous, which is

$$\lim_{x \rightarrow r} b^x = b^r \quad (1.10)$$

To differentiate the exponential function $f(x) = b^x$, we need to do it by the definition of differentiation that

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h} = \lim_{h \rightarrow 0} \frac{b^x(b^h - 1)}{h} \\ &= b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h} \end{aligned} \quad (1.11)$$

Note that we can rewrite the limit term as

$$\lim_{h \rightarrow 0} \frac{b^h - 1}{h} = \lim_{h \rightarrow 0} \frac{b^{(0+h)} - 1}{h} = f'(0) \quad (1.12)$$

So we can obtain

$$f'(x) = b^x \cdot f'(0) \quad (1.13)$$

For example, for $b = 2$, $f'(0) = \lim_{h \rightarrow 0} (2^h - 1)/h \approx 0.693$, while $b = 3$, $f'(0) = \lim_{h \rightarrow 0} (3^h - 1)/h \approx 1.099$. By these two examples, it is obvious that there exists

a number Ξ ($2 < \Xi < 3$), such that $f'(0) = 1$. We can define this number as $\Xi = e$ (called "Euler's Number", sometimes called "exponential"), which is

$$f'(0) = \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1 \quad (1.14)$$

So the derivative of the function $f(x) = e^x$ is

$$\boxed{f'(x) = e^x \cdot f'(0) = e^x} \quad (1.15)$$

We will not provide a detailed proof or explanation of how to approximate the value of e here. If you are really, really curious, please refer to *Stewart's Calculus*. In this book, we simply adopt the standard approximation: $e \approx 2.7182818$. When there is a composite function of exponential, for example $f(u) = e^u$, $u = \sin x$, we need to apply the chain rule

$$f'(u) = e^u \cdot u' = e^{\sin x} \cdot \cos x$$

After discussing the exponential function, we can continue to the logarithmic function

$$f(x) = \log_b a \quad (1.16)$$

where $b > 0, b \neq 1$ and $a > 0$. The logarithmic function and the exponential function are the inverse functions of each other, which means

$$f(x) = b^x, f^{-1}(b^x) = \log_b b^x = x$$

When we replace the base of the logarithm with the number $\Xi = e$, we call it **natural log** and rewrite it as

$$\boxed{\log_e x := \ln x} \quad (1.17)$$

After understanding the differentiation rules for exponential functions, we now turn to the differentiation rules for logarithmic functions. Let $f(x) = \ln x$, we need to find $f'(x)$. By the definition of the natural log, we obtain

$$e^{f(x)} = x \Rightarrow \frac{d}{dx} e^{f(x)} = e^{f(x)} f'(x) = 1 \quad (1.18)$$

Dividing both sides by $e^{f(x)}$, then we can get

$$f'(x) = \frac{1}{e^{f(x)}} = \frac{1}{x} \quad (1.19)$$

Note that we substituted $e^{f(x)} = x$ into the result, so

$$\boxed{\frac{d}{dx} \ln x = \frac{1}{x}} \quad (1.20)$$

By the way, sometimes, when the exponent of e becomes too long, we tend to write it as $\exp(f(x))$ for convenience.

Now, let's return to the differentiation of logarithmic functions with bases other than e . We can let $f(x) = b^x = e^{\ln b^x} = e^{x \ln b}$, then

$$f'(x) = \frac{d}{dx} e^{x \ln b} = e^{x \ln b} \cdot \frac{d}{dx} (x \ln b) = e^{x \ln b} \cdot \ln b \quad (1.21)$$

Substituting the equation $e^{x \ln b} = b^x$ into it, we can verify that

$$\boxed{\frac{d}{dx} b^x = b^x \cdot \ln b} \quad (1.22)$$

In addition, we can differentiate the logarithmic function $f(x) = \log_a x$ by the change of base formula, which is

$$\log_a x = \frac{\log_e x}{\log_e a} = \frac{\ln x}{\ln a} \quad (1.23)$$

So

$$\boxed{\frac{d}{dx} \log_a x = \frac{1}{\ln a} \frac{d}{dx} \ln x = \frac{1}{x \ln a}} \quad (1.24)$$

There is another type of the exponential function, which is the "variable exponent function". It's a function in which the variable appears both as the base and as the exponent. Consider the form

$$f(x) = x^x \quad (1.25)$$

In such a problem, if we want to get $f'(x)$, we should take the natural logarithm of both sides of the equation. That is

$$\ln f(x) = \ln x^x = x \ln x \quad (1.26)$$

In this way, we change the variable exponent function into a simple composite function. Then, differentiate both sides,

$$\frac{d}{dx} \ln f(x) = \frac{1}{f(x)} \cdot f'(x) = \frac{d}{dx} (x \ln x) = \ln x + x \cdot \frac{1}{x}$$

So it becomes

$$f'(x) = f(x)(\ln x + 1)$$

Substituting $f(x) = x^x$ into the result, we can get

$$\boxed{\frac{d}{dx} x^x = x^x (\ln x + 1)} \quad (1.27)$$

Example 1.3 Find the derivative of the following function

$$1. f(x) = 2^{3x^2 - x}$$

2. $f(x) = e^{x^{3x}}$

3. $f(x) = e^x x^{\sin x}$

Solution 1.3

1. We can directly differentiate this function by applying the chain rule to it, that is

$$\frac{d}{dx} 2^{3x^2-x} = 2^{3x^2-x} \cdot \ln(3x^2-x) \cdot \frac{d}{dx}(3x^2-x)$$

So the final answer is

$$f'(x) = (6x-1) \cdot \ln(3x^2-x) \cdot 2^{3x^2-x}$$

2. Let $f(x) = e^u$, and $u = x^{3x}$, the differentiation is

$$f'(x) = e^u \cdot u'$$

By $u = x^{3x}$, it's a variable exponent function, we need to take natural logarithm of both sides of the function, that is

$$\ln u = \ln(x^{3x}) = 3x \ln x \Rightarrow \frac{d}{dx} \ln u = \frac{1}{u} u' = 3 \ln x + 3$$

So $u' = u(3 \ln x + 3) = x^{3x}(3 \ln x + 3)$. Substituting this into $f'(x) = e^u \cdot u'$, we can finally get

$$f'(x) = e^{x^{3x}} \cdot x^{3x}(3 \ln x + 3)$$

3. the function consists of the function $g(x) = e^x$ and the function $h(x) = x^{\sin x}$, so the differentiation

$$f'(x) = g'(x)h(x) + g(x)h'(x) = \left(\frac{d}{dx} e^x\right) x^{\sin x} + e^x \left(\frac{d}{dx} x^{\sin x}\right)$$

The function $h(x)$ is a variable exponent function, so let $u = x^{\sin x}$ and take the natural log of both sides $\ln u = \ln x^{\sin x} = \sin x \cdot \ln x$. Then

$$\frac{d}{dx} \ln u = \frac{1}{u} u' = \frac{d}{dx} (\sin x \cdot \ln x) = \cos x \cdot \ln x + \frac{\sin x}{x}$$

So we get $u' = u(\cos x \cdot \ln x + \frac{\sin x}{x}) = x^{\sin x}(\cos x \cdot \ln x + \frac{\sin x}{x})$. Substituting this into $f'(x)$

$$f'(x) = e^x \sin x + e^x x^{\sin x} \left(\cos x \cdot \ln x + \frac{\sin x}{x} \right)$$

1.2.4 Implicit Differentiation

An implicit function is a non-standard form of a function on the two-dimensional plane. It is expressed as an equation involving both variables x and y , and we cannot easily determine which variable is a function of the other. This is why it is called an "implicit" function. A classic example is the equation of a circle or an ellipse, both of which are implicit functions, such as

$$x^2 + y^2 = r^2 \quad \text{or} \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

In this part, I'll show you how to get the derivative of y with respect to x . The key point is the **chain rule**. Consider the form

$$F(x, y) = f(x)g(y) = C \quad (1.28)$$

Where $F(x, y) = 0$ is the standard form of an implicit function. Differentiating both sides with respect to x , we get

$$\frac{d}{dx}(f(x)g(y)) = \left(\frac{d}{dx}f(x)\right)g(y) + f(x)\left(\frac{d}{dx}g(y)\right) = 0 \quad (1.29)$$

Note that $g(y)$ is a function with respect to y , so we need to apply the chain rule to it. That is

$$\frac{d}{dx}g(y) = \frac{dg(y)}{dy} \cdot \frac{dy}{dx} \quad (1.30)$$

Substituting (1.29) into (1.30), we obtain

$$\left(\frac{d}{dx}f(x)\right)g(y) + f(x)\left(\frac{dg(y)}{dy} \cdot \frac{dy}{dx}\right) = 0 \quad (1.31)$$

So the derivative of y is

$$\frac{dy}{dx} = \frac{-f'(x)g(y)}{f(x)g'(y)} \quad (1.32)$$

For example, we can use implicit differentiation to solve the problem of finding the tangent line to a circle, which is commonly encountered in high school mathematics. There is an equation of the circle $(x - 1)^2 + (y - 2)^2 = 4$, and we need to get dy/dx of it. First,

$$\frac{d}{dx}(x - 1)^2 + \frac{d}{dx}(y - 2)^2 = 0 \Rightarrow (2x - 2) + \left[\frac{d}{dy}(y - 2)^2 \cdot \frac{dy}{dx}\right] = 0$$

The differentiation $d/dy (y - 2)^2$ is $2y - 4$. Substituting it into the original equation,

$$(2x - 2) + (2y - 4) \frac{dy}{dx} = 0$$

So the final answer is

$$\frac{dy}{dx} = \frac{-2x + 2}{2y - 4}$$

The slope of the tangent line passing through the point $(2, \sqrt{3} + 2)$ is $dy/dx = (-2 \times 2 + 2)/(2 \times (\sqrt{3} + 2) - 4) = -1/\sqrt{3}$. There is another classic example. Consider an implicit function

$$y^7 - y^3 + x^3y^3 - 6x = 8$$

If we want to find dy/dx , we need to differentiate both sides

$$\begin{aligned} \frac{d}{dx}y^7 - \frac{d}{dx}y^3 + \frac{d}{dx}(x^3y^3) - \frac{d}{dx}6x &= 0 \\ \Rightarrow 7y^6 \frac{dy}{dx} - 3y^2 \frac{dy}{dx} + 3x^2y^3 + 2x^3y^2 \frac{dy}{dx} - 6 &= 0 \\ \Rightarrow \frac{dy}{dx}(7y^6 - 3y^2 + 2x^3y^2) &= 6 - 3x^2y^3 \\ \Rightarrow \frac{dy}{dx} &= \frac{6 - 3x^2y^3}{7y^6 - 3y^2 + 2x^3y^2} \end{aligned}$$

This brings us to the final answer. Below, I've created two challenging problems that integrate all the differentiation techniques covered in this book. If you can solve both of them independently, it means you've truly mastered the concepts, and it's also a great way to test whether you've been paying attention in class or reading this book carefully. :)

Example 1.4 Find the derivative dy/dx of the following implicit functions.

1. $y = 2^{x^y}$
2. $y^{\exp(\sin x)} = x^{\sin x}$

Solution 1.4

1. This implicit function is an variable exponent function, and it must be differentiated using logarithmic differentiation. That is

$$\ln y = \ln \left(2^{x^y} \right) = x^y \cdot \ln 2$$

Differentiating both sides, we get

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx}(x^y) \ln 2 + x^y \frac{d}{dx} \ln 2 = \frac{d}{dx}(x^y) \ln 2$$

Let $x^y = \eta$ then

$$\frac{d}{dx} \ln \eta = \frac{1}{\eta} \left(\frac{d}{dx} \eta \right) = \frac{d}{dx} (y \ln x) = \frac{dy}{dx} \ln x + \frac{y}{x}$$

Substituting this into $x^y = \eta$, we get

$$\frac{d}{dx} \eta = \frac{d}{dx} (x^y) = x^y \left(\frac{dy}{dx} \ln x + \frac{y}{x} \right)$$

Put this result back to the original equation

$$\frac{dy}{dx} = y \left(\frac{d}{dx} (x^y) \ln 2 \right) = y \left[x^y \left(\frac{dy}{dx} \ln x + \frac{y}{x} \right) \ln 2 \right]$$

By rearranging the equation, we obtain the final result that

$$\frac{dy}{dx} = \frac{\ln 2 \cdot y^2}{x - x \ln 2 \cdot y \cdot x^y \cdot \ln x}$$

2. This implicit function is a variable exponent function, too. So, it must be differentiated using logarithmic differentiation.

$$e^{\sin x} \ln y = \sin x \ln x$$

Differentiating both sides, we get

$$\begin{cases} \frac{d}{dx} (e^{\sin x} \ln y) = \cos x e^{\sin x} \ln y + e^{\sin x} \cdot \frac{1}{y} \frac{dy}{dx} \\ \frac{d}{dx} (\sin x \ln x) = \cos x \ln x + \sin x \cdot \frac{1}{x} \end{cases}$$

Combining them to get dy/dx

$$\frac{dy}{dx} = \frac{y \cos x (x \ln x - x \cdot e^{\sin x} \ln y) + y \sin x}{x e^{\sin x}}$$

1.3 Fundamental Techniques of Integration

1.3.1 Fundamental Theorem of Calculus

Consider the functions $F(x)$, $f(x)$, where $f, F : [a, b] \rightarrow \mathbb{R}$ and $a, b \in \mathbb{R}$, and they satisfy

$$F'(x) = f(x) \tag{1.33}$$

Then, we call $F(x)$ the antiderivative of $f(x)$. To find the antiderivative, we create a notation $\int$, which is

$$\int f(x) \, dx = F(x) + C \quad (1.34)$$

(1.34) means that if we know a function $f(x)$, its antiderivative is not unique since any two antiderivatives can differ by a constant. You can check this in the equation (1.33). By (1.33) and (1.34), we know that differentiation and integration are inverse operations. This book does not cover the definition of Riemann sums in detail. If you have forgotten it, please review your high school mathematics textbook.

To rigorously establish the connection between differentiation and integration, we introduce the **Fundamental Theorem of Calculus (FTC)**. This theorem consists of two parts. Let's begin with the first part (FTC I). The antiderivative $F(x)$ of $f(x)$ is defined to be

$$F(x) = \int_a^x f(t) \, dt \quad (1.35)$$

By the definition, the derivative of $F(x)$ is

$$F'(x) = \frac{d}{dx} \int_a^x f(t) \, dt = f(x) \quad (1.36)$$

We can extend it to a more general definition. When (1.35) becomes

$$F(x) = \int_a^{g(x)} f(t) \, dt \quad (1.37)$$

Then the derivative of $F(x)$ must be applied to the chain rule

$$F'(x) = \frac{d}{du} \frac{du}{dx} \int_a^u f(t) \, dt = f(u) \cdot \frac{du}{dx}$$

where $u = g(x)$, so the general form, it satisfy

$$F'(x) = \frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x)) \cdot g'(x) \quad (1.38)$$

In short, FTC I tells us that integrating a function gives an antiderivative, and differentiating it brings us back to the original function.

We now turn to the Second Fundamental Theorem of Calculus (FTC II), which is based on the assumption that FTC I holds. If $x \in [a, b]$, and $F'(x) = f(x)$, with $f(x)$ integrable over $[a, b]$, then we can formally link the operations of definite integration, differentiation, and antiderivatives. That is,

$$\int_a^b f(x) \, dx = F(x) \Big|_a^b = F(b) - F(a) \quad (1.39)$$

Simply put, the indefinite integral is a kind of "inverse operation" that represents the set of all antiderivatives of a function $f(x)$. The First Fundamental Theorem of Calculus (FTC I) describes the differentiability of the accumulation function $F(x)$, and shows that differentiating it recovers the original function $f(x)$. Finally, the Second Fundamental Theorem of Calculus (FTC II) states that the difference of the accumulation function at the endpoints a and b is equal to the area under its derivative $f(x)$ over the interval $[a, b]$, thereby connecting the concepts of differentiation, derivatives, antiderivatives, and Riemann sums.

Although both parts of the Fundamental Theorem of Calculus are deeply related, they have a logical order: FTC I must be established first. This theorem shows that the integral of a continuous function defines a differentiable function whose derivative recovers the integrand. Only after FTC I is proven can we then use this result to define and compute definite integrals using antiderivatives, which is what FTC II asserts. Therefore, FTC II depends logically on FTC I, not the other way around.

1.3.2 Integration by Substitution

The function $F(u)$ is the antiderivative of the function $f(u)$, which is

$$F'(u) = f(u) \quad (1.40)$$

Consider a composite function $F(g(x))$, by the chain rule, we obtain

$$\frac{d}{dx} [F(g(x))] = F'(g(x)) \cdot g'(x) \quad (1.41)$$

Apply FTC and let the function $g(x)$ be u , we can thus verify

$$\int F'(g(x)) \cdot g'(x) dx = \int \frac{d}{dx} [F(g(x))] dx = F(g(x)) + C = F(u) + C \quad (1.42)$$

Note that we can rewrite the $F(u) + C$ as an indefinite integral and by (1.40), we obtain

$$F(u) + C = \int F'(u) du = \int f(u) du \quad (1.43)$$

By (1.42) and (1.43), we can finally verify that

$$\boxed{\int f(g(x)) \cdot g'(x) dx = \int f(u) du} \quad (1.44)$$

This is the **substitution rule** in integration. Substitution requires that the substitution function $g(x)$ is differentiable and the resulting integral is well-defined. Improper substitutions can lead to incorrect results. So that you can quickly write down the equation

$$du = g'(x) dx$$

Note that if you are dealing with a definite integral, you must change the limits of integration to match the new variable u . For example

$$\int_a^b f(g(x)) \cdot g'(x) dx = \int_{g(a)}^{g(b)} f(u) du \quad (1.45)$$

Since $u = g(x)$, $a \rightarrow g(a)$, $b \rightarrow g(b)$. However, if you're feeling lazy (like I usually am), you can first compute the antiderivative in terms of u , and then substitute $u = g(x)$ back into the result. That way, you don't have to change the limits. For example, we want to evaluate the integral

$$\int \frac{1}{x^2} \sqrt{1 + \frac{1}{x}} dx$$

Let $u = 1 + x^{-1}$, then $du = -x^{-2}dx$, so the integral becomes

$$\int \sqrt{1 + \frac{1}{x}} \left(\frac{1}{x^2} dx \right) = \int \sqrt{u} (-du) = - \int \sqrt{u} du = -2u^{3/2} + C$$

substituting $u = 1 + x^{-1}$ back into the result, we can obtain

$$\int \frac{1}{x^2} \sqrt{1 + \frac{1}{x}} dx = \left(1 + \frac{1}{x} \right)^{-3/2} + C$$

Example 1.5 Evaluate the following integrals.

1. $\int \frac{(\ln x)^2}{3x} dx$
2. $\int_1^2 (x-1)e^{(x-1)^2} dx$

Solution 1.5

1. Let $\ln x = u$, so $du = d/dx(\ln x) dx = 1/x dx$, the integral becomes

$$\int \frac{(\ln x)^2}{3x} dx = \frac{1}{3} \int (\ln x)^2 \left(\frac{1}{x} dx \right) = \frac{1}{3} \int u^2 du = \frac{1}{9} u^3 + C$$

substituting $u = \ln x$ back into the result, we can obtain the answer

$$\int \frac{(\ln x)^2}{3x} dx = \frac{1}{9} (\ln x)^3 + C$$

2. Let $(x-1)^2 = u$, then $du = 2(x-1) dx$, the integral becomes

$$\int_1^2 (x-1)e^{(x-1)^2} dx = \int_1^2 e^{(x-1)^2} \left[\frac{1}{2} \cdot 2(x-1) dx \right] = \frac{1}{2} \int_0^1 e^u du = \frac{1}{2} e^u \Big|_0^1 = \frac{e-1}{2}$$

1.3.3 Integration by Parts

Next, we introduce what I consider to be one of the most important techniques in integration, **integration by parts**. When dealing with a complicated function, this method allows you to eliminate the interference of polynomial terms (see Chapter 4). Moreover, in Chapter 6 of Taylor's *Classical Mechanics*, the variational method (see Chapter 5) is also built upon this technique.

Consider two functions u, v with respect to x , which satisfy the rule of differentiation

$$\frac{d}{dx}(uv) = \left(\frac{d}{dx}u\right)v + u\left(\frac{d}{dx}v\right) \quad (1.46)$$

Which can be simplified to

$$\frac{d}{dx}(uv) = u'v + uv' \quad (1.47)$$

Then it becomes

$$d(uv) = u'v \, dx + uv' \, dx \quad (1.48)$$

Integrate both sides over the interval $[a, b]$, we obtain

$$\int_a^b d(uv) = uv \Big|_a^b = \int_a^b u'v \, dx + \int_a^b uv' \, dx \quad (1.49)$$

Bring any term from the right side of the equation to the left, we can finally get the formula

$$\boxed{\int_a^b uv' \, dx = uv \Big|_a^b - \int_a^b u'v \, dx} \quad (1.50)$$

Sometimes, we may write

$$\int_a^b u \, dv = uv \Big|_a^b - \int_a^b v \, du \quad (1.51)$$

Since $u' \, dx = du$, $v' \, dx = dv$.

When you encounter a product of several functions, first check whether it can be solved using substitution. If that doesn't work, try expanding or rewriting the expression to see if substitution becomes possible. If all else fails, it's time to consider using integration by parts. "*If once is not enough, do it twice.*". That's my guiding principle when dealing with such cases.

Example 1.6 Evaluate the following integral

$$\int x \ln x \, dx$$

Solution 1.6

Let $x = u'$, $\ln x = v$, by integration by parts, we obtain

$$\begin{aligned} \int x \ln x \, dx &= \frac{1}{2}x^2 \ln x - \int \frac{1}{2}x^2 \cdot \frac{1}{x} dx \\ &= \frac{1}{2}x^2 \ln x - \frac{1}{2} \int x \, dx \end{aligned}$$

So the final answer is

$$\int x \ln x \, dx = \frac{1}{4}x^2(2 \ln x - 1) + C$$

1.3.4 Exponential and Logarithmic Integral

In the section on the Fundamental Theorem of Calculus, we have already discussed that integration and differentiation are inverse operations. So, the first one

$$\frac{d}{dx} b^x = b^x \cdot \ln b \Leftrightarrow \int b^x \, dx = \frac{b^x}{\ln b} + C \quad (1.52)$$

The second term is

$$\frac{d}{dx} e^{ax} = ae^{ax} \Leftrightarrow \int e^{ax} \, dx = \frac{1}{a}e^{ax} + C \quad (1.53)$$

The most important one is the integral of $1/x$, which is

$$\frac{d}{dx} \ln x = \frac{1}{x} \Leftrightarrow \int \frac{1}{x} \, dx = \ln x + C \quad (1.54)$$

The equation (1.54) often appears in the integration by substitution. Consider an integral

$$\int \frac{g'(x)}{g(x)} \, dx \quad (1.55)$$

Let $u = g(x)$, then $du = g'(x)dx$, so the integral becomes

$$\int \frac{g'(x)}{g(x)} \, dx = \int \frac{1}{u} \, du = \ln u + C = \ln g(x) + C \quad (1.56)$$

For example,

$$\int \frac{x}{x^2 - 1} dx$$

Let $x^2 - 1 = u$, $du = 2x dx$, so

$$\int \frac{x}{x^2 - 1} dx = \int \frac{1}{2u} du = \frac{1}{2} \ln u + C = \frac{1}{2} \ln(x^2 - 1) + C$$

After introducing the three most fundamental types of integrals, we will now discuss the integration of logarithmic functions, beginning with the natural logarithm. Note that this requires the technique of integration by parts. If you've reached this point and still don't understand how integration by parts works, please turn back to the previous page and review it first.

The integral

$$\int \ln x \, dx \tag{1.57}$$

Let $u = x$, $v = \ln x$, so

$$\int \ln x \, dx = \int u' v \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - x + C \tag{1.58}$$

Next, we will deal with the general logarithmic function, where the base is not e . In such cases, we must use the change of base formula to convert all logarithmic expressions into natural logarithms ($\ln$).

The integral

$$\int \log_a g(x) \, dx = \int \frac{\ln g(x)}{\ln a} \, dx = \frac{1}{\ln a} \int \ln g(x) \, dx \tag{1.59}$$

Let $u = x$, $v = \ln g(x)$. Note that the derivative of v is

$$v' = \frac{1}{g(x)} \cdot g'(x) \tag{1.60}$$

So we apply the integration by parts to $\int \ln g(x) \, dx$, we can obtain

$$\int \ln g(x) \, dx = x \ln g(x) - \int x \cdot \frac{g'(x)}{g(x)} \, dx + C \tag{1.61}$$

So, the final result of the integral (1.59) becomes

$$\int \log_a g(x) \, dx = \frac{x \ln g(x)}{\ln a} - \int \frac{x}{\ln a} \cdot \frac{g'(x)}{g(x)} \, dx + C \tag{1.62}$$

The integral on the right-hand side of the equation depends on the form of $g(x)$.

Example 1.7 Evaluate the following integrals.

1. $\int \left(xe^x + \frac{\ln x}{x} + x2^x \right) dx$
2. $\int e^{-\beta x} \sin(bx) dx$

1. The integral can be decomposed to

$$\int x \cdot e^x dx + \int \frac{\ln x}{x} dx + \int x \cdot 2^x dx$$

We can let $x = u$, $e^x = v'$ of the first term, we obtain

$$\int x \cdot e^x dx = xe^x - \int e^x dx = e^x(x - 1) + C$$

In the second term, let $\ln x = u$, $v' = 1/x \Rightarrow v = \ln x$, we obtain

$$\int \frac{\ln x}{x} dx = (\ln x)^2 - \int \frac{\ln x}{x} dx \Rightarrow 2 \int \frac{\ln x}{x} dx = (\ln x)^2$$

So the final result of the second term is

$$\int \frac{\ln x}{x} dx = \frac{(\ln x)^2}{2}$$

In the final term, let $x = u$, $2^x = v' \Rightarrow v = 2^x / \ln 2$, we get

$$\int x \cdot 2^x dx = \frac{x \cdot 2^x}{\ln 2} - \int \frac{2^x}{\ln 2} dx = \frac{x \cdot 2^x}{\ln 2} - \frac{2^x}{(\ln 2)^2} + C$$

Combine these results together, we can thus get the answer to the problem

$$\int \left(xe^x + \frac{\ln x}{x} + x2^x \right) dx = e^x(x - 1) + \frac{(\ln x)^2}{2} + \frac{x \cdot 2^x}{\ln 2} - \frac{2^x}{(\ln 2)^2} + C$$

2. Let $e^{-\beta x} = u'$, $\sin(bx) = v$, so

$$\int e^{-\beta x} \sin(bx) dx = \frac{-1}{\beta} e^{-\beta x} \sin(bx) + \frac{b}{\beta} \int e^{-\beta x} \cos(bx) dx$$

Note that we can get $\sin(bx)$ when we apply integration by parts again, which is

$$\frac{b}{\beta} \int e^{-\beta x} \cos(bx) dx = -\frac{b}{\beta^2} e^{-\beta x} \cos(bx) - \frac{b^2}{\beta^2} \int e^{-\beta x} \sin(bx) dx$$

So we obtain

$$\begin{aligned}\int e^{-\beta x} \sin(bx) dx &= \frac{-1}{\beta} e^{-\beta x} \sin(bx) - \frac{b}{\beta^2} e^{-\beta x} \cos(bx) - \frac{b^2}{\beta^2} \int e^{-\beta x} \sin(bx) dx \\ \Rightarrow \frac{\beta^2 + b^2}{\beta^2} \int e^{-\beta x} \sin(bx) dx &= \frac{-1}{\beta} e^{-\beta x} \sin(bx) - \frac{b}{\beta^2} e^{-\beta x} \cos(bx)\end{aligned}$$

The final answer is

$$\int e^{-\beta x} \sin(bx) dx = -\frac{\beta}{\beta^2 + b^2} e^{-\beta x} \sin(bx) - \frac{b}{\beta^2 + b^2} e^{-\beta x} \cos(bx) + C$$

1.3.5 Trigonometric Substitution

Trigonometric integrals and trigonometric substitutions are among the most important techniques in integral calculus. In the analysis of physical systems, the appearance of trigonometric functions often implies information about the system's periodicity, trajectories, and underlying symmetries. Thus, understanding a physical system often begins with mastering these types of integrals. We will start with basic trigonometric integrals, which can be viewed as the reverse process of differentiation.

The basic trigonometric differentiations are $\sin$ and $\cos$. Which is

$$\boxed{\begin{cases} \int \sin \theta d\theta = -\cos \theta + C \\ \int \cos \theta d\theta = \sin \theta + C \end{cases}} \quad (1.63)$$

Other trigonometric integrals require certain techniques to evaluate. We begin with those involving $\tan \theta$,

$$\int \tan \theta d\theta = \int \frac{\sin \theta}{\cos \theta} d\theta$$

Let $\cos \theta = u$, $du = -\sin \theta d\theta$, so

$$\int \frac{\sin \theta}{\cos \theta} d\theta = -\int \frac{1}{u} du = -\ln u \Rightarrow \boxed{\int \tan \theta d\theta = -\ln \cos \theta + C} \quad (1.64)$$

And the next one is $\cot \theta$, which is

$$\int \cot \theta d\theta = \int \frac{\cos \theta}{\sin \theta} d\theta$$

Let $\sin \theta = u$, $du = \cos \theta d\theta$, so

$$\int \frac{\cos \theta}{\sin \theta} d\theta = \int \frac{1}{u} du = \ln u \Rightarrow \boxed{\int \cot \theta d\theta = \ln \sin \theta + C} \quad (1.65)$$

The third one, $\sec \theta$, is the most skillful one in the trigonometric integral.

$$\int \sec \theta d\theta = \int \frac{\sec \theta (\sec \theta + \tan \theta)}{\sec \theta + \tan \theta} d\theta$$

Let $\sec \theta + \tan \theta = u$, $du = (\tan \theta \sec \theta + \sec^2 \theta) d\theta$, so the integral can be written as

$$\begin{aligned} \int \frac{\sec \theta (\sec \theta + \tan \theta)}{\sec \theta + \tan \theta} d\theta &= \int \frac{1}{u} du = \ln u \\ &\Rightarrow \boxed{\int \sec \theta d\theta = \ln(\sec \theta + \tan \theta) + C} \end{aligned} \quad (1.66)$$

The final one is $\csc \theta$, we can use the former results to calculate the integral,

$$\begin{aligned} \int \csc \theta d\theta &= \int \frac{1}{\sin \theta} d\theta = \int \frac{\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} d\theta \\ &= \frac{1}{2} \left(\int \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} d\theta + \int \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} d\theta \right) \end{aligned}$$

Note that when $u = \cos(\theta/2)$, $du = -(1/2) \sin(\theta/2)$, so the coefficient $1/2$ disappeared. By the results (1.64) and (1.65), we can obtain

$$\boxed{\int \csc \theta d\theta = \int \frac{1}{\sin \theta} d\theta = \ln \tan \left(\frac{\theta}{2} \right) + C} \quad (1.67)$$

Trigonometric integrals do not have a unique form. You can express the answer using various trigonometric identities. Likewise, there is no single correct method for evaluating these integrals, as long as you choose an appropriate substitution or transformation, even involving special functions, your approach is valid. The method provided here is just one reference path. So if you see different solutions or final expressions elsewhere, don't panic!

After these trigonometric integrals, we can continue to deal with the trigonometric substitution. There are six forms of integrals, and every type of integral has its corresponding substitution method.

$$\left\{ \begin{array}{l} \int \frac{b}{\sqrt{a^2 - x^2}} dx \Rightarrow \text{Let } x = a \sin \theta \\ \int \frac{b}{\sqrt{a^2 + x^2}} dx \Rightarrow \text{Let } x = a \tan \theta \\ \int \frac{b}{\sqrt{x^2 - a^2}} dx \Rightarrow \text{Let } x = a \sec \theta \end{array} \right. \quad \left\{ \begin{array}{l} \int \frac{b}{a^2 - x^2} dx \Rightarrow \text{Let } x = a \sin \theta \\ \int \frac{b}{a^2 + x^2} dx \Rightarrow \text{Let } x = a \tan \theta \\ \int \frac{b}{x^2 - a^2} dx \Rightarrow \text{Let } x = a \sec \theta \end{array} \right. \quad (1.68)$$

1. The integral

$$\int \frac{b}{\sqrt{a^2 - x^2}} dx$$

Let $x = a \sin \theta$, $dx = a \cos \theta d\theta$, so the integral becomes

$$\int \frac{b}{\sqrt{a^2 - x^2}} dx = b \int \frac{a \cos \theta d\theta}{a \cos \theta} = b\theta + C$$

We're not done yet, recall that our original variable was x , so we need to express the result of the integral in terms of x . Since we made the substitution $x = a \sin \theta$, we can invert it using the inverse trigonometric function

$$x = a \sin \theta \quad \Rightarrow \quad \theta = \sin^{-1} \left(\frac{x}{a} \right) = \arcsin \left(\frac{x}{a} \right) \quad (1.69)$$

Substituting this back into our earlier result, we obtain the final answer

$$\int \frac{b}{\sqrt{a^2 - x^2}} dx = \frac{b}{a} \theta + C = b \arcsin \left(\frac{x}{a} \right) + C \quad (1.70)$$

2. The integral

$$\int \frac{b}{\sqrt{a^2 + x^2}} dx$$

Let $x = a \tan \theta$, $dx = a \sec^2 \theta d\theta$, so the integral becomes

$$\int \frac{b}{\sqrt{a^2 + x^2}} dx = b \int \frac{a \sec^2 \theta d\theta}{a \sec \theta} = b \int \sec \theta d\theta = b \ln(\sec \theta + \tan \theta) + C$$

Because $x = a \tan \theta$, so $\tan \theta = x/a$; $\sec \theta = \sqrt{a^2 + x^2}/a$, so we substitute this back into our earlier result, we obtain the final answer

$$\int \frac{b}{\sqrt{a^2 + x^2}} dx = b \ln(\sec \theta + \tan \theta) + C = b \ln(x + \sqrt{a^2 + x^2}) + C \quad (1.71)$$

The term $-b \ln a$ is absorbed into the constant of integration.

3. The integral

$$\int \frac{b}{\sqrt{x^2 - a^2}} dx$$

Remember the identity $\sec \theta - 1 = \tan \theta$, so we let $x = a \sec \theta$, $dx = a \sec \theta \tan \theta d\theta$, the integral becomes

$$\int \frac{b}{\sqrt{x^2 - a^2}} dx = b \int \frac{a \sec \theta \tan \theta d\theta}{a \tan \theta} = b \int \sec \theta d\theta = b \ln(\sec \theta + \tan \theta) + C$$

Because $x = a \sec \theta$, so $\sec \theta = x/a$; $\tan \theta = \sqrt{x^2 - a^2}/a$, so the final answer is familiar with the last one

$$\int \frac{b}{\sqrt{x^2 - a^2}} dx = b \ln(\sec \theta + \tan \theta) + C = b \ln(x + \sqrt{x^2 - a^2}) + C \quad (1.72)$$

4. The integral

$$\int \frac{b}{a^2 - x^2} dx$$

Let $x = a \sin \theta$, $dx = a \cos \theta d\theta$, so

$$\int \frac{b}{a^2 - x^2} dx = b \int \frac{a \cos \theta d\theta}{a^2 \cos^2 \theta} = \frac{b}{a} \int \sec \theta d\theta = \frac{b}{a} \ln(\sec \theta + \tan \theta) + C$$

Because $x = a \sin \theta$, so $\sec \theta = a/\sqrt{a^2 - x^2}$; $\tan \theta = x/\sqrt{a^2 - x^2}$, we can obtain

$$\int \frac{b}{a^2 - x^2} dx = \frac{b}{a} \ln(\sec \theta + \tan \theta) + C = \frac{b}{a} \ln \left(\frac{a + x}{\sqrt{a^2 - x^2}} \right) + C \quad (1.73)$$

5. The integral

$$\int \frac{b}{a^2 + x^2} dx$$

Let $x = a \tan \theta$, $dx = a \sec^2 \theta d\theta$, so

$$\int \frac{b}{a^2 + x^2} dx = b \int \frac{a \sec^2 \theta d\theta}{a^2 \sec^2 \theta} = \frac{b}{a} \int d\theta = \frac{b}{a} \theta + C$$

Where $\theta = \arctan(x/a)$, and we can obtain the final answer

$$\int \frac{b}{a^2 + x^2} dx = \frac{b}{a} \theta + C = \frac{b}{a} \arctan \left(\frac{x}{a} \right) + C \quad (1.74)$$

6. We have finally reached the final stage of trigonometric substitution. The integral

$$\int \frac{b}{x^2 - a^2} dx$$

Let $x = a \sec \theta$, $dx = a \tan \theta \sec \theta d\theta$, so

$$\int \frac{b}{x^2 - a^2} dx = b \int \frac{a \sec \theta \tan \theta d\theta}{a^2 \tan \theta} = \frac{b}{a} \int \sec \theta d\theta = \frac{b}{a} \ln(\sec \theta + \tan \theta) + C$$

Because $\sec \theta = x/a$, we get $\tan \theta = \sqrt{x^2 - a^2}/a$. The final answer

$$\int \frac{b}{x^2 - a^2} dx = \frac{b}{a} \ln(\sec \theta + \tan \theta) + C = \frac{b}{a} \ln(x + \sqrt{x^2 - a^2}) + C \quad (1.75)$$

The six forms above serve as standard templates. More complex expressions are often transformations or combinations of these cases. In later examples, you are encouraged to identify how they relate to these core types.

Example 1.8 Evaluate the following integrals.

1. $\int \frac{x^2}{(a^2 + x^2)^{3/2}} dx$
2. $\int \frac{1}{x^2 \sqrt{x^2 - a^2}} dx$

Solution 1.7

1. The integral

$$\int \frac{x^2}{(a^2 + x^2)^{3/2}} dx$$

Let $x = a \tan \theta$, $dx = a \sec^2 \theta d\theta$, so

$$\int \frac{x^2}{(a^2 + x^2)^{3/2}} dx = \int \frac{a^2 \tan^2 \theta \cdot a \sec^2 \theta d\theta}{a^3 \sec^3 \theta} = \int \frac{\tan^2 \theta d\theta}{\sec \theta} = \int \frac{\sin^2 \theta}{\cos \theta} d\theta$$

The identity $\sin^2 \theta = 1 - \cos^2 \theta$, we substitute this into the integral

$$\int \frac{\sin^2 \theta}{\cos \theta} d\theta = \int \frac{1 - \cos^2 \theta}{\cos \theta} d\theta = \int \sec \theta d\theta - \int \cos \theta d\theta = \ln(\sec \theta + \tan \theta) - \sin \theta + C$$

Because $x = a \tan \theta$, we can get $\sec \theta = \sqrt{x^2 + a^2}/a$; $\sin \theta = x/\sqrt{x^2 + a^2}$. We obtain

$$\int \frac{x^2}{(a^2 + x^2)^{3/2}} dx = \ln(\sqrt{x^2 + a^2} + x) - \frac{x}{\sqrt{x^2 + a^2}} + C$$

2. The integral

$$\int \frac{1}{x^2 \sqrt{x^2 - a^2}} dx$$

Let $x = a \sec \theta$, $dx = a \tan \theta \sec \theta d\theta$, so

$$\int \frac{1}{x^2 \sqrt{x^2 - a^2}} dx = \int \frac{a \tan \theta \sec \theta d\theta}{a^2 \sec^2 \theta \cdot a \tan \theta} = \frac{1}{a^2} \int \frac{1}{\sec \theta} d\theta = \frac{1}{a^2} \int \cos \theta d\theta = \frac{1}{a^2} \sin \theta + C$$

By the equation $x = a \sec \theta$ we get $\sin \theta = \sqrt{x^2 - a^2}/x$. So, the final answer is

$$\int \frac{1}{x^2 \sqrt{x^2 - a^2}} dx = \frac{1}{a^2} \sin \theta + C = \frac{1}{a^2} \cdot \frac{\sqrt{x^2 - a^2}}{x} + C$$

1.3.6 Hyperbolic Functions

Back in high school, we learned about the standard form of a hyperbola

$$x^2 - y^2 = 1 \quad (1.76)$$

In calculus, we extend this equation to define a new class of functions called the **hyperbolic functions**. If we look closely at the structure of this equation, it resembles that of the unit circle:

$$x^2 + y^2 = 1$$

The unit circle can be parametrized using angles and radii to describe points on the circle. This leads to the identity:

$$\cos^2 \theta + \sin^2 \theta = 1$$

Here, the angle θ is the key parameter in the circular parametrization. Now, we ask a natural question: Is there a similar kind of function and parameter that can describe points on the hyperbola? This motivates the definition of the hyperbolic functions, which satisfy the identity

$$\cosh^2 a - \sinh^2 a = 1 \quad (1.77)$$

This mirrors the structure of the circular identity and provides a way to parametrize the hyperbola. However, unlike the circular case where the parameter is an angle, the parameter a here is defined differently: it is **twice the area** enclosed by the ray from the origin, the hyperbola, and the x -axis, as illustrated in the figure below.

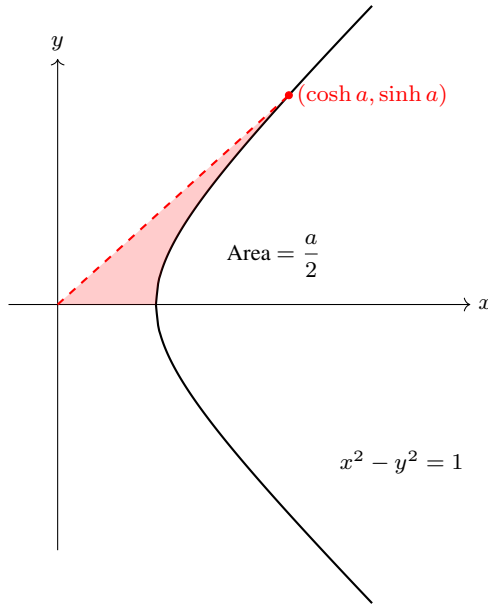


Fig. 1.1 The illustration of the hyperbolic function and its definition of parameter

Interestingly, the hyperbolic functions $\sinh x$ and $\cosh x$ were not originally defined using exponentials. They were first introduced to describe geometric properties of the hyperbola, analogous to how trigonometric functions describe the circle. Later, it was discovered that these functions could be neatly expressed in terms of exponentials

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2} \quad (1.78)$$

This remarkable identity unites the geometric and algebraic aspects of the hyperbolic functions. Then, we can define other hyperbolic functions with similar forms as the trigonometric functions, that is

$$\left\{ \begin{array}{l} \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \\ \coth x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \\ \operatorname{csch} x = \frac{1}{\sinh x} = \frac{1}{e^x - e^{-x}} \\ \operatorname{sech} x = \frac{1}{\cosh x} = \frac{1}{e^x + e^{-x}} \end{array} \right. \quad (1.79)$$

And the identities

$$\begin{cases} \tanh^2 x = 1 - \operatorname{sech}^2 x \\ \coth^2 x = 1 + \operatorname{csch}^2 x \end{cases} \quad (1.80)$$

At the same time, we can verify the differential relationships of the hyperbolic functions using their exponential definitions. Let's begin with the most fundamental ones

$$\begin{cases} \frac{d}{dx} \sinh x = \cosh x \\ \frac{d}{dx} \cosh x = \sinh x \end{cases} \quad (1.81)$$

Next, we consider the derivatives of the remaining hyperbolic functions, starting with $\tanh x$. We know that

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

To differentiate this expression, we apply the quotient rule

$$\begin{aligned} \frac{d}{dx} \tanh x &= \frac{d}{dx} \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) \\ &= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2} \\ &= 1 - \tanh^2 x = \operatorname{sech}^2 x \end{aligned} \quad (1.82)$$

Using the same method, we can derive the derivatives of the other hyperbolic functions as follows:

$$\begin{cases} \frac{d}{dx} \coth x = -\operatorname{csch}^2 x \\ \frac{d}{dx} \operatorname{csch} x = -\operatorname{csch} x \coth x \\ \frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \tanh x \end{cases} \quad (1.83)$$

However, the most important aspect in this section is to explore the inverse relationships among these functions. In most integration problems, we don't necessarily need to use hyperbolic substitution, in fact, trigonometric substitution is often more than sufficient.

But in certain physical problems, such as the classic catenary problem, where hyperbolic functions were first discovered, the result of the integration may initially appear messy. Yet by expressing the solution in terms of hyperbolic functions, the form becomes clean and elegant, and the overall behavior of the function is immediately clear at a glance.

The inverse function of $\sinh x$ is defined as $\operatorname{arsinh} x$. Note! it is not "arc" but "ar", where "ar" means area. So, we have

$$y = \sinh x = \frac{e^x - e^{-x}}{2} \Leftrightarrow x = \sinh y = \frac{e^y - e^{-y}}{2} \quad (1.84)$$

So, by (1.84), we can construct a quadratic equation in one variable

$$\begin{aligned} \frac{e^y - e^{-y}}{2} = x &\Rightarrow e^{2y} - 1 = 2xe^y \\ &\Rightarrow (e^y)^2 - 2x(e^y) - 1 = 0 \\ &\Rightarrow e^y = \frac{2x + \sqrt{4x^2 + 4}}{2} = x + \sqrt{x^2 + 1} \end{aligned}$$

Note that we should take the positive term of the quadratic equation because $e^y > 0$. So the inverse function

$$\operatorname{arsinh} x = \ln \left(x + \sqrt{x^2 + 1} \right) \quad (1.85)$$

Using the same method, we can derive the inverse function

$$\operatorname{arcosh} x = \ln \left(x + \sqrt{x^2 - 1} \right) \quad (1.86)$$

Do these two functions look familiar to you? That's right, they already appeared as solutions to the trigonometric substitution integrals (1.71) and (1.72). In fact, these two integrals can be re-expressed in terms of inverse hyperbolic functions

$$\boxed{\begin{cases} \int \frac{b}{\sqrt{x^2 + a^2}} dx = b \operatorname{arsinh}(ax) + C \\ \int \frac{b}{\sqrt{x^2 - a^2}} dx = b \operatorname{arcosh}(ax) + C \end{cases}} \quad (1.87)$$

It's amazing that we can unify these two strikingly similar integrals using hyperbolic functions, it's so cool! In addition, we can get the same results by letting $x = a \cosh u$ or $x = a \sinh u$. I'll show you this in the case $\operatorname{artanh} x$.

Now, let's turn to the last and most commonly encountered inverse hyperbolic function: $\operatorname{artanh} x$.

$$y = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \Leftrightarrow x = \tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}} \quad (1.88)$$

Simplify the expression and rewrite the inverse function in terms of x .

$$\begin{aligned} \frac{e^y - e^{-y}}{e^y + e^{-y}} = x &\Rightarrow e^y - e^{-y} = x(e^y + e^{-y}) \Rightarrow (1 - x)e^y = (1 + x)e^{-y} \\ &\Rightarrow e^{2y} = \frac{1 + x}{1 - x} \Rightarrow y = \frac{1}{2} \ln \left(\frac{1 + x}{1 - x} \right) \end{aligned}$$

So the inverse function

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \quad (1.89)$$

This result is, in fact, equivalent to equation (1.73). We will verify this in two parts. First, let's rewrite the form of equation (1.73). That is

$$\begin{aligned} \frac{b}{a} \ln \left(\frac{a+x}{\sqrt{a^2-x^2}} \right) &= \frac{b}{a} \left[\ln(a+x) - \frac{1}{2} \ln(a-x) - \frac{1}{2} \ln(a+x) \right] \\ &= \frac{b}{a} \cdot \frac{1}{2} \ln \left(\frac{a+x}{a-x} \right) \\ &= \frac{b}{a} \cdot \frac{1}{2} \ln \left(\frac{1+\frac{x}{a}}{1-\frac{x}{a}} \right) = \frac{b}{a} \operatorname{artanh} \left(\frac{x}{a} \right) \end{aligned}$$

The second method is the substitution, consider the integral

$$\int \frac{b}{a^2-x^2} dx$$

Let $x = a \tanh u$, $dx = a \operatorname{sech}^2 u \, du$. We can rewrite the integral

$$\int \frac{b}{a^2-x^2} dx = b \int \frac{a \operatorname{sech}^2 u \, du}{a^2 \operatorname{sech}^2 u} = \frac{b}{a} \int du = \frac{b}{a} u + C$$

By the equation $x = a \tanh u$, we obtain

$$u = \operatorname{artanh} \left(\frac{x}{a} \right)$$

So, we can verify this important result, which is

$$\boxed{\int \frac{b}{a^2-x^2} dx = \frac{b}{a} \operatorname{artanh} \left(\frac{x}{a} \right) + C} \quad (1.90)$$

For example, we have an integral

$$\int \frac{dx}{1-x^2/\omega^2} = \omega^2 \int \frac{dx}{\omega^2-x^2}$$

Thus, it returns to the familiar integral form given in equation (1.73). By (1.90), we obtain the answer to the integral

$$\int \frac{dx}{1-x^2/\omega^2} = \omega \operatorname{artanh} \left(\frac{x}{\omega} \right) + C$$

1.3.7 The Method of Partial Fraction

The form $\frac{N(x)}{D(x)}$, where $\deg(N) > \deg(D)$, can all be denoted by

$$\frac{N(x)}{D(x)} = \left(\sum_i C_i x^i \right) + \sum_i \frac{A_i}{(px + q)^i} + \sum_i \frac{B_i x + k_i}{(ax^2 + bx + c)^i} \quad (1.91)$$

Where $N(x)$, $D(x)$ are polynomials, and $A_j, C_j \in \mathbb{R}$. While we will not prove here why such a decomposition is always possible, this result comes from a fundamental theorem in algebra. Our focus will be on how to apply it effectively to compute integrals. If both the numerator and the denominator of a function are polynomials, we can decompose it into several simpler fractions and integrate them individually. This makes the original problem much easier to solve, this technique is known as the **method of partial fractions**.

For example, the fraction can be decomposed to partial fractions as shown below.

$$\frac{3x + 4}{(x - 1)(x^2 + 2x + 9)} = \frac{A}{x - 1} + \frac{Bx + C}{x^2 + 2x + 9}$$

Where $A, B, C \in \mathbb{R}$. After decomposing the expression into partial fractions, the most important step is to determine the constants in each term. Of course, you can expand both sides and compare coefficients, but here, I will use the *limit method* to eliminate unnecessary terms, making it easier to identify the constants directly. Consider a fraction

$$\frac{N(x)}{D(x)} = \frac{A}{ax + b} + \frac{B}{x + c} + \frac{C}{(x + c)^2} + \frac{dx + E}{dx^2 + ex + f} \quad (1.92)$$

Where $D(x) = (ax + b)(x + c)^2(dx^2 + ex + f)$. To solve for A , we multiply both sides of the equation by $ax + b$, that is

$$(ax + b) \frac{N(x)}{D(x)} = \frac{N(x)}{(x + c)^2(dx^2 + ex + f)} = A + (ax + b) \left[\frac{B}{x + c} + \frac{C}{(x + c)^2} + \frac{dx + E}{dx^2 + ex + f} \right]$$

The limit

$$\lim_{x \rightarrow -\frac{b}{a}} A + \lim_{x \rightarrow -\frac{b}{a}} (ax + b) \left[\frac{B}{x + c} + \frac{C}{(x + c)^2} + \frac{dx + E}{dx^2 + ex + f} \right] = A$$

The limit of the left-hand side is totally the same as the right-hand side, so we can obtain the constant A , which is

$$\lim_{x \rightarrow -\frac{b}{a}} \frac{N(x)}{(x + c)^2(dx^2 + ex + f)} = A \quad (1.93)$$

However, this method cannot be applied to solve for b because taking the limit would cause the term $C/(x + c)$ on the right-hand side to diverge. Instead, we must multiply

both sides by $(x + c)^2$ before taking the limit in order to isolate C . Following the same logic as before, we can verify that

$$\lim_{x \rightarrow -c} \frac{N(x)}{(ax + b)(dx^2 + ex + f)} = C \quad (1.94)$$

Suppose the factorization of $dx^2 + ex + f$ is $(\lambda x - \lambda' i)(\omega x - \omega' i)$, and we select one of the factors to take the limit. We apply the same method, we can get

$$\lim_{x \rightarrow \frac{\lambda' i}{\lambda}} \frac{N(x)}{(ax + b)(x + c)^2} = \left(\frac{D\lambda' i}{\lambda} + E \right) \quad (1.95)$$

If there are any remaining unknown constants, such as B , we can group them with the known constants and solve a system of equations. This allows us to determine all the constants involved.

Example 1.9 Evaluate the following integrals.

1. $\int \frac{x}{(x-1)(x-2)^2} dx$
2. $\int \frac{1}{x^3 + 1} dx$

Solution 1.8

1. First, apply the method of partial fraction

$$\int \frac{x}{(x-1)(x-2)^2} dx = \int \frac{A}{x-1} dx + \int \frac{B}{x-2} dx + \int \frac{C}{(x-2)^2} dx$$

Then we use the limit method to get A , C , that is

$$\begin{cases} A = \lim_{x \rightarrow 1} \frac{x}{(x-2)^2} = 1 \\ C = \lim_{x \rightarrow 2} \frac{x}{x-1} = 2 \end{cases}$$

By $A = 1$, $C = 2$, we can obtain the remaining constant $B = -1$. So we can start to calculate the integrals.

$$\int \frac{x}{(x-1)(x-2)^2} dx = \int \frac{1}{x-1} dx + \int \frac{-1}{x-2} dx + \int \frac{2}{(x-2)^2} dx$$

The final answer is

$$\int \frac{x}{(x-1)(x-2)^2} dx = \ln \left(\frac{x-1}{x-2} \right) + \frac{2}{x-2} + C$$

2. This problem is a classic example of the partial fraction. It looks simple, but it is actually complicated. Although this problem is difficult, it is also beautiful. It brings together all the integration techniques you've learned so far, which is why it is an important one.

We factorize the fraction to be

$$\int \frac{1}{x^3 + 1} dx = \int \frac{1}{(x+1)(x^2 - x + 1)} dx = \int \frac{A}{x+1} dx + \int \frac{Bx + C}{x^2 - x + 1} dx$$

Then we use the limit method to get A , C , that is

$$\begin{cases} A = \lim_{x \rightarrow -1} \frac{1}{x^2 - x + 1} = \frac{1}{3} \\ C = \lim_{x \rightarrow 0} \frac{1}{(x+1)(x^2 - x + 1)} - A = \frac{2}{3} \end{cases}$$

So the constant $B = -1/3$. Then, the integral becomes

$$\int \frac{A}{x+1} dx + \int \frac{Bx + C}{x^2 - x + 1} dx = \frac{1}{3} \int \frac{1}{x+1} dx - \frac{1}{3} \int \frac{x-2}{x^2 - x + 1} dx$$

We can start to calculate the integral

$$\begin{aligned} \frac{1}{3} \int \frac{1}{x+1} dx - \frac{1}{3} \int \frac{x-2}{x^2 - x + 1} dx &= \frac{1}{3} \ln(x+1) - \frac{1}{6} \int \frac{2x-4}{x^2 - x + 1} dx \\ &= \frac{1}{3} \ln(x+1) - \frac{1}{6} \int \frac{(2x-1) - 3}{x^2 - x + 1} dx \\ &= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2 - x + 1) - \frac{1}{6} \int \frac{-3}{x^2 - x + 1} dx \\ &= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2 - x + 1) + \frac{1}{2} \int \frac{1}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx \end{aligned}$$

Let $x - 1/2 = \sqrt{3}/2 \tan \theta$, $dx = \sqrt{3}/2 \sec^2 \theta d\theta$, and the remaining integral becomes

$$\int \frac{1}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx = \int \frac{\frac{\sqrt{3}}{2} \sec^2 \theta d\theta}{\frac{3}{4} \sec^2 \theta} = \frac{2}{\sqrt{3}} \theta + C$$

Because $x - 1/2 = \sqrt{3}/2 \tan \theta$, then

$$\theta = \arctan\left(\frac{2}{\sqrt{3}}x - \frac{1}{\sqrt{3}}\right)$$

So, the final answer is

$$\int \frac{1}{x^3 + 1} dx = \frac{1}{3} \ln(x + 1) - \frac{1}{6} \ln(x^2 - x + 1) + \frac{1}{\sqrt{3}} \arctan\left(\frac{2}{\sqrt{3}}x - \frac{1}{\sqrt{3}}\right) + C$$

1.4 Taylor Expansion

1.4.1 General Form of Taylor Expansion

Taylor expansion is one of the most important mathematical tools in physics. It allows us to represent all types of functional relationships as an infinite series of polynomials. Moreover, this method provides an explanation for linear approximation behavior in small neighborhoods. Many modern mathematical techniques have been developed based on this fundamental property.

As introduced in *Classical Dynamics* by Thornton and Marion, we define a definite integral as

$$I := \int_{x_0}^{x_0+h} f'(x) dx = f(x_0 + h) - f(x_0) \quad (1.96)$$

let $x = x_0 + h - t$, where t is a variable, so $dx = -dt$ (This is the integration by substitution). We rewrite the integral as

$$I = \int_h^0 f'(x_0 + h - t) (-dt) = \int_0^h f'(x_0 + h - t) dt \quad (1.97)$$

We need to apply the integration by parts to decompose the integral

$$\int_0^h f'(x_0 + h - t) dt = t \cdot f'(x_0 + h - t) \Big|_0^h + \int_0^h t \cdot f''(x_0 + h - t) dt \quad (1.98)$$

Note that we need to apply the chain rule to get the derivative of $f'(x_0 + h - t)$, which is

$$\frac{d}{dt} f'(x_0 + h - t) = f''(x_0 + h - t) \cdot (-1)$$

Therefore, the integration by parts in (1.98) results in a positive term. We will continue to use integration by parts to break down the integral expression.

$$\begin{aligned}
t \cdot f'(x_0 + h - t) \Big|_0^h + \int_0^h t \cdot f''(x_0 + h - t) dt &= hf'(x_0) + \frac{1}{2}t^2 f''(x_0 + h - t) \Big|_0^h + \dots \\
&= hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots + \frac{h^n}{n!} f^{(n)}(x_0)
\end{aligned}$$

Note that the notation $f^{(n)}(x)$ represents the n th derivative of the function. So, the original integral (1.96)

$$\begin{aligned}
I &= f(x_0 + h) - f(x_0) = hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots + \frac{h^n}{n!} f^{(n)}(x_0) \\
\Rightarrow f(x_0 + h) &= f(x_0) + hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots + \frac{h^n}{n!} f^{(n)}(x_0)
\end{aligned}$$

Here, x_0 represents the expansion point, and h denotes a small displacement from that point. This formulation makes it easier to introduce the concept of limits and the definition of derivatives (by considering $h \rightarrow 0$). Such an expansion is a **local approximation**, meaning that it describes the behavior of the function near x_0 , including its value, slope, and curvature. As more terms are included, the approximation becomes more accurate, and the closer x is to x_0 , the smaller the error.

Let $x_0 = 0$, $h = x - 0$, then I describes the behavior of the function near the point $x = 0$, it's called the **Maclaurin Expansion**.

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots \quad (1.99)$$

We can also write this as a series; it's called the Maclaurin series.

$$f(x) = \sum_i \frac{x^i f^{(i)}(0)}{i!} \quad (1.100)$$

In general, we can let $h = x - x_0$, then I describes the behavior of the function near the point x_0 , where $x_0 \in f$, it's called the **Taylor Expansion**. It's denoted by

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{(x - x_0)^2}{2!} f''(x_0) + \dots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + \dots \quad (1.101)$$

We can also write this as a series; it's called the Taylor series.

$$f(x) = \sum_i \frac{(x - x_0)^i f^{(i)}(x_0)}{i!} \quad (1.102)$$

There are three important expansions,

$$\begin{cases} \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \\ \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\ e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \end{cases} \quad (1.103)$$

In addition, we can expand any differentiable, continuous, and smooth function $f(x)$, including functions such as the natural logarithm. For example,

$$\begin{aligned} \ln(x_0 + (x - x_0)) &= \ln(x_0) + (x - x_0) \frac{d}{dx} \ln(x) \Big|_{x=x_0} + \frac{(x - x_0)^2}{2!} \frac{d^2 \ln(x)}{dx^2} \Big|_{x=x_0} + \cdots \\ &= \ln(x_0) + \frac{x - x_0}{x_0} - \frac{(x - x_0)^2}{2x_0^2} + \frac{(x - x_0)^3}{3x_0^3} + \cdots \end{aligned}$$

Note that $|x - x_0| < x_0$ since $\ln x, x > 0$. So, the Taylor series of $\ln(x)$ can be denoted by

$$\ln(x) = \ln(x_0) + \sum_i (-1)^{i+1} \frac{(x - x_0)^i}{i \cdot x_0^i} \quad (1.104)$$

Similarly, when $x_0 = 1$, $h = -x$, the function $\ln(1 - x)$ can be expanded into

$$\ln(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots = -\sum_i \frac{x^i}{i} \quad (1.105)$$

1.4.2 Linear Approximation

Taylor expansion allows us to approximate a function locally based on the choice of an expansion point x_0 . This property enables us to describe how the function behaves under small variations in the input variable. If we take $x_0 = x$, $h = \Delta x$, which means that in a more general sense, the variation of a function can be described according to its domain. That is

$$f(x_0 + h) = f(x + \Delta x) = f(x) + \Delta x f'(x) + \frac{(\Delta x)^2}{2!} f''(x) + \cdots \quad (1.106)$$

The function increment Δf is defined as

$$\Delta f = f(x + \Delta x) - f(x) = \Delta x f'(x) + \frac{(\Delta x)^2}{2!} f''(x) + \cdots \quad (1.107)$$

Note that if we take $\|\Delta\| \rightarrow 0$, in the case where two points are very close to each other, the behavior of the function resembles motion along the tangent line. This is why we call it a **Linear Approximation**. That is, the ratio of change $\Delta y / \Delta x$ remains nearly constant,

which corresponds to the slope you learned in high school. Since this approximates the tangent, the Taylor expansion only needs to retain the first-order derivative term (i.e., the slope of the tangent), and the higher-order terms can be neglected because the variation becomes extremely small. So, it becomes

$$\Delta f|_{\|\Delta\| \rightarrow 0} := df = f(x + dx) - f(x) = f'(x) dx + O((dx)^2) \quad (1.108)$$

We keep only the first-order term and ignore all higher-order terms, which are collectively denoted as $\mathcal{O}((dx)^2)$. So the linear approximation is

$$\boxed{df \approx f'(x) \cdot dx} \quad (1.109)$$

We can also generalize the Taylor expansion to n dimensions, which can be denoted by

$$f(\mathbf{x} + \mathbf{a}) = f(\mathbf{x}) + \sum_i \frac{\partial f}{\partial x_i} a_i + \frac{1}{2!} \sum_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} a_i a_j + \frac{1}{3!} \sum_{ijk} \frac{\partial^3 f}{\partial x_i \partial x_j \partial x_k} a_i a_j a_k + \cdots$$

It can be written in a general form

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \sum_{|\alpha|=1} \frac{1}{\alpha!} \partial^\alpha f(\mathbf{x}_0) \mathbf{h}^\alpha + R_n \quad (1.110)$$

Where $\mathbf{x}_0 = (x_1, x_2, \dots, x_n)$, $\mathbf{h} = (h_1, \dots, h_n)$, $|\alpha| = \alpha_1 + \cdots + \alpha_n$, and R_n is the remainder term of the expansion. The term $\partial^\alpha f$ is defined as

$$\partial^\alpha f := \frac{\partial^{|\alpha|} f}{\prod_i \partial^{\alpha_i} x_i}$$

In this book, we will not delve deeply into the Taylor expansion of multivariable functions, as it involves extensive mathematical background that cannot be fully covered at once. Those who are interested may refer to real analysis and advanced calculus for a more rigorous treatment. Multivariable functions will be introduced on the next page, and the Taylor expansion for multivariable cases will appear later in the context of total differentials.

1.5 Partial Differentiation

1.5.1 Functions of Several Variables

In real-world physics problems, very few quantities depend on a single variable. For instance, the temperature in a room varies with both position and time: $T(x, y, z, t)$. Similarly, the potential energy of a charged particle depends on its spatial coordinates: $V(x, y, z)$. These are all multivariable functions. To analyze these systems mathematically, we need to generalize our understanding of functions from a single variable to

multiple variables.

We start with two variables, there's a function $z = f(x, y)$. The domain has extended from the one-dimensional real line to the two-dimensional real plane, meaning that for every input from the domain, the function f maps it to a unique real number. The mapping relationship is illustrated as follows.

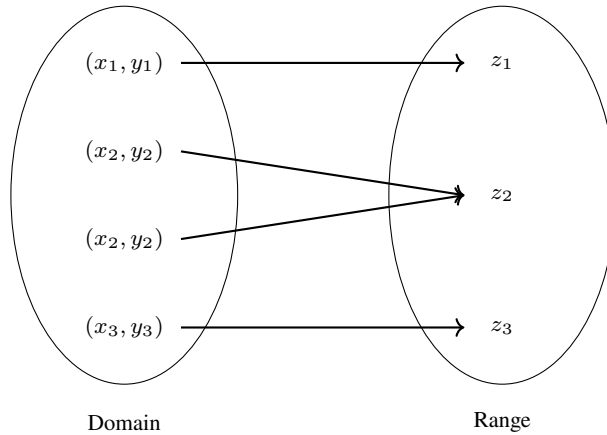


Fig. 1.2 The correspondence between the domain and the range of a multivariable function

In mathematics and physics, a function of two variables is typically defined as a mapping from $\mathbb{R}^2$ to $\mathbb{R}$, that is, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. In other words, each input pair (x, y) is assigned a unique real number $z = f(x, y)$. Geometrically, this defines a surface in three-dimensional space, determined by the coordinates (x, y, z) . When the function value is held constant as $f(x, y) = c$, the resulting set of points forms a *level curve* or *contour line*, which represents the intersection between the surface and the horizontal plane $z = c$. These contour lines are widely used to illustrate scalar fields in two dimensions.

Extending this idea, a function of three variables is a mapping $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, written as $\Phi = f(x, y, z)$. This defines a scalar field in three-dimensional space, where each point (x, y, z) is associated with a real-valued quantity Φ . Since we cannot visualize four-dimensional graphs, such functions are usually represented through *level surfaces*, that is, surfaces of constant function value $\Phi = c$. These level surfaces describe regions in space where the function maintains the same value and are frequently used in physics to represent quantities such as temperature, pressure, potential, and density fields.

More generally, we can extend the concept of functions to n -dimensional space. Suppose we have a function of n variables, $f(x_1, x_2, \dots, x_n)$. In physics, when the number of variables exceeds three, it is common to denote the input variables using indexed notation such as x_i for $i = 1, 2, \dots, n$. This notation will become especially important in later physical applications. Such a function is formally defined as a mapping

$$f : \mathbb{R}^n \rightarrow \mathbb{R},$$

which assigns a real number to each point in n -dimensional space. Due to the limitation of human visualization, it is no longer possible to represent these functions geometrically in a simple way when $n > 3$. Nevertheless, these high-dimensional functions play a crucial role in various branches of physics, particularly in the calculus of variations discussed in Chapter 5, where functionals often depend on multiple variables and their behavior in high-dimensional domains.

1.5.2 Partial Derivatives

In single-variable calculus, we define the derivative and differential using the limit of a small change in a single variable:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

As previously mentioned, in physics, we often need multivariable functions to accurately and rigorously describe physical phenomena. Therefore, we must find a way to generalize the idea of differentiation to three, four, or even n dimensions. This leads us to the concept of **partial derivatives**, which describe the rate of change of a function in the direction of a specific variable.

For a two-variable function $f(x, y)$, the partial derivatives are defined as:

$$\begin{cases} \frac{\partial f}{\partial x} = f_x(x, y) := \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \\ \frac{\partial f}{\partial y} = f_y(x, y) := \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h} \end{cases} \quad (1.111)$$

This can be interpreted geometrically. In three-dimensional space, taking the partial derivative with respect to x means that we fix the value of y , and study the behavior of f as x varies, yielding a result analogous to the slope of the tangent in the x -direction. This process isolates the rate of change along a single variable, treating all others as constants. The same logic applies for the partial derivative with respect to y . Additionally, when performing partial differentiation multiple times with respect to different variables, we define:

$$\begin{cases} f_{xy}(x, y) = \frac{\partial^2 f}{\partial x \partial y} \\ f_{yx}(x, y) = \frac{\partial^2 f}{\partial y \partial x} \end{cases} \quad (1.112)$$

For a function with three variables, say $f(x, y, z)$, the third-order mixed partial derivative is written as:

$$f_{xyz}(x, y, z) = \frac{\partial^3 f}{\partial x \partial y \partial z} \quad (1.113)$$

More generally, for an n -variable function $f(x_1, x_2, \dots, x_n)$, the partial derivative with respect to a specific variable x_i is defined as:

$$\frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h} \quad (1.114)$$

To illustrate the computation, consider the function $f(x, y) = xy^2 + y^5 + x^4y + x^6$. If we take the partial derivative with respect to x , we treat y as a constant and differentiate only with respect to x . This is an important idea: when computing a partial derivative, all other variables are treated as constants. Thus,

$$f_x = \frac{\partial f}{\partial x} = y^2 + 4x^3y + 6x^5$$

Note that the term y^5 is eliminated because it is treated as a constant during differentiation with respect to x . Similarly, the partial derivative with respect to y is

$$f_y = \frac{\partial f}{\partial y} = 2xy + 5y^4 + x^4$$

This procedure can be extended to functions of more variables. This is the essence of partial differentiation. In classical mechanics, electromagnetism, quantum mechanics, and beyond, we will no longer be satisfied with single-variable functions. We will frequently encounter multivariable systems that require partial derivatives. Therefore, mastering this concept early is essential. Since partial differentiation is not usually taught in high school, this section of the book focuses more on the theoretical understanding rather than problem-solving. If you're interested, you may consult a calculus textbook to find practice problems, none will be found here :3

To find the extrema of a multivariable function, such as $f(x, y)$, we follow the same logic as in the single-variable case: the slope of the tangent at an extremum must be zero. Therefore, for a function of two variables, the critical points where extrema may occur are those that satisfy

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0 \quad (1.115)$$

1.5.3 Total Derivative

Consider a set $\mathbf{x} = (x_1, x_2, \dots)$, $\mathbf{a} = (a_1, a_2, \dots)$ $f : \mathbb{R}^n \rightarrow \mathbb{R}$, and we've known that the Taylor series for functions of several variables

$$f(\mathbf{x} + \mathbf{a}) = f(\mathbf{x}) + \sum_i \frac{\partial f}{\partial x_i} a_i + \frac{1}{2!} \sum_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} a_i a_j + \frac{1}{3!} \sum_{ijk} \frac{\partial^3 f}{\partial x_i \partial x_j \partial x_k} a_i a_j a_k + \dots$$

When we consider $\mathbf{a} = \Delta\mathbf{x} = (\Delta x_1, \Delta x_2, \dots)$ a small variation in x , then the Taylor series becomes

$$f(\mathbf{x} + \Delta\mathbf{x}) = f(\mathbf{x}) + \sum_i \frac{\partial f}{\partial x_i} \Delta x_i + \frac{1}{2!} \sum_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} \Delta x_i \Delta x_j + \dots \quad (1.116)$$

The function increment is

$$\Delta f = f(\mathbf{x} + \Delta\mathbf{x}) - f(\mathbf{x}) = \sum_i \frac{\partial f}{\partial x_i} \Delta x_i + \frac{1}{2!} \sum_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} \Delta x_i \Delta x_j + \dots \quad (1.117)$$

While we take $\|\Delta\| \rightarrow 0 := d$ like what we did in 1-3, the higher-order terms can be neglected because the variation becomes extremely small. So, we only need to retain the first-order partial derivative term

$$df = f(\mathbf{x} + d\mathbf{x}) - f(\mathbf{x}) = \sum_i \frac{\partial f}{\partial x_i} dx_i + O(dx_1 dx_2 \dots) \quad (1.118)$$

So we define this approximation as the **Total Derivatives**, which can be denoted by

$$\boxed{df = \sum_i \frac{\partial f}{\partial x_i} dx_i} \quad (1.119)$$

In other words, the total differential is the linear approximation given by the first-order partial derivatives in the multivariable Taylor expansion. From a physical point of view, each variable of the function f undergoes a small variation, and the variation of each variable can be approximated as a movement along the tangent line with all other variables held constant. The total contribution df is then the accumulation of all these infinitesimal changes across each direction.

For example, there is a 3-dimensional function $f(x, y, z)$, which is what you'll often meet in Taylor's *Classical Mechanics*, then the total derivatives of the function f is

$$df = f(x + dx, y + dy, z + dz) - f(x, y, z) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz \quad (1.120)$$

1.5.4 The Chain Rule of Several Variables

We have discussed the chain rule in single-variable function $f(x)$. When the variable x becomes a function $g(x)$, we need to apply the chain rule to it because f depends on g ; in the meantime, g depends on x , so we should consider both functions. Similarly, consider a function $f(x, y, z)$, and x, y, z depend on the variable t , which can be denoted by

$$f(x(t), y(t), z(t)) \quad (1.121)$$

If we want to know the derivative of f with respect to t , then we need to "dig" the variable t out from each variable, that is

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \quad (1.122)$$

However, when the elements of f depend on two (or more) variables u, v , which is

$$f(x(u, v), y(u, v), z(u, v)) \quad (1.123)$$

Then the chain rule becomes

$$\begin{cases} \frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u} \\ \frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial v} \end{cases} \quad (1.124)$$

Then the total derivatives of f can be denoted by

$$df = \left(\frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u} \right) du + \left(\frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial v} \right) dv \quad (1.125)$$

That is to say, we must take into account how the variables u and v contribute to x, y, z . Since x, y, z in turn influence the overall behavior of the function f , we can say that u and v indirectly affect the function.

In general, consider a function $f(x_1(u_1, u_2, \dots, u_n), \dots, x_n(u_1, u_2, \dots, u_n))$, the total derivative of f can be generalized to

$$df = \sum_i \nabla f \cdot \frac{\partial \mathbf{r}}{\partial u_i} du_i \quad (1.126)$$

Where $\nabla = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)$ and $\mathbf{r} = (x_1, \dots, x_n)$, both of them are vector. This will be discussed in detail in the next chapter *introduction to vector analysis*.

““latex

1.5.5 Constrained Partial Differentials

In single-variable calculus, the chain rule is usually very direct. If

$$z = f(y), \quad y = g(x), \quad (1.127)$$

then $z = f(g(x))$, and the derivative is

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}. \quad (1.128)$$

If the inverse functions are well-defined, we also have

$$\frac{dx}{dz} \frac{dz}{dy} \frac{dy}{dx} = \frac{dx}{dx} = 1. \quad (1.129)$$

This result looks natural because the variables are connected in a one-dimensional chain:

$$x \longrightarrow y \longrightarrow z. \quad (1.130)$$

However, in multivariable calculus and thermodynamics, the situation is more subtle. A partial derivative such as

$$\left(\frac{\partial z}{\partial x} \right)_y \quad (1.131)$$

does not simply mean that z changes with respect to x . It means that z changes with respect to x while y is held fixed. The subscript is therefore part of the meaning of the derivative.

This is especially important when the variables are constrained by an equation. Suppose three variables x , y , and z are related by an equation of state

$$\eta(x, y, z) = 0. \quad (1.132)$$

Then x , y , and z are not independent. If two of them are chosen, the third one is determined locally. For example, we may write

$$z = z(x, y), \quad y = y(x, z), \quad x = x(y, z), \quad (1.133)$$

provided that the corresponding functions are well-defined.

The total differential of $z = z(x, y)$ is

$$dz = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy. \quad (1.134)$$

Here the notation means that

$$\left(\frac{\partial z}{\partial x} \right)_y \quad (1.135)$$

measures the change of z when x changes but y is fixed, while

$$\left(\frac{\partial z}{\partial y} \right)_x \quad (1.136)$$

measures the change of z when y changes but x is fixed.

From the same equation of state, we can also solve for $y = y(x, z)$. Then its total differential is

$$dy = \left(\frac{\partial y}{\partial x} \right)_z dx + \left(\frac{\partial y}{\partial z} \right)_x dz. \quad (1.137)$$

Substituting (1.137) into (1.134), we get

$$\begin{aligned}
dz &= \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x \left[\left(\frac{\partial y}{\partial x}\right)_z dx + \left(\frac{\partial y}{\partial z}\right)_x dz \right] \\
&= \left[\left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z \right] dx + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial z}\right)_x dz.
\end{aligned} \tag{1.138}$$

But the left hand side is just dz . Therefore, the coefficient of dz on the right hand side must be 1, and the coefficient of dx must be 0. Hence

$$\left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial z}\right)_x = 1, \tag{1.139}$$

and

$$\left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z = 0. \tag{1.140}$$

Equation (1.139) gives the reciprocal relation:

$$\boxed{\left(\frac{\partial y}{\partial z}\right)_x = \left[\left(\frac{\partial z}{\partial y}\right)_x \right]^{-1}}. \tag{1.141}$$

This is the constrained version of the familiar single-variable identity

$$\frac{dy}{dz} = \left(\frac{dz}{dy}\right)^{-1}. \tag{1.142}$$

The only difference is that we must specify what variable is being held fixed.

From (1.140), we get

$$\left(\frac{\partial z}{\partial x}\right)_y = - \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z. \tag{1.143}$$

Using the reciprocal relation

$$\left(\frac{\partial x}{\partial y}\right)_z = \left[\left(\frac{\partial y}{\partial x}\right)_z \right]^{-1}, \tag{1.144}$$

we can rewrite (1.143) as

$$\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z = - \left(\frac{\partial z}{\partial y}\right)_x. \tag{1.145}$$

Multiplying both sides by

$$\left(\frac{\partial y}{\partial z}\right)_x, \tag{1.146}$$

we obtain

$$\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x = - \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial z}\right)_x. \quad (1.147)$$

By (1.139), the right hand side is -1 . Therefore,

$$\boxed{\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x = -1}. \quad (1.148)$$

This is called the *triple product rule*.

Notice the minus sign. This is the part that makes constrained partial derivatives different from the ordinary chain rule. For ordinary single-variable derivatives, the chain rule gives

$$\frac{dx}{dz} \frac{dz}{dy} \frac{dy}{dx} = 1. \quad (1.149)$$

But for constrained partial derivatives, the cyclic product gives

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1. \quad (1.150)$$

The reason is that the variables are not arranged in a simple one-dimensional composition. They are tied together by a constraint surface

$$\eta(x, y, z) = 0. \quad (1.151)$$

Moving along this surface forces one variable to compensate for the change of another variable, and this compensation is what produces the negative sign.

Theorem 1.1 (Reciprocal Relation) *If x , y , and z are related by an equation of state, and the corresponding partial derivatives exist and are nonzero, then*

$$\boxed{\left(\frac{\partial y}{\partial z}\right)_x = \frac{1}{\left(\frac{\partial z}{\partial y}\right)_x}}. \quad (1.152)$$

Proof. When x is fixed, the equation of state reduces the relation between y and z to a one-variable relation. Hence locally we may write

$$z = z(y) \quad (1.153)$$

while holding x fixed. If this relation is locally invertible, then

$$y = y(z). \quad (1.154)$$

The ordinary inverse function rule gives

$$\frac{dy}{dz} = \left(\frac{dz}{dy}\right)^{-1}. \quad (1.155)$$

Restoring the fixed variable x , this becomes

$$\left(\frac{\partial y}{\partial z}\right)_x = \left[\left(\frac{\partial z}{\partial y}\right)_x\right]^{-1}. \quad (1.156)$$

□

Theorem 1.2 (Triple Product Rule) *If x , y , and z are related by an equation of state, then*

$$\boxed{\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1}. \quad (1.157)$$

Proof. Start from the total differential

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy. \quad (1.158)$$

Now move along a curve on which z is fixed. Then $dz = 0$, so

$$0 = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy. \quad (1.159)$$

Solving for dx/dy along this curve gives

$$\left(\frac{\partial x}{\partial y}\right)_z = -\frac{\left(\frac{\partial z}{\partial y}\right)_x}{\left(\frac{\partial z}{\partial x}\right)_y}. \quad (1.160)$$

Multiplying both sides of (1.160) by

$$\left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y \quad (1.161)$$

gives

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -\left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial z}\right)_x. \quad (1.162)$$

Using the reciprocal relation

$$\left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial z}\right)_x = 1, \quad (1.163)$$

we obtain

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1. \quad (1.164)$$

□

It is also useful to rewrite the triple product relation into alternative forms. From

$$\left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z = 0, \quad (1.165)$$

we have

$$\left(\frac{\partial z}{\partial x}\right)_y = - \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z. \quad (1.166)$$

Similarly,

$$\left(\frac{\partial x}{\partial y}\right)_z = - \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x. \quad (1.167)$$

These identities are often more useful than memorizing the cyclic product itself, because they allow us to convert one constrained partial derivative into two others.

Physical Meaning in Thermodynamics

The most common place where constrained partial derivatives appear in physics is thermodynamics. For a simple gas, the variables pressure P , volume V , and temperature T are related by an equation of state. For an ideal gas, this equation is

$$PV = nRT. \quad (1.168)$$

This equation means that P , V , and T are not independent. If P and T are known, then V is fixed. If V and T are known, then P is fixed. This is exactly the same structure as

$$\eta(x, y, z) = 0. \quad (1.169)$$

For example,

$$\left(\frac{\partial V}{\partial T}\right)_P \quad (1.170)$$

means the change of volume with respect to temperature while pressure is held fixed. For the ideal gas law,

$$V = \frac{nRT}{P}. \quad (1.171)$$

Therefore,

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{nR}{P}. \quad (1.172)$$

On the other hand,

$$\left(\frac{\partial V}{\partial P}\right)_T \quad (1.173)$$

means the change of volume with respect to pressure while temperature is held fixed. Since

$$V = \frac{nRT}{P}, \quad (1.174)$$

we get

$$\left(\frac{\partial V}{\partial P}\right)_T = -\frac{nRT}{P^2} = -\frac{V}{P}. \quad (1.175)$$

These two derivatives describe completely different physical experiments. In the first one, we heat the gas while keeping pressure fixed, so the gas expands. In the second one, we compress the gas while keeping temperature fixed, so the volume decreases.

Now let us verify the triple product rule for the ideal gas. We compute

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{nR}{V}, \quad (1.176)$$

$$\left(\frac{\partial T}{\partial V}\right)_P = \frac{P}{nR}, \quad (1.177)$$

and

$$\left(\frac{\partial V}{\partial P}\right)_T = -\frac{nRT}{P^2} = -\frac{V}{P}. \quad (1.178)$$

Multiplying them gives

$$\begin{aligned} \left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_P \left(\frac{\partial V}{\partial P}\right)_T &= \left(\frac{nR}{V}\right) \left(\frac{P}{nR}\right) \left(-\frac{V}{P}\right) \\ &= -1. \end{aligned} \quad (1.179)$$

Thus,

$$\boxed{\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_P \left(\frac{\partial V}{\partial P}\right)_T = -1}. \quad (1.180)$$

This example explains the physical meaning of the minus sign. If temperature increases at fixed volume, pressure increases. If volume increases at fixed pressure, temperature must also increase. But if pressure increases at fixed temperature, volume must decrease. One of the three responses is opposite in sign, so the product becomes negative.

The key lesson is that in constrained partial derivatives, the subscript tells us which physical process is being performed. Changing the subscript changes the experiment, so it also changes the derivative. This is why constrained partial derivatives are so important in thermodynamics, fluid mechanics, and statistical mechanics. ““

1.5.6 Multiple Integrals

We’ve learned that if we want to calculate the area under a curve $f(x)$, we have

$$A = \int_a^b f(x) \, dx$$

But, in three dimensions, when we want to calculate the area of a specific region, and the boundary of that region is no longer constant but varies with functions, we can no longer use single-variable integration as we did in two-dimensional problems. In such cases, we must introduce **double integrals**, also known as **area integrals**. Its general form is:

$$\iint_S f(x, y) \, da \quad (1.181)$$

The concept is analogous to the Riemann sum you are familiar with, the domain is infinitely partitioned into smaller rectangles, and the total area is obtained by summing the contributions of $f(x, y)$ over each small patch using a limiting process.

Similarly, we also have **triple integrals**, also known as **volume integrals**, which cut the volume into infinitesimal cubes and then sum them. The form is:

$$\iiint_V f(x, y, z) \, d\tau \quad (1.182)$$

Note that in some physics textbooks, the notation for multiple integrals is simplified as:

$$\iint_S f(x, y) \, da := \int_S f(x, y) \, da, \quad \iiint_V f(x, y, z) \, d\tau := \int_V f(x, y, z) \, d\tau$$

However, the meaning remains the same. In this book, we will use this shorthand notation in the next chapter. In some advanced physics, to determine what dimension we are discussing, the element can be denoted by

$$da \equiv d^2x, \quad d\tau \equiv d^3x. \quad (1.183)$$

In higher dimensions, it's more convenient to use $d^N x$ to represent N -dimension space. In addition, sometimes you may argue that it's ridiculous to put the element $d^N x$ right after the integral sign, but since you are a physics student, we can regard this as an operator like:

$$\int d^N x [\cdot] \equiv L[\cdot].$$

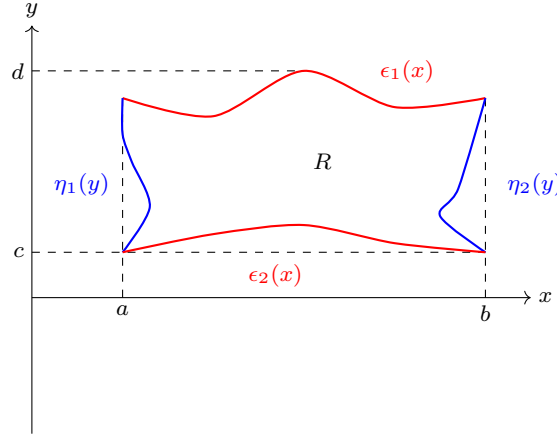


Fig. 1.3 Illustration of the integration region R with variable limits. The region can be described either by fixing x and integrating over y between $\epsilon_2(x)$ and $\epsilon_1(x)$, or by fixing y and integrating over x between $\eta_1(y)$ and $\eta_2(y)$.

To compute an area integral, let us take a closer look at how to actually perform the calculation. The figure above shows a region R in the xy -plane bounded by curved edges, rather than straight lines.

Let us start with the more familiar case: integrating with respect to x first. In this case, the integration bounds can no longer be expressed simply as constants a and b , since the left and right boundaries are given by curves $\eta_1(y)$ and $\eta_2(y)$. Therefore, the inner integral becomes

$$\int_{\eta_1(y)}^{\eta_2(y)} f(x, y) \, dx \quad (1.184)$$

We can interpret this as follows: at a fixed value of y , the function $f(x, y)$ becomes a single-variable function in x , and integrating it gives the contribution from a "vertical strip" at that specific y . This is the inner integral.

Then, by allowing y to vary from c to d , we add up all of these vertical strips to obtain the total area over the region R . This gives the full double integral:

$$\int_c^d \int_{\eta_1(y)}^{\eta_2(y)} f(x, y) \, dx \, dy \quad (1.185)$$

By the same logic, we can also reverse the order of integration. That is, we can first fix x and integrate with respect to y , then integrate with respect to x . The integral becomes:

$$\int_a^b \int_{\epsilon_1(x)}^{\epsilon_2(x)} f(x, y) \, dy \, dx \quad (1.186)$$

This naturally leads to the introduction of **Fubini's Theorem**. Let $D \subset \mathbb{R}^2$ be a bounded, measurable domain, and $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be integrable on D . Then we have

$$\int_D f(x, y) \, dx \, dy = \int_{x \in \pi_x(D)} \left(\int_{y \in D_x} f(x, y) \, dy \right) \, dx \quad (1.187)$$

where $D_x = \{y \mid (x, y) \in D\}$ represents the vertical slice of D at fixed x . In simpler terms, you are free to choose the order of integration, as long as you correctly determine the limits based on the geometry of the domain. Similarly, for volume integrals, we must identify three bounding surfaces, and the integral is written as:

$$\int_a^b \int_{\epsilon_1(x)}^{\epsilon_2(x)} \int_{\xi_1(x, y)}^{\xi_2(x, y)} f(x, y, z) \, dz \, dy \, dx \quad (1.188)$$

This also satisfies the property that the order of integration can be freely rearranged, provided the limits are properly adjusted. In terms of computation, the idea is very similar to how you handle partial derivatives. When integrating with respect to one variable, you treat the others as constants and perform integration accordingly. For instance, to evaluate the following volume integral:

$$\int_{-1}^1 \int_0^2 \int_0^{1-x^2} dz \, dy \, dx$$

you first integrate with respect to z , treating both x and y as constants, then proceed with y , and finally x . That is

$$\begin{aligned} \int_{-1}^1 \int_0^2 \int_0^{1-x^2} dz \, dy \, dx &= \int_{-1}^1 \int_0^2 \left(z \Big|_0^{1-x^2} \right) dy \, dx \\ &= \int_{-1}^1 \int_0^2 (1 - x^2) dy \, dx \\ &= \int_{-1}^1 y(1 - x^2) \Big|_0^2 dx \\ &= \int_{-1}^1 2 - 2x^2 \, dx = \frac{8}{3} \end{aligned}$$

1.5.7 Lagrange multipliers

We begin with the case of a two-variable function. In this section, we deal with the problem of a **constraint function**. A constraint function $g(x, y) = c$ (which is an *iso-contour*) restricts the domain of a function $f(x, y)$ whose extremum d we want to find.

That is, we are seeking the value $f(x, y) = d$ *subject to the condition* $g(x, y) = c$. In other words, $f(x, y) = d \in g(x, y)$, and d must be an extremum of f .

You can think of this like filtering college admission scores: we're looking for the highest score among applicants, but only among those who had perfect scores in math and science. This is the basic idea behind a constraint.

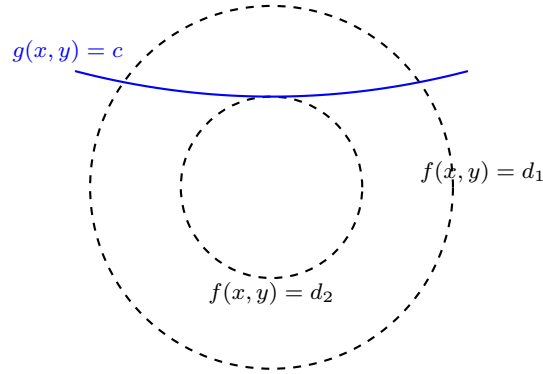


Fig. 1.4 A constraint curve $g(x, y) = c$ (blue) tangent to a level curve $f(x, y) = d$ (black dashed)

As illustrated above, we have contour lines $f(x, y) = d_1$, $f(x, y) = d_2$, etc. For the extremum to satisfy the constraint $g(x, y) = c$, the point must lie on both f and g , and more importantly, the contours must be **tangent** at the extremum point. As we mentioned earlier, the extremum of a multivariable function must satisfy

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$$

So this tangency condition is natural. Since f and g are tangent, their normal vectors must be **parallel** or anti-parallel. Here, we subtly introduce the concept of a normal vector and the gradient (see Chapter 2). The normal vector is given by

$$\nabla f = \mathbf{n} \quad (1.189)$$

where $\nabla = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)$. Returning to our earlier condition, since the gradients of f and g are parallel at the point of tangency, they differ by a scalar multiple. We define this scalar as λ , leading to

$$\nabla f = -\lambda \nabla g \quad (1.190)$$

This constant λ is called the **Lagrange multiplier**. In the two-variable case, this becomes

$$\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = -\lambda \left(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y} \right) \quad (1.191)$$

From this, we obtain the following system

$$\begin{cases} \frac{\partial f}{\partial x} + \lambda \frac{\partial g}{\partial x} = 0 \\ \frac{\partial f}{\partial y} + \lambda \frac{\partial g}{\partial y} = 0 \end{cases} \quad (1.192)$$

We now multiply the first equation by dx and the second by dy

$$\begin{cases} \frac{\partial f}{\partial x} dx + \lambda \frac{\partial g}{\partial x} dx = 0 \\ \frac{\partial f}{\partial y} dy + \lambda \frac{\partial g}{\partial y} dy = 0 \end{cases} \quad (1.193)$$

Integrating both and summing them yields

$$\begin{aligned} & \left(\int \frac{\partial f}{\partial x} dx \right) + \lambda \left(\int \frac{\partial g}{\partial x} dx \right) + \left(\int \frac{\partial f}{\partial y} dy \right) + \lambda \left(\int \frac{\partial g}{\partial y} dy \right) = C \\ \Rightarrow & \int \left(\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \right) + \lambda \int \left(\frac{\partial g}{\partial x} dx + \frac{\partial g}{\partial y} dy \right) \end{aligned}$$

Both are total differentials (df , dg), so we can write

$$f(x, y) + \lambda g(x, y) = C \quad (1.194)$$

This motivates the definition of a new function, the **Lagrangian**:

$$\mathcal{L}(x, y, \lambda) := f(x, y) + \lambda g(x, y) \quad (1.195)$$

Thus, the original problem of finding an extremum of $f(x, y)$ under the constraint $g(x, y) = c$ is transformed into the unconstrained extremum of the Lagrangian:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial y} = 0 \quad (1.196)$$

This method, **the method of Lagrange multipliers**, is crucial and will appear frequently in Lagrangian mechanics later.

The method can also be generalized to n -dimensional functions:

$$\mathcal{L}(x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_n) = f(x_1, \dots, x_n) + \sum_i \lambda_i g_i(x_1, \dots, x_n) \quad (1.197)$$

For example, if we must find the extremum of $f(x, y, z) = xyz > 0$ under the constraint $g(x, y, z) = 6x + 4y + 3z - 24 = 0$. Let the Lagrange multiplier $\lambda \in \mathbb{R}$, and define a function which depends on x , y , z and λ , that is

$$\mathcal{L}(x, y, z, \lambda) = xyz + \lambda(6x + 4y + 3z - 24)$$

So, to find the extremum of $\mathcal{L}$, we have

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x} = yz + 6\lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial y} = xz + 4\lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial z} = xy + 3\lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} = 6x + 4y + 3z - 24 = 0 \end{cases}$$

After calculating, we have $x = 4/3$, $y = 2$, $z = 8/3$, so the extremum of f under the constraint g is

$$xyz = \frac{64}{9}$$

1.5.8 Jacobian Matrix and Coordinate Transformation

Let $\Phi(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ denote the original coordinates in the reference coordinate system. We define a smooth transformation $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$, such that

$$\Phi(x_1, x_2, \dots, x_n) \xrightarrow{F} \Phi'(x'_1, x'_2, \dots, x'_n) \quad (1.198)$$

This means that each point of Φ in the original system is mapped to a new point of Φ' via the transformation F , which defines the new coordinate frame. We define matrices (see Chapter 6.) that

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad X' = \begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{bmatrix} \quad (1.199)$$

then we can use the matrices to describe the coordinates $\Phi(X)$ and $\Phi'(X')$. Rewrite (1.198) as

$$X \xrightarrow{F} X' \quad (1.200)$$

So, we can write down the relationship of function between the new coordinate and the old one, that is

$$X' = F(X) \Leftrightarrow X = F^{-1}(X') \quad (1.201)$$

Based on the relationship above, we can express the following equality

$$\Phi(X) = \Phi(F^{-1}(X)) \quad (1.202)$$

In other words, we can construct a new one-to-one function using the functional relationship between the old and new coordinates. This function takes the new coordinates as its domain, meaning that for each input in the new coordinate system, we obtain a unique output value. This mapping is entirely equivalent to the correspondence established by the new coordinates themselves. That is,

$$\Phi(X) = \Phi'(X') \quad (1.203)$$

From a physical standpoint, this implies that the physical quantity remains unchanged regardless of whether we express it in the old or the new coordinate system. Coordinates merely serve as a reference framework for describing physical quantities, they do not alter the physical essence of the quantity itself.

The transformation F defines a mapping from the original coordinate system to the new one. It consists of a collection of scalar-valued component functions, such that,

$$F(X) = \begin{bmatrix} f_1(X) \\ f_2(X) \\ \vdots \\ f_n(X) \end{bmatrix} \quad (1.204)$$

where $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$. So, in detail

$$\begin{aligned} (x_1, x_2, \dots, x_n) &\xrightarrow{f_1} x'_1 = f_1(x_1, x_2, \dots, x_n) \\ &\xrightarrow{f_2} x'_2 = f_2(x_1, x_2, \dots, x_n) \\ &\vdots \end{aligned} \quad (1.205)$$

Under the inverse transformation, this can be written as

$$\begin{aligned} (x'_1, x'_2, \dots, x'_n) &\xrightarrow{f_1^{-1}} x_1 = f_1^{-1}(x'_1, x'_2, \dots, x'_n) \\ &\xrightarrow{f_2^{-1}} x_2 = f_2^{-1}(x'_1, x'_2, \dots, x'_n) \\ &\vdots \end{aligned} \quad (1.206)$$

That is, according to equations (1.118) and (1.119), the new coordinates are functions of the original coordinates; conversely, the original coordinates can also be expressed as functions of the new ones, and it can be denoted by

$$\Phi'(x'_1, x'_2, \dots, x'_n) = \Phi(x_1(x'_1, x'_2, \dots, x'_n), x_2(x'_1, x'_2, \dots, x'_n), \dots, x_n(x'_1, x'_2, \dots, x'_n)) \quad (1.207)$$

Φ depends on $x_1, x_2, \dots$, and Φ' depends on $x'_1, x'_2, \dots$, so Φ also depends on $x'_1, x'_2, \dots$. We can naturally obtain the chain rule of the partial differentiation under the coordinate transformation,

$$\boxed{\frac{\partial \Phi'}{\partial x'_i} = \sum_j \frac{\partial \Phi}{\partial x_j} \frac{\partial x_j}{\partial x'_i}} \quad (1.208)$$

After discussing the chain rule under the coordinate transformation, we can further define a crucial matrix in such a transformation, the **Jacobian**. Consider a first-order infinitesimal variation in a Taylor expansion dx' , then

$$X' + dX' = \begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{bmatrix} + \begin{bmatrix} dx'_1 \\ dx'_2 \\ \vdots \\ dx'_n \end{bmatrix} \quad (1.209)$$

By (1.119), the latter term can be rewritten as the total differential of the function f_i , that is

$$dx' = \begin{bmatrix} dx'_1 \\ dx'_2 \\ \vdots \\ dx'_n \end{bmatrix} = \begin{bmatrix} df_1 \\ df_2 \\ \vdots \\ df_n \end{bmatrix} = \begin{bmatrix} \sum_i \frac{\partial f_1}{\partial x_i} dx_i \\ \sum_i \frac{\partial f_2}{\partial x_i} dx_i \\ \vdots \\ \sum_i \frac{\partial f_n}{\partial x_i} dx_i \end{bmatrix} \quad (1.210)$$

We can decompose the matrix into the product of two matrices.

$$\begin{bmatrix} \sum_i \frac{\partial f_1}{\partial x_i} dx_i \\ \sum_i \frac{\partial f_2}{\partial x_i} dx_i \\ \vdots \\ \sum_i \frac{\partial f_n}{\partial x_i} dx_i \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}_{n \times n} \begin{bmatrix} dx_1 \\ dx_2 \\ \vdots \\ dx_n \end{bmatrix}_{n \times 1}$$

Then, we define the $n \times n$ matrix as

$$J_F := \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \Rightarrow \det(J_F) := \frac{\partial(f_1, f_2, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)} \quad (1.211)$$

The Jacobian matrix defined here represents how an infinitesimal variation in the new coordinates, under the transformation F , corresponds to variations in the original coordinates. It describes the relative scaling and directional contribution of each element across the coordinate systems. We can write down

$$\boxed{dX' = J_F dX} \quad (1.212)$$

In other words, this matrix connects the rate of change between the two coordinate systems. We will explore the meaning and formal definition of this matrix in vector spaces more thoroughly and rigorously in Chapter 6.

In high school, we learned that if an area or volume is transformed by a matrix, then the resulting area or volume must be multiplied by the determinant of that matrix. For example, an area A is transformed by λ , then the resulting area A' becomes

$$A' = |\det(\lambda)|A \quad (1.213)$$

So do the Jacobian and the coordinate transformation. Under the coordinate transformation in two dimensions, if we wish to describe how variables in a different coordinate system contribute to the original Cartesian coordinate system, we must use the Jacobian matrix to connect the two systems.

Suppose we have a coordinate system $\Phi'(u, v)$, and changes in u and v correspond to changes in the actual Cartesian system $\Phi(x, y)$. To quantify how much change in x and y results from a variation in u and v , we need to map Φ' onto Φ , that is, we write $\Phi(x, y)$ as $\Phi(x(u, v), y(u, v))$. In this case, both u and v contribute to x and y , consistent with the relationship described on the previous page

$$\Phi = J \cdot \Phi' \quad (1.214)$$

At this point, a small area element $du dv$ in the Φ' system is mapped through the Jacobian transformation to the small area element $dx dy$ in the Φ system. As you learned in high school, this results in

$$dx dy = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv = |\det(J)| du dv = \left| \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \right| du dv \quad (1.215)$$

We call $dx dy$, $dv du$ the area element. So the area integrals become

$$\int_S dx dy = \int_\Omega \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv = \int_\Omega |\det(J)| du dv \quad (1.216)$$

For example, the polar coordinate $\Phi'(r, \theta)$ and the Cartesian system $\Phi(x, y)$. We've known the relationship between them

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

The relationship between the area elements of them is

$$dx \, dy = \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| dr \, d\theta = \left\| \begin{array}{cc} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{array} \right\| dr \, d\theta = r \cdot dr \, d\theta$$

So the integral must be rewritten as

$$\int_{y_1}^{y_2} \int_{\epsilon_1(x)}^{\epsilon_2(x)} dx \, dy = \int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} r \cdot dr \, d\theta \quad (1.217)$$

We can also apply the same logic to the volume elements. Take the spherical coordinates for example. The relationship between spherical coordinates $\Phi'(r, \theta, \phi)$ and Cartesian coordinate $\Phi(x, y, z)$ is

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases}$$

Where θ is the polar angle and ϕ is the azimuthal angle (see Chapter 1-6). The relationship between the volume elements of them is

$$dx \, dy \, dz = \left| \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} \right| dr \, d\theta \, d\phi = \left\| \begin{array}{ccc} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{array} \right\| dr \, d\theta \, d\phi = r^2 \sin \theta \cdot dr \, d\theta \, d\phi$$

The volume integral becomes

$$\int_V dx \, dy \, dz = \int r^2 \sin \theta \cdot dr \, d\theta \, d\phi \quad (1.218)$$

Sometimes, the determinant of the Jacobian matrix can be denoted by

$$\det(J) := J \left(\frac{\partial x_i}{\partial x'_j} \right)$$

and the integral after a coordinate transformation in N dimensions can be written as:

$$\int d^N x = \int \left| J \left(\frac{\partial x_i}{\partial x'_j} \right) \right| d^N x' \quad (1.219)$$

1.6 Elliptic Integrals in the Pendulum Problem

When dealing with complex physical behaviors, such as large-angle pendulum motion, celestial mechanics, and orbital curvature, it is often impossible to represent the solutions using elementary functions. Instead, we must introduce special functions and evaluate them using numerical methods or series expansions to approximate real-world values. One such integral is called the **elliptic integral**, which was originally developed in the context of computing the arc length of an ellipse. There are three types of elliptic integrals: the first kind, second kind, and third kind.

The elliptic integral of the first kind is commonly written as $F(k, \phi)$, where k is a parameter determined by the initial condition of the system, and ϕ is the upper limit of integration, also known as the amplitude. Its definition is:

$$F(k, \phi) = \int_0^\phi \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}, \quad k^2 < 1 \quad (1.220)$$

The elliptic integral of the second kind, denoted $E(k, \phi)$, corresponds to the original arc length formula of an ellipse and is defined as:

$$E(k, \phi) = \int_0^\phi \sqrt{1 - k^2 \sin^2 \theta} d\theta, \quad k^2 < 1 \quad (1.221)$$

The elliptic integral of the third kind, written as $\Pi(n, k, \phi)$, is the most complex among the three and often appears in advanced simulations such as space mission trajectory modeling. Its definition is:

$$\Pi(n, k, \phi) = \int_0^\phi \frac{d\theta}{(1 + n \sin^2 \theta) \sqrt{1 - k^2 \sin^2 \theta}} \quad (1.222)$$

In this chapter, we will focus solely on the first kind of elliptic integral to address the large-angle pendulum problem.

1.6.1 Gaussian Integral

The Gaussian integral I is

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx \quad (1.223)$$

It is an **improper integral** over an infinite domain. In this book, we will not go into the detailed theory of improper integrals. At this point, it is sufficient for us to know whether the integrand is convergent and integrable over the given interval. We observe that the function $\exp(-x^2)$ decays faster than any rational function as $|x| \rightarrow \infty$, so the integral converges.

If we want to calculate this integral, we need to use some methods. First, we have

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx = \int_{-\infty}^{\infty} e^{-y^2} dy \quad (1.224)$$

So, we multiply both integrals, and then we get

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \quad (1.225)$$

We transform the Cartesian coordinate to the polar coordinates, that is

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

By (1.217), the area element $dx dy$ becomes $r \cdot dr d\theta$. Substituting these into (1.225), we can obtain

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta \quad (1.226)$$

Let $u = r^2$, $du = 2r dr$, so the integral becomes

$$\int_0^{2\pi} \int_0^{\infty} \frac{1}{2} e^{-u} du d\theta = \int_0^{2\pi} \left. -\frac{1}{2} e^{-u} \right|_0^{\infty} d\theta = \frac{1}{2} \int_0^{2\pi} d\theta = \pi$$

Substituting this into (1.225), $I^2 = \pi$, so we can finally obtain the Gaussian integral

$$\boxed{I = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}} \quad (1.227)$$

1.6.2 Gamma Function

The Gamma function is introduced to extend the factorial function to noninteger values in a continuous manner. It's defined as

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt, \quad x > 0 \quad (1.228)$$

When $x = n, n \in \mathbb{N}$, it is

$$\Gamma(n) = \int_0^{\infty} t^{n-1} e^{-t} dt \quad (1.229)$$

The formula $\Gamma(n) = (n-1)!$ is important, where we define $0! = 1$. We can use integration by parts and the mathematical induction to analyze it. When $x = 1$ and $x = 2$, we get

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = -e^{-t} \Big|_0^{\infty} = 1 = 1!$$

We assume that the formula holds when $x = n$, $n \in \mathbb{N}$, which is

$$\Gamma(n) = \int_0^{\infty} t^{n-1} e^{-t} dt = (n-1)!$$

When $x = n+1$, it becomes

$$\begin{aligned} \Gamma(n+1) &= \int_0^{\infty} t^n e^{-t} dt \\ &= -t^n e^{-t} \Big|_0^{\infty} + n \int_0^{\infty} t^{n-1} e^{-t} dt \end{aligned}$$

Note that the limit $\lim_{t \rightarrow \infty} -t^n e^{-t} = 0$ because the growth rate of exponential functions is faster than that of polynomial functions. So

$$\Gamma(n+1) = n \int_0^{\infty} t^{n-1} e^{-t} dt = n(n-1)! = n!$$

By mathematical induction, we can verify that for all $x \in \mathbb{N}$, the formula $\Gamma(n) = (n-1)!$ holds. More specifically, you can generalize it to $\Gamma(x)$ in the same way. After that, we have two crucial properties of the Gamma function

$$\boxed{\begin{cases} \Gamma(n) = (n-1)!, & n \in \mathbb{N} \\ \Gamma(x+1) = x\Gamma(x), & \forall x \in \mathbb{R}^+ \end{cases}} \quad (1.230)$$

However, when x is a noninteger value, such as $x = 1/2$, $x = 1/3 \dots$, it may become the familiar form that we have discussed, the Gaussian integral. For example, $x = 1/2$, the Gamma function

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt \quad (1.231)$$

Let $u = \sqrt{t}$, $du = 1/2 \cdot t^{-1/2}$, so (1.231) becomes

$$\int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt = 2 \int_0^{\infty} e^{-u^2} du \quad (1.232)$$

(1.232) is the form of the Gaussian integral. Since the function e^{-x^2} is symmetric at $x = 0$, so

$$\int_0^\infty e^{-u^2} du = \frac{1}{2} \int_{-\infty}^\infty e^{-u^2} du = \frac{\sqrt{\pi}}{2}$$

Substituting it into (1.232), we can finally get

$$\Gamma\left(\frac{1}{2}\right) = 2 \int_0^\infty e^{-u^2} du = \sqrt{\pi} \quad (1.233)$$

When $x = 3/2$, by the formula (3.132), we obtain

$$\Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2} + 1\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2} \quad (1.234)$$

However, except for $x = 1/2$, which can be evaluated using the Gaussian integral, most other cases, such as $x = 1/3, 2/3, \dots$ can only be computed numerically.

We have learned binomial expansion in high school, which is

$$(1+x)^r = \sum_i \binom{r}{i} x^i y^{r-i}$$

We can use the Gamma function to generalize the binomial expansion. Which can be denoted by

$$(x+y)^r = \sum_i \frac{\Gamma(r+1)}{\Gamma(i+1)\Gamma(r-i+1)} x^i y^{r-i} \quad (1.235)$$

Although we have just introduced the generalized binomial theorem, note that if the value of $r-i+1$ becomes less than zero, the Gamma function involved will diverge and become undefined. That is, $(1-x)^{-r}$, if we differentiate it with respect to x , we get

$$\frac{d^n}{dx^n} (1-x)^{-r} = \prod_{i=0}^{n-1} (r+i) (1-x)^{-r-n}$$

Let $f(x) = (1-x)^{-r}$, we've known that

$$f(x) = \sum_{i=0} \frac{f^{(i)}(0)}{i!} x^i$$

So the Taylor series of $f(x) = (1-x)^{-r}$ is

$$(1-x)^{-r} = \sum_{i=0} \frac{\prod_{j=0}^{i-1} (r+j)}{i!} x^i = \sum_i \binom{r+i-1}{i} x^i = \sum_i \frac{(r+i-1)!}{i!(r-1)!} = \sum_i \frac{\Gamma(r+i)}{\Gamma(i+1)\Gamma(r)} x^i$$

Take $r = -1/2$ for example, we obtain

$$(1-x)^{-1/2} = \sum_i \frac{\Gamma\left(i + \frac{1}{2}\right)}{\Gamma(i+1)\Gamma\left(\frac{1}{2}\right)} x^i \quad (1.236)$$

Where $\Gamma(i+1) = i!$, $\Gamma(1/2) = \sqrt{\pi}$. To express these Gamma functions in terms of factorials, we can repeatedly apply the identity $\Gamma(x+1) = x\Gamma(x)$ to break them down.

Let's deal with $\Gamma\left(i + \frac{1}{2}\right)$, we can obtain

$$\begin{aligned} \Gamma\left(i + \frac{1}{2}\right) &= \Gamma\left(i - \frac{1}{2} + 1\right) \\ &= \left(i - \frac{1}{2}\right) \Gamma\left(i - \frac{1}{2}\right) \\ &= \left(i - \frac{1}{2}\right) \left(i - \frac{3}{2}\right) \Gamma\left(i - \frac{3}{2}\right) \\ &= \left(i - \frac{1}{2}\right) \left(i - \frac{3}{2}\right) \left(i - \frac{5}{2}\right) \cdots \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \end{aligned}$$

By factoring out i terms of $1/2$, the remaining product becomes a sequence of odd integers starting from $2i-1$ and decreasing by 2. That is,

$$\frac{1}{2^i} (2i-1)(2i-1) \cdots 3 \cdot 2$$

We can perform a common-denominator transformation to introduce the missing even-numbered terms in the denominator

$$\frac{(2i)!}{2^i \times i! \times 2^i} = \frac{(2i)!}{i!4^i}$$

So, $\Gamma\left(i + \frac{1}{2}\right)$ can be rewritten as

$$\Gamma\left(i + \frac{1}{2}\right) = \frac{(2i)!}{i!4^i} \sqrt{\pi} \quad (1.237)$$

We substitute this result (1.237) into (1.236), and we get the final expansion

$$(1-x)^{-1/2} = \sum_i \frac{\frac{(2i)!}{i!4^i} \sqrt{\pi}}{i! \sqrt{\pi}} x^i = \sum_i \frac{(2i)!}{(i!)^2 \times 4^i} x^i = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \cdots \quad (1.238)$$

1.6.3 Beta Function

To better understand the relationship between the integral and the factorial, I must introduce another special function, the Beta function. You may be surprised, or even shocked, by the beauty of this structure later.

Define the Beta function $B(p, q)$

$$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt \quad (1.239)$$

We can use this definition to combine the integrals and the Gamma function in complicated problems, such as the elliptic integral. To find the relationship between the Gamma function and the Beta function, we construct

$$\begin{aligned} \Gamma(p)\Gamma(q) &= \left(\int_0^\infty x^{p-1} e^{-x} dx \right) \left(\int_0^\infty y^{q-1} e^{-y} dy \right) \\ &= \int_0^\infty \int_0^\infty x^{p-1} y^{q-1} e^{-(x+y)} dx dy \end{aligned} \quad (1.240)$$

Let $x = ru$, $y = r(1-u)$, where $r \in [0, \infty)$, $u \in [0, 1]$. By the coordinate transformation equation

$$dx dy = \left| \frac{\partial(x, y)}{\partial(r, u)} \right| dr du = r dr du \quad (1.241)$$

So the integral (1.240) becomes

$$\int_0^\infty \int_0^\infty x^{p-1} y^{q-1} e^{-(x+y)} dx dy = \int_0^1 \int_0^\infty (ru)^{p-1} (r(1-u))^{q-1} e^{-r} r dr du \quad (1.242)$$

Rewriting the integrand and calculating this integral

$$\begin{aligned} \int_0^1 \int_0^\infty (ru)^{p-1} (r(1-u))^{q-1} e^{-r} r dr du &= \int_0^1 \int_0^\infty u^{p-1} (1-u)^{q-1} r^{p+q-1} e^{-r} dr du \\ &= \int_0^1 \left(\int_0^\infty r^{p+q-1} e^{-r} dr \right) u^{p-1} (1-u)^{q-1} du \end{aligned}$$

Note that the inner integral is the same as the definition of the Gamma function, and the outer integral is the same as the definition of the Beta function, so

$$\begin{aligned}
\int_0^1 \left(\int_0^\infty r^{p+q-1} e^{-r} dr \right) u^{p-1} (1-u)^{q-1} du &= \int_0^1 \Gamma(p+q) u^{p-1} (1-u)^{q-1} du \\
&= \Gamma(p+q) \int_0^1 u^{p-1} (1-u)^{q-1} du \\
&= \Gamma(p+q) \cdot B(p, q)
\end{aligned}$$

Substituting this into (1.240), we can obtain a crucial formula for the relationship between the Gamma and Beta functions. That is

$$B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} \quad (1.243)$$

1.6.4 Elliptic Integral and Pendulum

The motion of a simple pendulum is one of the most fundamental problems in classical mechanics, as it helps us understand the behavior of nonlinear physical systems. In high school, we typically focus on pendulum motion with small initial angles. However, as the angle becomes larger, the same method of analysis is no longer valid.

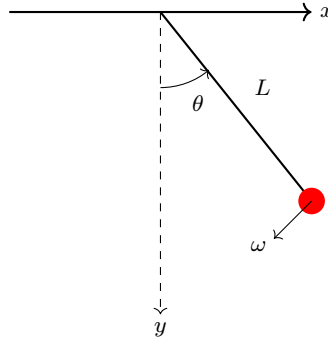


Fig. 1.5 Sample of a system constrained by rod

As shown in the above figure, the bob is released with the initial angle θ_0 and $v = L\dot{\theta} = 0$, so the kinetic energy of the bob is 0 at $\theta = \theta_0$, and the total energy and the potential energy are

$$\begin{cases} V = mgL(1 - \cos \theta) = 2mgL \sin^2 \left(\frac{\theta}{2} \right) \\ E = mgL(1 - \cos \theta_0) = 2mgL \sin^2 \left(\frac{\theta_0}{2} \right) \end{cases} \quad (1.244)$$

We use the difference between the potential energy and the total energy to express the kinetic energy

$$T = \frac{1}{2}mL^2\dot{\theta}^2 = E - V = 2mgL \left(\sin^2 \left(\frac{\theta_0}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) \right)$$

So

$$\dot{\theta} = \frac{d\theta}{dt} = 2\sqrt{\frac{g}{L}} \sqrt{\left(\sin^2 \left(\frac{\theta_0}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) \right)} \quad (1.245)$$

Here, we separate the variable (see Chapter 4-1), and we can easily solve the differential equation

$$\frac{1}{2}\sqrt{\frac{L}{g}} \left[\left(\sin^2 \left(\frac{\theta_0}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) \right) \right]^{-1/2} d\theta = dt$$

By integrating both sides, we obtain the time t it takes for the pendulum to swing from $\theta = \theta_0$ to $\theta = 0$, which corresponds to one-quarter of the total period T . That is,

$$T = 4t = 4 \int dt = 2\sqrt{\frac{L}{g}} \int_0^{\theta_0} \frac{d\theta}{\left[\left(\sin^2 \left(\frac{\theta_0}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) \right) \right]^{1/2}} \quad (1.246)$$

This is a typical elliptic integral of the first kind (1.220). Factor $\sin \theta_0/2$ out of the square root, it becomes

$$2\sqrt{\frac{L}{g}} \int_0^{\theta_0} \frac{d\theta}{\left[\left(\sin^2 \left(\frac{\theta_0}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) \right) \right]^{1/2}} = 2\sqrt{\frac{L}{g}} \int_0^{\theta_0} \frac{d\theta}{\sin \left(\frac{\theta_0}{2} \right) \left[1 - \left(\frac{\sin(\theta/2)}{\sin(\theta_0/2)} \right)^2 \right]^{1/2}} \quad (1.247)$$

Apply the integration by substitution, let

$$z = \frac{\sin(\theta/2)}{\sin(\theta_0/2)}; \quad k = \sin(\theta_0/2)$$

Differentiating this to get

$$dz = \frac{\cos(\theta/2)}{2\sin(\theta_0/2)} d\theta = \frac{\sqrt{1 - \sin^2(\theta/2)}}{2\sin(\theta_0/2)} d\theta = \frac{\sqrt{1 - k^2 z^2}}{2k} d\theta \Rightarrow d\theta = \frac{2k dz}{\sqrt{1 - k^2 z^2}}$$

substituting this into (1.247), we have

$$2\sqrt{\frac{L}{g}} \int_0^{\theta_0} \frac{d\theta}{\sin \left(\frac{\theta_0}{2} \right) \left[1 - \left(\frac{\sin(\theta/2)}{\sin(\theta_0/2)} \right)^2 \right]^{1/2}} = 2\sqrt{\frac{L}{g}} \int_0^1 \frac{2k dz}{k (1 - z^2)^{1/2} (1 - k^2 z^2)^{1/2}}$$

Remember to change the limits of integration after the substitution, $z(\theta_0) = 1$. So, the total period of the pendulum is

$$T = 4\sqrt{\frac{L}{g}} \int_0^1 [(1-z^2)(1-k^2z^2)]^{-1/2} dz \quad (1.248)$$

We can expand the term $(1-k^2z^2)^{-1/2}$ by (1.238)

$$(1-k^2z^2)^{-1/2} = \sum_i \frac{(2i)!}{(i!)^2 \times 4^i} (k^2z^2)^i = 1 + \frac{k^2z^2}{2} + \frac{3k^4z^4}{8} + \frac{5k^6z^6}{16} + \dots \quad (1.249)$$

Substituting (1.249) into (1.248), we can start to calculate the integral

$$\begin{aligned} T &= 4\sqrt{\frac{L}{g}} \int_0^1 \frac{1}{\sqrt{(1-z^2)}} \left(1 + \frac{k^2z^2}{2} + \frac{3k^4z^4}{8} + \frac{5k^6z^6}{16} + \dots \right) dz \\ &= 4\sqrt{\frac{L}{g}} \left(\int_0^1 \frac{dz}{\sqrt{(1-z^2)}} + \frac{k^2}{2} \int_0^1 \frac{z^2 dz}{\sqrt{(1-z^2)}} + \dots \right) \end{aligned}$$

Let $z = \sin \phi$, $dz = \cos \theta d\theta$, then the integral can be described as

$$4\sqrt{\frac{L}{g}} \left(\int_0^{\frac{\pi}{2}} d\theta + \frac{k^2}{2} \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta + \frac{3k^4}{8} \int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta + \frac{5k^6}{16} \int_0^{\frac{\pi}{2}} \sin^6 \theta d\theta + \dots \right)$$

We can observe that each term of the integral contains the integral of $\sin^{2n} \theta$ from 0 to $\pi/2$. Let $u = \sin^2 \theta$, $du = 2 \sin \theta \cos \theta d\theta$, so $d\theta = 1/2 \times u^{-1/2} \times (1-u)^{-1/2} du$. Substituting this into the integral, it becomes

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sin^{2n} \theta d\theta &= \frac{1}{2} \int_0^1 u^n \times u^{-1/2} \times (1-u)^{-1/2} du \\ &= \frac{1}{2} \int_0^1 u^{n-1/2} (1-u)^{-1/2} du \\ &= \frac{1}{2} \int_0^1 u^{(n+1/2)-1} (1-u)^{1/2-1} du \\ &= \frac{1}{2} B\left(n + \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2} \frac{\Gamma\left(n + \frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma(n+1)} \end{aligned}$$

We thus obtain

$$\int_0^{\frac{\pi}{2}} \sin^{2n} \theta \, d\theta = \frac{1}{2} \frac{\Gamma\left(n + \frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma(n+1)} = \frac{1}{2} \frac{\left(\frac{(2n)!}{n!4^n} \sqrt{\pi}\right) \sqrt{\pi}}{n!} = \frac{(2n)!}{(n!)^2 4^n} \cdot \frac{\pi}{2} \quad (1.250)$$

Each integral of the form $\int_0^{\pi/2} \sin^{2n} \theta \, d\theta$ yields a result of the form $c \times \pi/2$, where c is a constant. By comparing with equation (1.249), we observe that the coefficient of each $\frac{\pi}{2}$ exactly matches the coefficient of k^{2n} in the expansion of (1.249). Therefore,

$$\begin{aligned} T &= 4\sqrt{\frac{L}{g}} \left(\frac{\pi}{2} + \frac{k^2}{2} \cdot \frac{1}{2} \cdot \frac{\pi}{2} + \frac{3k^4}{8} \cdot \frac{3}{8} \cdot \frac{\pi}{2} + \frac{5k^6}{16} \cdot \frac{5}{16} \cdot \frac{\pi}{2} + \cdots \right) \\ &= 2\pi\sqrt{\frac{L}{g}} \left(1 + \frac{k^2}{4} + \frac{9k^4}{64} + \frac{25k^6}{512} + \cdots \right) = 2\pi\sqrt{\frac{L}{g}} \sum_i \left(\frac{(2i)!}{(i!)^2 4^i} \right)^2 k^{2i} \end{aligned} \quad (1.251)$$

By the Taylor expansion of $\sin x$ function, we can approximate the term $k = \sin(\theta_0/2)$, which is

$$k = \sin\left(\frac{\theta_0}{2}\right) = \frac{\theta_0}{2} - \frac{\theta_0^3}{48} + \frac{\theta_0^5}{3840} - \frac{\theta_0^7}{645120} + \cdots \quad (1.252)$$

So, the final result of the period of the simple pendulum is

$$T = 2\pi\sqrt{\frac{L}{g}} \sum_i \left(\frac{(2i)!}{(i!)^2 4^i} \right)^2 \left(\frac{\theta_0}{2} - \frac{\theta_0^3}{48} + \frac{\theta_0^5}{3840} - \frac{\theta_0^7}{645120} + \cdots \right)^{2i} \quad (1.253)$$

From this equation, we can clearly see that when the pendulum undergoes small-angle motion, the contribution of the angular terms approaches zero, resulting in an approximately linear periodic behavior.

$$T = 2\pi\sqrt{\frac{L}{g}} \Big|_{\theta_0 \rightarrow 0}$$

1.7 Acceleration in Three Coordinates

In mathematics and physics, the most commonly used coordinate systems can be categorized into four types: Cartesian, polar, cylindrical, and spherical coordinates. In this chapter, I will start with the unit vectors and their differential relationships in the Cartesian coordinates and use them to transform the unit vectors of the other three systems. By doing so, we will derive the relationships between the unit vectors in each coordinate system and further obtain the expressions for velocity and acceleration in these various coordinate frameworks.

1.7.1 Polar Coordinates

We begin with the polar coordinate system, which can be decomposed into the radial unit vector $\mathbf{e}_r$ and the angular (or tangential) unit vector $\mathbf{e}_\phi$, as shown in the figure below

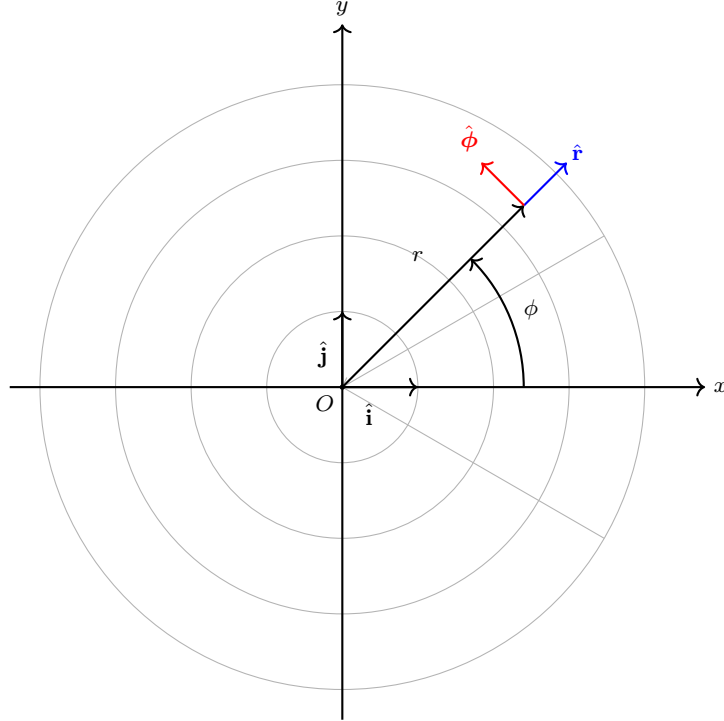


Fig. 1.6 Polar coordinate system with unit vectors $\mathbf{e}_r$ and $\mathbf{e}_\phi$, along with the Cartesian basis vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$.

We can first write down the relationship between the two coordinates, that is

$$\begin{cases} \mathbf{e}_r = \cos \phi \hat{\mathbf{i}} + \sin \phi \hat{\mathbf{j}} \\ \mathbf{e}_\phi = -\sin \phi \hat{\mathbf{i}} + \cos \phi \hat{\mathbf{j}} \end{cases} \quad (1.254)$$

The two unit vectors in polar coordinates depend only on the angle ϕ . Differentiating both unit vectors with respect to ϕ , we get

$$\begin{cases} \frac{d\mathbf{e}_r}{d\phi} = -\sin \phi \hat{\mathbf{i}} + \cos \phi \hat{\mathbf{j}} = \mathbf{e}_\phi \\ \frac{d\mathbf{e}_\phi}{d\phi} = -\cos \phi \hat{\mathbf{i}} - \sin \phi \hat{\mathbf{j}} = -\mathbf{e}_r \end{cases} \quad (1.255)$$

We can express the above equation as a second-order Taylor expansion approximation.

$$\begin{cases} d\mathbf{e}_r = d\phi \mathbf{e}_\phi \Rightarrow \dot{\mathbf{e}}_r = \dot{\phi} \mathbf{e}_\phi \\ d\mathbf{e}_\phi = -d\phi \mathbf{e}_r \Rightarrow \dot{\mathbf{e}}_\phi = -\dot{\phi} \mathbf{e}_r \end{cases} \quad (1.256)$$

then the position vector can be written as

$$\mathbf{r}(t) = r(t)\mathbf{e}_r \quad (1.257)$$

Differentiating the position vector, we can obtain

$$\dot{\mathbf{r}}(t) = \frac{d}{dt}(r(t)\mathbf{e}_r) = \dot{r}(t)\mathbf{e}_r + r(t)\dot{\mathbf{e}}_r$$

So the velocity is

$$\mathbf{v}(t) = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\phi} \mathbf{e}_\phi \quad (1.258)$$

Next, by differentiating the velocity vector once more, we obtain the acceleration

$$\begin{aligned} \mathbf{a}(t) = \dot{\mathbf{v}} = \ddot{\mathbf{r}} &= \frac{d}{dt}(\dot{r}\mathbf{e}_r + r\dot{\phi} \mathbf{e}_\phi) \\ &= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\phi}\mathbf{e}_\phi + r\ddot{\phi}\mathbf{e}_\phi + r\dot{\phi}\dot{\mathbf{e}}_\phi \end{aligned}$$

Substituting (1.256) into it, then

$$\ddot{\mathbf{r}} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\phi} \mathbf{e}_\phi + \dot{r}\dot{\phi}\mathbf{e}_\phi + r\ddot{\phi}\mathbf{e}_\phi - r\dot{\phi}^2 \mathbf{e}_r = (\ddot{r} - r\dot{\phi}^2)\mathbf{e}_r + (r\ddot{\phi} + 2\dot{r}\dot{\phi})\mathbf{e}_\phi$$

So, the final result of the acceleration in polar coordinates is

$$\mathbf{a} = \begin{cases} \text{radial : } (\ddot{r} - r\dot{\phi}^2)\mathbf{e}_r \\ \text{angular : } (r\ddot{\phi} + 2\dot{r}\dot{\phi})\mathbf{e}_\phi \end{cases} \quad (1.259)$$

When an object undergoes uniform circular motion, its radius(r) of motion remains constant, and its angular velocity($\dot{\phi}$) stays uniform. That is to say, $\dot{r}, \ddot{r} = 0$, and $\ddot{\phi} = 0$. The final result is familiar in high school, which is

$$\mathbf{a} = -r\dot{\phi}^2 \mathbf{e}_r = -r\omega^2 \mathbf{e}_r \quad (1.260)$$

The negative sign here indicates that the direction of acceleration always points opposite to the radial direction, that is, toward the center of the circle.

1.7.2 Cylindrical Coordinates

The second coordinate, the cylindrical system, can be decomposed into the radial unit vector $\mathbf{e}_r$, an angular (azimuthal) unit vector $\mathbf{e}_\phi$, and a vertical unit vector $\mathbf{e}_z$. As shown in figure

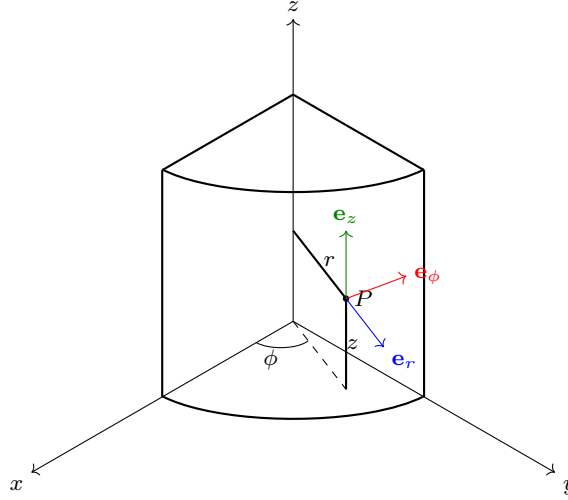


Fig. 1.7 Cylindrical coordinate system with unit vectors $\mathbf{e}_r$, $\mathbf{e}_\phi$, and $\mathbf{e}_z$.

We begin by rewriting these unit vectors in terms of the Cartesian basis vectors, which is

$$\begin{cases} \mathbf{e}_r = \cos \phi \hat{\mathbf{i}} + \sin \phi \hat{\mathbf{j}} \\ \mathbf{e}_\phi = -\sin \phi \hat{\mathbf{i}} + \cos \phi \hat{\mathbf{j}} \\ \mathbf{e}_z = \hat{\mathbf{k}} \end{cases} \quad (1.261)$$

We can also express the above equation as a second-order Taylor expansion approximation, as we did in the last analysis.

$$\begin{cases} d\mathbf{e}_r = \frac{d\mathbf{e}_r}{d\phi} d\phi = d\phi \mathbf{e}_\phi \Rightarrow \dot{\mathbf{e}}_r = \dot{\phi} \mathbf{e}_\phi \\ d\mathbf{e}_\phi = \frac{d\mathbf{e}_\phi}{d\phi} d\phi = -d\phi \mathbf{e}_r \Rightarrow \dot{\mathbf{e}}_\phi = -\dot{\phi} \mathbf{e}_r \\ d\mathbf{e}_z = \frac{d\mathbf{e}_z}{d\phi} d\phi = 0 \end{cases} \quad (1.262)$$

The result indicates that the unit vector $\mathbf{e}_z$ does not change with respect to the angle or the position; it remains $(0, 0, 1)$. The position vector is given by $\mathbf{r} = r\mathbf{e}_r + z\mathbf{e}_z$, since

$r\mathbf{e}_r$ can only describe vectors in the xy -plane. So, the velocity is

$$\mathbf{v} = \dot{\mathbf{r}} = \frac{d}{dt}(r\mathbf{e}_r + z\mathbf{e}_z) = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r + \dot{z}\mathbf{e}_z + z\dot{\mathbf{e}}_z \quad (1.263)$$

Substituting (1.262) into (1.263), we get

$$\dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\phi}\mathbf{e}_\phi + \dot{z}\mathbf{e}_z \quad (1.264)$$

Next, by differentiating the velocity vector once more, we obtain the acceleration

$$\begin{aligned} \mathbf{a} = \ddot{\mathbf{r}} &= \frac{d}{dt}(\dot{r}\mathbf{e}_r + r\dot{\phi}\mathbf{e}_\phi + \dot{z}\mathbf{e}_z) \\ &= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\phi}\mathbf{e}_\phi + r\ddot{\phi}\mathbf{e}_\phi + r\dot{\phi}\dot{\mathbf{e}}_\phi + \ddot{z}\mathbf{e}_z + \dot{z}\dot{\mathbf{e}}_z \\ &= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\phi}\mathbf{e}_\phi + \dot{r}\dot{\phi}\mathbf{e}_\phi + r\ddot{\phi}\mathbf{e}_\phi - r\dot{\phi}^2\mathbf{e}_r + \ddot{z}\mathbf{e}_z \\ &= (\ddot{r} - r\dot{\phi}^2)\mathbf{e}_r + (r\ddot{\phi} + 2\dot{r}\dot{\phi})\mathbf{e}_\phi + \ddot{z}\mathbf{e}_z \end{aligned} \quad (1.265)$$

So, the final result of the acceleration in cylindrical coordinates is

$$\mathbf{a} = \begin{cases} \text{radial : } (\ddot{r} - r\dot{\phi}^2)\mathbf{e}_r \\ \text{angular : } (r\ddot{\phi} + 2\dot{r}\dot{\phi})\mathbf{e}_\phi \\ \text{vertical : } \ddot{z}\mathbf{e}_z \end{cases} \quad (1.266)$$

1.7.3 Spherical Coordinates

The third coordinate system, the spherical coordinate system, can be decomposed into the radial unit vector $\mathbf{e}_r$, the polar (or colatitudinal) unit vector $\mathbf{e}_\theta$, and the azimuthal unit vector $\mathbf{e}_\phi$. As shown in the figure, these unit vectors form an orthonormal basis that changes direction depending on the position.

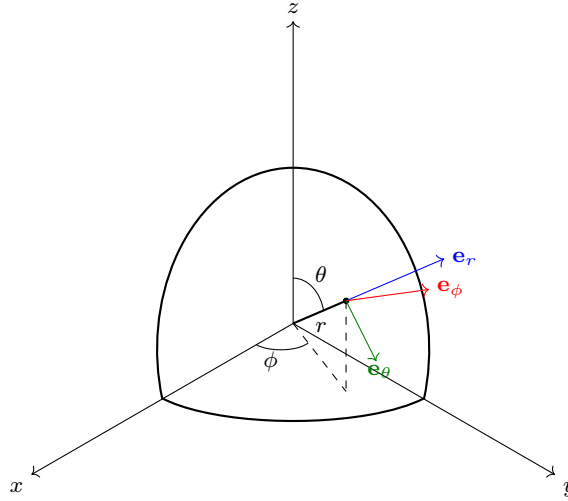


Fig. 1.8 Spherical coordinate system with unit vectors $\mathbf{e}_r$, $\mathbf{e}_\phi$, and $\mathbf{e}_\theta$.

Similarly, we begin by rewriting these unit vectors in terms of the Cartesian basis vectors, which is

$$\begin{cases} \mathbf{e}_r = \sin \theta \cos \phi \hat{\mathbf{i}} + \sin \theta \sin \phi \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}} \\ \mathbf{e}_\phi = -\sin \phi \hat{\mathbf{i}} + \cos \phi \hat{\mathbf{j}} \\ \mathbf{e}_\theta = \cos \theta \cos \phi \hat{\mathbf{i}} + \cos \theta \sin \phi \hat{\mathbf{j}} - \sin \theta \hat{\mathbf{k}} \end{cases} \quad (1.267)$$

Since this coordinate system involves both θ and ϕ as independent variables, we can consider the total differentials of the three unit vectors, which are given by

$$\begin{cases} d\mathbf{e}_r = \frac{\partial \mathbf{e}_r}{\partial \theta} d\theta + \frac{\partial \mathbf{e}_r}{\partial \phi} d\phi = d\theta \mathbf{e}_\theta + d\phi \sin \theta \mathbf{e}_\phi \\ d\mathbf{e}_\phi = \frac{\partial \mathbf{e}_\phi}{\partial \theta} d\theta + \frac{\partial \mathbf{e}_\phi}{\partial \phi} d\phi = (-\sin \theta \mathbf{e}_r - \cos \theta \mathbf{e}_\theta) d\phi \\ d\mathbf{e}_\theta = \frac{\partial \mathbf{e}_\theta}{\partial \theta} d\theta + \frac{\partial \mathbf{e}_\theta}{\partial \phi} d\phi = -d\theta \mathbf{e}_r + d\phi \cos \theta \mathbf{e}_\phi \end{cases} \quad (1.268)$$

The position vector is given by $\mathbf{r} = r\mathbf{e}_r$, so the velocity is

$$\mathbf{v} = \dot{\mathbf{r}} = \frac{d}{dt}(r\mathbf{e}_r) = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r \quad (1.269)$$

Substituting (1.268) into (1.269), we get

$$\dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + r\dot{\phi}\sin\theta\mathbf{e}_\phi \quad (1.270)$$

Next, by differentiating the velocity vector once more, we obtain the acceleration

$$\begin{aligned} \mathbf{a} = \ddot{\mathbf{r}} &= \frac{d}{dt}(\dot{r}\mathbf{e}_r) + \frac{d}{dt}(r\dot{\theta}\mathbf{e}_\theta) + \frac{d}{dt}(r\dot{\phi}\sin\theta\mathbf{e}_\phi) \\ &= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta + \dot{r}\dot{\phi}\sin\theta\mathbf{e}_\phi \\ &\quad + r\dot{\phi}\sin\theta\dot{\mathbf{e}}_\phi + r\dot{\phi}\left(\frac{d}{dt}\sin\theta\right)\mathbf{e}_\phi + r\dot{\phi}\sin\theta\dot{\mathbf{e}}_\phi \end{aligned} \quad (1.271)$$

Note that $\sin\theta$ is the function of θ , while θ is the function of t . By chain rule, we get

$$\left(\frac{d}{dt}\sin\theta\right) = \dot{\theta}\cos\theta \quad (1.272)$$

So, (1.271) becomes

$$\begin{aligned} \ddot{\mathbf{r}} &= \ddot{r}\mathbf{e}_r + (\dot{r}\dot{\theta}\mathbf{e}_\theta + \dot{r}\dot{\phi}\sin\theta\mathbf{e}_\phi) + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + (-r\dot{\theta}^2\mathbf{e}_r + r\dot{\theta}\dot{\phi}\cos\theta\mathbf{e}_\phi) \\ &\quad + \dot{r}\dot{\phi}\sin\theta\mathbf{e}_\phi + r\ddot{\phi}\sin\theta\mathbf{e}_\phi + r\dot{\theta}\dot{\phi}\cos\theta\mathbf{e}_\phi + (-r\dot{\phi}^2\sin^2\theta\mathbf{e}_r - r\dot{\phi}^2\sin\theta\cos\theta\mathbf{e}_\theta) \\ &= (\ddot{r} - r\dot{\theta}^2 - r\dot{\phi}^2\sin^2\theta)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\dot{\phi}^2\sin\theta\cos\theta)\mathbf{e}_\theta + (2\dot{r}\dot{\phi}\sin\theta + \\ &\quad 2r\dot{\theta}\dot{\phi}\cos\theta + r\ddot{\phi}\sin\theta)\mathbf{e}_\phi \end{aligned}$$

The final result of the acceleration in spherical coordinates is

$$\mathbf{a} = \begin{cases} \text{radial : } (\ddot{r} - r\dot{\theta}^2 - r\dot{\phi}^2\sin^2\theta)\mathbf{e}_r \\ \text{colatitudinal : } (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\dot{\phi}^2\sin\theta\cos\theta)\mathbf{e}_\theta \\ \text{azimuthal : } (2\dot{r}\dot{\phi}\sin\theta + 2r\dot{\theta}\dot{\phi}\cos\theta + r\ddot{\phi}\sin\theta)\mathbf{e}_\phi \end{cases} \quad (1.273)$$

Chapter 2

Introduction to Vector Analysis

Abstract They appear extensively in mechanics, electromagnetism, and many other branches of physics. Unlike the simple introduction to vectors and scalars in high school, this chapter will redefine and reconstruct our understanding of vectors through a more abstract yet rigorous algebraic approach. In addition, we will introduce powerful new operators and apply calculus directly to vector quantities. This is not just a technical extension, it is the key to formulating physical laws in a generalized way that is valid under all coordinate systems and boundary conditions, unlike the limited scenarios often seen in high school problems. This chapter might be one of the most challenging parts of this book, but believe me, if you push through, you'll find yourself in one of the most beautiful and exciting areas in mathematical physics...Maybe.

2.1 Basic Vector Operations

2.1.1 Algebra of Vector

Before moving on to the next concept, I'd like to introduce two notations you may not have seen before, the Levi-Civita symbol ϵ and the Kronecker delta δ . In this book, I'll not introduce these two notations in detail because they are related to tensor analysis and are far beyond the scope of first-year classical mechanics. Interested students are encouraged to explore it on their own. These notations are fundamental in expressing vector calculus identities and linear algebra, especially in index notation.

The Levi-Civita symbol ϵ represents the sign of a permutation. It equals $+1$ for even permutations, -1 for odd permutations, and 0 if any two indices are repeated, which can be denoted by

$$\epsilon_{ijk} = \begin{cases} +1 & \text{even permutations} \\ -1 & \text{odd permutations} \\ 0 & \text{repeated indices} \end{cases} \quad (2.1)$$

For example, starting from the number set $(1, 2, 3)$: to reach $(2, 1, 3)$, you need an odd permutation (one swap); to reach $(3, 1, 2)$, you need an even permutation (two swaps). That is

$$\epsilon_{213} = -1; \quad \epsilon_{312} = 1 \quad (2.2)$$

The other notation δ , called **Kronecker delta symbol**, can be denoted by

$$\delta_{ij} = \begin{cases} +1, & \text{When } i = j \\ 0, & \text{When } i \neq j \end{cases} \quad (2.3)$$

In three dimensions, the relationship between the two notations is given by

$$\epsilon_{ijk}\epsilon_{lmn} = \begin{vmatrix} \delta_{il} & \delta_{im} & \delta_{in} \\ \delta_{jl} & \delta_{jm} & \delta_{jn} \\ \delta_{kl} & \delta_{km} & \delta_{kn} \end{vmatrix} \quad (2.4)$$

This determinant form arises from the identity involving Levi-Civita and Kronecker delta symbols. The determinant indicates that there are 3×3 types of combinations of non-zero indices. We can generalize to an N -dimensional system, which is

$$\epsilon_{a_1 a_2 \dots a_n} \epsilon_{b_1 b_2 \dots b_n} = \begin{vmatrix} \delta_{a_1 b_1} & \delta_{a_1 b_2} & \dots & \delta_{a_1 b_n} \\ \delta_{a_2 b_1} & \delta_{a_2 b_2} & \dots & \delta_{a_2 b_n} \\ \vdots & \vdots & \ddots & \vdots \\ \delta_{a_n b_1} & \delta_{a_n b_2} & \dots & \delta_{a_n b_n} \end{vmatrix} \quad (2.5)$$

Under this relationship, we will focus on discussing one specific case.

$$\sum_k \epsilon_{ijk} \epsilon_{lmk} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl} \quad (2.6)$$

We will use this equation shortly. It won't be proved here, but if you're sharp, you might find it pretty obvious.

Let $\Phi : U \subset M \rightarrow \mathbb{R}^n$ be a coordinate chart¹, assigning coordinates

$$(x_1, x_2, \dots, x_n)$$

to each point in the space. Consider two quantities A_1 and A_2 , which are geometric objects defined independently of any coordinate system. Under a change of coordinates

$$x_i \mapsto x'_i(x_1, \dots, x_n),$$

we have the following classifications:

¹ A *coordinate chart* is a map $\Phi : U \subset M \rightarrow \mathbb{R}^n$ that assigns a coordinate tuple $(x_1, x_2, \dots, x_n)$ to each point $p \in U$. Different coordinate charts provide different numerical descriptions of the same geometric point.

- If the numerical value of A_1 remains unchanged,

$$A'_1 = A_1,$$

then A_1 is called a **scalar**.

- If the components of A_2 transform according to

$$A_{2i} = \sum_j T_{ij} A_{2j},$$

where T_{ij} are the components of a matrix representing the coordinate transformation, then A_2 is called a **vector**, i.e. an element of the vector space V . To see more details, you may refer to Chapter 7.

Thus, whether a quantity is a scalar or a vector is determined by its *transformation behavior under coordinate changes*, not by a particular coordinate representation. The unit vector $\mathbf{u}$ is defined to be a vector with the length $|\mathbf{u}| = 1$. We can also make all vectors into a unit vector by dividing its length, that is

$$\mathbf{u} = \frac{\mathbf{A}}{A} \quad (2.7)$$

where A is

$$A = |\mathbf{A}| = \sum_i A_i^2$$

Under the different coordinates, the unit vectors have different kinds of forms; they can be denoted by the following forms²

$$\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}; \quad \hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}; \quad \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \quad (2.8)$$

We have a N -dimensional coordinate $\Phi(x_1, x_2, \dots, x_n)$, it becomes $\Phi'(x'_1, x'_2, \dots, x'_n)$ after an orthogonal rotation(see Section 6.6). We define the cosine between the original coordinate and the new one as

$$\lambda_{ij} = \cos(x'_i, x_j) \quad (2.9)$$

Where (x'_i, x_j) is the angle of the intersection. So, the equation of the coordinate transformation is

$$x'_i = \sum_j \lambda_{ij} x_j \quad (2.10)$$

This means that the new coordinates depend on the projection of the original ones.

² I only list the orthogonal coordinates.

2.1.2 Vector Products

I'm going to introduce the most important products in vector analysis, the **Dot product** and the **Cross product**. First, the dot product of the two vectors $\mathbf{A}$, $\mathbf{B}$ is defined as

$$\boxed{\mathbf{A} \cdot \mathbf{B} := \sum_i A_i B_i} \quad (2.11)$$

A_i is the element of $\mathbf{A}$, while B_i is the element of $\mathbf{B}$. We can use this equation to define the cosine between the two by using both of their **direction cosine** (Thornton 2003), that is

$$\frac{\mathbf{A} \cdot \mathbf{B}}{A \cdot B} = \sum_i \frac{A_i}{A} \frac{B_i}{B} := \sum_i \Lambda_i^A \Lambda_i^B = \cos \theta \quad (2.12)$$

Where Λ is the direction cosine. The meaning of direction cosine is to use the cosine values to illustrate the **unit vector**, for instance, $\mathbf{A}/A = (\Lambda_1^A, \Lambda_2^A, \dots, \Lambda_n^A)$. The result is obvious. If we transform the coordinate from $\Phi(x_1, \dots, x_n)$ to $\Phi'(x'_1, \dots, x'_n)$, then the elements of the vector $\mathbf{A}$, $\mathbf{B}$ become

$$\begin{cases} A'_i \rightarrow \sum_j \lambda_{ij} A_j \\ B'_i \rightarrow \sum_k \lambda_{ik} B_k \end{cases} \quad (2.13)$$

So, the dot product becomes

$$\begin{aligned} \mathbf{A}' \cdot \mathbf{B}' &= \sum_i A'_i B'_i = \sum_i \left(\sum_j \lambda_{ij} A_j \right) \left(\sum_k \lambda_{ik} B_k \right) \\ &= \sum_{jk} \left(\sum_i \lambda_{ij} \lambda_{ik} \right) A_j B_k = \sum_{jk} \delta_{jk} A_j B_k = \sum_j A_j B_j \\ &= \mathbf{A} \cdot \mathbf{B} \end{aligned}$$

It's independent of the coordinate transformation, so we can verify that the dot product is a **scalar**.

The other vector product, the cross product of $\mathbf{A}$ with $\mathbf{B}$ is defined as

$$\boxed{(\mathbf{A} \times \mathbf{B})_i := \sum_{jk} \epsilon_{ijk} A_j B_k} \quad (2.14)$$

In three-dimensional coordinates, it's also denoted by (see chapter 6)

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} \quad (2.15)$$

If we transform the coordinate, like what we did in the dot product, the cross product becomes

$$\begin{aligned} (\mathbf{A}' \times \mathbf{B}')_i &= \sum_{jk} \epsilon'_{ijk} A'_j B'_k \\ &= \sum_{jk} \epsilon'_{ijk} \left(\sum_l \lambda_{jl} A_l \right) \left(\sum_m \lambda_{km} B_m \right) \\ &= \sum_{jklm} \epsilon'_{ijk} \lambda_{jl} \lambda_{km} A_l B_m \neq (\mathbf{A} \times \mathbf{B})_i \end{aligned}$$

It depends on the coordinate transformation, so the cross product is a **vector**.³ Because it's a vector, the cross product must have length denoted by $|\mathbf{A} \times \mathbf{B}|$. I'll focus on the three-dimensional coordinates.

The length of the cross product is

$$\begin{aligned} |\mathbf{A} \times \mathbf{B}|^2 &= \sum_i \left(\sum_{jk} \epsilon_{ijk} A_j B_k \right) \left(\sum_{lm} \epsilon_{ilm} A_l B_m \right) \\ &= \sum_{ijklm} \epsilon_{ijk} \epsilon_{ilm} A_j A_l B_k B_m = \sum_{jklm} (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) A_j A_l B_k B_m \\ &= \sum_j A_j^2 \sum_k B_k^2 - \left(\sum_j A_j B_j \right) \left(\sum_k A_k B_k \right) \end{aligned}$$

Note that $\sum_j A_j^2 \sum_k B_k^2$ is $A^2 B^2$, and $\left(\sum_j A_j B_j \right) \left(\sum_k A_k B_k \right) = (\mathbf{A} \cdot \mathbf{B})^2$. So, the final result is

$$|\mathbf{A} \times \mathbf{B}| = \sqrt{A^2 B^2 - (\mathbf{A} \cdot \mathbf{B})^2} = AB \sqrt{1 - \left(\sum_i \Lambda_i^A \Lambda_i^B \right)^2} = AB \sin \theta$$

That's the familiar form of the length of cross product, namely

$$\boxed{|\mathbf{A} \times \mathbf{B}| = AB \sin \theta} \quad (2.16)$$

³ Actually, the cross product is a *pseudovector* since ϵ_{ijk} is a psedotensor. (See Chapter 7)

2.1.3 Vector Identities

There are four fundamental vector identities presented below. In this section, I will verify each of them using algebraic methods rather than geometric intuition.

1. The definition of the cross product is

$$(\mathbf{B} \times \mathbf{D})_i = \sum_{jk} \epsilon_{ijk} B_j D_k \quad (2.17)$$

Because both $\mathbf{B} \times \mathbf{D}$ and $\mathbf{A}$ are vectors, the dot product of the two is

$$\begin{aligned} \mathbf{A} \cdot (\mathbf{B} \times \mathbf{D}) &= \sum_i A_i (\mathbf{B} \times \mathbf{D})_i = \sum_i A_i \left(\sum_{jk} \epsilon_{ijk} B_j D_k \right) \\ &= \sum_{ijk} \epsilon_{ijk} A_i B_j D_k \end{aligned} \quad (2.18)$$

At the same time, we can use the same way to get

$$\mathbf{D} \cdot (\mathbf{A} \times \mathbf{B}) = \sum_{ijk} \epsilon_{ijk} D_i A_j B_k \quad (2.19)$$

Note that the Levi-Civita symbol equals +1 for even permutations and -1 for odd permutations. The permutations of the indices also satisfy this rule, for example, $(i, j, k) \rightarrow (j, i, k)$ is an odd permutation, while $(i, j, k) \rightarrow (k, i, j)$ is an even permutation. So, we can get that

$$\sum_{ijk} \epsilon_{ijk} D_i A_j B_k = \sum_{ijk} \epsilon_{jki} A_j B_k D_i = \sum_{ijk} \epsilon_{ijk} A_i B_j D_k = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{D})$$

Thus, we can obtain the first identity

$$\boxed{\mathbf{A} \cdot (\mathbf{B} \times \mathbf{D}) = \mathbf{D} \cdot (\mathbf{A} \times \mathbf{B})} \quad (2.20)$$

This is what you've learned in high school; it represents the volume of a parallelepiped in three-dimensional space. We can also define this identity as

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{D}) = \mathbf{D} \cdot (\mathbf{A} \times \mathbf{B}) := \mathbf{ABD} \quad (2.21)$$

2. First, the cross product of $\mathbf{B}$ with $\mathbf{C}$ is

$$(\mathbf{B} \times \mathbf{C})_i = \sum_{jk} \epsilon_{ijk} B_j C_k \quad (2.22)$$

After that, the cross product of $\mathbf{A}$ with $\mathbf{B} \times \mathbf{C}$ is

$$\begin{aligned}
(\mathbf{A} \times (\mathbf{B} \times \mathbf{C}))_i &= \sum_{lm} \epsilon_{ilm} A_l (\mathbf{B} \times \mathbf{C})_m = \sum_{lm} \epsilon_{ilm} A_l \left(\sum_{jk} \epsilon_{mjk} B_j C_k \right) \\
&= \sum_{jklm} \epsilon_{ilm} \epsilon_{jkm} A_l B_j C_k
\end{aligned} \tag{2.23}$$

We need to use (2.6), so (2.23) becomes

$$\sum_{jklm} \epsilon_{ilm} \epsilon_{jkm} A_l B_j C_k = \sum_{jkl} (\delta_{ij} \delta_{lk} - \delta_{ik} \delta_{lj}) A_l B_j C_k \tag{2.24}$$

When $i = j$, $l = k$,

$$(\delta_{ij} \delta_{lk} - \delta_{ik} \delta_{lj}) A_l B_j C_k = A_l C_l B_i;$$

when $i = k$, $l = j$,

$$(\delta_{ij} \delta_{lk} - \delta_{ik} \delta_{lj}) A_l B_j C_k = -A_l B_l C_i.$$

Substituting these results into (2.24), we get

$$\sum_{jkl} (\delta_{ij} \delta_{lk} - \delta_{ik} \delta_{lj}) A_l B_j C_k = \left(\sum_l A_l C_l \right) B_i - \left(\sum_l A_l B_l \right) C_i \tag{2.25}$$

Thus, we verify that

$$(\mathbf{A} \times (\mathbf{B} \times \mathbf{C}))_i = \left(\sum_l A_l C_l \right) B_i - \left(\sum_l A_l B_l \right) C_i \tag{2.26}$$

Note that the elements in the brackets are the dot products, so we can finally claim that

$$\boxed{\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})} \tag{2.27}$$

which is also called the "Bac(k)-Cab Law".

3. First, the cross products are

$$\begin{cases} (\mathbf{A} \times \mathbf{B})_i = \sum_{jk} \epsilon_{ijk} A_j B_k \\ (\mathbf{C} \times \mathbf{D})_i = \sum_{lm} \epsilon_{ilm} C_l D_m \end{cases} \tag{2.28}$$

Then the dot product of $\mathbf{A} \times \mathbf{B}$ with $\mathbf{C} \times \mathbf{D}$ is

$$\begin{aligned}
(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) &= \sum_i \left(\sum_{jk} \epsilon_{ijk} A_j B_k \right) \left(\sum_{lm} \epsilon_{ilm} C_l D_m \right) \\
&= \sum_{jklm} \left(\sum_i \epsilon_{ijk} \epsilon_{ilm} \right) A_j B_k C_l D_m \\
&= \sum_{jklm} (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) A_j B_k C_l D_m
\end{aligned} \tag{2.29}$$

When $j = l$, $k = m$,

$$(\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) A_j B_k C_l D_m = A_j C_j B_k D_k;$$

when $j = m$, $k = l$,

$$(\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) A_j B_k C_l D_m = -A_j D_j B_k C_k.$$

Substituting these results into (2.29), we get

$$\begin{aligned}
\sum_{jklm} (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) A_j B_k C_l D_m &= \left(\sum_{jk} A_j C_j B_k D_k \right) - \left(\sum_{jk} A_j D_j B_k C_k \right) \\
&= \left(\sum_j A_j C_j \right) \left(\sum_k B_k D_k \right) - \left(\sum_j A_j D_j \right) \left(\sum_k B_k C_k \right) \\
&= (\mathbf{A} \cdot \mathbf{C}) (\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D}) (\mathbf{B} \cdot \mathbf{C})
\end{aligned}$$

So we can finally verify the identity

$$\boxed{(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})} \tag{2.30}$$

4. We can directly use the result of 2. That is

$$(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) = [(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{D}] \mathbf{C} - [(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}] \mathbf{D} \tag{2.31}$$

So, the final result is undeniable that

$$\boxed{(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{D}) \mathbf{C} - (\mathbf{A} \cdot \mathbf{C}) \mathbf{D}} \tag{2.32}$$

This is probably the easiest proof in this book. :) You need to cherish it!

2.2 Definition of Vector Differentiation

2.2.1 Vector Field and Curve

A vector function $\mathbf{A}$ is a function that assigns a unique vector to each point in its domain D , which lies in a plane or in space. The collection of all such output vectors forms the image of the vector function, which in physics is often called a **field**.

More formally, a vector function is a function from a domain $D \subset \mathbb{R}^n$ to $\mathbb{R}^m$, where each point $\mathbf{r} \in D$ corresponds to a unique vector $\mathbf{F}(\mathbf{r}) \in \mathbb{R}^m$.

In a three-dimensional system, a vector function $\mathbf{A}$ consists of three scalar functions along the three coordinate directions. That is

$$\mathbf{A}(x_1, x_2, x_3) = A_1(x_1, x_2, x_3)\mathbf{e}_1 + A_2(x_1, x_2, x_3)\mathbf{e}_2 + A_3(x_1, x_2, x_3)\mathbf{e}_3 \quad (2.33)$$

Where A_1, A_2, A_3 are scalar functions. So, in an N -dimensional system, it becomes

$$\mathbf{A}(x_1, x_2, \dots, x_n) = \sum_i A_i(x_1, \dots, x_n)\mathbf{e}_i \quad (2.34)$$

For example, a vector function $\mathbf{F}(x, y, z) = xyz^2\mathbf{e}_1 + x^3z\mathbf{e}_2 + (2y + 3x)\mathbf{e}_3$, the vector of the point $(2, 1, -1)$ is

$$\mathbf{F}(2, 1, -1) = 2\mathbf{e}_1 - 8\mathbf{e}_2 + 8\mathbf{e}_3$$

In addition, we can use the vector and parameter t to describe the graph in space. A parameter t is a controlling quantity used to label or describe the position or variation of a geometric object, such as a point, a curve, or a surface. While a parameter does not necessarily carry physical meaning by itself, varying the parameter allows us to trace out the entire shape step by step.

We start with a curve C . We can construct a vector function using a parameter as its domain. By assigning a unique vector to each value of the parameter and collecting all such vectors, we generate a curve in space. This method of describing a curve is called parametrization. It can be denoted as

$$C : \{\mathbf{r}(t) = x(t)\mathbf{e}_1 + y(t)\mathbf{e}_2 + z(t)\mathbf{e}_3, \quad t \in [a, b]\} \quad (2.35)$$

where $\mathbf{r}(t)$ is the parametrization of the curve. In mathematics, it is defined as

$$\mathbf{r}(t) : \mathbb{R} \supset I \rightarrow \mathbb{R}^n \quad (2.36)$$

More specifically, the idea of parametrization is that each parameter value corresponds to a point in space, and each point is associated with a direction. By smoothly connecting these directional changes, we form a winding curve, one that can be seen as composed of a sequence of varying tangent vectors along its path.

2.2.2 The Definition of Limit and Differentiation

After discussing the vector function, I will introduce the differentiation of a vector function in space. Starting with the single variable function $\mathbf{F}(t)$, the limit of it is defined as

$$\lim_{t \rightarrow t_0} \mathbf{F}(t) = \mathbf{L} \iff |\mathbf{F}(t) - \mathbf{L}| = 0, \quad t \rightarrow t_0 \quad (2.37)$$

It means that when t approaches t_0 , the angle of intersection of vector $\mathbf{F}$ and $\mathbf{L}$, in other words, the components of $\mathbf{F}$ approach those of $\mathbf{L}$. That is

$$\lim_{t \rightarrow t_0} F_i = L_i \quad (2.38)$$

So that the limit of a vector function is

$$\lim_{t \rightarrow t_0} \mathbf{F}(t) = \left(\lim_{t \rightarrow t_0} F_1(t), \lim_{t \rightarrow t_0} F_2(t), \lim_{t \rightarrow t_0} F_3(t) \right) = (L_1, L_2, L_3) \quad (2.39)$$

Assume that $\mathbf{F}(t)$ is continuous at $t = t_0$, when the limit exists, we can define the derivative of vector function as

$$\left. \frac{d\mathbf{F}}{dt} \right|_{t=t_0} = \lim_{\|\Delta t\| \rightarrow 0} \frac{\mathbf{F}(t_0 + \Delta t) - \mathbf{F}(t_0)}{\Delta t} \quad (2.40)$$

Geometrically, the derivative of a vector function represents the tangent vector to the curve traced out by $\mathbf{F}$. Expanding the components of $\mathbf{F}$, (2.40) becomes

$$\lim_{\|\Delta t\| \rightarrow 0} \frac{(F_1(t_0 + \Delta t) - F_1(t), F_2(t_0 + \Delta t) - F_2(t), F_3(t_0 + \Delta t) - F_3(t))}{\Delta t} \quad (2.41)$$

So, the derivative of the vector function $\mathbf{F}$ is

$$\left. \frac{d\mathbf{F}}{dt} \right|_{t=t_0} = \left(\left. \frac{dF_1}{dt} \right|_{t=t_0}, \left. \frac{dF_2}{dt} \right|_{t=t_0}, \left. \frac{dF_3}{dt} \right|_{t=t_0} \right) \quad (2.42)$$

For example, a position vector function $\mathbf{r}(t)$ ⁴ Then velocity and the acceleration of it is

$$\begin{cases} \mathbf{v}(t) = \frac{d\mathbf{r}(t)}{dt} := \dot{\mathbf{r}} = (\dot{x}, \dot{y}, \dot{z}) \\ \mathbf{a}(t) = \frac{d^2\mathbf{r}(t)}{dt^2} := \ddot{\mathbf{r}} = (\ddot{x}, \ddot{y}, \ddot{z}) \end{cases}$$

We can apply the same logic to multiple variables and vector functions. Consider a vector function $\mathbf{F}(x_1, x_2, \dots, x_n)$, the partial differentiation is

⁴ position vector function represents $\mathbf{r}(t) = (x(t), y(t), z(t))$, it's a fixed usage.

$$\frac{\partial \mathbf{F}}{\partial x_j} = \sum_i \frac{\partial F_i}{\partial x_j} \mathbf{e}_i \quad (2.43)$$

Similarly, the total differential is denoted by

$$\begin{aligned} d\mathbf{F} &= \left(\sum_i \frac{\partial F_1}{\partial x_i} dx_i, \sum_i \frac{\partial F_2}{\partial x_i} dx_i, \dots, \sum_i \frac{\partial F_n}{\partial x_i} dx_i \right) \\ &= \sum_{ij} \frac{\partial F_j}{\partial x_i} dx_i \mathbf{e}_j \end{aligned} \quad (2.44)$$

The meaning of this is that a small change in a vector field is the sum of small directional changes in each component. Note that the total differential of a vector field is still a vector.

In addition, let $\mathbf{r}(t)$ be a differentiable vector function defined with respect to time t . We can expand it around the point t_0 using a Taylor series, that is

$$\mathbf{r}(t_0 + dt) = \mathbf{r}(t_0) + \left. \frac{d\mathbf{r}}{dt} \right|_{t_0} dt + \frac{1}{2!} \left. \frac{d^2\mathbf{r}}{dt^2} \right|_{t_0} (dt)^2 + \dots \quad (2.45)$$

We take the approximation up to the second term. In fact, the second term can be regarded as the infinitesimal displacement $d\mathbf{r}$ along the curve, which is essentially the core idea of the total differential. To simplify the following proof, we can rewrite the second term as

$$\left. \frac{d\mathbf{r}}{dt} \right|_{t_0} dt := \Delta\mathbf{r} \big|_{\|\Delta\| \rightarrow 0} \quad (2.46)$$

Below are six important identities. I'll show you the proofs of (3) to (6) since (1), (2) are simple and obvious.

1. $\frac{d}{dt}(\mathbf{A}(t) \pm \mathbf{B}(t)) = \frac{d\mathbf{A}}{dt} \pm \frac{d\mathbf{B}}{dt}$
2. $\frac{d}{dt}(f(t)\mathbf{A}(t)) = \frac{df}{dt}\mathbf{A} + f\frac{d\mathbf{A}}{dt}$
3. $\frac{d}{dt}(\mathbf{A}(t) \cdot \mathbf{B}(t)) = \frac{d\mathbf{A}}{dt} \cdot \mathbf{B} + \mathbf{A} \cdot \frac{d\mathbf{B}}{dt}$

We expand the dot product of $\mathbf{A}$ with $\mathbf{B}$

$$\begin{aligned}
\frac{d}{dt}(\mathbf{A}(t) \cdot \mathbf{B}(t)) &= \lim_{\|\Delta\| \rightarrow 0} \frac{(\mathbf{A} + \Delta\mathbf{A}) \cdot (\mathbf{B} + \Delta\mathbf{B}) - \mathbf{A} \cdot \mathbf{B}}{\Delta t} \\
&= \lim_{\|\Delta\| \rightarrow 0} \frac{\mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \Delta\mathbf{B} + \mathbf{B} \cdot \Delta\mathbf{A} + \Delta\mathbf{A} \cdot \Delta\mathbf{B} - \mathbf{A} \cdot \mathbf{B}}{\Delta t} \\
&= \lim_{\|\Delta\| \rightarrow 0} \left(\mathbf{A} \cdot \frac{\Delta\mathbf{B}}{\Delta t} + \mathbf{B} \cdot \frac{\Delta\mathbf{A}}{\Delta t} + \frac{\Delta\mathbf{A} \cdot \Delta\mathbf{B}}{\Delta t} \right)
\end{aligned}$$

Since the third term is sufficiently small, we neglect it in the mathematical approximation, and thus obtain the final result that

$$\frac{d}{dt}(\mathbf{A}(t) \cdot \mathbf{B}(t)) = \frac{d\mathbf{A}}{dt} \cdot \mathbf{B} + \mathbf{A} \cdot \frac{d\mathbf{B}}{dt} \quad (2.47)$$

$$4. \frac{d}{dt}(\mathbf{A}(t) \times \mathbf{B}(t)) = \frac{d\mathbf{A}}{dt} \times \mathbf{B} + \mathbf{A} \times \frac{d\mathbf{B}}{dt}$$

Apply the same method to prove this, so we need to expand the cross product

$$\begin{aligned}
\frac{d}{dt}(\mathbf{A}(t) \times \mathbf{B}(t)) &= \lim_{\|\Delta\| \rightarrow 0} \frac{(\mathbf{A} + \Delta\mathbf{A}) \times (\mathbf{B} + \Delta\mathbf{B}) - \mathbf{A} \times \mathbf{B}}{\Delta t} \\
&= \lim_{\|\Delta\| \rightarrow 0} \frac{\mathbf{A} \times \mathbf{B} + \mathbf{A} \times \Delta\mathbf{B} + \mathbf{B} \times \Delta\mathbf{A} + \Delta\mathbf{A} \times \Delta\mathbf{B} - \mathbf{A} \times \mathbf{B}}{\Delta t} \\
&= \lim_{\|\Delta\| \rightarrow 0} \left(\mathbf{A} \times \frac{\Delta\mathbf{B}}{\Delta t} + \mathbf{B} \times \frac{\Delta\mathbf{A}}{\Delta t} + \frac{\Delta\mathbf{A} \times \Delta\mathbf{B}}{\Delta t} \right)
\end{aligned}$$

Since the third term is sufficiently small, we neglect it in the mathematical approximation, and thus obtain the final result that

$$\frac{d}{dt}(\mathbf{A}(t) \times \mathbf{B}(t)) = \frac{d\mathbf{A}}{dt} \times \mathbf{B} + \mathbf{A} \times \frac{d\mathbf{B}}{dt} \quad (2.48)$$

$$5. \frac{d}{dt}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} \times \frac{d\mathbf{C}}{dt} + \mathbf{A} \cdot \frac{d\mathbf{B}}{dt} \times \mathbf{C} + \frac{d\mathbf{A}}{dt} \cdot \mathbf{B} \times \mathbf{C}$$

Let the cross product $\mathbf{B} \times \mathbf{C}$ be the vector $\mathbf{r}$, use the result of (3), we can get

$$\frac{d}{dt}(\mathbf{A} \cdot \mathbf{r}) = \frac{d\mathbf{A}}{dt} \cdot \mathbf{r} + \mathbf{A} \cdot \frac{d\mathbf{r}}{dt} \quad (2.49)$$

Substituting $\mathbf{B} \times \mathbf{C} = \mathbf{r}$ into (2.49), we can finally get

$$\frac{d\mathbf{A}}{dt} \cdot \mathbf{B} \times \mathbf{C} + \mathbf{A} \cdot \frac{d(\mathbf{B} \times \mathbf{C})}{dt} = \frac{d\mathbf{A}}{dt} \cdot \mathbf{B} \times \mathbf{C} + \mathbf{A} \cdot \left(\frac{d\mathbf{B}}{dt} \times \mathbf{C} + \mathbf{B} \times \frac{d\mathbf{C}}{dt} \right)$$

So

$$\frac{d}{dt}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} \times \frac{d\mathbf{C}}{dt} + \mathbf{A} \cdot \frac{d\mathbf{B}}{dt} \times \mathbf{C} + \frac{d\mathbf{A}}{dt} \cdot \mathbf{B} \times \mathbf{C} \quad (2.50)$$

$$6. \frac{d}{dt}(\mathbf{A} \times (\mathbf{B} \times \mathbf{C})) = \mathbf{A} \times \left(\mathbf{B} \times \frac{d\mathbf{C}}{dt} \right) + \mathbf{A} \times \left(\frac{d\mathbf{B}}{dt} \times \mathbf{C} \right) + \frac{d\mathbf{A}}{dt} \times (\mathbf{B} \times \mathbf{C})$$

Similarly, let the cross product $\mathbf{B} \times \mathbf{C}$ be the vector $\mathbf{r}$, and use the result of (4), we can get

$$\frac{d}{dt}(\mathbf{A} \times \mathbf{r}) = \frac{d\mathbf{A}}{dt} \times \mathbf{r} + \mathbf{A} \times \frac{d\mathbf{r}}{dt} \quad (2.51)$$

Substituting $\mathbf{B} \times \mathbf{C} = \mathbf{r}$ into (2.51), we can finally get

$$\frac{d\mathbf{A}}{dt} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{A} \times \frac{d(\mathbf{B} \times \mathbf{C})}{dt} = \frac{d\mathbf{A}}{dt} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{A} \times \left(\frac{d\mathbf{B}}{dt} \times \mathbf{C} + \mathbf{B} \times \frac{d\mathbf{C}}{dt} \right)$$

So the final result is solved

$$\frac{d}{dt}(\mathbf{A} \times (\mathbf{B} \times \mathbf{C})) = \mathbf{A} \times \left(\mathbf{B} \times \frac{d\mathbf{C}}{dt} \right) + \mathbf{A} \times \left(\frac{d\mathbf{B}}{dt} \times \mathbf{C} \right) + \frac{d\mathbf{A}}{dt} \times (\mathbf{B} \times \mathbf{C}) \quad (2.52)$$

2.2.3 Directional Derivative

Consider an infinitesimal variation at $z = f(x, y) \in \mathbb{R}^3$ shown below,

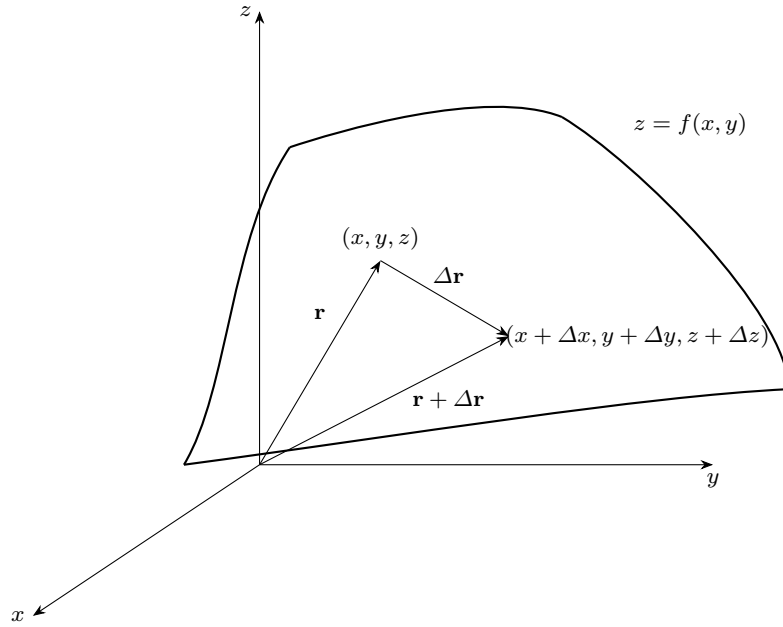


Fig. 2.1 Infinitesimal variation on the surface $z = f(x, y)$

In this figure, the original position vector $\mathbf{r} = (x, y, z)$ undergoes a small variation in the direction of $\Delta\mathbf{r}$. We regard this variation as a progression in a certain direction $\mathbf{u}$. Along this direction, the change in the function's value is equal to

$$\Delta f = f(x + \Delta x, y + \Delta y, z + \Delta z) - f(x, y, z) \quad (2.53)$$

By the linear approximation of a multivariable function via Taylor expansion (see Chapter 1), we get

$$\Delta f \approx \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y + \frac{\partial f}{\partial z} \Delta z \quad (2.54)$$

Dividing both sides by $\Delta r = |\Delta\mathbf{r}|$, that is

$$\frac{\Delta f}{\Delta r} = \frac{\partial f}{\partial x} \frac{\Delta x}{\Delta r} + \frac{\partial f}{\partial y} \frac{\Delta y}{\Delta r} + \frac{\partial f}{\partial z} \frac{\Delta z}{\Delta r} \quad (2.55)$$

As the point approaches infinitely close, the direction of variation becomes the tangent vector at the initial point. So, if we take $\|\Delta\| \rightarrow 0$, it becomes

$$\frac{df}{dr} = \lim_{\|\Delta\| \rightarrow 0} \frac{\Delta f}{\Delta r} \quad (2.56)$$

Then we can define df/ds as the **Directional Derivative** along the tangent vector $d\mathbf{r}/dr$. The directional derivative of f at a point in a direction $\mathbf{u}$ measures how fast the function changes in that direction, where $\mathbf{u} = d\mathbf{r}/dr$. More formally and rigorously,

the directional derivative is defined as

$$D_{\mathbf{u}}f(\mathbf{r}) = \lim_{\alpha \rightarrow 0} \frac{f(\mathbf{r} + \alpha \mathbf{u}) - f(\mathbf{r})}{\alpha} \quad (2.57)$$

Where $\mathbf{r} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ and $\mathbf{u} = (dx_1/dr, dx_2/dr, \dots, dx_n/dr)$ is the vector along the tangent direction. By the linear approximation of a multivariable function via Taylor expansion, we get

$$\begin{aligned} D_{\mathbf{u}}f(\mathbf{r}) &= \lim_{\alpha \rightarrow 0} \frac{f(\mathbf{r}) + \left(\frac{\partial f}{\partial x_1} \alpha \frac{dx_1}{dr} + \dots + \frac{\partial f}{\partial x_n} \alpha \frac{dx_n}{dr} \right) - f(\mathbf{r})}{\alpha} \\ &= \lim_{\alpha \rightarrow 0} \frac{\frac{\partial f}{\partial x_1} \alpha \frac{dx_1}{dr} + \dots + \frac{\partial f}{\partial x_n} \alpha \frac{dx_n}{dr}}{\alpha} \\ &= \frac{\partial f}{\partial x_1} \frac{dx_1}{dr} + \dots + \frac{\partial f}{\partial x_n} \frac{dx_n}{dr} \end{aligned}$$

So, this result can be written as the dot product of two vectors, that is

$$\begin{aligned} \frac{\partial f}{\partial x_1} \frac{dx_1}{dr} + \dots + \frac{\partial f}{\partial x_n} \frac{dx_n}{dr} &= \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right) \cdot \left(\frac{dx_1}{dr}, \frac{dx_2}{dr}, \dots, \frac{dx_n}{dr} \right) \\ &= \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right) \cdot \mathbf{u} \end{aligned}$$

At first glance, this may seem like a “strange” vector because it consists entirely of partial derivatives of f . However, this vector is of fundamental importance and is known as the **Gradient**. That is,

$$\left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right) = \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \dots, \frac{\partial}{\partial x_n} \right) f := \nabla f \quad (2.58)$$

The notation ∇ is a differential operator; it is

$$\nabla := \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \dots, \frac{\partial}{\partial x_n} \right) \quad (2.59)$$

We can use this operator to redefine the directional derivative as

$$D_{\mathbf{u}}f(\mathbf{r}) = \nabla f \cdot \mathbf{u} \quad (2.60)$$

It is given by the projection of the gradient ∇ onto the unit direction $\mathbf{u}$. This formula arises naturally from the first-order Taylor expansion of f , and geometrically, it tells us how much of the gradient’s influence is felt along direction $\mathbf{u}$.

Note that the vector $\mathbf{u}$ here refers to an arbitrary direction vector. When we previously referred to it as a "tangent vector," we did not mean the tangent vector to the level curve $f(x, y, z) = C$ specifically. Rather, $\mathbf{u}$ lies in the tangent plane of the surface $z = f(x, y)$, and represents a general direction along which one may move tangentially to the surface. The tangent vector to the level curve is simply one special case of such a direction.

2.2.4 Gradient

After introducing the directional derivatives, I'll show you the meaning. We know that the gradient can be denoted by (2.59), but we don't know why it could be described as a vector. We have discussed the coordinate transformation in Chapter 1, which is

$$\frac{\partial \Phi'}{\partial x'_i} = \sum_j \frac{\partial \Phi}{\partial x_j} \frac{\partial x_j}{\partial x'_i}$$

In such a transformation, we can write down the inverse transformation

$$x_j = \sum_k \lambda_{kj} x'_k \quad (2.61)$$

So, the partial differentiation of x_j can be denoted by

$$\frac{\partial x_j}{\partial x'_i} = \frac{\partial}{\partial x'_i} \left(\sum_k \lambda_{kj} x'_k \right) = \sum_k \lambda_{kj} \frac{\partial x'_k}{\partial x'_i} = \sum_k \lambda_{kj} \delta_{ki} \quad (2.62)$$

The result is now evident. Based on the property of the Kronecker delta, only the term with $i = k$ remains, while all others vanish. That is

$$\frac{\partial x_j}{\partial x'_i} = \lambda_{ij} \quad (2.63)$$

Therefore, we can substitute this result into the coordinate transformation equation to obtain

$$\frac{\partial \Phi'}{\partial x'_i} = \sum_j \lambda_{ij} \frac{\partial \Phi}{\partial x_j} \quad (2.64)$$

It follows the rule of vectors(see Chapter 2-1), the term $\partial \Phi / \partial x_j$ is the j -th component of the vector. Since the coordinate Φ is a scalar function, the term, $\partial / \partial x_j$ must be a vector. In this way, we can verify that the gradient ∇ is truly a vector, and it's defined as

$$\text{grad} = \nabla := \sum_i \frac{\partial}{\partial x_i} \mathbf{e}_i \quad (2.65)$$

Now, let us return to the concept of the total differential for a function of n variables $\Phi(x_1, \cdot, x_n)$. The total differential is

$$d\phi = \sum_i \frac{\partial \Phi}{\partial x_i} dx_i$$

Substituting the gradient into the total differential, we can rewrite the infinitesimal displacement of an n -variable function as

$$d\phi = \nabla \Phi \cdot d\mathbf{r} = |\nabla \Phi| |d\mathbf{r}| \cos \theta \quad (2.66)$$

where $d\mathbf{r} = (dx_1, dx_2, \dots, dx_n)$. The meaning of the vector $d\mathbf{r}$ shares a similarity with the direction vector $\mathbf{u}$ in directional derivatives: both describe a certain direction of variation. However, $d\mathbf{r}$ is not a unit vector, but rather a collection of infinitesimal displacements along each coordinate direction. Therefore, the dot product of the gradient and $d\mathbf{r}$ yields the total amount of change in the function. On the other hand, $\mathbf{u}$ is a unit vector, so the dot product with the gradient yields the rate of change in that specific direction.

The direction of $\nabla \Phi$ is fixed, and the vector $d\mathbf{r}$ is arbitrary. So, the infinitesimal displacements $d\phi$ have different values with different directions of $d\mathbf{r}$. If we choose $d\mathbf{r} \parallel \nabla \Phi$, the angle $\theta = 0$, so

$$\max(d\phi) = |\nabla \Phi| |d\mathbf{r}| \quad (2.67)$$

Equation (2.67) implies that the gradient of a function points in the direction of the greatest change at a given point. Therefore, by choosing a small displacement in this direction, one can ensure movement in the direction of the most rapid increase of the function. While we choose $d\mathbf{r} \perp \nabla \Phi$,

$$d\phi = |\nabla \Phi| |d\mathbf{r}| \cos \left(\frac{\pi}{2} \right) = 0$$

In other words, if we move along the direction tangent to the level curve at a given point, the function value clearly remains unchanged since we are staying on the same contour. This is similar to hiking along a trail that maintains a constant elevation; you will never reach the summit or descend to the base, because you are moving horizontally along the same height.

On the other hand, the gradient direction is naturally perpendicular to the tangent direction of the level curve. In the hiking analogy, this corresponds to walking directly uphill or downhill, moving in the direction of steepest ascent or descent, which brings you to the summit or base as quickly as possible.

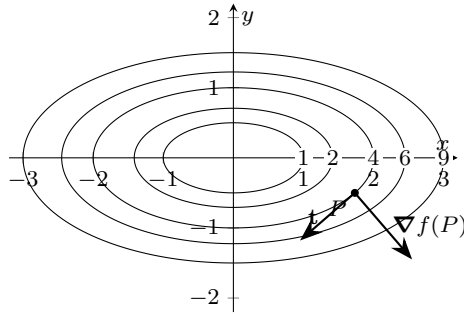


Fig. 2.2 Gradient points in the direction of steepest increase for $f(x, y) = x^2 + 4y^2$.

There is another operator I should introduce, the **Laplacian**. It is defined as

$$\nabla \cdot \nabla = \nabla^2 := \sum_i \frac{\partial^2}{\partial x_i^2} \quad (2.68)$$

When the Laplacian operates on a scalar(function), we have

$$\nabla^2 V = \sum_i \frac{\partial^2 V}{\partial x_i^2} = -\frac{\rho(\mathbf{r})}{\epsilon_0} \quad (2.69)$$

in electrodynamics. However, within the scope of undergraduate classical mechanics, this operator does not typically appear. Therefore, it will not be discussed in detail in this book.

2.2.5 Definition of Divergence and Curl

Gradient is an operator that describes the variation of a scalar function. To describe the vector field, we need to introduce another operator, **divergence** and **curl**.

First, consider a vector field $\mathbf{F} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots$, then its divergence $\text{div} \mathbf{F} = \nabla \cdot \mathbf{F}$ is defined as

$$\text{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \sum_i \frac{\partial F_i}{\partial x_i} \quad (2.70)$$

For example, $\mathbf{F} = (x^2 - y^2)\mathbf{e}_1 + (x^2 - z^2)\mathbf{e}_2 + (y^2 - z^2)\mathbf{e}_3$, the the divergence of this vector function is

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2 - y^2) + \frac{\partial}{\partial y}(x^2 - z^2) + \frac{\partial}{\partial z}(y^2 - z^2) = 2x - 2z$$

Second, consider a 3-dimensional vector field $\mathbf{F} = F_1 \mathbf{e}_1 + F_2 \mathbf{e}_2 + F_3 \mathbf{e}_3$, then its curl $\text{curl} \mathbf{F} = \nabla \times \mathbf{F}$ is defined as

$$\text{curl} \mathbf{F} = \nabla \times \mathbf{F} = \sum_{ijk} \epsilon_{ijk} \frac{\partial F_k}{\partial x_j} \mathbf{e}_i \quad (2.71)$$

At this point, I will not yet introduce the geometric meanings of divergence and curl, because they require vector integration in order to be properly explained. If you're too curious to wait, feel free to skip ahead to Gauss' Law and Stokes' Theorem!

Example 2.1 There is a vector function $\mathbf{F} = F_x \mathbf{e}_1 + F_y \mathbf{e}_2 + F_z \mathbf{e}_3$ and a scalar function f , prove that

$$\begin{cases} \nabla \times (\nabla f) = 0 \\ \nabla \cdot (\nabla \times \mathbf{F}) = 0 \end{cases} \quad (2.72)$$

Solution 2.1

1. The first equation is really **important**, we will check it later. By the definition of the curl and the gradient, we can get

$$\nabla \times (\nabla f) = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix}$$

So, it becomes

$$\left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \mathbf{e}_1 + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) \mathbf{e}_2 + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \mathbf{e}_3 = 0$$

2. it is obvious that

$$\nabla \cdot (\nabla \times \mathbf{F}) = \frac{\partial}{\partial x} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) = 0$$

2.2.6 Identities of Vector Operators

Before proving the multiple identities of gradient, divergence, and curl, I must first introduce a completely new rule of operation that you have likely never encountered before to make the proof clearer: the **Einstein summation convention**.

The core idea of this convention is to perform a summation over any index that appears twice in a single term. So, the summation must be redefined as

$$\sum_i A_i B_i := A_i B_i \quad (2.73)$$

and the partial differentiation is redefined as

$$\sum_i \frac{\partial}{\partial x_i} f(x_1, x_2, \dots, x_n) := \partial_i f(x_1, x_2, \dots, x_n) \quad (2.74)$$

Here, ∂_i is shorthand for $\partial/\partial x_i$. From now on, we will use this simplified notation. The dot product and cross product become

$$\begin{cases} \mathbf{A} \cdot \mathbf{B} = A_i B_i \\ (\mathbf{A} \times \mathbf{B})_k = \epsilon_{kij} A_i B_j \end{cases} \quad (2.75)$$

Now that we have defined the Einstein summation rule and simplified notation, we can re-express all our familiar vector operators in this unified index form

$$\begin{cases} (\nabla f)_i = \partial_i f \\ \nabla \cdot \mathbf{A} = \partial_i A_i \\ (\nabla \times \mathbf{A})_k = \epsilon_{kij} \partial_i A_j \end{cases} \quad (2.76)$$

And (2.6) can be denoted by

$$\epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl} \quad (2.77)$$

For example, the derivation of the length of the cross product can be rewritten as

$$\begin{aligned} |\mathbf{A} \times \mathbf{B}|^2 &= (\epsilon_{ijk} A_j B_k) (\epsilon_{ilm} A_l B_m) = \epsilon_{ijk} \epsilon_{ilm} A_j A_l B_k B_m \\ &= (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) A_j A_l B_k B_m = A_j^2 B_k^2 - (A_j B_j) (A_k B_k) \end{aligned}$$

After the introduction, I can show you 15 identities and prove them all.

1. $\nabla c = 0$, $c \in \mathbb{R}$

It is obvious that the differentiation of constant c equals 0, namely

$$\nabla c = \partial_i c \mathbf{e}_i = 0$$

2. $\nabla(cf) = c\nabla f$ where f is a function.

We directly expand the gradient that

$$\nabla(cf) = \partial_i cf \mathbf{e}_i = c \partial_i f \mathbf{e}_i = c \nabla f$$

3. $\nabla(f \pm g) = \nabla f \pm \nabla g$

$$\nabla(f \pm g) = \partial_i (f \pm g) \mathbf{e}_i = \partial_i f \mathbf{e}_i \pm \partial_i g \mathbf{e}_i = \nabla f \pm \nabla g$$

$$4. \nabla(fg) = f\nabla g + g\nabla f$$

$$\begin{aligned}\nabla(fg) &= \partial_i(fg)\mathbf{e}_i = (f\partial_i g + g\partial_i f)\mathbf{e}_i \\ &= f\partial_i g \mathbf{e}_i + g\partial_i f \mathbf{e}_i = f\nabla g + g\nabla f\end{aligned}$$

$$5. \nabla(f/g) = \frac{1}{g^2}(g\nabla f - f\nabla g)$$

$$\begin{aligned}\nabla\left(\frac{f}{g}\right) &= \frac{1}{g}\nabla f + f\nabla\left(\frac{1}{g}\right) = \frac{1}{g}\partial_i f \mathbf{e}_i + f\partial_i g^{-1} \mathbf{e}_i \\ &= \frac{1}{g}\partial_i f \mathbf{e}_i - f\frac{1}{g^2}\partial_i g \mathbf{e}_i \frac{1}{g}\nabla f - \frac{f}{g^2}\nabla g = \frac{1}{g^2}(g\nabla f - f\nabla g)\end{aligned}$$

$$6. \nabla(f(u)) = f'(u)\nabla u$$

$$\nabla(f(u)) = \partial_i f(u)\mathbf{e}_i = f'(u)\partial_i u \mathbf{e}_i = f'(u)\nabla u$$

The proofs of the above identities are similar to what you did in differentiation, so I don't want to stay here so long. The next proof is the climax of this section.

$$7. \nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} + (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{B} \times (\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla \times \mathbf{B})$$

First, we need to write down the expansion of the gradient term, which is

$$[\nabla(\mathbf{A} \cdot \mathbf{B})]_i = \partial_i(A_j B_j) = B_j \partial_i A_j + A_j \partial_i B_j \quad (2.78)$$

We have to use some skills to deal with terms $\partial_i A_j$ and $\partial_i B_j$. So,

$$\partial_i A_j = \frac{1}{2}(\partial_i A_j + \partial_j A_i) + \frac{1}{2}(\partial_i A_j - \partial_j A_i) \quad (2.79)$$

The term $\partial_i A_j - \partial_j A_i$ can be written as the combination of Kronecker delta symbols, that is

$$\begin{aligned}\partial_i A_j - \partial_j A_i &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\partial_l A_m \\ &= (\epsilon_{ijk}\epsilon_{lmk})\partial_l A_m \\ &= \epsilon_{ijk}(\epsilon_{lmk}\partial_l A_m)\end{aligned} \quad (2.80)$$

Note that the curl $(\nabla \times \mathbf{A})_k = \epsilon_{kij}\partial_i A_j$, so (2.80) can be rewritten as

$$\partial_i A_j - \partial_j A_i = \epsilon_{ijk}(\nabla \times \mathbf{A})_k \quad (2.81)$$

In the same way, we obtain

$$\partial_i B_j - \partial_j B_i = \epsilon_{ijk}(\nabla \times \mathbf{B})_k \quad (2.82)$$

Substituting (2.81) and (2.82) into (2.78), we get

$$\begin{aligned}
B_j \partial_i A_j + A_j \partial_i B_j &= \frac{1}{2}(\partial_i A_j + \partial_j A_i)B_j + \frac{1}{2}(\partial_i B_j + \partial_j B_i)A_j \\
&+ \frac{1}{2}[\epsilon_{ijk}B_j(\nabla \times \mathbf{A})_k] + \frac{1}{2}[\epsilon_{ijk}A_j(\nabla \times \mathbf{B})_k]
\end{aligned} \tag{2.83}$$

And we separate the term $B_j \partial_i A_j$ with $A_j \partial_i B_j$, we get the equations

$$\begin{aligned}
B_j \partial_i A_j &= \frac{1}{2}B_j \partial_i A_j + \frac{1}{2}B_j \partial_j A_i + \frac{1}{2}\epsilon_{ijk}B_j(\nabla \times \mathbf{A})_k \\
\Rightarrow B_j \partial_i A_j &= B_j \partial_j A_i + \epsilon_{ijk}B_j(\nabla \times \mathbf{A})_k
\end{aligned} \tag{2.84}$$

Similarly,

$$A_j \partial_i B_j = A_j \partial_j B_i + \epsilon_{ijk}A_j(\nabla \times \mathbf{B})_k \tag{2.85}$$

Substituting both into the beginning equation (2.78), we can get

$$[\nabla(\mathbf{A} \cdot \mathbf{B})]_i = (B_j \partial_j)A_i + (A_j \partial_j)B_i + \epsilon_{ijk}B_j(\nabla \times \mathbf{A})_k + \epsilon_{ijk}A_j(\nabla \times \mathbf{B})_k \tag{2.86}$$

We can finally get this "beautiful" result, and that is what we want

$$\boxed{\nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} + (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{B} \times (\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla \times \mathbf{B})} \tag{2.87}$$

This is the first major identity in this book that fully integrates the core principles of index notation, partial derivatives, symmetry and antisymmetry, and the Levi-Civita structure. By proving it purely algebraically, we cross the threshold from vector intuition into rigorous mathematical formulation.

1. $\nabla \cdot (c\mathbf{A}) = c\nabla \cdot \mathbf{A}$

$$\nabla \cdot (c\mathbf{A}) = \partial_i c A_i = c \partial_i A_i = c \nabla \cdot \mathbf{A}$$

2. $\nabla \cdot (\mathbf{A} \pm \mathbf{B}) = \nabla \cdot \mathbf{A} \pm \nabla \cdot \mathbf{B}$

$$\nabla \cdot (\mathbf{A} \pm \mathbf{B}) = \partial_i (A_i \pm B_i) = \partial_i A_i \pm \partial_i B_i = \nabla \cdot \mathbf{A} \pm \nabla \cdot \mathbf{B}$$

3. $\nabla \cdot (f\mathbf{A}) = f\nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla f$

$$\nabla \cdot (f\mathbf{A}) = \partial_i (f A_i) = f \partial_i A_i + A_i \partial_i f = f \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla f$$

1. $\nabla \times (c\mathbf{A}) = c\nabla \times \mathbf{A}$

$$\nabla \times (c\mathbf{A}) = \epsilon_{ijk} \partial_j c A_k = c \epsilon_{ijk} \partial_j A_k = c \nabla \times \mathbf{A}$$

2. $\nabla \times (\mathbf{A} \pm \mathbf{B}) = \nabla \times \mathbf{A} \pm \nabla \times \mathbf{B}$

$$\nabla \times (\mathbf{A} \pm \mathbf{B}) = \epsilon_{ijk} \partial_j (A_k \pm B_k) = \epsilon_{ijk} \partial_j A_k \pm \epsilon_{ijk} \partial_j B_k = \nabla \times \mathbf{A} \pm \nabla \times \mathbf{B}$$

$$3. \nabla \times (f\mathbf{A}) = f\nabla \times \mathbf{A} + \nabla f \times \mathbf{A}$$

$$\begin{aligned} \nabla \times (f\mathbf{A}) &= \epsilon_{ijk} \partial_j (fA_k) \mathbf{e}_i = \epsilon_{ijk} (f \partial_j A_k + A_k \partial_j f) \mathbf{e}_i \\ &= f \epsilon_{ijk} \partial_j A_k \mathbf{e}_i + (\epsilon_{ijk} \partial_j f) A_k \mathbf{e}_i = f \nabla \times \mathbf{A} + \nabla f \times \mathbf{A} \end{aligned}$$

$$4. \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B}$$

$$\begin{aligned} \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \partial_i \epsilon_{ijk} A_j B_k = \epsilon_{ijk} \partial (A_j B_k) \\ &= \epsilon_{ijk} B_k \partial_i A_j + \epsilon_{ijk} A_j \partial_i B_k \end{aligned} \quad (2.88)$$

We need to change the order of the indices of ϵ . That is, $\epsilon_{ijk} = \epsilon_{kij}$ because of the even permutation; $\epsilon_{ijk} = -\epsilon_{jik}$ because of the odd permutation. So (2.88) becomes

$$\epsilon_{ijk} B_k \partial_i A_j + \epsilon_{ijk} A_j \partial_i B_k = B_k (\epsilon_{kij} \partial_i A_j) - A_j (\epsilon_{jik} \partial_i B_k) \quad (2.89)$$

Note that the reason why we change the order of the indices is to get the curl term satisfying the expansion of it, so

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B} \quad (2.90)$$

$$5. \mathbf{A} \times (\nabla \times \mathbf{A}) = \frac{1}{2} \nabla A^2 - (\mathbf{A} \cdot \nabla) \mathbf{A}$$

We first expand the cross product term that

$$[\mathbf{A} \times (\nabla \times \mathbf{A})]_i = \epsilon_{ijk} A_j (\nabla \times \mathbf{A})_k = \epsilon_{ijk} A_j (\epsilon_{klm} \partial_l A_m) \quad (2.91)$$

Note that we have another structure of $\epsilon_{ijk} \epsilon_{klm}$, and it equals to $\epsilon_{ijk} \epsilon_{lmk}$, so it can be written as

$$\begin{aligned} \epsilon_{ijk} A_j (\epsilon_{klm} \partial_l A_m) &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_j \partial_l A_m \\ &= A_m \partial_i A_m - (A_l \partial_l) A_i \\ &= \frac{1}{2} \partial_i (A_m^2) - (A_l \partial_l) A_i \end{aligned} \quad (2.92)$$

That is a detail, $\partial_i (A_m \times A_m) = A_m \partial_i A_m + A_m \partial_i A_m$, so the term $A_m \partial_i A_m = 1/2 \partial_i (A_m \times A_m)$. After all, the result is

$$\mathbf{A} \times (\nabla \times \mathbf{A}) = \frac{1}{2} \nabla A^2 - (\mathbf{A} \cdot \nabla) \mathbf{A} \quad (2.93)$$

Where A is the length of the vector $\mathbf{A}$.

2.3 Vector Integral

2.3.1 Definition of Vector Integration

After becoming familiar with vector differentiation and vector operators, we are now ready to introduce one of the most important mathematical tools in vector analysis: vector integration. We begin with the fundamental theorem of calculus for vector fields, whose definition parallels that of scalar differentiation. The indefinite integral of the vector field $\mathbf{A}(t) = (A_1(t), A_2(t), A_3(t))$ is

$$\int \mathbf{A}(t)dt = \left(\int A_1(t)dt, \int A_2(t)dt, \int A_3(t)dt \right) \quad (2.94)$$

And the definite integral is defined as

$$\int_a^b \mathbf{A}(t)dt = \left(\int_a^b A_1(t)dt, \int_a^b A_2(t)dt, \int_a^b A_3(t)dt \right) \quad (2.95)$$

Let $\mathbf{F} = d/dt \mathbf{A}(t)$, so that $\mathbf{A}(t)$ is the antiderivative vector function of $\mathbf{F}$. By (2.94) and (2.95), we can apply the fundamental theorem of calculus to the functions, namely

$$\begin{cases} \int \mathbf{F}(t)dt = \int \frac{d\mathbf{A}(t)}{dt}dt = \mathbf{A}(t) + C \\ \int_a^b \mathbf{F}(t)dt = \mathbf{A}(b) - \mathbf{A}(a) \end{cases} \quad (2.96)$$

Vector integration also possesses integral identities analogous to the differential identities discussed earlier. In fact, one can derive similar results by simply integrating the differential identities on pages 66 and 67 in reverse. Therefore, we will not repeat those derivations here.

However, there is one important property that deserves special attention: integration by parts for vector fields. There are two vector functions $\mathbf{A}(t)$ and $\mathbf{B}(t)$, each of which depends on the parameter t . By (2.47), we have

$$\frac{d}{dt}(\mathbf{A}(t) \cdot \mathbf{B}(t)) = \frac{d\mathbf{A}(t)}{dt} \cdot \mathbf{B}(t) + \mathbf{A}(t) \cdot \frac{d\mathbf{B}(t)}{dt}$$

Multiply both sides by dt and integrate both sides with respect to t ,

$$\int d(\mathbf{A}(t) \cdot \mathbf{B}(t)) = \mathbf{A}(t) \cdot \mathbf{B}(t) = \int \dot{\mathbf{A}}(t) \cdot \mathbf{B}(t)dt + \int \mathbf{A}(t) \cdot \dot{\mathbf{B}}(t)dt$$

Thus, we obtain

$$\int \dot{\mathbf{A}}(t) \cdot \mathbf{B}(t)dt = \mathbf{A}(t) \cdot \mathbf{B}(t) - \int \mathbf{A}(t) \cdot \dot{\mathbf{B}}(t)dt \quad (2.97)$$

Similarly, the integration by parts formula involving the cross product also exhibits analogous properties

$$\int \dot{\mathbf{A}}(t) \times \mathbf{B}(t) dt = \mathbf{A}(t) \times \mathbf{B}(t) - \int \mathbf{A}(t) \times \dot{\mathbf{B}}(t) dt \quad (2.98)$$

For example, there is an integral

$$\int \mathbf{A} \times \ddot{\mathbf{A}} dt$$

Apply the integration by parts, we can evaluate the integral

$$\int \mathbf{A} \times \ddot{\mathbf{A}} dt = \mathbf{A} \times \dot{\mathbf{A}} - \int \dot{\mathbf{A}} \times \dot{\mathbf{A}} dt = \mathbf{A} \times \dot{\mathbf{A}} + \mathbf{C}$$

Where $\mathbf{C}$ is a constant vector.

2.3.2 Line Integral and Kinetic Energy

Back in calculus, we already encountered the equation of the length L of a curve $S : \{(x, y) | y = f(x), x \in \mathbb{R}\}$ in the two-dimensional plane

$$L = \int_a^b \sqrt{(dx)^2 + (dy)^2} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b \sqrt{1 + (f'(x))^2} dx$$

Now, we extend this concept to three-dimensional space. Since a curve is a one-dimensional object, it can still be described using a single parameter t , which determines the position of a point $(x(t), y(t), z(t))$ along the curve. Thus, a space curve(or Path) can be represented as a set of points in the following form

$$\mathcal{P} = \{ (x, y, z) \mid x = x(t), y = y(t), z = z(t), t \in [a, b] \} \quad (2.99)$$

Or

$$\mathcal{P} = \{ \mathbf{r}(t) = x(t)\mathbf{e}_1 + y(t)\mathbf{e}_2 + z(t)\mathbf{e}_3, a \leq t \leq b \} \quad (2.100)$$

So, the **arc length**($s(t)$) of the curve $\mathcal{P}$ is similar to the curve in the two-dimensional plane, which is

$$s(t) = \int_0^t \sqrt{(dx)^2 + (dy)^2 + (dz)^2} = \int_0^t \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Note that $\mathbf{r}'(t) = (dx/dt, dy/dt, dz/dt)$, so $\sqrt{(dx/dt)^2 + (dy/dt)^2 + (dz/dt)^2} = |\mathbf{dr}/dt|$. Then the curve length in the three-dimensional plane can be denoted by

$$s(t) = \int_0^t \left| \frac{d\mathbf{r}(t)}{dt} \right| dt = \int_0^t |\mathbf{r}'(t)| dt \quad (2.101)$$

In the physical perspective, the derivative of $\mathbf{r}(t)$ with respect to t means the velocity of a particle, and dt means infinitesimal time difference, so $|\mathbf{r}'(t)|dt$ represents the infinitesimal distance. The integral adds all the infinitesimal distances together, so we obtain the arc length. By FTC I, we have

$$\frac{ds(t)}{dt} = \frac{d}{dt} \int_0^t |\mathbf{r}'(t)| dt = |\mathbf{r}'(t)| \Rightarrow ds = |\mathbf{r}'(t)| dt \quad (2.102)$$

We call the notation ds as **arc length element**.

After understanding the concept of the arc length element, we can proceed to introduce one of the most important types of integrals in mechanics and electromagnetism - the **line integral**.

We begin with the first type of line integral, namely, the line integral of a scalar function. Let $\mathcal{P}$ be a piecewise smooth curve defined on the interval $[a, b]$, and let f be a continuous function defined on the curve $\mathcal{P}$, that is, the domain of f satisfies $D_f \subseteq \mathcal{P}$. Then the line integral of f along the curve $\mathcal{P}$ is given by

$$\int_{\mathcal{P}} f(x, y, z) ds = \int_{\mathcal{P}} f(\mathbf{r}(t)) |\mathbf{r}'(t)| dt \quad (2.103)$$

In other words, the line integral of a scalar function represents the total accumulation of function values along the curve. From a physical perspective, if $f(x, y, z)$ denotes the linear density of a rope, then the line integral of f over the curve gives the total mass of the rope.

For example, $\mathcal{P}$ is the upper semicircle of radius a , which is

$$\mathcal{P} = \{(x, y) | x = a \cos t, y = a \sin t, 0 \leq t \leq \pi\}$$

Then evaluate the line integral

$$\int_{\mathcal{P}} x^2 y ds$$

We need to compute the arc length element ds

$$ds = |\mathbf{r}'(t)| dt = a dt$$

So the line integral becomes

$$\int_{\mathcal{P}} x^2 y ds = \int_0^\pi (a^2 \cos^2 t) (a \sin t) a dt = \frac{2}{3} a^4$$

Now it is time for vector fields to take the stage! After discussing line integrals of scalar functions, we have seen that the key idea behind a line integral is to map the domain of the function onto a curve and then perform the integration *along* the curve $\mathcal{P}$. Similarly,

we can apply the same idea to vector-valued functions.

Vector fields were originally developed to address problems in physics, particularly the computation of work. In earlier physics courses, we encountered work calculations restricted to special cases, such as constant force along straight-line paths. However, such formulas could not be applied to compute the work done along arbitrary paths in space. To overcome this limitation, we developed the concept of the **line integral of a vector field**. There are a vector function $\mathbf{F} = (F_1, F_2, F_3)$ and a infinitesimal tangent vector $d\mathbf{r} = (dx, dy, dz)$ of the curve $\mathcal{P}$. The work done by force, or called line integral of a vector field is defined as

$$\int_{\mathcal{P}} \mathbf{F} \cdot d\mathbf{r} = \int_{\mathcal{P}} F_1 dx + F_2 dy + F_3 dz \quad (2.104)$$

In other words, we sum the dot product between the vector field and each infinitesimal displacement along the curve, according to the parameter domain of the curve. From a physical perspective, this corresponds to adding up the infinitesimal amounts of work done at each moment, resulting in the total work performed along the path.

We can rewrite (2.104) as

$$\int_{\mathcal{P}} F_1 dx + F_2 dy + F_3 dz = \int_{\mathcal{P}} \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt = \int_{\mathcal{P}} (F_1, F_2, F_3) \cdot \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right) dt$$

So, the generalization of the line integral of the vector function can be denoted by

$$\boxed{\int_{\mathcal{P}} \mathbf{F} \cdot d\mathbf{r} = \int_{\mathcal{P}} \mathbf{F} \cdot \mathbf{r}'(t) dt} \quad (2.105)$$

After understanding these theorems and rules, we now proceed to generalize the expression for kinetic energy along an arbitrary path. Consider a force given by $\mathbf{F} = m\mathbf{a} = m\ddot{\mathbf{r}}$, and let $\mathcal{P}$ be an arbitrary path parameterized with respect to time t . Assume that the curve $\mathcal{P}$ is smooth on its domain, and it is given by

$$\mathcal{P} = \{\mathbf{r}(t) = x(t), y(t), z(t), t_1 \leq t \leq t_2\}$$

Then the work done by the force $\mathbf{F}$ is the line integral (2.105), we substitute $\mathbf{F} = m\mathbf{a} = m\ddot{\mathbf{r}}$ into it, it becomes

$$W = \int_{t_1}^{t_2} m\ddot{\mathbf{r}} \cdot \dot{\mathbf{r}} dt \quad (2.106)$$

We need to apply the integration by parts, such as (2.98), we obtain

$$\begin{aligned} W &= \int_{t_1}^{t_2} m\ddot{\mathbf{r}} \cdot \dot{\mathbf{r}} dt \\ &= m(\dot{\mathbf{r}} \cdot \dot{\mathbf{r}}) \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} m\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} dt \end{aligned}$$

Since the integral $\int_{t_1}^{t_2} m\ddot{\mathbf{r}} \cdot \dot{\mathbf{r}} dt$ and $\int_{t_1}^{t_2} m\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} dt$ are the same, we let the integrals be I . Therefore

$$2I = m(\dot{\mathbf{r}} \cdot \dot{\mathbf{r}}) \Big|_{t_1}^{t_2} \Rightarrow W = I = \frac{1}{2}m|\dot{\mathbf{r}}|^2 \Big|_{t_1}^{t_2} = \frac{1}{2}m[v(t_2)]^2 - \frac{1}{2}m[v(t_1)]^2$$

So, we can finally get the **The work–energy theorem** in a rigorous way, that is

$$W_F = \int_{\mathcal{P}} \mathbf{F} \cdot d\mathbf{r} = \frac{1}{2}m[v(t_2)]^2 - \frac{1}{2}m[v(t_1)]^2 \quad (2.107)$$

Before going to the next part, I must introduce another integrating notation – the closed integral $\oint$. As long as the starting point and the ending point of the curve are the same, then we can rewrite the line integral (2.105) as

$$\oint_{\mathcal{P}} \mathbf{F} \cdot d\mathbf{r} \quad (2.108)$$

Moreover, when the line integral is taken along the boundary of a region, it can also be expressed as a closed line integral. Consider a continuous surface with boundary $\mathcal{S}$ in $\mathbb{R}^3$, its boundary can be denoted by $\partial\mathcal{S}$, the line integral becomes

$$\oint_{\partial\mathcal{S}} \mathbf{F} \cdot d\mathbf{r} \quad (2.109)$$

In a closed line integral, the curve $\mathcal{P}$ must have a specified direction, that is, we must define whether the path is traversed in a clockwise or counterclockwise manner. This orientation is usually indicated by the symbol $\oint$, which denotes integration along a *closed* curve with a given direction. For example, in the plane, a **positively oriented** curve typically means counterclockwise, while a negatively oriented one is clockwise. Although the exact orientation becomes very important in later theorems (such as Green's and Stokes' Theorem), for now, you only need to recognize this symbol and understand that direction matters when computing line integrals over closed curves.

2.3.3 Conservative Field and Potential Functions

There exists a continuous scalar function $V(x, y, z)$, by the properties of the gradient ∇ , we obtain the line integral of the gradient function of V , which is

$$\begin{aligned} \int_A^B \nabla V \cdot d\mathbf{r} &= \int_A^B \left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z} \right) \cdot d\mathbf{r} \\ &= \int_A^B \left(\frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz \right) \end{aligned}$$

Where A and B represent the position. Note that the term $\frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz$ is the total derivatives of the scalar function dv , that is

$$dV = \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz$$

So, the integral can be written as

$$\int_A^B \nabla V \cdot d\mathbf{r} = \int_A^B dv = V(B) - V(A) \quad (2.110)$$

By (2.110), we can observe that the line integral ∇V is independent of the path it traverses through. The function only depends on the initial and the final positions A and B . That is to say, when a vector function $\mathbf{F}$ can be written in the form ∇V ,⁵ the vector function $\mathbf{F}$ is called a **Conservative Vector Field**, and the scalar function V is the **Potential Function** of $\mathbf{F}$. It is similar to potential energy in textbooks.

There are some methods to tell whether the vector functions are conservative or not. By the identities (2.72), we've known that

$$\nabla \times \nabla f = \mathbf{0}$$

Where f is a scalar function. So, when a vector function $\mathbf{F}$ satisfy that

$$\nabla \times \mathbf{F} = \mathbf{0} \quad (2.111)$$

Then the vector function $\mathbf{F}$ is a conservative function since there exists a potential function V satisfying $\nabla V = \mathbf{F}$, so we obtain

$$\nabla \times \mathbf{F} = \nabla \times (\nabla V) = \mathbf{0} \quad (2.112)$$

On the other hand, when a line integral of a conservative field $\mathbf{F}$ along a closed curve, then the line integral is 0. It is given by

$$\oint_{\mathcal{P}} \mathbf{F} \cdot d\mathbf{r} := \int_{\mathcal{O}}^{\mathcal{O}} \mathbf{F} \cdot d\mathbf{r} = \int_{\mathcal{O}}^{\mathcal{O}} \nabla V \cdot d\mathbf{r} = V(\mathcal{O}) - V(\mathcal{O}) = 0 \quad (2.113)$$

This integral may be mentioned in Chapter 2.6. Back to the physical perspective, the work W done by force $\mathbf{F}$ is

$$W = \int_{\mathcal{P}} \mathbf{F} \cdot d\mathbf{r}$$

We find that if the work done by a force $\mathbf{F}$ depends only on the initial and final positions, and not on the path taken, then the force is called a **conservative force**. In such cases, we can define a type of energy, denoted by V , which allows us to express the work in terms of the force itself. This quantity is known as the **potential energy**. That is

⁵ You need to write down $\mathbf{F} = -\nabla V$ only when you are dealing with specific physical problems.

$$\int_{x_1}^{x_2} \mathbf{F} \cdot d\mathbf{r} := V(x_1) - V(x_2) \Leftrightarrow \boxed{\mathbf{F} = -\nabla V} \quad (2.114)$$

Note that, since potential energy is defined based on the work done by a conservative force, its value corresponds to the initial potential energy minus the final potential energy. Therefore, there is a negative sign difference between this definition and equation (2.114). In other words, the negative sign between conservative force and potential energy arises naturally from the definition.

So, we can thus verify the law of conservation of energy by this definition in vector analysis. Suppose an object is subjected only to conservative forces $\mathbf{F}$, by (2.107), we can first write down the work-energy theorem

$$W = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \frac{1}{2}m[v(t_2)]^2 - \frac{1}{2}m[v(t_1)]^2 \quad (2.115)$$

Then, by (2.114), we obtain

$$W = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = - \int_{\mathbf{r}_1}^{\mathbf{r}_2} \nabla V \cdot d\mathbf{r} = V(\mathbf{r}_1) - V(\mathbf{r}_2) \quad (2.116)$$

The work value (2.115) is equal to (2.116), we have

$$\begin{aligned} \frac{1}{2}m[v(t_2)]^2 - \frac{1}{2}m[v(t_1)]^2 &= V(\mathbf{r}(t_1)) - V(\mathbf{r}(t_2)) \\ \Rightarrow \frac{1}{2}m[v(t_1)]^2 + V(\mathbf{r}(t_1)) &= \frac{1}{2}m[v(t_2)]^2 + V(\mathbf{r}(t_2)) \end{aligned} \quad (2.117)$$

When the potential energy depends on time t , which is written as $V(\mathbf{r}, t)$, and $\mathbf{r} = (x_1, x_2, \dots, x_n)$. Let the trajectory be parameterized by time,

$$\mathcal{P} : t \in [t_1, t_2] \mapsto (\mathbf{r}(t), t)$$

so the total derivative of the potential energy becomes

$$dV \Big|_{\mathcal{P}} = (\partial_i V dx_i + \partial_t V dt)_{\mathcal{P}} = \nabla V \cdot d\mathbf{r} + \partial_t V dt \quad (2.118)$$

Therefore, we obtain

$$\nabla V \cdot d\mathbf{r} = dV \Big|_{\mathcal{P}} - \partial_t V dt$$

Since the work done by a conservative force depends only on the spatial displacement, we integrate only the spatial component:

$$\begin{aligned} W &= - \int_{\mathcal{P}} \nabla V \cdot d\mathbf{r} = - \left(\int_{\mathcal{P}} dV - \int_{t_1}^{t_2} \partial_t V dt \right) \\ &= V(\mathbf{r}(t_1), t_1) - V(\mathbf{r}(t_2), t_2) + \int_{t_1}^{t_2} \partial_t V dt \end{aligned}$$

So the conservation of energy must be rewritten as

$$\left[T_2 + V(\mathbf{r}(t_2), t_2) \right] - \left[T_1 + V(\mathbf{r}(t_1), t_1) \right] = \int_{t_1}^{t_2} \partial_t V dt$$

If you differentiate the total energy $E = T + V$ with respect to t , you may get the result of the power P

$$P = \frac{dE}{dt} = \partial_t V \quad (2.119)$$

This part only focuses on the mathematical principle of the relationship between energy and line integrals, I'll not discuss the physical meaning in detail. Those who are interested may refer to your own classical mechanics textbook.

2.3.4 Surface integral

Unlike the line integral, the surface integral is used to calculate the flux and the surface area in three dimensions. In this section, we focus on discussing surfaces in three-dimensional coordinates. In Cartesian coordinates $\mathbb{R}^3$, the surface is often written as $z = f(x, y)$, so the surface can be denoted by a set consisting of the elements of the function, which is

$$\mathcal{S} = \{(x, y, z) | z = f(x, y), f : \mathbb{R}^2 \rightarrow \mathbb{R}\} \quad (2.120)$$

Just like curves in space, surfaces can also be parameterized. However, we need two parameters to describe a surface since it is a two-dimensional structure. So, the surfaces can be rewritten as

$$\mathcal{S} = \{(x, y, z) | x = x(u, v), y = y(u, v), z = z(u, v)\} \quad (2.121)$$

To calculate the surface area or the surface integral of scalar functions, we need to understand the relationship between the area element after parametrization and the original surface element da .

To begin, let us first consider the case of a non-parameterized surface $z = f(x, y)$. There is a point $(x, y, f(x, y))$ on the surface, and we can obtain the tangent vector by taking the partial derivative $\mathbf{r}_x$, $\mathbf{r}_y$ of x and y .

$$\begin{aligned} \mathbf{r}_x &= (x + dx, y, f(x + dx, y)) - (x, y, f(x, y)) \\ &= (dx, 0, df) = \left(dx, 0, \frac{\partial f}{\partial x} dx \right) = \left(1, 0, \frac{\partial f}{\partial x} \right) dx \end{aligned} \quad (2.122)$$

Similarly, tangent vector $\mathbf{r}_y$ is

$$\mathbf{r}_y = \left(0, 1, \frac{\partial f}{\partial y} \right) dy \quad (2.123)$$

The area, which is also known as the surface element ds , enclosed by these two vectors $\mathbf{r}_x$ and $\mathbf{r}_y$ is

$$da = |\mathbf{r}_x \times \mathbf{r}_y| = \left| \left(1, 0, \frac{\partial f}{\partial x} \right) \times \left(0, 1, \frac{\partial f}{\partial y} \right) \right| dx dy$$

So

$$da = \sqrt{1 + \left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2} dx dy \quad (2.124)$$

If we consider a parameterized surface $S : \mathbf{r}(u, v) = (x(u, v), y(u, v), z(u, v))$, then the tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$ of the point $(x(u, v), y(u, v), z(u, v))$ satisfy

$$\begin{aligned} \mathbf{R}_u &= (x(u + du, v), y(u + du, v), z(u + du, v)) - (x(u, v), y(u, v), z(u, v)) \\ &= (dx, dy, dz) = \left(\frac{\partial x}{\partial u} du, \frac{\partial y}{\partial u} du, \frac{\partial z}{\partial u} du \right) = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right) du \end{aligned} \quad (2.125)$$

Then we define the tangent vector $\mathbf{r}_u$ as

$$\mathbf{r}_u = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right)$$

Similarly, tangent vector $\mathbf{r}_v$ is

$$\mathbf{r}_v = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right) \quad (2.126)$$

The surface element ds enclosed by these two vectors $\mathbf{r}_u$ and $\mathbf{r}_v$ is

$$da = |\mathbf{R}_u \times \mathbf{R}_v| = |\mathbf{r}_u \times \mathbf{r}_v| du dv = \left| \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right) \times \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right) \right| du dv \quad (2.127)$$

In general, we can also use Jacobian to describe a surface when we meet a higher-dimensional coordinate. While the full Jacobian determinant does not directly apply to a mapping from the parameterization $\mathbf{r}(u, v) : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and $\Omega \rightarrow \mathcal{S}$, we can still use the Jacobian-type structure to compute the surface area element. We define a Jacobian and its transpose matrix

$$J := \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{bmatrix}_{3 \times 2} \Rightarrow J^T = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{bmatrix}_{2 \times 3} \quad (2.128)$$

So, the matrix product of J^T and J is

$$\begin{aligned} \det(J^T J) &= \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{vmatrix} = \begin{vmatrix} |\mathbf{r}_u|^2 & \mathbf{r}_u \cdot \mathbf{r}_v \\ \mathbf{r}_v \cdot \mathbf{r}_u & |\mathbf{r}_v|^2 \end{vmatrix} \\ &= [|\mathbf{r}_u|^2 |\mathbf{r}_v|^2 - (\mathbf{r}_u \cdot \mathbf{r}_v)^2] (du dv)^2 = |\mathbf{r}_u \times \mathbf{r}_v|^2 (du dv)^2 \end{aligned}$$

Here, we illustrate only the case of a parametrized surface in three-dimensional space, so the surface element in space can be expressed using the Jacobian as

$$\boxed{da = \sqrt{\det(J^T J)} du dv} \quad (2.129)$$

In fact, this approach offers greater generality and consistency with the underlying geometric structure. This method of defining surface elements naturally extends to higher-dimensional volume integrals and manifold theory. In higher-dimensional spaces, it is no longer feasible to rely on geometric visualization to interpret the relationship between the original coordinates and their parametrized counterparts, and therefore, the Jacobian becomes an essential tool in capturing the local distortion of volume or area under the transformation.

After discussing the surface element in space, we can further apply this to integrals. Just as line integrals are used to compute the length of a curve, we can apply a similar method to compute the area of a surface. The only difference is that, since the surface is described by two variables, a double integral is required.⁶ So, the area of the surface S is

$$\boxed{\text{Surface} = \int_S da = \int_{\Omega} |\mathbf{r}_u \times \mathbf{r}_v| du dv = \int_{\Omega} \sqrt{\det(J^T J)} du dv} \quad (2.130)$$

We can now further extend the concept to the surface integral of a scalar function. Suppose a scalar function f is a continuous real-valued function defined on a surface S , then the surface integral of f over S is given by

$$\begin{aligned} \int_S f(x, y, z) da &= \int_{\Omega} f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| du dv \\ &= \int_{\Omega} f(\mathbf{r}(u, v)) \sqrt{\det(J^T J)} du dv \end{aligned} \quad (2.131)$$

⁶ To simplify the discussion, I'll just use $\int$ to represent all the integrals. So, you need to be aware of the integration elements and the limits of integration.

The final part involves the surface integral of a vector field. In this context, the surface element is treated as a vector-valued area element $d\mathbf{a}$, defined as the unit normal vector to the surface patch S , which means $d\mathbf{a} = \mathbf{n}da$. The unit normal vector $\mathbf{n}$ is

$$\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|} \quad (2.132)$$

By (2.132), the vector-valued area element $d\mathbf{a}$ becomes

$$d\mathbf{a} = \mathbf{n}da = \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|} \cdot |\mathbf{r}_u \times \mathbf{r}_v| du dv = \mathbf{r}_u \times \mathbf{r}_v du dv \quad (2.133)$$

Over this surface, suppose a vector field $\mathbf{F} = P\mathbf{e}_1 + Q\mathbf{e}_2 + R\mathbf{e}_3$ passes through the region. Then, the surface integral of $\mathbf{F}$ over S is

$$\int_S \mathbf{F} \cdot d\mathbf{a} = \int_S \mathbf{F} \cdot \mathbf{n}da = \int_\Omega \mathbf{F} \cdot \mathbf{r}_u \times \mathbf{r}_v du dv := \int_\Omega [\mathbf{F}, \mathbf{r}_u, \mathbf{r}_v] du dv \quad (2.134)$$

We can observe that the meaning of the surface integral of the vector function is to project the vector function $\mathbf{F}$ onto the direction of the vector-valued area element $d\mathbf{a}$. It describes the component $\mathbf{F} \cos \theta$ of the vector field passing through each surface element, then the integral means to add all the dot products of $\mathbf{F}$ and $d\mathbf{a}$. Overall, this integral represents one of the most important physical quantities in fluid mechanics and electromagnetism, namely, the **flux**. As illustrated in the figure below.

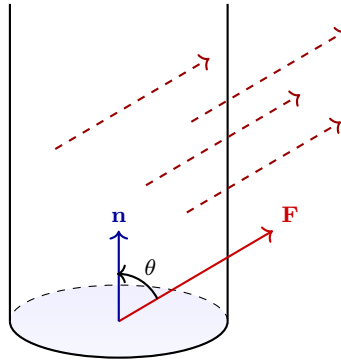


Fig. 2.3 Geometric illustration of the field through curve integrals (fluxes)

We take a classic physical problem for example. Calculate the flux(Φ) of the vector field $\mathbf{F} = q/r^3 \mathbf{r}$ through the spherical surface $S : x^2 + y^2 + z^2 = a^2$, where $\mathbf{r}$ denotes the position vector (x, y, z) . Since the outward unit normal vector on the sphere coincides with the radial direction of the position vector, we observe that $\mathbf{n} = \mathbf{r}/r$. So, the surface integral

$$\oint_S \mathbf{F} \cdot d\mathbf{a} = \oint_S \mathbf{F} \cdot \mathbf{n} da = \oint_S \frac{q}{r^2} da = \frac{q}{r^2} \oint_S da$$

Because the surface area of a sphere $S = 4\pi r^2$, so the flux(Φ) of the vector field through the spherical surface is

$$\frac{q}{r^2} \oint_S da = 4\pi q$$

2.4 Green's Theorem

Green's Theorem is one of the most fundamental and important components in all of vector analysis. It describes how the line integral of a vector field $\mathbf{F}$ over the boundary $\partial\Omega$ of a bounded region $\Omega \subset \mathbb{R}^2$ can be transformed into a double integral over the region Ω itself. Specifically, it generalizes the concept of the line integral introduced in equation (2.109) to a closed path surrounding a two-dimensional region and expresses the result in a remarkably elegant and simplified form.

Consider a region Ω in the plane defined by:

$$\Omega = \{ (x, y) \mid x \in [a, b], y \in [g_1(x), g_2(x)] \} \quad (2.135)$$

as illustrated in the figure below.

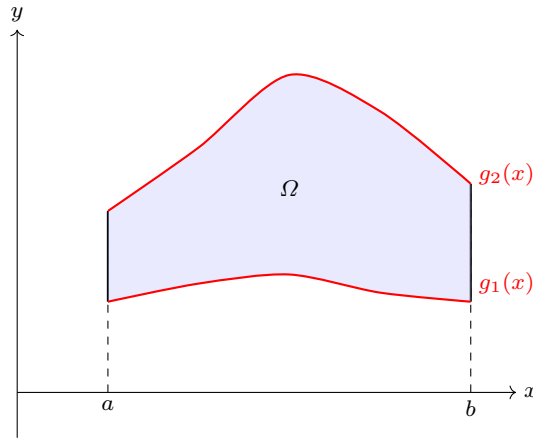


Fig. 2.4 Region Ω bounded by the curves $g_1(x)$ and $g_2(x)$, for use in Green's Theorem

We now consider a vector field $\mathbf{F} = (P(x, y), Q(x, y))$ passing through the region Ω , then the closed line integral can be written as

$$\oint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{r} = \oint_{\partial\Omega} P(x, y)dx + Q(x, y)dy \quad (2.136)$$

We can divide this line integral into four parts, corresponding to the four segments of the boundary. These are $(a, g_1(x)) \rightarrow (b, g_1(x))$; $(b, g_1(x)) \rightarrow (b, g_2(x))$; $(b, g_2(x)) \rightarrow (a, g_2(x))$; $(a, g_2(x)) \rightarrow (a, g_1(x))$. So, we can extend (2.136) to

$$\begin{aligned} \oint_{\partial\Omega} P(x, y)dx + Q(x, y)dy &= \int_a^b P(x, g_1(x))dx + \int_{g_1(x)}^{g_2(x)} Q(b, y)dy \\ &\quad + \int_b^a P(x, g_2(x))dx + \int_{g_2(x)}^{g_1(x)} Q(a, y)dy \end{aligned}$$

By rewriting the integrals of the P and Q terms with matching limits of integration, we can combine them into a single expression.

$$\begin{aligned} &\int_a^b P(x, g_1(x))dx + \int_{g_1(x)}^{g_2(x)} Q(b, y)dy + \int_b^a P(x, g_2(x))dx + \int_{g_2(x)}^{g_1(x)} Q(a, y)dy \\ &= \int_{g_1(x)}^{g_2(x)} [Q(b, y) - Q(a, y)]dy - \int_a^b [P(x, g_2(x)) - P(x, g_1(x))]dx \end{aligned}$$

Note that the terms $[Q(b, y) - Q(a, y)]$ and $[P(x, g_2(x)) - P(x, g_1(x))]$ can be rewritten as the integrals that

$$\begin{cases} Q(b, y) - Q(a, y) = \int_a^b \frac{\partial Q(x, y)}{\partial x} dx \\ P(x, g_2(x)) - P(x, g_1(x)) = \int_{g_1(x)}^{g_2(x)} \frac{\partial P(x, y)}{\partial y} dy \end{cases} \quad (2.137)$$

Substituting (2.137) into the original integrals, we obtain

$$\begin{aligned} &\int_{g_1(x)}^{g_2(x)} [Q(b, y) - Q(a, y)]dy - \int_a^b [P(x, g_2(x)) - P(x, g_1(x))]dx \\ &= \int_{g_1(x)}^{g_2(x)} \left(\int_a^b \frac{\partial Q(x, y)}{\partial x} dx \right) dy - \int_a^b \left(\int_{g_1(x)}^{g_2(x)} \frac{\partial P(x, y)}{\partial y} dy \right) dx \\ &= \int_{g_1(x)}^{g_2(x)} \int_a^b \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \end{aligned}$$

Here, we rewrite $dx dy$ as the area element da . Thus, combining the previous derivations and concepts, we have derived the most well-known Green's Theorem.

$$\oint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{r} = \int_{\Omega} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) da = \int_{\Omega} \left| \frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \right| \begin{vmatrix} P \\ Q \end{vmatrix} da \quad (2.138)$$

For example, let $\mathcal{P}$ be the boundary of the triangle enclosed by the x -axis, the y -axis, and the line $x + y = 1$. Then, evaluate the integral

$$\oint_{\mathcal{P}} xy^2 dx + (x^2 + y) dy$$

First, let $xy^2 = P$, $x^2 + y = Q$, so the integral becomes

$$\oint_{\mathcal{P}} P dx + Q dy = \int_{\Omega} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) da = \int_{\Omega} (2x - 2xy) dx dy$$

Next, we need to determine the range of the two variables, $0 \leq x \leq 1$ and $0 \leq y \leq 1 - x$. substituting it into the integral, we obtain

$$\begin{aligned} \int_{\Omega} (2x - 2xy) dx dy &= \int_0^1 \int_0^{1-x} (2x - 2xy) dy dx \\ &= \int_0^1 2x(1-x) - x(1-x)^2 dx = \frac{1}{4} \end{aligned}$$

Moreover, this theorem is closely related to the concept of area. The area of Ω is

$$\Omega = \int_{\Omega} da \quad (2.139)$$

If we take

$$\begin{cases} Q = x, P = 0 \\ Q = 0, P = -y \end{cases} \Rightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 \quad (2.140)$$

substituting (2.140) into (2.139) and combining it with Green's Theorem, we have

$$\Omega = \int_{\Omega} da = \oint_{\partial\Omega} x dy = \oint_{\partial\Omega} -y dx \quad (2.141)$$

We can use this property to evaluate the area of an ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

First, parameterize the ellipse

$$\mathbf{r}(\theta) = \{(x, y) | x = a \cos \theta, y = b \sin \theta, 0 \leq \theta \leq 2\pi\}$$

We take $\mathbf{F} = (x, 0)$, and $dy = b \cos \theta$ then we have

$$\begin{aligned}\Omega &= \oint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{r} \\ &= \oint_{\partial\Omega} x dy = \int_0^{2\pi} ab \cos^2 \theta d\theta = \pi ab\end{aligned}$$

2.5 Gauss's Divergence Theorem

2.5.1 The Derivation of the Theorem

The Divergence Theorem is a reformulation of Green's Theorem. In this book, I will only discuss the two-dimensional and three-dimensional cases. In Green's Theorem, we integrate a vector field along the direction of the curve, meaning we compute the line integral with respect to the tangent vector at each point. However, we now take a different perspective: instead of integrating along the direction of the curve $d\mathbf{r}$, we perform the integration using the outward-pointing normal vector $\mathbf{n}$ at each point on the curve. See the illustration below.

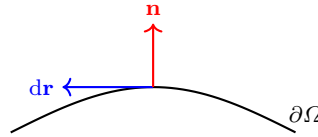


Fig. 2.5 A curve with the tangent vector $d\mathbf{r}$ and the outward normal vector $\mathbf{n}$.

The vector

$$d\mathbf{r} = \left(\frac{dx}{dr}, \frac{dy}{dr} \right), \quad ds = \mathbf{n} dr \quad (2.142)$$

Note that the normal vector we use here is always taken to point outward. In order to maintain this outward orientation consistently, we apply a rotation matrix that you have already encountered in high school: a 90° rotation to the right of the tangent vector $d\mathbf{r}$. This ensures that the resulting vector always represents the outward normal in the positive (counterclockwise) orientation. That is

$$\begin{bmatrix} \cos \frac{\pi}{2} & \sin \frac{\pi}{2} \\ -\sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{bmatrix} \begin{bmatrix} \frac{dx}{dr} \\ \frac{dy}{dr} \end{bmatrix} = \begin{bmatrix} \frac{dy}{dr} \\ -\frac{dx}{dr} \end{bmatrix} \quad (2.143)$$

So, the closed integral of the vector function $\mathbf{F} = (P, Q)$ with respect to the normal vector $\mathbf{n}$ is

$$\oint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{s} = \oint_{\partial\Omega} \mathbf{F} \cdot \mathbf{n} dr = \oint_{\partial\Omega} (P, Q) \cdot \left(\frac{dy}{dr}, -\frac{dx}{dr} \right) dr = \oint_{\partial\Omega} P dy - Q dx \quad (2.144)$$

Substituting (2.144) into Green's theorem, and we can obtain

$$\oint_{\partial\Omega} P dy - Q dx = \int_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) da = \int_{\Omega} \nabla \cdot \mathbf{F} da \quad (2.145)$$

The meaning of (2.145) is that the line integral of a vector field around a closed curve is equal to the area integral of the divergence of the vector field over the enclosed region. We can extend this idea to the three-dimensional case using the same logic.

If $\mathcal{V}$ is a three-dimensional region (volume), then its boundary $\partial\mathcal{V}$ is a two-dimensional manifold, which is the surface enclosing the volume. There's a volume $\mathcal{V} \subset \mathbb{R}^3$, and $\partial\mathcal{V}$ is positive. Consider $\mathcal{V} = [a, b] \times [c, d] \times [e, f]$ and the vector function $\mathbf{F}(P, Q, R)$, we start from the x -direction, the integral

$$\int \left(\int_a^b \frac{\partial P}{\partial x} dx \right) dy dz = \int [P(b, y, z) - P(a, y, z)] dy dz \quad (2.146)$$

On the other hand, the surface integral of $\mathbf{F}$ with respect to $\mathcal{S}_{x=a}$ is

$$\begin{aligned} \int_{\mathcal{S}_{x=a}} \mathbf{F} \cdot d\mathbf{a} &= \int_{\mathcal{S}_{x=a}} \mathbf{F} \cdot \mathbf{n} da \\ &= \int_{\mathcal{S}_{x=a}} (P, Q, R) \cdot (-1, 0, 0) da \\ &= \int_{\mathcal{S}_{x=a}} -P(a, y, z) da \end{aligned} \quad (2.147)$$

Note that we always take the normal vector outside, so the normal vector of the surface $\mathcal{S}_{x=a}$ is $(-1, 0, 0)$, and the normal vector of $\mathcal{S}_{x=b}$ is $(1, 0, 0)$. So,

$$\int_{\mathcal{S}_{x=b}} \mathbf{F} \cdot d\mathbf{a} = \int_{\mathcal{S}_{x=b}} (P, Q, R) \cdot (1, 0, 0) da = \int_{\mathcal{S}_{x=b}} P(b, y, z) da \quad (2.148)$$

At this moment, the normal vector of the surface $\mathcal{S}_x$ points in the same direction as the normal vector of the area element $dy dz$, so we rewrite the area element da of $\mathcal{S}_x$ as $dy dz$. So the surface integral over $\mathcal{S}_x$ is

$$\begin{aligned} \int_{\mathcal{S}_x} \mathbf{F} \cdot \mathbf{n} ds &= \int_{\mathcal{S}_{x=b}} P(b, y, z) dy dz - \int_{\mathcal{S}_{x=a}} P(a, y, z) dy dz \\ &= \int [P(b, y, z) - P(a, y, z)] dy dz \end{aligned} \quad (2.149)$$

substituting (2.149) into (2.146), we can finally get

$$\int_{\mathcal{V}} \frac{\partial P}{\partial x} d\tau = \int_{\mathcal{S}_x} \mathbf{F} \cdot \mathbf{n} da \quad (2.150)$$

Where $d\tau = dx dy dz$, apply the same method, we can also obtain

$$\begin{cases} \int_{\mathcal{V}} \frac{\partial Q}{\partial y} d\tau = \int_{\mathcal{S}_y} \mathbf{F} \cdot \mathbf{n} da \\ \int_{\mathcal{V}} \frac{\partial R}{\partial z} d\tau = \int_{\mathcal{S}_z} \mathbf{F} \cdot \mathbf{n} da \end{cases} \quad (2.151)$$

Combine (2.150) with (2.151), we have

$$\begin{aligned} \int_{\mathcal{V}} \frac{\partial P}{\partial x} d\tau + \int_{\mathcal{V}} \frac{\partial Q}{\partial y} d\tau + \int_{\mathcal{V}} \frac{\partial R}{\partial z} d\tau &= \int_{\mathcal{V}} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) d\tau \\ \int_{\mathcal{S}_x} \mathbf{F} \cdot \mathbf{n} da + \int_{\mathcal{S}_y} \mathbf{F} \cdot \mathbf{n} da + \int_{\mathcal{S}_z} \mathbf{F} \cdot \mathbf{n} da &= \int_{\partial\mathcal{V}} \mathbf{F} \cdot \mathbf{da} \end{aligned}$$

Note that we can rewrite the term

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = \nabla \cdot \mathbf{F}$$

Finally, based on the derivation above, we can generalize Gauss's divergence theorem to three dimensions, leading to a significant result

$$\boxed{\oint_{\partial\mathcal{V}} \mathbf{F} \cdot \mathbf{da} = \int_{\mathcal{V}} \nabla \cdot \mathbf{F} d\tau} \quad (2.152)$$

As we discussed before, the surface integral of a vector field represents the physical concept of flux. In the case of a closed volume $\mathcal{V}$, the surface integral corresponds to the total flux across all the faces enclosing the volume. This is equivalent to the volume integral of the divergence of the vector field over $\mathcal{V}$. In other words, the physical interpretation of divergence is that, when applied to a vector field passing through a closed volume, the total accumulation of divergence within the volume is equal to the net flux through the entire surface of the volume.

Put differently, the divergence represents the total flux density of a system. More precisely, the divergence theorem states that

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{F} d\tau = \lim_{\|\Delta\| \rightarrow 0} \sum \nabla \cdot \mathbf{F} \times \Delta\mathcal{V} = \oint_{\partial\mathcal{V}} \mathbf{F} \cdot \mathbf{da}$$

So

$$\nabla \cdot \mathbf{F} = \lim_{\|\Delta\| \rightarrow 0} \frac{1}{\Delta\mathcal{V}} \oint_{\partial\mathcal{V}} \mathbf{F} \cdot \mathbf{n} da \quad (2.153)$$

In summary, the divergence represents the **rate of flux change per unit volume**. Therefore, we can determine the type of physical system simply by examining the value of the divergence.

1. When $\nabla \cdot \mathbf{F}(\mathbf{r}_0) > 0$, the point $\mathbf{r}_0$ is called a **source point**.

By the local definition of divergence,

$$\nabla \cdot \mathbf{F}(\mathbf{r}_0) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\text{Vol}(B_\varepsilon)} \oint_{\partial B_\varepsilon} \mathbf{F} \cdot \hat{\mathbf{n}} \, da,$$

where B_ε is a small ball centered at $\mathbf{r}_0$ and $\hat{\mathbf{n}}$ is the outward unit normal vector on ∂B_ε .

Therefore, if $\nabla \cdot \mathbf{F}(\mathbf{r}_0) > 0$, then for sufficiently small ε ,

$$\oint_{\partial B_\varepsilon} \mathbf{F} \cdot \hat{\mathbf{n}} \, da > 0.$$

This means that the net outward flux through a small closed surface around $\mathbf{r}_0$ is positive. Hence, more field flows out of the small region than flows into it.

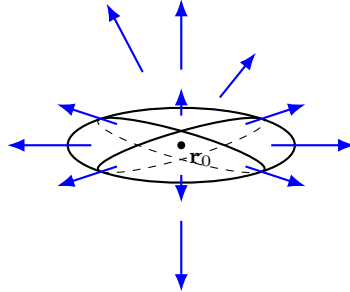


Fig. 2.6 A small spherical region with positive net outward flux representing positive divergence

2. When $\nabla \cdot \mathbf{F}(\mathbf{r}_0) < 0$, the point $\mathbf{r}_0$ is called a **sink point**.

By the local definition of divergence,

$$\nabla \cdot \mathbf{F}(\mathbf{r}_0) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\text{Vol}(B_\varepsilon)} \oint_{\partial B_\varepsilon} \mathbf{F} \cdot \hat{\mathbf{n}} \, da.$$

Therefore, if $\nabla \cdot \mathbf{F}(\mathbf{r}_0) < 0$, then for sufficiently small ε ,

$$\oint_{\partial B_\varepsilon} \mathbf{F} \cdot \hat{\mathbf{n}} \, da < 0.$$

This means that the net outward flux through a small closed surface around $\mathbf{r}_0$ is negative. Equivalently, more field flows into the small region than flows out of it.

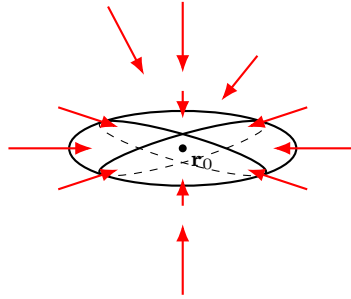


Fig. 2.7 A small spherical region with negative net outward flux representing negative divergence

3. When $\nabla \cdot \mathbf{F} = 0$ in a region, the vector field is called a **source-free field**.
 By the divergence theorem, for every closed surface $\partial\mathcal{V}$ enclosing a volume $\mathcal{V}$ inside this region,

$$\oint_{\partial\mathcal{V}} \mathbf{F} \cdot \hat{\mathbf{n}} \, da = \int_{\mathcal{V}} \nabla \cdot \mathbf{F} \, d\tau = 0.$$

Thus, the net flux through every closed surface in the region is zero. This does not mean that the vector field must vanish. It only means that the amount of field flowing into any closed region equals the amount flowing out.

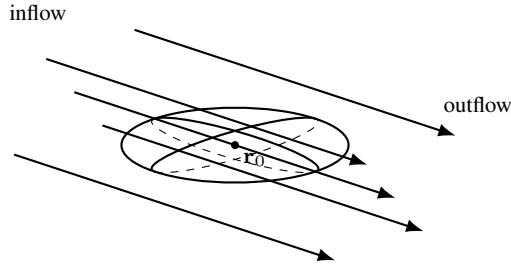


Fig. 2.8 A uniform flow through a small spherical region with zero net flux

2.5.2 Gauss's Law

2.5.2.1 Delta Function/Step Function and Distribution

In the theory of distributions, a distribution T is defined as a linear functional acting on a space of test functions ϕ . This action is denoted by

$$T : \phi \mapsto \langle T, \phi \rangle \in \mathbb{R} \quad (2.154)$$

where $\langle T, \cdot \rangle$ represents a functional. Formally, one may write

$$\mathcal{F}[f] = \langle T, f \rangle := \int_{\text{boundary}} T(x) f(x) \, dx \quad (2.155)$$

where the integral is understood in the sense of distributions. Two important examples of distributions are the *Dirac delta function* $\delta(x)$ and the *Heaviside step function* $\Theta(x)$ ⁷, which are informally defined by

$$\delta(x) = \begin{cases} \infty, & x = 0, \\ 0, & x \neq 0, \end{cases} \quad \Theta(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases} \quad (2.156)$$

The defining property of the delta distribution is

$$\boxed{\int_{-\infty}^{\infty} \delta(x) \, dx = 1} \quad (2.157)$$

More generally, since the delta distribution has support at a single point, the value of the integral remains invariant under any sufficiently small neighborhood:

$$\int_{-\epsilon}^{\epsilon} \delta(x) \, dx = 1 \quad \int_{a-\epsilon}^{a+\epsilon} \delta(x-a) \, dx = 1 \quad \forall \epsilon > 0. \quad (2.158)$$

Hence, the action of the delta distribution on a test function f is given by

$$\langle \delta_0, f \rangle = f(0) \quad \langle \delta_a, f \rangle = f(a) \quad (2.159)$$

In three dimensions, the Dirac delta function is defined as

$$\delta^3(\mathbf{r}) := \delta(x)\delta(y)\delta(z) \quad (2.160)$$

which satisfies

$$\int_{\mathbb{R}^3} \delta^3(\mathbf{r}) \, d^3x = 1 \quad (2.161)$$

A fundamental property of distributions is the relation between differentiation and test functions:

$$\langle T', \phi \rangle = -\langle T, \phi' \rangle \quad (2.162)$$

This follows immediately from integration by parts,

$$\int_{-\infty}^{\infty} T'(x) \phi(x) \, dx = T(x) \phi(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} T(x) \phi'(x) \, dx \quad (2.163)$$

Applying this result to the Heaviside function, we obtain

$$\langle \Theta', \phi \rangle = - \int_0^{\infty} \phi'(x) \, dx = \phi(0) = \langle \delta, \phi \rangle \quad (2.164)$$

⁷ Most of the books may write $H(x)$

and therefore

$$\boxed{\frac{d\Theta(x)}{dx} = \delta(x)} \quad (2.165)$$

2.5.2.2 Derivation of Gauss's Law

Our goal is to derive the fundamental identity

$$\boxed{\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^3(\mathbf{r})} \quad (2.166)$$

A naive computation suggests

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = \nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right) = 3r^{-3} - 3r^{-5} \mathbf{r} \cdot \mathbf{r} = 0 \quad (2.167)$$

which is valid only for $r \neq 0$. The singular behavior at the origin must therefore be treated separately.

To do so, we remove a small spherical region $\mathcal{B}(\mathbf{0}, \epsilon)$ of radius ϵ centered at the origin. Let $\mathcal{V}$ denote the original volume and $\mathcal{V}_\epsilon = \mathcal{V} \setminus \mathcal{B}(\mathbf{0}, \epsilon)$. Their boundaries satisfy

$$\partial \mathcal{V}_\epsilon = \partial \mathcal{V} + \partial \mathcal{B}(\mathbf{0}, \epsilon) \quad (2.168)$$

By the relation between the volumes, we have

$$\int_{\mathcal{V}} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) d\tau = \int_{\mathcal{V}_\epsilon} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) d\tau + \int_{\mathcal{B}(\mathbf{0}, \epsilon)} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) d\tau$$

Applying the divergence theorem, we find

$$\lim_{\epsilon \rightarrow 0} \int_{\mathcal{B}(\mathbf{0}, \epsilon)} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) d\tau = \lim_{\epsilon \rightarrow 0} \oint_{\partial \mathcal{B}(\mathbf{0}, \epsilon)} \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{a} = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon^2} 4\pi \epsilon^2 = 4\pi \quad (2.169)$$

Combining this result with the fact that the divergence vanishes for $r \neq 0$, we conclude

$$\int_{\mathcal{V}} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) d\tau = \int_{\mathbb{R}^3} 4\pi \delta^3(\mathbf{r}) d\tau \quad (2.170)$$

Hence,

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^3(\mathbf{r}) \quad (2.171)$$

More generally, for a shifted coordinate $\mathbf{r} = \mathbf{r}' + \mathbf{r}_0$ with $r = |\mathbf{r} - \mathbf{r}'|$, one obtains

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^3(\mathbf{r}) \quad (2.172)$$

For example, the continuous form of the gravitational field is as follows

$$\mathbf{g} = -G \int_{\mathcal{V}} d\tau' \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}}$$

We take the divergence of the gravitational field

$$\nabla \cdot \mathbf{g} = -G \int_{\mathcal{V}} d\tau' \rho(\mathbf{r}') \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = -G \int_{\mathcal{V}} d\tau' \rho(\mathbf{r}') \cdot 4\pi \delta^3(\mathbf{r})$$

Then we can obtain

$$\boxed{\nabla \cdot \mathbf{g} = -4\pi G \rho(\mathbf{r})}$$

And

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{g} d\tau = \oint \mathbf{g} \cdot d\mathbf{a} = - \int_{\mathcal{V}} 4\pi G \rho(\mathbf{r}') d\tau' = -4\pi G M_{\text{enclosed}}$$

Note that when $(0, 0, 0) \in \partial\mathcal{V}$, the surface area is denoted by

$$\frac{|\partial\mathcal{B}(\mathbf{0}, \epsilon) \cap \mathcal{V}|}{\epsilon^2} \rightarrow 2\pi \quad (2.173)$$

Then

$$\oint_{\partial\mathcal{V}} \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{a} = \oint_{\partial\mathcal{B}(\mathbf{0}, \epsilon)} \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{a} = 2\pi \quad (2.174)$$

2.6 Stokes' Theorem

For the sake of notational simplicity in the following proof, we rewrite the Cartesian coordinates (x, y, z) as the physicist's favorite form (x_1, x_2, x_3) , so that we can conveniently apply the Einstein summation convention later. To ensure clarity in the subsequent derivations, let us first review the relationship between the curl $\nabla \times$ and a vector field $\mathbf{F}(x_1, x_2, x_3) = (F_1, F_2, F_3)$, which is

$$\nabla \times \mathbf{F} = \left(\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3} \right) \mathbf{e}_1 + \left(\frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1} \right) \mathbf{e}_2 + \left(\frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2} \right) \mathbf{e}_3$$

So, the Green's theorem can be written as

$$\oint_{\mathcal{P}} F_1 dx_1 + F_2 dx_2 = \int_{\mathcal{S}} \nabla \times \mathbf{F} \cdot \mathbf{e}_3 da \quad (2.175)$$

This is the form of Green's theorem on the $x - y$ plane. Next, we aim to extend this theorem to a three-dimensional surface. To do so, we must first establish a way to describe the surface $\mathcal{S}$ and the properties of its boundary $\partial\mathcal{S}$.

The parametric form $\mathbf{r} : \Omega \rightarrow \mathcal{S} \subset \mathbb{R}^3$ of the surface $\mathcal{S}$ is

$$\mathcal{S} = \{\mathbf{r}(u, v) | x = x_1(u, v), x_2 = x_2(u, v), x_3 = x_3(u, v)\}$$

In addition, just as we did for curves, we can also parametrize $\ell : \mathcal{P} \rightarrow \Omega \subset \mathbb{R}^2$ the boundary $\partial\mathcal{S}$. That is

$$\partial\mathcal{S} = \{\ell(t) = \mathbf{r}(u(t), v(t)) | t \in [a, b]\}$$

Where $\mathbf{r}$ is the surface parametrization defined above. By the chain rule of several variables, we have

$$\frac{d\ell}{dt} = \frac{\partial\mathbf{r}}{\partial u} \frac{du}{dt} + \frac{\partial\mathbf{r}}{\partial v} \frac{dv}{dt} \Rightarrow d\ell = \frac{\partial\mathbf{r}}{\partial u} du + \frac{\partial\mathbf{r}}{\partial v} dv \quad (2.176)$$

Next, we evaluate the line integral of the vector field $\mathbf{F}$ along the surface boundary $\mathcal{S}$. In this case, the projection of the vector field is taken along the parameterized boundary $\partial\mathcal{S}$, that is, along $d\ell$. The closed integral

$$\oint_{\partial\mathcal{S}} \mathbf{F} \cdot d\ell = \oint_{\mathcal{P}} \mathbf{F}(\mathbf{r}(u, v)) \cdot \left(\frac{\partial\mathbf{r}}{\partial u} du + \frac{\partial\mathbf{r}}{\partial v} dv \right) = \oint_{\mathcal{P}} \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial u} \right) du + \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial v} \right) dv \quad (2.177)$$

By (2.138), we can rewrite the integral

$$\oint_{\mathcal{P}} \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial u} \right) du + \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial v} \right) dv = \int_{\Omega} \underbrace{\left[\frac{\partial}{\partial u} \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial v} \right) - \frac{\partial}{\partial v} \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial u} \right) \right]}_{\text{Antisymmetry}} du dv \quad (2.178)$$

According to the properties of differentiation and partial differentiation, we obtain

$$\begin{cases} \frac{\partial}{\partial u} \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial v} \right) = \frac{\partial\mathbf{F}}{\partial u} \cdot \frac{\partial\mathbf{r}}{\partial v} + \mathbf{F} \cdot \frac{\partial^2\mathbf{r}}{\partial v \partial u} = \frac{\partial\mathbf{F}}{\partial u} \cdot \frac{\partial\mathbf{r}}{\partial v} + \mathbf{F} \cdot \mathbf{r}_{vu} \\ \frac{\partial}{\partial v} \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial u} \right) = \frac{\partial\mathbf{F}}{\partial v} \cdot \frac{\partial\mathbf{r}}{\partial u} + \mathbf{F} \cdot \frac{\partial^2\mathbf{r}}{\partial u \partial v} = \frac{\partial\mathbf{F}}{\partial v} \cdot \frac{\partial\mathbf{r}}{\partial u} + \mathbf{F} \cdot \mathbf{r}_{uv} \end{cases}$$

Where $\mathbf{F} \cdot \mathbf{r}_{vu} = \mathbf{F} \cdot \mathbf{r}_{uv}$. Substituting this into (2.178), it becomes

$$\int_{\Omega} \left[\frac{\partial}{\partial u} \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial v} \right) - \frac{\partial}{\partial v} \left(\mathbf{F} \cdot \frac{\partial\mathbf{r}}{\partial u} \right) \right] du dv = \int_{\Omega} \left(\frac{\partial\mathbf{F}}{\partial u} \cdot \frac{\partial\mathbf{r}}{\partial v} - \frac{\partial\mathbf{F}}{\partial v} \cdot \frac{\partial\mathbf{r}}{\partial u} \right) du dv \quad (2.179)$$

Applying the Einstein summation convention, we can rewrite $\frac{\partial\mathbf{F}}{\partial u} \cdot \frac{\partial\mathbf{r}}{\partial v} - \frac{\partial\mathbf{F}}{\partial v} \cdot \frac{\partial\mathbf{r}}{\partial u}$ as

$$\begin{cases} \frac{\partial\mathbf{F}}{\partial u} \cdot \frac{\partial\mathbf{r}}{\partial v} = \partial_j F_i \frac{\partial x_j}{\partial u} \frac{\partial x_i}{\partial v} \\ \frac{\partial\mathbf{F}}{\partial v} \cdot \frac{\partial\mathbf{r}}{\partial u} = \partial_j F_i \frac{\partial x_j}{\partial v} \frac{\partial x_i}{\partial u} \end{cases} \Rightarrow \frac{\partial\mathbf{F}}{\partial u} \cdot \frac{\partial\mathbf{r}}{\partial v} - \frac{\partial\mathbf{F}}{\partial v} \cdot \frac{\partial\mathbf{r}}{\partial u} = \partial_j F_i \left(\frac{\partial x_j}{\partial u} \frac{\partial x_i}{\partial v} - \frac{\partial x_j}{\partial v} \frac{\partial x_i}{\partial u} \right)$$

Note that we should use the Kronecker delta and the Levi-Civita symbol again to unravel the structure of this; we have

$$\begin{aligned}
 \partial_j F_i \left(\frac{\partial x_j}{\partial u} \frac{\partial x_i}{\partial v} - \frac{\partial x_j}{\partial v} \frac{\partial x_i}{\partial u} \right) &= \partial_j F_i (\delta_{im} \delta_{jl} - \delta_{il} \delta_{jm}) \frac{\partial x_l}{\partial u} \frac{\partial x_m}{\partial v} \\
 &= \partial_j F_i (\epsilon_{ijk} \epsilon_{mlk}) \frac{\partial x_l}{\partial u} \frac{\partial x_m}{\partial v} = \epsilon_{ijk} \partial_j F_i \left(\epsilon_{mlk} \frac{\partial x_l}{\partial u} \frac{\partial x_m}{\partial v} \right) \\
 &= (-\epsilon_{kji} \partial_j F_i) \left(-\epsilon_{klm} \frac{\partial x_l}{\partial u} \frac{\partial x_m}{\partial v} \right) = (\nabla \times \mathbf{F})_k (\mathbf{r}_u \times \mathbf{r}_v)_k = \nabla \times \mathbf{F} \cdot \mathbf{r}_u \times \mathbf{r}_v
 \end{aligned}$$

Substituting this into (2.179), you can obtain a familiar integral

$$\int_{\Omega} \left(\frac{\partial \mathbf{F}}{\partial u} \cdot \frac{\partial \mathbf{r}}{\partial v} - \frac{\partial \mathbf{F}}{\partial v} \cdot \frac{\partial \mathbf{r}}{\partial u} \right) du dv = \int_{\Omega} \nabla \times \mathbf{F} \cdot \mathbf{r}_u \times \mathbf{r}_v du dv \quad (2.180)$$

Where $\mathbf{r}_u \times \mathbf{r}_v du dv = \mathbf{n} da = d\mathbf{a}$. So (2.180) becomes

$$\int_{\Omega} \nabla \times \mathbf{F} \cdot \mathbf{r}_u \times \mathbf{r}_v du dv = \int_S \nabla \times \mathbf{F} \cdot \mathbf{n} da = \int_S \nabla \times \mathbf{F} \cdot d\mathbf{a} \quad (2.181)$$

Combining (2.181) with (2.177), we obtain the final result, the **Stokes' Theorem**.

$$\boxed{\oint_{\partial S} \mathbf{F} \cdot d\boldsymbol{\ell} = \int_S \nabla \times \mathbf{F} \cdot d\mathbf{a}} \quad (2.182)$$

$\oint_{\partial S} \mathbf{F} \cdot d\boldsymbol{\ell}$ represents the line integral of the vector field along the boundary of the surface. The result reflects the total "amount of rotation" along the boundary of the surface. In other words, we can use the curl and the line integral over the boundary to describe the rotational behavior of the field, that is

$$|\nabla \times \mathbf{F}| = \lim_{\|\Delta\| \rightarrow 0} \frac{1}{S} \oint_{\partial S} \mathbf{F} \cdot d\boldsymbol{\ell} \quad (2.183)$$

The line integral represents the circulation of the vector field along the boundary of the surface. Its magnitude corresponds to the total projection of the curl of the vector field $\mathbf{F}$ onto the normal direction of each infinitesimal surface element $d\mathbf{S}$ over the surface.

We define the direction of $\nabla \times \mathbf{F}$ on $d\mathbf{a}$ to follow the **right-hand rule**, where the curled fingers represent the direction of rotation. We always define the counterclockwise direction as the positive orientation. This convention allows us to conveniently handle various problems related to electric current, magnetic fields, and other topics in electromagnetism. The direction of the curl is illustrated in the following diagram.

““latex

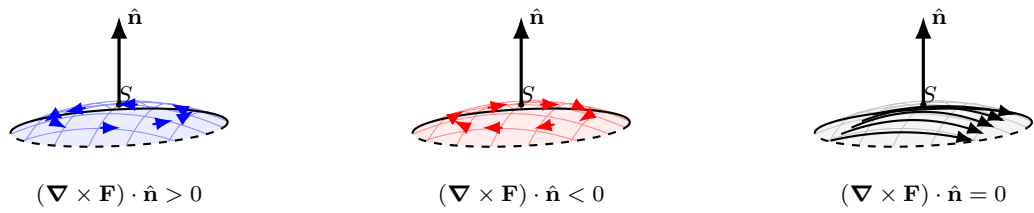


Fig. 2.9 Three curved-surface illustrations for Stokes' theorem. With the upward unit normal vector $\hat{\mathbf{n}}$ fixed, counterclockwise circulation gives positive curl flux, clockwise circulation gives negative curl flux, and a uniform flow gives zero curl flux.

““

In fact, every infinitesimal surface element on the surface satisfies Stokes' theorem. The projection of the curl onto the normal vector of each element represents the local direction and magnitude of rotation in that small region. However, the rotational effects in the interior cancel each other out, and after integrating over the entire surface, only the net circulation along the boundary remains, as illustrated below.

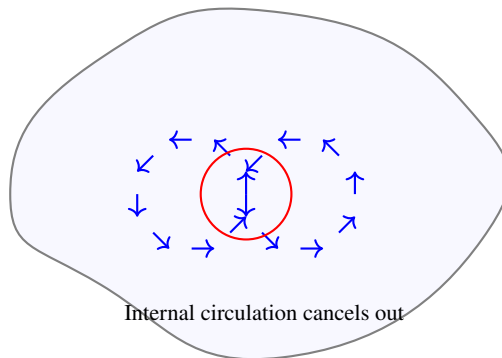


Fig. 2.10 Internal rotational effects cancel out, leaving only boundary circulation (Stokes' Theorem)

To combine Green's theorem, divergence theorem, and Stokes' theorem, you may refer to Appendix B.

Chapter 3

Complex Numbers in Physics

Abstract This chapter introduces one of the most important tools in both physics and engineering: the complex number system. Unlike real numbers, complex numbers cannot directly represent concrete physical quantities or magnitudes. However, their algebraic structure greatly simplifies many calculations in the real domain. In other words, complex numbers allow us to express various properties of a physical quantity, such as amplitude, phase, and periodicity, within a unified symbolic framework.

In electrical engineering, for example, complex numbers are used to simultaneously account for magnitude and phase in AC circuits. In mechanics, the use of complex numbers is generally more straightforward, often even simpler than problems encountered in high school physics. Therefore, this chapter will provide only a brief introduction to the essential tools and calculations involving complex numbers that are not typically covered in high school. The main focus will instead be placed on Fourier analysis.

You may consider this section a moment to catch your breath, a lighter, transitional chapter before moving on to more intense mathematical structures.

3.1 Euler's Formula and Complex Exponentials

3.1.1 Euler's Formula

Review the Taylor expansion of the exponential function e^x , we have

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots$$

We can observe that the expansion of e^x is related to the expansion of $\sin x$ and $\cos x$, that is

$$\begin{cases} \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \\ \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \end{cases}$$

In the series expansions of these two functions, we observe that their terms are structurally identical to those of e^x , with only differences in sign. This observation allows us to introduce the imaginary unit i as a unifying tool, enabling us to combine both functions into a single exponential expression. Let $f(x) = e^{ix}$, where $i = \sqrt{-1}$, we can obtain

$$\begin{aligned} f(x) = e^{ix} &= f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots \\ &= 1 + ix + \frac{x^2}{2!} \times i^2 + \frac{x^3}{3!} \times i^3 + \cdots \\ &= 1 + ix - \frac{x^2}{2!} - i\frac{x^3}{3!} + \frac{x^4}{4!} + i\frac{x^5}{5!} + \cdots \end{aligned}$$

We can factor i out of the terms, which is

$$e^{ix} = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots\right) + i \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right) \quad (3.1)$$

Substituting the expansion of $\cos x$ and $\sin x$ into (3.1), we can thus obtain the most important formula, the **Euler's formula**.

$$\boxed{e^{ix} = \cos x + i \sin x} \quad (3.2)$$

When $x = \theta$, we can also denote

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (3.3)$$

When $\theta = \pi$, then the exponential function becomes $e^{i\pi}$, by (3.3), we can obtain

$$e^{i\pi} + 1 = (\cos \pi + i \sin \pi) + 1 = -1 + 1 = 0 \quad (3.4)$$

Previously, in Equation (1.78), we used the exponential function e^x to define the hyperbolic functions $\sinh$ and $\cosh$. Now, we also observe that the ordinary trigonometric functions $\sin x$ and $\cos x$ share a similar connection with the exponential function. Thus, we can likewise express $\sin x$ and $\cos x$ in terms of e^{ix} using its properties. There are two functions e^{ix} and e^{-ix} , we have

$$\begin{cases} e^{ix} = \cos x + i \sin x \\ e^{-ix} = e^{i(-x)} = \cos(-x) + i \sin(-x) = \cos x - i \sin x \end{cases}$$

Subtracting the two equations

$$e^{ix} - e^{-ix} = 2i \sin x$$

So, the trigonometric function $\sin x$ is denoted by

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad (3.5)$$

Similarly, the function $\cos x$ is

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} \quad (3.6)$$

Dividing (3.5) by (3.6), we can also get the function $\tan x$ in terms of e^{ix}

$$\tan x = -\frac{i(e^{ix} - e^{-ix})}{e^{ix} + e^{-ix}} \quad (3.7)$$

After introducing the basic structure, we now arrive at an interesting mathematical phenomenon. In the real domain, we all know that the range of the sine function is bounded by

$$-1 \leq \sin x \leq 1 \quad \text{for } x \in \mathbb{R} \quad (3.8)$$

However, once we extend our domain beyond the real numbers, this limitation no longer holds. For example, in the complex domain, it is entirely possible for

$$\sin x = 2 \quad \text{for some } x = z \in \mathbb{C} \quad (3.9)$$

Substituting $\sin x = \sin z$ into (3.5), we have

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = 2 \Rightarrow e^{iz} - e^{-iz} = 4i \quad (3.10)$$

We multiply both sides of (3.10) by e^{iz} , and we can obtain a quadratic equation in terms of e^{iz} .

$$\begin{aligned} (e^{iz})^2 - 4i(e^{iz}) - 1 &= 0 \\ \Rightarrow e^{iz} &= \frac{4i \pm \sqrt{-16 + 4}}{2} = i(2 \pm \sqrt{3}) \end{aligned}$$

Taking the natural logarithm of both sides of the equation, we get

$$\ln e^{iz} = iz = \ln [i(2 \pm \sqrt{3})] = \ln i + \ln (2 \pm \sqrt{3})$$

Note that the term $\ln i$ is weird; we now consider this in the complex plane. Back in high school, we learned that the complex plane can be viewed as the real coordinate plane, with the y -axis replaced by the imaginary axis i . Therefore, we can also define an angular coordinate θ in the complex plane, ranging from $0 \leq \theta \leq 2\pi$. In the complex plane, any nonzero complex number z can be expressed in polar coordinates as

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta} \quad (3.11)$$

where $r = |z|$ is the modulus and $\theta = \arg(z)$ is the argument of the complex number. The illustration is shown below,

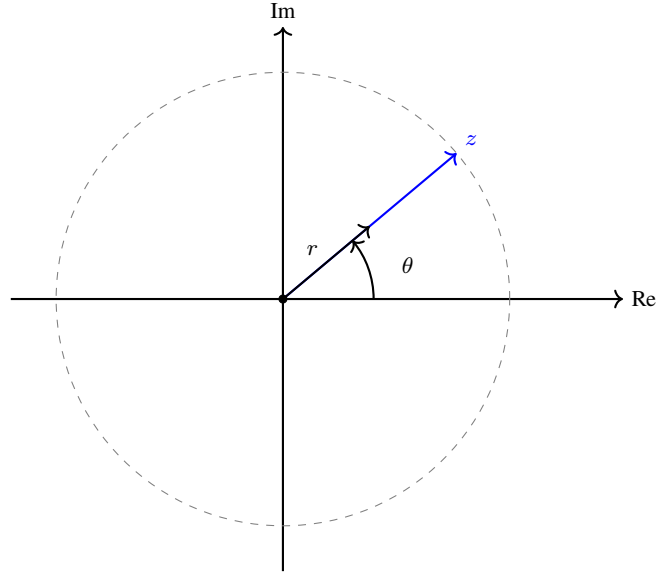


Fig. 3.1 Polar Representation of a Complex Number

So, by (3.11), we have

$$i = 1 \times \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = |i|e^{i\pi/2} = e^{i\pi/2} \quad (3.12)$$

Substituting (3.12) into $\ln i$, we obtain

$$iz = \ln i + \ln(2 \pm \sqrt{3}) = \ln e^{i\pi/2} + \ln(2 \pm \sqrt{3}) = \frac{\pi}{2}i + \ln(2 \pm \sqrt{3})$$

Dividing both sides by i , we can finally obtain the complex number making $\sin z = 2$, that is

$$z = \frac{\pi}{2} - i \ln(2 \pm \sqrt{3}) \quad (3.13)$$

So,

$$\sin \left(\frac{\pi}{2} - i \ln(2 \pm \sqrt{3}) \right) = 2 \quad (3.14)$$

In summary, the introduction of complex numbers allows us to resolve many problems that seem undefined or unreasonable within the real number system. Moreover, by working with the compact expression e^{ix} , we can significantly streamline calculations. In cases involving logarithmic functions, for example, the exponential can often be canceled, leaving only the exponent behind. In the next chapter, we will explore how

complex numbers are meaningfully applied in physics and why they play such a crucial role.

3.1.2 De Moivre's Theorem

Consider a complex number $z = r(\cos \theta + i \sin \theta)$, De Moivre's Theorem in high school can be denoted by

$$z^n = r^n (\cos(n\theta) + i \sin(n\theta)) \quad (3.15)$$

In the previous section, we discussed the exponential form of complex numbers in polar coordinates. We are now ready to prove De Moivre's Theorem in a remarkably elegant and concise way. Based on Equation (3.11), we rewrite the complex number z as:

$$z = re^{i\theta}$$

So, the n th power of z is

$$z^n = (re^{i\theta})^n = r^n (e^{i\theta})^n = r^n e^{i(n\theta)} \quad (3.16)$$

Substituting this into Euler's formula, we can obtain

$$\boxed{z^n = r^n e^{i(n\theta)} = r^n (\cos(n\theta) + i \sin(n\theta))} \quad (3.17)$$

We can apply the same logic to the $1/n$ th power of z , that is

$$z^{1/n} = r^{1/n} e^{i\theta/n} = \sqrt[n]{r} \left(\cos \frac{\theta}{n} + i \sin \frac{\theta}{n} \right) \quad (3.18)$$

Building on Equation , we can now use the complex exponential form to solve problems involving algebraic roots. Consider the equation

$$x^n = r$$

where we can rewrite $r \in \mathbb{C}$ as

$$r = |r|e^{i(\theta+2k\pi)} \quad k = 0, 1, \dots \quad (3.19)$$

since $e^{i(\theta+2k\pi)} = e^{i\theta} = e^{i(\theta+2\pi)} = \dots$, according to the fundamental theorem of algebra, the equation $x^n = r$ must have n roots, so

$$x = r^{1/n} = \left(|r|e^{i(\theta+2k\pi)} \right)^{1/n} = \sqrt[n]{|r|} \exp i \left(\frac{\theta + 2k\pi}{n} \right) \quad 0 \leq k \leq n-1 \quad (3.20)$$

For example, to compute the equation $x^3 = -8i$, we have

$$x^3 = -8i = 8 \exp i \left(-\frac{\pi}{2} + 2k\pi \right)$$

So,

$$x = \sqrt[3]{-8} = \sqrt[3]{8} \exp i \left(-\frac{\pi}{6} + \frac{2k\pi}{3} \right) \quad k = 0, 1, 2$$

That is,

$$x = \sqrt[3]{-8} = \sqrt[3]{8} e^{-i\pi/6}, \sqrt[3]{8} e^{i\pi/2}, \sqrt[3]{8} e^{i7\pi/6}$$

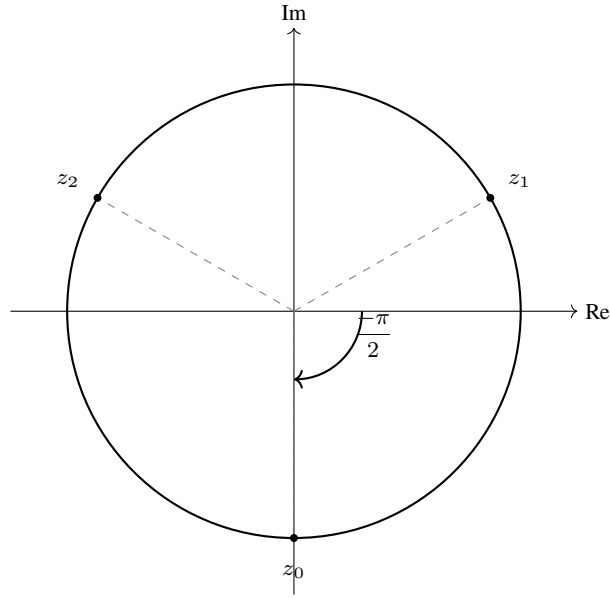


Fig. 3.2 The three complex roots of the equation $x^3 = -8i$ are evenly distributed on a circle of radius 2 in the complex plane, starting at an angle of $-\frac{\pi}{2}$.

Furthermore, consider two complex numbers $z_1 = r_1 e^{i\alpha} = r_1 (\cos \alpha + i \sin \alpha)$ and $z_2 = r_2 e^{i\beta} = r_2 (\cos \beta + i \sin \beta)$, then $z_1^n z_2^m$ and z_1^n / z_2^m is

$$z_1^n z_2^m = (r_1 e^{i\alpha})^n (r_2 e^{i\beta})^m = r_1^n r_2^m e^{i(n\alpha+m\beta)} \quad \frac{z_1^n}{z_2^m} = \frac{(r_1 e^{i\alpha})^n}{(r_2 e^{i\beta})^m} = \frac{r_1^n}{r_2^m} e^{i(n\alpha-m\beta)}$$

Therefore,

$$\begin{cases} z_1^n z_2^m = r_1^n r_2^m e^{i(n\alpha+m\beta)} = r_1^n r_2^m [\cos(n\alpha + m\beta) + i \sin(n\alpha + m\beta)] \\ \frac{z_1^n}{z_2^m} = \frac{r_1^n}{r_2^m} e^{i(n\alpha-m\beta)} = \frac{r_1^n}{r_2^m} [\cos(n\alpha - m\beta) + i \sin(n\alpha - m\beta)] \end{cases} \quad (3.21)$$

3.1.3 Trigonometric Identities

Back in high school, we learned many trigonometric identities, such as the double-angle, triple-angle, and sum and difference formulas. However, these identities were often used without formal proof. Now, with Euler's formula in hand, we are finally able to derive these identities in a much more elegant and straightforward way. We've known that Euler's formula satisfies

$$e^{ix} = \cos x + i \sin x$$

The sum of angles in terms of e^{ix} can be denoted by

$$e^{i(\alpha+\beta)} = (e^{i\alpha}) (e^{i\beta}) \quad (3.22)$$

Substituting $e^{i\alpha} = \cos \alpha + i \sin \alpha$ and $e^{i\beta} = \cos \beta + i \sin \beta$ into (3.22)

$$\begin{aligned} (e^{i\alpha}) (e^{i\beta}) &= (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) \\ &= \cos \alpha \cos \beta + i \sin \beta \cos \alpha + i \sin \alpha \cos \beta - \sin \alpha \sin \beta \\ &= (\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \beta \cos \alpha + \sin \alpha \cos \beta) \end{aligned} \quad (3.23)$$

By (3.22),

$$e^{i(\alpha+\beta)} = \cos(\alpha + \beta) + i \sin(\alpha + \beta) \quad (3.24)$$

Comparing (3.23) with (3.24), we obtain

$$\begin{cases} i \sin(\alpha + \beta) = i(\sin \beta \cos \alpha + \sin \alpha \cos \beta) \\ \cos(\alpha + \beta) = (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \end{cases}$$

So, we have proved that

$$\boxed{\begin{cases} \sin(\alpha + \beta) = (\sin \beta \cos \alpha + \sin \alpha \cos \beta) \\ \cos(\alpha + \beta) = (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \end{cases}} \quad (3.25)$$

Similarly, let $\beta = -\gamma$, then (3.25) becomes

$$\begin{cases} \sin(\alpha - \gamma) = (\sin(-\gamma) \cos \alpha + \sin \alpha \cos(-\gamma)) \\ \cos(\alpha - \gamma) = (\cos \alpha \cos(-\gamma) - \sin \alpha \sin(-\gamma)) \end{cases} \quad (3.26)$$

since $\sin(-\gamma) = -\sin(\gamma)$ and $\cos(-\gamma) = \cos(\gamma)$, we obtain

$$\begin{cases} \sin(\alpha - \gamma) = (\sin \alpha \cos \gamma - \sin \gamma \cos \alpha) \\ \cos(\alpha - \gamma) = (\cos \alpha \cos \gamma + \sin \alpha \sin \gamma) \end{cases} \quad (3.27)$$

Furthermore, since our method of proof relies on comparing the real and imaginary parts of a complex expression, the operation of extracting a specific part from a complex number is called **taking the real part** or **taking the imaginary part**. We denote this as follows

$$\begin{cases} \sin(\alpha \pm \beta) = \text{Im } e^{i(\alpha \pm \beta)} \\ \cos(\alpha \pm \beta) = \text{Re } e^{i(\alpha \pm \beta)} \end{cases} \quad (3.28)$$

3.2 Basic Complex Calculus

The advantage of rewriting the polar form of complex numbers into its exponential form becomes especially clear in this chapter. Trigonometric functions are inherently periodic, and when analyzing a physical system involving differentiation and integration, we often desire an expression that both preserves this periodicity and is convenient to manipulate mathematically. This is where the exponential function e^x becomes our most powerful tool. Because e^x remains unchanged in form under both differentiation and integration, we are free to repeatedly apply calculus operations without the expression collapsing or becoming overly complicated.

Consider a function $f(z) = z_0(\cos(\omega z + \theta) + i \sin(\omega z + \theta))$, where $f : \mathbb{C} \rightarrow \mathbb{C}$, then we differentiate it and integrate it

$$\begin{aligned} \frac{d}{dz} f(z) &= z_0(-\omega \sin(\omega z + \theta) + i\omega \cos(\omega z + \theta)) \\ &= i\omega z_0((\cos(\omega z + \theta) + i \sin(\omega z + \theta)) \\ &= i\omega f(z) \end{aligned} \quad (3.29)$$

Similarly,

$$\frac{d^2}{dz^2} f(z) = -\omega^2 f(z) \quad (3.30)$$

Applying Euler's formula, the function $f(z)$ can be rewritten as

$$f(z) = z_0 e^{i(\omega z + \theta)} \quad (3.31)$$

By Chapter 1, we can directly obtain the relationship between $f'(z)$ and $f(z)$, which is

$$\begin{cases} \frac{d}{dz} f(z) = i\omega z_0 e^{i(\omega z + \theta)} = i\omega z_0 e^{i(\omega z - \theta)} = i\omega f(z) \\ \frac{d^2}{dz^2} f(z) = \frac{d}{dz} i\omega z_0 e^{i(\omega z + \theta)} = (i\omega)^2 z_0 e^{i(\omega z + \theta)} = -\omega^2 f(z) \end{cases} \quad (3.32)$$

We verify this fact by differentiating $\sin(\omega z - \theta)$. Let $f(z) = e^{i(\omega z - \theta)}$. By (3.28), we can denote it as

$$\sin(\omega z - \theta) = \operatorname{Im} f(z) = \operatorname{Im} e^{i(\omega z - \theta)} \quad (3.33)$$

Differentiating both sides

$$\frac{d}{dz} \sin(\omega z - \theta) = \frac{d}{dz} \operatorname{Im} f(z) = \operatorname{Im} \frac{d}{dz} e^{i(\omega z - \theta)} = \operatorname{Im} (i\omega e^{i(\omega z - \theta)})$$

since

$$\operatorname{Im} (i\omega e^{i(\omega z - \theta)}) = \operatorname{Im} (i\omega \cos(\omega z - \theta) - \omega \sin(\omega z - \theta)) = \omega \cos(\omega z - \theta)$$

Hence, we have verified that

$$\frac{d}{dz} \sin(\omega z - \theta) = \omega \cos(\omega z - \theta) \quad (3.2.6)$$

as expected. In addition, we can express $\sin(\omega z - \theta)$ as the form of (3.5), that is

$$\sin(\omega z - \theta) = \frac{e^{i(\omega z - \theta)} - e^{-i(\omega z - \theta)}}{2i} \quad (3.34)$$

Differentiating this with respect to z ,

$$\begin{aligned} \frac{d}{dz} \frac{e^{i(\omega z - \theta)} - e^{-i(\omega z - \theta)}}{2i} &= \frac{i\omega e^{i(\omega z - \theta)} + i\omega e^{-i(\omega z - \theta)}}{2i} \\ &= \frac{\omega e^{i(\omega z - \theta)} + \omega e^{-i(\omega z - \theta)}}{2} \\ &= \omega \left[\frac{e^{i(\omega z - \theta)} + e^{-i(\omega z - \theta)}}{2} \right] \\ &= \omega \cos(\omega z - \theta) \end{aligned}$$

As before. The same logic applies to integration. Let $f(z) = z_0 e^{i\omega z}$, we obtain

$$\int f(z) dz = \int z_0 e^{i\omega z} dz = z_0 \int e^{i\omega z} dz = \frac{z_0}{i\omega} e^{i\omega z} + C = \frac{-z_0 i}{\omega} e^{i\omega z} + C \quad (3.35)$$

So we can verify that

$$\int f(z) dz = \frac{-i}{\omega} f(z) + C \quad (3.36)$$

This shows that even after integration, the exponential term $e^{i\omega z}$ remains, which allows us to easily retain information related to periodicity. In other words, by transforming the problem into an integral involving e^x , the entire computation becomes significantly simplified. This shows that even after integration, the exponential term $e^{i\omega z}$ remains, which allows us to easily retain information related to periodicity. In other words, by transforming the problem into an integral involving e^x , the entire computation becomes significantly simplified.

For example, the function z_1 and z_2 satisfy

$$\begin{cases} \frac{d}{dz} z_1 = \zeta z_2 \\ \frac{d}{dz} z_2 = -\zeta z_1 \end{cases} \quad (3.37)$$

Note that when the derivatives of two functions differ only by a minus sign and result in each other's form, we can safely assume a complex representation

$$\mathcal{Z} = z_1 + iz_2 \quad (3.38)$$

Then the differentiation of $\mathcal{Z}$ can be denoted by

$$\frac{d}{dz} \mathcal{Z} = \frac{d}{dz} (z_1 + iz_2) = \frac{d}{dz} z_1 + i \frac{d}{dz} z_2$$

Substituting (3.37) into (3.38), we get

$$\frac{d}{dz} \mathcal{Z} = \zeta z_2 - i\zeta z_1 = -i\zeta(z_1 + iz_2) = -i\zeta \mathcal{Z}$$

It becomes a first-order differential equation (See Chapter 4).

$$\begin{aligned} \frac{d\mathcal{Z}}{\mathcal{Z}} &= -i\zeta dz \Rightarrow \int \frac{d\mathcal{Z}}{\mathcal{Z}} = -i\zeta \int dz \\ &\Rightarrow \ln \mathcal{Z} = -i\zeta z + C \\ &\Rightarrow \mathcal{Z} = Ae^{-i\zeta z} \end{aligned} \quad (3.39)$$

where $A = e^C$. We can obtain A by the *initial condition* (See Chapter 4). So (3.39) can be rewritten as

$$z_1 + iz_2 = Ae^{-i\zeta z} = A \cos(\zeta z) - Ai \sin(\zeta z) \quad (3.40)$$

So, the functions z_1 and z_2 are

$$\begin{cases} z_1 = \operatorname{Re}(Ae^{-i\zeta z}) = A \cos(\zeta z) \\ z_2 = \operatorname{Im}(Ae^{-i\zeta z}) = -A \sin(\zeta z) \end{cases} \quad (3.41)$$

In summary, the introduction of Euler's formula allows us to handle differentiation, integration, and even differential equations in a more compact and elegant way. By working with the exponential form, we can extract the full behavior and periodicity of the function, and then isolate the real or imaginary part as needed. However, in physical systems, both the real and imaginary parts do not always correspond to physical quantities. In the next section, I will illustrate this with two classic examples in physics.

3.3 Real and Imaginary Parts: Physical Interpretation

3.3.1 Simple Harmonic Motion

This part involves second-order differential equations, which are covered in detail in Chapter 4.2. Therefore, we will not go into the full solution here. Instead, we will present only the final result and focus on explaining its physical meaning. Students who are interested in the full derivation are encouraged to study the differential equations in Chapter 4 first, and then return to this section to work through the solution independently. The equation of motion of SHM system is

$$m\ddot{x} = -kx \Rightarrow m\ddot{x} + kx = 0 \quad (3.42)$$

Let $\omega^2 := k/m$, and this second-order differential equation becomes

$$\ddot{x} + \omega^2 x = 0 \quad (3.43)$$

The solution to the ODE is

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t} \quad (3.44)$$

(3.44) can be expanded to

$$\begin{aligned} x(t) &= [C_1 \cos(\omega t) + iC_1 \sin(\omega t)] + [C_2 \cos(\omega t) - iC_2 \sin(\omega t)] \\ &= (C_1 + C_2) \cos(\omega t) + i(C_1 - C_2) \sin(\omega t) \end{aligned} \quad (3.45)$$

Because we don't let the function $x(t)$ be the complex number on purpose, we can deduce that $C_1, C_2 \in \mathbb{C}$. The initial condition $x(0) = x_0$ and $\dot{x}(0) = v_0$ can help us to determine C_1 and C_2 , and we substitute these into (3.44)

$$\begin{cases} x(0) = C_1 e^0 + C_2 e^0 = C_1 + C_2 = x_0 \\ \dot{x}(0) = i\omega C_1 e^0 - i\omega C_2 e^0 = i\omega(C_1 - C_2) = v_0 \end{cases}$$

So we can compute that

$$\begin{cases} C_1 = \frac{x_0}{2} - i\frac{v_0}{2\omega} \\ C_2 = \frac{x_0}{2} + i\frac{v_0}{2\omega} \end{cases} \quad (3.46)$$

Substituting (3.46) into (3.45), we obtain

$$x(t) = x_0 \cos(\omega t) + \frac{v_0}{\omega} \sin(\omega t) \quad (3.47)$$

The angle relationship is shown below

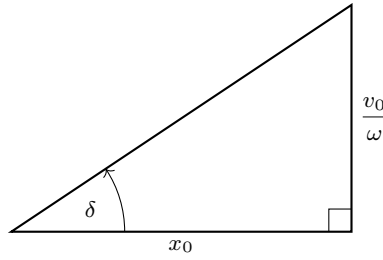


Fig. 3.3 A right triangle representing the initial condition decomposition of the solution $x(t) = x_0 \cos(\omega t) + \frac{v_0}{\omega} \sin(\omega t)$, where x_0 and $\frac{v_0}{\omega}$ are the projections onto cosine and sine components, respectively.

By the figure, we can rewrite (3.47) as

$$\begin{aligned} x(t) &= \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2} \left(\frac{x_0}{\sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2}} \cos(\omega t) + \frac{v_0/\omega}{\sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2}} \sin(\omega t) \right) \\ &= \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2} \cos(\omega t - \delta) = \sqrt{x_0^2 + \frac{mv_0^2}{k}} \cos(\omega t - \delta) \end{aligned} \quad (3.48)$$

The angle δ is

$$\delta = \arctan\left(\frac{v_0}{\omega x_0}\right) = \arctan\left(\frac{v_0}{x_0} \sqrt{\frac{m}{k}}\right) \quad (3.49)$$

It determines the initial phase diagram by the initial condition, including the initial position x_0 and the initial velocity v_0 . And the period of the system is given by the trigonometric function $\cos(\omega t - \delta)$, which is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \quad (3.50)$$

The key point here is that we did not deliberately choose to use complex numbers. What we've demonstrated is a result that naturally emerges from solving the differential equation itself. Therefore, we don't need to force ourselves to pick out the real or imaginary part, the solution already carries physical meaning as it is. Most importantly, from the very beginning, the exponential term $e^{i\omega t}$ directly reveals the periodic nature of the system's motion. In the next example, we will see a different situation, one where we intentionally introduce complex numbers in order to simplify the computation.

3.3.2 AC Circuit

In the AC system, both voltage and current are quantities that oscillate with time. The voltage can be denoted by

$$V(t) = V_0 \cos(\omega t) \quad (3.51)$$

Let $\mathcal{V}(t)$ be the complex number, which is

$$\mathcal{V}(t) = V_0 e^{i\omega t} \quad (3.52)$$

Under the series RLC circuit, the impedance Z of the system is

$$Z = R + i\omega L + \frac{1}{i\omega C} = R + i\left(\omega L - \frac{1}{\omega C}\right) \quad (3.53)$$

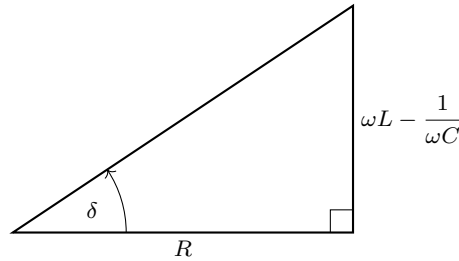


Fig. 3.4 A right triangle representing the impedance triangle in a series RLC circuit, where the angle δ corresponds to the phase angle between voltage and current.

As shown in the figure, the modulus of the complex number $|Z|$ is

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \quad (3.54)$$

So, the impedance can be rewritten as

$$Z = |Z| \left(\frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} + i \frac{\omega L - \frac{1}{\omega C}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \right) \quad (3.55)$$

$$:= |Z|(\cos \delta + i \sin \delta) = |Z|e^{i\delta}$$

We can now write Ohm's law in the complex domain. Let the current be the function $\mathcal{I}(t)$, which satisfies

$$\mathcal{I}(t) = \frac{\mathcal{V}(t)}{Z} = \frac{V_0 e^{i\omega t}}{|Z|e^{i\delta}} = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} e^{i(\omega t - \delta)} \quad (3.56)$$

Where δ means the **phase difference** between the voltage and the current. It is given by

$$\delta = \arctan \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right) = \arctan \left(\frac{\omega^2 CL - 1}{R\omega C} \right) \quad (3.57)$$

Through the above derivations, we can extract several key pieces of information: the magnitude and frequency of the voltage, the magnitude and frequency of the current, and the phase difference between them. However, it is important to note that in the complex representation, the quantities we can actually measure using instruments such as oscilloscopes are given by the real parts of these complex expressions. That is,

$$\begin{cases} V(t) = \text{Re}(\mathcal{V}(t)) = V_0 \cos(\omega t) \\ I(t) = \text{Re}(\mathcal{I}(t)) = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \cos(\omega t - \delta) \end{cases} \quad (3.58)$$

In this circuit system, we move all real-world quantities and physical values into the complex domain. As I mentioned earlier, in the complex domain, we can represent everything using exponential terms, which greatly simplifies the computation. (Trust me, you really don't want to abandon $e^{i\omega t}$ and go back to brute-force real-number trigonometry.)

Although the imaginary part cannot be directly measured in experiments, its presence is crucial for preserving phase information in the system. It's like a wedding, where the real part is the bride and groom, and the imaginary part is the best man. The best man is there to help the ceremony go smoothly, but in the end, no one really remembers who

he was. And honestly, we don't need to. (Apologies to anyone who has ever been a best man.)

3.4 Introduction to Fourier Series and Transform

The Fourier series expresses a periodic function as an infinite sum of sine and cosine terms. In contrast, the Fourier transform generalizes this idea to non-periodic functions by replacing the discrete sum with a continuous integral, effectively decomposing the signal into a continuous spectrum of frequency components.

3.4.1 Average Value of Sines and Cosines

Consider a function $f(x)$, $x \in [a, b]$, and we divide this interval into n subintervals. Then the average of $f(x)$ on $[a, b]$ can be denoted by

$$\langle f(x) \rangle := \frac{f(x_1) + f(x_2) + \cdots + f(x_n)}{n}$$

Let x_1, x_2 be Δx apart, then we have

$$\frac{[f(x_1) + f(x_2) + \cdots + f(x_n)]\Delta x}{n\Delta x}$$

When $\|\Delta\| \rightarrow 0$, the numerator can be written as definite integral, which is

$$\langle f(x) \rangle = \frac{[f(x_1) + f(x_2) + \cdots + f(x_n)]\Delta x}{n\Delta x} \Big|_{\|\Delta\| \rightarrow 0} = \frac{1}{b-a} \int_a^b f(x) dx \quad (3.59)$$

After introducing the average of a function, I'm going to introduce the periodic function next, such as $f(x) = \sin x$ and $f(x) = \cos x$, since this chapter focuses on the average over a period. A periodic function is defined as

$$f(x+T) = f(x) \quad (3.60)$$

Then the period of the function is T , or sometimes a multiple such as nT , depending on the frequencies involved. We know that $\sin x$ and $\cos x$ are periodic functions with period 2π . Therefore, combinations such as $\sin(nx) \sin(mx)$, $\cos(nx) \cos(mx)$, or $\sin(nx) \cos(mx)$ are also periodic, because the argument of each trigonometric function is shifted by integer multiples of 2π , which do not affect the function value. To illustrate

$$\sin[n(x+2\pi)] \sin[m(x+2\pi)] = \sin(nx+2n\pi) \sin(mx+2m\pi) = \sin(nx) \sin(mx)$$

since $\sin(\theta + 2\pi k) = \sin(\theta)$ for any integer k . Similarly,

$$\begin{cases} \cos[n(x+2\pi)] \cos[m(x+2\pi)] = \cos(nx) \cos(mx) \\ \sin[n(x+2\pi)] \cos[m(x+2\pi)] = \sin(nx) \cos(mx) \end{cases}$$

Although the combination $\sin(nx)\sin(mx)$, $\cos(nx)\cos(mx)$ or $\sin(nx)\cos(mx)$ may have a shorter period than 2π , in Fourier series we still integrate over the original period of the function, typically $[-\pi, \pi]$ or $[0, 2\pi]$. This is because the Fourier coefficients are defined relative to the period of the function being expanded, not the individual basis functions. We'll discuss this concept next part. So, we discuss the average values of them over a period 2π . that is

$$\langle f(x) \rangle = \begin{cases} \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx \\ \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx \\ \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(nx) \cos(mx) dx \end{cases} \quad (3.61)$$

By (3.5) and (3.6), we express all trigonometric functions in terms of complex exponentials via Euler's formula, so

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{e^{inx} - e^{-inx}}{2i} \right) \left(\frac{e^{imx} - e^{-imx}}{2i} \right) dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\frac{e^{i(n+m)x} - e^{i(n-m)x} - e^{i(m-n)x} + e^{-i(n+m)x}}{-4} \right] dx \end{aligned}$$

Then it becomes

$$\frac{-1}{8\pi} \left[\int_{-\pi}^{\pi} e^{i(n+m)x} dx - \int_{-\pi}^{\pi} e^{i(n-m)x} dx - \int_{-\pi}^{\pi} e^{i(m-n)x} dx + \int_{-\pi}^{\pi} e^{-i(n+m)x} dx \right] \quad (3.62)$$

When $m \neq n$, we have

$$\begin{aligned} &= \frac{-1}{8i(n+m)} e^{i(n+m)x} \Big|_{-\pi}^{\pi} + \frac{1}{8i(n-m)} e^{i(n-m)x} \Big|_{-\pi}^{\pi} + \frac{1}{8i(m-n)} e^{i(m-n)x} \Big|_{-\pi}^{\pi} \\ &\quad + \frac{1}{8i(n+m)} e^{-i(n+m)x} \Big|_{-\pi}^{\pi} \\ &= \frac{-\sin[(n+m)\pi]}{4} + \frac{\sin[(n-m)\pi]}{4} + \frac{\sin[(m-n)\pi]}{4} = 0 \end{aligned} \quad (3.63)$$

Since $m, n \in \mathbb{Z}$, $\sin(n \pm m)\pi = \sin k\pi = 0$. When $m = n \neq 0$, (3.62) becomes

$$\begin{aligned}
& \frac{-1}{8\pi} \left[\int_{-\pi}^{\pi} e^{i2nx} dx - \int_{-\pi}^{\pi} dx - \int_{-\pi}^{\pi} dx + \int_{-\pi}^{\pi} e^{-i2nx} dx \right] \\
&= \frac{-\sin(2n\pi)}{8\pi} + \frac{2\pi}{8\pi} + \frac{2\pi}{8\pi} = \frac{1}{2}
\end{aligned} \tag{3.64}$$

When $m = n = 0$, (3.62) becomes

$$\frac{-1}{8\pi} \left[\int_{-\pi}^{\pi} dx - \int_{-\pi}^{\pi} dx - \int_{-\pi}^{\pi} dx + \int_{-\pi}^{\pi} dx \right] = 0 \tag{3.65}$$

So, we can obtain the average value of sines over a period 2π in three different situations, which is

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx = \begin{cases} 0 & m \neq n \\ \frac{1}{2} & m = n \neq 0 \\ 0 & m = n = 0 \end{cases} \tag{3.66}$$

Apply the same logic to the cosines $\cos(nx) \cos(mx)$, we have

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx = \begin{cases} 0 & m \neq n \\ \frac{1}{2} & m = n \neq 0 \\ 1 & m = n = 0 \end{cases} \tag{3.67}$$

Similarly, for the product of sine and cosine functions, regardless of the values of m or n , the result will be a combination of terms similar to equation (3.63), involving complex exponentials like $e^{i(kx)}$ and $e^{-i(kx)}$ with nonzero frequencies. Since the average value of such oscillating exponential terms over a full period is zero, the average of the entire product function is also zero. That is,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(nx) \cos(mx) dx = 0 \tag{3.68}$$

3.4.2 Fourier Series in the Real Domain

Given a function $f(x)$ of period 2π , we can express the function as a combination of sine and cosine terms, since these are all periodic functions. By adjusting the size of each coefficient, we determine how much each periodic component contributes to the overall shape of $f(x)$. In this way, we aim to approximate the function as closely as

possible using these fundamental building blocks. Therefore, our goal is to write

$$\begin{aligned} f(x) = & A + a_1 \cos x + a_2 \cos(2x) + a_3 \cos(3x) + \cdots \\ & + b_1 \sin x + b_2 \sin(2x) + b_3 \sin(3x) + \cdots \end{aligned} \quad (3.69)$$

Where $A = a_0/2$, I'll show you the reason later. Let's start with the cosine term. When we want to get a_n , we multiply both sides by $\cos(nx)$, for example

$$\begin{aligned} f(x) \cos x = & A \cos x + a_1 \cos^2 x + a_2 \cos(2x) \cos x + \cdots \\ & + b_1 \sin x \cos x + b_2 \sin(2x) \cos x + \cdots \end{aligned}$$

Then the average value over the period of both sides is given by

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \cos x dx = & A \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos x dx + a_1 \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos^2 x dx + \cdots \\ & + b_1 \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin x \cos x dx + \cdots \end{aligned}$$

In the cosine terms, according to (3.67), the average value of the function is zero whenever $m \neq n$. Similarly, based on (3.68), the integral of the product of sine and cosine functions is always zero. This means that all terms vanish except the a_1 term (you can verify that the A term integrates to zero as well). Thus, we obtain

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \cos x dx = a_1 \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos^2 x dx = \frac{a_1}{2} \Rightarrow a_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos x dx \quad (3.70)$$

Similarly, we can verify

$$\left\{ \begin{aligned} a_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos x dx \\ a_2 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(2x) dx \\ &\vdots \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \end{aligned} \right\} \quad \left\{ \begin{aligned} b_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin x dx \\ b_2 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(2x) dx \\ &\vdots \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \end{aligned} \right. \quad (3.71)$$

What about A ? Because A can be seen as the coefficient of the term $\cos 0x$, we can integrate both sides directly

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx &= A \frac{1}{2\pi} \int_{-\pi}^{\pi} dx + a_1 \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos x dx + \cdots \\ &\quad + b_1 \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin x dx + \cdots \end{aligned}$$

According to (3.68), the integral of the products of sine and cosine functions is always zero ($\sin x = \sin 1x \cos 0x$ and so on). So

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = A \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos 0x dx = A \quad (3.72)$$

As we can see from Equation (3.72), its integral does not follow the same pattern as that of Equation (3.71), and therefore cannot be written in a unified formula. To make it consistent with the pattern, we need to multiply the result by 2. That is

$$2A = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \quad (3.73)$$

We define $2A := a_0$, then we can get $A = a_0/2$. Substituting $A = a_0/2$ into (3.69), we can thus write down the Fourier series as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)) \quad (3.74)$$

Where the coefficients a_n and b_n are denoted by

$$\begin{cases} a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \\ b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \end{cases} \quad (3.75)$$

Example 3.1 Express the following functions using the Fourier series.

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases}$$

Where $f(x + 2\pi) = f(x)$.

Solution 3.1

The coefficient a_n satisfies

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \cos(nx) \, dx + \int_0^{\pi} \cos(nx) \, dx \right]$$

We need to carefully examine the graph of the periodic function $f(x)$. If the function is not globally continuous, we must analyze it piecewise. For this example, we divide the interval into two parts, $[-\pi, 0]$ and $[0, \pi]$. Starting from a_n , when $n \neq 0$, we have

$$\begin{aligned} a_n &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \cos(nx) \, dx + \int_0^{\pi} \cos(nx) \, dx \right] \\ &= \frac{1}{\pi} \int_0^{\pi} \cos(nx) \, dx = \frac{1}{n\pi} \sin(nx) \Big|_0^{\pi} = 0 \end{aligned}$$

When $n = 0$, we have

$$\begin{aligned} a_n &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \cos(0x) \, dx + \int_0^{\pi} \cos(0x) \, dx \right] \\ &= \frac{1}{\pi} \int_0^{\pi} dx = 1 \end{aligned}$$

And the coefficient b_n satisfies

$$\begin{aligned} b_n &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \sin(nx) \, dx + \int_0^{\pi} \sin(nx) \, dx \right] = \frac{1}{\pi} \int_0^{\pi} \sin(nx) \, dx \\ &= \frac{-1}{n\pi} \cos(nx) \Big|_0^{\pi} = \frac{-1}{n\pi} [(-1)^n - 1] = \begin{cases} 0 & n = 2k, k \in \mathbb{N} \\ \frac{2}{n\pi} & n = 2k + 1 \end{cases} \end{aligned}$$

Substituting a_n, b_n into (3.75), we can finally get

$$f(x) = \frac{1}{2} + \frac{2}{\pi} \left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \frac{\sin 7x}{7} + \cdots \right)$$

Next, we can generalize the Fourier series to functions that have periods other than 2π . Given a function $f(x)$ of period $2L$, where $L \in \mathbb{R}$. That is

$$f(x + 2L) = f(x)$$

When analyzing this function, we need to change the interval for computing the average value of the periodic function from $[-\pi, \pi]$ to $[-L, L]$. That is

$$\langle f(x) \rangle = \frac{1}{2L} \int_{-L}^L f(x) \, dx$$

To expand $f(x)$, we use functions that have the same periodicity, namely a period of $2L$. This can be done by rewriting the trigonometric functions $\sin(nx)$ and $\cos(nx)$ as $\sin(n\pi x/L)$ and $\cos(n\pi x/L)$, respectively, which ensures that the resulting functions have period $2L$ since

$$\begin{cases} \sin\left[\frac{n\pi}{L}(x+2L)\right] = \sin\left(\frac{n\pi x}{L} + 2n\pi\right) = \sin\left(\frac{n\pi x}{L}\right) \\ \cos\left[\frac{n\pi}{L}(x+2L)\right] = \cos\left(\frac{n\pi x}{L} + 2n\pi\right) = \cos\left(\frac{n\pi x}{L}\right) \end{cases} \quad (3.76)$$

As a reminder, in high school we learned that the period of $\sin(nx)$ is $2\pi/n$. However, for the sake of convenience in calculation and analysis, we usually treat them as if they have period 2π , as long as their behavior repeats correctly within that interval. So, the Fourier series becomes

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right] \quad (3.77)$$

Where the coefficients a_n and b_n are changed into

$$\begin{cases} a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \\ b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \end{cases} \quad (3.78)$$

Similarly, these coefficients a_n and b_n determine how much of each sine or cosine wave contributes to the shape of $f(x)$. If we want to express

$$f(x) = \begin{cases} 0 & -L < x < 0 \\ 1 & 0 < x < L \end{cases}$$

Then the answer becomes

$$f(x) = \frac{1}{2} + \frac{2}{L} \left[\sin\left(\frac{\pi x}{L}\right) + \frac{1}{3} \sin\left(\frac{3\pi x}{L}\right) + \frac{1}{5} \sin\left(\frac{5\pi x}{L}\right) + \dots \right]$$

In general, we do not necessarily need to separate the coefficient a_0 from a_n . We can directly use

$$f(x) = \sum_{n=0}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

on the interval $[-\pi, \pi]$, or after rescaling the coordinate, we can write

$$f(x) = \sum_{n=0}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

on the interval $[-L, L]$. The reason this notation is often more convenient is that every coefficient can be treated as a projection of $f(x)$ onto a basis function. In this form, the coefficient is not guessed. It is obtained by multiplying both sides by the corresponding sine or cosine mode and then integrating over the whole interval.

For example, if we multiply the expansion by $\cos\left(\frac{m\pi x}{L}\right)$ and integrate from $-L$ to L , we obtain

$$\begin{aligned} \int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx &= \sum_{n=0}^{\infty} a_n \int_{-L}^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx \\ &\quad + \sum_{n=0}^{\infty} b_n \int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx. \end{aligned}$$

All the cross terms vanish by orthogonality. Therefore, only the term with $n = m$ survives, and we get

$$\int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx = a_m \int_{-L}^L \cos^2\left(\frac{m\pi x}{L}\right) dx.$$

Hence

$$a_m = \frac{\int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx}{\int_{-L}^L \cos^2\left(\frac{m\pi x}{L}\right) dx}.$$

Similarly,

$$b_m = \frac{\int_{-L}^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx}{\int_{-L}^L \sin^2\left(\frac{m\pi x}{L}\right) dx}.$$

This is the most geometric way to understand Fourier coefficients. The functions $\cos\left(\frac{n\pi x}{L}\right)$, $\sin\left(\frac{n\pi x}{L}\right)$ play the role of orthogonal directions in an infinite dimensional function space, and the Fourier coefficients are exactly the components of $f(x)$ along these directions.

3.4.2.1 Orthogonality of Sine and Cosine

The above derivation depends on the orthogonality of sine and cosine. To make this statement precise, we define the inner product of two real valued functions on $[-L, L]$ by

$$\langle f, g \rangle = \int_{-L}^L f(x)g(x) dx.$$

This is the direct analogue of the Euclidean dot product

$$\mathbf{u} \cdot \mathbf{v} = \sum_i u_i v_i.$$

The only difference is that the discrete sum over components is replaced by an integral over the continuous variable x .

Now define

$$C_n(x) = \cos\left(\frac{n\pi x}{L}\right), \quad S_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

Then the orthogonality relations can be written in the compact form

$$\langle C_n, C_m \rangle = \int_{-L}^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx = L(1 + \delta_{n0})\delta_{nm}, \quad n, m \geq 0,$$

The factor $1 + \delta_{n0}$ only records the special normalization of the constant mode. Indeed,

$$C_0(x) = 1, \quad \langle C_0, C_0 \rangle = \int_{-L}^L 1 dx = 2L,$$

while for $n \neq 0$,

$$\langle C_n, C_n \rangle = \int_{-L}^L \cos^2\left(\frac{n\pi x}{L}\right) dx = L.$$

For sine functions, we take $n, m \geq 1$, since $S_0(x) = 0$. Then

$$\langle S_n, S_m \rangle = \int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = L\delta_{nm}, \quad n, m \geq 1.$$

Also, every sine mode is orthogonal to every cosine mode,

$$\langle S_n, C_m \rangle = \int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx = 0, \quad n \geq 1, \quad m \geq 0.$$

These formulas are exactly the infinite dimensional version of $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$ in orthogonal coordinates. However, the Fourier basis above is orthogonal but not completely normalized. More precisely,

$$\langle C_n, C_m \rangle = \|C_n\|^2 \delta_{nm}, \quad \langle S_n, S_m \rangle = \|S_n\|^2 \delta_{nm},$$

where

$$\|C_n\|^2 = L(1 + \delta_{n0}), \quad \|S_n\|^2 = L.$$

This is why the Fourier coefficients are obtained by projection divided by the squared length of the corresponding basis function.

Therefore, if

$$f(x) = \sum_{n=0}^{\infty} a_n C_n(x) + \sum_{n=1}^{\infty} b_n S_n(x),$$

then taking the inner product with C_m gives

$$\langle f, C_m \rangle = \sum_{n=0}^{\infty} a_n \langle C_n, C_m \rangle + \sum_{n=1}^{\infty} b_n \langle S_n, C_m \rangle.$$

Using the orthogonality relations, every term vanishes except the $n = m$ cosine term. Hence

$$\langle f, C_m \rangle = a_m \|C_m\|^2.$$

Thus

$$a_m = \frac{\langle f, C_m \rangle}{\langle C_m, C_m \rangle} = \frac{\int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx}{\int_{-L}^L \cos^2\left(\frac{m\pi x}{L}\right) dx}, \quad m \geq 0.$$

Equivalently,

$$a_m = \frac{1}{L(1 + \delta_{m0})} \int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx.$$

Similarly, taking the inner product with S_m gives

$$\langle f, S_m \rangle = b_m \|S_m\|^2.$$

Therefore,

$$b_m = \frac{\langle f, S_m \rangle}{\langle S_m, S_m \rangle} = \frac{\int_{-L}^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx}{\int_{-L}^L \sin^2\left(\frac{m\pi x}{L}\right) dx}, \quad m \geq 1.$$

Equivalently,

$$b_m = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx.$$

Thus the Fourier series is a projection formula in function space. The sine and cosine modes are the basis directions, the inner product is the integral, and the coefficients a_m, b_m are the components of $f(x)$ along these directions. This is the precise mathematical meaning behind saying that a function can be decomposed into frequencies.

3.4.3 Complex Form of Fourier Series

Since we can express a function with period 2π as a sum of sine and cosine functions, it naturally follows from Euler's formula that we can also rewrite these sine and cosine terms using exponential functions of the form e^{ix} . This substitution not only simplifies the expression but also helps maintain the compactness and elegance of the overall formula. Given a function $f(x)$ of period 2π , we can express the function as

$$f(x) = c_0 + c_1 e^{ix} + c_{-1} e^{-ix} + c_2 e^{i2x} + c_{-2} e^{-i2x} + \dots \quad (3.79)$$

To determine c_0 , we need to compute the average value of $f(x)$ over the period

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = c_0 \frac{1}{2\pi} \int_{-\pi}^{\pi} dx + c_1 \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ix} dx + \dots \quad (3.80)$$

Note that when we integrate e^{ikx} over the interval from $-\pi$ to π , the result is always zero as long as $k \neq 0$. Because

$$\int_{-\pi}^{\pi} e^{ikx} dx = \left. \frac{1}{ik} e^{ikx} \right|_{-\pi}^{\pi} = \frac{1}{ik} (e^{ik\pi} - e^{-ik\pi}) = \frac{2 \sin(k\pi)}{ik} = 0 \quad (3.81)$$

So, (3.80) becomes

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = c_0 \frac{1}{2\pi} \int_{-\pi}^{\pi} dx + 0 = c_0 \frac{2\pi}{2\pi} = c_0 \quad (3.82)$$

Similarly, when we want to determine c_n , then we should multiply both sides by e^{-inx} since all terms other than e^{inx} multiplied by e^{-inx} can be rewritten in the form e^{ikx} with $k \neq 0$, their integrals from $-\pi$ to π will vanish. As a result, only the term c_n survives in the integral, that is

$$\begin{aligned} f(x) e^{-inx} &= c_0 + c_1 e^{ix} e^{-inx} + \dots + c_n e^{inx} e^{-inx} + \dots \\ \Rightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx &= 0 + \dots + c_n \frac{1}{2\pi} \int_{-\pi}^{\pi} dx + 0 + \dots = c_n \end{aligned} \quad (3.83)$$

Substituting these results into (3.79), we can express the periodic function using the Fourier series in the complex form, which is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \quad (3.84)$$

Where the coefficient c_n is denoted by

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \quad (3.85)$$

Similarly, if the periodic function $f(x)$ now has period $2L$, we can follow the structure of Equation (3.77) and rewrite e^{inx} as $e^{in\pi x/L}$. In this case, the Fourier series becomes

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L} \quad c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx \quad (3.86)$$

3.4.4 Convergence and the Gibbs Phenomenon

Consider the partial sums of a general Fourier trigonometric series $S_N(x)$, which means we take the sum from $n = 0$ to $n = N$ of a function $f(x)$ defined on $[-L, L]$. That is

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^N \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right] \quad (3.87)$$

We substitute (3.79) into (3.88), we have

$$\begin{aligned} S_N(x) &= \frac{1}{2L} \int_{-L}^L f(t) dt + \sum_{n=1}^N \left[\left(\frac{1}{L} \int_{-L}^L f(t) \cos\left(\frac{n\pi t}{L}\right) dt \right) \cos\left(\frac{n\pi x}{L}\right) \right. \\ &\quad \left. + \left(\frac{1}{L} \int_{-L}^L f(t) \sin\left(\frac{n\pi t}{L}\right) dt \right) \sin\left(\frac{n\pi x}{L}\right) \right] \\ &= \frac{1}{L} \int_{-L}^L \left[\frac{1}{2} + \sum_{n=1}^N \left(\cos\left(\frac{n\pi t}{L}\right) \cos\left(\frac{n\pi x}{L}\right) + \sin\left(\frac{n\pi t}{L}\right) \sin\left(\frac{n\pi x}{L}\right) \right) \right] f(t) dt \\ &= \frac{1}{L} \int_{-L}^L \left[\frac{1}{2} + \sum_{n=1}^N \left(\cos \frac{n\pi(t-x)}{L} \right) \right] f(t) dt \end{aligned}$$

Then we have

$$S_N(x) = \frac{1}{L} \int_{-L}^L D_N(t-x) f(t) dt \quad (3.88)$$

Where we define

$$D_N(x) = \frac{1}{2} + \sum_{n=1}^N \left(\cos \frac{n\pi x}{L} \right) \quad (3.89)$$

Then we called the function D_N **Dirichlet kernel**. This representation shows that the partial sum $S_N(x)$ is the convolution of the Dirichlet kernel with the function $f(x)$. Let $\pi x/L = \phi$, and we multiply $D_N(x)$ by $2 \sin(\phi/2)$, we have

$$\begin{aligned}
 2 \sin\left(\frac{\phi}{2}\right) D_N(x) &= 2 \sin\left(\frac{\phi}{2}\right) \left[\frac{1}{2} + \sum_{n=1}^N (\cos n\phi) \right] \\
 &= 2 \sin\left(\frac{\phi}{2}\right) \left[\frac{1}{2} + \cos \phi + \cos 2\phi + \cdots \right] \\
 &= \sin\left(\frac{\phi}{2}\right) + 2 \cos \phi \sin\left(\frac{\phi}{2}\right) + 2 \cos 2\phi \sin\left(\frac{\phi}{2}\right) + \cdots
 \end{aligned} \tag{3.90}$$

We need to use trigonometric identities that we learned in high school, that is

$$\begin{aligned}
 \sin\left(N\phi + \frac{\phi}{2}\right) - \sin\left(N\phi - \frac{\phi}{2}\right) &= \sin(N\phi) \cos\left(\frac{\phi}{2}\right) + \sin\left(\frac{\phi}{2}\right) \cos(N\phi) \\
 &\quad - \sin(N\phi) \cos\left(\frac{\phi}{2}\right) + \sin\left(\frac{\phi}{2}\right) \cos(N\phi) \\
 &= 2 \sin\left(\frac{\phi}{2}\right) \cos(N\phi)
 \end{aligned}$$

Substituting this into (3.90), we can write

$$\begin{aligned}
 2 \sin\left(\frac{\phi}{2}\right) D_N(x) &= \sin\left(\frac{\phi}{2}\right) + \left(\sin \frac{3\phi}{2} - \sin \frac{\phi}{2} \right) + \left(\sin \frac{5\phi}{2} - \sin \frac{3\phi}{2} \right) + \cdots \\
 &\quad + \left[\sin\left(N\phi + \frac{\phi}{2}\right) - \sin\left(N\phi - \frac{\phi}{2}\right) \right] \\
 &= \sin\left(N\phi + \frac{\phi}{2}\right) = \sin\left(N + \frac{1}{2}\right) \phi
 \end{aligned}$$

If $2 \sin\left(\frac{\phi}{2}\right) \neq 0$, we can obtain

$$D_N(x) = \frac{\sin\left(N + \frac{1}{2}\right) \phi}{2 \sin\left(\frac{\phi}{2}\right)} \quad \phi = \frac{\pi x}{L} \tag{3.91}$$

In the following, we will use the Dirichlet kernel to explain a notable phenomenon. We know that the more terms we include in the Fourier series, the more closely it approximates the target function. However, if you look carefully, you will note that near points of discontinuity, the approximation visibly overshoots the actual function value. This overshoot is referred to as an **overshoot error**. As we increase the number of terms, the oscillations become more concentrated near the discontinuity, but the magnitude of the overshoot does not vanish. This persistent behavior is known as the **Gibbs phenomenon**. As shown below

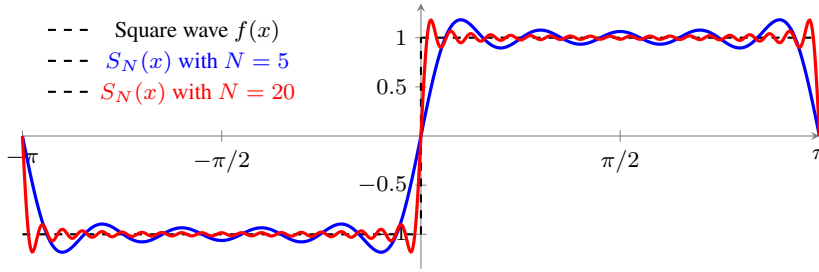


Fig. 3.5 Comparison of the Fourier series approximations of a square wave using $N = 5$ (blue) and $N = 20$ (red).

Consider a square wave function $f(x)$ of period $2L$, which satisfies

$$f(x) = \begin{cases} -1 & x \in (-L, 0) \\ 1 & x \in (0, L) \end{cases}$$

We apply the substitution rule to (3.88), let $\zeta = t - x$, then (3.88) becomes

$$S_N(x) = \frac{1}{L} \int_{-L-x}^{L-x} D_N(\zeta) f(\zeta + x) d\zeta \quad (3.92)$$

Because the function $f(x)$ has the same behavior over a period $2L$, we can shift the interval back to $[-L, L]$, that is

$$S_N(x) = \frac{1}{L} \int_{-L}^L D_N(\zeta) f(\zeta + x) d\zeta \quad (3.93)$$

We divide the interval into two parts, $[-L, 0]$ and $[0, L]$, then

$$S_N(x) = \frac{1}{L} \int_{-L}^0 D_N(\zeta) f(\zeta + x) d\zeta + \int_0^L D_N(\zeta) f(\zeta + x) d\zeta$$

Let the first integral $\zeta \rightarrow -\zeta$, then $D_N(\zeta) f(\zeta + x) d\zeta$ becomes $D_N(-\zeta) f(x - \zeta) (-d\zeta)$. We can observe (3.89) that the Dirichlet kernel is an even function, so $D_N(-\zeta) =$

$D_N(\zeta)$. We obtain

$$\begin{aligned} \frac{1}{L} \int_{-L}^0 D_N(\zeta) f(\zeta + x) d\zeta &= \frac{1}{L} \int_L^0 D_N(-\zeta) f(x - \zeta) (-d\zeta) \\ &= \frac{1}{L} \int_0^L -D_N(\zeta) f(x - \zeta) (-d\zeta) \\ &= \frac{1}{L} \int_0^L D_N(\zeta) f(x - \zeta) d\zeta \end{aligned}$$

Substituting this into $S_N(x)$, we can write down

$$S_N(x) = \frac{1}{L} \int_0^L [f(x - \zeta) + f(x + \zeta)] D_N(\zeta) d\zeta \quad (3.94)$$

To illustrate the Gibbs phenomenon of this square wave, we can narrow down the scope of analysis by focusing on the interval $x \in (0, L/2)$ and $\zeta \in [0, L]$. This allows us to examine only the amplitude extrema of the oscillations within this region. Due to the symmetry of the periodic function, if we can demonstrate the behavior in this interval, it implies that the same conclusion applies to the entire function.

Then we have three different values of the function $f(x - \zeta) + f(x + \zeta)$. When $f(x - \zeta) + f(x + \zeta) = 2$, it means that

$$x - \zeta > 0 \quad x + \zeta < L$$

Where $\zeta \geq 0$. Since $x \in (0, L/2)$, we can deduce that $x < L - x$, then we have

$$f(x - \zeta) + f(x + \zeta) = 2 \quad 0 \leq \zeta < x \quad (3.95)$$

Similarly,

$$\begin{cases} f(x - \zeta) + f(x + \zeta) = 2 & 0 \leq \zeta < x \\ f(x - \zeta) + f(x + \zeta) = 0 & x < \zeta < L - x \\ f(x - \zeta) + f(x + \zeta) = -2 & L - x < \zeta < L \end{cases} \quad (3.96)$$

Substituting the conditions of (3.96) into (3.94), we obtain

$$S_N(x) = \frac{2}{L} \int_0^x D_N(\zeta) d\zeta - \frac{2}{L} \int_{L-x}^L D_N(\zeta) d\zeta \quad (3.97)$$

Let $u = L - \zeta$, $d\zeta = -du$, and (3.97) becomes

$$\begin{aligned}
S_N(x) &= \frac{2}{L} \int_0^x D_N(u) du - \frac{2}{L} \int_x^0 D_N(L-u) (-du) \\
&= \frac{2}{L} \int_0^x D_N(u) du - \frac{2}{L} \int_0^x D_N(L-u) du
\end{aligned} \tag{3.98}$$

Since we are focusing on the region $x \in (0, L/2)$, the primary contribution to the partial sum $S_N(x)$ arises from the integration of $D_N(u)$ over the neighborhood of the origin, where the Dirichlet kernel is sharply peaked. The term $D_N(L-u)$, centered near $u = L$, lies far from the singularity and contributes only small oscillations. Owing to the localization property of D_N and its decay away from the origin, the second term becomes asymptotically negligible as $N \rightarrow \infty$, for any fixed $x \ll L$. And we have

$$D_N(x) \approx \frac{\sin \left[N \left(\frac{\pi x}{L} \right) \right]}{2 \left(\frac{\pi x}{2L} \right)} \Bigg|_{N \rightarrow \infty, x \rightarrow 0} = \frac{\sin \left[N \left(\frac{\pi x}{L} \right) \right]}{\left(\frac{\pi x}{L} \right)} \tag{3.99}$$

So (3.98) can be written as

$$S_N(x) = \frac{2}{L} \int_0^x \frac{\sin \left[N \left(\frac{\pi u}{L} \right) \right]}{\left(\frac{\pi u}{L} \right)} du \tag{3.100}$$

Let $\pi u/L = \xi$, then $du = L/\pi \cdot d\xi$, and substitute this into (3.100), we have

$$S_N(x) = \frac{2}{L} \int_0^{\pi x/L} \frac{\sin N\xi}{\xi} \frac{L}{\pi} d\xi = \frac{2}{\pi} \int_0^{\pi x/L} \frac{\sin N\xi}{\xi} d\xi \tag{3.101}$$

Differentiating both sides with respect to x , we write down

$$\frac{d}{dx} S_N(x) = \frac{2}{\pi} \frac{d}{dx} \int_0^{\pi x/L} \frac{\sin N\xi}{\xi} d\xi = \frac{2}{\pi} \frac{\sin N \left(\frac{\pi x}{L} \right)}{\left(\frac{\pi x}{L} \right)} \cdot \frac{\pi}{L} = \frac{2}{L} \frac{\sin N \left(\frac{\pi x}{L} \right)}{\left(\frac{\pi x}{L} \right)} = 0 \tag{3.102}$$

When the extremum occurs, $\sin(N\pi x/L) = 0$, so $N\pi x/L = m\pi$, $m \in \mathbb{Z}$. Take $m = 1$, which means that $x = L/N$. Substituting x into (3.101), we obtain

$$S_N \left(\frac{L}{N} \right) = \frac{2}{\pi} \int_0^{\pi/N} \frac{\sin N\xi}{\xi} d\xi \tag{3.103}$$

Let $t = N\xi$, so $d\xi = dt/N$. (3.103) becomes

$$\frac{2}{\pi} \int_0^{\pi} \frac{\sin t}{t/N} \frac{dt}{N} = \frac{2}{\pi} \int_0^{\pi} \frac{\sin t}{t} dt := \frac{2}{\pi} \text{Si}(\pi) \tag{3.104}$$

So, we can verify that

$$\max S_N(x)|_{x \rightarrow 0} = \frac{2}{\pi} \text{Si}(\pi) \approx 1.17897 \quad (3.105)$$

In mathematics, even if we take a large number of terms to approximate the original function, oscillatory behavior will still appear near the discontinuities. The maximum overshoot caused by this oscillation converges to a value approximately 17.9% higher than the actual function value at the jump.

3.4.5 Fourier Transform of a Non-periodic Function

In (3.86), we know that

$$f_L(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}, \quad c_n = \frac{1}{2L} \int_{-L}^L f_L(x) e^{-in\pi x/L} dx.$$

Here $f_L(x)$ is a function of period $2L$. We define the discrete angular frequency

$$\omega_n := \frac{n\pi}{L}.$$

Then c_n can be regarded as a function evaluated at the discrete frequency ω_n , that is

$$c(\omega_n) = \frac{1}{2L} \int_{-L}^L f_L(\xi) e^{-i\omega_n \xi} d\xi. \quad (3.106)$$

As previously mentioned, the coefficient c_n represents the strength of contribution of each complex exponential function with different frequencies or periods. In other words, the function $c(\omega_n)$, defined on the discrete frequency points ω_n , gives the contribution of each Fourier mode. Therefore, $c(\omega_n)$ can be regarded as the amplitude spectrum of $f_L(x)$ in the Fourier series sense (Felix Lee, 2009).

For periodic functions with a finite period, the frequencies are discrete, and the spacing between two adjacent frequencies is given by

$$\Delta\omega = \omega_n - \omega_{n-1} = \frac{\pi}{L}. \quad (3.107)$$

As $L \rightarrow \infty$, the frequency spacing becomes smaller and smaller. However, we should be careful here. The coefficient $c(\omega_n)$ itself does not directly become the continuous Fourier transform. Since

$$\frac{1}{2L} = \frac{\Delta\omega}{2\pi},$$

the coefficient $c(\omega_n)$ contains a factor $\Delta\omega$. Hence the quantity that approaches a continuous spectral density is

$$\frac{c(\omega_n)}{\Delta\omega} = \frac{1}{2\pi} \int_{-L}^L f_L(\xi) e^{-i\omega_n \xi} d\xi.$$

When $L \rightarrow \infty$, we say that $f_L(x) \rightarrow f(x)$ is a non-periodic function, that is

$$\lim_{L \rightarrow \infty} f_L(x) = f(x). \quad (3.108)$$

Furthermore, the spacing between the frequencies becomes

$$\Delta\omega = \frac{\pi}{L} \Big|_{L \rightarrow \infty} = d\omega \rightarrow 0. \quad (3.109)$$

Assume that $f(x)$ exists and satisfies

$$\int_{-\infty}^{\infty} |f(x)| dx < M < \infty, \quad (3.110)$$

where M is a finite constant. By (3.110), we can ensure that the improper integral is convergent and that the Fourier integral is well defined at least in the ordinary integrable sense. Then we can rewrite the Fourier series as

$$\begin{aligned} f(x) &= \lim_{L \rightarrow \infty} \sum_{n=-\infty}^{\infty} \left(\frac{1}{2L} \int_{-L}^L f_L(\xi) e^{-i\omega_n \xi} d\xi \right) e^{i\omega_n x} \\ &= \lim_{L \rightarrow \infty} \sum_{n=-\infty}^{\infty} \frac{\Delta\omega}{2\pi} e^{i\omega_n x} \left(\int_{-L}^L f_L(\xi) e^{-i\omega_n \xi} d\xi \right) \\ &= \lim_{L \rightarrow \infty} \sum_{n=-\infty}^{\infty} \frac{1}{2\pi} e^{i\omega_n x} \Delta\omega \left(\int_{-L}^L f_L(\xi) e^{-i\omega_n \xi} d\xi \right). \end{aligned}$$

Substituting $\Delta\omega = \pi/L$ and taking $L \rightarrow \infty$, the discrete frequencies become continuous. Thus the above summation becomes a Riemann sum in frequency space:

$$\begin{aligned} f(x) &= \frac{1}{2\pi} \lim_{\|\Delta\| \rightarrow 0} \sum_{n=-\infty}^{\infty} e^{i\omega_n x} \Delta\omega \left(\int_{-\infty}^{\infty} f(\xi) e^{-i\omega_n \xi} d\xi \right) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega x} \left(\int_{-\infty}^{\infty} f(\xi) e^{-i\omega \xi} d\xi \right) d\omega \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega x} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(\xi) e^{-i\omega \xi} d\xi \right] d\omega. \end{aligned}$$

So, we can obtain

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega x} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(\xi) e^{-i\omega \xi} d\xi \right] d\omega. \quad (3.111)$$

Then we define the conventional Fourier transform pair as

$$\begin{cases} \mathcal{F}[f(x)] = F(\omega) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\ \mathcal{F}^{-1}[F(\omega)] = f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega \end{cases} \quad (3.112)$$

Here, $f(x)$ is referred to as a function in the **time domain**, which represents the actual shape of the function as it varies with the input variable x . On the other hand, $F(\omega)$ lies in the **frequency domain**, representing the transformation of the original aperiodic function into its frequency components as a function of the continuous frequency variable ω . Since the frequency domain is continuous, we cannot precisely state that only finitely many specific frequencies constitute the original function. Instead, all $\omega \in (-\infty, \infty)$ may contribute to the reconstruction of $f(x)$. However, the value of $F(\omega)$ at each ω tells us the strength and phase contribution of that frequency to the construction of $f(x)$. By examining $F(\omega)$, we can determine which frequencies contribute the most to $f(x)$. For example, in the case of sound waves, this allows us to identify the dominant frequency components of a signal, helping us analyze the pitch or tonal quality of a sound. The following is a graph of the real experimental data.

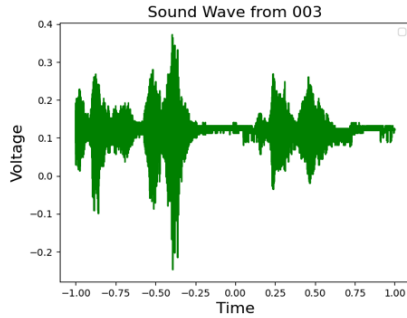


Fig. 3.6 Waveform of the sound signal detected from 003 (Godtone's Wife), converted via oscilloscope

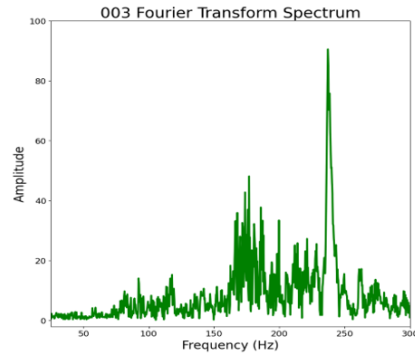


Fig. 3.7 Frequency domain amplitude spectrum of the sound waveform of 003 (Godtone's Wife) after Fourier transform

Consider an odd function $f(x)$, which means that $f(-x) = -f(x)$, and we expand Euler's formula in Equation (3.112).

$$\begin{aligned}
F(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) [\cos(\omega x) - i \sin(\omega x)] dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \cos(\omega x) dx - \frac{i}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \sin(\omega x) dx.
\end{aligned} \tag{3.113}$$

Since the function $\cos(\omega x)$ is an even function of x , the product $f(x) \cos(\omega x)$ is odd, that is

$$f(-x) \cos(-\omega x) = -f(x) \cos(\omega x). \tag{3.114}$$

Similarly, the function $\sin(\omega x)$ is an odd function of x , so the product $f(x) \sin(\omega x)$ is even, since

$$f(-x) \sin(-\omega x) = (-f(x))(-\sin(\omega x)) = f(x) \sin(\omega x). \tag{3.115}$$

Note that the integral of an odd function over a symmetric interval about the origin $x = 0$ is 0, which is

$$\int_{-\infty}^{\infty} f(x) \cos(\omega x) dx = 0. \tag{3.116}$$

Conversely, the integral of an even function over a symmetric interval about the origin $x = 0$ is twice the integral over $x > 0$, that is

$$\int_{-\infty}^{\infty} f(x) \sin(\omega x) dx = 2 \int_0^{\infty} f(x) \sin(\omega x) dx. \tag{3.117}$$

Substituting this into (3.113), we have

$$F(\omega) = \frac{-2i}{\sqrt{2\pi}} \int_0^{\infty} f(x) \sin(\omega x) dx = -i \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin(\omega x) dx. \tag{3.118}$$

Similarly, because changing ω into $-\omega$ changes the sign of $\sin(\omega x)$, we can verify that

$$F(-\omega) = -F(\omega).$$

Thus $F(\omega)$ is an odd function of ω . Then we have

$$\begin{aligned}
f(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega) [\cos(\omega x) + i \sin(\omega x)] d\omega \\
&= \frac{2i}{\sqrt{2\pi}} \int_0^{\infty} F(\omega) \sin(\omega x) d\omega \\
&= i \sqrt{\frac{2}{\pi}} \int_0^{\infty} F(\omega) \sin(\omega x) d\omega.
\end{aligned} \tag{3.119}$$

Substituting (3.118) into (3.119), we obtain

$$\begin{aligned}
f(x) &= i \sqrt{\frac{2}{\pi}} \int_0^{\infty} \left[-i \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(\xi) \sin(\omega \xi) d\xi \right] \sin(\omega x) d\omega \\
&= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sin(\omega x) \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(\xi) \sin(\omega \xi) d\xi \right] d\omega.
\end{aligned}$$

Then we can define a pair of **Fourier sine transforms** representing **odd** function $f(x)$, that is

$$\boxed{
\begin{cases}
\mathcal{F}_s[f(x)] = F_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin(\omega x) dx \\
\mathcal{F}_s^{-1}[F_s(\omega)] = f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s(\omega) \sin(\omega x) d\omega
\end{cases}
} \tag{3.120}$$

We can apply the same way to analyze an even function $f(x)$. If $f(-x) = f(x)$, we have

$$\begin{cases}
f(-x) \cos(-\omega x) = f(x) \cos(\omega x) & \text{even function} \\
f(-x) \sin(-\omega x) = -f(x) \sin(\omega x) & \text{odd function}
\end{cases}$$

So, the integral

$$\int_{-\infty}^{\infty} f(x) \cos(\omega x) dx = 2 \int_0^{\infty} f(x) \cos(\omega x) dx. \tag{3.121}$$

Similarly,

$$\int_{-\infty}^{\infty} f(x) \sin(\omega x) dx = 0. \tag{3.122}$$

Substituting these into (3.113), we can obtain

$$F(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos(\omega x) dx. \quad (3.123)$$

We can deduce that $F(-\omega) = F(\omega)$, so

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty F(\omega) \cos(\omega x) d\omega. \quad (3.124)$$

Then we can define a pair of **Fourier cosine transforms** representing even function $f(x)$, that is

$$\begin{cases} \mathcal{F}_c[f(x)] = F_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos(\omega x) dx \\ \mathcal{F}_c^{-1}[F_c(\omega)] = f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty F_c(\omega) \cos(\omega x) d\omega \end{cases} \quad (3.125)$$

3.4.6 Delta Function Identity

Consider a function which will diverge from $-\infty \rightarrow \infty$:

$$I = \int_{-\infty}^\infty e^{ik(x-x')} dk \rightarrow \infty. \quad (3.126)$$

In general, the integral is only valid in the sense of distribution. But first, we need to consider the integral:

$$I_\epsilon(x) = \lim_{\epsilon \rightarrow 0} \int_{-\infty}^\infty dk e^{ik(x-x')} e^{i\epsilon k^2}. \quad (3.127)$$

Consider the identity of Gaussian integral¹:

$$\int_{-\infty}^\infty dx e^{-ax^2+bx} = \sqrt{\frac{\pi}{a}} e^{b^2/4a}, \quad (3.128)$$

where $a = \epsilon$ and $b = i(x - x')$ in (3.127). Therefore:

$$I_\epsilon = \lim_{\epsilon \rightarrow 0} \sqrt{\frac{\pi}{\epsilon}} \exp \left[-\frac{(x - x')^2}{4\epsilon} \right] = \lim_{\epsilon \rightarrow 0} \sqrt{\frac{\pi}{\epsilon}} \frac{1}{\exp \left[(x - x')^2/4\epsilon \right]}. \quad (3.129)$$

Note that there must be some problems with the result of the integral I_ϵ :

¹ See *Mathematical Physics*.

- If $x = x'$, then the exponential term will be 1. Since $\epsilon \rightarrow 0$, so the term $\sqrt{\pi/\epsilon}$ will diverge.
- If $x \neq x'$, then the exponential term will decay to zero since the decay speed of exponential functions is far faster than the polynomials.

In general, we have a well-defined test function $\phi(x)$, the distribution says

$$\langle I_\epsilon, \phi \rangle = \int_{-\infty}^{\infty} dx \lim_{\epsilon \rightarrow 0} \sqrt{\frac{\pi}{\epsilon}} \exp \left[-\frac{(x-x')^2}{4\epsilon} \right] \phi(x) = \lim_{\epsilon \rightarrow 0} \sqrt{\frac{\pi}{\epsilon}} \int_{-\infty}^{\infty} dx e^{-\left(\frac{x-x'}{2\sqrt{\epsilon}}\right)^2} \phi(x)$$

Let $u = (x - x')/2\sqrt{\epsilon}$, then $2\sqrt{\epsilon} du = dx$, and $x = x' + 2u\sqrt{\epsilon}$. Therefore, we obtain:

$$\begin{aligned} \langle I_\epsilon, \phi \rangle_{\epsilon \rightarrow 0} &= \lim_{\epsilon \rightarrow 0} \sqrt{\frac{\pi}{\epsilon}} \int_{-\infty}^{\infty} 2\sqrt{\epsilon} du e^{-u^2} f(x' + 2u\sqrt{\epsilon}) \\ &= 2\sqrt{\pi} \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} du e^{-u^2} f(x' + 2u\sqrt{\epsilon}) = 2\sqrt{\pi} f(x') \underbrace{\int_{-\infty}^{\infty} du e^{-u^2}}_{=\sqrt{\pi}} = 2\pi f(x'). \end{aligned}$$

After the argument and evaluation, the integral I at the beginning has the meaning under the distribution. The result above shows that the distribution is equivalent to:

$$\langle I_\epsilon, \phi \rangle_{\epsilon \rightarrow 0} = \langle I, \phi \rangle = \langle 2\pi\delta, \phi \rangle = \phi(x'),$$

i.e.

$$\boxed{I = \int_{-\infty}^{\infty} dk e^{ik(x-x')} = 2\pi\delta(x-x').} \quad (3.130)$$

In 3-dimension, the integrating element k becomes $\mathbf{k} = (k_x, k_y, k_z)$, and the domain D_x becomes $D_{\mathbf{r}}$, where $\mathbf{r} = (x, y, z)$.² The integral is written as

$$I = \int_{\mathbb{R}^3} d^3\mathbf{k} e^{i(\mathbf{k} \cdot \mathbf{r})} = \int d^3\mathbf{k} e^{i(k_x x + k_y y + k_z z)}, \quad (3.131)$$

which is equal to

$$I = \int_{D_x} dx e^{ik_x x} \int_{D_y} dy e^{ik_y y} \int_{D_z} dz e^{ik_z z} = (2\pi)^3 \delta(x) \delta(y) \delta(z) = (2\pi)^3 \delta^3(\mathbf{r}). \quad (3.132)$$

More general, when $\mathbf{r} \rightarrow \mathbf{r}'$, the integral becomes

$$I = \int d^3\mathbf{k} e^{i\mathbf{k} \cdot \mathbf{r}'} = 8\pi^3 \delta(x-x') \delta(y-y') \delta(z-z') = 8\pi^3 \delta^3(\mathbf{r}'). \quad (3.133)$$

² From this argument, we can predict that this new integral can be considered to be the distribution under three bases, or directions.

3.4.7 Another Perspective of Fourier Transform and Representation

A well-defined (sometimes not, but I don't really care) function $f(x)$ can be denoted by an infinite periodic function³ with respect to the frequency k , this is called *Fourier Transformation*. The Fourier transform is defined as

$$f(x) = \int_{-\infty}^{\infty} \frac{dk}{\mathcal{N}_1} \tilde{f}(k) e^{ikx}, \quad \tilde{f}(k) = \int_{-\infty}^{\infty} \frac{dx'}{\mathcal{N}_2} f(x') e^{-ikx'}. \quad (3.134)$$

WLOG, substituting $\tilde{f}(k)$ into the $f(x)$ transformation, we must have

$$f(x) = \int_{-\infty}^{\infty} \frac{dk}{\mathcal{N}_1} \left[\int_{-\infty}^{\infty} \frac{dx'}{\mathcal{N}_2} f(x') e^{-ikx'} \right] e^{ikx}. \quad (3.135)$$

After the substitution, the equation must hold. Then we expand the integral:

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{dk}{\mathcal{N}_1} \left[\int_{-\infty}^{\infty} \frac{dx'}{\mathcal{N}_2} f(x') e^{-ikx'} \right] e^{ikx} &= \frac{1}{\mathcal{N}_1 \mathcal{N}_2} \int_{-\infty}^{\infty} dk \int_{-\infty}^{\infty} dx' f(x') e^{ik(x-x')} \\ &= \frac{1}{\mathcal{N}_1 \mathcal{N}_2} \int_{-\infty}^{\infty} dx' f(x') \int_{-\infty}^{\infty} dk e^{ik(x-x')} \\ &= \frac{1}{\mathcal{N}_1 \mathcal{N}_2} \int_{-\infty}^{\infty} dx' f(x') \cdot 2\pi \delta(x-x'), \end{aligned}$$

and the integral is $f(x)$, i.e.

$$f(x) = \frac{1}{\mathcal{N}_1 \mathcal{N}_2} \int_{-\infty}^{\infty} dx' f(x') \cdot 2\pi \delta(x-x') = \frac{2\pi}{\mathcal{N}_1 \mathcal{N}_2} f(x). \quad (3.136)$$

To make the equation hold, it's obvious that

$$\boxed{\mathcal{N}_1 \mathcal{N}_2 = 2\pi} \quad \text{1D.} \quad (3.137)$$

I'm going to generalize the result to 3-D. Consider a function of the position vector $f(\mathbf{r})$ can also be applied the Fourier transform:

$$f(\mathbf{r}) = \int \frac{d^3 \mathbf{k}}{\mathcal{N}_1} \tilde{f}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r})}, \quad \tilde{f}(\mathbf{k}) = \int \frac{d^3 \mathbf{r}'}{\mathcal{N}_2} f(\mathbf{r}') e^{-i(\mathbf{k} \cdot \mathbf{r}')}. \quad (3.138)$$

Similarly,

³ Continuous Fourier series

$$\begin{aligned}
f(\mathbf{r}) &= \frac{1}{\mathcal{N}_1 \mathcal{N}_2} \int d^3 \mathbf{k} \left[\int d^3 \mathbf{r}' f(\mathbf{r}') e^{-i(\mathbf{k} \cdot \mathbf{r}')} \right] e^{i(\mathbf{k} \cdot \mathbf{r})} \\
&= \frac{1}{\mathcal{N}_1 \mathcal{N}_2} \int d^3 \mathbf{r}' f(\mathbf{r}') \int d^3 \mathbf{k} e^{i[\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')]},
\end{aligned}$$

by (7.82), we obtain

$$f(\mathbf{r}) = \frac{1}{\mathcal{N}_1 \mathcal{N}_2} \int d^3 \mathbf{r}' f(\mathbf{r}') (2\pi)^3 \delta(\mathbf{r} - \mathbf{r}') = \frac{(2\pi)^3}{\mathcal{N}_1 \mathcal{N}_2} \int d^3 \mathbf{r}' f(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}'). \quad (3.139)$$

Therefore,

$$\boxed{\mathcal{N}_1 \mathcal{N}_2 = (2\pi)^3} \quad 3\text{D}. \quad (3.140)$$

Again, this can be generalized to N-dimensions. This transformation is also valid if the function becomes a vector function in N-dimensions:

$$\mathbf{F}(\mathbf{r}) = \frac{1}{\mathcal{N}_1} \int d^N \mathbf{k} \tilde{\mathbf{F}}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r})}, \quad \tilde{\mathbf{F}}(\mathbf{k}) = \frac{1}{\mathcal{N}_2} \int d^N \mathbf{r}' \mathbf{F}(\mathbf{r}') e^{-i(\mathbf{k} \cdot \mathbf{r}')}. \quad (3.141)$$

Since the exponential term is Euler's formula $e^{ix} = \cos x + i \sin x$, so it's ok to add some phase difference into the transformation without the validity of it, for example,

$$\mathbf{F}(\mathbf{r}) = \frac{1}{\mathcal{N}_1} \int d^N \mathbf{k} \tilde{\mathbf{F}}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r})} \rightarrow \mathbf{F}(\mathbf{r}) = \frac{1}{\mathcal{N}_1} \int d^N \mathbf{k} \tilde{\mathbf{F}}(\mathbf{k}) e^{i[\mathbf{k} \cdot \mathbf{r} \pm \omega(\mathbf{k})t]}. \quad (3.142)$$

3.4.8 Applications: DFT and FFT

In the previous subsection, we have seen that Fourier transformation can be understood as a representation of a function in terms of exponential modes. This is not only a beautiful algebraic rewriting. In fluid mechanics and atmospheric physics, it is one of the most direct ways to extract structure from data. The data may come from a thermometer recording temperature every second, a sonic anemometer recording wind velocity, a pressure sensor inside a turbulent boundary layer, or a numerical model storing the full velocity field $\mathbf{u}(\mathbf{r}, t)$ on a spatial grid. In all of these cases, the raw data are usually not a continuous function. They are finite lists of numbers.

We first consider the simplest case. Suppose that the observed signal is almost a single linear wave,

$$x(t) = A \cos(2\pi f_0 t + \phi). \quad (3.143)$$

Then the time series oscillates regularly. If the sampling is good enough, the frequency f_0 can be identified almost by looking at the graph. However, real atmospheric data are rarely so clean. In high Reynolds number flow, nonlinear advection transfers energy among scales. Large eddies, small eddies, wave packets, acoustic noise, gravity waves, instrument noise, and coherent structures may all appear in the same record. A more

realistic signal is therefore closer to

$$x(t) = \sum_{j=1}^M A_j \cos(2\pi f_j t + \phi_j) + \eta(t), \quad (3.144)$$

where $\eta(t)$ represents the unresolved part, stochastic fluctuation, measurement noise, or small scale turbulent residue. In the time domain, (3.144) may look completely messy. Still, if the dominant frequencies f_j are present, Fourier analysis can separate them into spectral modes. This is the reason why spectral decomposition is so useful in atmospheric observation and computational fluid dynamics.

To connect the continuous notation with real data, let

$$t_n = n\Delta t, \quad x_n = x(t_n), \quad n = 0, 1, \dots, N-1, \quad (3.145)$$

where Δt is the sampling interval. The discrete Fourier transform is defined by

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N}, \quad k = 0, 1, \dots, N-1. \quad (3.146)$$

The inverse transformation is

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{i2\pi kn/N}. \quad (3.147)$$

The index k is the discrete frequency mode. If the signal is sampled with interval Δt , then the total observation time is

$$T = N\Delta t, \quad (3.148)$$

and the frequency spacing is

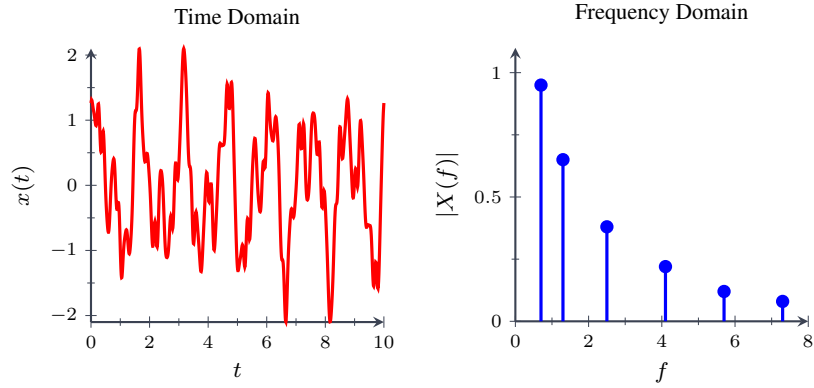
$$\Delta f = \frac{1}{T} = \frac{1}{N\Delta t}. \quad (3.149)$$

Thus the physical frequency corresponding to the k th mode is

$$f_k = k\Delta f = \frac{k}{N\Delta t}. \quad (3.150)$$

The meaning is direct. The sequence $\{x_n\}_{n=0}^{N-1}$ is the data in the time domain, and $\{X_k\}_{k=0}^{N-1}$ is the same data written in the frequency domain. The transformation does not create new information. It reorganizes the information into modes.

The following figure illustrates this idea. The left panel shows a signal made from several oscillatory components with different frequencies, amplitudes, and phases. It looks like a random and chaotic turbulent time series. The right panel shows the corresponding frequency content. The apparently messy signal is decomposed into several clean discrete modes.



A random looking turbulent time series and its discrete spectral decomposition. The left signal is constructed from several oscillatory modes, while the right panel displays the corresponding amplitudes in the frequency domain.

The essential point of Figure 3.4.8 is that complexity in physical space or time does not necessarily imply complexity in spectral space. In a stratified atmosphere, for example, an observed vertical velocity signal may contain a slowly varying synoptic component, mesoscale gravity wave activity, turbulent fluctuation, and local sensor noise. In the time domain these components are entangled. In the frequency domain, they can often be separated by scale. This gives us a systematic way to ask which part of the observed field is energetic, which part is coherent, and which part should be regarded as unresolved fluctuation.

From the viewpoint of linear algebra, the DFT in (3.146) is a matrix multiplication. Define

$$W_N = e^{-i2\pi/N}. \quad (3.151)$$

Then (3.146) becomes

$$X_k = \sum_{n=0}^{N-1} x_n W_N^{kn}. \quad (3.152)$$

Equivalently,

$$\begin{bmatrix} X_0 \\ X_1 \\ \vdots \\ X_{N-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & W_N & \cdots & W_N^{N-1} \\ 1 & W_N^2 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & \cdots & W_N^{(N-1)(N-1)} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}. \quad (3.153)$$

This matrix is constructed from the N th roots of unity on the complex plane. Geometrically, the DFT projects the data vector onto rotating directions uniformly distributed on the unit circle. Each row tests how much the data rotate like a particular mode. If the data contain that mode, the complex phases add constructively. If not, the phases cancel each other. This is why Fourier analysis can extract periodic structure from a signal that looks visually disordered.

However, the direct computation of (3.153) is expensive. For each k , we must sum over all $n = 0, 1, \dots, N - 1$. Since there are N different values of k , the total number of operations scales like

$$C_{\text{DFT}}(N) = O(N^2). \quad (3.154)$$

For small data this is acceptable. For atmospheric data this quickly becomes impossible. A single simulation may contain millions or billions of grid point values. A three dimensional velocity field may also be transformed at every time step. In that situation, the difference between N^2 and $N \log N$ is not a minor improvement. It changes the problem from practically impossible to computationally routine.

The fast Fourier transform is not a different mathematical transform. It computes the same X_k in (3.146), but it uses the symmetry and periodicity of the roots of unity to avoid repeated work. The standard Cooley and Tukey idea is to split the data into even and odd indices. Assume that N is even. Then

$$\begin{aligned} X_k &= \sum_{n=0}^{N-1} x_n W_N^{kn} \\ &= \sum_{m=0}^{N/2-1} x_{2m} W_N^{k(2m)} + \sum_{m=0}^{N/2-1} x_{2m+1} W_N^{k(2m+1)} \\ &= \sum_{m=0}^{N/2-1} x_{2m} W_{N/2}^{km} + W_N^k \sum_{m=0}^{N/2-1} x_{2m+1} W_{N/2}^{km}. \end{aligned} \quad (3.155)$$

Define the two shorter DFTs

$$E_k = \sum_{m=0}^{N/2-1} x_{2m} W_{N/2}^{km}, \quad O_k = \sum_{m=0}^{N/2-1} x_{2m+1} W_{N/2}^{km}. \quad (3.156)$$

Then

$$\boxed{X_k = E_k + W_N^k O_k}, \quad k = 0, 1, \dots, \frac{N}{2} - 1. \quad (3.157)$$

Using the identity

$$W_N^{k+N/2} = -W_N^k, \quad (3.158)$$

we also obtain

$$\boxed{X_{k+N/2} = E_k - W_N^k O_k}, \quad k = 0, 1, \dots, \frac{N}{2} - 1. \quad (3.159)$$

Equations (3.157) and (3.159) form the butterfly computation. Two outputs are obtained from the same pair E_k and O_k , differing only by the sign before the twiddle factor $W_N^k O_k$. Repeating this splitting recursively gives

$$C_{\text{FFT}}(N) = 2C_{\text{FFT}}\left(\frac{N}{2}\right) + O(N). \quad (3.160)$$

Therefore,

$$C_{\text{FFT}}(N) = O(N \log_2 N). \quad (3.161)$$

The distinction between DFT and FFT can be summarized as follows.

	DFT	FFT
Mathematical meaning	The discrete Fourier transform itself. It maps $\{x_n\}$ to $\{X_k\}$ by direct summation.	An algorithm for computing exactly the same DFT coefficients.
Main structure	Dense matrix multiplication by the Fourier matrix in (3.153).	Recursive factorization using roots of unity, even odd splitting, and butterfly computation.
Computational cost	$O(N^2)$.	$O(N \log N)$.
Physical role	Defines the spectral content of sampled data.	Makes spectral analysis possible for large scale simulations and observations.

Table 3.1 Comparison between the discrete Fourier transform and the fast Fourier transform.

In computational fluid dynamics, the advantage of FFT becomes even more important because many operations become simple in spectral space. For example, if a field $u(x)$ is represented by

$$u(x) = \sum_k \hat{u}_k e^{ikx}, \quad (3.162)$$

then differentiation becomes multiplication:

$$\frac{\partial u}{\partial x} = \sum_k ik \hat{u}_k e^{ikx}. \quad (3.163)$$

This is one of the reasons why spectral and pseudo spectral methods are so powerful. Spatial derivatives in the Navier Stokes equations can be computed with high accuracy, and the nonlinear term can be evaluated by transforming back and forth between physical space and spectral space.

For a three dimensional grid with N_x, N_y, N_z points, a separable Fourier transform gives the approximate computational cost

$$C_{3\text{DFFT}} = O(N_x N_y N_z [\log N_x + \log N_y + \log N_z]). \quad (3.164)$$

This scaling is crucial in large eddy simulation, numerical weather prediction, and high frequency observation of geostrophic or quasi geostrophic flows. When the data size increases, a direct DFT grows quadratically and rapidly becomes useless. The FFT grows almost linearly, up to a logarithmic factor. This is why modern atmospheric science cannot really be separated from FFT based computation.

Physically, the Fourier transform is a language of modes. Numerically, the FFT is the machine that allows this language to be used on large data. In ideal linear waves, the spectral peaks are sharp and easy to interpret. In turbulent atmospheric flows, the spectrum becomes broader, and the energy is distributed across many scales. Still, the idea remains the same. We take complicated data, decompose it into modes, and then interpret the physical meaning of these modes. In this sense, DFT gives the mathematical object, while FFT gives the practical computational route to obtain it.

Chapter 4

Ordinary Differential Equation

Abstract This chapter is about learning how to handle the equations of motion of physical systems mathematically. And honestly, it might be the most important chapter you'll come across in college. We are going to learn how to eliminate derivatives in equations to bring out the underlying structure, something you will see over and over again, not just in physics but also in electrical and civil engineering. If you are planning to survive and thrive in NCU Physics, the tools in this chapter are your best friends. So, please pay close attention; this one is worth it.

4.1 First-Order ODEs

4.1.1 Linearity of ODE and Differentiation

Before solving differential equations, we first need to know what kind of object an ODE is. A first-order ordinary differential equation usually relates an unknown function $y(x)$ to its derivative $y'(x)$ and the independent variable x . A general first-order ODE may be written as

$$F(x, y, y') = 0. \quad (4.1)$$

If the equation can be solved for y' , we may also write it as

$$y' = f(x, y). \quad (4.2)$$

This form is especially useful in physics because it tells us how the state changes once the present state is known.

Definition 4.1 (Linear First-Order ODE) A first-order ODE is called *linear* if it can be written as

$$y' + P(x)y = Q(x), \quad (4.3)$$

where $P(x)$ and $Q(x)$ are functions of the independent variable only.

The word linear means that the unknown function y and its derivative y' appear only to the first power. Terms such as y^2 , $(y')^2$, yy' , or $\sin y$ destroy linearity. This distinction is not only a matter of algebra. Linear equations allow superposition, and this is why they appear so naturally in small oscillations, RC circuits, heat conduction, and quantum mechanics.

To make this idea more precise, we introduce the differentiation operator

$$D := \frac{d}{dx}. \quad (4.4)$$

This operator takes a differentiable function and returns another function:

$$D[y] = \frac{dy}{dx}. \quad (4.5)$$

Therefore, differentiation is a map between function spaces. For example, if $C^1(I)$ is the space of continuously differentiable functions on an interval I , then

$$D : C^1(I) \rightarrow C^0(I). \quad (4.6)$$

The important point is that D is a linear map.

Definition 4.2 (Linear Operator) Let V and W be vector spaces. A map $T : V \rightarrow W$ is called a *linear operator* if for any $f, g \in V$ and any constants a, b ,

$$T[af + bg] = aT[f] + bT[g]. \quad (4.7)$$

Theorem 4.1 (Linearity of Differentiation) *The differentiation operator*

$$D = \frac{d}{dx} \quad (4.8)$$

is linear. That is, for differentiable functions f, g and constants a, b ,

$$D[af + bg] = aD[f] + bD[g]. \quad (4.9)$$

Proof. By the definition of differentiation,

$$D[af + bg] = \frac{d}{dx}[af(x) + bg(x)]. \quad (4.10)$$

Since constants can be pulled out of the derivative and the derivative of a sum is the sum of the derivatives, we have

$$\frac{d}{dx}[af(x) + bg(x)] = a \frac{df}{dx} + b \frac{dg}{dx}. \quad (4.11)$$

Thus

$$D[af + bg] = aD[f] + bD[g]. \quad (4.12)$$

Hence D is linear. $\square$

This is the point where differential equations start to connect with linear algebra. In linear algebra, a matrix A acts on a vector $\mathbf{x}$ and gives another vector:

$$A\mathbf{x} = \mathbf{b}. \quad (4.13)$$

In a linear ODE, a differential operator acts on a function y and gives another function:

$$L[y] = Q(x). \quad (4.14)$$

For a first-order linear ODE, the operator is

$$L = D + P(x), \quad (4.15)$$

so the equation

$$y' + P(x)y = Q(x) \quad (4.16)$$

can be written as

$$L[y] = Q(x). \quad (4.17)$$

Here L is linear because

$$\begin{aligned} L[ay_1 + by_2] &= D[ay_1 + by_2] + P(x)(ay_1 + by_2) \\ &= aD[y_1] + bD[y_2] + aP(x)y_1 + bP(x)y_2 \\ &= aL[y_1] + bL[y_2]. \end{aligned} \quad (4.18)$$

For a second-order linear ODE with variable coefficients,

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = Q(x), \quad (4.19)$$

we may define

$$L = a_2(x)D^2 + a_1(x)D + a_0(x). \quad (4.20)$$

Then the ODE becomes

$$L[y] = Q(x). \quad (4.21)$$

This notation is useful because it turns the ODE into an operator equation. The unknown object is still a function, but the structure is similar to a matrix equation.

Theorem 4.2 (Superposition Principle) *Let L be a linear differential operator. If*

$$L[y_1] = Q_1(x), \quad L[y_2] = Q_2(x), \quad (4.22)$$

then for any constants a, b ,

$$L[ay_1 + by_2] = aQ_1(x) + bQ_2(x). \quad (4.23)$$

Proof. Since L is linear,

$$L[ay_1 + by_2] = aL[y_1] + bL[y_2]. \quad (4.24)$$

Using

$$L[y_1] = Q_1(x), \quad L[y_2] = Q_2(x), \quad (4.25)$$

we obtain

$$L[ay_1 + by_2] = aQ_1(x) + bQ_2(x). \quad (4.26)$$

□

Definition 4.3 (Homogeneous and Nonhomogeneous ODE) Let L be a linear differential operator. An ODE written in the form

$$L[y] = 0 \quad (4.27)$$

is called a *homogeneous ODE*. An ODE written in the form

$$L[y] = Q(x), \quad Q(x) \neq 0, \quad (4.28)$$

is called a *nonhomogeneous ODE*.

The difference between homogeneous and nonhomogeneous equations is not just the right hand side. The homogeneous equation describes the natural behavior of the system without an external input. The nonhomogeneous equation describes the response of the system under an external source, force, or driving term.

For example, the equation

$$m\ddot{x} + kx = 0 \quad (4.29)$$

describes the free oscillation of a mass spring system. There is no external driving force. On the other hand,

$$m\ddot{x} + kx = F(t) \quad (4.30)$$

is nonhomogeneous, because the motion is driven by the external force $F(t)$.

Definition 4.4 (Homogeneous Solution and Particular Solution) Consider a nonhomogeneous linear ODE

$$L[y] = Q(x). \quad (4.31)$$

A function y_h is called a *homogeneous solution* if

$$L[y_h] = 0. \quad (4.32)$$

A function y_p is called a *particular solution* if

$$L[y_p] = Q(x). \quad (4.33)$$

With these definitions, the structure of a linear nonhomogeneous ODE becomes very clear. If y_p is one particular solution and y_h is any homogeneous solution, then

$$L[y_p + y_h] = L[y_p] + L[y_h] = Q(x) + 0 = Q(x). \quad (4.34)$$

Therefore, every solution of the nonhomogeneous equation has the form

$$\boxed{y = y_h + y_p}. \quad (4.35)$$

This formula is one of the most important ideas in linear ODEs. The particular solution gives one response to the forcing term, while the homogeneous solution gives the freedom left by the natural motion of the system.

In linear algebra, this is exactly the same idea as solving

$$A\mathbf{x} = \mathbf{b}. \quad (4.36)$$

One particular vector solves the nonhomogeneous equation, and the null space of A can be added to it. For differential equations, the null space is replaced by the solution space of

$$L[y] = 0. \quad (4.37)$$

The homogeneous equation has the same role as the null space in linear algebra. If y_1 and y_2 are homogeneous solutions, then

$$L[y_1] = 0, \quad L[y_2] = 0. \quad (4.38)$$

By linearity,

$$L[c_1y_1 + c_2y_2] = c_1L[y_1] + c_2L[y_2] = 0. \quad (4.39)$$

Therefore, homogeneous solutions form a vector space. (See [Appendix A](#))

4.1.2 Homogeneous Problems and Separable Equations

After introducing the operator notation, we first study the simplest homogeneous first-order linear ODE:

$$y' + P(x)y = 0. \quad (4.40)$$

Using the differentiation operator

$$D = \frac{d}{dx}, \quad (4.41)$$

this equation may be written as

$$(D + P(x))y = 0. \quad (4.42)$$

However, for this first-order case, the most direct method is separation of variables.

From (4.40), we have

$$\frac{dy}{dx} = -P(x)y. \quad (4.43)$$

If $y \neq 0$ on the interval we are considering, then we may divide both sides by y and multiply by dx :

$$\frac{dy}{y} = -P(x)dx. \quad (4.44)$$

Now integrate both sides:

$$\int \frac{1}{y} dy = - \int P(x) dx. \quad (4.45)$$

This gives

$$\ln |y| = - \int P(x) dx + C. \quad (4.46)$$

Exponentiating both sides, we obtain

$$|y| = e^C \exp \left(- \int P(x) dx \right). \quad (4.47)$$

Since the sign can be absorbed into the constant, the general solution is

$$\boxed{y = C \exp \left(- \int P(x) dx \right)}. \quad (4.48)$$

This formula also includes the zero solution by choosing $C = 0$. Therefore, (4.48) is the full homogeneous solution.

The method used above is called *separation of variables*. The main idea is to rewrite the differential equation so that all y dependent terms stay on one side and all x dependent terms stay on the other side:

$$g(y) dy = f(x) dx. \quad (4.49)$$

After that, both sides can be integrated:

$$\int g(y) dy = \int f(x) dx. \quad (4.50)$$

This method is simple, but it only works when the variables can actually be separated.

For example, consider

$$y' + ky = 0, \quad k > 0. \quad (4.51)$$

Here

$$P(x) = k. \quad (4.52)$$

Using (4.48), we get

$$y = C \exp \left(- \int k dx \right) = C e^{-kx}. \quad (4.53)$$

This is the standard exponential decay law. It appears in radioactive decay, RC circuits, linear damping, and Newton's law of cooling.

For instance, in an RC circuit without a battery, the charge $q(t)$ on the capacitor satisfies

$$R \frac{dq}{dt} + \frac{1}{C} q = 0. \quad (4.54)$$

Equivalently,

$$\frac{dq}{dt} + \frac{1}{RC}q = 0. \quad (4.55)$$

Thus

$$q(t) = q_0 e^{-t/(RC)}. \quad (4.56)$$

The homogeneous solution describes how the system relaxes naturally when there is no external source.

If a source term is added, for example

$$y' + P(x)y = Q(x), \quad (4.57)$$

then the equation becomes nonhomogeneous. The solution is no longer only the natural response. It must be written as

$$y = y_h + y_p. \quad (4.58)$$

Here y_h solves the homogeneous equation

$$y'_h + P(x)y_h = 0, \quad (4.59)$$

while y_p is any one solution of the full equation

$$y'_p + P(x)y_p = Q(x). \quad (4.60)$$

This distinction will be used repeatedly when we solve second-order equations by the descent method and the factorization method.

Example 4.1 Consider a rocket moving vertically upward in space (neglecting air resistance and gravity). Let $m(t)$ be the mass of the rocket at time t , and assume the rocket expels gas at a constant rate, so $m(t) = m_0 - kt$, where $k > 0$ is a constant. According to the rocket equation, the velocity $v(t)$ satisfies the equation

$$m(t) \frac{dv}{dt} + v \frac{dm}{dt} = 0$$

Solve for $v(t)$ as a function of t , and find the constant using the initial condition $v(0) = v_0$.

Solution 4.1

By the function $m(t) = m_0 - kt$, so that

$$\frac{dm(t)}{dt} = -k$$

Substituting this to the original ODE, we can get

$$(m_0 - kt) \frac{dv}{dt} - kv = 0 \Rightarrow \frac{dv}{v} = \frac{k}{m_0 - kt} dt$$

Let $m_0 - kt = u$, $du = -k dt$, and substitute this into the equation and integrate both sides,

$$\int \frac{dv}{v} = - \int \frac{1}{u} du$$

So we obtain

$$\ln v = - \ln u + \text{const} \Rightarrow v = \frac{C}{u} = \frac{C}{m_0 - kt}$$

By the initial condition, $v(0) = v_0$, the constant $C = m_0 v_0$. So, the final answer is

$$v(t) = \frac{m_0 v_0}{m_0 - kt}$$

4.1.3 Integrating Factor

Consider an ODE $P(x, y)dx + Q(x, y)dy = 0$, we can reduce the equation to the form of

$$\mathbf{F}(P(x_i), Q(x_j)) \cdot d\mathbf{r} = 0 \quad (4.61)$$

Which implies the meaning of vector field in the ODE. So the geometric meaning of an ODE is the projection of a vector field $\mathbf{F}$ onto the tangent vector $d\mathbf{r}$ to derive the equipotential surfaces while the ODE can be solved.

If we can find a scalar function ϕ which satisfies that

$$d\phi = Pdx + Qdy \quad (4.62)$$

or the equation

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad (4.63)$$

Then the original ODE $Pdx + Qdy = 0$ is called an **exact differential equation**. In other words, the vector field $\mathbf{F} = (P, Q)$ in the exact differential equation can be written as $\mathbf{F} = \nabla\phi$ where ϕ is the potential function of $\mathbf{F}$, which is a conservative vector field then we can solve the ODE naturally because the line integral

$$\int_A^x \mathbf{F} \cdot d\mathbf{r} = \int_A^x \nabla\phi \cdot d\mathbf{r} = \phi(x) - \phi(A) \quad (4.64)$$

where we often define the constant term $-\phi(A)$ to be C . But if the differential equation $Pdx + Qdy = 0$ is not an exact ODE, we cannot integrate it directly. However, if there exists a function μ such that $\mu Pdx + \mu Qdy = 0$ be an exact ODE, then the function μ is called **integrating factor**.

That is to say, as long as the integrating factor exists, the ODE can be solved or called "integrable." Let's verify the facts in detail. If the differential equation $\mathbf{F} \cdot d\mathbf{r} = 0$ is

integrable, then the integrating factor can make it be

$$\mu \mathbf{F} = \nabla \phi \quad (4.65)$$

We know that the curl of the gradient is 0, so

$$\nabla \times \nabla \phi = \nabla \times (\mu \mathbf{F}) = 0 \quad (4.66)$$

(4.66) can be expanded to

$$\nabla \times (\mu \mathbf{F}) = \mu \nabla \times \mathbf{F} + \nabla \mu \times \mathbf{F} = 0$$

$$\Rightarrow \nabla \times \mathbf{F} = -\frac{\nabla \mu \times \mathbf{F}}{\mu}, \mu \neq 0$$

After that, the dot product between the original vector field can be denoted by

$$\mathbf{F} \cdot \nabla \times \mathbf{F} = -\frac{1}{\mu} [\mathbf{F} \cdot (\nabla \mu \times \mathbf{F})] = 0 \quad (4.67)$$

So, when (4.67) holds, it shows that $\exists \mu$ such that the differential equation $Pdx + Qdy = 0$, $\mathbf{F} = (P, Q)$ is integrable and solvable.

Now I am going to show an example of the integrating factor by solving the first-order linear ODE, which is the following "main character." Consider an ODE

$$y' + P(x)y = Q(x) \quad (4.68)$$

We need to separate the variable that

$$dy + P(x)ydx = Q(x)dx$$

Assume that this ODE is integrable and solvable, we multiply both sides by $\mu(x)$. Which means that

$$\mu(x)dy + \mu(x)P(x)ydx = d(\mu(x)y) \quad (4.69)$$

The total derivative of $d(\mu(x)y)$ satisfies the differential rule, which is

$$d(\mu(x)y) = \mu(x)dy + \mu'(x)ydx$$

Compare the coefficient of ydx with (4.69), we have

$$\mu'(x) = \mu(x)P(x)$$

it is obvious that we can take

$$\boxed{\mu = \exp \left(\int P(x)dx \right)} \quad (4.70)$$

Back to the original ODE, we multiplied a factor μ in the whole equation, so

$$\mu dy + \mu P(x)y dx = d(\mu y) = \mu Q(x) dx$$

So that

$$\int d(\mu y) = \int \mu Q(x) dx \Rightarrow \mu y = \int \mu Q(x) dx + C'$$

We can finally get the familiar solution

$$y = \exp\left(-\int P(x) dx\right) \left(\int \left[\exp\left(\int P(x) dx\right) Q(x)\right] dx + C'\right) \quad (4.71)$$

That is why we should solve the homogeneous term of an ODE, and we need to put it back into the original equation to get the final answer. The point is **INTEGRABLE**.

In general, the solution can be written as

$$y(x) = \frac{\mu(0)y(0)}{\mu(x)} + \frac{1}{\mu(x)} \int_0^x \mu(s)G(s) ds \quad (4.72)$$

Example 4.2 Consider a radioactive decay chain: Ra-226 decays into Rn-222, and Rn-222 itself further decays into other elements. Let $N_{\text{Ra}}(t)$ and $N_{\text{Rn}}(t)$ denote the number of Ra-226 and Rn-222 atoms at time t , respectively.

Assume that Ra-226 decays at a constant rate λ_1 , Rn-222 decays at a rate λ_2 . Initially, $N_{\text{Ra}}(t) \approx N_0$ (since Ra-226 decays very slowly compared to Rn-222, its number remains nearly constant over the timescale considered). Solve for $N_{\text{Rn}}(t)$ with initial condition $N_{\text{Rn}}(0) = 0$.

Solution 4.2

The key to this problem lies in understanding the physical meaning behind the decay equation. Before solving for the function $N_{\text{Rn}}(t)$, we must first consider the sequence of decay. As R_a decays into R_n , and R_n further decays into another atom, the quantity $N_{\text{Rn}}(t)$ is simultaneously affected by both the decay of R_a (which produces R_n) and the decay of R_n itself (which reduces the amount of R_n). In other words, $N_{\text{Rn}}(t)$ is governed by a balance between production and loss. The differential equation must therefore include a source term from R_a and a sink term from the decay of R_n .

Let's start with the $N_{\text{Ra}}(t)$, the decay function is shown below

$$\frac{dN_{\text{Ra}}(t)}{dt} = -\lambda_1 N_{\text{Ra}}(t)$$

The solution is

$$N_{\text{Ra}}(t) = N_0 e^{-\lambda_1 t}$$

Since the initial condition $N_{\text{Ra}}(0) = N_0$. In the same time, the decay function of $N_{\text{Rn}}(t)$ is

$$\frac{dN_{Rn}(t)}{dt} = \lambda_1 N_{Ra}(t) - \lambda_2 N_{Rn}(t)$$

Note that we must be careful to put $\lambda_1 N_{Ra}(t)$. So it becomes a nonhomogeneous ODE. The integrating factor of the equation is

$$\mu = e^{\lambda_2 t}$$

multiply the factor at both sides, and substitute $N_{Ra}(t) = N_0 e^{-\lambda_1 t}$ in the equation, we can solve that

$$\begin{aligned} \int d(N_{Rn} e^{\lambda_2 t}) &= \int \lambda_1 N_0 e^{-\lambda_1 t} \cdot e^{\lambda_2 t} dt \\ &= \lambda_1 N_0 \int e^{(\lambda_2 - \lambda_1)t} dt \\ &= \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} e^{(\lambda_2 - \lambda_1)t} + C \end{aligned}$$

and we can calculate that

$$N_{Rn}(t) = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + C e^{-\lambda_2 t}$$

At time $t = 0$, $N_{Rn}(0) = 0$, so the constant $C = \frac{-\lambda_1 N_0}{\lambda_2 - \lambda_1}$, substituting C into the result, so the final solution to the N_{Rn} is

$$N_{Rn} = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t})$$

4.1.4 The Bernoulli Equation

The differential equation

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \quad n \neq 0, 1, \quad (4.73)$$

is called the **Bernoulli equation**. Here P and Q are functions of x . The equation is nonlinear when written in terms of y , but it can be turned into a linear first-order ODE by a clever substitution.

Define

$$u = y^{1-n}. \quad (4.74)$$

By the chain rule,

$$u' = (1 - n)y^{-n}y'. \quad (4.75)$$

Multiplying the Bernoulli equation by $(1 - n)y^{-n}$ gives

$$(1 - n)y^{-n}y' + (1 - n)P(x)y^{1-n} = (1 - n)Q(x). \quad (4.76)$$

Therefore the equation becomes the linear ODE

$$u' + (1 - n)P(x)u = (1 - n)Q(x). \quad (4.77)$$

This is the key point. The original nonlinear equation for y is transformed into a linear equation for u . Using the integrating factor method, we obtain

$$u = y^{1-n} = \exp \left[- \int (1 - n)P(x)dx \right] \left(\int \exp \left[\int (1 - n)P(x)dx \right] (1 - n)Q(x)dx + C \right). \quad (4.78)$$

After solving for u , we substitute $u = y^{1-n}$ back to recover y .

Example 4.3 Consider a horizontal pipe through which an incompressible fluid is flowing with constant velocity v_0 . Due to nonlinear friction (such as turbulent effects and viscous resistance), the pressure $p(x)$ along the length of the pipe is not constant, and its spatial rate of change satisfies the differential equation:

$$\frac{dp}{dx} = -kp^2$$

where $k > 0$ is a constant describing the resistance per unit length, and $p(x)$ is the pressure at position x . The pressure at the inlet of the pipe $x = 0$ is p_0 . Solve the equation using the Bernoulli method and find the particular solution for $p(x)$.

Solution 4.3

To solve the problem, we need to let $u = p^{1-2} = p^{-1}$. The derivative of u is

$$u' = -p^{-2} \frac{dp}{dx}$$

Multiply both sides of the ODE by $-p^{-2}$, and substitute this in the original ODE, and it becomes

$$-p^{-2} \frac{dp}{dx} = -p^{-2} \times (-kp^2) \Rightarrow du = k dx$$

Integrate both sides, and we can get the solution to u

$$u = \frac{1}{p(x)} = \int k dx = kx + C$$

So, the pressure at position x is

$$p(x) = \frac{1}{kx + C}$$

We've known that $p(0) = p_0$, substitute it into $p(x)$, so the constant $C = 1/p_0$. The final solution is

$$p(x) = \frac{1}{kx + \frac{1}{p_0}} = \frac{p_0}{1 + kp_0x}$$

This result shows that the pressure decreases nonlinearly along the pipe due to nonlinear friction and becomes asymptotically small as $x \rightarrow \infty$.

4.1.5 Homogeneous Equation

A homogeneous equation in polynomials is slightly different from a homogeneous differential equation in the context of mathematics. A polynomial equation is said to be **homogeneous** of degree n if every term in the polynomial has the same total degree. It can be written as

$$\sum_{j=0} a_j x^j y^{n-j} = 0 \quad (4.79)$$

In such equations, since all terms have the same degree n , we can factor the entire expression by either x^n or y^n , allowing us to rewrite the equation as a function of y/x or x/y . For instance:

$$\frac{1}{x^n} \sum_j a_j x^j y^{n-j} = \sum_j \left(\frac{y}{x}\right)^{n-j} a_j := f\left(\frac{y}{x}\right) \quad (4.80)$$

In general, as long as an expression can be written in the form $x^n f(y/x)$, it is called a **homogeneous** expression.

To simplify the equation, we may introduce the substitution

$$u := \frac{y}{x} \Rightarrow y = xu \quad (4.81)$$

With this substitution, the function becomes $f(u)$, and the differential equation can now be simplified and solved more directly.

Example 4.4 Solve the ODE $x^2 dy + (y^2 - xy)dx = 0$

Solution 4.4

First, divide the equation by x^2 , and it becomes

$$dy + \left[\left(\frac{y}{x}\right)^2 - \frac{y}{x} \right] dx = 0$$

let $y/x = u$, then $y = xu$. By the rule of differentiation, we have

$$dy = d(xu) = xdu + udx$$

So, the equation becomes

$$xdu + udx + (u^2 - u)dx = xdu + u^2dx = 0$$

We change the original equation into the separable form by substitution, so

$$-\frac{1}{u^2}du = \frac{1}{x}dx \Rightarrow \int -\frac{1}{u^2}du = \int \frac{1}{x}dx \Rightarrow \frac{1}{u} = \ln(x) + C$$

Substituting back $y/x = u$ back into the expression, we obtain the general solution to the original ODE

$$x = y(\ln x + C)$$

This example illustrates how a seemingly complicated nonlinear first-order equation can be reduced to a separable form by identifying the homogeneous structure and using an appropriate substitution.

4.2 Second-Order ODEs

4.2.1 Homogeneous Problems

After calculating the first-order ODE, I'm going to show the second-order ODE next. We just begin with the homogeneous problem and focus on the method of solving it. The homogeneous second-order ODE means that the right-hand side (RHS) is 0, as shown in (4.82).

Consider the homogeneous equation

$$y'' + ay' + by = 0 \tag{4.82}$$

We can introduce a new operator D , it's defined as

$$D := \frac{d}{dx}; \quad D^2 = \frac{d^2}{dx^2} \tag{4.83}$$

Then substitute this into the original equation to be

$$(D^2 + aD + b)y = 0 \tag{4.84}$$

Factor the polynomial, and we get

$$(D - \alpha)(D - \beta)y = 0 \tag{4.85}$$

Where $\alpha\beta = b$, $-(\alpha + \beta) = a$. There are two different situations about the relationship between the constant α and β .

When $\alpha \neq \beta$, we let $u = (D - \beta)y$ and substitute it into (4.85), it becomes

$$(D - \alpha)u = u' - \alpha u = 0 \quad (4.86)$$

Note that we have transformed the original second-order ODE into a first-order one. We will discuss this technique in more detail in the next section. So, separate the variable

$$\frac{du}{u} = \alpha dx \Rightarrow u = C_1 e^{\alpha x}$$

Put the result back to $u = (D - \beta)y$, it becomes a first-order inhomogeneous ODE, that is

$$y' - \beta y = C e^{\alpha x} \quad (4.87)$$

The integrating factor $\mu = e^{-\beta x}$, then

$$d(e^{-\beta x} y) = C e^{(\alpha - \beta)x} dx \Rightarrow e^{-\beta x} y = \frac{C}{\alpha - \beta} e^{(\alpha - \beta)x} + C_1$$

So the final solution to y is

$$\boxed{y = A e^{\alpha x} + B e^{\beta x}, \alpha \neq \beta} \quad (4.88)$$

Where $A = \frac{C}{\alpha - \beta}$, $B = C_1$.

The other situation is $\alpha = \beta$. Let $(D - \alpha)y = u$, and we've had the solution to u in (4.87), then substituting u into $(D - \alpha)y = u$, we get

$$y' - \alpha y = C e^{\alpha x} \quad (4.89)$$

The integrating factor $\mu = e^{-\alpha x}$, then

$$d(e^{-\alpha x} y) = C e^{(\alpha - \alpha)x} dx = C dx \Rightarrow e^{-\alpha x} y = Cx + C_1$$

So the final solution to y is

$$\boxed{y = (Ax + B)e^{\alpha x}, \alpha = \beta} \quad (4.90)$$

Where $A = C$, $B = C_1$. These solutions to the ODEs are called the **general solution**, the complementary solution(y_c) to inhomogeneous ODEs, and sometimes the homogeneous solution(y_h).

4.2.2 Nonhomogeneous Problems and The Operator Descent Method

Another form of second-order ODEs is the nonhomogeneous one. We can use similar ways to solve homogeneous problems as you did in the last part. Consider a second-order ODE shown below, with RHS not 0.

$$y'' + ay' + by = f(x) \quad (4.91)$$

We define the derivative notation $D := \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$, so (3.1) can be written as

$$(D - \alpha)(D - \beta)y = f(x) \quad (4.92)$$

where $\alpha\beta = b$, $-(\alpha + \beta) = a$. Next, let the arbitrary one of the $(D - \cdot)y$ be a new variable u , for example

$$(D - \beta)y := u \quad (4.93)$$

Substituting it into (4.92), it becomes

$$(D - \alpha)u = u' - \alpha u = f(x) \quad (4.94)$$

The original problem becomes a first-order nonhomogeneous ODE problem, such a method is called **the operator descent method**.

By Section 2, we have known that the integrating factor

$$\mu = e^{-\alpha x} \Rightarrow d(e^{-\alpha x}u) = e^{-\alpha x}f(x)dx$$

Integrate both sides, and we can get

$$e^{-\alpha x}u = \int e^{-\alpha x}f(x)dx + C \Rightarrow u = e^{\alpha x} \int e^{-\alpha x}f(x)dx + Ce^{\alpha x}$$

Note that we haven't done it yet; this is just a solution to the new variable u , but what we want is y . So we need to put it back to (4.93), that is

$$(D - \beta)y = u = e^{\alpha x} \int e^{-\alpha x}f(x)dx + Ce^{\alpha x} \quad (4.95)$$

We can find that it becomes a first-order problem again, which is

$$y' - \beta y = e^{\alpha x} \int e^{-\alpha x}f(x)dx + Ce^{\alpha x}$$

The integrating factor of this is

$$\mu = e^{-\beta x} \Rightarrow d(e^{-\beta x}y) = e^{(\alpha-\beta)x} \int e^{-\alpha x}f(x)dx + Ce^{(\alpha-\beta)x}$$

Use the same way to get y

$$\begin{aligned}
e^{-\beta x} y &= \int \left[e^{(\alpha-\beta)x} \int e^{-\alpha x} f(x) dx + C e^{(\alpha-\beta)x} \right] dx \\
\Rightarrow y &= e^{\beta x} \int \left[e^{(\alpha-\beta)x} \int e^{-\alpha x} f(x) dx \right] dx + C_1 e^{\alpha x} + C_2 e^{\beta x}
\end{aligned} \tag{4.96}$$

Where $C_1 = C/(\alpha - \beta)$ and C_2 is the constant of the integral of RHS. Note the different terms in (4.96), the first term of the right is called the **particular solution** y_p , and the result of the integral depends on the form of $f(x)$ in the origin, so it's "particular"; the second and third term of the right is called **complementary solution** y_c of the ODE, which is the general solution of the homogeneous equation. So

$$\begin{cases} y_p = e^{\beta x} \int \left[e^{(\alpha-\beta)x} \int e^{-\alpha x} f(x) dx \right] dx \\ y_c = C_1 e^{\alpha x} + C_2 e^{\beta x} \end{cases} \tag{4.97}$$

In this method, we no longer need to memorize the particular solution to different kinds of ODEs, we just need to solve the first-order problem twice. In this way, we won't be afraid of any other complicated RHS, and as long as it is integrable, we can undoubtedly use the same method. Unlike the method of undetermined coefficients, which requires case-by-case memory, this unified structure emphasizes transparency over memorization.

4.3 Generalization to Arbitrary Right-Hand Side Functions

We've learned the unified method for solving the ODEs, so I will deal with some of the classical and difficult problems of ODEs. For example, $f(x) = \sin ax$, $f(x) = e^{cx}$, $f(x) = \ln x$, and so on. Remember, **you don't need to memorize any particular solution in this book.**

1. **The Exponential Term** The ODE in the exponential term is

$$y'' - (\alpha + \beta)y' + \alpha\beta y = ae^{\gamma x}.$$

This equation can be written as

$$(D - \alpha)(D - \beta)y = ae^{\gamma x}.$$

Let $u = (D - \beta)y$. Then we first solve

$$u' - \alpha u = ae^{\gamma x}.$$

The integrating factor is $e^{-\alpha x}$, so

$$d(e^{-\alpha x} u) = ae^{(\gamma-\alpha)x} dx \Rightarrow u = e^{\alpha x} \int ae^{(\gamma-\alpha)x} dx + Ce^{\alpha x}.$$

Substituting this into $u = (D - \beta)y$, we get

$$y = e^{\beta x} \int e^{(\alpha-\beta)x} \left(\int a e^{(\gamma-\alpha)x} dx \right) dx + \text{terms from constants of integration.}$$

The terms from constants of integration give the complementary solution. Therefore we only need to track the particular solution carefully.

There are two possible sources of special behavior. The first is that γ may equal one of the roots α, β . The second is that α and β may be equal. Thus we have the following complete classification.

(1) $\alpha \neq \beta, \gamma \neq \alpha$, and $\gamma \neq \beta$:

In this case, no exponential term becomes 1 during the integration:

$$\begin{aligned} y_p &= e^{\beta x} \int e^{(\alpha-\beta)x} \left[\int a e^{(\gamma-\alpha)x} dx \right] dx = e^{\beta x} \int e^{(\alpha-\beta)x} \left[\frac{a}{\gamma-\alpha} e^{(\gamma-\alpha)x} \right] dx \\ &= \frac{a}{\gamma-\alpha} e^{\beta x} \int e^{(\gamma-\beta)x} dx = \frac{a}{(\gamma-\alpha)(\gamma-\beta)} e^{\gamma x}. \end{aligned}$$

Hence

$$\boxed{y = A e^{\alpha x} + B e^{\beta x} + \frac{a}{(\gamma-\alpha)(\gamma-\beta)} e^{\gamma x}} \quad (4.98)$$

Equivalently, one may write

$$\boxed{y = D e^{\gamma x} + A e^{\alpha x} + B e^{\beta x}}, \quad D = \frac{a}{(\gamma-\alpha)(\gamma-\beta)}.$$

(2) $\alpha \neq \beta$ and $\gamma = \alpha$:

Now $e^{(\gamma-\alpha)x} = 1$, so the first integral gives a linear term:

$$\begin{aligned} y_p &= e^{\beta x} \int e^{(\alpha-\beta)x} \left(\int a dx \right) dx \\ &= e^{\beta x} \int a x e^{(\alpha-\beta)x} dx. \end{aligned}$$

We compute the remaining integral by integration by parts:

$$\begin{aligned} \int a x e^{(\alpha-\beta)x} dx &= \frac{a}{\alpha-\beta} x e^{(\alpha-\beta)x} - \frac{a}{\alpha-\beta} \int e^{(\alpha-\beta)x} dx \\ &= \frac{a}{\alpha-\beta} x e^{(\alpha-\beta)x} - \frac{a}{(\alpha-\beta)^2} e^{(\alpha-\beta)x}. \end{aligned}$$

Hence

$$y_p = \left[\frac{a}{\alpha - \beta} x - \frac{a}{(\alpha - \beta)^2} \right] e^{\alpha x}.$$

The constant multiple of $e^{\alpha x}$ is already part of the complementary solution, so it can be absorbed into $Ae^{\alpha x}$. Therefore

$$\boxed{y = Ae^{\alpha x} + Be^{\beta x} + \frac{a}{\alpha - \beta} xe^{\alpha x}} \quad (4.99)$$

Equivalently, this has the form

$$\boxed{y = (A + Cx)e^{\alpha x} + Be^{\beta x}}, \quad C = \frac{a}{\alpha - \beta}.$$

(3) $\alpha \neq \beta$ and $\gamma = \beta$:

This case is symmetric to the previous one. Since the forcing term has the same exponential as $e^{\beta x}$, the particular solution must contain $xe^{\beta x}$. Directly checking,

$$(D - \alpha)(D - \beta)(xe^{\beta x}) = (\beta - \alpha)e^{\beta x}.$$

Therefore

$$y_p = \frac{a}{\beta - \alpha} xe^{\beta x}.$$

Hence

$$\boxed{y = Ae^{\alpha x} + Be^{\beta x} + \frac{a}{\beta - \alpha} xe^{\beta x}} \quad (4.100)$$

Equivalently,

$$\boxed{y = Ae^{\alpha x} + (B + Cx)e^{\beta x}}, \quad C = \frac{a}{\beta - \alpha}.$$

(4) $\alpha = \beta$ and $\gamma \neq \alpha$:

Let $\lambda = \alpha = \beta$. Then the equation becomes

$$(D - \lambda)^2 y = ae^{\gamma x}.$$

Since

$$(D - \lambda)^2 e^{\gamma x} = (\gamma - \lambda)^2 e^{\gamma x},$$

a particular solution is

$$y_p = \frac{a}{(\gamma - \lambda)^2} e^{\gamma x}.$$

The complementary solution for the repeated root λ is

$$y_c = (A + Bx)e^{\lambda x}.$$

Hence

$$\boxed{y = (A + Bx)e^{\alpha x} + \frac{a}{(\gamma - \alpha)^2}e^{\gamma x}} \quad (4.101)$$

(5) $\alpha = \beta = \gamma$:

Let $\lambda = \alpha = \beta = \gamma$. Then

$$(D - \lambda)^2 y = ae^{\lambda x}.$$

Since the forcing exponential is exactly the repeated homogeneous exponential, the particular solution must contain $x^2 e^{\lambda x}$. Indeed,

$$(D - \lambda)^2 (x^2 e^{\lambda x}) = 2e^{\lambda x}.$$

Therefore

$$y_p = \frac{a}{2} x^2 e^{\lambda x}.$$

Together with the complementary solution, we obtain

$$\boxed{y = \left(A + Bx + \frac{a}{2} x^2 \right) e^{\alpha x}} \quad (4.102)$$

Equivalently, after redefining the coefficient of x^2 ,

$$\boxed{y = e^{\alpha x} (Ax^2 + Bx + D)}.$$

2. **The Trigonometric Term:** The ODE with a trigonometric forcing term can be written as

$$y'' - (\alpha + \beta)y' + \alpha\beta y = af(bx),$$

where f is either sin or cos. In this part, I will use $f(bx) = \sin(bx)$ as the main example. The case $f(bx) = \cos(bx)$ follows exactly the same logic. The equation becomes

$$y'' - (\alpha + \beta)y' + \alpha\beta y = a \sin(bx).$$

Using the operator notation, we may write it as

$$(D - \alpha)(D - \beta)y = a \sin(bx).$$

Let

$$u = (D - \alpha)y.$$

Then the equation becomes

$$(D - \beta)u = a \sin(bx),$$

or equivalently,

$$u' - \beta u = a \sin(bx).$$

The integrating factor is

$$\mu = e^{-\beta x}.$$

Hence

$$d(e^{-\beta x}u) = ae^{-\beta x} \sin(bx) dx.$$

Therefore,

$$u = ae^{\beta x} \int e^{-\beta x} \sin(bx) dx + Ce^{\beta x}.$$

Now we compute the integral

$$I_s = \int e^{-\beta x} \sin(bx) dx.$$

Using integration by parts twice, we get

$$I_s = -\frac{1}{\beta} e^{-\beta x} \sin(bx) + \frac{b}{\beta} \int e^{-\beta x} \cos(bx) dx.$$

Let

$$I_c = \int e^{-\beta x} \cos(bx) dx.$$

Again, integration by parts gives

$$I_c = -\frac{1}{\beta} e^{-\beta x} \cos(bx) - \frac{b}{\beta} I_s.$$

Substituting this expression for I_c back into the formula for I_s , we obtain

$$I_s = -\frac{1}{\beta} e^{-\beta x} \sin(bx) + \frac{b}{\beta} \left[-\frac{1}{\beta} e^{-\beta x} \cos(bx) - \frac{b}{\beta} I_s \right].$$

Thus

$$\left(1 + \frac{b^2}{\beta^2}\right) I_s = -\frac{1}{\beta} e^{-\beta x} \sin(bx) - \frac{b}{\beta^2} e^{-\beta x} \cos(bx).$$

Therefore,

$$\int e^{-\beta x} \sin(bx) dx = -\frac{\beta}{\beta^2 + b^2} e^{-\beta x} \sin(bx) - \frac{b}{\beta^2 + b^2} e^{-\beta x} \cos(bx). \quad (4.103)$$

Similarly,

$$\int e^{-\beta x} \cos(bx) dx = \frac{b}{\beta^2 + b^2} e^{-\beta x} \sin(bx) - \frac{\beta}{\beta^2 + b^2} e^{-\beta x} \cos(bx). \quad (4.104)$$

Substituting (4.103) into the expression for u , we get

$$u = -\frac{a\beta}{\beta^2 + b^2} \sin(bx) - \frac{ab}{\beta^2 + b^2} \cos(bx) + Ce^{\beta x}.$$

Since

$$u = (D - \alpha)y,$$

we now solve

$$y' - \alpha y = -\frac{a\beta}{\beta^2 + b^2} \sin(bx) - \frac{ab}{\beta^2 + b^2} \cos(bx) + Ce^{\beta x}.$$

The integrating factor is $e^{-\alpha x}$. Thus

$$d(e^{-\alpha x}y) = \left[-\frac{a\beta}{\beta^2 + b^2} e^{-\alpha x} \sin(bx) - \frac{ab}{\beta^2 + b^2} e^{-\alpha x} \cos(bx) + Ce^{(\beta-\alpha)x} \right] dx.$$

Integrating both sides gives

$$\begin{aligned} y &= -\frac{a\beta e^{\alpha x}}{\beta^2 + b^2} \int e^{-\alpha x} \sin(bx) dx - \frac{abe^{\alpha x}}{\beta^2 + b^2} \int e^{-\alpha x} \cos(bx) dx \\ &\quad + \frac{C}{\beta - \alpha} e^{\beta x} + C_1 e^{\alpha x}. \end{aligned}$$

Using the same integral formulas as (4.103) and (4.104), with β replaced by α , we obtain

$$\begin{aligned} y &= \frac{a(\alpha\beta - b^2)}{(\alpha^2 + b^2)(\beta^2 + b^2)} \sin(bx) + \frac{ab(\alpha + \beta)}{(\alpha^2 + b^2)(\beta^2 + b^2)} \cos(bx) \\ &\quad + C_1 e^{\alpha x} + C_2 e^{\beta x}. \end{aligned} \tag{4.105}$$

Therefore, the particular solution has the form

$$y_p = A \sin(bx) + B \cos(bx).$$

The complete solution can be written as

$$\boxed{y = A \sin(bx) + B \cos(bx) + D e^{\alpha x} + E e^{\beta x}}. \tag{4.106}$$

Here A and B are determined by the forcing term and the parameters α, β, b , while D and E are determined by the initial or boundary conditions.

The reason this form is natural is that the space spanned by $\sin(bx)$ and $\cos(bx)$ is closed under differentiation:

$$\begin{cases} \frac{d}{dx} \sin(bx) = b \cos(bx), & \frac{d^2}{dx^2} \sin(bx) = -b^2 \sin(bx), \\ \frac{d}{dx} \cos(bx) = -b \sin(bx), & \frac{d^2}{dx^2} \cos(bx) = -b^2 \cos(bx). \end{cases}$$

Thus, when the differential operator $(D - \alpha)(D - \beta)$ acts on

$$A \sin(bx) + B \cos(bx),$$

the result is still a linear combination of $\sin(bx)$ and $\cos(bx)$. This matches the form of the right hand side. For this reason, in the method of undetermined coefficients, we may directly try

$$y_p = A \sin(bx) + B \cos(bx)$$

for a sinusoidal forcing term.

The cosine case

$$y'' - (\alpha + \beta)y' + \alpha\beta y = a \cos(bx)$$

is handled in the same way. The particular solution is again a linear combination of $\sin(bx)$ and $\cos(bx)$. The coefficients are different, but the structure of the solution is the same.

In addition, we can use complex analysis to solve the problem. The reason is Euler's formula

$$e^{ibx} = \cos(bx) + i \sin(bx). \quad (4.107)$$

Therefore, trigonometric forcing terms can often be treated as the real or imaginary part of a complex exponential. This is useful because exponential functions are eigenfunctions of the differential operator $D = d/dx$.

Suppose the original equation contains the forcing term $a \sin(bx)$, where $a, b \in \mathbb{R}$. Instead of solving the trigonometric equation directly, we first solve the complex-valued equation

$$(D - \alpha)(D - \beta)Y = ae^{ibx}. \quad (4.108)$$

Here Y is a complex-valued function of the real variable x , so

$$Y : \mathbb{R} \rightarrow \mathbb{C}, \quad Y(x) = Y_R(x) + iY_I(x), \quad (4.109)$$

where Y_R and Y_I are real-valued functions:

$$Y_R, Y_I : \mathbb{R} \rightarrow \mathbb{R}. \quad (4.110)$$

Since the coefficients of the operator $(D - \alpha)(D - \beta)$ are real, we have

$$\begin{aligned} (D - \alpha)(D - \beta)Y &= (D - \alpha)(D - \beta)(Y_R + iY_I) \\ &= (D - \alpha)(D - \beta)Y_R + i(D - \alpha)(D - \beta)Y_I. \end{aligned} \quad (4.111)$$

On the other hand,

$$ae^{ibx} = a \cos(bx) + ia \sin(bx). \quad (4.112)$$

Comparing the real and imaginary parts gives

$$\begin{cases} (D - \alpha)(D - \beta)Y_R = a \cos(bx), \\ (D - \alpha)(D - \beta)Y_I = a \sin(bx). \end{cases} \quad (4.113)$$

Thus, if the original forcing term is $a \sin(bx)$, we may solve the complex equation (4.108) and then take the imaginary part:

$$y_p(x) = \operatorname{Im} Y(x). \quad (4.114)$$

If the original forcing term is $a \cos(bx)$, then we take the real part:

$$y_p(x) = \operatorname{Re} Y(x). \quad (4.115)$$

The subscript p means that this gives a particular solution.

Mathematically speaking, the reason this method works is the linearity of the differential operator with real coefficients. For any complex-valued function $Y = Y_R + iY_I$, we have

$$\begin{cases} (D - \alpha)(D - \beta) \operatorname{Re}(Y) = \operatorname{Re} [(D - \alpha)(D - \beta)Y], \\ (D - \alpha)(D - \beta) \operatorname{Im}(Y) = \operatorname{Im} [(D - \alpha)(D - \beta)Y]. \end{cases} \quad (4.116)$$

This guarantees that we may apply the differential operator to the complex function first and extract the real or imaginary part at the end. The main advantage is that exponential functions are usually easier to handle than sine and cosine functions. I will walk through a full example in Section 4.5.

For example, let

$$(D - \alpha)(D - \beta)y = a \sin(bx). \quad (4.117)$$

We solve

$$(D - \alpha)(D - \beta)Y = ae^{ibx}. \quad (4.118)$$

Since

$$(D - \alpha)(D - \beta)e^{ibx} = (ib - \alpha)(ib - \beta)e^{ibx}, \quad (4.119)$$

we may choose a particular solution of the form

$$Y_p(x) = \frac{a}{(ib - \alpha)(ib - \beta)}e^{ibx}, \quad (4.120)$$

provided that

$$(ib - \alpha)(ib - \beta) \neq 0. \quad (4.121)$$

Therefore, a particular solution to the sine-forced equation is

$$y_p(x) = \operatorname{Im} \left[\frac{a}{(ib - \alpha)(ib - \beta)}e^{ibx} \right]. \quad (4.122)$$

Similarly, a particular solution to the cosine-forced equation is

$$y_p(x) = \operatorname{Re} \left[\frac{a}{(ib - \alpha)(ib - \beta)}e^{ibx} \right]. \quad (4.123)$$

After this particular solution is found, the complete real solution is obtained by adding the homogeneous solution:

$$y(x) = y_h(x) + y_p(x), \quad (4.124)$$

where

$$(D - \alpha)(D - \beta)y_h = 0. \quad (4.125)$$

In addition, if the right-hand side has several forcing terms, for example

$$(D - \alpha)(D - \beta)y = C_1 \sin(\omega_1 x) + C_2 \cos(\omega_2 x), \quad (4.126)$$

we can solve the two forced equations separately:

$$\begin{cases} (D - \alpha)(D - \beta)y_{p,1} = C_1 \sin(\omega_1 x), \\ (D - \alpha)(D - \beta)y_{p,2} = C_2 \cos(\omega_2 x). \end{cases} \quad (4.127)$$

Adding the two equations gives

$$\begin{aligned} (D - \alpha)(D - \beta)(y_{p,1} + y_{p,2}) &= (D - \alpha)(D - \beta)y_{p,1} + (D - \alpha)(D - \beta)y_{p,2} \\ &= C_1 \sin(\omega_1 x) + C_2 \cos(\omega_2 x). \end{aligned} \quad (4.128)$$

Therefore,

$$y_p = y_{p,1} + y_{p,2} \quad (4.129)$$

is a particular solution of (4.126). The complete solution is

$$y = y_h + y_{p,1} + y_{p,2}. \quad (4.130)$$

This is precisely the idea behind the **superposition principle**. For a linear differential equation, the response to a sum of inputs is the sum of the responses to each input.

This principle is especially useful in physical systems with periodic or oscillatory forcing. For example, a driven harmonic oscillator may experience several sinusoidal external forces at different frequencies. Because the equation is linear, we can solve the response to each frequency separately and then add the results. This is one of the reasons complex exponentials are so common in mechanics, circuits, waves, and quantum mechanics.

3. **The Polynomial Term:** The ODE with the polynomial term can be written as

$$y'' - (\alpha + \beta)y' + \alpha\beta y = ae^{\gamma x} P_n(x),$$

where $P_n(x)$ is a polynomial of degree n . Equivalently,

$$(D - \alpha)(D - \beta)y = ae^{\gamma x} P_n(x).$$

In this part, we first consider the non-resonant case

$$\gamma \neq \alpha, \quad \gamma \neq \beta.$$

The key observation is that applying D to $e^{\gamma x}Q(x)$ gives

$$D(e^{\gamma x}Q(x)) = e^{\gamma x}(D + \gamma)Q(x).$$

Therefore,

$$(D - \alpha)(D - \beta)(e^{\gamma x}Q(x)) = e^{\gamma x}(D + \gamma - \alpha)(D + \gamma - \beta)Q(x).$$

Because the right hand side is $ae^{\gamma x}P_n(x)$, it is natural to try a particular solution of the form

$$y_p(x) = ae^{\gamma x}Q_n(x),$$

where $Q_n(x)$ is also a polynomial of degree n . Substituting this into the ODE gives

$$ae^{\gamma x}(D + \gamma - \alpha)(D + \gamma - \beta)Q_n(x) = ae^{\gamma x}P_n(x).$$

After canceling $ae^{\gamma x}$, we only need to solve the polynomial equation

$$(D + \gamma - \alpha)(D + \gamma - \beta)Q_n(x) = P_n(x). \quad (4.131)$$

This is much simpler than integrating the exponential polynomial term repeatedly. The differential operator in (4.131) maps polynomials to polynomials. Also, since

$$\gamma - \alpha \neq 0, \quad \gamma - \beta \neq 0,$$

the highest degree term of $Q_n(x)$ will not disappear. Hence $Q_n(x)$ has the same degree as $P_n(x)$:

$$\deg Q_n = \deg P_n = n.$$

To determine $Q_n(x)$ explicitly, write

$$P_n(x) = p_n x^n + p_{n-1} x^{n-1} + \cdots + p_1 x + p_0$$

and assume

$$Q_n(x) = q_n x^n + q_{n-1} x^{n-1} + \cdots + q_1 x + q_0.$$

Then substitute $Q_n(x)$ into (4.131) and compare the coefficients of equal powers of x . This determines $q_n, q_{n-1}, \dots, q_0$ one by one.

For example, if

$$P_1(x) = px + q,$$

we try

$$Q_1(x) = Ax + B.$$

Then

$$(D + \gamma - \alpha)(D + \gamma - \beta)(Ax + B) = px + q.$$

Let

$$c_\alpha = \gamma - \alpha, \quad c_\beta = \gamma - \beta.$$

Since

$$(D + c_\alpha)(D + c_\beta) = D^2 + (c_\alpha + c_\beta)D + c_\alpha c_\beta,$$

and $Q_1''(x) = 0$, we get

$$(c_\alpha + c_\beta)A + c_\alpha c_\beta(Ax + B) = px + q.$$

Comparing coefficients gives

$$c_\alpha c_\beta A = p,$$

and

$$(c_\alpha + c_\beta)A + c_\alpha c_\beta B = q.$$

Thus

$$A = \frac{p}{c_\alpha c_\beta}, \quad B = \frac{q - (c_\alpha + c_\beta)A}{c_\alpha c_\beta}.$$

Therefore, the particular solution is

$$y_p(x) = ae^{\gamma x}(Ax + B).$$

In general, the polynomial $Q_n(x)$ is obtained by coefficient comparison. Therefore, the full solution in the non-resonant case is

$$\boxed{y(x) = Ae^{\alpha x} + Be^{\beta x} + ae^{\gamma x}Q_n(x), \quad \deg Q_n = n.} \quad (4.132)$$

This result is the same form that we expect from the method of undetermined coefficients. The exponential factor $e^{\gamma x}$ is preserved, while the polynomial part is changed from $P_n(x)$ into another polynomial $Q_n(x)$ of the same degree.

If $\gamma = \alpha$ or $\gamma = \beta$, then resonance occurs. In that case, the trial solution must be multiplied by an extra factor of x . For example, if $\gamma = \alpha$ and $\gamma \neq \beta$, then the particular solution has the form

$$y_p(x) = ae^{\gamma x}xQ_n(x).$$

If $\gamma = \alpha = \beta$, then the particular solution has the form

$$y_p(x) = ae^{\gamma x}x^2Q_n(x).$$

These cases follow the same idea as the exponential forcing term. The extra power of x appears because the forcing term overlaps with the homogeneous solution.

4.4 Other Methods for Second-Order Equation

4.4.1 Differential Equations Involving Only y

1. When the differential equation satisfies

$$f(y, y', y'') = 0 \quad (4.133)$$

That is, the function is independent of x . Then we let

$$\boxed{y'(x) = p(y(x))} \quad (4.134)$$

By the chain rule, we can verify that

$$\boxed{y''(x) = \frac{d}{dy}p(y) \cdot y'(x) = p(y) \cdot \frac{dp(y)}{dy}} \quad (4.135)$$

Then (4.133) becomes

$$f\left(y, p, p \frac{dp}{dy}\right) = 0 \quad (4.136)$$

This method is effective in reducing a second-order differential equation that is independent of the variable x into a first-order linear or nonlinear differential equation, which can then be solved using techniques discussed earlier. Thus, it enables us to indirectly recover the function $y(x)$.

The most famous example is the equation of motion of a damped oscillation with a quadratic drag term ($-bv^2$).

$$m\ddot{x} + b(\dot{x})^2 + kx = 0 \quad (4.137)$$

Let $\dot{x} = p(x(t))$, by (4.135), $\ddot{x} = p \cdot \frac{dp}{dx}$. The equation of motion becomes

$$mp \frac{dp}{dx} + bp^2 + kx = 0 \Rightarrow \frac{dp}{dx} + 2\beta p + \omega_0^2 x p^{-1} = 0 \quad (4.138)$$

Where $2\beta := b/m$ and $\omega_0^2 := k/m$. After that, this differential equation becomes a classical Bernoulli equation.

$$\begin{cases} \frac{dp}{dx} + 2\beta p = -\omega_0^2 x p^{-1} \\ \frac{dy}{dx} + P(x)y = Q(x)y^n \quad (\text{Bernoulli equation}) \end{cases} \quad (4.139)$$

Then you can let $z = p^2$ to solve this differential equation.

Example 4.5 A particle moves vertically under the influence of a force that is inversely proportional to the square of its distance from the origin. The force acting on the particle is given by

$$F(y) = -\frac{k}{y^2}, \quad \text{where } k > 0.$$

Assuming the particle starts at position $y_0 > 0$ and is initially moving downward, find the relationship between the particle's velocity v and its position y .

Solution 4.5

From Newton's second law, we obtain

$$ma = -\frac{k}{y^2}, \quad a = \frac{dv}{dt}.$$

Since $v = dy/dt$, the chain rule gives

$$\frac{dv}{dt} = \frac{dv}{dy} \frac{dy}{dt} = v \frac{dv}{dy}.$$

Hence

$$v \frac{dv}{dy} = -\frac{k}{my^2}.$$

Separating the variables gives

$$v \, dv = -\frac{k}{m} y^{-2} dy.$$

Therefore

$$\frac{1}{2}v^2 = \frac{k}{my} + C.$$

If the initial velocity at $y = y_0$ is $v_0 < 0$, then

$$\frac{1}{2}v_0^2 = \frac{k}{my_0} + C.$$

Thus

$$v^2 = v_0^2 + \frac{2k}{m} \left(\frac{1}{y} - \frac{1}{y_0} \right).$$

Because the particle is moving downward, we choose the negative branch:

$$v(y) = -\sqrt{v_0^2 + \frac{2k}{m} \left(\frac{1}{y} - \frac{1}{y_0} \right)}.$$

This is the velocity-position relation. It is essentially an energy conservation statement written in the language of ODEs.

2. When the differential equation satisfies

$$y'' + f(y) = 0 \Rightarrow y'' = -f(y) \quad (4.140)$$

Then we must multiply both sides by y' , that is

$$\boxed{y' \cdot y'' = y' \cdot f(y)} \quad (4.141)$$

The reason why we must do this is to solve the differential equation for y' , which is

$$\begin{aligned} y' y'' = -f(y) y' &\Rightarrow y' \frac{dy'}{dx} = -f(y) \frac{dy}{dx} \Rightarrow y' dy' = -f(y) dy \\ &\Rightarrow \int y' dy' = \int -f(y) dy \Rightarrow \frac{1}{2} y'^2 = - \int f(y) dy + C \\ &\Rightarrow y' = \sqrt{-2 \int f(y) dy + C} \end{aligned}$$

Such differential equations most commonly appear in mechanics when we need to find the velocity-position relationship. In these cases, there is no need to indirectly solve for $y(t)$; instead, the velocity term can be obtained directly. This makes it more convenient to analyze the phase diagram and observe the trajectory of the system's motion.

4.4.2 Cauchy Equation

Differential equations of the form

$$a_2 x^2 y'' + a_1 x y' + a_0 y = f(x) \quad (4.142)$$

are called **Cauchy equation**. Being more convenient to identify such equations, we can verify that each term of the equation satisfies

$$a_n x^n D^n[y] \quad (4.143)$$

Where $D = d/dx$. In such problems, we can reduce the differential equation to a linear equation without x . That is, we should change the variable x to z where

$$x = e^z \Leftrightarrow z = \ln x \quad (4.144)$$

After changing the variable, the linear term $x \frac{dy}{dx}$ can be written as

$$x \frac{dy}{dx} = x \cdot \frac{dy}{dz} \cdot \frac{dz}{dx}$$

Note that $z = \ln x$, so the derivative of z with respect to x is

$$\frac{dz}{dx} = \frac{d \ln x}{dx} = \frac{1}{x}$$

Then we obtain

$$x \cdot \frac{dy}{dz} \cdot \frac{dz}{dx} = x \frac{dy}{dz} \cdot \frac{1}{x} = \frac{dy}{dz}$$

So, the first important conclusion is

$$\boxed{x \frac{dy}{dx} = \frac{dy}{dz}} \quad (4.145)$$

By (4.145), the equation can be also denoted by

$$\frac{dy}{dx} = \frac{1}{x} \frac{dy}{dz} \quad (4.146)$$

We can use the above conclusion to change the quadratic term with respect to z , that is

$$\begin{aligned} x^2 \frac{d^2 y}{dx^2} &= x^2 \frac{d}{dx} \left(\frac{dy}{dx} \right) = x^2 \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dz} \right) \\ &= x^2 \left[-\frac{1}{x^2} \frac{dy}{dz} + \frac{1}{x} \left(\frac{d}{dx} \frac{dy}{dz} \right) \right] \\ &= x^2 \left[-\frac{1}{x^2} \frac{dy}{dz} + \frac{1}{x} \left(\frac{d}{dz} \frac{dz}{dx} \frac{dy}{dz} \right) \right] \\ &= x^2 \left[-\frac{1}{x^2} \frac{dy}{dz} + \frac{1}{x^2} \frac{d^2 y}{dz^2} \right] = \frac{d^2 y}{dz^2} - \frac{dy}{dz} \end{aligned}$$

So, the second important conclusion is

$$\boxed{x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{dz^2} - \frac{dy}{dz}} \quad (4.147)$$

Substituting (4.145) and (4.147) into (4.142), we can obtain a normal second-order differential equation

$$a_2 \frac{d^2 y}{dz^2} + (a_1 - a_2) \frac{dy}{dz} + a_0 y = f(e^z) \quad (4.148)$$

You can see the example in the next chapter.

4.4.3 A Complicated Differential Equation with One of Its Solutions

When we meet a second-order linear homogeneous equation of the form

$$y'' + f(x)y' + g(x)y = 0, \quad (4.149)$$

and one nonzero solution $u(x)$ is already known, we can construct the second solution by the method of reduction of order. The idea is to write

$$y = u(x)v(x). \quad (4.150)$$

Here $v(x)$ is the unknown function. Differentiating gives

$$y' = u'v + uv', \quad y'' = u''v + 2u'v' + uv''.$$

Substituting (4.150) into (4.149), we get

$$u''v + 2u'v' + uv'' + f(u'v + uv') + guv = 0.$$

Collecting terms gives

$$uv'' + (2u' + fu)v' + (u'' + fu' + gu)v = 0.$$

Since u is already a solution of (4.149), it satisfies

$$u'' + fu' + gu = 0.$$

Therefore the equation reduces to

$$uv'' + (2u' + fu)v' = 0.$$

Let $w = v'$. Then

$$uw' + (2u' + fu)w = 0.$$

This is a first-order homogeneous ODE. Separating variables, we obtain

$$\frac{dw}{w} = - \left(\frac{2u'}{u} + f \right) dx.$$

Hence

$$w = v' = C \exp \left[- \int f(x) dx \right] \frac{1}{u^2}.$$

Integrating once more gives

$$v = C \int \frac{\exp \left[-\int f(x) dx \right]}{u(x)^2} dx + C_2.$$

Thus a second independent solution is

$$y_2 = u(x) \int \frac{\exp \left[-\int f(x) dx \right]}{u(x)^2} dx. \quad (4.151)$$

The complete general solution is

$$y = C_1 u(x) + C_2 u(x) \int \frac{\exp \left[-\int f(x) dx \right]}{u(x)^2} dx. \quad (4.152)$$

This formula explains why knowing one solution is already powerful. The original second-order equation is reduced to two integrations, and the second independent solution is recovered from the first one.

4.5 The Examples

Example 4.6 Solve the ODE $y'' - 3y' + 2y = \sin x$

Solution 4.6

First, rewrite the problem as

$$(D - 1)(D - 2)y = \sin x$$

Let $u = (D - 2)y$, the equation becomes $u' - u = \sin x$, so the solution to u is

$$u = e^x \int e^{-x} \sin x dx = -\frac{1}{2} \sin x - \frac{1}{2} \cos x + Ce^x$$

substituting it into $u = (D - 2)y$, we get

$$\begin{aligned} y' - 2y &= -\frac{1}{2} \sin x - \frac{1}{2} \cos x + Ce^x \\ \Rightarrow y &= e^{2x} \int e^{-2x} \left(-\frac{1}{2} \sin x - \frac{1}{2} \cos x + Ce^x \right) dx \end{aligned}$$

we can easily get the final solution, that is

$$y = \frac{1}{10} \sin x + \frac{3}{10} \cos x + C_1 e^x + C_2 e^{2x}$$

Example 4.7 Solve the ODE $y'' - 2y' + y = \sin 2x + \cos 3x$

Solution 4.7

In this problem, I'm going to use the superposition principle like what I did in Section 4.2. We break it into two separate ODEs, which is

$$\begin{cases} (D-1)(D-1)y_1 = \sin 2x \\ (D-1)(D-1)y_2 = \cos 3x \end{cases}$$

We solve the first ODE first, let $u = (D-1)y_1$, then

$$u' - u = \sin 2x \Rightarrow e^{-x}u = \int e^{-x} \sin 2x \, dx$$

So u equals

$$u = -\frac{1}{5} \sin 2x - \frac{2}{5} \cos 2x + Ce^x$$

substituting it into $u = (D-1)y_1$, we get

$$\begin{aligned} y_1 &= e^x \int e^{-x} \left(-\frac{1}{5} \sin 2x - \frac{2}{5} \cos 2x + Ce^x \right) dx \\ &= \frac{-3}{25} \sin 2x + \frac{4}{25} \cos 2x + (Cx + C_1)e^x \end{aligned}$$

Similarly,

$$y_2 = -\frac{3}{50} \sin 3x - \frac{2}{25} \cos 3x + (C_2x + C_3)e^x$$

Solution 4.8

By equation (4.129), the solution to the original ODE is

$$y = y_1 + y_2 = \frac{-3}{25} \sin 2x + \frac{4}{25} \cos 2x - \frac{3}{50} \sin 3x - \frac{2}{25} \cos 3x + (Ax + B)e^x$$

Example 4.8 Solve the ODE $y'' - 4y' + 3y = e^{5x} \cos 2x$

Here, I want to use the complex method to solve the problem. The ODE can be rewritten as

$$(D-1)(D-3)Y = e^{5x} \times e^{2ix}$$

Because $e^{2ix} = \cos 2x + i \sin 2x$, we need to take the real part of Y , which is a complex-valued function. By (4.116), we got

$$(D - 1)(D - 3)\operatorname{Re}(Y) = \operatorname{Re}[(D - 1)(D - 3)Y]$$

So, we need to solve the ODE

$$(D - 1)(D - 3)Y = e^{(5+2i)x}$$

By the result of (4.98), the final solution to the ODE of Y is

$$Y = \frac{1}{(5 + 2i - 1)(5 + 2i - 3)} e^{(5+2i)x} + C_1 e^x + C_2 e^{3x}$$

The complex term $(4 + 2i)(2 + 2i) = 4 + 12i$, so

$$\frac{1}{4 + 12i} = \frac{4 - 12i}{(4 + 12i)(4 - 12i)} = \frac{1 - 3i}{40}$$

Expand Y to get

$$\begin{aligned} Y &= \frac{1 - 3i}{40} e^{(5+2i)x} + C_1 e^x + C_2 e^{3x} \\ &= e^{5x} \left[\frac{1 - 3i}{40} (\cos 2x + i \sin 2x) \right] + C_1 e^x + C_2 e^{3x} \\ &= e^{5x} \left[\frac{3 \sin 2x + \cos 2x}{40} + \frac{i(\sin 2x - 3 \cos 2x)}{40} \right] + C_1 e^x + C_2 e^{3x} \end{aligned}$$

Solution 4.9

To get the solution to the original ODE, we must remember to take the real part of Y , so we can get the answer

$$y = \operatorname{Re}(Y) = \frac{3}{40} e^{5x} \sin 2x + \frac{1}{40} e^{5x} \cos 2x + C_1 e^x + C_2 e^{3x}$$

Example 4.9 Solve the ODE $xy'' = y' + (y')^3$

In such a problem, we must let $y' = p(x)$. Therefore, we can rewrite the equation into

$$x \frac{dp}{dx} = p + p^3 = p(p^2 + 1)$$

Separate the variables, and we obtain

$$\frac{dp}{p(p^2+1)} = \frac{dx}{x} \Leftrightarrow \int \frac{dp}{p(p^2+1)} = \int \frac{dx}{x}$$

By the method of partial fractions, the left-hand integral can be written as

$$\int \frac{1}{p} dp - \int \frac{p}{p^2+1} dp = \ln x + C$$

Then

$$\begin{aligned} \ln p - \frac{1}{2} \ln(p^2+1) &= \ln \left(\frac{p}{\sqrt{p^2+1}} \right) = \ln x + C \Rightarrow \frac{p}{\sqrt{p^2+1}} = kx, \quad k = e^C \\ \Rightarrow p^2 = \left(\frac{dy}{dx} \right)^2 &= \frac{k^2 x^2}{1 - k^2 x^2} \Rightarrow \frac{dy}{dx} = \pm \frac{kx}{\sqrt{1 - k^2 x^2}} \Rightarrow \int dy = \pm \int \frac{kx}{\sqrt{1 - k^2 x^2}} dx \end{aligned}$$

Note that the original problem becomes a trigonometric substitution. Let $kx = \sin \theta$, then $dx = \frac{1}{k} \cos \theta d\theta$. Substituting this into the integral, we get

$$y = \mp \frac{1}{k} \cos \theta + a \Rightarrow y - a = \mp \frac{1}{k} \cos \theta, \quad a = \text{const}$$

Because we have let $x = \frac{1}{k} \sin \theta$, by squaring them and adding, we obtain

$$x^2 + (y - a)^2 = \left(\frac{1}{k} \right)^2 \sin^2 \theta + \left(\frac{1}{k} \right)^2 \cos^2 \theta = \frac{1}{k^2}$$

It's an equation of a circle. Note that when y is a constant, it's obvious that a constant is a solution to the differential equation. So, the final solutions to the differential equation are

$$x^2 + (y - a)^2 = \frac{1}{k^2} \quad \text{or} \quad y = \text{const}$$

Example 4.10 Solve the ODE $2yy'' = (y')^2$

Solution 4.10

Let $y' = p$, by chain rule, $y'' = p \frac{dp}{dy}$. Then we can rewrite the equation into

$$2yp \frac{dp}{dy} = p^2 \Rightarrow 2y \frac{dp}{dy} = p$$

By separating the variables, we can obtain

$$\frac{dp}{p} = \frac{1}{2} \frac{dy}{y} \Leftrightarrow \int \frac{dp}{p} = \int \frac{1}{2} \frac{dy}{y}$$

Then the solution to p is

$$p = Cy^{1/2}$$

Substituting $y' = p$ into the equation

$$\frac{dy}{dx} = Cy^{1/2} \Rightarrow \int y^{-1/2} dy = \int C dx$$

Then the final solution to the ODE is

$$2\sqrt{y} = Cx + D$$

Where C, D are constants.

Example 4.11 The ODE $y'' = \frac{2}{y}$, find y' depending on y

Solution 4.11

Multiplying both sides by y' , the equation becomes

$$y'y'' = \frac{2}{y}y'$$

It equals

$$y' \frac{dy'}{dx} = \frac{2}{y} \frac{dy}{dx} \Rightarrow \int y' dy' = 2 \int \frac{dy}{y}$$

Therefore, we can obtain the solution to y' , that is

$$\begin{aligned} \frac{1}{2}y'^2 &= 2 \ln y + C \\ \Rightarrow y' &= \sqrt{4 \ln y + 2C} \end{aligned}$$

Example 4.12 Solve the ODE $x^2y'' - 3xy' + 4y = 6x^2 \ln x$

Solution 4.12

This is one of the Cauchy equations, and we let $x = e^z$. By (4.148), we obtain

$$\frac{d^2y}{dz^2} + (-3 - 1)\frac{dy}{dz} + 4y = 6(e^z)^2 \ln e^z = 6ze^{2z}$$

Then we get a second-order differential equation with respect to z , that is

$$y'' - 4y' + 4y = (D - 2)(D - 2)y = 6ze^{2z}$$

where $D = d/dz$. Let $(D - 2)y = u$, we can know that

$$u' - 2u = 6ze^{2z} \Rightarrow e^{-2z}u = 6 \int e^{-2z}(ze^{2z})dz = 6 \int z dz$$

The solution to u is

$$u = (D - 2)y = 3z^2e^{2z} + C_1e^{2z}$$

Similarly, the solution to y is

$$y = e^{2z} \int 3z^2 dz + e^{2z} \int C_1 dz = z^3e^{2z} + (C_1z + C_2)e^{2z}$$

Substituting $z = \ln x$ into this solution, we can finally get

$$y = x^2 [(\ln x)^3 + C_1 \ln x + C_2]$$

Example 4.13 Solve the ODE $x^2(2 - x)y'' + 2xy' - 2y = 0$, given that $u(x) = x$ is a solution

Solution 4.13

First, let the solution y be

$$y = u(x)v(x) = xv$$

Substituting $y = xv$ into the ODE, we obtain

$$\begin{aligned} x^2(2 - x) \frac{d^2}{dx^2}(xv) + 2x \frac{d}{dx}(xv) - 2xv &= x^2(2 - x)(2v' + xv'') + 2x(v + xv') - 2xv \\ &= (6x^2 - 2x^3)v' + (2x^3 - x^4)v'' \\ &= (6x^2 - 2x^3)v' + (2x^3 - x^4) \frac{dv'}{dx} = 0 \end{aligned}$$

Then we write down the differential equation for v'

$$\begin{aligned} (6x^2 - 2x^3)v' + (2x^3 - x^4) \frac{dv'}{dx} &= 0 \\ \Rightarrow \int \frac{dv'}{v'} &= \int \frac{2x^3 - 6x^2}{2x^3 - x^4} dx \\ \Rightarrow \ln v' &= \ln \left(\frac{2 - x}{x^3} \right) + C \\ \Rightarrow v' &= C_1 \left(\frac{2 - x}{x^3} \right) \end{aligned}$$

Solving the ODE again, we obtain

$$\frac{dv}{dx} = C_1 \left(\frac{2-x}{x^3} \right) \Rightarrow \int dv = v = C_1 \int \left(\frac{2-x}{x^3} \right) dx = C_1(x^{-1} - x^{-2}) + C_2$$

Substituting v into $y = xv$, we can finally obtain the solution to the question

$$y = C_1(1 - x^{-1}) + C_2x$$

4.6 Series Solution to ODE

Many differential equations in physics cannot be solved by elementary functions. This does not mean that the equation has no useful solution. Very often, the solution is represented as an infinite series. The central idea is to replace the unknown function by a local expansion and then let the differential equation determine the coefficients one by one.

This method is especially natural because differential equations are local equations. They compare the value of a function and its derivatives near one point. Therefore, when the coefficients of the equation are sufficiently smooth near that point, the solution can often be understood from its Taylor series.

4.6.1 Ordinary Points and Power Series Solutions

Consider a second-order linear ODE written in the standard form

$$y'' + P(x)y' + Q(x)y = R(x). \quad (4.153)$$

A point x_0 is called an ordinary point if $P(x)$, $Q(x)$, and $R(x)$ are analytic near x_0 . In this case, it is reasonable to look for a solution of the form

$$y(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n. \quad (4.154)$$

The derivatives are

$$y'(x) = \sum_{n=1}^{\infty} n a_n(x - x_0)^{n-1}, \quad y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n(x - x_0)^{n-2}. \quad (4.155)$$

After substituting these three series into the ODE, we rewrite every term as a series in the same power $(x - x_0)^n$. Since a power series is zero for all x near x_0 only when every

coefficient is zero, we can compare coefficients of equal powers. This gives algebraic equations for a_n .

The first two coefficients a_0 and a_1 usually remain arbitrary for a second-order ODE. This is exactly what we expect, because a second-order ODE needs two initial conditions. All higher coefficients are then determined recursively.

Example 4.14 Solve $y'' + y = 0$ by the power series method around $x_0 = 0$.

Solution 4.14 Assume

$$y = \sum_{n=0}^{\infty} a_n x^n.$$

Then

$$y'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}.$$

Let $m = n - 2$. Then

$$y'' = \sum_{m=0}^{\infty} (m+2)(m+1)a_{m+2}x^m.$$

The equation $y'' + y = 0$ becomes

$$\sum_{m=0}^{\infty} [(m+2)(m+1)a_{m+2} + a_m] x^m = 0.$$

Therefore,

$$a_{m+2} = -\frac{a_m}{(m+2)(m+1)}.$$

The even coefficients are determined by a_0 , and the odd coefficients are determined by a_1 . Hence

$$y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \right).$$

Thus

$$y = a_0 \cos x + a_1 \sin x.$$

This recovers the familiar solution from the series structure alone.

Example 4.15 Find the recurrence relation for the Airy-type equation

$$y'' - xy = 0$$

near $x = 0$.

Solution 4.15 Let

$$y = \sum_{n=0}^{\infty} a_n x^n, \quad y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n.$$

Also,

$$xy = \sum_{n=0}^{\infty} a_n x^{n+1} = \sum_{n=1}^{\infty} a_{n-1} x^n.$$

The equation becomes

$$2a_2 + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} - a_{n-1}] x^n = 0.$$

Therefore,

$$a_2 = 0, \quad a_{n+2} = \frac{a_{n-1}}{(n+2)(n+1)}, \quad n \geq 1.$$

Again, a_0 and a_1 are free constants. The equation determines the rest of the series from them.

4.6.2 Singular Points and the Frobenius Method

The power series method above works cleanly at ordinary points. However, many important physical equations have singular points. For example, equations in polar, cylindrical, or spherical coordinates often contain terms like $1/r$ or $1/r^2$. The origin then becomes singular because the coordinate system itself is singular there.

For an equation of the form

$$y'' + P(x)y' + Q(x)y = 0, \tag{4.156}$$

we call x_0 a regular singular point if

$$(x - x_0)P(x) \quad \text{and} \quad (x - x_0)^2 Q(x) \tag{4.157}$$

are analytic near x_0 . At a regular singular point, an ordinary power series may be too restrictive. We instead try the Frobenius form

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}, \quad a_0 \neq 0. \tag{4.158}$$

The number r is not chosen in advance. It is determined by the lowest-power term of the differential equation. The algebraic equation for r is called the indicial equation.

This extra exponent r is important. It allows the solution to behave like a non-integer power near the singular point. This is why the Frobenius method can handle many equations that ordinary Taylor series cannot.

Example 4.16 Solve the Euler equation

$$x^2 y'' + xy' - \alpha^2 y = 0$$

by the Frobenius idea near $x = 0$.

Solution 4.16 Try the simplest Frobenius form $y = x^r$. Then

$$y' = rx^{r-1}, \quad y'' = r(r-1)x^{r-2}.$$

Substitution gives

$$x^2 r(r-1)x^{r-2} + xrx^{r-1} - \alpha^2 x^r = 0.$$

Therefore,

$$[r(r-1) + r - \alpha^2] x^r = 0,$$

so the indicial equation is

$$r^2 - \alpha^2 = 0.$$

Thus

$$r = \pm\alpha.$$

The two independent solutions are

$$y_1 = x^\alpha, \quad y_2 = x^{-\alpha}.$$

This example shows how the leading power of the solution is selected by the singular structure of the ODE.

4.6.3 Physical Meaning of Series Solutions

Series solutions are useful because many physical laws are local. Near a point, the behavior of a system can often be approximated by the first few terms of a power series. In mechanics, this is the logic behind small oscillations. If $V(x)$ is a potential energy function and x_0 is an equilibrium point, then

$$V(x) = V(x_0) + V'(x_0)(x - x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \cdots \quad (4.159)$$

At equilibrium, $V'(x_0) = 0$, so

$$V(x) = V(x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \cdots \quad (4.160)$$

Keeping only the quadratic term gives the harmonic oscillator. Keeping higher-order terms gives anharmonic corrections. In this sense, the harmonic oscillator is the first local approximation to many stable mechanical systems.

The same idea appears in quantum mechanics. Near a minimum of a potential, the Schrodinger equation often becomes approximately the quantum harmonic oscillator. The series expansion of the potential determines the leading energy levels and the higher-order corrections. In electromagnetic theory, series expansions also appear when one studies fields near a point or expands a potential in powers of distance. The multipole expansion in electrostatics is one of the most important examples.

4.7 Laplace Transform

The Laplace transform changes a function of time into a function of a complex variable s . It is especially useful for initial value problems, because differentiation in time becomes multiplication by s after the transform. Initial conditions are automatically carried into the transformed equation. You are recommended to look up the table of Laplace transform for the inverse transform $\mathcal{L}^{-1}$.

Definition 4.5 (Laplace Transform) Let $f(t)$ be defined for $t \geq 0$. Its Laplace transform is

$$\mathcal{L}\{f(t)\}(s) = F(s) = \int_0^{\infty} e^{-st} f(t) dt, \quad (4.161)$$

provided the integral converges.

The inverse Laplace transform is denoted by

$$f(t) = \mathcal{L}^{-1}\{F(s)\}(t). \quad (4.162)$$

In applications, one usually transforms the ODE into an algebraic equation, solves for $F(s)$, and then transforms back to obtain $f(t)$.

4.7.1 Basic Properties

The Laplace transform is linear. If a and b are constants, then

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s). \quad (4.163)$$

The transform of derivatives is

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0), \quad (4.164)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0). \quad (4.165)$$

More generally,

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0). \quad (4.166)$$

This is the main reason why the Laplace transform is powerful for initial value problems. Some basic transforms are

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad \mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad (4.167)$$

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}, \quad \mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}. \quad (4.168)$$

The shift rule is

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a). \quad (4.169)$$

The convolution of two functions is defined by

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau. \quad (4.170)$$

Its transform is

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s). \quad (4.171)$$

This rule is the mathematical form behind Green's functions and impulse responses.

Example 4.17 Solve

$$y'' + \omega_0^2 y = 0, \quad y(0) = y_0, \quad y'(0) = v_0.$$

Solution 4.17 Taking the Laplace transform gives

$$s^2 Y(s) - sy_0 - v_0 + \omega_0^2 Y(s) = 0.$$

Therefore,

$$Y(s) = \frac{sy_0 + v_0}{s^2 + \omega_0^2}.$$

Using the inverse transform,

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + \omega_0^2}\right\} = \cos(\omega_0 t), \quad \mathcal{L}^{-1}\left\{\frac{1}{s^2 + \omega_0^2}\right\} = \frac{1}{\omega_0} \sin(\omega_0 t),$$

we obtain

$$y(t) = y_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0} \sin(\omega_0 t).$$

Example 4.18 Solve the damped equation

$$y'' + 2\beta y' + \omega_0^2 y = 0, \quad y(0) = y_0, \quad y'(0) = v_0.$$

Solution 4.18 Taking the Laplace transform gives

$$s^2Y - sy_0 - v_0 + 2\beta(sY - y_0) + \omega_0^2Y = 0.$$

Thus

$$Y(s) = \frac{(s + 2\beta)y_0 + v_0}{s^2 + 2\beta s + \omega_0^2}.$$

Complete the square in the denominator:

$$s^2 + 2\beta s + \omega_0^2 = (s + \beta)^2 + (\omega_0^2 - \beta^2).$$

For the underdamped case $\omega_0 > \beta$, define

$$\omega_d = \sqrt{\omega_0^2 - \beta^2}.$$

Then

$$Y(s) = \frac{y_0(s + \beta) + (v_0 + \beta y_0)}{(s + \beta)^2 + \omega_d^2}.$$

Therefore,

$$y(t) = e^{-\beta t} \left[y_0 \cos(\omega_d t) + \frac{v_0 + \beta y_0}{\omega_d} \sin(\omega_d t) \right].$$

4.7.2 Step Functions, Impulses, and Physical Interpretation

The Laplace transform is also useful for discontinuous forcing. The Heaviside step function is

$$\Theta(t - a) = \begin{cases} 0, & t < a, \\ 1, & t \geq a. \end{cases} \quad (4.172)$$

It turns on a force at time a . The Dirac delta function $\delta(t - a)$ represents an ideal impulse at time a . Its Laplace transform is

$$\mathcal{L}\{\delta(t - a)\} = e^{-as}. \quad (4.173)$$

For a particle, an impulse changes momentum. If a force is modeled by

$$F(t) = I\delta(t - a), \quad (4.174)$$

then the total impulse is

$$\int_{a-}^{a+} F(t) dt = I. \quad (4.175)$$

The Laplace transform handles this kind of singular forcing without separately solving the motion before and after the impulse.

In electrical circuits, the same idea appears in switching. For an RLC circuit, the charge $q(t)$ may satisfy

$$Lq'' + Rq' + \frac{1}{C}q = V(t). \quad (4.176)$$

After taking the Laplace transform, this becomes an algebraic equation in $Q(s)$:

$$\left(Ls^2 + Rs + \frac{1}{C}\right)Q(s) = V(s) + Lsq(0) + Lq'(0) + Rq(0). \quad (4.177)$$

Thus the differential equation is converted into an impedance-like relation. This is why the frequency-domain method in circuits works so cleanly.

Table of Common Laplace Transforms

No.	$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$	Condition
L1	1	$\frac{1}{s}$	$\Re(s) > 0$
L2	e^{-at}	$\frac{1}{s+a}$	$\Re(s+a) > 0$
L3	$\sin at$	$\frac{a}{s^2 + a^2}$	$\Re s > \Im a $
L4	$\cos at$	$\frac{s}{s^2 + a^2}$	$\Re s > \Im a $
L5	t^k	$\frac{k!}{s^{k+1}}$	$\Re s > 0$
L6	$t^k e^{-at}$	$\frac{k!}{(s+a)^{k+1}}$	$\Re(s+a) > 0$
L7	$\frac{e^{-at} - e^{-bt}}{b-a}$	$\frac{1}{(s+a)(s+b)}$	$\Re(s+a) > 0$
L8	$\frac{ae^{-at} - be^{-bt}}{a-b}$	$\frac{s}{(s+a)(s+b)}$	$\Re(s+a) > 0$
L9	$\sinh at$	$\frac{a}{s^2 - a^2}$	$\Re s > \Re a $
L10	$\cosh at$	$\frac{s}{s^2 - a^2}$	$\Re s > \Re a $
L11	$t \sin at$	$\frac{2as}{(s^2 + a^2)^2}$	$\Re s > \Im a $

No.	$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$	Condition
L12	$t \cos at$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$	$\Re s > \Im a $
L13	$e^{-at} \sin bt$	$\frac{b}{(s+a)^2 + b^2}$	$\Re(s+a) > \Im b $
L14	$e^{-at} \cos bt$	$\frac{s+a}{(s+a)^2 + b^2}$	$\Re(s+a) > \Im b $
L15	$1 - \cos at$	$\frac{a^2}{s(s^2 + a^2)}$	$\Re s > \Im a $
L16	$at - \sin at$	$\frac{a^3}{s^2(s^2 + a^2)}$	$\Re s > \Im a $
L17	$\sin at - at \cos at$	$\frac{2a^3}{(s^2 + a^2)^2}$	$\Re s > \Im a $
L18	$e^{-at}(1 - at)$	$\frac{s}{(s+a)^2}$	$\Re(s+a) > 0$
L19	$\frac{\sin at}{t}$	$\arctan\left(\frac{a}{s}\right)$	$\Re s > \Im a $
L20	$\frac{1}{t} \sin at \cos bt$	$\frac{1}{2} \left[\arctan\left(\frac{a+b}{s}\right) + \arctan\left(\frac{a-b}{s}\right) \right]$	$\Re s > 0$
L21	$\frac{e^{-at} - e^{-bt}}{t}$	$\ln\left(\frac{s+b}{s+a}\right)$	$\Re(s+a) > 0$
L22	$1 - \operatorname{erf}\left(\frac{a}{2\sqrt{t}}\right)$	$\frac{1}{s} e^{-a\sqrt{s}}$	$\Re s > 0$
L23	$J_0(at)$	$(s^2 + a^2)^{-1/2}$	$\Re s > \Im a $
L24	$\Theta(t-a)$	$\frac{1}{s} e^{-sa}$	$\Re s > 0$
L25	$\Theta(t-a) - \Theta(t-b)$	$\frac{e^{-as} - e^{-bs}}{s}$	all s
L26	square wave	$\frac{1}{s} \tanh\left(\frac{as}{2}\right)$	$\Re s > 0$
L27	$\delta(t-a)$	e^{-sa}	—
L28	$g(t-a)\Theta(t-a)$	$e^{-sa}G(s)$	—

No.	$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$	Condition
L29	$e^{-at}g(t)$	$G(s+a)$	—
L30	$g(at)$	$\frac{1}{a}G\left(\frac{s}{a}\right)$	—
L31	$\frac{g(t)}{t}$	$\int_s^\infty G(u) \, du$	—
L32	$t^n g(t)$	$(-1)^n \frac{d^n G}{ds^n}$	—
L33	$\int_0^t g(\tau) \, d\tau$	$\frac{1}{s}G(s)$	—
L34	$g * h$	$G(s)H(s)$	—

Useful Properties of the Laplace Transform

1. Laplace Transform Integral:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) \, dt$$

2. Transforms of Derivatives:

$$\mathcal{L}\{y'\} = sY - y_0$$

$$\mathcal{L}\{y''\} = s^2Y - sy_0 - y'_0$$

$$\mathcal{L}\{y'''\} = s^3Y - s^2y_0 - sy'_0 - y''_0$$

$$\vdots$$

$$\mathcal{L}\{y^{(n)}\} = s^nY - s^{n-1}y_0 - s^{n-2}y'_0 - \cdots - y_0^{(n-1)}$$

4.8 Sturm-Liouville Theorem

4.8.1 Sturm-Liouville Form

A Sturm-Liouville problem is a special eigenvalue problem for a second-order differential operator. It usually has the form

$$-\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + q(x)y = \lambda w(x)y, \quad a < x < b, \quad (4.178)$$

with suitable boundary conditions. Here $p(x) > 0$ and $w(x) > 0$ on the interval. The function $w(x)$ is called the weight function.

This equation can be written symbolically as

$$Ly = \lambda wy, \quad (4.179)$$

where

$$L = -\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + q(x). \quad (4.180)$$

The essential point is that, under appropriate boundary conditions, this operator behaves like a symmetric matrix in infinite-dimensional space.

Typical boundary conditions are

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0, \quad \beta_1 y(b) + \beta_2 y'(b) = 0. \quad (4.181)$$

They are chosen so that boundary terms vanish when integration by parts is used. This is the infinite-dimensional analogue of the condition that a matrix is symmetric.

4.8.2 Self-Adjointness and the Weighted Inner Product

Define the weighted inner product

$$\langle f, g \rangle_w = \int_a^b f(x)^* g(x) w(x) dx. \quad (4.182)$$

A Sturm-Liouville operator is self-adjoint when

$$\langle f, Lg \rangle = \langle Lf, g \rangle \quad (4.183)$$

under the chosen boundary conditions. More explicitly, one has the Lagrange identity

$$fLg - gLf = -\frac{d}{dx} [p(x) (fg' - gf')] \quad (4.184)$$

for real functions f and g . After integrating,

$$\int_a^b (fLg - gLf)dx = -\left[p(x)(fg' - gf')\right]_a^b. \quad (4.185)$$

For boundary conditions that make the right-hand side vanish, the operator is symmetric. This is the reason why its eigenfunctions have orthogonality properties.

4.8.3 Eigenfunctions and Orthogonality

If y_n satisfies

$$Ly_n = \lambda_n wy_n, \quad (4.186)$$

then y_n is called an eigenfunction and λ_n is called an eigenvalue.

Theorem 4.3 (Orthogonality of Eigenfunctions) Suppose y_m and y_n are eigenfunctions of the same Sturm-Liouville problem with eigenvalues λ_m and λ_n . If $\lambda_m \neq \lambda_n$, then

$$\int_a^b w(x)y_m(x)y_n(x)dx = 0. \quad (4.187)$$

Proof. Since

$$Ly_m = \lambda_m wy_m, \quad Ly_n = \lambda_n wy_n,$$

we multiply the first equation by y_n and the second by y_m , then subtract:

$$y_n Ly_m - y_m Ly_n = (\lambda_m - \lambda_n)wy_m y_n.$$

Integrating from a to b gives

$$\int_a^b (y_n Ly_m - y_m Ly_n)dx = (\lambda_m - \lambda_n) \int_a^b wy_m y_n dx.$$

Using the Lagrange identity, the left-hand side is a boundary term:

$$\int_a^b (y_n Ly_m - y_m Ly_n)dx = -\left[p(x)(y_n y'_m - y_m y'_n)\right]_a^b.$$

For Sturm-Liouville boundary conditions, this boundary term vanishes. Hence

$$(\lambda_m - \lambda_n) \int_a^b wy_m y_n dx = 0.$$

Since $\lambda_m \neq \lambda_n$, we obtain

$$\int_a^b w(x)y_m(x)y_n(x)dx = 0.$$

□

Theorem 4.4 (Reality of Eigenvalues) *For a regular Sturm-Liouville problem with real coefficients and self-adjoint boundary conditions, the eigenvalues are real.*

Proof. Let y be an eigenfunction with eigenvalue λ :

$$Ly = \lambda wy.$$

Multiply by y^* and integrate:

$$\int_a^b y^* Ly dx = \lambda \int_a^b w |y|^2 dx.$$

Because L is self-adjoint, the quantity $\int_a^b y^* Ly dx$ is real. Also, since $w(x) > 0$ and y is not the zero function,

$$\int_a^b w |y|^2 dx > 0.$$

Therefore, λ must be real. $\square$

4.8.4 Expansion in Normal Modes

The eigenfunctions of a regular Sturm-Liouville problem form the natural basis for expanding functions on the interval. If $\{y_n\}$ is an orthogonal set with respect to the weight $w(x)$, then a function $f(x)$ can be expanded formally as

$$f(x) = \sum_{n=1}^{\infty} c_n y_n(x), \quad (4.188)$$

where

$$c_n = \frac{\int_a^b f(x) y_n(x) w(x) dx}{\int_a^b y_n(x)^2 w(x) dx}. \quad (4.189)$$

This is the same idea as writing a vector as a linear combination of orthogonal basis vectors, except that the vector now is a function.

A physical example is the vibrating string. The wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < L, \quad (4.190)$$

with fixed ends

$$u(0, t) = u(L, t) = 0 \quad (4.191)$$

leads to the spatial eigenvalue problem

$$-X'' = \lambda X, \quad X(0) = X(L) = 0. \quad (4.192)$$

The eigenfunctions are

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2. \quad (4.193)$$

Thus the string motion is a sum of normal modes. Each mode oscillates independently. This is why Sturm-Liouville theory is not only a theorem about ODEs. It is also the mathematical language of normal modes in mechanics, electromagnetism, heat conduction, and quantum mechanics.

4.9 Special Function

Special functions are named functions that arise as solutions of important differential equations. They are not artificial objects. They appear because the same geometric structures occur repeatedly in physics. Cartesian coordinates lead naturally to sines, cosines, and exponentials. Cylindrical coordinates lead to Bessel functions. Spherical coordinates lead to Legendre polynomials and spherical harmonics.

The usual route is separation of variables. A partial differential equation is written in a coordinate system adapted to the symmetry of the problem. After separation, each coordinate part satisfies an ODE. The solutions of these ODEs are the special functions.

4.9.1 Bessel Function

Bessel functions appear naturally when differential equations are written in cylindrical coordinates. The standard Bessel equation is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0. \quad (4.194)$$

Equivalently,

$$y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right)y = 0. \quad (4.195)$$

The point $x = 0$ is a regular singular point. Therefore, the Frobenius method is the natural method.

We assume a solution of the form

$$y = \sum_{s=0}^{\infty} a_s x^{s+r}, \quad a_0 \neq 0. \quad (4.196)$$

Then

$$y' = \sum_{s=0}^{\infty} a_s (s+r) x^{s+r-1}, \quad y'' = \sum_{s=0}^{\infty} a_s (s+r)(s+r-1) x^{s+r-2}. \quad (4.197)$$

Substituting these into (4.194), we obtain

$$\sum_{s=0}^{\infty} a_s [(s+r)(s+r-1) + (s+r) - \nu^2] x^{s+r} + \sum_{s=0}^{\infty} a_s x^{s+r+2} = 0. \quad (4.198)$$

Since

$$(s+r)(s+r-1) + (s+r) = (s+r)^2, \quad (4.199)$$

the equation becomes

$$\sum_{s=0}^{\infty} a_s [(s+r)^2 - \nu^2] x^{s+r} + \sum_{s=0}^{\infty} a_s x^{s+r+2} = 0. \quad (4.200)$$

The lowest power comes from the first summation with $s = 0$. Thus the indicial equation is

$$a_0(r^2 - \nu^2) = 0. \quad (4.201)$$

Since $a_0 \neq 0$, we have

$$r^2 - \nu^2 = 0, \quad r = \pm \nu. \quad (4.202)$$

For $s \geq 2$, we shift the second summation by replacing s with $s - 2$. Then the coefficient of x^{s+r} gives

$$a_s [(s+r)^2 - \nu^2] + a_{s-2} = 0. \quad (4.203)$$

Hence the recurrence relation is

$$a_s = -\frac{a_{s-2}}{(s+r)^2 - \nu^2}. \quad (4.204)$$

The coefficient of x^{r+1} gives

$$a_1 [(1+r)^2 - \nu^2] = 0. \quad (4.205)$$

For the regular branch $r = \nu$, this gives $a_1 = 0$ in the ordinary case. Thus all odd coefficients vanish, and the solution is generated by the even coefficients.

Now take

$$r = \nu. \quad (4.206)$$

Then (4.204) becomes

$$a_s = -\frac{a_{s-2}}{s(s+2\nu)}. \quad (4.207)$$

For even indices $s = 2k$, we have

$$a_{2k} = -\frac{a_{2k-2}}{(2k)(2k+2\nu)} = -\frac{a_{2k-2}}{4k(k+\nu)}. \quad (4.208)$$

Applying this repeatedly gives

$$a_{2k} = \frac{(-1)^k a_0}{4^k k! (\nu + 1)(\nu + 2) \cdots (\nu + k)}. \quad (4.209)$$

Using the Gamma function,

$$(\nu + 1)(\nu + 2) \cdots (\nu + k) = \frac{\Gamma(\nu + k + 1)}{\Gamma(\nu + 1)}, \quad (4.210)$$

we obtain

$$a_{2k} = \frac{(-1)^k a_0 \Gamma(\nu + 1)}{2^{2k} k! \Gamma(\nu + k + 1)}. \quad (4.211)$$

If we choose

$$a_0 = \frac{1}{2^\nu \Gamma(\nu + 1)}, \quad (4.212)$$

then the regular solution becomes

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}. \quad (4.213)$$

This is called the Bessel function of the first kind.

For the other root

$$r = -\nu, \quad (4.214)$$

we obtain another Frobenius solution. When ν is not an integer, this second solution is proportional to $J_{-\nu}(x)$. Thus for noninteger ν , the general solution may be written as

$$y = AJ_\nu(x) + BJ_{-\nu}(x). \quad (4.215)$$

When $\nu = n$ is an integer, the two roots differ by an integer. The second Frobenius solution is no longer independent in the simple form J_{-n} . In that case, the second independent solution is usually taken to be

$$Y_n(x), \quad (4.216)$$

called the Bessel function of the second kind. In unified notation, the general solution is written as

$$y = AJ_\nu(x) + BY_\nu(x). \quad (4.217)$$

The function J_ν is usually regular at the origin, while Y_ν is singular there. For example,

$$J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 4^2} - \cdots, \quad J_1(x) = \frac{x}{2} - \frac{x^3}{2^2 4} + \cdots. \quad (4.218)$$

However, $Y_0(x)$ has a logarithmic singularity at the origin. Therefore, in physical problems that include $x = 0$, one often removes Y_ν by imposing regularity at the origin.

There are two useful recurrence relations:

$$J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_{\nu}(x), \quad (4.219)$$

and

$$J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_{\nu}(x). \quad (4.220)$$

These identities can be derived from the series form (4.213) by differentiating term by term and comparing coefficients.

4.9.1.1 Example: Circular Membrane

Consider the wave equation on a circular membrane:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u. \quad (4.221)$$

In polar coordinates,

$$\nabla^2 u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}. \quad (4.222)$$

We use separation of variables:

$$u(r, \theta, t) = R(r)\Theta(\theta)T(t). \quad (4.223)$$

Substituting this into the wave equation gives

$$R\Theta T'' = c^2 \left[\Theta T \frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + RT \frac{1}{r^2} \frac{d^2 \Theta}{d\theta^2} \right]. \quad (4.224)$$

Dividing by $c^2 R\Theta T$, we get

$$\frac{T''}{c^2 T} = \frac{1}{Rr} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \frac{1}{r^2 \Theta} \frac{d^2 \Theta}{d\theta^2}. \quad (4.225)$$

Multiplying by r^2 gives

$$r^2 \frac{T''}{c^2 T} = \frac{r}{R} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \frac{1}{\Theta} \frac{d^2 \Theta}{d\theta^2}. \quad (4.226)$$

The angular part must be independent of r and t . Therefore, we set

$$\frac{1}{\Theta} \frac{d^2 \Theta}{d\theta^2} = -m^2. \quad (4.227)$$

This gives

$$\Theta'' + m^2 \Theta = 0. \quad (4.228)$$

Since the membrane is circular, the angular function must satisfy

$$\Theta(\theta + 2\pi) = \Theta(\theta). \quad (4.229)$$

Thus

$$m = 0, 1, 2, \dots \quad (4.230)$$

For the time dependence, we set

$$T'' + \omega^2 T = 0. \quad (4.231)$$

Then the remaining radial equation becomes

$$r^2 R'' + r R' + \left(\frac{\omega^2}{c^2} r^2 - m^2 \right) R = 0. \quad (4.232)$$

Let

$$k = \frac{\omega}{c}, \quad x = kr. \quad (4.233)$$

Then the radial equation becomes

$$x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} + (x^2 - m^2) R = 0. \quad (4.234)$$

This is Bessel's equation of order m . Therefore,

$$R(r) = A J_m(kr) + B Y_m(kr). \quad (4.235)$$

If the membrane includes the center $r = 0$, then $Y_m(kr)$ is rejected because it is singular at the origin. Thus

$$R(r) = A J_m(kr). \quad (4.236)$$

If the edge of the membrane is fixed at $r = a$, then

$$u(a, \theta, t) = 0. \quad (4.237)$$

This gives

$$J_m(ka) = 0. \quad (4.238)$$

Let $j_{m,s}$ be the s th positive zero of J_m . Then

$$k_{m,s} = \frac{j_{m,s}}{a}, \quad \omega_{m,s} = c k_{m,s} = \frac{c j_{m,s}}{a}. \quad (4.239)$$

Therefore, the normal modes of the circular membrane are

$$u_{m,s}(r, \theta, t) = J_m\left(\frac{j_{m,s}}{a} r\right) (A_m \cos m\theta + B_m \sin m\theta) (C_{m,s} \cos \omega_{m,s} t + D_{m,s} \sin \omega_{m,s} t). \quad (4.240)$$

The zeros of Bessel functions determine the allowed frequencies. This is why the vibration modes of a circular drumhead are not described by ordinary sine waves in the radial direction.

Bessel functions also appear in electromagnetic waves inside cylindrical waveguides. The electric and magnetic fields satisfy the Helmholtz equation after assuming harmonic time dependence. In cylindrical coordinates, the radial part again becomes a Bessel equation. The conducting wall imposes boundary conditions such as

$$J_m(ka) = 0 \quad \text{or} \quad J'_m(ka) = 0, \quad (4.241)$$

depending on whether the mode is TM or TE. Therefore, cylindrical geometry turns Maxwell's equations into Bessel equations.

4.9.1.2 Non-truncation of Bessel Series

The recurrence relation for the regular Bessel solution is

$$a_{2k} = -\frac{a_{2k-2}}{4k(k+\nu)}. \quad (4.242)$$

For $\nu \geq 0$, the denominator does not become infinite or force a coefficient to vanish at some finite k . Therefore, once a_0 is chosen, the recurrence keeps producing infinitely many nonzero coefficients. Thus $J_\nu(x)$ is usually an infinite power series, not a polynomial.

This is one of the main differences between Bessel functions and Legendre polynomials. Bessel functions are selected by regularity and boundary conditions, but the Frobenius series itself usually does not terminate. The boundary condition selects the zeros of J_ν , not a finite polynomial degree.

4.9.2 Legendre Polynomial

Legendre polynomials appear naturally in spherical coordinates. The Legendre differential equation is

$$(1-x^2)y'' - 2xy' + \ell(\ell+1)y = 0. \quad (4.243)$$

For nonnegative integers ℓ , the regular solutions are the Legendre polynomials $P_\ell(x)$.

To see where the polynomial comes from, we first consider the more general equation

$$(1-x^2)y'' - 2xy' + \lambda y = 0. \quad (4.244)$$

Assume a power series solution about $x = 0$:

$$y = \sum_{j=0}^{\infty} a_j x^j. \quad (4.245)$$

Then

$$y' = \sum_{j=1}^{\infty} j a_j x^{j-1}, \quad y'' = \sum_{j=2}^{\infty} j(j-1) a_j x^{j-2}. \quad (4.246)$$

Substituting these into (4.244), we obtain

$$\sum_{j=2}^{\infty} j(j-1) a_j x^{j-2} - \sum_{j=2}^{\infty} j(j-1) a_j x^j - 2 \sum_{j=1}^{\infty} j a_j x^j + \lambda \sum_{j=0}^{\infty} a_j x^j = 0. \quad (4.247)$$

Shift the first summation by replacing j with $j+2$. Then

$$\sum_{j=0}^{\infty} (j+2)(j+1) a_{j+2} x^j - \sum_{j=0}^{\infty} j(j-1) a_j x^j - 2 \sum_{j=0}^{\infty} j a_j x^j + \lambda \sum_{j=0}^{\infty} a_j x^j = 0. \quad (4.248)$$

Combining the middle terms,

$$j(j-1) + 2j = j(j+1), \quad (4.249)$$

we get

$$\sum_{j=0}^{\infty} [(j+2)(j+1) a_{j+2} + (\lambda - j(j+1)) a_j] x^j = 0. \quad (4.250)$$

Therefore,

$$(j+2)(j+1) a_{j+2} + (\lambda - j(j+1)) a_j = 0. \quad (4.251)$$

The recurrence relation is

$$a_{j+2} = \frac{j(j+1) - \lambda}{(j+2)(j+1)} a_j. \quad (4.252)$$

If

$$\lambda = \ell(\ell+1), \quad (4.253)$$

where ℓ is a nonnegative integer, then

$$a_{j+2} = \frac{j(j+1) - \ell(\ell+1)}{(j+2)(j+1)} a_j. \quad (4.254)$$

When $j = \ell$, the numerator becomes zero:

$$\ell(\ell+1) - \ell(\ell+1) = 0. \quad (4.255)$$

Thus

$$a_{\ell+2} = 0, \quad (4.256)$$

and all later coefficients of the same parity vanish. Therefore, the series terminates if we choose the parity that reaches $j = \ell$. This produces a polynomial of degree ℓ . After normalization by

$$P_\ell(1) = 1, \quad (4.257)$$

we obtain the Legendre polynomial $P_\ell(x)$.

The first few are

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1), \quad (4.258)$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x), \quad P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3). \quad (4.259)$$

A compact formula for generating them is Rodrigues' formula:

$$P_\ell(x) = \frac{1}{2^\ell \ell!} \frac{d^\ell}{dx^\ell} (x^2 - 1)^\ell. \quad (4.260)$$

They also satisfy the recurrence relation

$$(\ell + 1)P_{\ell+1}(x) = (2\ell + 1)xP_\ell(x) - \ell P_{\ell-1}(x). \quad (4.261)$$

4.9.2.1 Orthogonality of Legendre Polynomials

We now prove the orthogonality relation in a direct way. The Legendre equation may be written in self adjoint form as

$$\frac{d}{dx} \left[(1 - x^2) \frac{dP_\ell}{dx} \right] + \ell(\ell + 1)P_\ell = 0. \quad (4.262)$$

Assume that $P_n(x)$ is another Legendre polynomial satisfying

$$\frac{d}{dx} \left[(1 - x^2) \frac{dP_n}{dx} \right] + n(n + 1)P_n = 0. \quad (4.263)$$

Multiply (4.262) by $P_n(x)$ and integrate from -1 to 1 :

$$\int_{-1}^1 P_n(x) \left\{ \frac{d}{dx} \left[(1 - x^2) \frac{dP_\ell}{dx} \right] + \ell(\ell + 1)P_\ell(x) \right\} dx = 0. \quad (4.264)$$

Using integration by parts,

$$\begin{aligned} & \int_{-1}^1 P_n(x) \frac{d}{dx} \left[(1 - x^2) \frac{dP_\ell}{dx} \right] dx \\ &= P_n(x) (1 - x^2) \frac{dP_\ell}{dx} \Big|_{-1}^1 - \int_{-1}^1 (1 - x^2) \frac{dP_n}{dx} \frac{dP_\ell}{dx} dx. \end{aligned} \quad (4.265)$$

The boundary term vanishes because

$$1 - x^2 = 0 \quad \text{at} \quad x = \pm 1. \quad (4.266)$$

Therefore, (4.264) becomes

$$-\int_{-1}^1 (1-x^2) \frac{dP_n}{dx} \frac{dP_\ell}{dx} dx + \ell(\ell+1) \int_{-1}^1 P_n(x) P_\ell(x) dx = 0. \quad (4.267)$$

Now multiply (4.263) by $P_\ell(x)$ and integrate from -1 to 1 . By the same calculation, we obtain

$$-\int_{-1}^1 (1-x^2) \frac{dP_\ell}{dx} \frac{dP_n}{dx} dx + n(n+1) \int_{-1}^1 P_\ell(x) P_n(x) dx = 0. \quad (4.268)$$

Subtracting (4.268) from (4.267), the derivative terms cancel. Thus

$$[\ell(\ell+1) - n(n+1)] \int_{-1}^1 P_n(x) P_\ell(x) dx = 0. \quad (4.269)$$

If $\ell \neq n$, then

$$\ell(\ell+1) - n(n+1) \neq 0. \quad (4.270)$$

Therefore,

$$\int_{-1}^1 P_n(x) P_\ell(x) dx = 0, \quad \ell \neq n. \quad (4.271)$$

It remains to compute the normalization integral when $\ell = n$. Using Rodrigues' formula (4.260),

$$\int_{-1}^1 [P_\ell(x)]^2 dx = \frac{1}{2^{2\ell}(\ell!)^2} \int_{-1}^1 \frac{d^\ell}{dx^\ell} (x^2-1)^\ell \frac{d^\ell}{dx^\ell} (x^2-1)^\ell dx. \quad (4.272)$$

Let

$$W(x) = (x^2-1)^\ell. \quad (4.273)$$

Then (4.272) becomes

$$\int_{-1}^1 [P_\ell(x)]^2 dx = \frac{1}{2^{2\ell}(\ell!)^2} \int_{-1}^1 W^{(\ell)}(x) W^{(\ell)}(x) dx. \quad (4.274)$$

We integrate by parts ℓ times. All boundary terms vanish because $W(x) = (x^2-1)^\ell$ has a zero of order ℓ at $x = 1$ and $x = -1$. Thus

$$\int_{-1}^1 W^{(\ell)}(x) W^{(\ell)}(x) dx = (-1)^\ell \int_{-1}^1 W(x) W^{(2\ell)}(x) dx. \quad (4.275)$$

Since the highest degree term of $W(x) = (x^2-1)^\ell$ is $x^{2\ell}$, we have

$$W^{(2\ell)}(x) = (2\ell)!. \quad (4.276)$$

Therefore,

$$\int_{-1}^1 W^{(\ell)}(x)W^{(\ell)}(x)dx = (-1)^\ell (2\ell)! \int_{-1}^1 (x^2 - 1)^\ell dx. \quad (4.277)$$

Since

$$(-1)^\ell (x^2 - 1)^\ell = (1 - x^2)^\ell, \quad (4.278)$$

we get

$$\int_{-1}^1 [P_\ell(x)]^2 dx = \frac{(2\ell)!}{2^{2\ell}(\ell!)^2} \int_{-1}^1 (1 - x^2)^\ell dx. \quad (4.279)$$

The integrand is even, so

$$\int_{-1}^1 (1 - x^2)^\ell dx = 2 \int_0^1 (1 - x^2)^\ell dx. \quad (4.280)$$

Let

$$t = x^2, \quad dx = \frac{1}{2} t^{-1/2} dt. \quad (4.281)$$

Then

$$2 \int_0^1 (1 - x^2)^\ell dx = 2 \int_0^1 (1 - t)^\ell \frac{1}{2} t^{-1/2} dt = \int_0^1 t^{-1/2} (1 - t)^\ell dt. \quad (4.282)$$

This is the beta function:

$$\int_0^1 t^{-1/2} (1 - t)^\ell dt = B\left(\frac{1}{2}, \ell + 1\right) = \frac{\Gamma\left(\frac{1}{2}\right) \Gamma(\ell + 1)}{\Gamma\left(\ell + \frac{3}{2}\right)}. \quad (4.283)$$

Since

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \Gamma(\ell + 1) = \ell!, \quad (4.284)$$

and

$$\Gamma\left(\ell + \frac{3}{2}\right) = \frac{(2\ell + 1)!}{2^{2\ell+1} \ell!} \sqrt{\pi}, \quad (4.285)$$

we obtain

$$B\left(\frac{1}{2}, \ell + 1\right) = \frac{2^{2\ell+1} (\ell!)^2}{(2\ell + 1)!}. \quad (4.286)$$

Substituting this into (4.279), we get

$$\int_{-1}^1 [P_\ell(x)]^2 dx = \frac{(2\ell)!}{2^{2\ell} (\ell!)^2} \frac{2^{2\ell+1} (\ell!)^2}{(2\ell + 1)!} = \frac{2}{2\ell + 1}. \quad (4.287)$$

Combining (4.271) and (4.287), we have

$$\boxed{\int_{-1}^1 P_n(x) P_\ell(x) dx = \frac{2}{2\ell + 1} \delta_{n\ell}.} \quad (4.288)$$

This orthogonality is one of the main reasons Legendre polynomials are useful in spherical boundary value problems.

4.9.2.2 Legendre Polynomial in Laplace's Equation

In spherical coordinates, Laplace's equation is

$$\nabla^2 \Phi = 0, \quad (4.289)$$

where

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}. \quad (4.290)$$

For axially symmetric problems, there is no ϕ dependence. We set

$$\Phi(r, \theta) = R(r)\Theta(\theta). \quad (4.291)$$

Substitution gives

$$\frac{1}{r^2} \Theta \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{r^2 \sin \theta} R \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0. \quad (4.292)$$

Multiplying by $r^2/(R\Theta)$ gives

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0. \quad (4.293)$$

The first term depends only on r , and the second term depends only on θ . Therefore, both must be constants. We choose the separation constant as

$$\ell(\ell + 1). \quad (4.294)$$

Then

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \ell(\ell + 1), \quad (4.295)$$

and

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \ell(\ell + 1)\Theta = 0. \quad (4.296)$$

Let

$$x = \cos \theta. \quad (4.297)$$

Then

$$\frac{d}{d\theta} = \frac{dx}{d\theta} \frac{d}{dx} = -\sin \theta \frac{d}{dx}. \quad (4.298)$$

Therefore,

$$\sin \theta \frac{d\Theta}{d\theta} = -\sin^2 \theta \frac{dy}{dx} = -(1-x^2) \frac{dy}{dx}. \quad (4.299)$$

Also,

$$\frac{1}{\sin \theta} \frac{d}{d\theta} = -\frac{d}{dx}. \quad (4.300)$$

Thus the angular equation becomes

$$\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \ell(\ell+1)y = 0, \quad (4.301)$$

which is exactly Legendre's equation.

The radial equation is

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \ell(\ell+1)R = 0. \quad (4.302)$$

Assume

$$R = r^s. \quad (4.303)$$

Then

$$\frac{d}{dr} \left(r^2 \frac{d}{dr} r^s \right) = \frac{d}{dr} (s r^{s+1}) = s(s+1)r^s. \quad (4.304)$$

Thus

$$s(s+1) = \ell(\ell+1). \quad (4.305)$$

The two roots are

$$s = \ell, \quad s = -(\ell+1). \quad (4.306)$$

Therefore,

$$R(r) = A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}. \quad (4.307)$$

The electrostatic potential in an axially symmetric region can be expanded as

$$\Phi(r, \theta) = \sum_{\ell=0}^{\infty} \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}} \right) P_\ell(\cos \theta). \quad (4.308)$$

The term $\ell = 0$ is the monopole term, $\ell = 1$ is the dipole term, $\ell = 2$ is the quadrupole term, and so on. This is one of the most important examples showing how ordinary differential equations enter electromagnetism.

4.9.2.3 Truncation of Legendre Series

The key recurrence relation is

$$a_{j+2} = \frac{j(j+1) - \ell(\ell+1)}{(j+2)(j+1)} a_j. \quad (4.309)$$

When $j = \ell$, the numerator vanishes. Therefore, the recurrence stops and the series becomes a polynomial if the correct parity is chosen. This is the truncation mechanism of Legendre polynomials.

This is different from the Bessel case. For Bessel functions, the recurrence relation is

$$a_{2k} = -\frac{a_{2k-2}}{4k(k+\nu)}. \quad (4.310)$$

There is no numerator that becomes zero and stops the series. Thus Bessel functions are usually infinite series, while Legendre functions become polynomials when $\lambda = \ell(\ell+1)$ and ℓ is a nonnegative integer.

4.9.3 Spherical Harmonics

Spherical harmonics are the angular eigenfunctions of the Laplacian in spherical coordinates. They appear when the full angular dependence in both θ and ϕ is kept.

The angular part of the Laplacian is

$$\nabla_{\Omega}^2 = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2}. \quad (4.311)$$

We look for eigenfunctions of the angular Laplacian:

$$\nabla_{\Omega}^2 Y(\theta, \phi) = -\ell(\ell+1)Y(\theta, \phi). \quad (4.312)$$

Use separation of variables:

$$Y(\theta, \phi) = \Theta(\theta)\Phi(\phi). \quad (4.313)$$

Substituting into (4.312) gives

$$\frac{\Phi}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{\Theta}{\sin^2 \theta} \frac{d^2 \Phi}{d\phi^2} = -\ell(\ell+1)\Theta\Phi. \quad (4.314)$$

Dividing by $\Theta\Phi$ and multiplying by $\sin^2 \theta$, we obtain

$$\frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \ell(\ell+1) \sin^2 \theta + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = 0. \quad (4.315)$$

The last term depends only on ϕ , while the other terms depend only on θ . Therefore, we set

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -m^2. \quad (4.316)$$

This gives

$$\Phi'' + m^2 \Phi = 0. \quad (4.317)$$

Thus

$$\Phi(\phi) = Ae^{im\phi} + Be^{-im\phi}. \quad (4.318)$$

The function must be single valued on the sphere:

$$\Phi(\phi + 2\pi) = \Phi(\phi). \quad (4.319)$$

Therefore, $m \in \mathbb{Z}$. The remaining equation for Θ is

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left[\ell(\ell + 1) - \frac{m^2}{\sin^2 \theta} \right] \Theta = 0. \quad (4.320)$$

Let

$$x = \cos \theta. \quad (4.321)$$

Then (4.320) becomes

$$(1 - x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \left[\ell(\ell + 1) - \frac{m^2}{1 - x^2} \right] \Theta = 0. \quad (4.322)$$

This is the associated Legendre equation. Its regular solutions are the associated Legendre functions $P_\ell^m(x)$. They are related to Legendre polynomials by

$$P_\ell^m(x) = (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x), \quad m \geq 0, \quad (4.323)$$

up to a conventional sign depending on the chosen definition. Regularity at the poles $\theta = 0, \pi$ requires

$$\ell = 0, 1, 2, \dots, \quad |m| \leq \ell. \quad (4.324)$$

The spherical harmonics are therefore

$$Y_\ell^m(\theta, \phi) = N_{\ell m} P_\ell^m(\cos \theta) e^{im\phi}, \quad (4.325)$$

where $N_{\ell m}$ is a normalization constant. A common convention is

$$N_{\ell m} = (-1)^m \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - m)!}{(\ell + m)!}}. \quad (4.326)$$

With this convention,

$$\int_0^{2\pi} \int_0^\pi Y_\ell^m(\theta, \phi)^* Y_{\ell'}^{m'}(\theta, \phi) \sin \theta d\theta d\phi = \delta_{\ell\ell'} \delta_{mm'}. \quad (4.327)$$

This orthogonality comes from the same Sturm Liouville structure as Legendre polynomials. The ϕ part gives

$$\int_0^{2\pi} e^{-im\phi} e^{im'\phi} d\phi = 2\pi \delta_{mm'}. \quad (4.328)$$

For fixed m , the θ part is a Sturm Liouville problem with weight $\sin \theta$. Therefore, eigenfunctions with different ℓ are orthogonal:

$$\int_0^\pi P_\ell^m(\cos \theta) P_{\ell'}^m(\cos \theta) \sin \theta d\theta = 0, \quad \ell \neq \ell'. \quad (4.329)$$

Combining the θ and ϕ parts gives (4.327).

Since spherical harmonics form a complete orthonormal basis on the sphere, a sufficiently nice angular function can be expanded as

$$f(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} c_{\ell m} Y_\ell^m(\theta, \phi). \quad (4.330)$$

The coefficient is obtained by projecting f onto Y_ℓ^m :

$$c_{\ell m} = \int_0^{2\pi} \int_0^\pi Y_\ell^m(\theta, \phi)^* f(\theta, \phi) \sin \theta d\theta d\phi. \quad (4.331)$$

This is the spherical analogue of Fourier series.

4.9.3.1 Quantum Mechanics

In quantum mechanics, spherical harmonics describe the angular part of the wave function for a central potential. For a central potential,

$$V = V(r), \quad (4.332)$$

the time independent Schrodinger equation is

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V(r) \psi = E \psi. \quad (4.333)$$

Because the potential depends only on r , the equation can be separated as

$$\psi(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^m(\theta, \phi). \quad (4.334)$$

The angular momentum operators satisfy

$$L^2 = -\hbar^2 \nabla_\Omega^2, \quad L_z = -i\hbar \frac{\partial}{\partial \phi}. \quad (4.335)$$

Using

$$\nabla_\Omega^2 Y_\ell^m = -\ell(\ell+1) Y_\ell^m, \quad (4.336)$$

we obtain

$$L^2 Y_\ell^m = \hbar^2 \ell(\ell+1) Y_\ell^m. \quad (4.337)$$

Also,

$$L_z Y_\ell^m = -i\hbar \frac{\partial}{\partial \phi} [N_{\ell m} P_\ell^m(\cos \theta) e^{im\phi}] = \hbar m Y_\ell^m. \quad (4.338)$$

Thus Y_ℓ^m are not merely mathematical functions. They encode the possible angular momentum states of a quantum particle in a central force field. The quantum number ℓ determines the magnitude of angular momentum, while m determines the z component of angular momentum.

In classical electromagnetism, spherical harmonics play a similar role in angular decomposition. For a localized charge distribution, the electrostatic potential can be expanded into multipoles. When the system is not axially symmetric, ordinary Legendre polynomials are not enough, and the full spherical harmonics are needed:

$$\Phi(r, \theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \left(A_{\ell m} r^\ell + \frac{B_{\ell m}}{r^{\ell+1}} \right) Y_\ell^m(\theta, \phi). \quad (4.339)$$

Thus the same angular functions that describe the hydrogen atom also describe the angular structure of electromagnetic fields around localized sources.

4.9.3.2 Comparison Between Bessel Functions and Legendre Polynomials

Both Bessel functions and Legendre polynomials arise from separation of variables, but their series structures are different.

For Bessel functions, the Frobenius recurrence relation is

$$a_{2k} = -\frac{a_{2k-2}}{4k(k+\nu)}. \quad (4.340)$$

This recurrence usually never stops. The result is an infinite series:

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+\nu+1)} \left(\frac{x}{2}\right)^{2k+\nu}. \quad (4.341)$$

In physical problems, boundary conditions select discrete zeros of J_ν , such as

$$J_m(ka) = 0. \quad (4.342)$$

Thus the quantization comes from the boundary condition.

For Legendre polynomials, the power series recurrence relation is

$$a_{j+2} = \frac{j(j+1) - \ell(\ell+1)}{(j+2)(j+1)} a_j. \quad (4.343)$$

When $j = \ell$, the numerator becomes zero. Thus the series terminates and becomes a polynomial. Therefore, the quantization of ℓ is tied to regularity on the sphere and the truncation of the series.

So the main difference is that, Legendre series can truncate into polynomials, while Bessel series usually remain infinite. This difference comes directly from their recurrence relations.

4.10 Applications: Driven Oscillation and RLC Circuit

4.10.1 Driven Oscillation

After such long explanations and demonstrations, we can finally begin to solve the most important second-order nonhomogeneous differential equation in a classical physical system, the driven oscillation.

The equation of motion of a driven oscillation system is

$$m\ddot{x} + b\dot{x} + kx = F(t) \quad (4.344)$$

where $b\dot{x}$ is the drag term, kx is the force model of an oscillating object, and $F(t)$ is a periodic external driven force. We rewrite the problem as

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos(\omega t) \quad (4.345)$$

Where $2\beta := b/m$; $\omega_0^2 := k/m$, and the driven term ω means the driven frequency. To simplify the calculation, I'll use the complex function to solve this ODE. Factor the operator to (5.2), and it becomes

$$\left[D - \left(-\beta + \sqrt{\beta^2 - \omega_0^2} \right) \right] \left[D - \left(-\beta - \sqrt{\beta^2 - \omega_0^2} \right) \right] X = f_0 e^{i\omega t} \quad (4.346)$$

Because $e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$, we need to take the real part of X ,

To simplify the equation, I define the term $\sqrt{\beta^2 - \omega_0^2} = \gamma$, let $[D - (-\beta - \gamma)] X = u$, then

$$\begin{aligned} [D - (-\beta + \gamma)] u &= f_0 e^{i\omega t} \\ \Rightarrow e^{(\beta-\gamma)t} u &= f_0 \int e^{[(\beta-\gamma+i\omega)t]} dt \\ \Rightarrow u &= \frac{f_0}{(\beta + i\omega - \gamma)} e^{i\omega t} + C e^{(-\beta+\gamma)t} \end{aligned}$$

substituting the result to $(D - (-\beta - \gamma))X = u$, then

$$e^{(\beta+\gamma)t} X = \frac{f_0}{[i\omega + (\beta - \gamma)]} \int e^{(i\omega+\beta+\gamma)t} dt + C \int e^{2\gamma t} dt$$

So the solution X is

$$\begin{aligned}
X(t) &= \frac{f_0}{[i\omega + (\beta - \gamma)][i\omega + (\beta + \gamma)]} e^{i\omega t} + C_1 e^{(-\beta + \gamma)t} + C_2 e^{(-\beta - \gamma)t} \\
&= \frac{f_0}{(\omega_0^2 - \omega^2) + 2i\omega\beta} e^{i\omega t} + e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t}) \\
&= \frac{f_0 [(\omega_0^2 - \omega^2) - 2i\omega\beta]}{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2} e^{i\omega t} + e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t})
\end{aligned} \tag{4.347}$$

The result of (4.347) has not been done yet, we must take the real part of the complex-valued function $X(t)$, that is

$$x(t) = \text{Re}(X(t)) = \frac{f_0 [(\omega_0^2 - \omega^2) \cos(\omega t) + 2\omega\beta \sin(\omega t)]}{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2} + e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t}) \tag{4.348}$$

To simplify (4.348), we apply a trigonometric combination, then we can obtain a really beautiful result

$$x(t) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}} \cos(\omega t - \delta) + e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t}) \tag{4.349}$$

where the angle δ is

$$\delta = \tan^{-1} \left(\frac{2\omega\beta}{\omega_0^2 - \omega^2} \right) \tag{4.350}$$

Note that the homogeneous solution $e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t})$, which is the decay term of the damping system in physical meaning, will disappear when the time $t \gg 1/\beta$. In mathematics, it is

$$\lim_{t \rightarrow \infty} e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t}) = 0 \tag{4.351}$$

It means that the long-term behavior $x_l(t)$ depends on the external driven term, that is,

$$x_l(t) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}} \cos(\omega t - \delta) \tag{4.352}$$

The amplitude A of the oscillating system is

$$A = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}} \tag{4.353}$$

In physics, ω_0 is called **natural frequency**, while the driven frequency ω approaches the specific frequency ω_R , the amplitude of the system reaches its highest point, and the phenomenon is called **resonance**. Once the resonance happens, the amplitude satisfies

$$\left. \frac{dA}{d\omega} \right|_{\omega=\omega_R} = 0 \tag{4.354}$$

Then we can compute that

$$\begin{aligned} \left. \frac{dA}{d\omega} \right|_{\omega=\omega_R} &= \frac{-f_0 \cdot \frac{d}{d\omega} [(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2]^{1/2}}{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2} \bigg|_{\omega=\omega_R} \\ &= \frac{-f_0(4\omega^3 - 4\omega\omega_0^2 + 8\omega\beta^2)}{2[(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2]^{3/2}} \bigg|_{\omega=\omega_R} = 0 \end{aligned}$$

So, the resonance frequency is

$$\omega_R = \sqrt{\omega_0^2 - 2\beta^2}$$

Substituting this into (4.353), we can compute the highest amplitude (A_{max}), that is

$$A_{max} = \frac{f_0}{2\beta\sqrt{\omega_0^2 - \beta^2}} \approx \frac{f_0}{2\beta\omega_0} \bigg|_{\beta \ll \omega_0}$$

This report focuses primarily on the analytical form of the solution to the driven oscillation problem, which is one of the most important second-order nonhomogeneous differential equations in classical physics. The main objective is to derive the general solution using the operator descent method and to understand how the solution behaves with respect to the natural frequency ω_0 and the driving frequency ω .

A more detailed analysis of the physical interpretation, such as energy transfer, resonance bandwidth, or quality factor, is beyond the scope of this report and will be left to future study or more advanced physics courses. In this work, I concentrate on solving the equation rigorously and extracting the mathematical structure behind the solution.

4.10.2 RLC Circuit

Another well-known and classic example of a second-order nonhomogeneous differential equation is the RLC circuit. Consider a series RLC circuit driven by an external voltage source $V(t) = V_0 \sin(\omega t)$. The current flows through a resistor R , a capacitor C , and an inductor L connected in series.

In this case, we can use the relationship between electric charge and time to formulate a second-order differential equation. Let $q(t)$ be the charge on the capacitor. By applying Kirchhoff's voltage law (KVL), we obtain:

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = V_0 \sin(\omega t) \quad (4.355)$$

This equation describes the dynamic behavior of the charge $q(t)$ in the circuit under the influence of the time-varying driving voltage.

First, rewrite the problem as

$$\ddot{q} + 2\beta \dot{q} + \omega_0^2 q = \varepsilon \sin(\omega t) \quad (4.356)$$

Where $2\beta = R/L$, $\omega_0^2 = 1/(LC)$ and $\varepsilon = V_0/L$. Factor the operator to this, and it becomes

$$\left[D - \left(-\beta + \sqrt{\beta^2 - \omega_0^2} \right) \right] \left[D - \left(-\beta - \sqrt{\beta^2 - \omega_0^2} \right) \right] Q = \varepsilon e^{i\omega t} \quad (4.357)$$

In the same way, I use the complex function to solve this ODE, which is like the driven oscillating problem, as long as we remember to take the imaginary part of the solution. Define $\sqrt{\beta^2 - \omega_0^2} = \gamma$ and $[D - (-\beta - \gamma)]Q = u$, then

$$\begin{aligned} [D - (-\beta + \gamma)] u &= \varepsilon e^{i\omega t} \\ \Rightarrow e^{(\beta - \gamma)t} u &= \varepsilon \int e^{[(\beta - \gamma + i\omega)t]} dt \\ \Rightarrow u &= \frac{\varepsilon}{(\beta + i\omega - \gamma)} e^{i\omega t} + C e^{(-\beta + \gamma)t} \end{aligned}$$

substituting the result to $(D - (-\beta - \gamma))Q = u$, then

$$e^{(\beta + \gamma)t} Q = \frac{\varepsilon}{[i\omega + (\beta - \gamma)]} \int e^{(i\omega + \beta + \gamma)t} dt + C \int e^{2\gamma t} dt$$

So the solution Q is

$$\begin{aligned}
Q(t) &= \frac{\varepsilon}{[i\omega + (\beta - \gamma)][i\omega + (\beta + \gamma)]} e^{i\omega t} + C_1 e^{(-\beta + \gamma)t} + C_2 e^{(-\beta - \gamma)t} \\
&= \frac{\varepsilon}{(\omega_0^2 - \omega^2) + 2i\omega\beta} e^{i\omega t} + e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t}) \\
&= \frac{\varepsilon [(\omega_0^2 - \omega^2) - 2i\omega\beta]}{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2} e^{i\omega t} + e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t})
\end{aligned} \tag{4.358}$$

The result of (4.358) has not been done yet, we must take the imaginary part of the complex-valued function $Q(t)$, that is

$$q(t) = \text{Im}(Q(t)) = \frac{\varepsilon [(\omega_0^2 - \omega^2) \sin(\omega t) - 2\omega\beta \cos(\omega t)]}{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2} + e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t}) \tag{4.359}$$

To simplify (4.359), we apply a trigonometric combination, and then we can obtain a really beautiful result

$$q(t) = \frac{V_0/L}{\sqrt{(1/LC - \omega^2)^2 + (R\omega/L)^2}} \sin(\omega t - \delta) + e^{-\beta t} (C_1 e^{\gamma t} + C_2 e^{-\gamma t}) \tag{4.360}$$

where the angle δ is

$$\delta = \tan^{-1} \left(\frac{R\omega/L}{1/LC - \omega^2} \right) \tag{4.361}$$

This angle represents the phase difference between the voltage and the charge (or current).

Similarly, in the time-dependent behavior of the current, as time progresses, the transient terms decay, and only the steady-state response remains.

$$q(t) = \frac{V_0/L}{\sqrt{(1/LC - \omega^2)^2 + (R\omega/L)^2}} \sin(\omega t - \delta) \tag{4.362}$$

Typically, in circuit experiments or analysis reports, we are more interested in the current as the main physical quantity. As a result, we need to differentiate the charge function once more to get the current as a function of time, that is

$$I = \dot{q}(t) = \frac{\omega V_0}{L\sqrt{(1/LC - \omega^2)^2 + (R\omega/L)^2}} \cos(\omega t - \delta) \tag{4.363}$$

If you are unsure whether the result is correct, you can check the dimensional consistency to avoid potentially damaging the circuit.

Although the RLC circuit and mechanical oscillation share the same mathematical form, I deliberately chose a different way to extract the solution, not for you to blindly mimic steps, but to help you realize this: mathematics is only a medium; what truly determines how we solve the equation is the physics behind it.

Chapter 5

Introduction to Partial Differential Equations

Abstract Partial differential equations describe the relation between a function of several variables and its partial derivatives. They are the natural language of continuum physics. Heat flow, waves, electrostatic potentials, gravitational potentials, fluid motion, and quantum wave functions are all governed by partial differential equations. In this chapter, we introduce the three classical model equations: the heat diffusion equation, the wave equation, and the Laplace and Poisson equations. The goal is not only to state the equations, but also to derive them from physical principles and solve representative examples in detail.

5.1 Basic Ideas of Partial Differential Equations

A partial differential equation, usually abbreviated as PDE, is an equation involving an unknown function of several independent variables and its partial derivatives. For example, if

$$u = u(x, t), \quad (5.1)$$

then an equation involving $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial t}$, $\frac{\partial^2 u}{\partial x^2}$, or $\frac{\partial^2 u}{\partial t^2}$ is a PDE. In physics, the independent variables are often space and time, while the unknown function represents a physical field. For instance, $u(x, t)$ may represent temperature, displacement, pressure, electric potential, or probability amplitude.

The main difference between an ordinary differential equation and a partial differential equation is the number of independent variables. An ODE describes how a quantity changes along one variable, while a PDE describes how a field changes in several directions at the same time. This is why PDEs are so important in continuum mechanics and field theory.

5.1.1 Order and Linearity

The order of a PDE is the highest order of partial derivative appearing in the equation. For example,

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad (5.2)$$

is a second order PDE because the highest derivative is $\frac{\partial^2 u}{\partial x^2}$. It is first order in time and second order in space.

A PDE is called linear if the unknown function and all of its derivatives appear linearly. A general linear second order PDE in two variables has the form

$$A(x, y) \frac{\partial^2 u}{\partial x^2} + 2B(x, y) \frac{\partial^2 u}{\partial x \partial y} + C(x, y) \frac{\partial^2 u}{\partial y^2} + D(x, y) \frac{\partial u}{\partial x} + E(x, y) \frac{\partial u}{\partial y} + F(x, y)u = G(x, y). \quad (5.3)$$

If $G = 0$, the equation is homogeneous. If $G \neq 0$, the equation is nonhomogeneous.

Linearity is extremely useful because it gives the superposition principle. If u_1 and u_2 are solutions of a homogeneous linear PDE, then

$$u = c_1 u_1 + c_2 u_2 \quad (5.4)$$

is also a solution. Physically, this means that two small waves can be added, two electrostatic potentials can be added, and two normal modes can be superposed.

5.1.2 Three Classical Types of Second Order PDEs

For a second order PDE in two variables, the highest derivative terms are

$$A \frac{\partial^2 u}{\partial x^2} + 2B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2}. \quad (5.5)$$

The quantity

$$\Delta_{PDE} = B^2 - AC \quad (5.6)$$

classifies the local type of the equation.

If

$$B^2 - AC < 0, \quad (5.7)$$

then the equation is called elliptic. The prototype is Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \quad (5.8)$$

Elliptic equations usually describe equilibrium states, such as steady temperature, electrostatic potential, and gravitational potential.

If

$$B^2 - AC = 0, \quad (5.9)$$

then the equation is called parabolic. The prototype is the heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}. \quad (5.10)$$

Parabolic equations describe diffusion, smoothing, and irreversible evolution.

If

$$B^2 - AC > 0, \quad (5.11)$$

then the equation is called hyperbolic. The prototype is the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}. \quad (5.12)$$

Hyperbolic equations describe propagation of signals with finite speed.

5.1.3 Initial and Boundary Conditions

A PDE alone usually does not determine a unique physical solution. We also need initial conditions, boundary conditions, or both.

For an evolution equation, an initial condition specifies the state at the initial time. For the heat equation, one usually gives

$$u(x, 0) = f(x). \quad (5.13)$$

For the wave equation, because it is second order in time, one usually gives both initial displacement and initial velocity:

$$u(x, 0) = f(x), \quad \frac{\partial u}{\partial t}(x, 0) = g(x). \quad (5.14)$$

Boundary conditions describe how the solution behaves at the edge of the domain. Common examples are:

$$\text{Dirichlet condition: } u = \text{given value on the boundary}, \quad (5.15)$$

$$\text{Neumann condition: } \frac{\partial u}{\partial n} = \text{given normal derivative on the boundary}, \quad (5.16)$$

$$\text{Robin condition: } au + b \frac{\partial u}{\partial n} = \text{given value on the boundary}. \quad (5.17)$$

Here $\partial u / \partial n$ means the derivative in the outward normal direction. In heat conduction, the Dirichlet condition fixes the temperature at the boundary, while the Neumann condition fixes the heat flux. In electrostatics, the Dirichlet condition fixes the potential on a conductor, while the Neumann condition fixes the normal component of the electric field.

5.2 Heat Diffusion Equation

The heat equation is the prototype of a parabolic PDE. It describes how temperature spreads out in a medium. The essential physical idea is simple. Heat flows from hot regions to cold regions, and this flow gradually smooths out temperature differences.

5.2.1 Derivation in One Dimension

Consider a thin rod lying along the x axis. Let

$$u(x, t) \tag{5.18}$$

be the temperature at position x and time t . Let ρ be the mass density, c_p the specific heat, and K the thermal conductivity. We examine a small interval $[x, x + \Delta x]$.

The thermal energy stored inside this interval is approximately

$$E(t) = \rho c_p A \Delta x u(x, t), \tag{5.19}$$

where A is the cross sectional area of the rod. More precisely, if the temperature is not constant inside the interval, then

$$E(t) = \int_x^{x+\Delta x} \rho c_p A u(s, t) ds. \tag{5.20}$$

The rate of change of this energy is

$$\frac{dE}{dt} = \int_x^{x+\Delta x} \rho c_p A \frac{\partial u}{\partial t}(s, t) ds. \tag{5.21}$$

Fourier's law of heat conduction states that the heat flux is proportional to the negative temperature gradient:

$$q = -K \frac{\partial u}{\partial x}. \tag{5.22}$$

The minus sign means that heat flows from high temperature to low temperature. The heat entering through the left end per unit time is

$$Aq(x, t) = -KA \frac{\partial u}{\partial x}(x, t), \tag{5.23}$$

while the heat leaving through the right end per unit time is

$$Aq(x + \Delta x, t) = -KA \frac{\partial u}{\partial x}(x + \Delta x, t). \tag{5.24}$$

Thus the net heat entering the interval per unit time is

$$Aq(x, t) - Aq(x + \Delta x, t) = KA \left[\frac{\partial u}{\partial x}(x + \Delta x, t) - \frac{\partial u}{\partial x}(x, t) \right]. \quad (5.25)$$

Energy conservation gives

$$\int_x^{x+\Delta x} \rho c_p A \frac{\partial u}{\partial t}(s, t) ds = KA \left[\frac{\partial u}{\partial x}(x + \Delta x, t) - \frac{\partial u}{\partial x}(x, t) \right]. \quad (5.26)$$

Divide by $A\Delta x$:

$$\frac{1}{\Delta x} \int_x^{x+\Delta x} \rho c_p \frac{\partial u}{\partial t}(s, t) ds = K \frac{\frac{\partial u}{\partial x}(x + \Delta x, t) - \frac{\partial u}{\partial x}(x, t)}{\Delta x}. \quad (5.27)$$

Taking the limit $\Delta x \rightarrow 0$, we obtain

$$\rho c_p \frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2}. \quad (5.28)$$

Therefore,

$$\boxed{\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}}, \quad \alpha = \frac{K}{\rho c_p}. \quad (5.29)$$

The constant α is called the thermal diffusivity. A larger α means that temperature differences disappear more quickly.

5.2.2 Derivation in Three Dimensions

In three dimensions, the temperature is

$$u = u(\mathbf{x}, t), \quad \mathbf{x} = (x, y, z). \quad (5.30)$$

Fourier's law becomes

$$\mathbf{q} = -K \nabla u, \quad (5.31)$$

where $\mathbf{q}$ is the heat flux vector. Energy conservation in a small volume says

$$\rho c_p \frac{\partial u}{\partial t} + \nabla \cdot \mathbf{q} = 0. \quad (5.32)$$

Substituting Fourier's law gives

$$\rho c_p \frac{\partial u}{\partial t} - \nabla \cdot (K \nabla u) = 0. \quad (5.33)$$

If K is constant, this becomes

$$\rho c_p \frac{\partial u}{\partial t} = K \nabla^2 u. \quad (5.34)$$

Thus the heat equation in three dimensions is

$$\boxed{\frac{\partial u}{\partial t} = \alpha \nabla^2 u}. \quad (5.35)$$

Here

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \quad (5.36)$$

is the Laplacian of u .

5.2.3 Solving the Heat Equation on a Finite Rod

Consider a rod of length L with both ends held at zero temperature:

$$u(0, t) = 0, \quad u(L, t) = 0. \quad (5.37)$$

The initial temperature is

$$u(x, 0) = f(x). \quad (5.38)$$

We want to solve

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < L, \quad t > 0. \quad (5.39)$$

We use separation of variables. Assume

$$u(x, t) = X(x)T(t). \quad (5.40)$$

Then

$$\frac{\partial u}{\partial t} = X(x)T'(t), \quad \frac{\partial^2 u}{\partial x^2} = X''(x)T(t). \quad (5.41)$$

Substituting into the heat equation gives

$$XT' = \alpha X''T. \quad (5.42)$$

Divide by αXT :

$$\frac{T'}{\alpha T} = \frac{X''}{X}. \quad (5.43)$$

The left hand side depends only on t , while the right hand side depends only on x . Therefore, both sides must be a constant. Since the boundary condition is homogeneous, we choose

$$\frac{X''}{X} = -\lambda. \quad (5.44)$$

Then

$$X'' + \lambda X = 0, \quad T' + \alpha \lambda T = 0. \quad (5.45)$$

The boundary conditions give

$$X(0) = 0, \quad X(L) = 0. \quad (5.46)$$

We now solve the eigenvalue problem

$$X'' + \lambda X = 0, \quad X(0) = X(L) = 0. \quad (5.47)$$

If $\lambda = 0$, then $X = Ax + B$. The condition $X(0) = 0$ gives $B = 0$, and $X(L) = 0$ gives $A = 0$. This is only the trivial solution.

If $\lambda < 0$, write $\lambda = -\mu^2$. Then

$$X = Ae^{\mu x} + Be^{-\mu x}. \quad (5.48)$$

The boundary conditions again force $A = B = 0$. Hence no nontrivial solution occurs.

If $\lambda > 0$, write $\lambda = k^2$. Then

$$X = A \cos kx + B \sin kx. \quad (5.49)$$

The condition $X(0) = 0$ gives $A = 0$. The condition $X(L) = 0$ gives

$$B \sin kL = 0. \quad (5.50)$$

For a nontrivial solution, $B \neq 0$, so

$$\sin kL = 0. \quad (5.51)$$

Thus

$$kL = n\pi, \quad n = 1, 2, 3, \dots \quad (5.52)$$

Therefore,

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi x}{L}\right). \quad (5.53)$$

The time equation becomes

$$T'_n + \alpha \left(\frac{n\pi}{L}\right)^2 T_n = 0, \quad (5.54)$$

so

$$T_n(t) = C_n \exp \left[-\alpha \left(\frac{n\pi}{L}\right)^2 t \right]. \quad (5.55)$$

Therefore, each separated solution is

$$u_n(x, t) = C_n \sin\left(\frac{n\pi x}{L}\right) \exp \left[-\alpha \left(\frac{n\pi}{L}\right)^2 t \right]. \quad (5.56)$$

By superposition, the general solution satisfying the boundary conditions is

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \exp \left[-\alpha \left(\frac{n\pi}{L}\right)^2 t \right]. \quad (5.57)$$

The coefficients are determined from the initial condition:

$$f(x) = u(x, 0) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right). \quad (5.58)$$

Thus b_n are the Fourier sine coefficients:

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (5.59)$$

5.2.4 Example: A Single Sine Temperature Profile

Suppose

$$f(x) = T_0 \sin\left(\frac{\pi x}{L}\right). \quad (5.60)$$

Then only the $n = 1$ mode is present, so

$$u(x, t) = T_0 \sin\left(\frac{\pi x}{L}\right) \exp\left[-\alpha \left(\frac{\pi}{L}\right)^2 t\right]. \quad (5.61)$$

The shape remains a sine function, but its amplitude decays exponentially. This is the simplest illustration of diffusion. High frequency modes decay even faster because the decay rate is proportional to n^2 :

$$\propto \left(\frac{n\pi}{L}\right)^2. \quad (5.62)$$

Physically, sharp temperature variations are smoothed out more quickly than broad temperature variations.

5.2.5 Energy Decay

The heat equation is irreversible. One mathematical sign of irreversibility is the decay of the integral of u^2 . Assume $u(0, t) = u(L, t) = 0$. Define

$$E(t) = \frac{1}{2} \int_0^L u(x, t)^2 dx. \quad (5.63)$$

Then

$$\frac{dE}{dt} = \int_0^L u \frac{\partial u}{\partial t} dx = \alpha \int_0^L u \frac{\partial^2 u}{\partial x^2} dx. \quad (5.64)$$

Integrating by parts gives

$$\int_0^L u \frac{\partial^2 u}{\partial x^2} dx = u \frac{\partial u}{\partial x} \Big|_0^L - \int_0^L \frac{\partial u^2}{\partial x} dx. \quad (5.65)$$

The boundary term vanishes because $u(0, t) = u(L, t) = 0$. Hence

$$\frac{dE}{dt} = -\alpha \int_0^L \frac{\partial u^2}{\partial x} dx \leq 0. \quad (5.66)$$

This shows that the heat equation smooths and dissipates. It does not preserve the size of the initial disturbance.

5.3 Wave Equation

The wave equation is the prototype of a hyperbolic PDE. It describes propagation of disturbances with finite speed. The simplest form is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}. \quad (5.67)$$

Unlike the heat equation, the wave equation is time reversible. A wave can propagate, reflect, and interfere without automatically smoothing out.

5.3.1 Derivation from a Vibrating String

Consider a string stretched along the x axis. Let

$$u(x, t) \quad (5.68)$$

be the vertical displacement of the string. Let T be the tension and ρ the linear mass density. We assume the displacement is small, so the slope $\frac{\partial u}{\partial x}$ is small.

Take a small segment between x and $x + \Delta x$. The vertical force on the left end is approximately

$$-T \frac{\partial u}{\partial x}(x, t), \quad (5.69)$$

while the vertical force on the right end is approximately

$$T \frac{\partial u}{\partial x}(x + \Delta x, t). \quad (5.70)$$

Thus the net vertical force is

$$T \left[\frac{\partial u}{\partial x}(x + \Delta x, t) - \frac{\partial u}{\partial x}(x, t) \right]. \quad (5.71)$$

The mass of the segment is approximately

$$\rho \Delta x. \quad (5.72)$$

Newton's second law gives

$$\rho \Delta x \frac{\partial^2 u}{\partial t^2}(x, t) \approx T \left[\frac{\partial u}{\partial x}(x + \Delta x, t) - \frac{\partial u}{\partial x}(x, t) \right]. \quad (5.73)$$

Divide by Δx :

$$\rho \frac{\partial^2 u}{\partial t^2}(x, t) \approx T \frac{\frac{\partial u}{\partial x}(x + \Delta x, t) - \frac{\partial u}{\partial x}(x, t)}{\Delta x}. \quad (5.74)$$

Taking the limit $\Delta x \rightarrow 0$, we obtain

$$\rho \frac{\partial^2 u}{\partial t^2} = T \frac{\partial^2 u}{\partial x^2}. \quad (5.75)$$

Therefore,

$$\boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}}, \quad c = \sqrt{\frac{T}{\rho}}. \quad (5.76)$$

The constant c is the wave speed. A larger tension or smaller mass density gives a faster wave.

5.3.2 D'Alembert Solution on the Infinite Line

Consider the wave equation on the whole line:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad -\infty < x < \infty. \quad (5.77)$$

Introduce new variables

$$\xi = x - ct, \quad \eta = x + ct. \quad (5.78)$$

Using the chain rule,

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta}, \quad (5.79)$$

while

$$\frac{\partial}{\partial t} = -c \frac{\partial}{\partial \xi} + c \frac{\partial}{\partial \eta}. \quad (5.80)$$

Thus

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial \xi^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}, \quad (5.81)$$

and

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial \xi^2} - 2c^2 \frac{\partial^2 u}{\partial \xi \partial \eta} + c^2 \frac{\partial^2 u}{\partial \eta^2}. \quad (5.82)$$

Substituting into $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ gives

$$c^2 \frac{\partial^2 u}{\partial \xi^2} - 2c^2 \frac{\partial^2 u}{\partial \xi \partial \eta} + c^2 \frac{\partial^2 u}{\partial \eta^2} = c^2 \frac{\partial^2 u}{\partial \xi^2} + 2c^2 \frac{\partial^2 u}{\partial \xi \partial \eta} + c^2 \frac{\partial^2 u}{\partial \eta^2}. \quad (5.83)$$

Therefore,

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0. \quad (5.84)$$

Integrating with respect to ξ and η , we obtain

$$u(\xi, \eta) = F(\xi) + G(\eta). \quad (5.85)$$

Returning to x and t , the general solution is

$$\boxed{u(x, t) = F(x - ct) + G(x + ct)}. \quad (5.86)$$

The term $F(x - ct)$ travels to the right with speed c , and $G(x + ct)$ travels to the left with speed c .

If the initial conditions are

$$u(x, 0) = f(x), \quad \frac{\partial u}{\partial t}(x, 0) = g(x), \quad (5.87)$$

then D'Alembert's formula is

$$\boxed{u(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds}. \quad (5.88)$$

This formula shows finite propagation speed. The value of $u(x, t)$ depends only on the initial data inside the interval $[x - ct, x + ct]$.

5.3.3 Finite String with Fixed Ends

Now consider a string of length L fixed at both ends:

$$u(0, t) = 0, \quad u(L, t) = 0. \quad (5.89)$$

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}. \quad (5.90)$$

Use separation of variables:

$$u(x, t) = X(x)T(t). \quad (5.91)$$

Substituting gives

$$XT'' = c^2 X''T. \quad (5.92)$$

Divide by $c^2 XT$:

$$\frac{T''}{c^2 T} = \frac{X''}{X} = -\lambda. \quad (5.93)$$

Hence

$$X'' + \lambda X = 0, \quad T'' + c^2 \lambda T = 0. \quad (5.94)$$

The boundary conditions give the same spatial eigenfunctions as the heat equation:

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2. \quad (5.95)$$

The time equation becomes

$$T_n'' + \left(\frac{n\pi c}{L}\right)^2 T_n = 0. \quad (5.96)$$

Thus

$$T_n(t) = A_n \cos\left(\frac{n\pi c}{L}t\right) + B_n \sin\left(\frac{n\pi c}{L}t\right). \quad (5.97)$$

Therefore,

$$u(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{n\pi c}{L}t\right) + B_n \sin\left(\frac{n\pi c}{L}t\right) \right] \sin\left(\frac{n\pi x}{L}\right). \quad (5.98)$$

Given

$$u(x, 0) = f(x), \quad \frac{\partial u}{\partial t}(x, 0) = g(x), \quad (5.99)$$

we get

$$f(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right), \quad (5.100)$$

so

$$A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (5.101)$$

Also,

$$g(x) = \sum_{n=1}^{\infty} B_n \frac{n\pi c}{L} \sin\left(\frac{n\pi x}{L}\right), \quad (5.102)$$

so

$$B_n = \frac{2}{n\pi c} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (5.103)$$

5.3.4 Example: Plucked String Released from Rest

Suppose the string is released from rest:

$$g(x) = 0. \quad (5.104)$$

Then all $B_n = 0$. If the initial shape is a single normal mode,

$$f(x) = A \sin\left(\frac{2\pi x}{L}\right), \quad (5.105)$$

then only the $n = 2$ coefficient is nonzero. The solution is

$$u(x, t) = A \cos\left(\frac{2\pi c}{L}t\right) \sin\left(\frac{2\pi x}{L}\right). \quad (5.106)$$

This mode oscillates without decay. The heat equation damps modes exponentially, while the ideal wave equation preserves energy and produces periodic oscillation.

5.3.5 Energy Conservation for the Wave Equation

For a fixed string, define the energy

$$E(t) = \frac{1}{2} \int_0^L \left(\frac{\partial u}{\partial t}^2 + c^2 \frac{\partial u}{\partial x}^2 \right) dx. \quad (5.107)$$

Then

$$\frac{dE}{dt} = \int_0^L \left(\frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial t^2} + c^2 \frac{\partial u}{\partial x} \frac{\partial^2 u}{\partial x \partial t} \right) dx. \quad (5.108)$$

Using $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$,

$$\frac{dE}{dt} = c^2 \int_0^L \left(\frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \frac{\partial^2 u}{\partial x \partial t} \right) dx. \quad (5.109)$$

Integrating the first term by parts,

$$\int_0^L \frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial x^2} dx = \left. \frac{\partial u}{\partial t} \frac{\partial u}{\partial x} \right|_0^L - \int_0^L \frac{\partial^2 u}{\partial t \partial x} \frac{\partial u}{\partial x} dx. \quad (5.110)$$

Therefore,

$$\frac{dE}{dt} = c^2 \left. \frac{\partial u}{\partial t} \frac{\partial u}{\partial x} \right|_0^L. \quad (5.111)$$

For fixed ends, $u(0, t) = u(L, t) = 0$, so $\frac{\partial u}{\partial t}(0, t) = \frac{\partial u}{\partial t}(L, t) = 0$. Hence

$$\frac{dE}{dt} = 0. \quad (5.112)$$

The ideal wave equation conserves energy. This is the mathematical reason why it describes reversible propagation rather than diffusion.

5.4 General Solution to Wave Functions

5.4.1 Superposition of Waves

Consider a wave function $\phi(x, t)$ that satisfies the wave equation:

$$\frac{\partial^2 \phi(x, t)}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \phi(x, t)}{\partial t^2}. \quad (5.113)$$

This can also be rewritten as

$$\left(\frac{\partial^2}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \phi(x, t) = \left(\frac{\partial}{\partial x} - \frac{1}{c} \frac{\partial}{\partial t} \right) \left(\frac{\partial}{\partial x} + \frac{1}{c} \frac{\partial}{\partial t} \right) \phi = 0, \quad (5.114)$$

where we define

$$\left(\frac{\partial}{\partial x} + \frac{1}{c} \frac{\partial}{\partial t} \right) \phi(x, t) = \Psi(x, t). \quad (5.115)$$

We use the change of variables $x_{\pm} = x \pm ct$, or say, 'let'. Then we can write the origin variables in terms of x_+ and x_- :

$$\begin{cases} x_+ = x + ct \\ x_- = x - ct \end{cases} \Rightarrow \begin{cases} x = \frac{x_+ + x_-}{2} \\ t = \frac{x_+ - x_-}{2c} \end{cases}$$

therefore we obtain

$$\frac{\partial}{\partial x_{\pm}} = \frac{\partial x}{\partial x_{\pm}} \frac{\partial}{\partial x} + \frac{\partial t}{\partial x_{\pm}} \frac{\partial}{\partial t} = \frac{1}{2} \left(\frac{\partial}{\partial x} \pm \frac{1}{c} \frac{\partial}{\partial t} \right),$$

since x, t are the function of $x_{\pm}$. In this situation, the function $\Psi(x, t)$ can be expressed as the function of x_+, x_- naturally, that is, $\Psi(x_+, x_-)$. Substituting (5.115) into (5.114), we get

$$\left(\frac{\partial}{\partial x} - \frac{1}{c} \frac{\partial}{\partial t} \right) \Psi(x_+, x_-) = 2 \frac{\partial}{\partial x_-} \Psi(x_+, x_-) = 0. \quad (5.116)$$

This differential equation indicates that the function Ψ is independent of x_- , so

$$\underbrace{\left(\frac{\partial}{\partial x} + \frac{1}{c} \frac{\partial}{\partial t}\right)}_{2\partial x_+} \phi(x_+, x_-) = \Psi(x_+) \Rightarrow 2 \frac{\partial}{\partial x_+} \phi(x_+, x_-) - \Psi(x_+) = 0, \quad (5.117)$$

it's trivial that ϕ is also independent of x_- , so one of the solutions to ϕ can be written as $\phi_+(x_+)$. We can also let

$$\left(\frac{\partial}{\partial x} - \frac{1}{c} \frac{\partial}{\partial t}\right) \phi(x_+, x_-) = \Psi_2(x_+, x_-),$$

which gives another solution to $\phi = \phi_-(x_-)$. Therefore, the general solution to the wave function can always be denoted by

$$\phi(x, t) = \phi_+(x_+) + \phi_-(x_-). \quad (5.118)$$

Check:

$$4\partial_{x_+}\partial_{x_-}[\phi_+(x_+) + \phi_-(x_-)] = 4\partial_{x_+}[\partial_{x_-}\phi_+(x_+)] + 4\partial_{x_-}[\partial_{x_+}\phi_-(x_-)] = 0 \quad \checkmark$$

The solution indicates that a wave can be represented as the [superposition of other waves](#).

5.4.2 Complex Solution in 1D

The most familiar wave function is the sinusoidal wave; the following argument shows why. Since we have the PDE (5.114), we can use separation of variables, which means that we let

$$\phi(x, t) = X(x)T(t), \quad (5.119)$$

then substituting (5.119) into (5.114), we have

$$\underbrace{\frac{X''(x)}{X}}_{-k^2} - \frac{1}{c^2} \underbrace{\frac{T''(t)}{T}}_{-\omega^2} = 0 \Rightarrow \begin{cases} X(x) = A \cos(kx) + B \sin(kx) \\ T(t) = A' \cos(\omega t) + B' \sin(\omega t), \end{cases}$$

where $\omega^2 = c^2 k^2$, and $\omega = \pm ct$, (See Chapter. 4 to solve this.) By Euler's formula,

$$\begin{aligned} A \cos(kx) + B \sin(kx) &= \sqrt{A^2 + B^2} (\cos kx \cos \delta + \sin kx \sin \delta) \\ &\equiv A \cos(kx - \delta) = \operatorname{Re} \left[A e^{i(kx - \delta)} \right] \\ &= \operatorname{Re} \left[(A e^{i\delta}) e^{ikx} \right] \equiv \operatorname{Re} \left[\tilde{A}_1 e^{ikx} \right] \quad (\tilde{A}_1 \in \mathbb{C}), \end{aligned} \quad (5.120)$$

similarly

$$A' \cos(\omega t) + B' \sin(\omega t) = \operatorname{Re} \left[\tilde{A}_2 e^{-i\omega t} \right]. \quad (5.121)$$

Therefore, the general solution with the complex number can be denoted by the complex form:

$$\phi(x, t) = \operatorname{Re} \left[\sum_k \tilde{A} e^{i(kx - \omega t)} \right] \rightarrow \operatorname{Re} \left[\int dk \tilde{A} e^{i(kx - \omega t)} \right] \quad \text{Continuous,} \quad (5.122)$$

since by the general solution to the wave function, we know that the wave can be superposed by waves. In the 1D case, there are just two directions of propagation, left or right.

On the other hand, the Fourier transform tells us that the wave solution can be denoted by:

$$\tilde{\phi}_+(x_+) = \tilde{\phi}_+(x + ct) = \int dk \tilde{A}_1(k) e^{ik(x+ct)} = \int dk \tilde{A}_1(k) e^{i(kx - \omega t)}. \quad (5.123)$$

Note that $\omega = \pm ck$, and meanwhile,

$$\tilde{\phi}_-(x_-) = \tilde{\phi}_-(x - ct) = \int dk \tilde{A}_2(k) e^{ik(x-ct)} = \int dk \tilde{A}_2(k) e^{i(kx - \omega t)}, \quad (5.124)$$

by (5.123) and (5.124), the general solution is

$$\tilde{\phi}(x, t) = \tilde{\phi}_+(x_+) + \tilde{\phi}_-(x_-) = \int dk \tilde{A}(k) e^{i(kx - \omega t)}. \quad (5.125)$$

Same as (5.122), but we still need to remember to take the real part when considering the real physical waves.

5.4.3 Solution in 3D and the Wave Vector

When we generalize the solution to 3D waves, it's more complicated but still has a good version of the superposition of waves. This time, the wave function becomes $\phi(x, y, z, t)$, and can also be denoted by $\phi(\mathbf{r}, t)$ where $\mathbf{r} = (x, y, z)$ is a position vector. By separation, let

$$\phi = X(x)Y(y)Z(z)T(t), \quad (5.126)$$

it leads to

$$\underbrace{\frac{X''(x)}{X}}_{-k_x^2} + \underbrace{\frac{Y''(y)}{Y}}_{-k_y^2} + \underbrace{\frac{Z''(z)}{Z}}_{-k_z^2} - \frac{1}{c^2} \underbrace{\frac{T''(t)}{T}}_{-\omega^2} = 0. \quad (5.127)$$

Then the solution to the wave function is the linear combination of each direction and time:

$$\phi(\mathbf{r}, t) = \begin{pmatrix} \sin(k_x x) \\ \cos(k_x x) \end{pmatrix} \begin{pmatrix} \sin(k_y y) \\ \cos(k_y y) \end{pmatrix} \begin{pmatrix} \sin(k_z z) \\ \cos(k_z z) \end{pmatrix} \begin{pmatrix} \sin(\omega t) \\ \cos(\omega t) \end{pmatrix}. \quad (5.128)$$

We define a new quantity $\mathbf{k} = (k_x, k_y, k_z)$, which is called **wave vector**¹. In this case, ω also depends on k , which is

$$\omega^2(k) = c^2(k_x^2 + k_y^2 + k_z^2) \Rightarrow \omega = \pm c|\mathbf{k}| = \pm ck. \quad (5.129)$$

Each wave vector $\mathbf{k}$ represents a different mode of propagation. For the wave vector $\mathbf{k}_i = (k_{x_i}, k_{y_i}, k_{z_i})$, the solution is:

$$\tilde{\phi}(\mathbf{r}, t) = \tilde{A}_i(\mathbf{k}) \left(e^{ik_{x_i}x} \right) \left(e^{ik_{y_i}y} \right) \left(e^{ik_{z_i}z} \right) \left(e^{-i\omega t} \right) = \tilde{A}_i(\mathbf{k}) e^{i(k_{x_i}x + k_{y_i}y + k_{z_i}z - \omega t)},$$

where $k_{x_i}x + k_{y_i}y + k_{z_i}z$ can be denoted by:

$$k_{x_i}x + k_{y_i}y + k_{z_i}z = \mathbf{k}_i \cdot \mathbf{r}, \quad (5.130)$$

so it can be rewritten as

$$\tilde{\phi}(\mathbf{r}, t) = \tilde{A}_i(\mathbf{k}) e^{i(\mathbf{k}_i \cdot \mathbf{r} - \omega t)}. \quad (5.131)$$

This means that the wave propagates along the direction $\hat{\mathbf{k}}$ since if

$$\mathbf{k}_i \cdot \mathbf{r} - \omega t = \text{const} \Rightarrow \mathbf{r}(t) = \mathbf{r}_0 + \frac{\omega}{k_i^2} \mathbf{k}_i t = \mathbf{r}_0 + \frac{\omega}{k_i} t \hat{\mathbf{k}}_i.$$

In addition, the form of (5.130) is a plane function in $\mathbb{R}^3$. Therefore, the expression $\mathbf{k} \cdot \mathbf{r} - \omega t$ represents the **phase information** of a wave. This implies that at any given instant in time t , the set of all position vectors $\mathbf{r}$ whose projection onto the direction of propagation ($\mathbf{k}$) results in a constant value defines a geometric plane. On this specific plane, at time t , the phase of the wave is uniform (identical). Because the surfaces of constant phase are parallel planes, this method of three-dimensional propagation is known as a **plane wave**. We've known that waves can be represented as the superposition of all the continuous waves in 3D space along each direction of propagation, hence the general solution to the 3D wave function is:

$$\tilde{\phi}(\mathbf{r}, t) = \sum_{\mathbf{k}} \tilde{A}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \Rightarrow \tilde{\phi}(\mathbf{r}, t) = \int d\mathbf{k} \tilde{A}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}. \quad (5.132)$$

5.4.4 Electromagnetic Wave Equation

In vacuum, Maxwell's equations are

¹ In 1D, it's called wave number, which represents how many phases the wave will propagate when changing the position x .

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0 \quad (5.133)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (5.134)$$

Taking the curl of Faraday's law gives

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}). \quad (5.135)$$

Using

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}, \quad (5.136)$$

and $\nabla \cdot \mathbf{E} = 0$, we get

$$-\nabla^2 \mathbf{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (5.137)$$

Therefore,

$$\boxed{\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0}. \quad (5.138)$$

This is a wave equation with speed

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}. \quad (5.139)$$

The same derivation gives

$$\boxed{\nabla^2 \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0}. \quad (5.140)$$

Thus electromagnetic waves are solutions of Maxwell's equations. The general solution to the electric wave equation can be denoted by the linear combination of each direction $\hat{\mathbf{k}}$, which is similar to (5.132) in the continuous space:

$$\mathbf{E}(\mathbf{r}, t) = \text{Re} \left[\int d^3 \mathbf{k} \tilde{\mathbf{E}}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega(\mathbf{k})t)} \right], \quad (5.141)$$

similarly, the general solution to the magnetic wave can also be denoted by:

$$\mathbf{B}(\mathbf{r}, t) = \text{Re} \left[\int d^3 \mathbf{k} \tilde{\mathbf{B}}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega(\mathbf{k})t)} \right]. \quad (5.142)$$

I'll use the complex vector to represent the wave since the direction is the most important thing in this chapter. By Gauss's Law in source-free electric field $\nabla \cdot \mathbf{E} = 0$, (5.141) is applied the divergence to be:

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \nabla \cdot \int d^3\mathbf{k} \tilde{\mathbf{E}}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega(\mathbf{k})t)} = \int d^3\mathbf{k} \nabla \cdot \left[\tilde{\mathbf{E}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right] \\
&= \int d^3\mathbf{k} (\nabla \cdot \tilde{\mathbf{E}}) e^{i(\dots)} + \int d^3\mathbf{k} \tilde{\mathbf{E}} \cdot \nabla e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \\
&= \int d^3\mathbf{k} i(\tilde{\mathbf{E}} \cdot \mathbf{k}) e^{i(\dots)} = 0,
\end{aligned}$$

Therefore, we can find the important conclusion:

$$\mathbf{k} \cdot \tilde{\mathbf{E}} = 0 \equiv \hat{\mathbf{k}} \cdot \mathbf{E} = 0, \quad \forall \hat{\mathbf{k}}_i. \quad (5.143)$$

So, the electric field is **perpendicular** to the direction of the electric wave which propagates along $\hat{\mathbf{k}}$. Similarly, the curl of the electric field is

$$\begin{aligned}
\nabla \times \mathbf{E} &= \nabla \times \int d^3\mathbf{k} \tilde{\mathbf{E}}(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega(\mathbf{k})t)} = \int d^3\mathbf{k} \nabla \times \left[\tilde{\mathbf{E}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right] \\
&= \underbrace{\int d^3\mathbf{k} (\nabla \times \tilde{\mathbf{E}}) e^{i(\dots)}}_{=0 \text{ } (\tilde{\mathbf{E}} \text{ is } \mathbf{k} \text{ dependence})} - \int d^3\mathbf{k} \tilde{\mathbf{E}} \times \nabla e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \\
&= - \int d^3\mathbf{k} i(\tilde{\mathbf{E}} \times \mathbf{k}) e^{i(\dots)} = \int d^3\mathbf{k} ik(\hat{\mathbf{k}} \times \tilde{\mathbf{E}}) e^{i(\dots)},
\end{aligned}$$

by Faraday's law, we have

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = - \int d^3\mathbf{k} \tilde{\mathbf{B}} \left[\frac{\partial}{\partial t} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right] = \int d^3\mathbf{k} i\omega \tilde{\mathbf{B}} e^{i(\dots)}.$$

By comparing the results of curl and Faraday's law, we finally get:

$$\omega \tilde{\mathbf{B}} = k(\hat{\mathbf{k}} \times \tilde{\mathbf{E}}) \Rightarrow \tilde{\mathbf{B}} = \frac{k}{\omega}(\hat{\mathbf{k}} \times \tilde{\mathbf{E}}) = \frac{1}{c}(\hat{\mathbf{k}} \times \tilde{\mathbf{E}}). \quad (5.144)$$

When we take a real part, we can conclude that

$$\begin{cases} \hat{\mathbf{k}} \cdot \mathbf{E}(\mathbf{r}, t) = 0 & \mathbf{E} \perp \hat{\mathbf{k}} \\ \mathbf{B}(\mathbf{r}, t) = \frac{1}{c}(\hat{\mathbf{k}} \times \mathbf{E}) & \mathbf{B} \perp \mathbf{E} \perp \hat{\mathbf{k}}. \end{cases} \quad (5.145)$$

Therefore, once an electric field is given, the corresponding magnetic field is also be determined by the relation (5.144).

5.5 Laplace Equation

Laplace's equation and Poisson's equation (Section 5.6) are the prototypes of elliptic PDEs. They describe equilibrium fields. Unlike the heat and wave equations, there is no time variable in the simplest form. The goal is to find a spatial field satisfying the equation and the boundary conditions.

5.5.1 Laplace Equation

Laplace's equation is

$$\boxed{\nabla^2 u = 0}. \quad (5.146)$$

A solution of Laplace's equation is called a harmonic function. In two dimensions,

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}, \quad (5.147)$$

so Laplace's equation is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \quad (5.148)$$

One way to understand the equation is through steady heat conduction. The heat equation is

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u. \quad (5.149)$$

At steady state, the temperature no longer changes with time, so

$$\frac{\partial u}{\partial t} = 0. \quad (5.150)$$

Therefore,

$$\nabla^2 u = 0. \quad (5.151)$$

Thus Laplace's equation describes equilibrium temperature when there are no internal heat sources. In electrostatics, the electric field is related to the electric potential by

$$\mathbf{E} = -\nabla\Phi. \quad (5.152)$$

Gauss's law is

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (5.153)$$

Substituting $\mathbf{E} = -\nabla\Phi$ gives

$$-\nabla \cdot (\nabla \Phi) = \frac{\rho}{\epsilon_0}. \quad (5.154)$$

Hence

$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0}. \quad (5.155)$$

In a charge-free region, $\rho = 0$, so

$$\nabla^2 \Phi = 0. \quad (5.156)$$

This is Laplace's equation for the electrostatic potential.

5.5.2 3-D Problem

We now solve a three-dimensional Laplace problem in a rectangular box. Let

$$0 < x < c, \quad 0 < y < a, \quad 0 < z < b. \quad (5.157)$$

We want to determine the potential $V(x, y, z)$ satisfying Laplace's equation

$$\nabla^2 V = 0, \quad (5.158)$$

that is

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0. \quad (5.159)$$

The six boundary conditions are

1. $V = 0$ when $y = 0$,
2. $V = 0$ when $y = a$,
3. $V = 0$ when $z = 0$,
4. $V = V_0$ when $z = b$,
5. $V = 0$ when $x = 0$,
6. $V = 0$ when $x = c$.

Physically, this is an electrostatic boundary value problem in a rectangular box. Five faces are grounded, and the top face is held at the constant potential V_0 . The uniqueness theorem tells us that once these six boundary values are fixed, there is at most one harmonic function inside the box. Thus if we construct a solution satisfying the equation and the boundary conditions, that solution is the physical solution.

We use separation of variables:

$$V(x, y, z) = X(x)Y(y)Z(z). \quad (5.160)$$

Substituting into Laplace's equation gives

$$X''(x)Y(y)Z(z) + X(x)Y''(y)Z(z) + X(x)Y(y)Z''(z) = 0. \quad (5.161)$$

Dividing by $X(x)Y(y)Z(z)$ gives

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} + \frac{Z''(z)}{Z(z)} = 0. \quad (5.162)$$

The first term depends only on x , the second only on y , and the third only on z . Therefore, each term must be a constant. Since the boundary conditions in the x and y directions are homogeneous Dirichlet conditions, we choose

$$\frac{X''(x)}{X(x)} = -\lambda, \quad \frac{Y''(y)}{Y(y)} = -\mu, \quad \frac{Z''(z)}{Z(z)} = \lambda + \mu. \quad (5.163)$$

Then (5.162) is automatically satisfied.

The equations are

$$X'' + \lambda X = 0, \quad Y'' + \mu Y = 0, \quad Z'' - (\lambda + \mu)Z = 0. \quad (5.164)$$

Their general solutions are

$$\begin{cases} X(x) = A \sin(\sqrt{\lambda}x) + B \cos(\sqrt{\lambda}x), \\ Y(y) = C \sin(\sqrt{\mu}y) + D \cos(\sqrt{\mu}y), \\ Z(z) = E e^{\sqrt{\lambda+\mu}z} + F e^{-\sqrt{\lambda+\mu}z}. \end{cases} \quad (5.165)$$

The boundary conditions $V = 0$ at $x = 0$ and $x = c$ imply

$$X(0) = 0, \quad X(c) = 0. \quad (5.166)$$

Thus $B = 0$, and

$$\sin(\sqrt{\lambda}c) = 0. \quad (5.167)$$

Hence

$$\sqrt{\lambda} = \frac{n\pi}{c}, \quad n \in \mathbb{N}. \quad (5.168)$$

Similarly, the boundary conditions $V = 0$ at $y = 0$ and $y = a$ imply

$$Y(0) = 0, \quad Y(a) = 0, \quad (5.169)$$

so $D = 0$, and

$$\sqrt{\mu} = \frac{m\pi}{a}, \quad m \in \mathbb{N}. \quad (5.170)$$

Define

$$\gamma_{nm} = \sqrt{\left(\frac{n\pi}{c}\right)^2 + \left(\frac{m\pi}{a}\right)^2} = \pi \sqrt{\left(\frac{n}{c}\right)^2 + \left(\frac{m}{a}\right)^2}. \quad (5.171)$$

The condition $V = 0$ at $z = 0$ gives

$$Z(0) = 0, \quad (5.172)$$

so $E + F = 0$. Therefore,

$$Z(z) = K_{nm} \sinh(\gamma_{nm}z). \quad (5.173)$$

The separated modes are

$$V_{nm}(x, y, z) = K_{nm} \sinh(\gamma_{nm}z) \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right). \quad (5.174)$$

By linearity, the full solution is a superposition:

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} K_{nm} \sinh(\gamma_{nm}z) \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right). \quad (5.175)$$

The remaining boundary condition is imposed at $z = b$:

$$V(x, y, b) = V_0. \quad (5.176)$$

Thus

$$V_0 = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} K_{nm} \sinh(\gamma_{nm}b) \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right). \quad (5.177)$$

This is a double Fourier sine series for the constant function V_0 on the rectangle $0 < x < c, 0 < y < a$.

To determine K_{nm} , multiply both sides of (5.177) by

$$\sin\left(\frac{n'\pi x}{c}\right) \sin\left(\frac{m'\pi y}{a}\right) \quad (5.178)$$

and integrate over the top face:

$$\begin{aligned} & \int_0^c \int_0^a V_0 \sin\left(\frac{n'\pi x}{c}\right) \sin\left(\frac{m'\pi y}{a}\right) dy dx \\ &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} K_{nm} \sinh(\gamma_{nm}b) \int_0^c \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{n'\pi x}{c}\right) dx \int_0^a \sin\left(\frac{m\pi y}{a}\right) \sin\left(\frac{m'\pi y}{a}\right) dy. \end{aligned}$$

Using orthogonality,

$$\int_0^c \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{n'\pi x}{c}\right) dx = \frac{c}{2} \delta_{nn'}, \quad (5.179)$$

and

$$\int_0^a \sin\left(\frac{m\pi y}{a}\right) \sin\left(\frac{m'\pi y}{a}\right) dy = \frac{a}{2} \delta_{mm'}, \quad (5.180)$$

we get

$$\int_0^c \int_0^a V_0 \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right) dy dx = K_{nm} \sinh(\gamma_{nm}b) \frac{ac}{4}. \quad (5.181)$$

Therefore,

$$K_{nm} = \frac{4}{ac \sinh(\gamma_{nm}b)} \int_0^c \int_0^a V_0 \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right) dy dx. \quad (5.182)$$

Since V_0 is constant,

$$\begin{aligned} K_{nm} &= \frac{4V_0}{ac \sinh(\gamma_{nm}b)} \left[\int_0^c \sin\left(\frac{n\pi x}{c}\right) dx \right] \left[\int_0^a \sin\left(\frac{m\pi y}{a}\right) dy \right] \\ &= \frac{4V_0}{ac \sinh(\gamma_{nm}b)} \left[\frac{c}{n\pi} (1 - \cos n\pi) \right] \left[\frac{a}{m\pi} (1 - \cos m\pi) \right] \\ &= \frac{4V_0(1 - \cos n\pi)(1 - \cos m\pi)}{nm\pi^2 \sinh(\gamma_{nm}b)}. \end{aligned} \quad (5.183)$$

Thus

$$K_{nm} = 0 \quad \text{if } n \text{ or } m \text{ is even,} \quad (5.184)$$

and

$$K_{nm} = \frac{16V_0}{nm\pi^2 \sinh(\gamma_{nm}b)} \quad \text{if } n \text{ and } m \text{ are both odd.} \quad (5.185)$$

Substituting this into (5.175), we obtain

$$V(x, y, z) = \frac{16V_0}{\pi^2} \sum_{\substack{n,m=1 \\ n,m \text{ odd}}}^{\infty} \frac{1}{nm} \frac{\sinh(\gamma_{nm}z)}{\sinh(\gamma_{nm}b)} \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right). \quad (5.186)$$

This formula satisfies the five grounded faces automatically. On the top face, the double sine series reconstructs the constant boundary value V_0 inside the rectangle. At the edges of the top face there are jumps in the idealized boundary data, so the Fourier series converges to the appropriate boundary average there, exactly as in ordinary Fourier theory.

5.5.3 Spherical Problem

The rectangular box problem shows how Cartesian boundary conditions select sine functions in the bounded directions and hyperbolic functions in the remaining direction. When the boundary is a sphere, the natural coordinates are spherical coordinates, and the same separation idea produces Legendre polynomials instead of sine functions. Here we consider the axisymmetric case, so the potential is independent of the azimuthal angle ϕ :

$$V = V(r, \theta). \quad (5.187)$$

In a charge-free region, the electrostatic potential satisfies Laplace's equation

$$\nabla^2 \Phi = 0. \quad (5.188)$$

For an axisymmetric function in spherical coordinates, this becomes

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) = 0. \quad (5.189)$$

We use separation of variables:

$$V(r, \theta) = R(r)\Theta(\theta). \quad (5.190)$$

Substituting this into (5.189), we get

$$\frac{\Theta}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{R}{r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0. \quad (5.191)$$

Multiplying by $r^2/(R\Theta)$ gives

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0. \quad (5.192)$$

The first term depends only on r , while the second term depends only on θ . Thus each term must be a constant. We choose the separation constant in the form $\ell(\ell+1)$, because this is the value that gives regular angular solutions on the sphere:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \ell(\ell+1), \quad (5.193)$$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \ell(\ell+1)\Theta = 0. \quad (5.194)$$

We first solve the angular equation. Let

$$x = \cos \theta. \quad (5.195)$$

Then

$$\frac{d}{d\theta} = \frac{dx}{d\theta} \frac{d}{dx} = -\sin \theta \frac{d}{dx}. \quad (5.196)$$

If $\Theta(\theta) = y(x)$, then

$$\sin \theta \frac{d\Theta}{d\theta} = -\sin^2 \theta \frac{dy}{dx} = -(1-x^2) \frac{dy}{dx}. \quad (5.197)$$

Also,

$$\frac{1}{\sin \theta} \frac{d}{d\theta} = -\frac{d}{dx}. \quad (5.198)$$

Therefore, (5.194) becomes

$$\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \ell(\ell+1)y = 0. \quad (5.199)$$

Equivalently,

$$(1-x^2) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + \ell(\ell+1)y = 0. \quad (5.200)$$

This is Legendre's differential equation. The regular solutions on $-1 \leq x \leq 1$ are the Legendre polynomials:

$$\Theta(\theta) = P_\ell(\cos \theta), \quad \ell = 0, 1, 2, \dots \quad (5.201)$$

Next, the radial equation is

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \ell(\ell+1)R = 0. \quad (5.202)$$

Assume

$$R(r) = r^s. \quad (5.203)$$

Then

$$\frac{dR}{dr} = sr^{s-1}, \quad r^2 \frac{dR}{dr} = sr^{s+1}, \quad (5.204)$$

so

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = s(s+1)r^s. \quad (5.205)$$

Substituting this into (5.202) gives

$$[s(s+1) - \ell(\ell+1)]r^s = 0. \quad (5.206)$$

Hence

$$s(s+1) = \ell(\ell+1), \quad (5.207)$$

with two roots

$$s = \ell, \quad s = -(\ell+1). \quad (5.208)$$

Therefore,

$$R(r) = A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}. \quad (5.209)$$

Combining the radial and angular parts, the general axisymmetric solution is

$$V(r, \theta) = \sum_{\ell=0}^{\infty} \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}} \right) P_\ell(\cos \theta). \quad (5.210)$$

This is the spherical analogue of the Fourier sine series in the rectangular box problem. The role of the sine functions is now played by $P_\ell(\cos \theta)$.

5.5.3.1 Interior Spherical Dirichlet Problem

Suppose the potential is specified on the surface of a hollow sphere of radius R :

$$V(R, \theta) = V_0(\theta). \quad (5.211)$$

We want to find the potential inside the sphere, where $0 \leq r < R$. Since the potential must remain finite at the origin, the terms

$$\frac{B_\ell}{r^{\ell+1}} \quad (5.212)$$

must be rejected. Thus

$$B_\ell = 0, \quad (5.213)$$

for all ℓ , and the interior solution has the form

$$V(r, \theta) = \sum_{\ell=0}^{\infty} A_\ell r^\ell P_\ell(\cos \theta). \quad (5.214)$$

At $r = R$, the boundary condition gives

$$V_0(\theta) = \sum_{\ell=0}^{\infty} A_\ell R^\ell P_\ell(\cos \theta). \quad (5.215)$$

To determine A_ℓ , multiply both sides by $P_k(\cos \theta) \sin \theta$ and integrate from 0 to π :

$$\int_0^\pi V_0(\theta) P_k(\cos \theta) \sin \theta \, d\theta = \sum_{\ell=0}^{\infty} A_\ell R^\ell \int_0^\pi P_\ell(\cos \theta) P_k(\cos \theta) \sin \theta \, d\theta. \quad (5.216)$$

Using the orthogonality relation

$$\int_0^\pi P_\ell(\cos \theta) P_k(\cos \theta) \sin \theta \, d\theta = \frac{2}{2\ell + 1} \delta_{\ell k}, \quad (5.217)$$

we get

$$\int_0^\pi V_0(\theta) P_\ell(\cos \theta) \sin \theta \, d\theta = A_\ell R^\ell \frac{2}{2\ell + 1}. \quad (5.218)$$

Hence

$$A_\ell = \frac{2\ell + 1}{2R^\ell} \int_0^\pi V_0(\theta) P_\ell(\cos \theta) \sin \theta \, d\theta. \quad (5.219)$$

Substituting (5.219) into the interior solution gives

$$V_{\text{in}}(r, \theta) = \sum_{\ell=0}^{\infty} \left[\frac{2\ell + 1}{2} \left(\frac{r}{R} \right)^\ell \int_0^\pi V_0(\theta') P_\ell(\cos \theta') \sin \theta' \, d\theta' \right] P_\ell(\cos \theta). \quad (5.220)$$

Here θ' is a dummy integration variable. The outside angle θ is the variable where we evaluate the solution.

5.5.3.2 Exterior Spherical Dirichlet Problem

Now suppose the same surface potential is specified at $r = R$, but we want the solution outside the sphere, where $r > R$. We also require

$$V(r, \theta) \rightarrow 0 \quad \text{as} \quad r \rightarrow \infty. \quad (5.221)$$

The terms $A_\ell r^\ell$ grow at infinity, so we must set

$$A_\ell = 0 \quad (5.222)$$

for all ℓ . Thus the exterior solution has the form

$$V(r, \theta) = \sum_{\ell=0}^{\infty} \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos \theta). \quad (5.223)$$

At $r = R$, the boundary condition gives

$$V_0(\theta) = \sum_{\ell=0}^{\infty} \frac{B_\ell}{R^{\ell+1}} P_\ell(\cos \theta). \quad (5.224)$$

Again multiply both sides by $P_k(\cos \theta) \sin \theta$ and integrate from 0 to π . Using orthogonality, we obtain

$$\int_0^\pi V_0(\theta) P_\ell(\cos \theta) \sin \theta \, d\theta = \frac{B_\ell}{R^{\ell+1}} \frac{2}{2\ell+1}. \quad (5.225)$$

Therefore,

$$B_\ell = \frac{2\ell+1}{2} R^{\ell+1} \int_0^\pi V_0(\theta) P_\ell(\cos \theta) \sin \theta \, d\theta. \quad (5.226)$$

The exterior solution is

$$V_{\text{out}}(r, \theta) = \sum_{\ell=0}^{\infty} \left[\frac{2\ell+1}{2} \left(\frac{R}{r} \right)^{\ell+1} \int_0^\pi V_0(\theta') P_\ell(\cos \theta') \sin \theta' \, d\theta' \right] P_\ell(\cos \theta). \quad (5.227)$$

5.5.3.3 Examples

If the surface potential is a constant,

$$V_0(\theta) = V_c, \quad (5.228)$$

then only the $\ell = 0$ term survives, because $P_0(\cos \theta) = 1$ and every higher Legendre polynomial is orthogonal to the constant mode. The interior solution is

$$V_{\text{in}}(r, \theta) = V_c, \quad (5.229)$$

and the exterior solution is

$$V_{\text{out}}(r, \theta) = V_c \frac{R}{r}. \quad (5.230)$$

This is the potential of a sphere held at a constant potential V_c .

As another example, let

$$V_0(\theta) = V_c \cos \theta = V_c P_1(\cos \theta). \quad (5.231)$$

Then only the $\ell = 1$ term survives. For the inside region,

$$V_{\text{in}}(r, \theta) = V_c \frac{r}{R} \cos \theta. \quad (5.232)$$

For the outside region,

$$V_{\text{out}}(r, \theta) = V_c \frac{R^2}{r^2} \cos \theta. \quad (5.233)$$

Thus a single Legendre mode on the boundary produces a single separated mode in the solution. If the boundary potential is a finite sum

$$V_0(\theta) = \sum_{\ell=0}^N c_\ell P_\ell(\cos \theta), \quad (5.234)$$

then the solution contains only finitely many terms. If $V_0(\theta)$ is a general function, then the solution is an infinite Legendre series.

This problem is the spherical version of the rectangular 3-D box problem. In the rectangular box, the boundary data on a plane are expanded in a double sine series. In the spherical problem, the boundary data on a sphere are expanded in Legendre polynomials. For a fully three-dimensional boundary potential $V_0(\theta, \phi)$, the Legendre polynomials must be replaced by spherical harmonics $Y_\ell^m(\theta, \phi)$.

5.5.4 Mean Value Property and Maximum Principle

Harmonic functions are functions satisfying Laplace's equation

$$\nabla^2 u = 0. \quad (5.235)$$

They are very rigid functions. Although a harmonic function may look locally complicated, its value at a point is controlled by the surrounding values. This is the content of the mean value property. It also leads directly to the maximum principle and the uniqueness theorem for the Dirichlet problem.

Theorem 5.1 (Mean Value Property) *Let $u \in C^2(B_R(\mathbf{0}))$ be harmonic in the disk*

$$B_R(\mathbf{0}) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < R^2\}. \quad (5.236)$$

Then for every $0 < r < R$,

$$u(\mathbf{0}) = \frac{1}{2\pi} \int_0^{2\pi} u(r \cos \theta, r \sin \theta) d\theta. \quad (5.237)$$

Proof. Define the average of u on the circle of radius r by

$$M(r) = \frac{1}{2\pi} \int_0^{2\pi} u(r \cos \theta, r \sin \theta) d\theta. \quad (5.238)$$

We prove that $M(r)$ is independent of r . Differentiating with respect to r , we get

$$M'(r) = \frac{1}{2\pi} \int_0^{2\pi} \frac{\partial u}{\partial r}(r, \theta) d\theta. \quad (5.239)$$

On the circle $\partial B_r(\mathbf{0})$, the outward normal derivative is the radial derivative:

$$\frac{\partial u}{\partial n} = \frac{\partial u}{\partial r}. \quad (5.240)$$

Since the line element on the circle is

$$ds = r d\theta, \quad (5.241)$$

we have

$$\int_0^{2\pi} \frac{\partial u}{\partial r}(r, \theta) d\theta = \frac{1}{r} \int_{\partial B_r(\mathbf{0})} \frac{\partial u}{\partial n} ds. \quad (5.242)$$

By the divergence theorem,

$$\int_{\partial B_r(\mathbf{0})} \frac{\partial u}{\partial n} ds = \int_{B_r(\mathbf{0})} \nabla \cdot (\nabla u) dA. \quad (5.243)$$

Since

$$\nabla \cdot (\nabla u) = \nabla^2 u = 0, \quad (5.244)$$

we obtain

$$M'(r) = \frac{1}{2\pi r} \int_{B_r(\mathbf{0})} \nabla^2 u dA = 0. \quad (5.245)$$

Thus $M(r)$ is constant in r . Taking $r \rightarrow 0$, continuity gives

$$\lim_{r \rightarrow 0} M(r) = u(\mathbf{0}). \quad (5.246)$$

Hence

$$M(r) = u(\mathbf{0}) \quad (5.247)$$

for every $0 < r < R$, which proves (5.237). $\square$

The same statement holds for any center $\mathbf{x}_0$ as long as the disk stays inside the domain:

$$u(\mathbf{x}_0) = \frac{1}{2\pi} \int_0^{2\pi} u(\mathbf{x}_0 + r(\cos \theta, \sin \theta)) d\theta. \quad (5.248)$$

The intuitive meaning is that a harmonic function cannot rise above its surroundings or sink below its surroundings at an interior point. Its value is always an average of nearby boundary values.

Theorem 5.2 (Weak Maximum Principle) *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain, and let*

$$u \in C^2(\Omega) \cap C^0(\overline{\Omega}) \quad (5.249)$$

satisfy

$$\nabla^2 u = 0 \quad \text{in } \Omega. \quad (5.250)$$

Then

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u, \quad (5.251)$$

and

$$\min_{\overline{\Omega}} u = \min_{\partial\Omega} u. \quad (5.252)$$

Proof. We first prove the maximum statement. For $\varepsilon > 0$, define

$$u_\varepsilon(\mathbf{x}) = u(\mathbf{x}) + \varepsilon|\mathbf{x}|^2. \quad (5.253)$$

Then

$$\nabla^2 u_\varepsilon = \nabla^2 u + \varepsilon \nabla^2 |\mathbf{x}|^2. \quad (5.254)$$

Since

$$\nabla^2 |\mathbf{x}|^2 = 2n, \quad (5.255)$$

and $\nabla^2 u = 0$, we obtain

$$\nabla^2 u_\varepsilon = 2n\varepsilon > 0. \quad (5.256)$$

Suppose that u_ε has a maximum at some interior point $\mathbf{x}_0 \in \Omega$. At an interior maximum, the Hessian matrix is negative semidefinite. Therefore,

$$\nabla^2 u_\varepsilon(\mathbf{x}_0) = \sum_{i=1}^n \frac{\partial^2 u_\varepsilon}{\partial x_i^2}(\mathbf{x}_0) \leq 0. \quad (5.257)$$

This contradicts $\nabla^2 u_\varepsilon = 2n\varepsilon > 0$. Hence u_ε cannot attain its maximum in the interior. Thus

$$\max_{\overline{\Omega}} u_\varepsilon = \max_{\partial\Omega} u_\varepsilon. \quad (5.258)$$

Let

$$R = \max_{\overline{\Omega}} |\mathbf{x}|^2. \quad (5.259)$$

For any $\mathbf{x} \in \overline{\Omega}$,

$$u(\mathbf{x}) \leq u_\varepsilon(\mathbf{x}) \leq \max_{\partial\Omega} u_\varepsilon \leq \max_{\partial\Omega} u + \varepsilon R. \quad (5.260)$$

Taking $\varepsilon \rightarrow 0$, we get

$$u(\mathbf{x}) \leq \max_{\partial\Omega} u. \quad (5.261)$$

Since this holds for every $\mathbf{x} \in \overline{\Omega}$,

$$\max_{\overline{\Omega}} u \leq \max_{\partial\Omega} u. \quad (5.262)$$

The opposite inequality is automatic because $\partial\Omega \subset \overline{\Omega}$. Therefore,

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u. \quad (5.263)$$

For the minimum statement, apply the maximum result to $-u$. Since $\nabla^2(-u) = 0$, we obtain

$$\min_{\overline{\Omega}} u = \min_{\partial\Omega} u. \quad (5.264)$$

□

The weak maximum principle says that a harmonic function cannot create a new maximum or minimum inside the region. The largest and smallest values must already be present on the boundary. This is why boundary data are physically decisive for Laplace's equation.

Theorem 5.3 (Strong Maximum Principle for Harmonic Functions) *Let $\Omega \subset \mathbb{R}^n$ be a connected domain, and let u be harmonic in Ω . If u attains a maximum or a minimum at an interior point of Ω , then u is constant in Ω .*

Proof. We prove the maximum case. The minimum case follows by applying the same argument to $-u$.

Assume that u attains its maximum value M at an interior point $\mathbf{x}_0 \in \Omega$:

$$u(\mathbf{x}_0) = M, \quad u(\mathbf{x}) \leq M \quad \text{for all } \mathbf{x} \in \Omega. \quad (5.265)$$

Choose a small ball $B_r(\mathbf{x}_0) \subset \Omega$. By the mean value property,

$$u(\mathbf{x}_0) = \frac{1}{|\partial B_r|} \int_{\partial B_r(\mathbf{x}_0)} u(\mathbf{x}) dS. \quad (5.266)$$

Since $u(\mathbf{x}_0) = M$, this becomes

$$M = \frac{1}{|\partial B_r|} \int_{\partial B_r(\mathbf{x}_0)} u(\mathbf{x}) dS. \quad (5.267)$$

But $u(\mathbf{x}) \leq M$ on the sphere. The only way the average of values no larger than M can equal M is that

$$u(\mathbf{x}) = M \quad (5.268)$$

for every $\mathbf{x} \in \partial B_r(\mathbf{x}_0)$. Since this holds for all sufficiently small r , we obtain

$$u(\mathbf{x}) = M \quad (5.269)$$

in a neighborhood of $\mathbf{x}_0$.

Now define

$$A = \{\mathbf{x} \in \Omega : u(\mathbf{x}) = M\}. \quad (5.270)$$

The argument above shows that A is open. Since u is continuous, A is also closed in Ω . It is nonempty because $\mathbf{x}_0 \in A$. Since Ω is connected, the only nonempty subset that is both open and closed in Ω is Ω itself. Therefore,

$$A = \Omega. \quad (5.271)$$

Hence

$$u(\mathbf{x}) = M \quad \text{for all } \mathbf{x} \in \Omega. \quad (5.272)$$

Thus u is constant. $\square$

The strong maximum principle is stronger than the weak one. The weak maximum principle says that the maximum is controlled by the boundary. The strong maximum principle says that if a nonconstant harmonic function reaches an extreme value inside the domain, then the function must actually be constant. Thus a nonconstant harmonic function has no isolated interior maximum or minimum.

Theorem 5.4 (Uniqueness Theorem for the Dirichlet Problem) *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain, and let*

$$u, v \in C^2(\Omega) \cap C^0(\overline{\Omega}) \quad (5.273)$$

solve the same Dirichlet problem

$$\nabla^2 u = 0, \quad \nabla^2 v = 0 \quad \text{in } \Omega, \quad (5.274)$$

with the same boundary condition $u = v$ on $\partial\Omega$. Then $u = v$ in Ω .

Proof. Let $w = u - v$. Since the Laplacian is linear,

$$\nabla^2 w = \nabla^2 u - \nabla^2 v = 0. \quad (5.275)$$

Thus w is harmonic in Ω . On the boundary,

$$w = u - v = 0. \quad (5.276)$$

By the weak maximum principle,

$$\max_{\overline{\Omega}} w = \max_{\partial\Omega} w = 0. \quad (5.277)$$

Hence

$$w(\mathbf{x}) \leq 0 \quad \text{for all } \mathbf{x} \in \overline{\Omega}. \quad (5.278)$$

Applying the weak maximum principle to $-w$, we also obtain

$$\max_{\overline{\Omega}} (-w) = \max_{\partial\Omega} (-w) = 0. \quad (5.279)$$

Thus $-w(\mathbf{x}) \leq 0$, so $w(\mathbf{x}) \geq 0$. Combining the two inequalities gives

$$w(\mathbf{x}) = 0 \quad \text{for all } \mathbf{x} \in \Omega. \quad (5.280)$$

Therefore, $u = v$ in Ω . □

This theorem is one of the most important results in boundary value problems. It says that for Laplace's equation, if a solution exists, then the boundary values determine it uniquely. The boundary condition is therefore not a small technical add-on. It selects the physical solution.

There is also an energy proof of the same uniqueness theorem when the boundary is sufficiently smooth. Let $w = u - v$. Then $w = 0$ on $\partial\Omega$ and $\nabla^2 w = 0$ in Ω . Using Green's first identity,

$$\int_{\Omega} |\nabla w|^2 dV = \int_{\partial\Omega} w \frac{\partial w}{\partial n} dS - \int_{\Omega} w \nabla^2 w dV. \quad (5.281)$$

The boundary term vanishes because $w = 0$ on $\partial\Omega$, and the volume term vanishes because $\nabla^2 w = 0$. Thus

$$\int_{\Omega} |\nabla w|^2 dV = 0. \quad (5.282)$$

Since $|\nabla w|^2 \geq 0$, this implies

$$\nabla w = \mathbf{0} \quad \text{in } \Omega. \quad (5.283)$$

Therefore, w is constant in each connected component of Ω . Because $w = 0$ on the boundary, this constant must be zero. Hence $u = v$. This gives another way to understand uniqueness. If two solutions have the same boundary values, their difference has zero Dirichlet energy, so it cannot contain any nontrivial variation inside the domain.

5.5.4.1 Electrostatics

In electrostatics, the electric field is related to the potential by

$$\mathbf{E} = -\nabla\Phi. \quad (5.284)$$

In a charge-free region, $\rho = 0$. Gauss's law gives

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} = 0. \quad (5.285)$$

Using $\mathbf{E} = -\nabla\Phi$, we obtain

$$\nabla^2\Phi = 0. \quad (5.286)$$

Thus the electrostatic potential is harmonic in every empty region.

The maximum principle says that Φ cannot have an isolated maximum or minimum inside a charge-free region unless Φ is constant. Therefore, the potential extrema are controlled by the boundary, such as conductors or prescribed potentials. This is closely related to Earnshaw's theorem. A charged particle cannot be held in a stable electrostatic equilibrium in an empty region using only static electric fields, because a stable equilibrium would require a local minimum of potential energy in the interior.

The uniqueness theorem gives an even stronger statement. Suppose two electrostatic potentials Φ_1 and Φ_2 satisfy

$$\nabla^2\Phi_1 = 0, \quad \nabla^2\Phi_2 = 0 \quad (5.287)$$

inside the same charge-free region, and suppose that they have the same potential on the boundary:

$$\Phi_1 = \Phi_2 \quad \text{on} \quad \partial\Omega. \quad (5.288)$$

Then the uniqueness theorem implies

$$\Phi_1 = \Phi_2 \quad \text{in} \quad \Omega. \quad (5.289)$$

Consequently,

$$\mathbf{E}_1 = -\nabla\Phi_1 = -\nabla\Phi_2 = \mathbf{E}_2. \quad (5.290)$$

So the electric field is also unique. This is why solving electrostatic boundary value problems is meaningful. Once the boundary potential is fixed, there is only one physically possible electrostatic potential inside.

5.5.4.2 Steady Heat Conduction

The same mathematics appears in steady heat conduction. If $T(\mathbf{x})$ is the temperature and there are no internal heat sources, then the steady heat equation is

$$\nabla^2 T = 0. \quad (5.291)$$

Therefore, T is harmonic. The maximum principle says that the hottest and coldest points of the body must occur on the boundary, unless the temperature is constant everywhere. There cannot be an isolated hot spot inside the material without an internal heat source.

The uniqueness theorem also says that if the boundary temperature is fixed, then the steady temperature distribution inside the material is unique. If two proposed temperature fields satisfy the same steady heat equation and the same boundary temperature, then their difference is harmonic and vanishes on the boundary. By the maximum

principle, the difference must vanish everywhere. Thus the two temperature fields are actually the same solution.

5.6 Poisson's Equation

Poisson's equation is

$$\boxed{\nabla^2 u = f}. \quad (5.292)$$

It differs from Laplace's equation by the source term f . In physics, this source term represents charge density, mass density, or heat generation. In Newtonian gravity, the gravitational field $\mathbf{g}$ is related to the gravitational potential Φ by

$$\mathbf{g} = -\nabla\Phi. \quad (5.293)$$

The gravitational Gauss law is

$$\nabla \cdot \mathbf{g} = -4\pi G\rho, \quad (5.294)$$

where ρ is the mass density. Substituting $\mathbf{g} = -\nabla\Phi$ gives

$$\nabla \cdot (-\nabla\Phi) = -4\pi G\rho. \quad (5.295)$$

Therefore,

$$\boxed{\nabla^2\Phi = 4\pi G\rho}. \quad (5.296)$$

This is Poisson's equation for Newtonian gravity (If you forget about it, see Section 2.5.2.2). For electrostatics, we similarly obtain

$$\boxed{\nabla^2\Phi = -\frac{\rho}{\epsilon_0}}. \quad (5.297)$$

The signs are different because electric charge can be positive or negative, while gravitational mass is conventionally positive and the gravitational force is attractive.

5.6.1 Gravitational Potential of a Uniform Sphere

Consider a sphere of radius R with constant mass density ρ_0 . The total mass is

$$M = \frac{4\pi R^3}{3}\rho_0. \quad (5.298)$$

By spherical symmetry, the potential depends only on r :

$$\Phi = \Phi(r). \quad (5.299)$$

The Laplacian of a radial function is

$$\nabla^2 \Phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right). \quad (5.300)$$

Inside the sphere, $0 < r < R$, Poisson's equation is

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 4\pi G \rho_0. \quad (5.301)$$

Multiplying by r^2 gives

$$\frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 4\pi G \rho_0 r^2. \quad (5.302)$$

Integrating once,

$$r^2 \frac{d\Phi}{dr} = \frac{4\pi G \rho_0}{3} r^3 + C_1. \quad (5.303)$$

To keep $d\Phi/dr$ finite at $r = 0$, we need $C_1 = 0$. Thus

$$\frac{d\Phi}{dr} = \frac{4\pi G \rho_0}{3} r. \quad (5.304)$$

Integrating again,

$$\Phi_{\text{in}}(r) = \frac{2\pi G \rho_0}{3} r^2 + C_2. \quad (5.305)$$

Outside the sphere, $r > R$, there is no mass density, so

$$\nabla^2 \Phi = 0. \quad (5.306)$$

The radial equation is

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 0. \quad (5.307)$$

Integrating twice gives

$$\Phi_{\text{out}}(r) = -\frac{GM}{r} + C_3. \quad (5.308)$$

If we choose $\Phi \rightarrow 0$ as $r \rightarrow \infty$, then $C_3 = 0$.

We determine C_2 by requiring continuity of the potential at $r = R$:

$$\Phi_{\text{in}}(R) = \Phi_{\text{out}}(R). \quad (5.309)$$

This gives

$$\frac{2\pi G \rho_0}{3} R^2 + C_2 = -\frac{GM}{R}. \quad (5.310)$$

Using

$$M = \frac{4\pi R^3}{3} \rho_0, \quad (5.311)$$

we have

$$\frac{2\pi G\rho_0}{3}R^2 = \frac{GM}{2R}. \quad (5.312)$$

Thus

$$C_2 = -\frac{3GM}{2R}. \quad (5.313)$$

The final result is

$$\Phi(r) = \begin{cases} -\frac{GM}{2R} \left(3 - \frac{r^2}{R^2} \right), & 0 \leq r \leq R, \\ -\frac{GM}{r}, & r \geq R. \end{cases} \quad (5.314)$$

The gravitational field is

$$\mathbf{g} = -\nabla\Phi. \quad (5.315)$$

Inside the sphere, the magnitude is proportional to r :

$$g(r) = \frac{GM}{R^3}r. \quad (5.316)$$

Outside the sphere, it is the inverse square law:

$$g(r) = \frac{GM}{r^2}. \quad (5.317)$$

This example shows how Poisson's equation reproduces Newtonian gravity.

5.6.2 Poisson's Equation and Green's Function in Electrostatics

Green's functions provide a systematic way to solve linear nonhomogeneous PDEs. For Poisson's equation, the main idea is very physical. We first find the response produced by a point source, and then we add up the responses from all sources by integration.

In electrostatics, Coulomb's law gives the electric field produced by a continuous charge distribution:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \rho(\mathbf{r}') \frac{\hat{\mathbf{r}}}{r^2} d\tau'. \quad (5.318)$$

Gauss's law says

$$\nabla \cdot \mathbf{E}(\mathbf{r}) = \frac{\rho(\mathbf{r})}{\epsilon_0}. \quad (5.319)$$

In electrostatics, the electric field is conservative:

$$\nabla \times \mathbf{E} = \mathbf{0}. \quad (5.320)$$

Therefore, we can introduce an electrostatic potential $V(\mathbf{r})$ such that

$$\mathbf{E} = -\nabla\Phi. \quad (5.321)$$

Substituting this into Gauss's law gives

$$\nabla \cdot (-\nabla \Phi) = \frac{\rho}{\epsilon_0}. \quad (5.322)$$

Since

$$\nabla \cdot (\nabla \Phi) = \nabla^2 \Phi, \quad (5.323)$$

we obtain

$$\nabla^2 \Phi(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon_0}. \quad (5.324)$$

This is Poisson's equation in electrostatics. If there is no charge density in the region, then $\rho = 0$, and Poisson's equation reduces to Laplace's equation:

$$\nabla^2 \Phi = 0. \quad (5.325)$$

Now we solve (5.324) by using a Green's function. More generally, suppose we want to solve

$$\mathcal{L}u(\mathbf{x}) = f(\mathbf{x}), \quad (5.326)$$

where $\mathcal{L}$ is a linear differential operator. A Green's function is the response of the operator to a point source. For the Laplacian in three dimensional free space, we choose $G(\mathbf{r}, \mathbf{r}')$ to satisfy

$$\nabla_{\mathbf{r}}^2 G(\mathbf{r}, \mathbf{r}') = -\delta^3(\mathbf{r} - \mathbf{r}'). \quad (5.327)$$

With this convention, if

$$\nabla^2 u(\mathbf{r}) = f(\mathbf{r}), \quad (5.328)$$

then the solution can be written as

$$u(\mathbf{r}) = - \int G(\mathbf{r}, \mathbf{r}') f(\mathbf{r}') d\tau'. \quad (5.329)$$

Indeed,

$$\begin{aligned} \nabla_{\mathbf{r}}^2 u(\mathbf{r}) &= - \int \nabla_{\mathbf{r}}^2 G(\mathbf{r}, \mathbf{r}') f(\mathbf{r}') d\tau' = - \int [-\delta^3(\mathbf{r} - \mathbf{r}')] f(\mathbf{r}') d\tau' \\ &= \int \delta^3(\mathbf{r} - \mathbf{r}') f(\mathbf{r}') d\tau' = f(\mathbf{r}). \end{aligned} \quad (5.330)$$

The central identity in three dimensions is

$$\nabla_{\mathbf{r}}^2 \left(\frac{1}{r} \right) = -4\pi \delta^3(\mathbf{r}). \quad (5.331)$$

Therefore,

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi r} \quad (5.332)$$

satisfies

$$\nabla_{\mathbf{r}}^2 G(\mathbf{r}, \mathbf{r}') = -\delta^3(\mathbf{r} - \mathbf{r}'). \quad (5.333)$$

Let us briefly justify the identity (5.331). For $\mathbf{r} \neq \mathbf{r}'$, we have $\mathbf{r} \neq 0$, and the function $1/\mathbf{r}$ satisfies

$$\nabla^2 \left(\frac{1}{\mathbf{r}} \right) = 0. \quad (5.334)$$

The singular behavior is concentrated only at $\mathbf{r} = \mathbf{r}'$. To determine its strength, integrate over a small ball B_a centered at $\mathbf{r}'$:

$$\int_{B_a} \nabla^2 \left(\frac{1}{\mathbf{r}} \right) d\tau = \int_{\partial B_a} \nabla \left(\frac{1}{\mathbf{r}} \right) \cdot d\mathbf{S}. \quad (5.335)$$

Since

$$\nabla \left(\frac{1}{\mathbf{r}} \right) = -\frac{\hat{\mathbf{r}}}{\mathbf{r}^2}, \quad (5.336)$$

and on the sphere ∂B_a we have $\mathbf{r} = a$ and $d\mathbf{S} = \hat{\mathbf{r}} a^2 d\Omega$, it follows that

$$\int_{\partial B_a} \nabla \left(\frac{1}{\mathbf{r}} \right) \cdot d\mathbf{S} = \int_{\partial B_a} \left(-\frac{\hat{\mathbf{r}}}{a^2} \right) \cdot \hat{\mathbf{r}} a^2 d\Omega = - \int_{\partial B_a} d\Omega = -4\pi. \quad (5.337)$$

Thus the distribution must be

$$\nabla^2 \left(\frac{1}{\mathbf{r}} \right) = -4\pi \delta^3(\mathbf{r}), \quad (5.338)$$

which proves (5.331). Now return to electrostatics. Poisson's equation is

$$\nabla^2 \Phi(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon_0}. \quad (5.339)$$

Comparing this with

$$\nabla^2 u = f, \quad (5.340)$$

we identify

$$f(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon_0}. \quad (5.341)$$

Using (5.329), we get

$$V(\mathbf{r}) = - \int G(\mathbf{r}, \mathbf{r}') \left[-\frac{\rho(\mathbf{r}')}{\epsilon_0} \right] d\tau' = \frac{1}{\epsilon_0} \int G(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d\tau'. \quad (5.342)$$

Substituting the free space Green's function (5.332), we finally obtain

$$\boxed{V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{\mathbf{r}} d\tau'}. \quad (5.343)$$

This is the familiar Coulomb potential produced by a continuous charge distribution.

The same method also works for Newtonian gravity. The gravitational potential satisfies

$$\nabla^2 \Phi = 4\pi G \rho. \quad (5.344)$$

Here

$$f(\mathbf{r}) = 4\pi G \rho(\mathbf{r}). \quad (5.345)$$

Thus

$$\Phi(\mathbf{r}) = - \int G(\mathbf{r}, \mathbf{r}') 4\pi G \rho(\mathbf{r}') d\tau'. \quad (5.346)$$

Using

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi r}, \quad (5.347)$$

we obtain

$$\Phi(\mathbf{r}) = -G \int \frac{\rho(\mathbf{r}')}{r} d\tau'. \quad (5.348)$$

This is exactly the Newtonian gravitational potential of a continuous mass distribution.

The important point is that Green's function is the inverse of a differential operator in the same way that a matrix inverse solves a linear algebra problem. For a matrix equation

$$A\mathbf{x} = \mathbf{b}, \quad (5.349)$$

we formally write

$$\mathbf{x} = A^{-1}\mathbf{b}. \quad (5.350)$$

For a PDE such as

$$\nabla^2 u = f, \quad (5.351)$$

the Green's function plays the role of the inverse Laplacian. It converts a source distribution into the corresponding potential field.

Chapter 6

Introduction to Linear Algebra

Abstract In linear algebra, the main goal is to analyze the structure of matrices, vector spaces in n dimensions, and linear equations. Among these, matrices are the most important concept throughout the entire subject. To better understand matrices, I want to start with the basic formula for the inverse matrix. I will use only algebra and rigorous conditions to make the proof clearer than what is typically found in textbooks on physical mathematics. By doing so, we can explore a different version of the underlying structure. To keep the content concise, **I will skip most of the concepts and interpretations already covered in high school and instead focus on the higher-level mathematical structures.**

6.1 Linear Map

Before studying matrices, it is useful to first understand what a matrix is trying to represent. A matrix is not only an array of numbers. In finite dimensional linear algebra, a matrix is the coordinate representation of a linear map. Therefore, the more fundamental object is the map itself.

A physical law often has the form of a map. It takes one state as input and produces another state as output. For example, a time evolution law sends an initial state to a later state, while a force law sends a configuration to a force. If the same input could lead to two different outputs under the same rule, then the rule would not define a function. In physics, this would mean that prediction is not unique.

6.1.1 Mappings Between Vector Spaces

Let V and W be two vector spaces over the same field $\mathbb{F}$. A map

$$T : V \rightarrow W \tag{6.1}$$

assigns to each vector $\mathbf{v} \in V$ exactly one vector $T(\mathbf{v}) \in W$. The word exactly is important. A map must be well defined. If one input vector gives more than one output vector, then the assignment is not a function.

Definition 6.1 (Map Between Vector Spaces) Let V and W be vector spaces over the same field $\mathbb{F}$. A map from V to W is a rule

$$T : V \rightarrow W \quad (6.2)$$

such that for every $\mathbf{v} \in V$, there exists a unique element $T(\mathbf{v}) \in W$ assigned to $\mathbf{v}$.

The following diagram shows the basic idea of a map. Vectors in the domain V are sent to vectors in the codomain W .

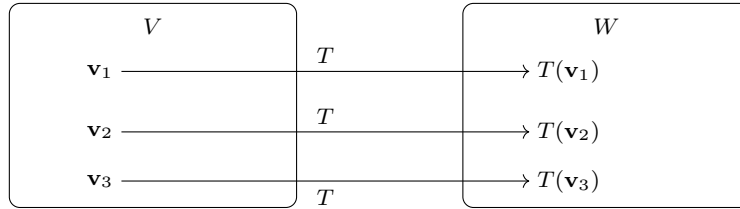


Fig. 6.1 A map $T : V \rightarrow W$ sends each vector in V to a unique vector in W .

This idea is already present in elementary mechanics. If the state of a particle at time $t = 0$ is denoted by $\mathbf{x}(0)$, then a deterministic equation of motion should define a map

$$\mathbf{x}(0) \mapsto \mathbf{x}(t). \quad (6.3)$$

If the same initial state could evolve into two different states at the same time t , then the time evolution would fail to be a function. This would destroy deterministic prediction.

6.1.2 Definition of Linearity

A general map may bend, fold, or distort vector space in many ways. A linear map is a very special map. It respects the two basic operations of a vector space, namely vector addition and scalar multiplication.

Definition 6.2 (Linear Map) Let V and W be vector spaces over the same field $\mathbb{F}$. A map $T : V \rightarrow W$ is called a linear map if for all $\mathbf{u}, \mathbf{v} \in V$ and for all $a, b \in \mathbb{F}$,

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v}). \quad (6.4)$$

Equivalently, T is linear if it satisfies

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(c\mathbf{u}) = cT(\mathbf{u}) \quad (6.5)$$

for all $\mathbf{u}, \mathbf{v} \in V$ and $c \in \mathbb{F}$.

The first condition says that the map preserves superposition. The second condition says that the map preserves scaling. Combining them gives the more compact formula

$$T\left(\sum_{i=1}^n c_i \mathbf{v}_i\right) = \sum_{i=1}^n c_i T(\mathbf{v}_i). \quad (6.6)$$

This equation is the main reason why linear maps are powerful. Once we know how a linear map acts on a basis, we know how it acts on every vector.

There are also some immediate consequences. Since

$$T(\mathbf{0}) = T(0\mathbf{v}) = 0T(\mathbf{v}) = \mathbf{0}, \quad (6.7)$$

every linear map sends the zero vector to the zero vector. Moreover,

$$T(-\mathbf{v}) = -T(\mathbf{v}). \quad (6.8)$$

Therefore, a map that does not send the origin to the origin cannot be linear.

A simple mechanical example is Hooke's law for a multidimensional spring system. Let $\mathbf{x} \in \mathbb{R}^n$ be the displacement from equilibrium, and let $K \in \mathbb{R}^{n \times n}$ be the stiffness matrix. The restoring force is

$$\mathbf{F}(\mathbf{x}) = -K\mathbf{x}. \quad (6.9)$$

The map $\mathbf{x} \mapsto \mathbf{F}(\mathbf{x})$ is linear. If two displacements are superposed, the forces are also superposed. This is exactly the small oscillation approximation used in mechanics.

6.1.3 Linear Maps and Matrices

In finite dimensional vector spaces, every linear map can be represented by a matrix after bases are chosen. The matrix is not the map itself. It is the coordinate description of the map with respect to chosen bases.

Let V be an n dimensional vector space with basis

$$\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}, \quad (6.10)$$

and let W be an m dimensional vector space with basis

$$\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}. \quad (6.11)$$

For a linear map $T : V \rightarrow W$, each basis vector $\mathbf{v}_i$ is mapped into W , so it can be expanded in the basis $\mathcal{C}$ as

$$T(\mathbf{v}_i) = \sum_{j=1}^m a_{ji} \mathbf{w}_j. \quad (6.12)$$

The coefficients a_{ji} form an $m \times n$ matrix

$$[T]_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}. \quad (6.13)$$

The i th column of this matrix is the coordinate vector of $T(\mathbf{v}_i)$ in the basis $\mathcal{C}$.

If

$$\mathbf{v} = \sum_{i=1}^n x_i \mathbf{v}_i, \quad (6.14)$$

then by linearity,

$$T(\mathbf{v}) = \sum_{i=1}^n x_i T(\mathbf{v}_i). \quad (6.15)$$

Therefore the coordinate vectors satisfy

$$[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{v}]_{\mathcal{B}}. \quad (6.16)$$

In the standard bases of $\mathbb{R}^n$ and $\mathbb{R}^m$, this becomes the familiar formula

$$T(\mathbf{x}) = A\mathbf{x}, \quad (6.17)$$

where

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}. \quad (6.18)$$

Thus the columns of A are the images of the standard basis vectors.

This matrix point of view is extremely useful in electromagnetism. For a linear, homogeneous, anisotropic medium, the displacement field and electric field are related by

$$\mathbf{D} = \epsilon \mathbf{E}, \quad (6.19)$$

where ϵ is a matrix called the permittivity tensor. This is a linear map from the electric field vector to the displacement field vector. If the medium is isotropic, then ϵ becomes a scalar multiple of the identity matrix I_n ,

$$\mathbf{D} = \epsilon I_n \mathbf{E} = \epsilon \mathbf{E}. \quad (6.20)$$

If the medium is anisotropic, the matrix can mix different components of the field. This is why a field pointing in one direction can produce a displacement field with components in other directions.

6.1.4 Physical Meaning of Linearity and Uniqueness

The language of linear maps is not only notation. It tells us what kind of physical structure is being preserved. Linearity preserves superposition. Injectivity preserves distinguishability of states. Surjectivity tells us whether the chosen target space is fully reached. Bijectivity gives a reversible correspondence between input and output.

In electrostatics, Gauss's law gives

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (6.21)$$

One may view the divergence operator as a linear map

$$\mathbf{E} \mapsto \nabla \cdot \mathbf{E}. \quad (6.22)$$

This map is not injective because adding a divergence-free field does not change the charge density. If

$$\nabla \cdot \mathbf{E}_0 = 0, \quad (6.23)$$

then

$$\nabla \cdot (\mathbf{E} + \mathbf{E}_0) = \nabla \cdot \mathbf{E}. \quad (6.24)$$

Therefore, the same charge density does not determine the whole electric field unless more physical conditions are imposed, such as boundary conditions and the electrostatic condition $\nabla \times \mathbf{E} = \mathbf{0}$. Without this extra data, the solution is not unique, and the physical field is not completely determined.

Thermal physics also gives a natural linear map. For heat conduction with constant thermal diffusivity κ , the temperature field satisfies

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T. \quad (6.25)$$

The solution operator sends the initial temperature profile to the later temperature profile,

$$T(\mathbf{x}, 0) \mapsto T(\mathbf{x}, t). \quad (6.26)$$

Formally, this can be written as

$$T(t) = e^{t\kappa\nabla^2} T(0). \quad (6.27)$$

This operator is linear. However, heat diffusion damps short wavelength information. In Fourier modes,

$$e^{i\mathbf{k}\cdot\mathbf{x}} \mapsto e^{-\kappa|\mathbf{k}|^2 t} e^{i\mathbf{k}\cdot\mathbf{x}}. \quad (6.28)$$

Large $|\mathbf{k}|$ modes decay very fast. This explains why the backward heat problem is unstable. Small errors at a later time may correspond to huge errors at an earlier time. Physically, diffusion destroys detailed thermal information in a practical sense.

In quantum mechanics, the state vector $\psi(t)$ of a closed system evolves by the Schrodinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi. \quad (6.29)$$

When H is time independent, the solution is

$$\psi(t) = U(t)\psi(0), \quad U(t) = e^{-iHt/\hbar}. \quad (6.30)$$

The map $\psi(0) \mapsto \psi(t)$ is linear. If H is Hermitian, then $U(t)$ is unitary, so

$$U(t)^*U(t) = I. \quad (6.31)$$

Therefore the evolution of a closed quantum system is bijective. The inverse map exists and is

$$U(t)^{-1} = U(t)^*. \quad (6.32)$$

If one initial quantum state could produce two different final states, then time evolution would not be a function, and determinism would fail. If two different initial states were mapped to the same final state, then injectivity would fail, and reversibility together with quantum information preservation would be lost. This is why linearity and unitarity are not only mathematical conveniences. They express basic physical requirements of closed quantum evolution.

6.2 Basic Rules for Matrices

6.2.1 Matrix Multiplication

An $m \times n$ **matrix** is a rectangular array consisting of m rows and n columns. The entries in a matrix can be numbers, symbols, or even functions.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n} \quad (6.33)$$

Let $A \in \mathbb{R}^{m \times n}$ be an m -by- n matrix. The ordered tuple $(a_{11}, a_{12}, \dots, a_{1n})$ is called the first row of the matrix, while $(a_{1j}, a_{2j}, \dots, a_{mj})$ is called the j -th column.

If we want to refer to the entry in the i -th row and j -th column, we write:

$$[A]_{ij} = a_{ij} \quad (6.34)$$

Moreover, there is a special matrix called the **transpose** of A , denoted A^T , which switches the roles of rows and columns. The transpose of A is defined as:

$$A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ \vdots & \vdots & \cdots & \vdots \\ a_{1j} & a_{2j} & \cdots & a_{mj} \\ \vdots & \vdots & \cdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}_{n \times m} \quad (6.35)$$

This definition implies the identity:

$$[A^T]_{ij} = [A]_{ji} = a_{ji} \quad (6.36)$$

So the matrix $A^T \in \mathbb{R}^{n \times m}$ is an n -by- m matrix.

In linear algebra, we consider a vector $\mathbf{v}$ to be a column matrix¹, such as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad (6.37)$$

In the context of matrices, the definition of the inner product becomes

$$\mathbf{v} \cdot \mathbf{u} = \sum_i v_i u_i \Rightarrow \mathbf{v}^T \mathbf{u} = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad (6.38)$$

Suppose that we have vectors $\mathbf{a}_1^T = (a_{11} \ a_{12} \ \cdots \ a_{1m})$, $\cdots$, $\mathbf{a}_n^T = (a_{n1} \ a_{n2} \ \cdots \ a_{nm}) \in \mathbb{R}^{1 \times m}$, then we define a n -by- m matrix A as

$$A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix}_{n \times m} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \quad (6.39)$$

Additionally, we have vectors $\mathbf{b}_1 = (b_{11} \ b_{21} \ \cdots \ b_{m1})^T$, $\cdots$, $\mathbf{b}_n = (b_{1n} \ b_{2n} \ \cdots \ b_{mn})^T \in \mathbb{R}^{n \times 1}$ then we define a m -by- n matrix B as

¹ You can also consider it to be a row matrix as long as you have a proper definition.

$$B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_n]_{m \times n} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & a_{nm} \end{bmatrix} \quad (6.40)$$

Now that we have defined these two matrices, we can define their product using the dot product formulation. That is

$$AB = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_n \end{bmatrix} [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_n] := \begin{bmatrix} (\mathbf{a}_1 \cdot \mathbf{b}_1) & \cdots & (\mathbf{a}_1 \cdot \mathbf{b}_n) \\ (\mathbf{a}_2 \cdot \mathbf{b}_1) & \cdots & (\mathbf{a}_2 \cdot \mathbf{b}_n) \\ \vdots & \ddots & \vdots \\ (\mathbf{a}_n \cdot \mathbf{b}_1) & \cdots & (\mathbf{a}_n \cdot \mathbf{b}_n) \end{bmatrix}_{n \times n} \quad (6.41)$$

By the property of the dot product, we obtain

$$\begin{bmatrix} (\mathbf{a}_1 \cdot \mathbf{b}_1) & \cdots & (\mathbf{a}_1 \cdot \mathbf{b}_n) \\ (\mathbf{a}_2 \cdot \mathbf{b}_1) & \cdots & (\mathbf{a}_2 \cdot \mathbf{b}_n) \\ \vdots & \ddots & \vdots \\ (\mathbf{a}_n \cdot \mathbf{b}_1) & \cdots & (\mathbf{a}_n \cdot \mathbf{b}_n) \end{bmatrix}_{n \times n} = \begin{bmatrix} \sum_k a_{1k} b_{k1} & \cdots & \sum_k a_{1k} b_{kn} \\ \sum_k a_{2k} b_{k1} & \cdots & \sum_k a_{2k} b_{kn} \\ \vdots & \ddots & \vdots \\ \sum_k a_{nk} b_{k1} & \cdots & \sum_k a_{nk} b_{kn} \end{bmatrix}_{n \times n} \quad (6.42)$$

That is, the matrix multiplication can be rewritten as

$$\boxed{[AB]_{ij} = \sum_k a_{ik} b_{kj}} \quad (6.43)$$

We need to ensure that the number of columns in the first matrix matches the number of rows in the second matrix, so that each element can be properly paired during multiplication. So, $A \in \mathbb{R}^{n \times k}$, $B \in \mathbb{R}^{k \times m} \Rightarrow AB \in \mathbb{R}^{n \times m}$.

6.2.2 Linear Transformation

We have explored operations such as vector addition and scalar multiplication. These operations are not limited to pure mathematics; they are, at their core, a language of transformation. In linear algebra, we naturally ask: "Is there a rule that systematically sends each vector to another, while preserving the underlying structure of the vector

space?” This is the essence of a **linear transformation**.

In detail, let V, W be vector spaces defined over the same field $\mathbb{F}$, a mapping $T : V \rightarrow W$ is called a linear transformation. In other words, we use the mapping rule T to send elements of V into elements of W . Conceptually, this is like a high school student transforming into a university student after taking an entrance exam, still a student (element of a vector space), but now in a different context (space). For all vectors $\mathbf{u}, \mathbf{v} \in V$ and the scalar $c \in \mathbb{F}$, we have

$$\begin{cases} T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \\ T(c\mathbf{u}) = cT(\mathbf{u}) \end{cases}$$

Moreover, every linear transformation between finite-dimensional vector spaces is equivalent to matrix multiplication. That is, for any linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, there exists a unique matrix $A \in \mathbb{R}^{m \times n}$ that completely describes its behavior, such that for all $\mathbf{x} \in \mathbb{R}^n$,

$$T(\mathbf{x}) = A\mathbf{x} \quad (6.44)$$

In n dimensions $\mathbb{R}^n$, a vector $\mathbf{x} \in \mathbb{R}^n$ can be written as

$$\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n = \sum_i x_i\mathbf{e}_i \quad (6.45)$$

By (6.2.2), the linear transformation T satisfies

$$T(\mathbf{x}) = T\left(\sum_i x_i\mathbf{e}_i\right) = \sum_i x_i T(\mathbf{e}_i) \quad (6.46)$$

For each standard basis vector $\mathbf{e}_i$ in the n -dimensional space, the linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ maps it to $T(\mathbf{e}_i)$. We define $\mathbf{a}_i := T(\mathbf{e}_i) \in \mathbb{R}^m$, which represents the image of the i -th basis vector under the transformation. Then, we can construct a matrix $A = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n) \in \mathbb{R}^{m \times n}$ whose columns are precisely the images of the standard basis vectors. Then, for any $\mathbf{x} = (x_1 \ \cdots \ x_n)^T$, we have

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{a}_1 x_1 + \cdots + \mathbf{a}_n x_n = \sum_i x_i \mathbf{a}_i = T(\mathbf{x}) \quad (6.47)$$

In other words, any linear transformation can be written as the action of a matrix A , and this matrix A is unique since the standard basis spans the space $\mathbb{R}^n$, the action of T on $\{\mathbf{e}_i\}$ uniquely determines the transformation.

6.2.3 Cofactor of a Matrix

There is a given $m \times n$ matrix A

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & \cdots & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & \cdots & \vdots \end{bmatrix}_{m \times n}$$

We define a determinant M_{ij} as the determinant of the matrix obtained by removing the i -th row and the j -th column from the original matrix A . This M_{ij} is called the **minor** of A .

Furthermore, we define a new matrix C , constructed from the minors M_{ij} by multiplying each element with $(-1)^{i+j}$, where i and j are the row and column indices. That is, $C_{ij} = (-1)^{i+j} M_{ij}$. This matrix C is called the *cofactor matrix* of A . So, the cofactor is defined as

$$\text{Cofactor} := (-1)^{i+j} M_{ij} = C_{ij} \quad (6.48)$$

To calculate the determinant of a $n \times n$ matrix A , we follow the **cofactor expansion method**, also known as **Laplace expansion**.

6.3 Definition of Determinants

6.3.1 Permutations and Sign of a Permutation

To simplify the representation of permutations in N -dimensions, we introduce a new notation: the generalized permutation $\sigma(i)$.

Given a number set $(1, 2, \dots, n)$, there are $n!$ possible permutations, each corresponding to a different σ , for example

$$\begin{aligned} \sigma_1 &= (1, 2, \dots, n) \\ \sigma_2 &= (1, n, \dots, 2) \\ &\vdots \\ \sigma_n &= (n, n-1, \dots, 3) \end{aligned}$$

This symbol can be interpreted as a "method of assigning seats" or a specific arrangement of elements. Every different σ means a different arrangement. This notation can also be written as a function $\sigma(i)$, which means that we need to find out an arbitrary element in the i -th row. For instance, there is a permutation $\sigma_i = (3, 2, 1)$, the output values of the function are

$$\sigma(1) = 3$$

$$\sigma(2) = 1$$

$$\sigma(3) = 2$$

In this interpretation, the input variable i can be thought of as the "seat number in the classroom," and the output $\sigma(i)$ indicates which "student" is assigned to that seat, that is, which number from the original set is placed in position i .

Like any other function, a permutation $\sigma(i)$ has an inverse function, denoted by $\sigma^{-1}(i)$. The inverse permutation allows us to find the "seat number" given a specific "student."

For example, if the permutation is $\sigma = (3, 2, 1)$, then its inverse permutation is:

$$\sigma^{-1}(1) = 3$$

$$\sigma^{-1}(2) = 2$$

$$\sigma^{-1}(3) = 1$$

So, we can readily verify the following identity:

$$\sigma(\sigma^{-1}(i)) = i \quad (6.49)$$

We can define another two permutation notations τ and π , the relationship between σ , τ and π is

$$\pi(i) = \sigma \circ \tau = \sigma(\tau(i)) \Leftrightarrow \sigma(i) = \pi \circ \tau^{-1} = \pi(\tau^{-1}(i)) \quad (6.50)$$

In this chain of permutations, τ represents the first "seat arrangement," while σ represents the second arrangement based on the first one. That is, $\sigma \circ \tau$ means we first select elements using the permutation τ , and then apply σ to the result. The final permutation π is thus the total effect of both steps. In other words, we can deduce the second permutation σ from the initial arrangement τ and the resulting permutation. This relationship is quite natural and evident. This property will be widely used in linear algebra.

The final part of the permutation is to define a new function $\text{sgn}(\sigma)$ to substitute the Levi-Civita symbol because it can clearly represent the permutations, which is $+1$ or -1 , without writing a lot of indices. The function $\text{sgn}(\sigma)$ is defined to be

$$\text{sgn}(\sigma) = \begin{cases} +1, & \text{When } \sigma \text{ is an even permutation} \\ -1, & \text{When } \sigma \text{ is an odd permutation} \end{cases} \quad (6.51)$$

Where $\sigma \in S_n$. Here, S_n denotes the symmetric group of degree n , which contains all possible $n!$ permutations of the set $\{1, 2, \dots, n\}$.

In addition, there is a beautiful property:

$$\text{sgn}(\sigma_1 \circ \sigma_2) = \text{sgn}(\sigma_1) \cdot \text{sgn}(\sigma_2) \quad (6.52)$$

This is because the total parity of the composed permutation $\sigma_1 \circ \sigma_2$ depends on the parities of the two individual permutations. If one is odd and the other is even, their composition is odd; if both are odd or both are even, the composition is even. Therefore, the sign of the composition can be expressed as the product of their individual signs.

By the symmetry of the sign function, the parity of a permutation σ is the same as that of its inverse σ^{-1} . After all, both the arrangement of seats and the assignment of students remain unchanged in terms of parity. This property can be expressed as

$$\text{sgn}(\sigma) = \text{sgn}(\sigma^{-1}) \quad (6.53)$$

6.3.2 Determinant of Order N

We select any row(i -th) of the matrix and use the corresponding cofactors from each column (j -th). That is,

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = a_{i1}(-1)^{i+1}M_{i1} + \cdots + a_{in}(-1)^{i+n}M_{in} \quad (6.54)$$

So the determinant of A based on the j -th row can be written as

$$\det(A) = \sum_j a_{ij}(-1)^{i+j}M_{ij} = \sum_j a_{ij}C_{ij} \quad (6.55)$$

For example, a given 3×3 matrix shown below,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow \det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

However, this method has a limitation: when the matrix has a lot of elements, we can't just use the cofactor $(-1)^{i+j}C_{ij}$ to describe the structure of the minor determinant. Therefore, we must introduce the more general form of the determinant using the sign

of a permutation, which is

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_i a_{i, \sigma(i)} \quad (6.56)$$

This way, all possible permutations and the ± 1 cofactor are included in the $\text{sgn}(\sigma)$. That's why it was important to introduce these notations earlier. Furthermore, the transpose of A merely switches the order of rows and columns, which has no effect on the value of the determinant. So,

$$\det(A) = \det(A^T) \quad (6.57)$$

6.3.3 Determinant Basic Formula

After such a long introduction, we can finally try to deal with one of the most important formulas in linear algebra, the **multiplicative property of determinants**.

There are two different given $n \times n$ matrices $[A]_{ij} = a_{ij}$ and $[B]_{ij} = b_{ij}$; both of them have the non-zero determinant. We define a new matrix λ to be

$$\lambda = AB \Rightarrow \lambda_{ij} = \sum_k a_{ik} b_{kj} \quad (6.58)$$

So, we can substitute (6.58) into the determinant of λ , which is

$$\det(\lambda) = \det(AB) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_i \lambda_{i, \sigma(i)} = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_i \left(\sum_k a_{i, k} b_{k, \sigma(i)} \right) \quad (6.59)$$

We redefine the index k to be $\tau(i)$, which means we select the $\tau(i)$ -th column in the k -th row, so the product of the summation can be rewritten as

$$\prod_i \left(\sum_k a_{i, k} b_{k, \sigma(i)} \right) = \sum_{\tau \in S_n} \left(\prod_i a_{i, \tau(i)} b_{\tau(i), \sigma(i)} \right) \quad (6.60)$$

This is because we need to consider all possible selections of $\tau(i)$. Multiplying everything together after summing over all possibilities is equivalent to listing every possible product first and then summing over all possible arrangements of $\tau(i)$. This way, we account for all possible cases thoroughly. Due to space limitations, I won't list them all here. However, if you have time, I encourage you to try listing them out and verifying the result on your own.

After that, we redefine $\sigma(i)$ is a double permutation

$$\sigma(i) = \pi \circ \tau = \pi(\tau(i)) \quad (6.61)$$

as shown in (6.50) and (6.52). So (6.59) becomes

$$\begin{aligned}
 \det(\lambda) = \det(AB) &= \sum_{\pi(\tau(i)) \in S_n} \operatorname{sgn}(\sigma) \sum_{\tau \in S_n} \left(\prod_i a_{i, \tau(i)} b_{\tau(i), \pi(\tau(i))} \right) \\
 &= \sum_{\pi \in S_n} \operatorname{sgn}(\pi \circ \tau) \sum_{\tau \in S_n} \left(\prod_i a_{i, \tau(i)} b_{\tau(i), \pi(\tau(i))} \right) \quad (6.62) \\
 &= \sum_{\pi, \tau \in S_n} \operatorname{sgn}(\pi) \operatorname{sgn}(\tau) \left(\prod_i a_{i, \tau(i)} b_{\tau(i), \pi(\tau(i))} \right)
 \end{aligned}$$

Note that the product $\prod_i b_{\tau(i), \pi(\tau(i))}$ is equivalent to $\prod_i b_{i, \sigma(i)}$ because the permutations τ, σ, π are one-to-one (Just the rearrangement). So, we can rewrite the product of the fourth equality of (6.62) into

$$\prod_i a_{i, \tau(i)} b_{\tau(i), \pi(\tau(i))} = \left(\prod_i a_{i, \tau(i)} \right) \left(\prod_i b_{\tau(i), \pi(\tau(i))} \right) \quad (6.63)$$

Substituting (6.63) into (6.62), we have

$$\begin{aligned}
 \det(AB) &= \sum_{\pi, \tau \in S_n} \operatorname{sgn}(\pi) \operatorname{sgn}(\tau) \left(\prod_i a_{i, \tau(i)} b_{\tau(i), \pi(\tau(i))} \right) \\
 &= \sum_{\pi, \tau \in S_n} \operatorname{sgn}(\pi) \operatorname{sgn}(\tau) \left(\prod_i a_{i, \tau(i)} \right) \left(\prod_i b_{\tau(i), \pi(\tau(i))} \right) = \det(A) \det(B) \quad (6.64)
 \end{aligned}$$

This term is exactly the expansion of $\det(A)$, with $\tau \in S_n$ and the product over $a_{i, \tau(i)}$, weighted by $\operatorname{sgn}(\tau)$. The same logic applies to $\det(B)$. We can also verify that the total terms of $\det(AB)$ are that

$$\text{Number of total terms} = \underbrace{n!}_{\tau(i) \text{ permutations}} \times \underbrace{n!}_{\pi(i) \text{ permutations}}$$

Note that we have already applied the permutation τ to the column indices of a , which means the order $\tau(i)$ has been fixed when passed into the row indices of b to match the column permutation of b . Therefore, the degree of freedom for τ in the indices of b has already been constrained.

That's the beautiful proof of the classical formula in linear algebra, which can be discussed by analyzing its algebraic structure. The final conclusion, the determinant of the product of two matrices equals the product of the determinants of each matrix, that is

$$\boxed{\det(AB) = \det(A) \det(B)} \quad (6.65)$$

We can use the same method to verify the reverse product $\det(BA) = \det(A) \det(B)$, you can try it by yourself.

6.4 Inverse Matrix

6.4.1 Meaning of the Identity Matrix

We have to find a matrix that multiplies another matrix A , the result is still A . We have an obvious equation that the matrix A satisfies

$$\sum_j \delta_{ij} A_{jk} = A_{ik} \quad ; \quad \sum_j A_{kj} \delta_{ji} = A_{ki} \quad (6.66)$$

Where A_{ik} , A_{ki} are the elements of A . Note that the rules of matrix multiplication A , B is

$$[AB]_{ik} = \sum_j A_{ij} B_{jk} \quad (6.67)$$

So, by (6.66), we define the identity matrix by $[I_n]_{ij} = \delta_{ij}$. And the equation

$$\det(AB) = \det(A) \det(B) \quad (6.68)$$

(6.68) implies that $\det(I_n A) = \det(I_n) \det(A) = \det(A)$, so $\det(I_n) = 1$. If this feels abstract now, don't worry; this Kronecker delta definition will naturally show up again when we derive the inverse matrix formula using cofactors.

6.4.2 Formula for the Inverse Matrix

After discussing the meaning and the definition of the identity matrix, we may be back to the original problem. We need to find a $[A^{-1}]_{jk} = \xi_{jk}$ satisfying

$$[AA^{-1}]_{ik} = \sum_j a_{ij} \xi_{jk} = \delta_{ik} \quad (6.69)$$

Where a_{ij} is the element of A , and ξ_{jk} is the element of A^{-1} . By the definition of the cofactor $(-1)^{i+j} M_{ij}$, the determinant of A can be denoted by

$$\det(A) = \sum_j (-1)^{k+j} a_{kj} M_{kj} \quad (6.70)$$

Where k is an arbitrary row of A . Note that this is not the matrix multiplication because of the order of the indices. If we change the index of a , it becomes

$$\sum_j (-1)^{k+j} a_{ij} M_{kj} = \det(A') \quad (6.71)$$

$$A' = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & \cdots & \vdots \\ a_{i1} & a_{i2} & \ddots & a_{in} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix} \quad (6.72)$$

By the properties of determinants, if a matrix has two identical rows or columns, then its determinant is zero. So (6.70) becomes

$$\det(A') = \sum_j (-1)^{k+j} a_{ij} M_{kj} = 0 \quad (6.73)$$

Combining (6.69) and (6.72), equation (6.70) can be rewritten as

$$\sum_j (-1)^{k+j} a_{ij} M_{kj} = \delta_{ik} \det(A) \quad (6.74)$$

Dividing both sides by $\det(A)$, (6.74) will become

$$\sum_j a_{ij} \left[\frac{1}{\det(A)} (-1)^{k+j} M_{kj} \right] := \sum_j a_{ij} \xi_{jk} = \delta_{ik} = [I_n]_{ik} \quad (6.75)$$

So the matrix $\xi_{jk} = 1/\det(A) \times (-1)^{k+j} M_{kj}$. In this equation, the indices between the two matrices are reversed, indicating that a transposition has been applied. We've known that $[A^T]_{ij} = [A]_{ji}$, so

$$\xi_{jk} = (-1)^{k+j} M_{kj} = [(-1)^{j+k} M_{jk}]^T \quad (6.76)$$

Note that the transposition doesn't affect the plus and minus signs of the cofactors since $(-1)^{k+j} = (-1)^{j+k}$. Substituting (6.76) into (6.75),

$$\sum_j a_{ij} [(-1)^{j+k} M_{jk}]^T = [A\xi]_{ik} = [I_n]_{ik} \quad (6.77)$$

And then we can get the final solution to the proof and verify that

$$\xi_{ij} = [A^{-1}]_{ij} = \frac{1}{\det(A)}(-1)^{i+j}M_{ij}^T = \frac{1}{\det(A)}C_{ij}^T \quad (6.78)$$

The proof for $A^{-1}A$ follows in the same way and will not be repeated here. This can be also denoted by

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A). \quad (6.79)$$

Example 6.1 Given the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

Find A^{-1}

Solution 6.1

We need to find the cofactors of each place. That

$$1^{st} \text{ row : } \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1; \quad - \begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix} = -2; \quad \begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix} = 0$$

$$2^{nd} \text{ row : } - \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} = 1; \quad \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} = 0; \quad - \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = -1$$

$$3^{rd} \text{ row : } \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1; \quad - \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1; \quad \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1$$

So the matrix C consisting of these cofactors is

$$C = \begin{bmatrix} 1 & -2 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix}$$

The determinant of the original matrix A is

$$\det(A) = 1 \times 1 - 2 \times 0 + 1 \times 0 = 1$$

So the final answer is

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 1 & -2 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 1 & -1 \\ -2 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

6.5 Linear Dependence and Independence

6.5.1 Linear Relations in a System of Equations

After introducing the fundamental formulas and operations, we now enter one of the core concepts in linear algebra: **linear dependence and independence**.

Consider a set of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \in \mathbb{R}^n$. If there exists a set of scalars $\eta_1, \eta_2, \dots, \eta_n$, not all zero, such that

$$\sum_i \eta_i \mathbf{x}_i = \mathbf{0} \quad (6.80)$$

then the vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ are said to be **linearly dependent**. The meaning of linear dependence is that a nontrivial linear combination exists that produces the zero vector. Geometrically, this implies that these vectors can only span a lower-dimensional subspace. In other words, although each $\mathbf{x}_i \in \mathbb{R}^n$, if they are linearly dependent, they do not span the entire $\mathbb{R}^n$, but rather some $\mathbb{R}^k$, where $k < n$.

We now examine the structure of a linear system represented by $A\mathbf{x} = \mathbf{b}$, where $A \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{b} \in \mathbb{R}^m$, that is

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \Rightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

where $\mathbf{a}_i = (a_{i1}, \dots, a_{in})$. We now focus on the determinant of the matrix A alone, we've gotten

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{vmatrix} = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_i a_{i, \sigma(i)}$$

If the vectors that form the matrix A satisfy the condition of linear dependence defined in (6.80), we can rewrite it as:

$$\sum_{i \neq j} \eta_i \mathbf{a}_i = \eta_j \mathbf{a}_j \quad (6.81)$$

This implies that the j th vector can be written as a linear combination of the others. By performing row operations (such as those taught in high school) we can reduce the j th row to all zeros. Which is

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{vmatrix} \quad (6.82)$$

According to the properties of the determinant discussed earlier, this means $\det(A) = 0$. In other words, if $\det(A) = 0$, then the column vectors of A are linearly dependent, and the image of the linear transformation degenerates into a lower-dimensional structure. From the perspective of solving linear systems, a zero determinant means that we cannot uniquely solve for all variables. In $\mathbb{R}^2$, this may correspond to two equations that are either parallel or identical, failing to intersect at a single point. Similarly, in $\mathbb{R}^3$, three equations with linearly dependent normal vectors may represent planes that are coplanar, colinear, or even disjoint, resulting in an unsolvable or underdetermined system.

Consider a set of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \in \mathbb{R}^n$. If there exists a set of scalars $\eta_1, \eta_2, \dots, \eta_n$, such that

$$\sum_i \eta_i \mathbf{x}_i = \mathbf{0} \Rightarrow \eta_1, \eta_2, \dots, \eta_n = 0 \quad (6.83)$$

then the vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ are said to be **linearly independent**. The meaning of linear independence is that there exists no nontrivial linear combination of the vectors that results in the zero vector.

Similarly, consider a system of linear equations $A\mathbf{x} = \mathbf{b}$. If the column vectors of A are linearly independent, then the following condition

$$\sum_{i \neq j} \eta_i \mathbf{a}_i \neq \eta_j \mathbf{a}_j \quad (6.84)$$

That is, no vector in the system can be expressed as a linear combination of the others. This ensures that the determinant $\det(A)$ is nonzero. According to the properties of the inverse matrix discussed earlier, any matrix with a nonzero determinant has a unique inverse A^{-1} . Therefore, the solution to the system can be written as

$$A\mathbf{x} = \mathbf{b} \quad \Leftrightarrow \quad \mathbf{x} = A^{-1}\mathbf{b} \quad (6.85)$$

which guarantees the existence and uniqueness of the solution, fully determining the solution space.

6.5.2 Wronskian and Abel's Identity

We've discussed the linear relationship between the vectors. In addition, we can also use the previous method to determine the linear independence among functions. Consider an n th-order differential equation as an example

$$y^{(n)} + P_{n-1}(x)y^{(n-1)} + \cdots + P_1(x)y' + P_0(x)y = 0 \quad (6.86)$$

The **solution space** of this differential equation is expected to be n -dimensional. Suppose that $y_1(x), y_2(x), \dots, y_n(x)$ are the solutions to the differential equation. To determine whether they are dependent or not, we express the solution as a linear combination

$$\eta_1 y_1(x) + \eta_2 y_2(x) + \cdots + \eta_n y_n(x) = 0 \quad (6.87)$$

Where $\eta_i \in \mathbb{R}$. Assume that y is $(n-1)$ -times differentiable. We have

$$\begin{aligned} & \frac{d}{dx}[\eta_1 y_1(x)] + \frac{d}{dx}[\eta_2 y_2(x)] + \cdots + \frac{d}{dx}[\eta_n y_n(x)] \\ & \Rightarrow \eta_1 y_1'(x) + \eta_2 y_2'(x) + \cdots + \eta_n y_n'(x) = 0 \end{aligned} \quad (6.88)$$

Similarly, we obtain

$$\begin{cases} \eta_1 y_1'(x) + \eta_2 y_2'(x) + \cdots + \eta_n y_n'(x) = 0 \\ \eta_1 y_1''(x) + \eta_2 y_2''(x) + \cdots + \eta_n y_n''(x) = 0 \\ \vdots \\ \eta_1 y_1^{(n-1)}(x) + \eta_2 y_2^{(n-1)}(x) + \cdots + \eta_n y_n^{(n-1)}(x) = 0 \end{cases}$$

We define $Y = (\mathbf{y}_i(x), \mathbf{y}_i'(x), \dots, \mathbf{y}_i^{(n-1)}(x))^T$, and η is the column matrix. So, the matrix multiplication

$$\begin{bmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (6.89)$$

Here, we define the determinant

$$W(y_1, y_2, \dots, y_n)(x) := \begin{vmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{vmatrix} \quad (6.90)$$

We call the determinant $W(y_1, y_2, \dots, y_n)(x)$ as **Wronskian**. If the determinant $W(x) \neq 0$, it means that the inverse matrix of Y exists, so (6.89) satisfies

$$Y\eta = \mathbf{0} \Leftrightarrow \eta = Y^{-1}\mathbf{0} = \mathbf{0} \quad (6.91)$$

By (6.83), we can verify that $y_1(x), y_2(x), \dots, y_n(x)$ are linearly independent. In other words, the solutions of this differential equation span an n -dimensional solution space $\mathbb{R}^n$. Conversely, if the Wronskian $W(x) = 0$, then $y_1(x), y_2(x), \dots, y_n(x)$ are linearly dependent, indicating that the solution space is degenerate and does not satisfy the general solution structure.

Moreover, differentiate the Wronskian $W(x) = 0$ with respect to x ,

$$\frac{d}{dx}W(x) = \frac{d}{dx} \det(Y) \quad (6.92)$$

We define $\langle A^{j \neq i}, A^i \rangle$ as the determinant of the matrix $A \in \mathbb{R}^{n \times n}$ where only the entries (A^i) in the i th row are differentiated, while all other entries $(A^{j \neq i})$ remain unchanged. So, we have

$$\frac{d}{dx} \det(A) = \sum_i \langle A^{j \neq i}, A^i \rangle \quad (6.93)$$

This is because the determinant is computed by summing over all possible permutations of the matrix elements, as given by

$$\frac{d}{dx} \det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \frac{d}{dx} \prod_i a_{i, \sigma(i)}. \quad (6.94)$$

According to the properties of differentiation, each product under a given permutation can be decomposed into n separate terms, just like

$$\frac{d}{dx}[a_1(x)a_2(x)a_3(x)] = a_1'(x)a_2(x)a_3(x) + a_1(x)a_2'(x)a_3(x) + a_1(x)a_2(x)a_3'(x).$$

Therefore, we only need to differentiate the entries of a single row while keeping all other entries unchanged. By summing the resulting determinants over all rows, we obtain

$$\begin{aligned} \frac{d}{dx}W(x) &= \sum_i \langle Y^{j \neq i}, Y^i \rangle \\ &= \begin{vmatrix} y_1' & y_2' & \cdots & y_n' \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix} + \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1'' & y_2'' & \cdots & y_n'' \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix} + \cdots \\ &\quad + \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ y_1^{(n)} & y_2^{(n)} & \cdots & y_n^{(n)} \end{vmatrix} \end{aligned}$$

Since the first $n - 1$ terms contain two identical rows, we can apply the properties of the determinant to conclude that

$$\begin{vmatrix} y_1' & y_2' & \cdots & y_n' \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix} = \begin{vmatrix} y_1' & y_2' & \cdots & y_n' \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix} = 0$$

Apply the same logic to the first $n - 1$ term; only the last term remains, we can get

$$\frac{d}{dx}W(x) = \sum_i \langle Y^{j \neq i}, Y^i \rangle = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ y_1^{(n)} & y_2^{(n)} & \cdots & y_n^{(n)} \end{vmatrix} \quad (6.95)$$

Because $y_1, y_2, \dots, y_n$ are the solutions to the ODE, so $y_1^{(n)}, y_2^{(n)}, \dots, y_n^{(n)}$ satisfy

$$y_i^{(n)} = -P_{n-1}(x)y_i^{(n-1)} - \cdots - P_1(x)y'_i - P_0(x)y_i \quad (6.96)$$

for $1 \leq i \leq n$. That is to say, $y_1^{(n)}, y_2^{(n)}, \dots, y_n^{(n)}$ can be written as the linear combination of the first $n-1$ terms. By performing row operations on the matrix, we can get

$$\begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ y_1^{(n)} & y_2^{(n)} & \cdots & y_n^{(n)} \end{vmatrix} = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ -P_{n-1}y_1^{(n-1)} & -P_{n-1}y_2^{(n-1)} & \cdots & -P_{n-1}y_n^{(n-1)} \end{vmatrix}$$

$$= -P_{n-1}(x) \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

$$= -P_{n-1}(x)W(x)$$

Substituting $-P_{n-1}(x)W(x)$ into the original differentiation, we obtain a first-order differential equation for $W(x)$.

$$\begin{aligned}
\frac{d}{dx}W(x) &= -P_{n-1}(x)W(x) \\
\Rightarrow \frac{dW(x)}{W(x)} &= -P_{n-1}(x)dx \\
\Rightarrow W(x) &= C \cdot \exp\left(-\int P_{n-1}(x)dx\right)
\end{aligned} \tag{6.97}$$

The initial condition at the point $x = x_0$, the Wronskian is $W(x_0)$, so we have

$$\Rightarrow W(x_0) = C \cdot \exp\left(-\int_{x_0}^{x_0} P_{n-1}(x)dx\right) = C \tag{6.98}$$

Substituting (6.98) into (6.97),

$$\boxed{W(x) = W(x_0) \cdot \exp\left(-\int_{x_0}^x P_{n-1}(x)dx\right)} \tag{6.99}$$

This is called **Abel's Identity**. Suppose that $W(x)$ is defined on the interval $I \subseteq \mathbb{R}$, when $W(x) = 0$ on I , then $y_1, y_2, \dots, y_n$ are linearly dependent, the solution space degenerates; when $W(x_0) \neq 0$ for any $x_0 \in I$, then $y_1, y_2, \dots, y_n$ are linearly independent, the solutions of this differential equation span an n -dimensional solution space $\mathbb{R}^n$ since

$$\exp\left(-\int_{x_0}^{x_0} P_{n-1}(x)dx\right) > 0, W(x_0) \neq 0 \Rightarrow W(x) \neq 0, \forall x \in I$$

If $W \neq 0$, then the set of functions is linearly independent. However, it is important to note that if we take an arbitrary set of functions $\{f_1, \dots, f_n\} \in C^{n-1}(I)$ defined on an interval I , then this condition is not a necessary and sufficient condition. In other words, we cannot conclude that the functions are linearly dependent if $W = 0$, unless we have the additional assumption that these functions come from the solution space of a linear differential equation.

This means that, even if the functions are locally linearly dependent on every subinterval $I' \subset I$, they may not be globally linearly dependent over the entire interval I . For example,

$$f_1(x) = x^2, \quad f_2(x) = x|x| = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$$

are linearly dependent on $x > 0$ and also on $x < 0$, which implies $W(x) = 0$. However, there does not exist a constant $c \in \mathbb{R}$ such that $f_2(x) = cf_1(x)$ holds for all $x \in I$. Thus, they are not globally linearly dependent.

In summary, only when the functions are known to be solutions of a linear differential equation (so that Abel's identity holds) does the condition $W = 0$ become both neces-

sary and sufficient for linear dependence. Otherwise, we can only use $W \neq 0$ to confirm linear independence.

6.6 Orthogonal Matrices

Before discussing diagonalization, we need one clean fact about changes of orthonormal coordinates. The main idea is simple: an orthogonal matrix may rotate or reflect a vector, but it does not change lengths or angles.

Throughout this section, vectors in $\mathbb{R}^n$ are column vectors and the standard inner product is

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}. \quad (6.100)$$

Definition 6.3 (Orthogonal Matrix) Let $Q \in M_{n \times n}(\mathbb{R})$. We call Q an **orthogonal matrix** if

$$Q^T Q = Q Q^T = I_n. \quad (6.101)$$

Equivalently, $Q^{-1} = Q^T$.

The word “orthogonal” here is not decorative. It says exactly that the columns, and also the rows, form an orthonormal basis of $\mathbb{R}^n$.

Theorem 6.1 For $Q \in M_{n \times n}(\mathbb{R})$, the following statements are equivalent:

- (a) Q is orthogonal;
- (b) the columns of Q form an orthonormal basis of $\mathbb{R}^n$;
- (c) the rows of Q form an orthonormal basis of $\mathbb{R}^n$.

Proof. Write $Q = (\mathbf{q}_1 \ \mathbf{q}_2 \ \cdots \ \mathbf{q}_n)$. Then

$$(Q^T Q)_{ij} = \mathbf{q}_i^T \mathbf{q}_j = \langle \mathbf{q}_i, \mathbf{q}_j \rangle.$$

Thus $Q^T Q = I_n$ if and only if $\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij}$, which is exactly the statement that the columns are orthonormal. Since there are n of them in $\mathbb{R}^n$, they form a basis. The same argument applied to $Q Q^T$ gives the corresponding statement for the rows. $\square$

Theorem 6.2 (Preservation of Inner Products) If Q is orthogonal, then for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\langle Q\mathbf{x}, Q\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle. \quad (6.102)$$

Consequently, $\|Q\mathbf{x}\| = \|\mathbf{x}\|$, and the angle between two nonzero vectors is preserved.

Proof. By the definition of the standard inner product,

$$\langle Q\mathbf{x}, Q\mathbf{y} \rangle = (Q\mathbf{x})^T (Q\mathbf{y}) = \mathbf{x}^T Q^T Q \mathbf{y} = \mathbf{x}^T \mathbf{y}.$$

Taking $\mathbf{y} = \mathbf{x}$ gives preservation of the norm. The usual formula for the cosine of the angle then gives preservation of angles. $\square$

Two quick consequences will be useful later. First,

$$1 = \det(I_n) = \det(Q^T Q) = \det(Q)^2, \quad \text{so} \quad \det(Q) = \pm 1. \quad (6.103)$$

Second, if Q_1 and Q_2 are orthogonal, then

$$(Q_1 Q_2)^T (Q_1 Q_2) = Q_2^T Q_1^T Q_1 Q_2 = I_n,$$

so their product is again orthogonal. In other words, successive changes of orthonormal coordinates are still changes of orthonormal coordinates.

Over $\mathbb{C}$, transpose must be replaced by conjugate transpose. We write

$$A^* = \overline{A}^T, \quad \langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i \overline{y_i} = \mathbf{y}^* \mathbf{x}. \quad (6.104)$$

This follows the convention of the textbook: the inner product is linear in its first argument. A complex matrix U satisfying $U^* U = U U^* = I_n$ is called **unitary**. It plays exactly the same role over $\mathbb{C}$ that an orthogonal matrix plays over $\mathbb{R}$.

6.7 Introduction to Eigenproblems

6.7.1 Definition of Eigenvalues and Eigenvectors

When a system is written component by component, the equations can look heavily coupled. The useful question is whether there are special directions that the linear transformation does not mix with the others. Along such a direction, the transformation only multiplies a vector by a scalar. That is the whole idea behind an eigenvector.

Definition 6.4 (Eigenvalue and Eigenvector) Let T be a linear operator on a vector space V over a field $\mathbb{F}$. A nonzero vector $\mathbf{v} \in V$ is an **eigenvector** of T if there exists $\lambda \in \mathbb{F}$ such that

$$T(\mathbf{v}) = \lambda \mathbf{v}. \quad (6.105)$$

The scalar λ is the **eigenvalue** corresponding to $\mathbf{v}$.

For $A \in M_{n \times n}(\mathbb{F})$, a nonzero vector $\mathbf{v} \in \mathbb{F}^n$ is an eigenvector of A if

$$A\mathbf{v} = \lambda \mathbf{v}. \quad (6.106)$$

Equivalently, $\mathbf{v}$ is an eigenvector of the left-multiplication operator $L_A(\mathbf{x}) = A\mathbf{x}$.

There is one small point that is easy to say incorrectly: an eigenvalue does not correspond to a unique eigenvector. If $\mathbf{v}$ is an eigenvector, then $c\mathbf{v}$ is another one for every nonzero scalar c . The natural object is therefore the entire eigenspace, not one particular representative.

Definition 6.5 (Eigenspace) Let λ be an eigenvalue of a linear operator T on V . The **eigenspace** corresponding to λ is

$$E_\lambda = \{\mathbf{x} \in V : T(\mathbf{x}) = \lambda\mathbf{x}\} = N(T - \lambda I). \quad (6.107)$$

For a matrix $A \in M_{n \times n}(\mathbb{F})$,

$$E_\lambda = N(A - \lambda I_n). \quad (6.108)$$

Thus E_λ contains the zero vector together with all eigenvectors corresponding to λ .

Definition 6.6 (Characteristic Polynomial) For $A \in M_{n \times n}(\mathbb{F})$, the polynomial

$$f_A(t) = \det(A - tI_n) \quad (6.109)$$

is called the **characteristic polynomial** of A .

Theorem 6.3 (Characteristic Equation) Let $A \in M_{n \times n}(\mathbb{F})$. A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A if and only if

$$f_A(\lambda) = \det(A - \lambda I_n) = 0. \quad (6.110)$$

Proof. The scalar λ is an eigenvalue if and only if there is a nonzero $\mathbf{v}$ such that

$$(A - \lambda I_n)\mathbf{v} = \mathbf{0}.$$

This happens if and only if $A - \lambda I_n$ has a nontrivial null space, which is equivalent to saying that $A - \lambda I_n$ is not invertible. For a square matrix, noninvertibility is equivalent to zero determinant. $\square$

The polynomial $f_A(t)$ has degree n and leading coefficient $(-1)^n$. Therefore A has at most n distinct eigenvalues. Over $\mathbb{C}$, the fundamental theorem of algebra says that f_A has exactly n roots when multiplicity is counted. Over $\mathbb{R}$, however, the polynomial may fail to split into linear factors. This is why a real matrix need not have even one real eigenvalue.

If T is a linear operator on an n -dimensional vector space V and β is an ordered basis for V , we define the characteristic polynomial of T by

$$f_T(t) = \det([T]_\beta - tI_n).$$

This definition does not depend on the chosen basis. Indeed, two matrix representations of the same operator are similar, and similar matrices have the same characteristic polynomial.

Definition 6.7 (Algebraic and Geometric Multiplicity) Suppose λ is an eigenvalue and

$$f_T(t) = (t - \lambda)^m g(t), \quad g(\lambda) \neq 0.$$

The integer m is the **algebraic multiplicity** of λ . The integer

$$\dim(E_\lambda)$$

is its **geometric multiplicity**.

These two multiplicities answer different questions. Algebraic multiplicity counts how often λ appears as a root of the characteristic polynomial. Geometric multiplicity counts how many linearly independent eigenvectors can actually be chosen for that eigenvalue.

Theorem 6.4 *Let T be a linear operator on a finite-dimensional vector space, and let λ have algebraic multiplicity m . Then*

$$1 \leq \dim(E_\lambda) \leq m. \quad (6.111)$$

Proof. Because λ is an eigenvalue, E_λ contains a nonzero vector, so $\dim(E_\lambda) \geq 1$. Put $p = \dim(E_\lambda)$, choose a basis $\mathbf{v}_1, \dots, \mathbf{v}_p$ for E_λ , and extend it to an ordered basis β for V . Since $T(\mathbf{v}_i) = \lambda \mathbf{v}_i$ for $1 \leq i \leq p$, the matrix of T has the block form

$$[T]_\beta = \begin{bmatrix} \lambda I_p & B \\ 0 & C \end{bmatrix}.$$

Hence

$$f_T(t) = \det([T]_\beta - tI) = (\lambda - t)^p \det(C - tI).$$

Thus $(t - \lambda)^p$ divides $f_T(t)$, and therefore $p \leq m$. $\square$

Theorem 6.5 (Eigenvectors from Distinct Eigenvalues) *Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of a linear operator T . If $\mathbf{v}_i$ is an eigenvector corresponding to λ_i , then*

$$\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$$

is linearly independent.

Proof. We use induction on k . The case $k = 1$ follows because an eigenvector is nonzero. Assume the result holds for $k - 1$, and suppose

$$a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k = \mathbf{0}.$$

Apply $T - \lambda_k I$ to both sides. Since $(T - \lambda_k I)\mathbf{v}_k = \mathbf{0}$, we get

$$a_1(\lambda_1 - \lambda_k)\mathbf{v}_1 + \dots + a_{k-1}(\lambda_{k-1} - \lambda_k)\mathbf{v}_{k-1} = \mathbf{0}.$$

By the induction hypothesis and the distinctness of the eigenvalues, $a_1 = \dots = a_{k-1} = 0$. The original relation then gives $a_k \mathbf{v}_k = \mathbf{0}$, so $a_k = 0$. $\square$

We keep the existing matrix example, but now write it using the convention above. Let

$$A = \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix}.$$

Its characteristic polynomial is

$$f_A(t) = \det(A - tI_2) = (5 - t)(2 - t) - 4 = (t - 1)(t - 6).$$

Thus $\lambda_1 = 1$ and $\lambda_2 = 6$. The corresponding eigenspaces are

$$E_1 = N(A - I_2) = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}, \quad E_6 = N(A - 6I_2) = \operatorname{span} \left\{ \begin{bmatrix} 2 \\ -1 \end{bmatrix} \right\}.$$

Notice what the answer really gives us: not two special vectors, but two one-dimensional eigenspaces.

6.7.2 Diagonalization and its Meaning

Diagonalization is a change-of-basis problem. We are not changing the linear operator itself; we are looking for coordinates in which its action is as simple as possible.

Definition 6.8 (Diagonalizable Operator and Matrix) Let T be a linear operator on a finite-dimensional vector space V . We call T **diagonalizable** if there is an ordered basis β for V such that $[T]_\beta$ is diagonal.

A matrix $A \in M_{n \times n}(\mathbb{F})$ is called **diagonalizable** if the operator L_A is diagonalizable. Equivalently, there exist an invertible matrix Q and a diagonal matrix D such that

$$Q^{-1}AQ = D.$$

Theorem 6.6 (Eigenbasis Criterion) A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if V has an ordered basis consisting of eigenvectors of T .

Proof. Suppose $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $[T]_\beta = D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. The j th column of D is the coordinate vector of $T(\mathbf{v}_j)$, so

$$T(\mathbf{v}_j) = \lambda_j \mathbf{v}_j.$$

Thus every vector in β is an eigenvector. Conversely, if $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ is an eigenbasis with $T(\mathbf{v}_j) = \lambda_j \mathbf{v}_j$, then the j th column of $[T]_\beta$ has only one nonzero entry, namely λ_j in its j th position. Hence $[T]_\beta$ is diagonal. $\square$

In matrix language, let

$$Q = (\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n), \quad D = \operatorname{diag}(\lambda_1, \dots, \lambda_n). \quad (6.112)$$

Then all eigenvalue equations can be collected into one equation:

$$AQ = QD. \quad (6.113)$$

The columns of Q form a basis exactly when Q is invertible. In that case,

$$A = QDQ^{-1}, \quad (6.114)$$

or, equivalently,

$$Q^{-1}AQ = D. \quad (6.115)$$

This is the precise meaning of a similarity transformation. The matrix A acts in the original coordinates, while D is the matrix of the same linear operator in the eigenbasis.

More explicitly, if

$$\mathbf{x} = q_1 \mathbf{v}_1 + \cdots + q_n \mathbf{v}_n = Q\mathbf{q}, \quad (6.116)$$

then

$$A\mathbf{x} = AQ\mathbf{q} = QD\mathbf{q}.$$

So in the $\mathbf{q}$ -coordinates, the i th component is simply multiplied by λ_i . This is the part worth remembering: diagonalization removes the mixing between different coordinate directions.

Theorem 6.7 (Test for Diagonalization) *Let T be a linear operator on an n -dimensional vector space V , and suppose its characteristic polynomial splits over $\mathbb{F}$. Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T , with algebraic multiplicities $m_1, \dots, m_k$. Then T is diagonalizable if and only if*

$$\dim(E_{\lambda_i}) = m_i \quad \text{for every } i. \quad (6.117)$$

When these equalities hold, the union of bases of the eigenspaces is an eigenbasis for V .

Proof. Suppose first that T is diagonalizable. Choose an eigenbasis. In the corresponding diagonal matrix, λ_i occurs on the diagonal exactly m_i times, because the characteristic polynomial is the product of the diagonal factors. Therefore the basis contains m_i linearly independent vectors in E_{λ_i} . Since the geometric multiplicity is at most the algebraic multiplicity, we obtain $\dim(E_{\lambda_i}) = m_i$.

Conversely, choose a basis β_i for each E_{λ_i} . If a linear combination of vectors in $\beta_1 \cup \cdots \cup \beta_k$ is zero, group the terms belonging to each eigenspace and write the relation as

$$\mathbf{w}_1 + \cdots + \mathbf{w}_k = \mathbf{0}, \quad \mathbf{w}_i \in E_{\lambda_i}.$$

The nonzero $\mathbf{w}_i$ would be eigenvectors belonging to distinct eigenvalues, so the preceding theorem forces every $\mathbf{w}_i$ to be zero. Each β_i is linearly independent, so all coefficients vanish. Hence the union is linearly independent. Its number of vectors is

$$\sum_{i=1}^k \dim(E_{\lambda_i}) = \sum_{i=1}^k m_i = n.$$

It is therefore an eigenbasis for V , and T is diagonalizable. $\square$

One useful sufficient condition now follows immediately: an operator on an n -dimensional vector space with n distinct eigenvalues is diagonalizable. The converse is false. For example, the identity operator is diagonalizable but has only one eigenvalue.

So “distinct eigenvalues” is a convenient sufficient condition, not the definition of diagonalizability.

For an actual calculation, we first compute $f_A(t) = \det(A - tI_n)$ and check that it splits over the field being used. We then find each eigenspace $E_\lambda = N(A - \lambda I_n)$. Whenever an eigenvalue is repeated, the important check is whether $\dim(E_\lambda) = n - \text{rank}(A - \lambda I_n)$ equals its algebraic multiplicity. If all the dimensions are sufficient, bases of the eigenspaces are placed into the columns of Q , and the diagonal form is $Q^{-1}AQ = D$. If those dimensions do not add up to n , the process stops there. Choosing different scalar multiples of the same eigenvectors cannot repair the missing directions.

There is one especially important case in physics.

Theorem 6.8 (Real Spectral Theorem) *Let $A \in M_{n \times n}(\mathbb{R})$. The matrix A is real symmetric if and only if there exist an orthogonal matrix Q and a real diagonal matrix D such that*

$$Q^T A Q = D, \quad \text{equivalently} \quad A = Q D Q^T. \quad (6.118)$$

The proof is the real version of the spectral theorem discussed in the next subsection. The columns of Q are an orthonormal eigenbasis. This is stronger than ordinary diagonalizability because the inverse of the change-of-basis matrix is simply its transpose.

For the inertia tensor $\mathcal{I}$ of a rigid body, the rotational kinetic energy is

$$T_{\text{rot}} = \frac{1}{2} \omega^T \mathcal{I} \omega.$$

Since $\mathcal{I}$ is real symmetric, choose an orthogonal Q such that

$$Q^T \mathcal{I} Q = D = \text{diag}(I_1, I_2, I_3).$$

Writing $\omega = Q\Omega$ gives

$$T_{\text{rot}} = \frac{1}{2} \Omega^T D \Omega = \frac{1}{2} (I_1 \Omega_1^2 + I_2 \Omega_2^2 + I_3 \Omega_3^2).$$

The mixed terms disappear because the coordinates are now measured along the principal axes. The same mechanism appears in small oscillations: after the relevant symmetric matrices are reduced to an eigenbasis, the new coordinates describe independent normal modes.

6.7.3 Diagonalizing Hermitian Matrices (Optional)

To discuss complex matrices rigorously, we need the standard inner product on $\mathbb{C}^n$ already introduced above:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i \overline{y_i} = \mathbf{y}^* \mathbf{x}. \quad (6.119)$$

Again, this convention is linear in the first argument and conjugate-linear in the second. The associated norm is $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$.

Definition 6.9 (Adjoint, Hermitian, Normal, and Unitary Matrices) For $A \in M_{n \times n}(\mathbb{C})$, its conjugate transpose is

$$A^* = \overline{A}^T.$$

We call A

- (a) **Hermitian** or **self-adjoint** if $A = A^*$;
- (b) **normal** if $AA^* = A^*A$;
- (c) **unitary** if $A^*A = AA^* = I_n$.

For a real matrix, $A^* = A^T$, so Hermitian reduces to symmetric and unitary reduces to orthogonal.

For instance,

$$H = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}$$

is Hermitian because $H^* = H$. The off-diagonal entries are complex conjugates of one another, while every diagonal entry is real.

Theorem 6.9 (Reality of Hermitian Eigenvalues) *Every eigenvalue of a Hermitian matrix is real.*

Proof. Let $H\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Since $H = H^*$ and $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^*\mathbf{y} \rangle$, we have

$$\begin{aligned} \lambda \langle \mathbf{v}, \mathbf{v} \rangle &= \langle H\mathbf{v}, \mathbf{v} \rangle \\ &= \langle \mathbf{v}, H\mathbf{v} \rangle = \langle \mathbf{v}, \lambda\mathbf{v} \rangle = \overline{\lambda} \langle \mathbf{v}, \mathbf{v} \rangle. \end{aligned}$$

Because $\langle \mathbf{v}, \mathbf{v} \rangle > 0$, it follows that $\lambda = \overline{\lambda}$, hence $\lambda \in \mathbb{R}$. □

Theorem 6.10 (Orthogonality of Distinct Eigenspaces) *Let H be Hermitian. Eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Proof. Suppose $H\mathbf{v}_1 = \lambda_1\mathbf{v}_1$ and $H\mathbf{v}_2 = \lambda_2\mathbf{v}_2$, where $\lambda_1 \neq \lambda_2$. By the preceding theorem both eigenvalues are real, and

$$\lambda_1 \langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \langle H\mathbf{v}_1, \mathbf{v}_2 \rangle = \langle \mathbf{v}_1, H\mathbf{v}_2 \rangle = \lambda_2 \langle \mathbf{v}_1, \mathbf{v}_2 \rangle.$$

Therefore $(\lambda_1 - \lambda_2) \langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$, so $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$. □

Distinct eigenspaces are automatically orthogonal. Inside an eigenspace belonging to a repeated eigenvalue, we may apply the Gram–Schmidt process to choose an orthonormal basis without leaving that eigenspace. This is the step that the informal statement “normalize the eigenvectors” often hides. Normalization alone is not enough when a repeated eigenspace has dimension greater than one; orthogonalization may also be required.

Theorem 6.11 (Spectral Theorem for Hermitian Matrices) For $H \in M_{n \times n}(\mathbb{C})$, the following are equivalent:

- (a) H is Hermitian;
- (b) there exist a unitary matrix U and a real diagonal matrix D such that

$$U^* H U = D. \quad (6.120)$$

Equivalently, $H = U D U^*$. The columns of U form an orthonormal basis of eigenvectors of H , and the diagonal entries of D are their corresponding eigenvalues.

Proof. Assume first that H is Hermitian. We prove the result by induction on n . The case $n = 1$ is immediate. Since the characteristic polynomial splits over $\mathbb{C}$, H has an eigenvalue λ_1 and a unit eigenvector $\mathbf{u}_1$. The eigenvalue is real by the preceding theorem.

Let $W = \text{span}\{\mathbf{u}_1\}^\perp$. If $\mathbf{x} \in W$, then

$$\langle H\mathbf{x}, \mathbf{u}_1 \rangle = \langle \mathbf{x}, H\mathbf{u}_1 \rangle = \langle \mathbf{x}, \lambda_1 \mathbf{u}_1 \rangle = 0.$$

Thus $H\mathbf{x} \in W$, so W is invariant under H . The restriction $H|_W$ is again Hermitian. By the induction hypothesis, W has an orthonormal basis $\mathbf{u}_2, \dots, \mathbf{u}_n$ consisting of eigenvectors of $H|_W$. Hence $\mathbf{u}_1, \dots, \mathbf{u}_n$ is an orthonormal eigenbasis of $\mathbb{C}^n$.

Let

$$U = (\mathbf{u}_1 \cdots \mathbf{u}_n), \quad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Then $HU = UD$ and $U^*U = I_n$, so $U^*HU = D$.

Conversely, suppose $H = UDU^*$, where U is unitary and D is real diagonal. Then $D^* = D$, and therefore

$$H^* = (UDU^*)^* = UD^*U^* = UDU^* = H.$$

Thus H is Hermitian. □

The textbook proves the slightly more general complex spectral theorem: a complex matrix is normal if and only if it is unitarily equivalent to a diagonal matrix. Hermitian matrices are automatically normal, and the extra Hermitian condition forces every diagonal entry to be real. So the final formula

$$H = UDU^*, \quad U^{-1} = U^*,$$

is not merely ordinary diagonalization. It is diagonalization by an orthonormal change of coordinates, which is exactly why it is so useful in quantum mechanics.

6.8 Applications to the Eigen Problem in Physical Systems

6.8.1 Coupled Spring System

As emphasized throughout this text, the goal is to address the mathematical structures behind problems in mechanics, rather than to delve into physical intuition or diagrams. In the classical coupled spring problem, we consider three springs with identical spring constants k , and two point masses of equal mass m situated between them. Let $x_1(t)$ and $x_2(t)$ denote the displacements of the two masses from equilibrium. The total potential energy of the system is given by

$$V = k(x_1^2 - x_1x_2 + x_2^2) \quad (6.121)$$

Then the equations of motion of both point masses are

$$\begin{cases} m\ddot{x}_1 = -\frac{\partial V}{\partial x_1} = -2kx_1 + kx_2 \\ m\ddot{x}_2 = -\frac{\partial V}{\partial x_2} = -2kx_2 + kx_1 \end{cases} \quad (6.122)$$

Let the matrix $\ddot{\mathbf{X}} = (\ddot{x}_1 \quad \ddot{x}_2)^T$ and $\mathbf{X} = (x_1 \quad x_2)^T$, so (6.122) can be written as

$$m \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} = K \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Leftrightarrow m\ddot{\mathbf{X}} = -kA\mathbf{X} \quad (6.123)$$

where the matrix $-kA$ is

$$-kA = \begin{bmatrix} -2k & k \\ k & -2k \end{bmatrix} \Rightarrow A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

By viewing the coupled elements as part of a unified system, we can analyze the second-order linear differential equation in matrix form more easily. This process is known as "decoupling" the system. Therefore, by Chapter 4, we can assume that the solution to the differential equation is

$$\mathbf{X} = \mathbf{v}e^{i\omega t} \quad (6.124)$$

Substituting this into (6.123), we obtain

$$-m\omega^2 \mathbf{v}e^{i\omega t} = -kA\mathbf{v}e^{i\omega t} \Rightarrow A\mathbf{v} = \frac{m\omega^2}{k} \mathbf{v} := \lambda \mathbf{v} \quad (6.125)$$

The question becomes a classic eigen-problem. That is,

$$(A - \lambda I_2) \mathbf{v} = 0 \quad (6.126)$$

Since $\mathbf{v} \neq 0$, we have

$$\det(A - \lambda I_2) = \begin{vmatrix} 2 - \lambda & -1 \\ -1 & 2 - \lambda \end{vmatrix} = \lambda^2 - 4\lambda + 3 = 0 \quad (6.127)$$

Then we can get two eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 3$, also, we can obtain their corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$. We obtain the frequency

$$\omega_1 = \sqrt{\frac{k\lambda_1}{m}} = \sqrt{\frac{k}{m}} \quad \omega_2 = \sqrt{\frac{k\lambda_2}{m}} = \sqrt{\frac{3k}{m}} \quad (6.128)$$

Then we diagonalize the matrix A

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} (\mathbf{v}_1 \quad \mathbf{v}_2) = (\mathbf{v}_1 \quad \mathbf{v}_2) \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \quad (6.129)$$

So (6.126) can be rewritten as

$$\begin{cases} \begin{bmatrix} 2 - 1 & -1 \\ -1 & 2 - 1 \end{bmatrix} \mathbf{v}_1 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} = 0 \\ \begin{bmatrix} 2 - 3 & -1 \\ -1 & 2 - 3 \end{bmatrix} \mathbf{v}_2 = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix} = 0 \end{cases} \quad (6.130)$$

By (6.130), the eigenvectors of the matrix A are

$$\mathbf{v}_1 = \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix} = \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (6.131)$$

From this, we obtain the motion states of the system, which are also referred to as its **modes**. The values ω_1 and ω_2 represent the frequencies associated with the **normal modes** of the system. Hence,

$$\mathbf{X}(t) = C_1 \mathbf{v}_1 \cos\left(\sqrt{\frac{k}{m}}t - \phi_1\right) + C_2 \mathbf{v}_2 \cos\left(\sqrt{\frac{3k}{m}}t - \phi_2\right) \quad (6.132)$$

We expand the result of (6.132)

$$\begin{aligned} \mathbf{X}(t) = & A_1 \mathbf{v}_1 \cos\left(\sqrt{\frac{k}{m}}t\right) + B_1 \mathbf{v}_1 \sin\left(\sqrt{\frac{k}{m}}t\right) \\ & + A_2 \mathbf{v}_2 \cos\left(\sqrt{\frac{3k}{m}}t\right) + B_2 \mathbf{v}_2 \sin\left(\sqrt{\frac{3k}{m}}t\right) \end{aligned} \quad (6.133)$$

Where the constants A_1, A_2, B_1, B_2 are

$$\begin{cases} A_1 = C_1 \cos \phi_1, B_1 = C_1 \sin \phi_1 \\ A_2 = C_2 \cos \phi_2, B_2 = C_2 \sin \phi_2 \end{cases}$$

To determine the behavior of the system, we need to consider the initial conditions. Let the initial position and velocity of the system be

$$\begin{cases} \mathbf{X}(0) := \mathbf{X}_0 = A_1 \mathbf{v}_1 + A_2 \mathbf{v}_2 \\ \dot{\mathbf{X}}(0) := \dot{\mathbf{X}}_0 = \sqrt{\frac{k}{m}} B_1 \mathbf{v}_1 + \sqrt{\frac{3k}{m}} B_2 \mathbf{v}_2 \end{cases} \quad (6.134)$$

To determine these constants, we recall the projection formula taught in high school, often referred to as the “length of the orthogonal projection.” That is

$$\begin{cases} A_1 = \frac{\mathbf{v}_1^T \mathbf{X}_0}{\|\mathbf{v}_1\|^2} \\ A_2 = \frac{\mathbf{v}_2^T \mathbf{X}_0}{\|\mathbf{v}_2\|^2} \end{cases} \quad (6.135)$$

Similarly,

$$B_1 = \frac{\mathbf{v}_1^T \dot{\mathbf{X}}_0}{\sqrt{k/m} \|\mathbf{v}_1\|^2} \quad B_2 = \frac{\mathbf{v}_2^T \dot{\mathbf{X}}_0}{\sqrt{3k/m} \|\mathbf{v}_2\|^2} \quad (6.136)$$

From equations (6.135) and (6.136), we observe that each coefficient is proportional to the projection of the initial condition onto the corresponding eigenvector, that is

$$A_i \propto \mathbf{v}_i^T \mathbf{X}_0, \quad B_i \propto \mathbf{v}_i^T \dot{\mathbf{X}}_0 \quad (6.137)$$

Hence, the relative magnitudes of these projections determine which normal mode is more strongly excited. For example, if $|\mathbf{v}_1^T \mathbf{X}_0| \gg |\mathbf{v}_2^T \mathbf{X}_0|$, the system is initially displaced mainly along the direction of $\mathbf{v}_1$, and the motion will closely follow the first mode.

While it is possible to analyze a three-mass coupled system in full generality, the resulting characteristic equation becomes algebraically complicated, and no new conceptual structure emerges. Therefore, we focus our efforts on the two-mass coupled system, which already encapsulates all fundamental features of modal decomposition, eigenvalue-based decoupling, orthogonality of modes, and projection of initial conditions. Higher-dimensional systems follow the same structure, and the essential ideas extend naturally to continuous systems such as the vibrating string.

6.8.2 Infinite Potential Well

A particle of mass m is confined in a one-dimensional region $x \in [0, L]$, where the potential outside this region is infinitely large. The potential function $V(x)$ is defined as:

$$V(x) = \begin{cases} 0, & 0 < x < L \\ \infty, & x \leq 0 \text{ or } x \geq L \end{cases} \quad (6.138)$$

Since the particle cannot exist outside the potential well, the wavefunction $\psi(x)$ (which represents the probability amplitude) must satisfy the boundary conditions

$$\psi(0) = \psi(L) = 0 \quad (6.139)$$

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x) \quad (6.140)$$

By the condition (6.138), the potential function $V(x) = 0$, and let $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$, then (6.140) becomes

$$\hat{H}\psi = E\psi \quad (6.141)$$

Comparing (6.141) and (6.125), we observe that

$$\begin{cases} \hat{H}\psi = E\psi \\ A\mathbf{v} = \lambda\mathbf{v} \end{cases} \quad (6.142)$$

Here, $\hat{H}$ is a differential operator, meaning it *acts* on the probability wavefunction ψ , while A is a linear transformation matrix that acts on a vector $\mathbf{v}$. Similarly, when the wavefunction ψ is acted upon by $\hat{H}$, the result differs by a constant factor E ; likewise, when the vector $\mathbf{v}$ is acted upon by A , the result differs by a constant λ . In this quantum system, the wavefunction ψ is referred to as an **eigenfunction**, and the energy E is the corresponding **eigenvalue**.

To solve such a problem, let $\lambda^2 = 2mE/\hbar^2$, then (6.141) becomes

$$\frac{d^2\psi}{dx^2} + \lambda^2\psi = 0 \quad (6.143)$$

It becomes a second-order differential equation. The solution to this is

$$\psi(x) = C_1 \sin(\lambda x) + C_2 \cos(\lambda x) \quad (6.144)$$

By the initial condition,

$$\begin{cases} \psi(0) = C_1 \sin(\lambda x) + C_2 \cos(\lambda x) = 0 \Rightarrow B = 0 \\ \psi(L) = C_1 \sin(L\lambda) = 0 \Rightarrow L\lambda = n\pi \quad n \in \mathbb{N} \end{cases} \quad (6.145)$$

So the constant λ is equal to

$$\lambda = \frac{n\pi}{L} \Rightarrow \psi(x) = C_1 \sin\left(\frac{n\pi x}{L}\right) \quad (6.146)$$

Substituting this into $\lambda^2 = 2mE/\hbar^2$, we can obtain the energy E , that is, the eigenvalue

$$E = \frac{n^2 \hbar^2 \pi^2}{2mL^2} \quad (6.147)$$

In quantum mechanics, we normalize the wavefunction so that the total probability within the potential well is equal to 1. As for the deeper physical interpretation, I'll leave that to Professor Kuo. :)

$$\begin{aligned} \int_0^L |\psi_n(x)|^2 dx &= C_1^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = C_1^2 \int_0^L \left[\frac{1 - \cos\left(\frac{n\pi x}{L}\right)}{2} \right] dx \\ &= C_1^2 \int_0^L \frac{1}{2} dx - \frac{1}{2} \int_0^L \cos\left(\frac{n\pi x}{L}\right) dx = \frac{LC_1^2}{2} - \frac{1}{2} \sin\left(\frac{n\pi x}{L}\right) \Big|_0^L = \frac{LC_1^2}{2} = 1 \end{aligned}$$

So, we obtain

$$C_1 = \sqrt{\frac{2}{L}} \quad (6.148)$$

Substituting this into the eigenfunction $\psi(x)$, we can obtain the final solution that

$$\begin{cases} \psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \\ E = \frac{n^2 \hbar^2 \pi^2}{2mL^2} \end{cases} \quad n \in \mathbb{N} \quad (6.149)$$

Much like the eigenvectors in a coupled spring system, these sinusoidal eigenfunctions represent the natural modes of the quantum system.

6.9 Curvilinear Coordinates

If you find this section difficult, you may first read Chapter 7 to better understand the core idea of this section.

6.9.1 General Definition of Curvilinear Coordinates

In Chapter 1.5.8, we've discussed the Jacobian matrix and coordinate transformation. In this section, we focus on the transformation from Cartesian coordinates to curvilinear coordinates in a three-dimensional system. Our goal is for this transformation to be one-to-one, and we have

$$(x_1, x_2, x_3) \leftrightarrow (u_1, u_2, u_3) \quad (6.150)$$

Therefore, the Jacobian between the two coordinate systems represents a bijective structure, which means that they are both linearly independent. We can verify

$$\det(J) = \frac{\partial(x_1, x_2, x_3)}{\partial(u_1, u_2, u_3)} = \begin{vmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_2} & \frac{\partial x_1}{\partial u_3} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_2} & \frac{\partial x_2}{\partial u_3} \\ \frac{\partial x_3}{\partial u_1} & \frac{\partial x_3}{\partial u_2} & \frac{\partial x_3}{\partial u_3} \end{vmatrix} = \frac{\partial \mathbf{r}}{\partial u_1} \cdot \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right) \neq 0 \quad (6.151)$$

Where $\mathbf{r} = (x_1, x_2, x_3)$. Similarly,

$$\det(J^{-1}) = \frac{\partial(u_1, u_2, u_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} & \frac{\partial u_1}{\partial x_3} \\ \frac{\partial u_2}{\partial x_1} & \frac{\partial u_2}{\partial x_2} & \frac{\partial u_2}{\partial x_3} \\ \frac{\partial u_3}{\partial x_1} & \frac{\partial u_3}{\partial x_2} & \frac{\partial u_3}{\partial x_3} \end{vmatrix} = \nabla u_1 \cdot (\nabla u_2 \times \nabla u_3) \neq 0 \quad (6.152)$$

According to the properties of matrix multiplication and inverse matrices,

$$J^{-1}J = \begin{bmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_2} & \frac{\partial x_1}{\partial u_3} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_2} & \frac{\partial x_2}{\partial u_3} \\ \frac{\partial x_3}{\partial u_1} & \frac{\partial x_3}{\partial u_2} & \frac{\partial x_3}{\partial u_3} \end{bmatrix} \begin{bmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} & \frac{\partial u_1}{\partial x_3} \\ \frac{\partial u_2}{\partial x_1} & \frac{\partial u_2}{\partial x_2} & \frac{\partial u_2}{\partial x_3} \\ \frac{\partial u_3}{\partial x_1} & \frac{\partial u_3}{\partial x_2} & \frac{\partial u_3}{\partial x_3} \end{bmatrix} \quad (6.153)$$

$$\Rightarrow (J^{-1}J)_{ij} = \sum_k \frac{\partial u_i}{\partial x_k} \frac{\partial x_k}{\partial u_j} = \frac{\partial u_i}{\partial u_j} = \delta_{ij}$$

We can therefore redefine the gradient in the new coordinate system such that it satisfies

$$\boxed{\nabla u_i \cdot \frac{\partial \mathbf{r}}{\partial u_j} = \delta_{ij}} \quad (6.154)$$

Here, we must introduce a new symbol and matrix, referred to as **Riemannian metric**. It is defined as

$$g_{ij} := \frac{\partial \mathbf{r}}{\partial u_i} \cdot \frac{\partial \mathbf{r}}{\partial u_j} \quad (6.155)$$

The matrix form of the Riemannian metric $\mathbf{g}$ can be written as

$$\mathbf{g} = \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{bmatrix} \quad (6.156)$$

where we can denote $\mathbf{g}_{ij} = g_{ij}$. Its inverse matrix $\mathbf{g}^{-1}$ satisfies that

$$\mathbf{g}_{ij}^{-1} := g^{ij} \quad (6.157)$$

That is,

$$\mathbf{g}\mathbf{g}^{-1} = I \Rightarrow (\mathbf{g}\mathbf{g}^{-1})_{ij} = \sum_k g_{ik} g^{kj} = \delta_{ij} \quad (6.158)$$

To better define the directional gradient in the new coordinate system, we make use of the properties stated in equation (6.158). Let $\nabla u_i = \sum_k C_k \frac{\partial \mathbf{r}}{\partial u_k}$, by (6.154) we have

$$\nabla u_i \cdot \frac{\partial \mathbf{r}}{\partial u_j} = \sum_k C_k \left(\frac{\partial \mathbf{r}}{\partial u_k} \cdot \frac{\partial \mathbf{r}}{\partial u_j} \right) = \delta_{ij}$$

By the definition of (6.155) and the property (6.158), we can obtain

$$\nabla u_i \cdot \frac{\partial \mathbf{r}}{\partial u_j} = \sum_k C_k g_{kj} = \delta_{ij} \Rightarrow C_k = g^{ik} \quad (6.159)$$

Then the gradient can be denoted by

$$\boxed{\nabla u_i = \sum_k g^{ik} \frac{\partial \mathbf{r}}{\partial u_k}} \quad (6.160)$$

Due to the independence of $\{\partial \mathbf{r} / \partial u_i\}$ and $\{u_i\}$, we can use the set of the tangent vector of u_i to construct the basis vectors $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, which satisfies

$$\text{span}\{\mathbf{e}_1, \dots, \mathbf{e}_n\} = \mathbb{R}^n \quad (6.161)$$

We define

$$\mathbf{e}_i = \frac{1}{h_i} \frac{\partial \mathbf{r}}{\partial u_i} \quad h_i = \left| \frac{\partial \mathbf{r}}{\partial u_i} \right| \quad (6.162)$$

Next, we can use this set of basis vectors in the new coordinate system to derive various elements introduced in Chapter 2, such as the differential arc length and surface area elements. Because the transformation ensures the bijective structure, we can verify that the vector $\mathbf{r}$ can be denoted by

$$\mathbf{r}(x_1, x_2, x_3) = \mathbf{r}(x_1(u_1, u_2, u_3), x_2(u_1, u_2, u_3), x_3(u_1, u_2, u_3)) \quad (6.163)$$

Applying the chain rule to the vector $\mathbf{r}$ gives

$$\begin{cases} \frac{\partial \mathbf{r}}{\partial u_1} = \frac{\partial \mathbf{r}}{\partial x_1} \frac{\partial x_1}{\partial u_1} + \frac{\partial \mathbf{r}}{\partial x_2} \frac{\partial x_2}{\partial u_1} + \frac{\partial \mathbf{r}}{\partial x_3} \frac{\partial x_3}{\partial u_1} \\ \frac{\partial \mathbf{r}}{\partial u_2} = \frac{\partial \mathbf{r}}{\partial x_1} \frac{\partial x_1}{\partial u_2} + \frac{\partial \mathbf{r}}{\partial x_2} \frac{\partial x_2}{\partial u_2} + \frac{\partial \mathbf{r}}{\partial x_3} \frac{\partial x_3}{\partial u_2} \\ \frac{\partial \mathbf{r}}{\partial u_3} = \frac{\partial \mathbf{r}}{\partial x_1} \frac{\partial x_1}{\partial u_3} + \frac{\partial \mathbf{r}}{\partial x_2} \frac{\partial x_2}{\partial u_3} + \frac{\partial \mathbf{r}}{\partial x_3} \frac{\partial x_3}{\partial u_3} \end{cases} \quad (6.164)$$

Then the total derivative of the vector $\mathbf{r}$ can be written as

$$\begin{aligned} d\mathbf{r} &= \frac{\partial \mathbf{r}}{\partial u_1} du_1 + \frac{\partial \mathbf{r}}{\partial u_2} du_2 + \frac{\partial \mathbf{r}}{\partial u_3} du_3 \\ &= h_1 \left(\frac{1}{h_1} \frac{\partial \mathbf{r}}{\partial u_1} \right) du_1 + h_2 \left(\frac{1}{h_2} \frac{\partial \mathbf{r}}{\partial u_2} \right) du_2 + h_3 \left(\frac{1}{h_3} \frac{\partial \mathbf{r}}{\partial u_3} \right) du_3 \\ &= \sum_i h_i du_i \mathbf{e}_i \end{aligned} \quad (6.165)$$

The length element ds is

$$\begin{aligned} ds^2 &= d\mathbf{r} \cdot d\mathbf{r} = \left(\frac{\partial \mathbf{r}}{\partial u_1} du_1 + \frac{\partial \mathbf{r}}{\partial u_2} du_2 + \frac{\partial \mathbf{r}}{\partial u_3} du_3 \right) \cdot \left(\frac{\partial \mathbf{r}}{\partial u_1} du_1 + \frac{\partial \mathbf{r}}{\partial u_2} du_2 + \frac{\partial \mathbf{r}}{\partial u_3} du_3 \right) \\ &= \left(\sum_i \frac{\partial \mathbf{r}}{\partial u_i} du_i \right) \cdot \left(\sum_j \frac{\partial \mathbf{r}}{\partial u_j} du_j \right) = \sum_{ij} \frac{\partial \mathbf{r}}{\partial u_i} \cdot \frac{\partial \mathbf{r}}{\partial u_j} du_i du_j \end{aligned}$$

Substituting the Riemannian metric into the equation above, we can thus obtain

$$ds^2 = \sum_{ij} g_{ij} du_i du_j \quad (6.166)$$

Then the length element can be written as

$$ds = \sqrt{\sum_{ij} g_{ij} du_i du_j} \quad (6.167)$$

As you did in the Cartesian coordinates, the surface area element is

$$\begin{aligned}
 ds_i &= \left| \frac{\partial \mathbf{r}}{\partial u_j} \times \frac{\partial \mathbf{r}}{\partial u_k} \right| du_j du_k \\
 &= \sqrt{\left(\frac{\partial \mathbf{r}}{\partial u_j} \times \frac{\partial \mathbf{r}}{\partial u_k} \right) \cdot \left(\frac{\partial \mathbf{r}}{\partial u_j} \times \frac{\partial \mathbf{r}}{\partial u_k} \right)} du_j du_k \\
 &= \sqrt{\left(\frac{\partial \mathbf{r}}{\partial u_j} \cdot \frac{\partial \mathbf{r}}{\partial u_j} \right) \left(\frac{\partial \mathbf{r}}{\partial u_k} \cdot \frac{\partial \mathbf{r}}{\partial u_k} \right) - \left(\frac{\partial \mathbf{r}}{\partial u_j} \cdot \frac{\partial \mathbf{r}}{\partial u_k} \right)^2} du_j du_k
 \end{aligned}$$

We obtain

$$ds_i = \sqrt{g_{jj}g_{kk} - g_{jk}^2} du_j du_k \quad (6.168)$$

where $\{i, j, k\}$ are permutations of the elements $\{1, 2, 3\}$.

6.9.2 Orthogonal Form of Curvilinear Coordinates

In orthogonal coordinate systems, such as spherical or cylindrical coordinates, the basis (or tangent) vectors are orthogonal to each other. Which is

$$g_{ij} = \frac{\partial \mathbf{r}}{\partial u_j} \cdot \frac{\partial \mathbf{r}}{\partial u_j} = h_i h_j \delta_{ij} \quad (6.169)$$

We can also write

$$\mathbf{e}_i \cdot \mathbf{e}_j = \frac{1}{h_i} \frac{\partial \mathbf{r}}{\partial u_j} \cdot \frac{1}{h_j} \frac{\partial \mathbf{r}}{\partial u_j} = \delta_{ij} \quad (6.170)$$

By (6.169), the Riemannian metric $\mathbf{g}$ satisfies

$$(\mathbf{g})_{ik} = g_{ik} = h_i h_j \delta_{ij} \quad (6.171)$$

Then the matrix form of the Riemannian metric becomes

$$\mathbf{g} = \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{bmatrix} = \begin{bmatrix} g_{11} & 0 & 0 \\ 0 & g_{22} & 0 \\ 0 & 0 & g_{33} \end{bmatrix} = \begin{bmatrix} h_1^2 & 0 & 0 \\ 0 & h_2^2 & 0 \\ 0 & 0 & h_3^2 \end{bmatrix} \quad (6.172)$$

By equation (6.158), the inverse matrix of the Riemannian metric satisfies

$$(\mathbf{g}\mathbf{g}^{-1})_{ij} = \sum_k g_{ik} g^{kj} = \sum_k h_i h_k \delta_{ik} g^{kj} = h_i^2 g^{ij} = \delta_{ij} \quad (6.173)$$

Then we obtain

$$g^{ij} = \frac{1}{h_i^2} \delta_{ij} \quad (6.174)$$

Substituting this into (6.160), the gradient becomes

$$\nabla u_i = \sum_k g^{ik} \frac{\partial \mathbf{r}}{\partial u_k} = g^{ii} \frac{\partial \mathbf{r}}{\partial u_i} = \frac{1}{h_i^2} \frac{\partial \mathbf{r}}{\partial u_i} = \frac{1}{h_i} \mathbf{e}_i \quad (6.175)$$

By (6.151), the Jacobian can be expressed as the product of the scale factors associated with the three tangent vectors

$$\det(J) = \frac{\partial \mathbf{r}}{\partial u_i} \cdot \left(\frac{\partial \mathbf{r}}{\partial u_j} \times \frac{\partial \mathbf{r}}{\partial u_k} \right) = \pm h_i h_j h_k \quad (6.176)$$

The length of the gradient in (6.175) is

$$|\nabla u_i| = \frac{1}{h_i^2} \left| \frac{\partial \mathbf{r}}{\partial u_i} \right| = \frac{1}{h_i} = \frac{h_j h_k}{h_i h_j h_k}$$

Substituting (6.176) into it, we can finally obtain the gradient in the orthogonal coordinates

$$\nabla u_i = \frac{1}{J} \left(\frac{\partial \mathbf{r}}{\partial u_j} \times \frac{\partial \mathbf{r}}{\partial u_k} \right) \quad (6.177)$$

Since

$$\nabla u_i \parallel \frac{\partial \mathbf{r}}{\partial u_j} \times \frac{\partial \mathbf{r}}{\partial u_k}$$

Here, all configurations implicitly follow the right-hand rule, but we will not delve into the orientation details. Readers are advised to pay careful attention to the direction conventions. Furthermore, since the basis vectors are mutually orthogonal, we can easily project the differential arc length onto each basis direction, that is

$$ds_i = d\mathbf{r} \cdot \mathbf{e}_i = h_i du_i \quad (6.178)$$

Therefore, the total arc length element is simply the combined contribution from all three coordinate directions

$$ds^2 = h_1^2 du_1^2 + h_2^2 du_2^2 + h_3^2 du_3^2 \quad (6.179)$$

Note that this projection-based approach for identifying the contribution along each coordinate direction is only valid in orthogonal coordinate systems. In the general curvilinear coordinates, the basis vectors are not necessarily mutually orthogonal, meaning that each projection may involve contributions from other directions due to the angles between the basis vectors. This is precisely why the Riemannian metric must be introduced to properly account for all directional contributions through the inner products of the basis vectors.

Finally, due to the orthogonality of the coordinate system, where the tangent vectors are mutually perpendicular, the expression for the surface element becomes significantly simpler. We have

$$dS_i = \left| \frac{\partial \mathbf{r}}{\partial u_j} \times \frac{\partial \mathbf{r}}{\partial u_k} \right| du_j du_k = h_j h_k du_j du_k \quad (6.180)$$

6.9.3 Gradient in Orthogonal Curvilinear Coordinates

Just like a direction vector, the function $f(x_1, x_2, x_3)$ can be denoted by

$$f(x_1(u_1, u_2, u_3), x_2(u_1, u_2, u_3), x_3(u_1, u_2, u_3))$$

The gradient of the function f is

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \frac{\partial f}{\partial x_3} \right)$$

Since we are working within an orthogonal coordinate system, the gradient can be directly projected onto each basis vector $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ to express its components along each coordinate direction, that is

$$\begin{aligned} \nabla f \cdot \mathbf{e}_1 &= \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \frac{\partial f}{\partial x_3} \right) \cdot \frac{1}{h_1} \frac{\partial \mathbf{r}}{\partial u_1} \\ &= \frac{1}{h_1} \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \frac{\partial f}{\partial x_3} \right) \cdot \left(\frac{\partial x_1}{\partial u_1}, \frac{\partial x_2}{\partial u_1}, \frac{\partial x_3}{\partial u_1} \right) \\ &= \frac{1}{h_1} \left(\frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial u_1} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial u_1} + \frac{\partial f}{\partial x_3} \frac{\partial x_3}{\partial u_1} \right) \end{aligned}$$

By the Cartesian expression for the gradient, we can rewrite the dot product as

$$\nabla f \cdot \mathbf{e}_1 = \frac{1}{h_1} \frac{\partial f}{\partial u_1} \quad (6.181)$$

Similarly,

$$\begin{cases} \nabla f \cdot \mathbf{e}_2 = \frac{1}{h_2} \frac{\partial f}{\partial u_2} \\ \nabla f \cdot \mathbf{e}_3 = \frac{1}{h_3} \frac{\partial f}{\partial u_3} \end{cases} \quad (6.182)$$

In this way, we have identified the components of the gradient along the three basis vectors, so the gradient in orthogonal curvilinear coordinates can be denoted by

$$\nabla f = \left(\frac{1}{h_1} \frac{\partial f}{\partial u_1}, \frac{1}{h_2} \frac{\partial f}{\partial u_2}, \frac{1}{h_3} \frac{\partial f}{\partial u_3} \right) = \sum_i \frac{1}{h_i} \frac{\partial f}{\partial u_i} \mathbf{e}_i \quad (6.183)$$

In curvilinear coordinates, the gradient operator is rewritten as

$$\nabla = \left(\frac{1}{h_1} \frac{\partial}{\partial u_1}, \frac{1}{h_2} \frac{\partial}{\partial u_2}, \frac{1}{h_3} \frac{\partial}{\partial u_3} \right) \quad (6.184)$$

The following presents the gradient operator in commonly used curvilinear coordinate systems.

1. Cylindrical Coordinates:

The position vector $\mathbf{r} = (r \cos \phi, r \sin \phi, z)$, then the tangent vectors and their scale factors are

$$\begin{cases} \frac{\partial \mathbf{r}}{\partial r} = (\cos \phi, \sin \phi, 0) \Rightarrow h_r = \left| \frac{\partial \mathbf{r}}{\partial r} \right| = 1 \\ \frac{\partial \mathbf{r}}{\partial \phi} = (-r \sin \phi, r \cos \phi, 0) \Rightarrow h_\phi = \left| \frac{\partial \mathbf{r}}{\partial \phi} \right| = r \\ \frac{\partial \mathbf{r}}{\partial z} = (0, 0, 1) \Rightarrow h_z = \left| \frac{\partial \mathbf{r}}{\partial z} \right| = 1 \end{cases}$$

Substituting these into (6.183), we can finally obtain

$$\nabla f = \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi + \frac{\partial f}{\partial z} \mathbf{e}_z \quad (6.185)$$

2. Spherical Coordinates:

The position vector $\mathbf{r} = (r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta)$, then the tangent vectors and their scale factors are

$$\begin{cases} \frac{\partial \mathbf{r}}{\partial r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \Rightarrow h_r = \left| \frac{\partial \mathbf{r}}{\partial r} \right| = 1 \\ \frac{\partial \mathbf{r}}{\partial \theta} = (r \cos \theta \cos \phi, r \cos \theta \sin \phi, -r \sin \theta) \Rightarrow h_\theta = \left| \frac{\partial \mathbf{r}}{\partial \theta} \right| = r \\ \frac{\partial \mathbf{r}}{\partial \phi} = (-r \sin \theta \sin \phi, r \sin \theta \cos \phi, 0) \Rightarrow h_\phi = \left| \frac{\partial \mathbf{r}}{\partial \phi} \right| = r \sin \theta \end{cases}$$

Substituting these into (6.183), we can finally obtain

$$\nabla f = \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi \quad (6.186)$$

6.9.4 Divergence in Orthogonal Coordinate Systems

Consider a vector field $\mathbf{F} = (F_1, F_2, F_3)$ in the orthogonal coordinate systems, which can be denoted by

$$\mathbf{F} = (F_1(u_1, u_2, u_3), F_2(u_1, u_2, u_3), F_3(u_1, u_2, u_3)) \quad (6.187)$$

Then the divergence of $\mathbf{F}$ becomes

$$\nabla \cdot \mathbf{F} = \nabla \cdot (F_1 \mathbf{e}_1) + \nabla \cdot (F_2 \mathbf{e}_2) + \nabla \cdot (F_3 \mathbf{e}_3) \quad (6.188)$$

By (6.175), we've obtained

$$\begin{cases} \nabla u_1 = \frac{1}{h_1} \mathbf{e}_1 \Rightarrow \mathbf{e}_1 = h_1 \nabla u_1 \\ \nabla u_2 = \frac{1}{h_2} \mathbf{e}_2 \Rightarrow \mathbf{e}_2 = h_2 \nabla u_2 \\ \nabla u_3 = \frac{1}{h_3} \mathbf{e}_3 \Rightarrow \mathbf{e}_3 = h_3 \nabla u_3 \end{cases} \quad (6.189)$$

Since $\mathbf{e}_1 = \mathbf{e}_2 \times \mathbf{e}_3$, by (6.189), we can verify that

$$\mathbf{e}_1 = h_2 h_3 (\nabla u_2 \times \nabla u_3) \Leftrightarrow \frac{1}{h_2 h_3} \mathbf{e}_1 = \nabla u_2 \times \nabla u_3 \quad (6.190)$$

Then we can discuss the first term on the right side of (6.188) through (6.190), we have

$$\begin{aligned} \nabla \cdot (F_1 \mathbf{e}_1) &= \nabla \cdot [F_1 h_2 h_3 (\nabla u_2 \times \nabla u_3)] \\ &= \nabla (F_1 h_2 h_3) \cdot (\nabla u_2 \times \nabla u_3) + F_1 h_2 h_3 \nabla \cdot (\nabla u_2 \times \nabla u_3) \\ &= \nabla (F_1 h_2 h_3) \cdot (\nabla u_2 \times \nabla u_3) \\ &= \nabla (F_1 h_2 h_3) \cdot \frac{1}{h_2 h_3} \mathbf{e}_1 \end{aligned}$$

Substituting the gradient into the equation, we can express that

$$\left[\frac{1}{h_1} \frac{\partial (F_1 h_2 h_3)}{\partial u_1} \mathbf{e}_1 + \frac{1}{h_2} \frac{\partial (F_1 h_2 h_3)}{\partial u_2} \mathbf{e}_2 + \frac{1}{h_3} \frac{\partial (F_1 h_2 h_3)}{\partial u_3} \mathbf{e}_3 \right] \cdot \frac{1}{h_2 h_3} \mathbf{e}_1 \quad (6.191)$$

Since $\mathbf{e}_1 \cdot \mathbf{e}_2 = 0$, $\mathbf{e}_1 \cdot \mathbf{e}_3 = 0$, we can obtain the finally result

$$\nabla \cdot (F_1 \mathbf{e}_1) = \frac{1}{h_1 h_2 h_3} \frac{\partial (F_1 h_2 h_3)}{\partial u_1} \quad (6.192)$$

Similarly,

$$\begin{cases} \nabla \cdot (F_2 \mathbf{e}_2) = \frac{1}{h_1 h_2 h_3} \frac{\partial(F_2 h_1 h_3)}{\partial u_2} \\ \nabla \cdot (F_3 \mathbf{e}_3) = \frac{1}{h_1 h_2 h_3} \frac{\partial(F_3 h_1 h_2)}{\partial u_3} \end{cases} \quad (6.193)$$

So the divergence in orthogonal curvilinear coordinates can be denoted by

$$\nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(F_1 h_2 h_3)}{\partial u_1} + \frac{\partial(F_2 h_1 h_3)}{\partial u_2} + \frac{\partial(F_3 h_1 h_2)}{\partial u_3} \right] \quad (6.194)$$

1. Cylindrical Coordinates:

$$\nabla \cdot \mathbf{F} = \frac{1}{r} \frac{\partial(r F_r)}{\partial r} + \frac{1}{r} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} \quad (6.195)$$

2. Spherical Coordinates:

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial(r^2 F_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(F_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi} \quad (6.196)$$

6.9.5 Curl in Orthogonal Coordinate Systems

Consider a vector field $\mathbf{F}(F_1, F_2, F_3)$, as before, we begin by discussing the first term $\nabla \times (F_1 \mathbf{e}_1)$. By (6.189), we substitute $\mathbf{e}_1 = h_1 \nabla u_1$ into the curl, we obtain

$$\begin{aligned} \nabla \times (F_1 \mathbf{e}_1) &= \nabla \times (F_1 h_1 \nabla u_1) \\ &= \nabla(F_1 h_1) \times \nabla u_1 + (F_1 h_1) \nabla \times \nabla u_1 \\ &= \nabla(F_1 h_1) \times \nabla u_1 \end{aligned}$$

Substituting the gradient into the equation, we can express that

$$\left(\frac{1}{h_1} \frac{\partial(F_1 h_1)}{\partial u_1} \mathbf{e}_1 + \frac{1}{h_2} \frac{\partial(F_1 h_1)}{\partial u_2} \mathbf{e}_2 + \frac{1}{h_3} \frac{\partial(F_1 h_1)}{\partial u_3} \mathbf{e}_3 \right) \times \frac{1}{h_1} \mathbf{e}_1 \quad (6.197)$$

Therefore, the curl

$$\nabla \times (F_1 \mathbf{e}_1) = \frac{1}{h_1 h_3} \frac{\partial(F_1 h_1)}{\partial u_3} \mathbf{e}_2 - \frac{1}{h_1 h_2} \frac{\partial(F_1 h_1)}{\partial u_2} \mathbf{e}_3 \quad (6.198)$$

Similarly,

$$\begin{cases} \nabla \times (F_2 \mathbf{e}_2) = \frac{1}{h_1 h_2} \frac{\partial(F_2 h_2)}{\partial u_1} \mathbf{e}_3 - \frac{1}{h_2 h_3} \frac{\partial(F_2 h_2)}{\partial u_3} \mathbf{e}_1 \\ \nabla \times (F_3 \mathbf{e}_3) = \frac{1}{h_2 h_3} \frac{\partial(F_3 h_3)}{\partial u_2} \mathbf{e}_1 - \frac{1}{h_1 h_3} \frac{\partial(F_3 h_3)}{\partial u_1} \mathbf{e}_2 \end{cases} \quad (6.199)$$

Based on the calculations above, we can express the curl using a determinant, written as follows

$$\nabla \times \mathbf{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{e}_1 & h_2 \mathbf{e}_2 & h_3 \mathbf{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ F_1 h_1 & F_2 h_2 & F_3 h_3 \end{vmatrix} \quad (6.200)$$

1. Cylindrical Coordinates:

The curl in the cylindrical coordinates is

$$\nabla \times \mathbf{F} = \frac{1}{r} \begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\phi & \mathbf{e}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_r & rF_\phi & F_z \end{vmatrix}$$

Expanding this determinant, we obtain

$$\begin{aligned} \nabla \times \mathbf{F} = & \frac{1}{r} \left(\frac{\partial F_z}{\partial \phi} - r \frac{\partial F_\phi}{\partial z} \right) \mathbf{e}_r + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r} \right) \mathbf{e}_\phi \\ & + \frac{1}{r} \left(\frac{\partial(rF_\phi)}{\partial r} - \frac{\partial F_r}{\partial \phi} \right) \mathbf{e}_z \end{aligned} \quad (6.201)$$

2. Spherical Coordinates:

The curl in the cylindrical coordinates is

$$\nabla \times \mathbf{F} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & r \sin \theta \mathbf{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ F_r & rF_\theta & r \sin \theta F_\phi \end{vmatrix}$$

Expanding this determinant, we obtain

$$\begin{aligned}
\nabla \times \mathbf{F} = & \frac{1}{r \sin \theta} \left[\frac{\partial(\sin \theta F_\phi)}{\partial \theta} - \frac{\partial F_\theta}{\partial \phi} \right] \mathbf{e}_r \\
& + \frac{1}{r \sin \theta} \left[\frac{\partial F_r}{\partial \phi} - \sin \theta \frac{\partial(r F_\phi)}{\partial r} \right] \mathbf{e}_\theta \\
& + \frac{1}{r} \left[\frac{\partial(r F_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right] \mathbf{e}_\phi
\end{aligned} \tag{6.202}$$

Chapter 7

Introduction to Tensor

Abstract In classical mechanics, the formal introduction of tensors is undoubtedly one of the most bizarre tools you will encounter. Nevertheless, you have actually been exposed to tensors since elementary school (no one ever told you so). In Chapter 10 of classical mechanics, the **inertia tensor** will be introduced explicitly. However, based on previous teaching experience, many students still finish the semester with little to no concrete understanding of what a tensor really is. Therefore, in order to make Chapter 10 as smooth as possible for everyone, I have decided to include a very brief introduction to the basic definitions and properties of tensors. If you are interested, you are strongly encouraged to consult more specialized textbooks for a deeper treatment. This note will also include concrete examples of the inertia tensor discussed in Chapter 10 to help you develop a more intuitive and structural understanding. Finally, please note that throughout this note, we will exclusively use the Einstein summation convention. If you are not familiar with this notation, please refer to Chapter 2.

Some Conventions in this Chapter

1. The component of a (contravariant) vector in 3-D Euclidean space: x^i, v^i
2. The component of a (contravariant) vector in 4-D (Euclidean) Minkowski time space: x^μ, η^μ
3. Vector representation in 3-D space: $\mathbf{x}, \mathbf{v}$
4. Vector representation in 4-D space: $\vec{x}^\mu, \vec{\eta}^\mu$
5. Vector representation in matrix form in 4-D space: x^μ, η^μ
6. **Einstein Summation Convention** is used in this chapter.
7. The inner (dot) products in 3-D and 4-D are

$$\begin{cases} \text{3-D : } \langle \mathbf{x}, \mathbf{y} \rangle = x^i \delta_{ij} y^j = x^i y^i \\ \text{4-D : } \langle \vec{x}^\mu, \vec{y}^\mu \rangle = x^\mu \delta^\nu_\mu y_\nu = x^\mu y_\mu = -x^0 y^0 + x^i y^i \end{cases}$$

7.1 Why Tensors Appear

The first thing to understand is that a tensor is not merely a strange object with many indices. A tensor is a mathematical object whose components change in a controlled way when we change coordinates. The object itself is geometric, but the numbers used to describe it depend on the basis chosen by the observer. For example, a vector $\mathbf{A}$ in three dimensional Euclidean space can be written as

$$\mathbf{A} = A_i \mathbf{e}_i. \quad (7.1)$$

The numbers A_i are the components of the vector. If we rotate the coordinate axes, the vector itself does not change, but its components do. This is the basic philosophy behind tensor notation.

A tensor is a geometric object whose components obey a definite transformation rule under a change of coordinates.

The order of a tensor is determined by the number of indices it carries. A scalar has no index, a vector has one index, and a rank two tensor has two indices. More generally, a tensor with p upper indices and q lower indices is called a tensor of type (p, q) .

$$\left\{ \begin{array}{ll} \text{Scalar} & f \\ \text{Vector} & A^i, \quad A_i \\ \text{Rank two tensor} & T^{ij}, \quad T_{ij}, \quad T^i_j \\ \text{General tensor} & T^{i_1 \dots i_p}_{j_1 \dots j_q} \end{array} \right. \quad (7.2)$$

A scalar is the simplest tensor because its value does not change under coordinate transformations. Examples are mass, temperature, kinetic energy, and the squared length $\mathbf{A} \cdot \mathbf{A}$. Vectors are the next level. Examples include position, velocity, force, angular momentum, and electric field. Rank two tensors appear naturally whenever a linear map sends one vector to another vector. The inertia tensor is one of the most important examples in classical mechanics.

7.2 Einstein Summation Convention

The Einstein summation convention is a notation that removes summation signs when an index is repeated once upstairs and once downstairs. (It doesn't matter whether up or down in normal Euclidean space) In this chapter, repeated indices are summed over automatically.

For a three-dimensional vector,

$$A^i B_i = \sum_{i=1}^3 A^i B_i = A^1 B_1 + A^2 B_2 + A^3 B_3. \quad (7.3)$$

$$A_i B_i = \sum_{i=1}^3 A_i B_i = A_1 B_1 + A_2 B_2 + A_3 B_3. \quad (7.4)$$

$$A^i B^i = \sum_{i=1}^3 A^i B^i = A^1 B^1 + A^2 B^2 + A^3 B^3. \quad (7.5)$$

For a four vector in Minkowski space:

$$A^\mu B_\mu = \sum_{\mu=0}^3 A^\mu B_\mu = A^0 B_0 + A^1 B_1 + A^2 B_2 + A^3 B_3. \quad (7.6)$$

In this note, Latin indices i, j, k, ℓ usually run through 1, 2, 3, while Greek indices μ, ν, ρ, σ usually run through 0, 1, 2, 3.

There are two important types of indices. A free index appears on both sides of an equation and is not summed. A dummy index is summed and can be renamed freely. For example,

$$C^i = T^i_j A^j \quad (7.7)$$

has free index i and dummy index j . The same equation can also be written as

$$C^i = T^i_k A^k. \quad (7.8)$$

The letter used for a dummy index has no meaning by itself. By contrast,

$$C^i = T^j_i A^j \quad (7.9)$$

is a different equation because the free index has changed position inside the tensor.

A repeated dummy index should appear exactly twice in one term. Expressions such as $A_i B_i C_i$ are not valid Einstein notation unless a special convention is explicitly stated.

A typical contraction looks like

$$T^{\mu\nu} Q_\nu{}^{\alpha\beta} = \Gamma^{\mu\alpha\beta}. \quad (7.10)$$

The index ν is summed over. The remaining indices μ, α, β are free indices, so the result is a rank three tensor. For a more complicated example,

$$A^{\mu\nu} B_\mu{}^\alpha C_\nu{}^\beta D_{\alpha\beta} E_\lambda \quad (7.11)$$

has dummy indices μ, ν, α, β and free index λ . Therefore, the whole expression is a covariant vector.

7.3 Euclidean Space and the Kronecker Delta

In ordinary three-dimensional Euclidean space, the metric tensor is the Kronecker delta

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \quad (7.12)$$

It measures ordinary length and ordinary angle. The inner product of two vectors is

$$\mathbf{A} \cdot \mathbf{B} = A^i \delta_{ij} B^j = A^i B_i = A_i B^i. \quad (7.13)$$

In Cartesian coordinates, raising and lowering an index does not change the numerical value of the component because

$$A_i = \delta_{ij} A^j = A^i. \quad (7.14)$$

This is why in elementary vector calculus we usually do not distinguish A_i and A^i . The distinction becomes essential in curvilinear coordinates and in Minkowski space.

The squared length of a vector may be written in several equivalent ways:

$$|\mathbf{A}|^2 = A_i \delta_{ij} A_j = A_i A_i = A^i A^i = A^i A_i. \quad (7.15)$$

In three dimensional Cartesian coordinates, all these forms give the same number. However, the expression with δ_{ij} is the most structurally honest one because it shows that the metric is being used.

The Kronecker delta also acts as the identity matrix:

$$\delta^i_j A^j = A^i, \quad \delta_{ij} A^j = A_i, \quad \delta^{ij} A_j = A^i. \quad (7.16)$$

7.4 Antisymmetrized Pseudo Tensor

The Levi Civita symbol ϵ_{ijk} , which is also called the **Antisymmetrized tensor** is defined by

$$\epsilon_{ijk} = \begin{cases} +1, & (i, j, k) \text{ is an even permutation of } (1, 2, 3), \\ -1, & (i, j, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0, & \text{two indices are equal.} \end{cases} \quad (7.17)$$

It is completely antisymmetric, meaning that exchanging any two indices changes the sign:

$$\epsilon_{ijk} = -\epsilon_{jik} = -\epsilon_{ikj}. \quad (7.18)$$

The Levi Civita symbol is called a pseudo tensor because it changes sign under an orientation reversing transformation. For a proper rotation $R \in SO(3)$, we have $\det R = 1$, and

$$R_{ia}R_{jb}R_{kc}\epsilon_{abc} = \epsilon_{ijk}. \quad (7.19)$$

For an improper orthogonal transformation with $\det R = -1$,

$$R_{ia}R_{jb}R_{kc}\epsilon_{abc} = -\epsilon_{ijk}. \quad (7.20)$$

This is why cross products are axial vectors or pseudo vectors. Under reflection, an ordinary vector changes differently from a pseudo vector.

Antisymmetrization is a compact way to extract the antisymmetric part of a tensor. For two indices,

$$\boxed{T_{ij} = \frac{1}{2}(T_{ij} - T_{ji}).} \quad (7.21)$$

The symmetric part is

$$T_{ij} = \frac{1}{2}(T_{ij} + T_{ji}). \quad (7.22)$$

For a vector field A_i , the antisymmetric derivative is

$$\partial_i A_j = \frac{1}{2}(\partial_i A_j - \partial_j A_i). \quad (7.23)$$

In three dimensions, this antisymmetric rank two object is equivalent to a pseudo vector:

$$B^k = \epsilon^{kij} \partial_i A_j. \quad (7.24)$$

This is the index form of

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (7.25)$$

In general, sometimes you can denote:

$$\mathbf{B} = \hat{x}_i \epsilon^{ijk} \partial_j A_k = \hat{x}^i \epsilon_{ijk} \partial^j A^k$$

The cross product $\mathbf{A} \times \mathbf{B}$ will be regarded as a vector only in 3D coordinates. The determinant of a 3×3 rotational matrix A^i_j can be denoted by

$$\det A = \epsilon_{ijk} A^1_i A^2_j A^3_k \Rightarrow \epsilon_{\alpha\beta\gamma} \det A = \epsilon_{ijk} A^\alpha_i A^\beta_j A^\gamma_k$$

In the Cartesian coordinate system, the matrix is orthogonal, so $\det A = \pm 1$. We obtain

$$\epsilon_{\alpha\beta\gamma} = (\det A) A_\alpha^i A_\beta^j A_\gamma^k \epsilon_{ijk}$$

So it is not a tensor; we call it **pseudotensor** in the Cartesian coordinate system. The definitions of the cross product and curl are given by

$$\begin{cases} \mathbf{A} \times \mathbf{B} = \hat{x}^i \epsilon_{ijk} A^j B^k \\ \nabla \times \mathbf{A} = \hat{x}^i \epsilon_{ijk} \partial^j A^k \end{cases}$$

Under a coordinate transformation $\hat{x}'^i = R^i_j \hat{x}^j$, $A'^i = R^i_j A^j$, $B'^i = R^i_j B^j$, we compute

$$\mathbf{A}' \times \mathbf{B}' = \hat{x}'^i \epsilon'_{ijk} A'^j B'^k.$$

Using the transformation rule for the Levi-Civita tensor, $\epsilon'_{ijk} = (\det R) R_i^\alpha R_j^\beta R_k^\gamma \epsilon_{\alpha\beta\gamma}$, we substitute all expressions to obtain

$$\begin{aligned} \mathbf{A}' \times \mathbf{B}' &= (\det R) R^i_\sigma R_i^\alpha R_j^\beta R_k^\gamma R^j_s R^k_t \hat{x}^\sigma \epsilon_{\alpha\beta\gamma} A^s B^t \\ &= (\det R) \delta_\sigma^\alpha \delta_s^\beta \delta_t^\gamma \hat{x}^\sigma \epsilon_{\alpha\beta\gamma} A^s B^t = (\det R) \hat{x}^\alpha \epsilon_{\alpha\beta\gamma} A^\beta B^\gamma \end{aligned}$$

Note that the orthogonality gives the Kronecker delta, hence the final result is

$$\mathbf{A}' \times \mathbf{B}' = (\det R) \hat{x}^\alpha \epsilon_{\alpha\beta\gamma} A^\beta B^\gamma = (\det R) \mathbf{A} \times \mathbf{B}. \quad (7.26)$$

7.5 Tensor Transformation Laws

Let the coordinates transform linearly as

$$x'^i = R^i_j x^j. \quad (7.27)$$

A contravariant vector transforms in the same way:

$$A'^i = R^i_j A^j. \quad (7.28)$$

A covariant vector transforms with the inverse transformation:

$$A'_i = (R^{-1})^j_i A_j. \quad (7.29)$$

For a rotation matrix, $R^{-1} = R^T$, so in Euclidean Cartesian coordinates this distinction is often hidden.

A rank two contravariant tensor transforms as

$$T'^{ij} = R^i_k R^j_\ell T^{k\ell}. \quad (7.30)$$

A rank two covariant tensor transforms as

$$T'_{ij} = (R^{-1})^k_i (R^{-1})^\ell_j T_{k\ell}. \quad (7.31)$$

A mixed tensor transforms as

$$T'^i_j = R^i_k (R^{-1})^\ell_j T^k_\ell. \quad (7.32)$$

For a general coordinate transformation $x^i \mapsto x'^i(x)$, the transformation laws are written with Jacobians:

$$A'^i = \frac{\partial x'^i}{\partial x^j} A^j, \quad A'_i = \frac{\partial x^j}{\partial x'^i} A_j. \quad (7.33)$$

A general tensor of type (p, q) transforms as

$$T'^{i_1 \dots i_p}_{j_1 \dots j_q} = \frac{\partial x'^{i_1}}{\partial x^{a_1}} \dots \frac{\partial x'^{i_p}}{\partial x^{a_p}} \frac{\partial x^{b_1}}{\partial x'^{j_1}} \dots \frac{\partial x^{b_q}}{\partial x'^{j_q}} T^{a_1 \dots a_p}_{b_1 \dots b_q}. \quad (7.34)$$

The reason tensor notation is powerful is that tensor equations are coordinate-independent. If

$$A^i = B^i \quad (7.35)$$

is true in one coordinate system, then it is true in every coordinate system. The same is true for

$$T^i_j = S^i_j. \quad (7.36)$$

7.6 The Inertia Tensor

In rigid body motion, the kinetic energy due to rotation is

$$T_{\text{rot}} = \frac{1}{2} I_{ij} \omega_i \omega_j. \quad (7.37)$$

Here ω_i are the components of angular velocity, and I_{ij} is the inertia tensor.

For point masses m_a at positions $\mathbf{r}_a$, the inertia tensor is

$$I_{ij} = \sum_a m_a (r_a^2 \delta_{ij} - x_{a,i} x_{a,j}). \quad (7.38)$$

For a continuous body,

$$I_{ij} = \int \rho(\mathbf{r}) (r^2 \delta_{ij} - x_i x_j) d^3x. \quad (7.39)$$

This formula looks strange at first, but it comes directly from the velocity of a point in a rotating rigid body:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}. \quad (7.40)$$

The rotational kinetic energy is

$$\begin{aligned}
T_{\text{rot}} &= \frac{1}{2} \sum_a m_a (\boldsymbol{\omega} \times \mathbf{r}_a) \cdot (\boldsymbol{\omega} \times \mathbf{r}_a) \\
&= \frac{1}{2} \sum_a m_a \epsilon_{ijk} \omega_j x_{a,k} \epsilon_{ilm} \omega_l x_{a,m} \\
&= \frac{1}{2} \sum_a m_a (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) \omega_j \omega_l x_{a,k} x_{a,m} \\
&= \frac{1}{2} \sum_a m_a (r_a^2 \omega_j \omega_j - \omega_j x_{a,j} \omega_l x_{a,l}) \\
&= \frac{1}{2} \left[\sum_a m_a (r_a^2 \delta_{jj} - x_{a,j} x_{a,j}) \right] \omega_j \omega_j.
\end{aligned} \tag{7.41}$$

Therefore,

$$T_{\text{rot}} = \frac{1}{2} I_{ij} \omega_i \omega_j. \tag{7.42}$$

The inertia tensor has several important mathematical properties.

1. **Symmetry.** Since $r^2 \delta_{ij} - x_i x_j = r^2 \delta_{ji} - x_j x_i$, we have

$$I_{ij} = I_{ji}. \tag{7.43}$$

2. **Positive semidefiniteness.** For any angular velocity ω_i ,

$$\omega_i I_{ij} \omega_j = \sum_a m_a |\boldsymbol{\omega} \times \mathbf{r}_a|^2 \geq 0. \tag{7.44}$$

Thus the rotational kinetic energy is always nonnegative.

3. **Orthogonal diagonalization.** Since I_{ij} is real and symmetric, there exists an orthogonal matrix R such that

$$R I R^T = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}. \tag{7.45}$$

The directions of the eigenvectors are called the principal axes, and the eigenvalues I_1, I_2, I_3 are called principal moments of inertia.

4. **Angular momentum relation.** The angular momentum of a rigid body is

$$L_i = I_{ij} \omega_j. \tag{7.46}$$

In general, $\mathbf{L}$ is not parallel to $\boldsymbol{\omega}$ unless the rotation is along a principal axis.

The tensor transformation law for the inertia tensor under a rotation is

$$I'_{ij} = R_{ik} R_{j\ell} I_{k\ell}. \quad (7.47)$$

In matrix notation, this is written as

$$I' = R I R^T. \quad (7.48)$$

This is exactly the transformation law of a rank two covariant tensor in Euclidean space. The parallel axis theorem can also be written tensorially. If the origin is shifted by a vector a_i from the center of mass, then

$$I'_{ij} = I_{ij}^{\text{cm}} + M(a^2 \delta_{ij} - a_i a_j). \quad (7.49)$$

This formula is cleaner than three separate scalar formulas because it keeps the full tensor structure.

7.7 Rotations as Matrices and Antisymmetric Generators

A rotation matrix R in three dimensional Euclidean space satisfies

$$R^T R = I, \quad \det R = 1. \quad (7.50)$$

The condition $R^T R = I$ means that inner products are preserved:

$$\mathbf{A}' \cdot \mathbf{B}' = A'_i B'_i = R_{ij} A_j R_{ik} B_k = (R^T R)_{jk} A_j B_k = A_j B_j.$$

For an infinitesimal rotation,

$$R_{ij} = \delta_{ij} + \Omega_{ij}, \quad (7.51)$$

where Ω_{ij} is very small. Using $R^T R = I$ gives

$$(\delta_{ki} + \Omega_{ki})(\delta_{kj} + \Omega_{kj}) = \delta_{ij}. \quad (7.52)$$

Keeping only first-order terms,

$$\Omega_{ij} + \Omega_{ji} = 0. \quad (7.53)$$

Therefore,

$$\Omega_{ij} = -\Omega_{ji}. \quad (7.54)$$

The generator of an infinitesimal rotation is antisymmetric.

In three dimensions, every antisymmetric rank two tensor is equivalent to a pseudo vector. We can write

$$\Omega_{ij} = -\epsilon_{ijk} \theta_k. \quad (7.55)$$

The infinitesimal change of a vector is then

$$\delta A_i = \Omega_{ij} A_j = -\epsilon_{ijk} \theta_k A_j. \quad (7.56)$$

Equivalently,

$$\delta \mathbf{A} = \boldsymbol{\theta} \times \mathbf{A}. \quad (7.57)$$

This is the mathematical origin of why angular velocity is naturally represented as a pseudo vector in three dimensions.

7.8 From Euclidean Space to Minkowski Space

In special relativity (to comprehend more physics, see Appendix D), space and time are combined into a four dimensional object called spacetime. The spacetime coordinate is written as

$$\vec{x}^\mu = (x^0, x^1, x^2, x^3) = (ct, x, y, z). \quad (7.58)$$

The metric of Minkowski space is

$$\eta = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (7.59)$$

Thus

$$\eta = \text{diag}(-1, 1, 1, 1). \quad (7.60)$$

The spacetime interval is

$$ds^2 = dx^\mu \eta_{\mu\nu} dx^\nu = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \quad (7.61)$$

This is the relativistic analog of the Euclidean distance formula

$$d\ell^2 = dx^i \delta_{ij} dx^j. \quad (7.62)$$

In Euclidean Cartesian coordinates, raising and lowering indices usually does not visibly change signs. In Minkowski space, it does. For a four vector $A^\mu = (A^0, A^1, A^2, A^3)$,

$$A_\mu = \eta_{\mu\nu} A^\nu. \quad (7.63)$$

Therefore,

$$\vec{A}_\mu = (-A^0, A^1, A^2, A^3). \quad (7.64)$$

The inner product of two 4-vectors is

$$A_\mu B^\mu = \eta_{\mu\nu} A^\mu B^\nu = -A^0 B^0 + A^1 B^1 + A^2 B^2 + A^3 B^3. \quad (7.65)$$

For the spacetime coordinate,

$$x_\mu x^\mu = -c^2 t^2 + x^2 + y^2 + z^2. \quad (7.66)$$

This quantity is invariant under Lorentz transformations. The four derivatives are

$$\vec{\partial}_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right). \quad (7.67)$$

Raising the index gives

$$\vec{\partial}^\mu = \eta^{\mu\nu} \partial_\nu = \left(-\frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right). \quad (7.68)$$

7.9 Lorentz Transformations

A Lorentz transformation is a linear transformation

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu \quad (7.69)$$

that preserves the Minkowski interval:

$$dx^\mu \eta_{\mu\nu} dx^\nu = dx'^\mu \eta_{\mu\nu} dx'^\nu. \quad (7.70)$$

Substituting $x'^\mu = \Lambda^\mu{}_\alpha x^\alpha$ gives

$$\begin{aligned} dx'^\mu \eta_{\mu\nu} dx'^\nu &= (\Lambda^\mu{}_\alpha dx^\alpha) \eta_{\mu\nu} (\Lambda^\nu{}_\beta dx^\beta) \\ &= dx^\alpha \Lambda_\alpha{}^\mu \eta_{\mu\nu} \Lambda^\nu{}_\beta dx^\beta = dx^\alpha \eta_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta dx^\beta \end{aligned}$$

For this to be true for all x^α , we need

$$\boxed{\eta_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta = \eta_{\alpha\beta}.} \quad (7.71)$$

In matrix form,

$$\boxed{\Lambda^T \eta \Lambda = \eta.} \quad (7.72)$$

For a boost along the x direction,

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}}. \quad (7.73)$$

The coordinate transformation is

$$t' = \gamma \left(t - \frac{vx}{c^2} \right), \quad x' = \gamma(x - vt), \quad y' = y, \quad z' = z. \quad (7.74)$$

In conclusion, contravariant four vectors transform as

$$A'^{\mu} = \Lambda^{\mu}_{\nu} A^{\nu}. \quad (7.75)$$

Covariant four vectors transform as

$$A'_{\mu} = (\Lambda^{-1})^{\nu}_{\mu} A_{\nu}. \quad (7.76)$$

Rank two tensors transform according to their index structure:

$$T'^{\mu\nu} = \Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta} T^{\alpha\beta}; \quad (7.77)$$

$$T'_{\mu\nu} = (\Lambda^{-1})^{\alpha}_{\mu} (\Lambda^{-1})^{\beta}_{\nu} T_{\alpha\beta}; \quad (7.78)$$

$$T'^{\mu}_{\nu} = \Lambda^{\mu}_{\alpha} (\Lambda^{-1})^{\beta}_{\nu} T^{\alpha}_{\beta}. \quad (7.79)$$

The main difference between Euclidean rotations and Lorentz transformations is the metric they preserve. Rotations preserve δ_{ij} , while Lorentz transformations preserve $\eta_{\mu\nu}$.

7.10 Four Velocity and Four Momentum

The proper time τ is defined by

$$c^2 d\tau^2 = -ds^2 = -dx^{\mu}(\tau) \eta_{\mu\nu} dx^{\nu} \quad (7.80)$$

since

$$ds^2 = -c^2 t^2 + dx^2 + dy^2 + dz^2 < 0 \quad (7.81)$$

for the particle at the proper time.

Therefore, the proper time is:

$$\begin{aligned}
d\tau &= \frac{1}{c} \sqrt{-ds^2} = \frac{1}{c} \sqrt{c^2 dt^2 - dx^2 - dy^2 - dz^2} \\
&= dt \sqrt{1 - \frac{1}{c^2} \left[\underbrace{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2}_{v^2(t)} \right]} = dt \sqrt{1 - \frac{v^2}{c^2}},
\end{aligned}$$

where we define:

$$\gamma(v(t)) \equiv \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \geq 1, \quad \gamma^{-1} = \sqrt{1 - \frac{v^2}{c^2}} \leq 1 \quad (7.82)$$

Hence,

$$d\tau = \gamma^{-1} dt \quad (7.83)$$

$$\frac{d\tau}{dt} = \gamma^{-1} = \sqrt{1 - \frac{v^2}{c^2}} \quad (7.84)$$

$$\frac{dt}{d\tau} = \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \quad (7.85)$$

So, when people have different velocities, the time they feel will be different, so the mechanism must be modified:

1. **Velocity η :** The original velocity $v^i(t)$ is:

$$x^i(t) \mapsto v^i(t) = \frac{dx^i(t)}{dt}.$$

With respect to t , we need to introduce the Lorentz matrices Λ . Since $dx^0 = cdt$, we have

$$dt = \frac{1}{c} dx^0 \longrightarrow dt = \frac{1}{c} \Lambda^0_{\alpha} dx^{\alpha}$$

hence we obtain

$$v^{\mu}(t) = \frac{dx^{\mu}}{dt} = \frac{\Lambda^{\mu}_{\nu} dx^{\nu}}{\frac{1}{c} \Lambda^0_{\alpha} dx^{\alpha}} = \frac{\Lambda^{\mu}_{\nu} dx^{\nu}/dt}{\frac{1}{c} \Lambda^0_{\alpha} dx^{\alpha}/dt} = \frac{\Lambda^{\mu}_{\nu} v^{\nu}(t)}{\frac{1}{c} \Lambda^0_{\alpha} v^{\alpha}(t)}. \quad (7.86)$$

Now, with proper time, we defined the velocity η to be the differentiation of displacement with respect to the proper time:

$$x^{\mu}(\tau) \mapsto \boxed{\eta^{\mu}(\tau) = \frac{dx^{\mu}}{d\tau}}. \quad (7.87)$$

In addition, by (7.82), the relation between η and v is:

$$\begin{cases} \eta^0 = \frac{dx^0}{d\tau} = \frac{cdt}{d\tau} = c\gamma = \frac{c}{\sqrt{1-v^2/c^2}} \\ \eta^i = \frac{dx^i}{d\tau} = \frac{dx^i}{dt} \frac{dt}{d\tau} = \gamma v^i(t) = \frac{v^i}{\sqrt{1-v^2/c^2}}. \end{cases} \quad (7.88)$$

In Minkowski space, the inner product $\eta^\mu \eta_\mu$ is:

$$\eta^\mu \eta_\mu = \frac{dx^\mu}{d\tau} \underbrace{\eta_{\mu\nu}}_{\text{metric}} \frac{dx^\nu}{d\tau} = \frac{dx^\mu \eta_{\mu\nu} dx^\nu}{d\tau^2} = \frac{ds^2}{-\frac{1}{c^2} ds^2} = -c^2, \quad (7.89)$$

this yields:

$$\eta^\mu \eta_\mu + c^2 = 0, \quad (7.90)$$

and note that

$$\eta^\mu \eta_\mu = -(\eta^0)^2 + (\eta^i)^2. \quad (7.91)$$

Hence, the final result can be written as

$$\eta^i \eta^i + c^2 = (\eta^0)^2 \quad (7.92)$$

2. **Acceleration** α : Similarly,

$$\begin{cases} a^i(t) = \frac{dv^i}{dt} = \frac{d^2 x^i}{dt^2} \\ \alpha^\mu(\tau) = \frac{d\eta^\mu}{d\tau} = \frac{d^2 x^\mu}{d\tau^2}. \end{cases} \quad (7.93)$$

3. **Momentum** $\mathcal{P}$: Similarly,

$$\begin{cases} P^i(t) = m \frac{dx^i}{dt} \\ \mathcal{P}^\mu(\tau) = m \eta^\mu(\tau) = m \frac{dx^\mu}{d\tau}. \end{cases} \quad (7.94)$$

4. **Force** K : Similarly,

$$\begin{cases} F^i(t) = m a^i = \frac{dP^i}{dt} \\ K^\mu(\tau) = m \alpha^\mu(\tau) = \frac{d\mathcal{P}^\mu}{d\tau}. \end{cases} \quad (7.95)$$

5. **Energy**: By the modified momentum $\mathcal{P}$, we have:

$$\mathcal{P}^\mu = m\eta^\mu \Rightarrow \begin{cases} \mathcal{P}^0 = m\eta^0 = m\gamma(v(t))c \\ \mathcal{P}^i = m\eta^i = m\gamma v^i. \end{cases} \quad (7.96)$$

We can use Taylor expansion on $\mathcal{P}$ with respect to $v(t)$, i.e.

$$m\eta^0 = mc\gamma(v) = mc \left(1 + \frac{1}{2} \frac{v^2}{c^2} + \dots \right) = \frac{1}{c} \underbrace{\left(mc^2 + \frac{1}{2}mv^2 + \dots \right)}_{\substack{\text{Kinetic E} \\ =E}}$$

Therefore

$$\mathcal{P}^0 = \frac{E}{c}. \quad (7.97)$$

The inner product of momentum equals

$$\mathcal{P}^\mu \mathcal{P}_\mu = m^2 \eta^\mu \eta_\mu = -m^2 c^2 \quad (7.98)$$

by (7.90). Therefore

$$\boxed{\mathcal{P}^\mu \mathcal{P}_\mu + m^2 c^2 = 0} \quad (7.99)$$

Since $\mathcal{P}^\mu \mathcal{P}_\mu = -(\mathcal{P}^0)^2 + (\mathcal{P}^i)^2$, and $\mathcal{P}^0$ is given by (7.97). Hence

$$-(\mathcal{P}^0)^2 + (\mathcal{P}^i)^2 + m^2 c^2 = -\left(\frac{E}{c}\right)^2 + \mathcal{P}^i \mathcal{P}^i + m^2 c^2$$

which gives the energy in SR:

$$E^2 = |\mathbf{P}|^2 c^2 + m^2 c^4. \quad (7.100)$$

Note that the term $|\mathbf{P}|$ is the spatial term in Minkowski space; in convention, $\mathbf{P}$ is spatial, while P^μ includes the time term. The equation (7.100) is called **On-Shell Condition**. When the momentum of a particle is small enough, the mass-energy equation comes out:

$$E = mc^2. \quad (7.101)$$

We can substituting (7.97) into the modified force, we obtain

$$K^\mu = \frac{d\mathcal{P}^\mu}{d\tau}, \quad K^0 = \frac{d}{d\tau} \mathcal{P}^0 = \frac{d}{d\tau} \frac{E}{c} = \frac{1}{c} \frac{dE}{d\tau} = \frac{1}{c} \gamma \frac{dE}{dt}. \quad (7.102)$$

This is called **Relativistic Force**.

6. **Angular Momentum $\mathcal{L}$:** Angular momentum in classical mechanics is defined in 3D Euclidean space:

$$\mathbf{L} = \mathbf{r} \times \mathbf{P} = \hat{x}^i \epsilon_{ijk} x^j P^k. \quad (7.103)$$

Since the cross product will only be considered as a vector in 3D space, so in 4D Minkowski space, we need to redefine the angular momentum as

$$\mathcal{L}^{\mu\nu} \equiv x^\mu \mathcal{P}^\nu - x^\nu \mathcal{P}^\mu = \epsilon^{\mu\nu\lambda} \mathcal{L}_\lambda. \quad (7.104)$$

The angular velocity is similar to the result.

7.11 Gauge Transformation and Field Strength

The electromagnetic potential is naturally combined into a four-vector:

$$\vec{A}^\mu = \left(\frac{V}{c}, \mathbf{A} \right). \quad (7.105)$$

With the metric convention $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$,

$$\vec{A}_\mu = \left(-\frac{V}{c}, \mathbf{A} \right). \quad (7.106)$$

The electromagnetic field strength tensor is defined by

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu. \quad (7.107)$$

It is automatically antisymmetric:

$$F^{\mu\nu} = -F^{\nu\mu}. \quad (7.108)$$

In matrix form, using the convention above,

$$F^{\mu\nu} = \begin{pmatrix} 0 & E^1/c & E^2/c & E^3/c \\ -E^1/c & 0 & B_3 & -B_2 \\ -E^2/c & -B_3 & 0 & B_1 \\ -E^3/c & B_2 & -B_1 & 0 \end{pmatrix}. \quad (7.109)$$

Equivalently,

$$F^{0i} = \frac{E^i}{c}, \quad F^{ij} = \epsilon^{ijk} B_k. \quad (7.110)$$

A gauge transformation changes the potential without changing the physical fields:

$$A^\mu \mapsto A'^\mu = A^\mu + \partial^\mu \lambda. \quad (7.111)$$

Under this transformation,

$$\begin{aligned}
F'^{\mu\nu} &= \partial^\mu A'^\nu - \partial^\nu A'^\mu \\
&= \partial^\mu (A^\nu + \partial^\nu \lambda) - \partial^\nu (A^\mu + \partial^\mu \lambda) \\
&= \partial^\mu A^\nu - \partial^\nu A^\mu + \partial^\mu \partial^\nu \lambda - \partial^\nu \partial^\mu \lambda.
\end{aligned} \tag{7.112}$$

Since partial derivatives commute,

$$\partial^\mu \partial^\nu \lambda = \partial^\nu \partial^\mu \lambda. \tag{7.113}$$

Therefore,

$$F'^{\mu\nu} = F^{\mu\nu}. \tag{7.114}$$

The key lesson is that A^μ is not uniquely determined by the physical fields. The tensor $F^{\mu\nu}$ is gauge invariant and contains the measurable electric and magnetic fields.

7.12 Maxwell Equations in Minkowski Space

The charge density and current density are combined into the four current:

$$\vec{J}^\mu = (c\rho, J^1, J^2, J^3). \tag{7.115}$$

The charge conservation law

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0 \tag{7.116}$$

can be written as

$$\boxed{\partial_\mu J^\mu = 0.} \tag{7.117}$$

The inhomogeneous Maxwell equations are

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}. \tag{7.118}$$

These two equations combine into the tensor equation

$$\boxed{\partial_\nu F^{\mu\nu} = \mu_0 J^\mu.} \tag{7.119}$$

We can use this to verify the continuity equation (7.117):

$$\partial_\mu (\partial_\nu F^{\mu\nu}) = \partial_\mu \mu_0 J^\mu = 0 \Rightarrow \partial_\mu J^\mu = 0, \checkmark$$

where ∂_μ and ∂_ν are symmetric, and $F^{\mu\nu}$ is antisymmetric, hence the result gives 0. To see the relation between (7.119) and the Maxwell's equations, take $\mu = 0$:

$$\partial_\nu F^{0\nu} = \partial_i F^{0i} = \partial_i \left(\frac{E^i}{c} \right) = \frac{1}{c} \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0 c}.$$

Since $\mu_0 c^2 = 1/\epsilon_0$, we have

$$\frac{\rho}{\epsilon_0 c} = \mu_0 c \rho = \mu_0 J^0.$$

Thus the $\mu = 0$ component gives Gauss's law. For $\mu = i$,

$$\partial_\nu F^{i\nu} = \partial_0 F^{i0} + \partial_j F^{ij} = \frac{1}{c} \frac{\partial}{\partial t} \left(-\frac{E^i}{c} \right) + \partial_j (\epsilon^{ijk} B_k) = -\frac{1}{c^2} \frac{\partial E^i}{\partial t} + (\nabla \times \mathbf{B})^i.$$

Thus

$$(\nabla \times \mathbf{B})^i - \frac{1}{c^2} \frac{\partial E^i}{\partial t} = \mu_0 J^i.$$

The homogeneous Maxwell equations are

$$\begin{cases} \nabla \cdot \mathbf{B} = \partial_i B_i = \partial_1 B_1 + \partial_2 B_2 + \partial_3 B_3 = \partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \Rightarrow \nabla \times \left(\frac{\mathbf{E}}{c} \right) = -\partial_0 \mathbf{B} \end{cases}$$

The Faraday's law in the form of tensor gives that

$$\epsilon_{ijk} \partial_j F_{k0} = -\partial_0 B_i \xrightarrow{\times \epsilon_{imn}} \epsilon_{imn} \epsilon_{ijk} \partial_j F_{k0} = (\delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}) \partial_j F_{k0} = -\partial_0 \epsilon_{imn} B_i,$$

which leads to

$$\partial_m F_{n0} - \partial_n F_{m0} = -\partial_0 (\epsilon_{ilm} B_i) \Rightarrow \partial_j F_{k0} - \partial_k F_{j0} = -\partial_0 (\epsilon_{ijk} B_i)$$

By the matrix (7.109), the red part can be rewritten as

$$\partial_j F_{k0} - \partial_k F_{j0} = -\partial_0 F_{jk} \quad (7.120)$$

Hence:

$$\partial_0 F_{jk} + \partial_j F_{k0} \underbrace{-\partial_k F_{j0}}_{=+\partial_k F_{0j}} = 0 \Rightarrow \partial_j F_{k0} + \partial_k F_{0j} + \partial_0 F_{jk} = 0 \quad (7.121)$$

Then we can conclude that the divergence of $\mathbf{B}$ and the Faraday's law are the permutations of the indices of ∂ and $F_{\mu\nu}$, then they are encoded in the Bianchi identity:

$$\boxed{\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0.} \quad (7.122)$$

This identity follows directly from the definition $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Using antisymmetrization, the same statement becomes

$$\partial_\lambda F_{\mu\nu} = 0.$$

The dual field strength tensor is defined as

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}. \quad (7.123)$$

Then the homogeneous Maxwell equations can also be written as

$$\partial_\nu \tilde{F}^{\mu\nu} = 0. \quad (7.124)$$

If you are lazy enough, you can also rewrite the Maxwell's equation in the **differential form**(see Appendix. [B](#)):

$$\begin{aligned} d \wedge *F &= *j \\ d \wedge F &= 0. \end{aligned} \quad (7.125)$$

Chapter 8

Introduction to Calculus of Variations

Abstract In this chapter, we will study the basic ideas of the calculus of variations. In ordinary calculus, we ask where a function reaches its maximum or minimum. In variational calculus, we ask a deeper question: which path, function, or field makes a certain quantity stationary?

This method is very important in analytical mechanics, quantum mechanics, dynamical systems, and classical field theory. It will also help you understand the later topics in Chapter 7 and Chapter 13 of the textbook. **This chapter is extremely hardcore**, so you do not need to understand every detail at the first reading. The main goal is to see how ordinary calculus develops into one of the central languages of theoretical physics.

Most of this chapter (about 80%) is based on my notes from I. M. Gelfand's *Calculus of Variations*. The remaining parts are supplemented by notes from Professor Kuo at National Central University and Professor Li Lin from the Department of Physics at National Chung Hsing University.

8.1 Definition of a Functional

8.1.1 Functional

Definition 8.1 (Functional) A functional takes functions from its domain as inputs and assigns a number to each of them.

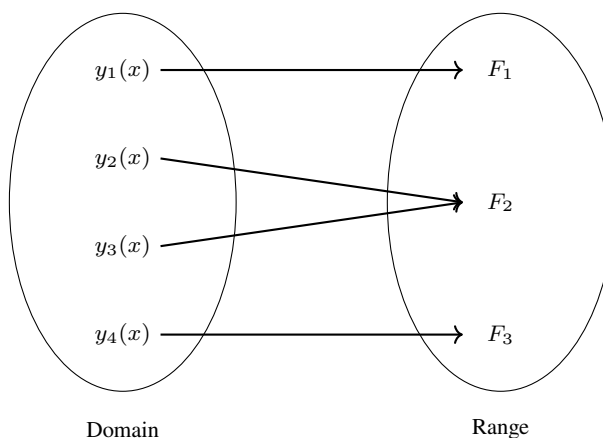


Fig. 8.1 The correspondence between the domain and the range of a functional

For example, $J[y(x), y'(x), x] = \int_a^b dx f(y(x), y'(x), x)$ is a classical functional often used to solve problems in analytical mechanics.

8.1.2 Extremum of Functional

We aim to find the function $y(x)$ that makes the functional attain an extremum. This is done by evaluating how the functional changes under small variations of $y(x)$ and requiring that the first variation vanishes. The method for finding such an extremum is known as the calculus of variations.

8.2 Analytic Foundations of the Variational Method

Before deriving any differential equation, we should make the class of allowed functions precise. This may look slightly formal at first, but it prevents a common mistake: an extremum is meaningful only after we decide which competitors are allowed and what “nearby” means.

8.2.1 Admissible Functions and Variations

Consider the functional

$$J[y] = \int_a^b F(x, y(x), y'(x)) dx. \quad (8.1)$$

For fixed endpoint data $y(a) = A$ and $y(b) = B$, a convenient admissible set is

$$\mathcal{A} = \{y \in C^1([a, b]) : y(a) = A, y(b) = B\}. \quad (8.2)$$

The corresponding space of admissible variations is

$$\mathcal{V} = \{\eta \in C^1([a, b]) : \eta(a) = \eta(b) = 0\}. \quad (8.3)$$

If $y \in \mathcal{A}$ and $\eta \in \mathcal{V}$, then

$$y_\alpha = y + \alpha\eta \quad (8.4)$$

remains in $\mathcal{A}$ for every sufficiently small real number α . The parameter α measures the size of the variation, while η specifies its shape. Keeping these two roles separate makes the logic much cleaner.

The usual C^1 norm is

$$\|y\|_{C^1} = \max_{x \in [a, b]} |y(x)| + \max_{x \in [a, b]} |y'(x)|. \quad (8.5)$$

Thus, a local minimum in the C^1 sense means that there exists $\varepsilon > 0$ such that

$$J[y] \leq J[z] \quad \text{whenever} \quad z \in \mathcal{A}, \quad \|z - y\|_{C^1} < \varepsilon. \quad (8.6)$$

Later we will call this a weak minimum. The word “weak” here belongs to the classical terminology of the calculus of variations. It should not be confused with weak convergence in a Sobolev space.

► **Why fixed endpoints force zero endpoint variations.**

Suppose $y_\alpha(a) = A$ for every α . Differentiating this identity with respect to α gives

$$\left. \frac{\partial y_\alpha(a)}{\partial \alpha} \right|_{\alpha=0} = \eta(a) = 0. \quad (8.7)$$

The same argument gives $\eta(b) = 0$. Notice that nothing forces $\eta'(a)$ or $\eta'(b)$ to vanish for a first order functional. This small point becomes important when we discuss natural boundary conditions and higher order functionals.

8.2.2 Gâteaux and Fréchet Derivatives

For an ordinary function, the derivative records the first order change along the number line. For a functional, there are infinitely many possible directions.

Definition 8.2 (First variation and Gâteaux derivative) Let

$$y_\alpha = y + \alpha\eta. \quad (8.8)$$

The first variation of J at y in the direction η is the Gâteaux derivative

$$\delta J[y; \eta] := DJ[y](\eta) = \left. \frac{d}{d\alpha} J[y + \alpha\eta] \right|_{\alpha=0}, \quad (8.9)$$

provided the limit exists. The variation of the path itself is the tangent vector to the one parameter family:

$$\delta y := \left. \frac{\partial y_\alpha}{\partial \alpha} \right|_{\alpha=0} = \eta, \quad \delta y' = \eta' = \frac{d}{dx}(\delta y). \quad (8.10)$$

Thus δ extracts the coefficient of the first order term and does not contain a factor $d\alpha$. The actual change produced by a small parameter α is

$$J[y + \alpha\eta] - J[y] = \alpha \delta J[y; \eta] + O(\alpha^2). \quad (8.11)$$

When the arguments are clear, we abbreviate $\delta J[y; \eta]$ as δJ .

Definition 8.3 (Fréchet derivative) The stronger Fréchet derivative asks for a bounded linear functional L_y satisfying

$$J[y + h] - J[y] - L_y(h) = o(\|h\|_{C^1}) \quad \text{as} \quad \|h\|_{C^1} \rightarrow 0. \quad (8.12)$$

If (8.12) holds, then every directional derivative exists and

$$\delta J[y; \eta] = DJ[y](\eta) = L_y(\eta). \quad (8.13)$$

The converse is not automatic. Directional derivatives may exist in every direction without fitting together continuously. In most elementary mechanical examples, the integrand is smooth enough that the derivative is indeed Fréchet differentiable.

Theorem 8.1 (First derivative of an integral functional) Assume F , $\frac{\partial F}{\partial y}$, and $\frac{\partial F}{\partial y'}$ are continuous on an open set containing

$$\{(x, y(x), y'(x)) : x \in [a, b]\}. \quad (8.14)$$

Then, for every $\eta \in C^1([a, b])$ and sufficiently small α ,

$$\delta J[y; \eta] = DJ[y](\eta) = \int_a^b \left[\frac{\partial F}{\partial y}(x, y, y')\eta + \frac{\partial F}{\partial y'}(x, y, y')\eta' \right] dx. \quad (8.15)$$

Proof. Define

$$\Phi(\alpha, x) = F(x, y(x) + \alpha\eta(x), y'(x) + \alpha\eta'(x)). \quad (8.16)$$

The ordinary chain rule gives

$$\begin{aligned} \frac{\partial \Phi}{\partial \alpha} &= \frac{\partial F}{\partial y}(x, y + \alpha\eta, y' + \alpha\eta')\eta \\ &\quad + \frac{\partial F}{\partial y'}(x, y + \alpha\eta, y' + \alpha\eta')\eta'. \end{aligned} \quad (8.17)$$

All functions involved are continuous on a compact set when α is restricted to a small closed interval. Therefore $\frac{\partial \Phi}{\partial \alpha}$ is uniformly bounded there, and differentiation under the integral sign is justified. Evaluating at $\alpha = 0$ gives (8.15). $\square$

This proof identifies the exact assumption hidden in the informal phrase “move the derivative through the integral.” We do not need heavy measure theory here. Continuity on a compact set is enough.

8.2.3 The Fundamental Lemma of the Calculus of Variations

The next lemma converts an integral identity into a pointwise equation. It is the logical heart of the Euler Lagrange derivation.

Theorem 8.2 (Fundamental lemma of the calculus of variations) *Let $g \in C([a, b])$. If*

$$\int_a^b g(x)\eta(x) \, dx = 0 \quad (8.18)$$

for every $\eta \in C_0^\infty(a, b)$, then

$$g(x) = 0 \quad \text{for every } x \in [a, b]. \quad (8.19)$$

Proof. Assume instead that $g(x_0) > 0$ at some interior point x_0 . By continuity, there are numbers $r > 0$ and $m > 0$ such that

$$g(x) \geq m \quad \text{for } x \in [x_0 - r, x_0 + r] \subset (a, b). \quad (8.20)$$

Choose a smooth bump function η supported in this interval, with $\eta \geq 0$ and $\eta \not\equiv 0$. Then

$$\int_a^b g(x)\eta(x) \, dx \geq m \int_{x_0-r}^{x_0+r} \eta(x) \, dx > 0, \quad (8.21)$$

contradicting (8.18). The case $g(x_0) < 0$ is handled by the same argument. Hence g vanishes in the interior. Continuity gives the endpoint values as well. $\square$

An explicit bump function is useful to keep in mind:

$$\eta(x) = \begin{cases} \exp\left[-\frac{1}{1 - ((x - x_0)/r)^2}\right], & |x - x_0| < r, \\ 0, & |x - x_0| \geq r. \end{cases} \quad (8.22)$$

It is smooth, nonnegative, and completely localized. In plain language, the lemma says that arbitrary local probes can detect any nonzero continuous function.

► **A useful vector version.**

If continuous functions g_i satisfy

$$\int_a^b \sum_{i=1}^N g_i(x) \eta_i(x) dx = 0 \quad (8.23)$$

for every independent choice $\eta_i \in C_0^\infty(a, b)$, then each g_i is zero. To prove this, set all variations except one equal to zero and apply the scalar lemma.

8.2.4 Stationarity, Minima, and the Role of the Second Variation

If y is a local minimum and $\alpha \mapsto J[y + \alpha\eta]$ is differentiable, then the ordinary one variable necessary condition gives

$$\delta J[y; \eta] = 0 \quad \text{for every admissible } \eta. \quad (8.24)$$

Thus every differentiable local minimum is stationary. The reverse statement is false. A stationary curve may be a minimum, a maximum, or a saddle point.

The analogy with several variable calculus is exact. For a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the equation $\nabla f = 0$ is only the first test. One then studies the Hessian. For a functional, the first variation replaces the gradient and the second variation replaces the Hessian.

Definition 8.4 (Second variation) Suppose the second directional derivative exists. The second variation in the direction η is

$$\delta^2 J[y; \eta] := \left. \frac{d^2}{d\alpha^2} J[y + \alpha\eta] \right|_{\alpha=0}. \quad (8.25)$$

Accordingly,

$$J[y + \alpha\eta] = J[y] + \alpha \delta J[y; \eta] + \frac{\alpha^2}{2} \delta^2 J[y; \eta] + o(\alpha^2). \quad (8.26)$$

If y is a local minimum, then

$$\delta^2 J[y; \eta] \geq 0 \quad \text{for every admissible } \eta. \quad (8.27)$$

This condition is necessary, but it is still not generally sufficient. A zero second variation can leave the decision to higher order terms. Later sections will sharpen this statement through the Legendre condition, Jacobi's equation, conjugate points, and coercivity.

8.3 Euler-Lagrange Equation and Lagrangian & Hamiltonian

8.3.1 Euler-Lagrange Equation without Constraints

To test whether a path makes the functional stationary, introduce the one parameter family

$$y_\alpha(x) = y(x) + \alpha\eta(x), \quad J[y] \mapsto J[y_\alpha] = J[y + \alpha\eta]. \quad (8.28)$$

According to (8.10), $\delta y = \eta$ and $\delta y' = \eta'$.

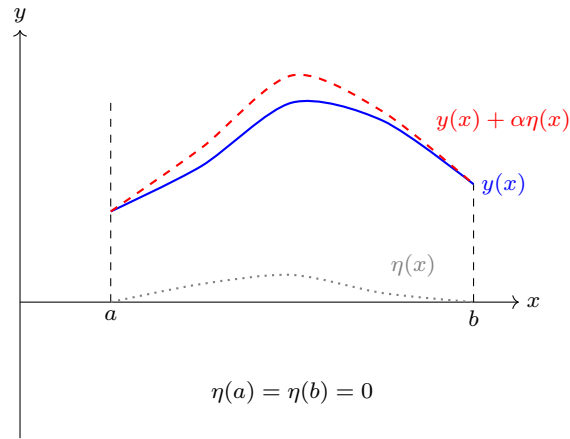


Fig. 8.2 Illustration of Perturbation Function in Calculus of Variations

where α is a small parameter, and $\eta(x)$ is an arbitrary function that is differentiable to first order and satisfies $\eta(a) = \eta(b) = 0$. It is called the perturbation function. When the extremum occurs at the path $y(x)$, the first variation of the functional vanishes. Consider the bivariate function $f(y(x), y'(x))$ which depends on y, y' , the functional J is defined to be the integral of f , that is

$$J[y(x) + \alpha\eta(x)] := J(\alpha) = \int_a^b dx f(y(x) + \alpha\eta(x), y'(x) + \alpha\eta'(x)) \quad (8.29)$$

When the extremum occurs at the path $y(x)$, the auxiliary one-variable function $J(\alpha)$ must have vanishing first derivative at $\alpha = 0$. In other words, $J(0) = J[y]$ is the extremum value and

$$\left. \frac{dJ}{d\alpha} \right|_{\alpha=0} = 0.$$

We still need to find what this condition implies for the original bivariate function and for y .

To do this, we take the Taylor expansion of the functional $J(\alpha)$ and obtain

$$J(\alpha \rightarrow 0) = J(0) + \left. \frac{dJ}{d\alpha} \right|_{\alpha=0} \cdot \alpha + O(\alpha^2) \quad (8.30)$$

Here α is a variable that approaches 0. At an extremum, the derivative of the functional with respect to α vanishes at $\alpha = 0$. Therefore,

$$\begin{aligned} \left. \frac{dJ}{d\alpha} \right|_{\alpha=0} &= \frac{d}{d\alpha} \left[\int_a^b dx f(y(x) + \alpha\eta(x), y'(x) + \alpha\eta'(x)) \right] \bigg|_{\alpha=0} \\ &= \int_a^b dx \left\{ \frac{\partial f}{\partial (y(x) + \alpha\eta(x))} \cdot \frac{d}{d\alpha} (y(x) + \alpha\eta(x)) \right. \\ &\quad \left. + \frac{\partial f}{\partial (y'(x) + \alpha\eta'(x))} \cdot \frac{d}{d\alpha} (y'(x) + \alpha\eta'(x)) \right\} \end{aligned} \quad (8.31)$$

The two derivatives with respect to α are $\frac{d}{d\alpha} (y(x) + \alpha\eta(x)) = \eta(x)$ and $\frac{d}{d\alpha} (y'(x) + \alpha\eta'(x)) = \eta'(x)$. Setting $\alpha = 0$, equation (8.31) becomes

$$\left. \frac{dJ}{d\alpha} \right|_{\alpha=0} = \int_a^b dx \left[\frac{\partial f}{\partial y(x)} \cdot \eta(x) + \frac{\partial f}{\partial y'(x)} \cdot \eta'(x) \right] \quad (8.32)$$

Use integration by parts, the integral

$$\int_a^b dx \frac{\partial f}{\partial y'(x)} \cdot \eta'(x) = \left. \frac{\partial f}{\partial y'(x)} \eta(x) \right|_a^b - \int_a^b dx \frac{d}{dx} \left[\frac{\partial f}{\partial y'(x)} \right] \cdot \eta(x)$$

Combining this with (8.32), we get

$$\begin{aligned} \left. \frac{dJ}{d\alpha} \right|_{\alpha=0} &= \int_a^b dx \frac{\partial f}{\partial y(x)} \cdot \eta(x) - \int_a^b dx \frac{d}{dx} \left[\frac{\partial f}{\partial y'(x)} \right] \cdot \eta(x) \\ &= \int_a^b dx \left[\frac{\partial f}{\partial y(x)} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'(x)} \right) \right] \cdot \eta(x) = 0 \end{aligned}$$

Because $\eta(x)$ is an arbitrary function that is differentiable to first order, the equation must satisfy

$$\frac{\partial f}{\partial y(x)} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'(x)} \right) = 0 \quad (8.33)$$

Equivalently, one may write this equation without displaying the arguments:

$$\boxed{\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0.}$$

This is called the **Euler-Lagrange equation**. Thus, any sufficiently smooth function $y(x)$ that makes the functional stationary must satisfy the Euler-Lagrange equation for all $x \in [a, b]$.

There is another useful form. For $J[y(x), y'(x), x] = \int_a^b dx f(y(x), y'(x), x)$, differentiate f with respect to x to obtain

$$\frac{df}{dx} = \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y'} \frac{dy'}{dx} + \frac{\partial f}{\partial x} \frac{dx}{dx}$$

Using $\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$, we can combine the first two terms and write

$$\frac{df}{dx} = \frac{d}{dx} \left(\frac{\partial f}{\partial y'} y' \right) + \frac{\partial f}{\partial x} \Rightarrow \frac{d}{dx} \left(f - \frac{\partial f}{\partial y'} y' \right) = \frac{\partial f}{\partial x}$$

If the function f is independent of x , which means $\frac{\partial f}{\partial x} = 0$, we can finally get the second form of the Euler-Lagrange equation, denoted by

$$\boxed{f - \frac{\partial f}{\partial y'} y' = \text{const}} \quad (8.34)$$

where $f = f(y, y')$.

This result extends to the N variable functional $J[y_1(x), y_1'(x), \dots, y_N(x), y_N'(x)] = \int_a^b dx f(y_1(x), y_1'(x), \dots, y_N(x), y_N'(x))$. Suppose the functions $y_i(x)$, where $i = 1, \dots, N$, make the functional stationary. Varying each function gives

$$y_i(x) \rightarrow y_i(x) + \alpha \eta_i(x); \quad y_i'(x) \rightarrow y_i'(x) + \alpha \eta_i'(x) \quad (8.35)$$

so we may write J as $J(\alpha) = \int_a^b dx f(y_i + \alpha \eta_i, y_i' + \alpha \eta_i')$, where $i = 1, \dots, N$. Since

$\left. \frac{dJ}{d\alpha} \right|_{\alpha=0} = 0$, we have

$$\begin{aligned} \left. \frac{dJ}{d\alpha} \right|_{\alpha=0} &= \frac{d}{d\alpha} \int_a^b dx f(y_i + \alpha \eta_i, y_i' + \alpha \eta_i') \Big|_{\alpha=0} \\ &= \int_a^b dx \sum_i \left(\frac{\partial f}{\partial (y_i + \alpha \eta_i)} \frac{d(y_i + \alpha \eta_i)}{d\alpha} + \frac{\partial f}{\partial (y_i' + \alpha \eta_i')} \frac{d(y_i' + \alpha \eta_i')}{d\alpha} \right) \Big|_{\alpha=0} \\ &= \int_a^b dx \sum_i \left(\frac{\partial f}{\partial y_i} \eta_i + \frac{\partial f}{\partial y_i'} \eta_i' \right) = \int_a^b dx \sum_i \left(\frac{\partial f}{\partial y_i} - \frac{d}{dx} \frac{\partial f}{\partial y_i'} \right) \eta_i = 0 \end{aligned}$$

Because the functions η_i are arbitrary and mutually independent, we obtain

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx} \frac{\partial f}{\partial y'_i} = 0, \quad i = 1, \dots, N \quad (8.36)$$

We now use the convention fixed in (8.9) and (8.10) throughout the rest of the chapter. The symbol δ differentiates the chosen one parameter family at $\alpha = 0$:

$$\delta J[y; \eta] = \left. \frac{d}{d\alpha} J[y + \alpha\eta] \right|_{\alpha=0}, \quad \delta y = \eta, \quad \delta y' = \eta'. \quad (8.37)$$

It follows that

$$\delta f = \left. \frac{d}{d\alpha} f(y + \alpha\eta, y' + \alpha\eta', x) \right|_{\alpha=0} = \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial y'} \delta y'. \quad (8.38)$$

The parameter α appears only when we write the actual nearby curve or the finite first order change. In particular,

$$J[y + \alpha\eta] = J[y] + \alpha \delta J[y; \eta] + O(\alpha^2). \quad (8.39)$$

The boundary conditions for the variation functions are

$$\delta y(a) = \delta y(b) = 0, \quad \eta(a) = \eta(b) = 0.$$

For a first-order variational problem, the fixed endpoints impose conditions on the variation of the path itself. They do not require $\delta y'(a)$ and $\delta y'(b)$ to vanish. The variation of the functional J is then

$$\begin{aligned} \delta J[y; \eta] &= \delta \int_a^b f(y, y', x) dx \\ &= \int_a^b \delta f(y, y', x) dx = \int_a^b \left(\frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial y'} \delta y' \right) dx \\ &= \int_a^b \frac{\partial f}{\partial y} \delta y dx + \int_a^b \frac{\partial f}{\partial y'} \delta y' dx \\ &= \int_a^b \frac{\partial f}{\partial y} \delta y dx + \left(\frac{\partial f}{\partial y'} \delta y \right) \Big|_a^b - \int_a^b \frac{d}{dx} \frac{\partial f}{\partial y'} \delta y dx \\ &= \int_a^b \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) \delta y dx \end{aligned} \quad (8.40)$$

At a stationary path, the variation of the functional must vanish in every admissible direction, so $\delta J[y; \eta] = 0$ and the result is the same as before. The term $\frac{\partial f}{\partial y'} \delta y \Big|_a^b$ vanishes because of the boundary conditions.

8.3.2 A Precise Euler Lagrange Theorem

We now collect the calculation into a theorem. This is useful because it separates the regularity assumptions from the conclusion.

Theorem 8.3 (Euler–Lagrange theorem) *Let $F \in C^2$ on an open set containing the graph of $(x, y(x), y'(x))$, and let*

$$J[y] = \int_a^b F(x, y, y') \, dx \quad (8.41)$$

on the admissible class $\mathcal{A} \subset C^1([a, b])$ with fixed endpoints. If $y \in C^2([a, b])$ is a local extremum of J , then

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0} \quad \text{on } (a, b). \quad (8.42)$$

Proof. Fix any $\eta \in C_0^\infty(a, b)$ and define

$$\varphi(\alpha) = J[y + \alpha\eta]. \quad (8.43)$$

Because y is a local extremum, $\alpha = 0$ is a local extremum of the ordinary function φ . Hence $\varphi'(0) = 0$. By (8.15),

$$0 = \int_a^b \left(\frac{\partial F}{\partial y} \eta + \frac{\partial F}{\partial y'} \eta' \right) \, dx. \quad (8.44)$$

Integration by parts gives

$$0 = \left[\frac{\partial F}{\partial y'} \eta \right]_a^b + \int_a^b \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta \, dx. \quad (8.45)$$

The boundary term is zero because η has compact support in (a, b) . The expression in square brackets is continuous, so the fundamental lemma implies (8.42). $\square$

The theorem gives a necessary condition only. To see why, consider

$$J[y] = \int_a^b -y'(x)^2 \, dx, \quad y(a) = y(b) = 0. \quad (8.46)$$

The zero function satisfies the Euler Lagrange equation, but

$$J[\alpha\eta] = -\alpha^2 \int_a^b \eta'(x)^2 \, dx < 0 \quad (8.47)$$

for every nonzero admissible η . Thus $y = 0$ is a strict maximum, not a minimum.

► **Weak form.**

If y is not known to be twice differentiable, the integration by parts step may not be legitimate in the classical sense. The stationary condition can then be kept in the form

$$\int_a^b \left(\frac{\partial F}{\partial y} \eta + \frac{\partial F}{\partial y'} \eta' \right) dx = 0 \quad \text{for all } \eta \in C_0^\infty(a, b). \quad (8.48)$$

Equation (8.48) is the weak Euler Lagrange equation. Whenever $\frac{\partial F}{\partial y'}$ is continuously differentiable, it reduces to the classical equation. The weak form is not a different physical law. It is the same stationary condition written without assuming derivatives that may not yet exist.

8.3.3 Free Endpoints and Natural Boundary Conditions

The boundary term should never be discarded automatically. It vanishes for fixed endpoints, but it becomes a new equation when an endpoint is free.

Suppose $y(a) = A$ is fixed while $y(b)$ is free. Admissible variations satisfy

$$\eta(a) = 0, \quad \eta(b) \text{ is arbitrary.} \quad (8.49)$$

The complete first variation is

$$\delta J[y; \eta] = \int_a^b \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta dx + \left[\frac{\partial F}{\partial y'} \eta \right]_a^b. \quad (8.50)$$

First choose variations supported away from the endpoints. The fundamental lemma gives the Euler Lagrange equation in the interior. Once the interior term is zero, (8.50) reduces to

$$\frac{\partial F}{\partial y'}(b) \eta(b) = 0. \quad (8.51)$$

Since $\eta(b)$ is arbitrary,

$$\frac{\partial F}{\partial y'}(b) = 0. \quad (8.52)$$

This is the natural boundary condition at a free right endpoint. If both endpoint values are free, then

$$\frac{\partial F}{\partial y'}(a) = 0, \quad \frac{\partial F}{\partial y'}(b) = 0. \quad (8.53)$$

The practical meaning is simple. The interior equation comes from variations localized inside the interval. Natural boundary conditions come from the remaining independent variations at the boundary. They are two parts of one calculation.

8.3.4 Moving Endpoints and Transversality

Now allow the terminal point itself to move. Let the original endpoint be $(b, y(b))$, and let its displacement be $(\Delta x, \Delta y)$. The change of the function at the old coordinate b is not Δy because part of the vertical displacement comes from sliding along the original tangent. To first order,

$$\delta y(b) = \Delta y - y'(b)\Delta x. \quad (8.54)$$

The moving upper limit contributes $F(b, y, y')\Delta x$. Combining it with the usual endpoint term gives

$$\delta J|_b = F\Delta x + \frac{\partial F}{\partial y'}(\Delta y - y'\Delta x) = \frac{\partial F}{\partial y'}\Delta y + \left(F - y'\frac{\partial F}{\partial y'}\right)\Delta x. \quad (8.55)$$

If both Δx and Δy are independent, stationarity gives two conditions:

$$\frac{\partial F}{\partial y'} = 0, \quad F - y'\frac{\partial F}{\partial y'} = 0. \quad (8.56)$$

Usually the endpoint is constrained to lie on a curve $G(x, y) = 0$. The allowed displacement must then satisfy

$$\frac{\partial G}{\partial x}\Delta x + \frac{\partial G}{\partial y}\Delta y = 0. \quad (8.57)$$

Equivalently, the two coefficients in (8.55) must be proportional to the normal covector of the terminal curve. Therefore there is a number λ such that

$$F - y'\frac{\partial F}{\partial y'} = \lambda\frac{\partial G}{\partial x}, \quad \frac{\partial F}{\partial y'} = \lambda\frac{\partial G}{\partial y}. \quad (8.58)$$

This is the transversality condition. It tells us how an extremal must meet a boundary curve. When $G(x, y) = y - \psi(x)$, eliminating λ gives

$$F + (\psi' - y')\frac{\partial F}{\partial y'} = 0. \quad (8.59)$$

8.3.5 Corner Conditions for Piecewise Smooth Extremals

An extremal may be continuous but only piecewise differentiable. Suppose a possible corner occurs at $x = c$, and split the functional as

$$J[y] = \int_a^c F(x, y, y') dx + \int_c^b F(x, y, y') dx. \quad (8.60)$$

Apply the endpoint formula to the two pieces. The displacement of the corner is common to both pieces, but the right piece contributes with the opposite orientation. Stationarity

for arbitrary Δx_c and Δy_c gives

$$\left[\frac{\partial F}{\partial y'} \right]_{c^-}^{c^+} = 0, \quad \left[F - y' \frac{\partial F}{\partial y'} \right]_{c^-}^{c^+} = 0. \quad (8.61)$$

These are the Weierstrass Erdmann corner conditions. The canonical momentum and the Hamiltonian quantity must remain continuous across an unconstrained corner. A visible kink is therefore impossible unless the integrand is singular, the corner is externally constrained, or the two conditions happen to allow different slopes.

8.3.6 Euler-Lagrange Equation under Constraints

Consider an N variable functional

$$J[y_1(x), y_1'(x), \dots, y_N(x), y_N'(x)] = \int_a^b dx f(y_1, y_1', \dots, y_N, y_N', x),$$

with M constraints among the coordinates, where $1 \leq M \leq N - 1$. The number of degrees of freedom¹ is then $N - M$.

Write the M constraints as

$$\begin{cases} g_1(y_1, y_1', \dots, y_N, y_N', x) = 0 \\ g_2(y_1, y_1', \dots, y_N, y_N', x) = 0 \\ \vdots \\ g_M(y_1, y_1', \dots, y_N, y_N', x) = 0 \end{cases}$$

The variation must now respect these constraints. If

$$y_j(x) \rightarrow y_j(x) + \alpha \eta_j(x), \quad y_j'(x) \rightarrow y_j'(x) + \alpha \eta_j'(x),$$

then the first-order variation of the k th constraint is

$$\left. \frac{dg_k}{d\alpha} \right|_{\alpha=0} = \sum_j \left(\frac{\partial g_k}{\partial y_j} \eta_j + \frac{\partial g_k}{\partial y_j'} \eta_j' \right) \quad (8.62)$$

The function $\eta_j(x)$ is not completely arbitrary anymore, because admissible variations must be tangent to the constraint surface. A convenient way to keep the calculation systematic is to introduce Lagrange multipliers $\lambda_k(x)$ and define the augmented integrand

¹ Degrees of freedom (DOF) refer to the number of independent parameters that define a system's state or motion. Constraints reduce the available DOF.

$$\tilde{f}(y_j, y'_j, \lambda_k, x) := f(y_j, y'_j, x) + \sum_{k=1}^M \lambda_k(x) g_k(y_j, y'_j, x). \quad (8.63)$$

Then the constrained variational problem can be written as an unconstrained variational problem for

$$\tilde{J}[y_j, \lambda_k] = \int_a^b \tilde{f}(y_j, y'_j, \lambda_k, x) dx.$$

The variation with respect to λ_k gives back the original constraints $g_k = 0$. The variation with respect to y_j gives

$$\delta \tilde{J} = \int_a^b dx \sum_j \left(\frac{\partial \tilde{f}}{\partial y_j} \delta y_j + \frac{\partial \tilde{f}}{\partial y'_j} \delta y'_j \right) = \int_a^b dx \sum_j \left[\frac{\partial \tilde{f}}{\partial y_j} - \frac{d}{dx} \left(\frac{\partial \tilde{f}}{\partial y'_j} \right) \right] \delta y_j.$$

The boundary term vanishes because the endpoints are fixed. Therefore, for each $j = 1, \dots, N$, the constrained Euler-Lagrange equation is

$$\frac{\partial \tilde{f}}{\partial y_j} - \frac{d}{dx} \left(\frac{\partial \tilde{f}}{\partial y'_j} \right) = 0 \quad (8.64)$$

or equivalently

$$\frac{\partial f}{\partial y_j} + \sum_{k=1}^M \lambda_k \frac{\partial g_k}{\partial y_j} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'_j} + \sum_{k=1}^M \lambda_k \frac{\partial g_k}{\partial y'_j} \right) = 0 \quad (8.65)$$

Notice that $\lambda_k(x)$ stays inside the total derivative if g_k depends on y'_j . This is important, because λ_k may also depend on x .

Most physical or mathematical problems we meet at this level use holonomic constraints, which depend on y_j but not on y'_j . In that common case, $g_k = g_k(y_j, x)$ and $\partial g_k / \partial y'_j = 0$. Then (8.65) reduces to

$$\frac{\partial f}{\partial y_j} + \sum_{k=1}^M \lambda_k \frac{\partial g_k}{\partial y_j} = \frac{d}{dx} \frac{\partial f}{\partial y'_j} \quad (8.66)$$

8.3.7 Integral Constraints and the Isoperimetric Multiplier

An integral constraint has a different form from a pointwise constraint. Suppose we want to extremize

$$J[y] = \int_a^b F(x, y, y') dx \quad (8.67)$$

subject to

$$K[y] = \int_a^b G(x, y, y') dx = K_0. \quad (8.68)$$

The multiplier is now a constant, not an arbitrary function of x . To see this carefully, choose two test functions η_1 and η_2 and consider

$$y_{\alpha,\beta} = y + \alpha\eta_1 + \beta\eta_2. \quad (8.69)$$

Define the ordinary functions

$$\mathcal{J}(\alpha, \beta) = J[y_{\alpha,\beta}], \quad \mathcal{K}(\alpha, \beta) = K[y_{\alpha,\beta}]. \quad (8.70)$$

If the derivative of the constraint is not identically zero, choose η_2 so that

$$\frac{\partial \mathcal{K}}{\partial \beta}(0, 0) \neq 0. \quad (8.71)$$

The implicit function theorem then gives a function $\beta = \beta(\alpha)$ for which $\mathcal{K}(\alpha, \beta(\alpha)) = K_0$. Since y is stationary under all such constrained curves,

$$\left. \frac{d}{d\alpha} \mathcal{J}(\alpha, \beta(\alpha)) \right|_{\alpha=0} = 0. \quad (8.72)$$

The ordinary two variable multiplier theorem says that there is a constant λ satisfying

$$DJ[y](\eta) - \lambda DK[y](\eta) = 0 \quad \text{for every admissible } \eta. \quad (8.73)$$

Therefore y is stationary for the augmented functional

$$\tilde{J}[y] = \int_a^b (F - \lambda G) dx, \quad (8.74)$$

and satisfies

$$\frac{\partial(F - \lambda G)}{\partial y} - \frac{d}{dx} \left[\frac{\partial(F - \lambda G)}{\partial y'} \right] = 0. \quad (8.75)$$

Why is λ constant here? The constraint supplies one number, namely $K[y] = K_0$, so it introduces one scalar multiplier. A pointwise constraint $g(x, y, y') = 0$ supplies one condition at every x , so it introduces a multiplier function $\lambda(x)$. This distinction is easy to forget and is worth stating explicitly.

If $DK[y](\eta) = 0$ for every η , the constraint itself is stationary and the previous multiplier argument becomes degenerate. One then uses constants λ_0 and λ_1 , not both zero, and writes

$$\frac{\partial(\lambda_0 F + \lambda_1 G)}{\partial y} - \frac{d}{dx} \left[\frac{\partial(\lambda_0 F + \lambda_1 G)}{\partial y'} \right] = 0. \quad (8.76)$$

The usual case has $\lambda_0 \neq 0$, after which we divide by λ_0 and recover (8.75). The Dido problem later in this chapter is the standard geometric example.

8.4 Applications of the Calculus of Variations

8.4.1 No One Has Told Me Why a Straight Line is the Shortest Distance!

From elementary school through high school, every math teacher tells us that the shortest distance between two points is a straight line. We have always believed it and taken it for granted. Yet few people stop to question the claim, and teachers rarely prove it. In mathematics and physics, intuition alone is not enough. So, fellow physics students, let us uncover the reason behind this familiar statement.

Our proof will stay within Euclidean geometry, which is the setting we already know. The first postulate of Euclidean geometry states that "a straight line segment can be drawn joining any two points." This is where we begin. Consider the arc length element

$$ds = \sqrt{(dx)^2 + (dy)^2} \quad (8.77)$$

From this element, we construct the functional $J[x, y, y']$

$$L = \int_{x_0}^{x_1} ds = \int_{x_0}^{x_1} \sqrt{(dx)^2 + (dy)^2} = \int_{x_0}^{x_1} \sqrt{1 + y'^2} dx \quad (8.78)$$

Here y' is the derivative of y with respect to x . Thus the functional is $J[x, y, y'] = \int_{x_0}^{x_1} f(x, y, y') dx$, where $f(x, y, y') = \sqrt{1 + y'^2}$. A stationary path of this functional must satisfy

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = \frac{\partial \sqrt{1 + y'^2}}{\partial y} - \frac{d}{dx} \frac{\partial \sqrt{1 + y'^2}}{\partial y'} = 0 \quad (8.79)$$

Substituting the function $f(x, y, y')$ into (8.79), we obtain

$$\frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0 \Rightarrow \frac{y'}{\sqrt{1 + y'^2}} = c \quad (8.80)$$

where c is a constant. Rearranging (8.80) gives an algebraic relation between y' and c

$$y'^2 = \frac{c^2}{1 - c^2} \Rightarrow y' = \pm \frac{|c|}{\sqrt{1 - c^2}} := a \quad (8.81)$$

By separating the variables, we have

$$dy = a dx \Rightarrow y = \int a dx = ax + b \quad (8.82)$$

where b is another constant. We have shown that every stationary path is a straight line. The positivity of the second variation then confirms that this stationary path gives the minimum length.

8.4.2 Fermat's Principle

Fermat's principle states that light travels along the path that minimizes the total travel time. In high school, we encountered the familiar equation

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

which is known as the law of refraction. At that time, we proved this equation using geometric arguments and the relationship between incident and refracted angles. However, upon closer inspection, this explanation merely describes the *apparent phenomenon* of how light bends at the interface rather than the *underlying reason* why the bending occurs. In fact, this problem lies within the scope of variational methods and functionals.

We can construct a functional T

$$T[x, y, y', v(x, y)] = \int_{x_1}^{x_2} \frac{ds}{v(x, y)} = \int_{x_1}^{x_2} \frac{\sqrt{1 + y'^2}}{v(x, y)} dx \quad (8.83)$$

Here v is the speed of light and may depend on position. At a stationary travel time, the functional satisfies

$$\frac{\partial \left(\sqrt{1 + y'^2} / v(x, y) \right)}{\partial y} - \frac{d}{dx} \frac{\partial \left(\sqrt{1 + y'^2} / v(x, y) \right)}{\partial y'} = 0 \quad (8.84)$$

If $v(x, y) = v \in \mathbb{R}$ is constant, the refractive index is uniform and the problem reduces to the previous shortest path problem. In other words, light travels in a straight line inside a homogeneous medium.

Now consider an interface at $y = 0$, with refractive index n_1 and speed $v_1 \in \mathbb{R}$ for $y > 0$, and refractive index n_2 and speed $v_2 \in \mathbb{R}$ for $y < 0$. Suppose the point of incidence is $(x, 0)$. The speed is constant inside each homogeneous medium, so each part of the light path is a straight line. The functional $T[x, y, y', v(x, y)]$ therefore reduces to an ordinary function $T(x)$. This is a **finite-dimensional problem**, with path lengths

$$\Delta S_1 = \sqrt{(x - x_1)^2 + y_1^2}, \Delta S_2 = \sqrt{(x - x_2)^2 + y_2^2} \quad (8.85)$$

We then have

$$T(x) = \frac{\Delta S_1}{v_1} + \frac{\Delta S_2}{v_2} = \frac{\sqrt{(x - x_1)^2 + y_1^2}}{v_1} + \frac{\sqrt{(x - x_2)^2 + y_2^2}}{v_2} \quad (8.86)$$

At a stationary value of the total time, we obtain

$$\begin{aligned}\frac{dT(x)}{dx} &= \frac{d}{dx} \frac{\sqrt{(x-x_1)^2 + y_1^2}}{v_1} + \frac{d}{dx} \frac{\sqrt{(x-x_2)^2 + y_2^2}}{v_2} \\ &= \frac{1}{v_1} \frac{x-x_1}{\sqrt{(x-x_1)^2 + y_1^2}} + \frac{1}{v_2} \frac{x-x_2}{\sqrt{(x-x_2)^2 + y_2^2}} = 0\end{aligned}$$

Assume $x_1 < x < x_2$ and multiply both sides by c , the speed of light in vacuum. Then

$$\frac{1}{v_1} \frac{x-x_1}{\sqrt{(x-x_1)^2 + y_1^2}} + \frac{1}{v_2} \frac{x-x_2}{\sqrt{(x-x_2)^2 + y_2^2}} = \frac{c}{v_1} \frac{x-x_1}{\sqrt{(x-x_1)^2 + y_1^2}} - \frac{c}{v_2} \frac{x_2-x}{\sqrt{(x-x_2)^2 + y_2^2}} = 0$$

Therefore,

$$\frac{c}{v_1} \frac{x-x_1}{\sqrt{(x-x_1)^2 + y_1^2}} = \frac{c}{v_2} \frac{x_2-x}{\sqrt{(x-x_2)^2 + y_2^2}} \quad (8.87)$$

Define c/v_1 and c/v_2 as the refractive indices

$$\frac{c}{v_1} := n_1, \quad \frac{c}{v_2} := n_2 \quad (8.88)$$

The term $(x-x_1)/\sqrt{(x-x_1)^2 + y_1^2}$ is the sine of the angle θ_1 between the incident ray and the normal to the interface. Similarly, $(x_2-x)/\sqrt{(x-x_2)^2 + y_2^2}$ is the sine of the angle θ_2 between the refracted ray and the normal. Rewriting (8.87), we recover the familiar law of refraction

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (8.89)$$

8.4.3 Cycloid Problem

A particle of mass m is subject to a uniform gravitational field g . It starts from rest at the point (x_1, y_1) and moves to a lower point (x_2, y_2) , where $0 \leq y_1 < y_2$. Find the path of the particle that minimizes the time taken for the motion. Set $(x_1, y_1) = (0, 0)$. To build the time functional, first find the speed v

$$\frac{1}{2}mv^2 = mgy \Rightarrow v = \sqrt{2gy} \quad (8.90)$$

The infinitesimal displacement is $ds = \sqrt{dx^2 + dy^2}$, and the corresponding travel time is $dt = ds/v$. Therefore the total time is

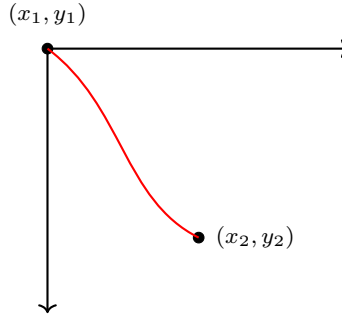


Fig. 8.3 A Diagram of the Position and Trajectory Between Two Points

$$\begin{aligned}
 T &= \int dt = \int \frac{ds}{v} = \int \frac{\sqrt{dx^2 + dy^2}}{\sqrt{2gy}} \\
 \Rightarrow T &= \frac{1}{\sqrt{2g}} \int_{y_1}^{y_2} \frac{\sqrt{1 + \left(\frac{dx}{dy}\right)^2}}{\sqrt{y}} dy
 \end{aligned} \tag{8.91}$$

Thus the total time is a functional with integrand $f(x(y), x'(y), y) = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} / \sqrt{y}$. A stationary travel time must satisfy the Euler-Lagrange equation.

$$\frac{\partial f}{\partial x(y)} - \frac{d}{dy} \frac{\partial f}{\partial x'(y)} = 0 \tag{8.92}$$

Since $\frac{\partial f}{\partial x(y)} = 0$, we have $\frac{d}{dy} \frac{\partial f}{\partial x'(y)} = 0$, so $\frac{\partial f}{\partial x'(y)} = C$. Without loss of generality, let $C = \frac{1}{\sqrt{2a}} > 0$. Then

$$\frac{\partial f}{\partial x'(y)} = \frac{x'(y)}{\sqrt{y(1 + x'(y)^2)}} = \frac{1}{\sqrt{2a}} \Rightarrow x'(y) = \pm \frac{y}{\sqrt{2ay - y^2}} \tag{8.93}$$

Separating the variables gives

$$dx = \pm \frac{y}{\sqrt{2ay - y^2}} dy \Rightarrow x = \int_0^y \frac{u}{\sqrt{2au - u^2}} du \tag{8.94}$$

where $x(y = 0) = 0$. Let

$$u = a(1 - \cos \theta), \quad du = a \sin \theta d\theta.$$

Then the equation becomes

$$\begin{aligned}
x &= \int_0^\theta \frac{a(1 - \cos \theta') a \sin \theta'}{\sqrt{2a^2(1 - \cos \theta') - a^2(1 - \cos \theta')^2}} d\theta' \\
&= a \int_0^\theta (1 - \cos \theta') d\theta' = a(\theta - \sin \theta).
\end{aligned} \tag{8.95}$$

At the same time, the substitution gives

$$y = a(1 - \cos \theta).$$

After calculation, we obtain the brachistochrone trajectory

$$\begin{cases} x = a(\theta - \sin \theta) \\ y = a(1 - \cos \theta) \end{cases} \tag{8.96}$$

Here θ is the curve parameter, and a is determined by the endpoint condition. This curve is called a **cycloid** in the brachistochrone problem.

8.4.4 Dido Problem

Among all planar curves of fixed length L that connect two points on the x -axis and enclose area above the axis, determine the shape of the curve that maximizes the enclosed area.

This problem is named after Queen Dido of Carthage, who was granted as much land as could be enclosed by a single piece of oxhide. By cutting the oxhide into thin strips and forming a semicircular boundary along a straight shoreline, she maximized the area, a physical realization of a variational principle.

This problem has the length constraint

$$\int_{x_1}^{x_2} \sqrt{1 + (y'(x))^2} dx = L \tag{8.97}$$

On the other hand, the enclosed area above the x -axis is

$$A = \int_{x_1}^{x_2} y(x) dx \tag{8.98}$$

The function y is subject to the constraint in (8.97), so this is a constrained variational problem. Introduce a Lagrange multiplier λ and combine (8.97) with (8.98) to define

$$A[y, y', \lambda] = \int_{x_1}^{x_2} \left(y + \lambda \sqrt{1 + y'^2} \right) dx = \int_{x_1}^{x_2} \tilde{f}(y, y', \lambda) dx \tag{8.99}$$

Because $\delta A[y] + \delta L = \delta A[y, y', \lambda] = 0$, the function $\tilde{f}(y, y', \lambda)$ satisfies

$$\frac{\partial \tilde{f}}{\partial y} - \frac{d}{dx} \frac{\partial \tilde{f}}{\partial y'} = \frac{\partial (y + \lambda \sqrt{1 + y'^2})}{\partial y} - \frac{d}{dx} \frac{\partial (y + \lambda \sqrt{1 + y'^2})}{\partial y'} = 0 \quad (8.100)$$

We obtain

$$1 - \frac{d}{dx} \frac{\lambda y'}{\sqrt{1 + y'^2}} = 0 \Rightarrow d \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = \frac{1}{\lambda} dx \quad (8.101)$$

Integrating both sides,

$$\frac{y'}{\sqrt{1 + y'^2}} = \frac{x}{\lambda} + C \Rightarrow \left(\frac{y'}{\sqrt{1 + y'^2}} \right)^2 = \left(\frac{x}{\lambda} + C \right)^2 \quad (8.102)$$

Rearranging (8.102) gives the differential equation

$$\frac{dy}{dx} = \frac{\left(\frac{x}{\lambda} + C \right)}{\sqrt{1 - \left(\frac{x}{\lambda} + C \right)^2}} \Rightarrow y = \int \frac{\left(\frac{x}{\lambda} + C \right)}{\sqrt{1 - \left(\frac{x}{\lambda} + C \right)^2}} dx \quad (8.103)$$

Let $x/\lambda + C = \sin \theta$, so $dx = \lambda \cos \theta$. Substituting into (8.103) gives

$$y = \int \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} \lambda \cos \theta d\theta = -\lambda \cos \theta + C' \quad (8.104)$$

Combining $x/\lambda + C = \sin \theta$ with (8.104), we obtain the parametric equations for the extremal curve in the Dido problem

$$\begin{cases} x = \lambda \sin \theta - \lambda C \\ y = -\lambda \cos \theta + C' \end{cases} \quad (8.105)$$

It is an arc of a circle. It becomes a semicircle only under the special boundary condition in which the two endpoints form the diameter of the circle. Although this form differs from the standard parametric representation of a circle, it is essentially the result of applying a clockwise rotation matrix followed by a translation

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} \lambda \cos \theta \\ \lambda \sin \theta \end{bmatrix} + \begin{bmatrix} -\lambda C \\ C' \end{bmatrix} = \begin{bmatrix} \lambda \sin \theta - \lambda C \\ -\lambda \cos \theta + C' \end{bmatrix} \quad (8.106)$$

Note that it is not necessary to express θ in terms of inverse trigonometric functions. As long as x and y are related through $\sin \theta$ and $\cos \theta$, the physical nature of the problem ensures that the solution retains the periodic and geometric properties of trigonometric motion.

8.5 The Canonical Form of the Euler Equation and Hamiltonian

8.5.1 The Legendre Transformation

The Legendre transformation is the bridge from a velocity description to a momentum description. The idea is easiest to see for a strictly convex function $f(\xi)$.

Definition 8.5 (Legendre transform) Define the new variable

$$p = \frac{df}{d\xi}. \quad (8.107)$$

If $d^2f/d\xi^2 > 0$, then p determines ξ locally and we may regard ξ as a function of p . The Legendre transform of f is

$$f^*(p) = p\xi - f(\xi). \quad (8.108)$$

At first sight, the right hand side still contains ξ . The point is that $\xi = \xi(p)$ is already fixed by $p = df/d\xi$. Taking the total differential gives

$$df^* = \xi dp + p d\xi - \frac{df}{d\xi} d\xi = \xi dp, \quad (8.109)$$

and hence

$$\frac{df^*}{dp} = \xi, \quad \frac{d^2f^*}{dp^2} = \left(\frac{d^2f}{d\xi^2} \right)^{-1} \quad (8.110)$$

The transformation exchanges the variable with the slope of the original graph. Applying it once more returns the original function, as long as the convexity and invertibility assumptions hold.

For several variables, let $f = f(v_1, \dots, v_N)$ and define

$$p_i = \frac{\partial f}{\partial v_i}, \quad f^*(p_1, \dots, p_N) = \sum_{i=1}^N p_i v_i - f(v_1, \dots, v_N). \quad (8.111)$$

The transformation is locally valid when the Hessian matrix

$$\left[\frac{\partial^2 f}{\partial v_i \partial v_j} \right]$$

is nonsingular. This condition is the precise mathematical reason why the velocities can be replaced by the momenta.

8.5.2 Canonical Form and Hamilton's Equations

Consider the functional

$$J[y] = \int_a^b F(x, y, y') dx. \quad (8.112)$$

Definition 8.6 (Canonical momentum and Hamiltonian) Introduce the canonical momentum

$$p = \frac{\partial F}{\partial y'}. \quad (8.113)$$

Suppose $\partial^2 F / \partial y'^2 \neq 0$, so this equation can be solved locally for $y' = y'(x, y, p)$. The Hamiltonian associated with F is the partial Legendre transform

$$\mathcal{H}(x, y, p) = py' - F(x, y, y'). \quad (8.114)$$

The shortest derivation of the canonical equations comes from taking one total differential. From (8.114),

$$\begin{aligned} d\mathcal{H} &= y' dp + p dy' - \frac{\partial F}{\partial x} dx - \frac{\partial F}{\partial y} dy - \frac{\partial F}{\partial y'} dy' \\ &= y' dp - \frac{\partial F}{\partial x} dx - \frac{\partial F}{\partial y} dy. \end{aligned} \quad (8.115)$$

The two terms containing dy' cancel because $p = \partial F / \partial y'$. We can now read off

$$\frac{\partial \mathcal{H}}{\partial p} = y', \quad \frac{\partial \mathcal{H}}{\partial y} = -\frac{\partial F}{\partial y}, \quad \frac{\partial \mathcal{H}}{\partial x} = -\frac{\partial F}{\partial x}. \quad (8.116)$$

The Euler-Lagrange equation says $p' = \partial F / \partial y$. Therefore the original second order equation becomes two first order equations

$$y' = \frac{\partial \mathcal{H}}{\partial p}, \quad p' = -\frac{\partial \mathcal{H}}{\partial y} \quad (8.117)$$

This is the canonical form of the Euler equation. Nothing physical has changed. We have only replaced one second order equation for y with a first order system for (y, p) .

For analytical mechanics, set $x = t$, $y_i = q_i$, and $F = \mathcal{L}(q_i, \dot{q}_i, t)$. Then

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}, \quad \mathcal{H}(q_i, p_i, t) = \sum_{i=1}^N p_i \dot{q}_i - \mathcal{L}(q_i, \dot{q}_i, t), \quad (8.118)$$

and the canonical equations become

$$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i} \quad (8.119)$$

These are Hamilton's equations.

8.5.3 First Integrals and Cyclic Variables

The canonical form makes several conservation laws almost automatic. Along a solution of (8.117),

$$\frac{d\mathcal{H}}{dx} = \frac{\partial\mathcal{H}}{\partial x} + \frac{\partial\mathcal{H}}{\partial y}y' + \frac{\partial\mathcal{H}}{\partial p}p' = \frac{\partial\mathcal{H}}{\partial x}. \quad (8.120)$$

Thus $\mathcal{H}$ is constant whenever it has no explicit dependence on x . In mechanics, this is the usual conservation of energy when $\mathcal{L}$ has no explicit time dependence.

There is another simple case. If F does not depend on y , then

$$p' = \frac{\partial F}{\partial y} = 0. \quad (8.121)$$

The momentum $p = \partial F / \partial y'$ is conserved. In mechanics, such a coordinate is called a cyclic coordinate.

8.5.4 Canonical Transformations and the Hamilton-Jacobi Equation

A transformation $(q_i, p_i) \mapsto (Q_i, P_i)$ is canonical if it preserves the form of Hamilton's equations. A practical way to guarantee this is to require that the two action integrands differ only by an exact differential

$$\sum_i p_i dq_i - \mathcal{H} dt = - \sum_i Q_i dP_i - \mathcal{H}^* dt + dS. \quad (8.122)$$

Because an exact differential contributes only endpoint terms to the action, it does not change the equations of motion.

For a generating function $S(q_i, P_i, t)$, comparison of the coefficients in (8.122) gives

$$p_i = \frac{\partial S}{\partial q_i}, \quad Q_i = \frac{\partial S}{\partial P_i}, \quad \mathcal{H}^* = \mathcal{H} + \frac{\partial S}{\partial t}. \quad (8.123)$$

The clever choice is to make the new Hamiltonian vanish. Then the new canonical variables are constant and S satisfies

$$\mathcal{H}\left(q_i, \frac{\partial S}{\partial q_i}, t\right) + \frac{\partial S}{\partial t} = 0 \quad (8.124)$$

This is the Hamilton-Jacobi equation. It converts the problem of integrating Hamilton's ordinary differential equations into a first order partial differential equation for one generating function S .

8.5.5 The Action as a Function of the Endpoint

The Hamilton Jacobi equation becomes much less mysterious when the action itself is treated as a function of the terminal point. Fix an initial point (t_0, q_{i0}) . For every terminal point (t, q_i) sufficiently close to it, assume there is a unique classical path joining the two points. Define

$$S(q_i, t) = \int_{t_0}^t \mathcal{L}(q_i, \dot{q}_i, \tau) d\tau, \quad (8.125)$$

where the integral is evaluated along that classical path.

We now vary only the terminal point. Since the path already satisfies the Euler Lagrange equations, all interior terms vanish. The endpoint formula gives

$$\delta S = \sum_i p_i \delta q_i - \mathcal{H} \delta t, \quad (8.126)$$

where

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}, \quad \mathcal{H} = \sum_i p_i \dot{q}_i - \mathcal{L} \quad (8.127)$$

are evaluated at the terminal point. On the other hand, the ordinary total differential of $S(q_i, t)$ is

$$\delta S = \sum_i \frac{\partial S}{\partial q_i} \delta q_i + \frac{\partial S}{\partial t} \delta t. \quad (8.128)$$

The endpoint displacements are independent, so comparison with (8.126) yields

$$\frac{\partial S}{\partial q_i} = p_i, \quad \frac{\partial S}{\partial t} = -\mathcal{H}. \quad (8.129)$$

Eliminating the momenta gives

$$\frac{\partial S}{\partial t} + \mathcal{H}\left(q_i, \frac{\partial S}{\partial q_i}, t\right) = 0. \quad (8.130)$$

Thus the Hamilton Jacobi equation is simply the differential statement of how the classical action changes when its endpoint moves.

8.5.6 Complete Integrals and Jacobi's Theorem

Definition 8.7 (Complete integral) A complete integral of the Hamilton Jacobi equation is a solution

$$S = S(q_1, \dots, q_N, \alpha_1, \dots, \alpha_N, t) \quad (8.131)$$

that depends on N independent parameters α_i and satisfies the nondegeneracy condition

$$\det \left[\frac{\partial^2 S}{\partial q_i \partial \alpha_j} \right] \neq 0. \quad (8.132)$$

An additive constant is irrelevant because only derivatives of S enter the equation.

Theorem 8.4 (Jacobi's theorem) *Let S be a complete integral satisfying (8.132). Define constants β_i by*

$$\frac{\partial S}{\partial \alpha_i} = \beta_i, \quad i = 1, \dots, N. \quad (8.133)$$

Then these N implicit equations determine $q_i(t)$ locally. With

$$p_i(t) = \frac{\partial S}{\partial q_i}, \quad (8.134)$$

the functions $(q_i(t), p_i(t))$ satisfy Hamilton's equations.

Proof. Differentiate (8.133) with respect to time:

$$0 = \frac{d}{dt} \left(\frac{\partial S}{\partial \alpha_i} \right) = \frac{\partial^2 S}{\partial t \partial \alpha_i} + \sum_j \frac{\partial^2 S}{\partial q_j \partial \alpha_i} \dot{q}_j. \quad (8.135)$$

Differentiate the Hamilton Jacobi equation with respect to α_i . Since $\mathcal{H}$ depends on α_i only through $p_j = \frac{\partial S}{\partial q_j}$,

$$\frac{\partial^2 S}{\partial t \partial \alpha_i} + \sum_j \frac{\partial \mathcal{H}}{\partial p_j} \frac{\partial^2 S}{\partial q_j \partial \alpha_i} = 0. \quad (8.136)$$

Subtracting (8.136) from (8.135) gives

$$\sum_j \frac{\partial^2 S}{\partial q_j \partial \alpha_i} \left(\dot{q}_j - \frac{\partial \mathcal{H}}{\partial p_j} \right) = 0. \quad (8.137)$$

The matrix in (8.132) is invertible, so

$$\dot{q}_j = \frac{\partial \mathcal{H}}{\partial p_j}. \quad (8.138)$$

Next differentiate (8.134) with respect to time:

$$\dot{p}_i = \frac{\partial^2 S}{\partial t \partial q_i} + \sum_j \frac{\partial^2 S}{\partial q_i \partial q_j} \dot{q}_j. \quad (8.139)$$

Differentiate the Hamilton Jacobi equation with respect to q_i :

$$\frac{\partial^2 S}{\partial t \partial q_i} + \frac{\partial \mathcal{H}}{\partial q_i} + \sum_j \frac{\partial \mathcal{H}}{\partial p_j} \frac{\partial^2 S}{\partial q_i \partial q_j} = 0. \quad (8.140)$$

Using (8.138) in (8.139), the terms containing the second derivatives of S cancel, leaving

$$\dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i}. \quad (8.141)$$

Both Hamilton equations have been recovered. $\square$

Jacobi's theorem explains the real power of the method. Solving one first order partial differential equation with a complete set of parameters can produce the full $2N$ parameter family of solutions of Hamilton's ordinary differential equations. The constants α_i and β_i play the role of new canonical variables.

8.5.7 Time Independent Hamiltonians

If $\mathcal{H}$ has no explicit time dependence, seek a separated solution

$$S(q_i, t) = W(q_i) - Et. \quad (8.142)$$

The Hamilton Jacobi equation becomes

$$\mathcal{H}\left(q_i, \frac{\partial W}{\partial q_i}\right) = E. \quad (8.143)$$

The function W is Hamilton's characteristic function. Since

$$p_i = \frac{\partial W}{\partial q_i}, \quad (8.144)$$

equation (8.143) is often separable when the original equations of motion are not.

For one degree of freedom with

$$\mathcal{H}(q, p) = \frac{p^2}{2m} + V(q), \quad (8.145)$$

we obtain

$$\frac{1}{2m} \left(\frac{dW}{dq} \right)^2 + V(q) = E. \quad (8.146)$$

Therefore

$$W(q, E) = \pm \int_{q_0}^q \sqrt{2m(E - V(s))} \, ds. \quad (8.147)$$

Jacobi's theorem gives

$$\frac{\partial S}{\partial E} = \frac{\partial W}{\partial E} - t = \beta, \quad (8.148)$$

or

$$t + \beta = \pm \int_{q_0}^q \frac{m \, ds}{\sqrt{2m(E - V(s))}}. \quad (8.149)$$

This is exactly the usual quadrature obtained from energy conservation. The Hamilton Jacobi method has not changed the physics. It has organized the integration into the geometry of the action.

8.5.8 Canonical Transformations and the Lagrange Bracket Test

Suppose the old variables are functions of new variables and time,

$$q_i = q_i(Q_j, P_j, t), \quad p_i = p_i(Q_j, P_j, t). \quad (8.150)$$

To preserve Hamilton's equations for every Hamiltonian, the transformation must preserve the canonical two form. In coordinates this requires

$$\begin{aligned} \sum_i \left(\frac{\partial q_i}{\partial Q_j} \frac{\partial p_i}{\partial Q_k} - \frac{\partial q_i}{\partial Q_k} \frac{\partial p_i}{\partial Q_j} \right) &= 0, \\ \sum_i \left(\frac{\partial q_i}{\partial P_j} \frac{\partial p_i}{\partial P_k} - \frac{\partial q_i}{\partial P_k} \frac{\partial p_i}{\partial P_j} \right) &= 0, \\ \sum_i \left(\frac{\partial q_i}{\partial Q_j} \frac{\partial p_i}{\partial P_k} - \frac{\partial q_i}{\partial P_k} \frac{\partial p_i}{\partial Q_j} \right) &= \delta_{jk}. \end{aligned} \quad (8.151)$$

These relations are the coordinate version of the canonical transformation condition used in the source notes. A generating function automatically satisfies them because taking one more exterior derivative of (8.122) removes the exact term dS .

For one degree of freedom, (8.151) reduces to

$$\frac{\partial(q, p)}{\partial(Q, P)} = 1. \quad (8.152)$$

Thus a canonical transformation preserves oriented phase area. This gives a quick test in two dimensional phase space, although a generating function remains the safest construction in higher dimensions.

Ok, you've done your part by studying this chapter, which is mentioned in Taylor's textbook and Prof. Kuo's course. I sincerely recommend you stop here to protect your wonderful freshman year. But if you enjoy being abused by knowledge, then go on. But if you're not a freshman and you still quit? Shame on you.

8.6 The Second Variation

The Euler-Lagrange equation only tells us that the first variation vanishes. It does not tell us whether the stationary path is a minimum, a maximum, or neither. The second variation is the next test, just as the second derivative is in ordinary calculus.

Let

$$J[y] = \int_a^b F(x, y, y') \, dx, \quad y_\alpha = y + \alpha\eta, \quad \eta(a) = \eta(b) = 0. \quad (8.153)$$

Using (8.9) and (8.25), the expansion about $\alpha = 0$ is

$$J[y_\alpha] = J[y] + \alpha \delta J[y; \eta] + \frac{\alpha^2}{2} \delta^2 J[y; \eta] + o(\alpha^2). \quad (8.154)$$

Along an extremal, $\delta J = 0$. The sign of the leading quadratic term is controlled by

$$\delta^2 J[y; \eta] = \int_a^b \left[\frac{\partial^2 F}{\partial y'^2} \eta'^2 + 2 \frac{\partial^2 F}{\partial y \partial y'} \eta \eta' + \frac{\partial^2 F}{\partial y^2} \eta^2 \right] dx. \quad (8.155)$$

All derivatives of F in this section are evaluated along the candidate extremal unless stated otherwise.

8.6.1 Quadratic Functional and Legendre Condition

The mixed term in (8.155) can be simplified by integration by parts. Define

$$P(x) = \frac{\partial^2 F}{\partial y'^2}, \quad Q(x) = \frac{\partial^2 F}{\partial y^2} - \frac{d}{dx} \left(\frac{\partial^2 F}{\partial y \partial y'} \right). \quad (8.156)$$

Since $\eta(a) = \eta(b) = 0$,

$$\delta^2 J[y; \eta] = \int_a^b (P\eta'^2 + Q\eta^2) \, dx. \quad (8.157)$$

This is a quadratic functional of the variation η . A necessary condition for a weak minimum is

$$P(x) = \frac{\partial^2 F}{\partial y'^2} \geq 0 \quad \text{for all } x \in [a, b]. \quad (8.158)$$

This is Legendre's necessary condition. The reason is local. If $P < 0$ near some point, we can choose a narrow variation with a large derivative in that region. The negative $P\eta'^2$ term then dominates the bounded $Q\eta^2$ term, so the second variation becomes negative.

The stronger assumption

$$\frac{\partial^2 F}{\partial y'^2} > 0 \quad (8.159)$$

is called the strengthened Legendre condition. It is important for sufficient conditions, but it is not sufficient by itself. The $Q\eta^2$ term can still make the full integral negative.

8.6.2 Proof of Legendre's Necessary Condition

The local argument behind Legendre's condition is short enough that we should prove it. Assume that a stationary curve y gives a weak minimum and that

$$P(x_0) = \frac{\partial^2 F}{\partial y'^2}(x_0, y(x_0), y'(x_0)) < 0 \quad (8.160)$$

at some interior point x_0 . By continuity, there are $h_0 > 0$ and $\gamma > 0$ such that

$$P(x) \leq -\gamma \quad \text{for } |x - x_0| \leq h_0. \quad (8.161)$$

For $0 < h < h_0$, take the tent variation

$$\eta_h(x) = \begin{cases} h - |x - x_0|, & |x - x_0| \leq h, \\ 0, & |x - x_0| > h. \end{cases} \quad (8.162)$$

It is piecewise continuously differentiable and can be smoothed near its three corners without changing the following estimates. Almost everywhere on its support,

$$\eta'_h(x)^2 = 1, \quad 0 \leq \eta_h(x) \leq h. \quad (8.163)$$

Let M be an upper bound for $|Q|$ near x_0 . The quadratic functional satisfies

$$\delta^2 J[y; \eta_h] = \int_{x_0-h}^{x_0+h} (P\eta_h'^2 + Q\eta_h^2) dx \leq -2\gamma h + M \int_{x_0-h}^{x_0+h} \eta_h^2 dx. \quad (8.164)$$

A direct calculation gives

$$\int_{x_0-h}^{x_0+h} \eta_h^2 dx = \frac{2h^3}{3}. \quad (8.165)$$

Therefore

$$\delta^2 J[y; \eta_h] \leq -2\gamma h + \frac{2M}{3} h^3 < 0 \quad (8.166)$$

for sufficiently small h . This contradicts the necessary condition $\delta^2 J \geq 0$. Hence

$$\frac{\partial^2 F}{\partial y'^2} \geq 0 \quad (8.167)$$

along every weak minimizing extremal. $\square$

The proof also explains why the coefficient of η'^2 receives special attention. By shrinking the support, the derivative term is of order h , while the term containing η^2 is only of order h^3 . Localized variations therefore isolate the highest derivative part of the quadratic functional.

8.6.3 The Accessory Variational Problem

The quadratic functional

$$\mathcal{Q}[\eta] = \int_a^b (P\eta'^2 + Q\eta^2) dx \quad (8.168)$$

is itself a variational problem. Vary η by $\eta + \varepsilon\zeta$, where $\zeta(a) = \zeta(b) = 0$. Then

$$\left. \frac{d}{d\varepsilon} \mathcal{Q}[\eta + \varepsilon\zeta] \right|_{\varepsilon=0} = 2 \int_a^b (P\eta'\zeta' + Q\eta\zeta) dx. \quad (8.169)$$

After integration by parts,

$$\delta\mathcal{Q}[\eta; \zeta] = 2 \int_a^b \left[Q\eta - \frac{d}{dx}(P\eta') \right] \zeta dx. \quad (8.170)$$

The fundamental lemma gives

$$\frac{d}{dx}(P\eta') - Q\eta = 0. \quad (8.171)$$

This is precisely the Jacobi equation. It is called the accessory equation because it is the Euler Lagrange equation of the second variation rather than of the original functional.

There is also a direct relation to neighboring extremals. Let $y(x, s)$ be a smooth one parameter family of solutions of the original Euler Lagrange equation, and define

$$u(x) = \left. \frac{\partial y(x, s)}{\partial s} \right|_{s=0}. \quad (8.172)$$

Differentiate the Euler Lagrange equation with respect to s . After collecting terms and using equality of mixed partial derivatives, one obtains

$$\frac{d}{dx}(Pu') - Qu = 0. \quad (8.173)$$

Thus Jacobi fields are infinitesimal separations between neighboring extremals. This is why their zeros detect the intersection or focusing of an extremal family.

8.6.4 Jacobi's Necessary Condition and Conjugate Points

To study the complete quadratic functional, vary (8.157) with respect to η . Its Euler-Lagrange equation is

$$\frac{d}{dx}(Pu') - Qu = 0. \quad (8.174)$$

This is the Jacobi equation. A point $c > a$ is conjugate to a if there is a nonzero solution of (8.174) satisfying

$$u(a) = u(c) = 0. \quad (8.175)$$

The geometric meaning is useful. Nearby extremals start to focus when a Jacobi field vanishes again. At that point, the original extremal begins to lose its local minimizing property.

Assume $P > 0$. If the second variation is positive for every nonzero admissible variation, then there can be no point conjugate to a in $(a, b]$. Otherwise a Jacobi field vanishing at a and at a conjugate point can be used to construct a variation for which the quadratic functional fails to be positive.

Jacobi's necessary condition for a weak minimum is usually stated in the slightly weaker form

$$\text{There is no point conjugate to } a \text{ in } (a, b). \quad (8.176)$$

If b itself is conjugate to a , the second variation may be zero in a nontrivial direction, so strict minimality is lost.

8.6.5 Proof of Jacobi's Necessary Condition

Assume the strengthened Legendre condition $P > 0$ and suppose, for contradiction, that $c \in (a, b)$ is conjugate to a . Then there is a nonzero Jacobi field u such that

$$u(a) = u(c) = 0. \quad (8.177)$$

Define the piecewise variation

$$\eta(x) = \begin{cases} u(x), & a \leq x \leq c, \\ 0, & c < x \leq b. \end{cases} \quad (8.178)$$

It is continuous, satisfies the endpoint conditions, and is piecewise C^1 . Since u satisfies the Jacobi equation,

$$\begin{aligned} \mathcal{Q}[\eta] &= \int_a^c (Pu'^2 + Qu^2) dx \\ &= [Pu'u]_a^c - \int_a^c u [(Pu')' - Qu] dx = 0. \end{aligned} \quad (8.179)$$

Zero alone does not yet contradict nonnegativity. The decisive point is that η has a corner at c . Choose a smooth admissible function ζ with $\zeta(c) \neq 0$. The bilinear form associated with $\mathcal{Q}$ is

$$\mathcal{B}(v, w) = \int_a^b (Pv'w' + Qvw) dx. \quad (8.180)$$

Using the Jacobi equation on $[a, c]$ and integrating by parts,

$$\mathcal{B}(\eta, \zeta) = P(c)u'(c)\zeta(c). \quad (8.181)$$

A nonzero solution of a second order linear equation cannot satisfy both $u(c) = 0$ and $u'(c) = 0$, because uniqueness would force $u \equiv 0$. Hence the right hand side of (8.181) is nonzero for a suitable ζ .

Now use the exact quadratic identity

$$\mathcal{Q}[\eta + \varepsilon\zeta] = \mathcal{Q}[\eta] + 2\varepsilon\mathcal{B}(\eta, \zeta) + \varepsilon^2\mathcal{Q}[\zeta]. \quad (8.182)$$

Choose the sign of ε opposite to the sign of $\mathcal{B}(\eta, \zeta)$ and take $|\varepsilon|$ sufficiently small. Because (8.179) gives $\mathcal{Q}[\eta] = 0$, the linear term dominates and

$$\mathcal{Q}[\eta + \varepsilon\zeta] < 0. \quad (8.183)$$

This contradicts the necessary nonnegativity of the second variation. Therefore no point of (a, b) can be conjugate to a . $\square$

If b itself is conjugate to a , the Jacobi field is already an admissible variation and gives $\mathcal{Q}[u] = 0$. This does not contradict a nonstrict minimum, but it rules out a positive definite second variation. That is the reason for the different endpoint conventions in the necessary and sufficient versions of Jacobi's condition.

8.6.6 Sufficient Condition for a Weak Extremum

Suppose y satisfies the Euler-Lagrange equation, $P > 0$, and there is no point conjugate to a in $(a, b]$. Let u solve the Jacobi equation with $u(a) = 0$ and $u'(a) \neq 0$. The absence of conjugate points means $u(x) \neq 0$ for $a < x \leq b$.

The key identity is

$$P\eta'^2 + Q\eta^2 = Pu^2 \left[\frac{d}{dx} \left(\frac{\eta}{u} \right) \right]^2 + \frac{d}{dx} \left(P \frac{u'}{u} \eta^2 \right). \quad (8.184)$$

To verify the identity, write $\eta = uv$. Then

$$\eta' = u'v + uv'. \quad (8.185)$$

Using $(Pu')' = Qu$, we find

$$\begin{aligned}
P\eta'^2 + Q\eta^2 &= Pu^2v'^2 + 2Puuv'v' + Pu'^2v^2 + Qu^2v^2 \\
&= Pu^2v'^2 + \frac{d}{dx} (Puuv'^2).
\end{aligned} \tag{8.186}$$

Since $v = \eta/u$, this is exactly (8.184). At b , the boundary term is zero because $\eta(b) = 0$. Near a , both u and η vanish linearly, so

$$P \frac{u'}{u} \eta^2 = O(x - a) \rightarrow 0. \tag{8.187}$$

The identity may therefore be integrated first over $[a + \varepsilon, b]$ and then passed to the limit $\varepsilon \rightarrow 0$. We obtain

$$\delta^2 J[y; \eta] = \int_a^b Pu^2 \left[\frac{d}{dx} \left(\frac{\eta}{u} \right) \right]^2 dx > 0 \tag{8.188}$$

for every nonzero admissible η . This gives the standard sufficient condition

y satisfies the Euler-Lagrange equation,

$$\frac{\partial^2 F}{\partial y'^2} > 0, \quad \implies y \text{ is a strict weak minimum.} \tag{8.189}$$

and no point in $(a, b]$ is conjugate to a

There is one more logical step between a positive second variation and a genuine weak minimum. We include it because this is exactly where a formal proof can become too quick. Under the absence of conjugate points, one may choose a Jacobi solution w that has no zero on the closed interval $[a, b]$. Repeating the preceding identity gives

$$\mathcal{Q}[\eta] = \int_a^b Pw^2 \left[\frac{d}{dx} \left(\frac{\eta}{w} \right) \right]^2 dx. \tag{8.190}$$

Because P and w^2 have positive minima on the compact interval, (8.190), together with the endpoint conditions, yields a constant $C > 0$ such that

$$\mathcal{Q}[\eta] \geq C \int_a^b (\eta'^2 + \eta^2) dx. \tag{8.191}$$

The third derivatives of F are bounded in a sufficiently small tube around the extremal. Taylor's theorem then gives

$$J[y + \eta] - J[y] = \frac{1}{2} \mathcal{Q}[\eta] + R[\eta], \tag{8.192}$$

$$|R[\eta]| \leq C_1 \|\eta\|_{C^1} \int_a^b (\eta'^2 + \eta^2) dx. \tag{8.193}$$

Choose $\|\eta\|_{C^1} < C/(4C_1)$. Combining this estimate with (8.191) gives

$$J[y + \eta] - J[y] \geq \frac{C}{4} \int_a^b (\eta'^2 + \eta^2) dx > 0 \quad (8.194)$$

for every nonzero admissible variation. This completes the sufficiency proof and shows exactly how the strengthened Legendre condition controls the Taylor remainder.

8.6.7 Several Dependent Variables

For N dependent variables, write the variation as a column vector η . Under the usual symmetry assumption on the mixed derivative matrix, the second variation takes the matrix form

$$\delta^2 J = \int_a^b (\eta'^T P \eta' + \eta^T Q \eta) dx, \quad (8.195)$$

where

$$P_{ij} = \frac{\partial^2 F}{\partial y'_i \partial y'_j}, \quad Q_{ij} = \frac{\partial^2 F}{\partial y_i \partial y_j} - \frac{d}{dx} \left(\frac{\partial^2 F}{\partial y_i \partial y'_j} \right). \quad (8.196)$$

Legendre's condition now requires the matrix P to be positive semidefinite. The strengthened condition requires it to be positive definite.

The Jacobi equation becomes the linear system

$$\frac{d}{dx} (P \mathbf{u}') - Q \mathbf{u} = 0. \quad (8.197)$$

Let $U(x)$ be the matrix whose columns are the solutions satisfying $U(a) = 0$ and $U'(a) = I$. A point c is conjugate to a when

$$\det U(c) = 0. \quad (8.198)$$

This is the direct generalization of a Jacobi field vanishing twice. The sufficient condition is unchanged in spirit. The candidate must satisfy the Euler-Lagrange equations, P must be positive definite, and there must be no conjugate point in $(a, b]$.

8.7 Sufficient Conditions for a Strong Extremum

8.7.1 Weak and Strong Extrema

Definition 8.8 (Weak and strong extrema) A weak neighborhood controls both the curve and its slope. A strong neighborhood controls only the curve itself. In symbols, a weak minimum compares y with curves satisfying

$$\|y^* - y\|_\infty + \|y^{*'} - y'\|_\infty < \varepsilon, \quad (8.199)$$

whereas a strong minimum only requires

$$\|y^* - y\|_\infty < \varepsilon. \quad (8.200)$$

A nearby curve can remain very close to y while oscillating rapidly, so its slope may be far from y' . This is why a strong minimum needs a condition that works for finite changes of slope. The second variation only sees infinitesimal slope changes.

8.7.2 A Field of Extremals and Hilbert's Invariant Integral

Definition 8.9 (Field of extremals) A one parameter family $y = \phi(x, \alpha)$ is called a field of extremals in a region if every member is an extremal and exactly one member passes through each point of the region. The condition

$$\frac{\partial \phi}{\partial \alpha} \neq 0 \quad (8.201)$$

prevents neighboring extremals from intersecting locally. The field assigns a slope $p(x, y)$ to every point by taking the slope of the unique extremal through that point.

For this field, define the line integral

$$I^*[\gamma] = \int_\gamma \left[F(x, y, p) - p \frac{\partial F}{\partial y'}(x, y, p) \right] dx + \frac{\partial F}{\partial y'}(x, y, p) dy. \quad (8.202)$$

The Euler-Lagrange equation for the field makes the one form in (8.202) exact. Hence the value of I^* depends only on the endpoints of γ . This is Hilbert's invariant integral.

If γ itself follows the field, then $dy = p dx$ and (8.202) reduces to

$$I^*[\gamma] = \int_\gamma F(x, y, y') dx = J[\gamma]. \quad (8.203)$$

This lets us compare the extremal with any nearby curve having the same endpoints.

8.7.3 The Weierstrass Excess Function

Definition 8.10 (Weierstrass excess function) The Weierstrass excess function is

$$E(x, y, p, q) = F(x, y, q) - F(x, y, p) - (q - p) \frac{\partial F}{\partial y'}(x, y, p). \quad (8.204)$$

Geometrically, fix (x, y) and view F as a function of the slope. Then E is the vertical difference between $F(q)$ and the tangent line to F at p . Therefore $E \geq 0$ for every q precisely when F lies above that tangent line.

Let y be an extremal embedded in a field with slope $p(x, y)$. Hilbert's invariant integral gives the exact identity

$$J[y^*] - J[y] = \int_a^b E(x, y^*, p(x, y^*), y^{*'}) \, dx. \quad (8.205)$$

This is the main proof. If $E \geq 0$ throughout the field for every finite trial slope, then the right hand side is nonnegative even when $y^{*'}$ is far from y' . Thus y gives a strong minimum.

The sufficient conditions can be summarized as

$$\begin{aligned} &y \text{ is an extremal embedded in a field,} \\ &E(x, y, p, q) \geq 0 \text{ for every finite } q \implies y \text{ is a strong minimum.} \end{aligned} \quad (8.206)$$

Conversely, a strong minimum must satisfy Weierstrass' necessary condition

$$E(x, y, y', q) \geq 0 \quad \text{for every finite } q. \quad (8.207)$$

For q close to p , Taylor's theorem gives

$$E(x, y, p, q) = \frac{1}{2}(q - p)^2 \frac{\partial^2 F}{\partial y'^2}(x, y, p + \theta(q - p)), \quad 0 < \theta < 1. \quad (8.208)$$

So the Weierstrass condition contains the Legendre condition as its local, small slope limit.

8.7.4 Why Hilbert's Integral is Path Independent

The path independence used above is not a guess. It follows directly from the Euler Lagrange equation for the extremal field. Let $p = p(x, y)$ denote the slope assigned by the field and define

$$\begin{aligned} A(x, y) &= F(x, y, p) - p \frac{\partial F}{\partial y'}(x, y, p), \\ B(x, y) &= \frac{\partial F}{\partial y'}(x, y, p). \end{aligned} \quad (8.209)$$

Hilbert's integral is the line integral of

$$A(x, y) \, dx + B(x, y) \, dy. \quad (8.210)$$

In a simply connected region, this one form is exact if

$$\frac{\partial A}{\partial y} = \frac{\partial B}{\partial x}. \quad (8.211)$$

Here the derivatives of A and B include the dependence of p on (x, y) . A direct chain rule calculation gives

$$\begin{aligned}\frac{\partial A}{\partial y} &= \frac{\partial F}{\partial y} - p \frac{\partial^2 F}{\partial y \partial y'} - p \frac{\partial^2 F}{\partial y'^2} \frac{\partial p}{\partial y}, \\ \frac{\partial B}{\partial x} &= \frac{\partial^2 F}{\partial x \partial y'} + \frac{\partial^2 F}{\partial y'^2} \frac{\partial p}{\partial x}.\end{aligned}\tag{8.212}$$

Along an extremal of the field, $dy/dx = p(x, y)$. Hence the derivative of the field slope along that extremal is

$$\frac{dp}{dx} = \frac{\partial p}{\partial x} + p \frac{\partial p}{\partial y}.\tag{8.213}$$

Apply the Euler Lagrange equation to the field:

$$\frac{\partial F}{\partial y} = \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = \frac{\partial^2 F}{\partial x \partial y'} + p \frac{\partial^2 F}{\partial y \partial y'} + \frac{\partial^2 F}{\partial y'^2} \left(\frac{\partial p}{\partial x} + p \frac{\partial p}{\partial y} \right).\tag{8.214}$$

Substituting (8.214) into the first line of (8.212) leaves exactly the second line. Thus (8.211) holds.

Consequently there is a local function $S(x, y)$ such that

$$dS = A dx + B dy.\tag{8.215}$$

The symbol d here is the exterior derivative. Therefore

$$I^*[\gamma] = S(x_1, y_1) - S(x_0, y_0)\tag{8.216}$$

depends only on the endpoints. This completes the proof of Hilbert's invariant integral.

8.7.5 Derivation of the Weierstrass Comparison Formula

Let y be the member of the extremal field joining the fixed endpoints, and let y^* be any sufficiently nearby piecewise C^1 competitor. Because Hilbert's integral is path independent,

$$I^*[y^*] = I^*[y].\tag{8.217}$$

Along the field extremal y , its actual slope equals the field slope, so

$$I^*[y] = J[y].\tag{8.218}$$

Along the comparison curve y^* , denote the field slope by

$$p^*(x) = p(x, y^*(x))\tag{8.219}$$

and the actual comparison slope by

$$q^*(x) = y^{*'}(x). \quad (8.220)$$

The Hilbert integrand along y^* is

$$\begin{aligned} & F(x, y^*, p^*) - p^* \frac{\partial F}{\partial y'}(x, y^*, p^*) + q^* \frac{\partial F}{\partial y'}(x, y^*, p^*) \\ &= F(x, y^*, p^*) + (q^* - p^*) \frac{\partial F}{\partial y'}(x, y^*, p^*). \end{aligned} \quad (8.221)$$

Subtract this expression from the actual integrand $F(x, y^*, q^*)$. By definition, the difference is

$$E(x, y^*, p^*, q^*). \quad (8.222)$$

Using (8.217), we obtain the exact identity

$$\begin{aligned} J[y^*] - J[y] &= J[y^*] - I^*[y^*] \\ &= \int_a^b E(x, y^*, p^*, q^*) \, dx. \end{aligned} \quad (8.223)$$

No Taylor expansion has been used. The comparison slope q^* may differ from p^* by a finite amount. This is exactly why the Weierstrass function can handle strong neighborhoods.

8.7.6 Proof of Weierstrass' Necessary Condition

Suppose y gives a strong minimum but, at some interior point x_0 , there is a finite slope q such that

$$E(x_0, y(x_0), y'(x_0), q) < 0. \quad (8.224)$$

By continuity, the excess remains negative in a small neighborhood of the corresponding arguments. We construct a needle competitor that replaces a short part of the original curve by two line segments. On the first segment its slope is approximately q ; the second segment returns to the original curve so that the endpoints of the replacement match.

Let the width of the replacement interval be h . Its vertical deviation from y is of order h , so the competitor converges uniformly to y as $h \rightarrow 0$. Its slope, however, makes a finite jump. A first order expansion of the change in the functional gives

$$J[y_h] - J[y] = h E(x_0, y(x_0), y'(x_0), q) + o(h). \quad (8.225)$$

The $o(h)$ term contains the slow variation of x , y , and the field slope across the short interval. By (8.224), the right hand side of (8.225) is negative for sufficiently small h . Yet y_h is arbitrarily close to y in the uniform norm. This contradicts strong minimality. Therefore

$$E(x, y, y', q) \geq 0 \quad \text{for every finite } q. \quad (8.226)$$

This proof shows the hierarchy of the conditions. The Legendre test uses infinitesimal changes of slope and examines the second derivative at $q = y'$. The Weierstrass test allows a finite slope q inside an arbitrarily narrow interval. Strong minimality must survive both tests.

8.7.7 Convexity in the Slope Variable

Fix (x, y) and view $v \mapsto F(x, y, v)$ as an ordinary one variable function. The inequality

$$F(x, y, q) \geq F(x, y, p) + (q - p) \frac{\partial F}{\partial y'}(x, y, p) \quad (8.227)$$

for all p and q is precisely the supporting tangent characterization of convexity. Thus

$$E(x, y, p, q) \geq 0 \quad \text{for all } p, q \quad (8.228)$$

if and only if F is convex in its slope variable.

If

$$\frac{\partial^2 F}{\partial y'^2} \geq m > 0 \quad (8.229)$$

in a neighborhood of the field, Taylor's theorem with integral remainder gives the quantitative estimate

$$\begin{aligned} E(x, y, p, q) &= (q - p)^2 \int_0^1 (1 - s) \frac{\partial^2 F}{\partial y'^2}(x, y, p + s(q - p)) \, ds \\ &\geq \frac{m}{2} (q - p)^2. \end{aligned} \quad (8.230)$$

Combining (8.230) with the comparison formula yields

$$J[y^*] - J[y] \geq \frac{m}{2} \int_a^b (y^{*'} - p(x, y^*))^2 \, dx. \quad (8.231)$$

Equality forces the comparison curve to follow the same extremal field. With fixed endpoints and local uniqueness, that curve is y itself. This gives a clean strict version of the strong minimum theorem.

8.8 Variational Problems Involving Multiple Integrals

The independent variable is now a point $x = (x^1, \dots, x^n)$ in a region $\Omega \subset \mathbb{R}^n$. For one unknown function $u(x)$, consider

$$J[u] = \int_{\Omega} F \left(x, u, \frac{\partial u}{\partial x^1}, \dots, \frac{\partial u}{\partial x^n} \right) d^n x. \quad (8.232)$$

This is the natural setting for continuum mechanics and classical field theory. The ordinary derivative y' is replaced by the gradient of a field.

8.8.1 Variation of a Functional Defined on a Fixed Region

Let $u \mapsto u + \alpha\eta$, with $\eta = 0$ on $\partial\Omega$. The first variation is

$$\delta J = \int_{\Omega} \left[\frac{\partial F}{\partial u} \eta + \sum_{i=1}^n \frac{\partial F}{\partial(\partial u / \partial x^i)} \frac{\partial \eta}{\partial x^i} \right] d^n x. \quad (8.233)$$

Integrating each derivative term by parts and using the divergence theorem gives

$$\begin{aligned} \delta J = \int_{\Omega} \left[\frac{\partial F}{\partial u} - \sum_{i=1}^n \frac{\partial}{\partial x^i} \left(\frac{\partial F}{\partial(\partial u / \partial x^i)} \right) \right] \eta d^n x \\ + \int_{\partial\Omega} \eta \sum_{i=1}^n n_i \frac{\partial F}{\partial(\partial u / \partial x^i)} dS. \end{aligned} \quad (8.234)$$

For fixed boundary values, the surface term vanishes. Since η is arbitrary inside Ω , the Euler-Lagrange equation is

$$\frac{\partial F}{\partial u} - \sum_{i=1}^n \frac{\partial}{\partial x^i} \left(\frac{\partial F}{\partial(\partial u / \partial x^i)} \right) = 0. \quad (8.235)$$

If the boundary value of u is free, the surface term in (8.234) gives the natural boundary condition

$$\sum_{i=1}^n n_i \frac{\partial F}{\partial(\partial u / \partial x^i)} = 0 \quad \text{on } \partial\Omega. \quad (8.236)$$

8.8.2 Equations of Motion for a Continuous System

Take a stretched string with transverse displacement $u(x, t)$, linear density ρ , and tension T . Its Lagrangian density is

$$\mathcal{L} = \frac{1}{2} \rho \left(\frac{\partial u}{\partial t} \right)^2 - \frac{1}{2} T \left(\frac{\partial u}{\partial x} \right)^2. \quad (8.237)$$

The action is a double integral over space and time

$$S[u] = \int_{t_0}^{t_1} \int_0^\ell \mathcal{L} \, dx \, dt. \quad (8.238)$$

Substitution into (8.235) gives

$$\rho \frac{\partial^2 u}{\partial t^2} - T \frac{\partial^2 u}{\partial x^2} = 0. \quad (8.239)$$

The wave equation is therefore an Euler-Lagrange equation for an infinite dimensional mechanical system. A continuous string can be viewed as the limit of more and more coupled oscillators.

For several fields $\phi_a(x)$ on spacetime, the same calculation gives

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0. \quad (8.240)$$

Equation (8.240) is the basic equation of motion in classical field theory.

8.8.3 Variation of a Functional Defined on a Variable Region

If the integration region moves, two effects occur at once. The field changes and the boundary of the region changes. Let the infinitesimal coordinate displacement be ξ^i and let the field variation at fixed coordinates be η . Then the first variation has the compact form

$$\begin{aligned} \delta J = & \int_\Omega \left[\frac{\partial F}{\partial u} - \partial_i \left(\frac{\partial F}{\partial (\partial_i u)} \right) \right] \eta \, d^n x \\ & + \int_{\partial\Omega} \left[n_i \frac{\partial F}{\partial (\partial_i u)} \eta + F \xi^i n_i \right] dS. \end{aligned} \quad (8.241)$$

The first line gives the field equation. The second line records the work done by changing boundary data and moving the boundary itself.

The one dimensional formula is especially useful. If an endpoint moves by $(\Delta x, \Delta y)$, the endpoint term is

$$\frac{\partial F}{\partial y'} \Delta y + \left(F - y' \frac{\partial F}{\partial y'} \right) \Delta x. \quad (8.242)$$

If the endpoint is constrained to lie on $y = \psi(x)$, then $\Delta y = \psi'(x) \Delta x$. Equation (8.242) gives the transversality condition

$$F + (\psi' - y') \frac{\partial F}{\partial y'} = 0. \quad (8.243)$$

This condition tells the extremal how to meet a movable boundary.

8.8.4 Several Fields on a Fixed Region

Let $u^a(x)$, where $a = 1, \dots, N$, be several dependent fields and consider

$$J[u^a] = \int_{\Omega} F(x^i, u^a, \partial_i u^a) \, d^n x. \quad (8.244)$$

Take independent variations

$$u^a \mapsto u^a + \alpha \eta^a. \quad (8.245)$$

The first variation is

$$\delta J = \int_{\Omega} \sum_a \left[\frac{\partial F}{\partial u^a} \eta^a + \sum_i \frac{\partial F}{\partial (\partial_i u^a)} \partial_i \eta^a \right] d^n x. \quad (8.246)$$

Apply the divergence theorem to each derivative term:

$$\begin{aligned} \delta J = \int_{\Omega} \sum_a \left[\frac{\partial F}{\partial u^a} - \sum_i \partial_i \left(\frac{\partial F}{\partial (\partial_i u^a)} \right) \right] \eta^a \, d^n x \\ + \int_{\partial \Omega} \sum_a \sum_i n_i \frac{\partial F}{\partial (\partial_i u^a)} \eta^a \, dS. \end{aligned} \quad (8.247)$$

Since the interior variations are independent, the field equations are

$$\frac{\partial F}{\partial u^a} - \sum_i \partial_i \left(\frac{\partial F}{\partial (\partial_i u^a)} \right) = 0, \quad a = 1, \dots, N. \quad (8.248)$$

On a free part of the boundary, the independent boundary variations give

$$\sum_i n_i \frac{\partial F}{\partial (\partial_i u^a)} = 0. \quad (8.249)$$

Dirichlet data fix u^a and therefore set $\eta^a = 0$. Natural boundary data instead fix the normal flux in (8.249). Mixed boundary value problems simply use these two choices on different portions of $\partial \Omega$.

8.8.5 Electrostatic Potential as a Variational Problem

The simplest bridge to field theory is electrostatics. For a prescribed charge density $\rho(\mathbf{x})$, define

$$J[\phi] = \int_{\Omega} \left[\frac{\epsilon_0}{2} \|\nabla \phi\|^2 - \rho \phi \right] d^3 x. \quad (8.250)$$

Take $\phi \mapsto \phi + \alpha \eta$. Then

$$\begin{aligned}
\delta J &= \int_{\Omega} (\epsilon_0 \nabla \phi \cdot \nabla \eta - \rho \eta) d^3x \\
&= \int_{\partial\Omega} \epsilon_0 \eta \nabla \phi \cdot \mathbf{n} dS - \int_{\Omega} (\epsilon_0 \nabla^2 \phi + \rho) \eta d^3x.
\end{aligned} \tag{8.251}$$

For fixed potential at the boundary, $\eta = 0$ on $\partial\Omega$. The fundamental lemma gives

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}. \tag{8.252}$$

This is Poisson's equation. In a charge free region it reduces to Laplace's equation.

The second variation is

$$\delta^2 J[\phi; \eta] = \epsilon_0 \int_{\Omega} \|\nabla \eta\|^2 d^3x. \tag{8.253}$$

It is strictly positive for every nonzero variation with zero boundary values. Therefore a solution of Poisson's equation is not merely stationary. It minimizes (8.250). If two solutions had the same Dirichlet boundary data, their difference w would satisfy $\nabla^2 w = 0$ and $w = 0$ on the boundary. Multiplying by w and integrating gives

$$\int_{\Omega} \|\nabla w\|^2 d^3x = 0, \tag{8.254}$$

so $w = 0$. The variational structure therefore gives uniqueness almost for free.

If the boundary potential is free, the surface term instead gives

$$\nabla \phi \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega. \tag{8.255}$$

This is the natural Neumann condition for zero prescribed normal electric field.

8.8.6 The Vibrating Membrane

Let $u(x, y, t)$ be the transverse displacement of a membrane with surface mass density ρ and isotropic tension T . Its Lagrangian density is

$$\mathcal{L} = \frac{\rho}{2} \left(\frac{\partial u}{\partial t} \right)^2 - \frac{T}{2} \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right]. \tag{8.256}$$

Varying the spacetime action and integrating by parts in t , x , and y gives

$$\delta S = - \int \left[\rho \frac{\partial^2 u}{\partial t^2} - T \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \right] \eta dx dy dt, \tag{8.257}$$

where the boundary terms vanish for fixed initial, final, and edge data. Hence

$$\rho \frac{\partial^2 u}{\partial t^2} - T \nabla^2 u = 0. \quad (8.258)$$

The one dimensional string and the two dimensional membrane are the same variational mechanism in different spatial dimensions.

8.8.7 Second Variation for a Multiple Integral

For a scalar field u , write $u_i = \partial_i u$. Along a stationary field, the second variation is

$$\delta^2 J[u; \eta] = \int_{\Omega} \left[\frac{\partial^2 F}{\partial u^2} \eta^2 + 2 \sum_i \frac{\partial^2 F}{\partial u \partial u_i} \eta \partial_i \eta + \sum_{i,j} \frac{\partial^2 F}{\partial u_i \partial u_j} \partial_i \eta \partial_j \eta \right] d^n x. \quad (8.259)$$

The analogue of the Legendre condition concerns the symmetric matrix

$$A^{ij} = \frac{\partial^2 F}{\partial u_i \partial u_j}. \quad (8.260)$$

If there were a point x_0 and a vector ξ with

$$\sum_{i,j} A^{ij}(x_0) \xi_i \xi_j < 0, \quad (8.261)$$

one could choose a rapidly varying localized perturbation whose gradient points mainly in the ξ direction. The highest derivative term in (8.259) would then dominate and make the second variation negative. Thus a necessary condition for a minimum is

$$\sum_{i,j} A^{ij}(x) \xi_i \xi_j \geq 0 \quad \text{for every } \xi \in \mathbb{R}^n. \quad (8.262)$$

This is the multidimensional Legendre condition. Positive definiteness of A^{ij} is the strengthened form and is closely related to ellipticity of the Euler Lagrange equation.

For several fields u^a , the highest derivative coefficient has four indices,

$$A_{ab}^{ij} = \frac{\partial^2 F}{\partial(\partial_i u^a) \partial(\partial_j u^b)}. \quad (8.263)$$

Localized rank one perturbations give the necessary condition

$$\sum_{a,b,i,j} A_{ab}^{ij} \zeta^a \zeta^b \xi_i \xi_j \geq 0 \quad (8.264)$$

for every field vector ζ^a and spatial covector ξ_i . This is the natural several field version of the Legendre test.

8.9 Noether's Theorem in Mathematics and Field Theory

Noether's theorem connects two ideas that at first seem unrelated. A continuous invariance of a functional produces a first integral of the Euler Lagrange equations. For a functional of one independent variable, the first integral is constant along every extremal. For a functional involving several independent variables, the corresponding statement is a local conservation law expressed by a divergence-free current.

The word “invariant” must be used locally. In the one-dimensional case, the functional must be invariant on every subinterval. For a multiple integral, it must be invariant on every sufficiently regular subregion. This local requirement is what allows a global equality of integrals to become a pointwise differential relation.

8.9.1 Invariant Functionals with One Independent Variable

Consider the functional

$$J[\mathbf{y}] = \int_{x_0}^{x_1} F(x, \mathbf{y}, \mathbf{y}') \, dx, \quad \mathbf{y} = (y^1, \dots, y^m). \quad (8.265)$$

Let a one parameter family of transformations be given by

$$\begin{aligned} x^* &= \Phi(x, \mathbf{y}, \mathbf{y}'; \alpha), \\ y^{*a} &= \Psi^a(x, \mathbf{y}, \mathbf{y}'; \alpha), \quad a = 1, \dots, m, \end{aligned} \quad (8.266)$$

where the value $\alpha = 0$ gives the identity transformation. Its infinitesimal generators are

$$\tau = \left. \frac{\partial \Phi}{\partial \alpha} \right|_{\alpha=0}, \quad \xi^a = \left. \frac{\partial \Psi^a}{\partial \alpha} \right|_{\alpha=0}. \quad (8.267)$$

Thus

$$x^* = x + \alpha\tau + o(\alpha), \quad y^{*a} = y^a + \alpha\xi^a + o(\alpha). \quad (8.268)$$

When the independent variable moves, ξ^a is not the change of the graph measured at a fixed value of x . The corresponding vertical variation is

$$Q^a = \xi^a - y'^a \tau. \quad (8.269)$$

The functions Q^a are often called the characteristics of the infinitesimal transformation.

Definition 8.11 (Invariance of a one-dimensional functional) The functional (8.265) is invariant under the family (8.266) if, for every admissible curve and every subinterval $[x_0, x_1]$,

$$\int_{x_0}^{x_1^*} F\left(x^*, \mathbf{y}^*, \frac{d\mathbf{y}^*}{dx^*}\right) dx^* = \int_{x_0}^{x_1} F(x, \mathbf{y}, \mathbf{y}') \, dx. \quad (8.270)$$

More generally, the transformation is a divergence symmetry if the first-order change of the integrand and measure is a total derivative

$$\delta(F dx) = \frac{dB}{dx} dx \quad (8.271)$$

for some function B . Exact invariance corresponds to $B = 0$.

Define the Euler Lagrange expressions and the canonical momenta by

$$\mathcal{E}_a(F) = \frac{\partial F}{\partial y^a} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'^a} \right), \quad p_a = \frac{\partial F}{\partial y'^a}. \quad (8.272)$$

A direct chain rule and one integration by parts give the identity

$$\delta(F dx) = \left[\sum_{a=1}^m \mathcal{E}_a(F) Q^a + \frac{d}{dx} \left(\sum_{a=1}^m p_a Q^a + F\tau \right) \right] dx. \quad (8.273)$$

This formula is valid before the Euler Lagrange equations are imposed.

Theorem 8.5 (Noether's theorem for one independent variable) *Suppose the functional (8.265) is invariant up to the total derivative in (8.271). Then every extremal satisfies*

$$\frac{d}{dx} \left[\sum_{a=1}^m p_a Q^a + F\tau - B \right] = 0. \quad (8.274)$$

Equivalently,

$$\sum_{a=1}^m p_a \xi^a + \left(F - \sum_{a=1}^m p_a y'^a \right) \tau - B = \text{const.} \quad (8.275)$$

Proof. Subtract (8.271) from (8.273). This gives

$$\sum_{a=1}^m \mathcal{E}_a(F) Q^a + \frac{d}{dx} \left[\sum_{a=1}^m p_a Q^a + F\tau - B \right] = 0. \quad (8.276)$$

Along an extremal, $\mathcal{E}_a(F) = 0$ for every a . The quantity in square brackets is therefore constant. Substituting $Q^a = \xi^a - y'^a \tau$ gives (8.275). $\square$

Two familiar first integrals are immediate consequences. If F is independent of one coordinate y^b , the translation $y^{*b} = y^b + \alpha$ has $\tau = 0$, $\xi^a = \delta^a_b$, and $B = 0$. Noether's theorem gives

$$p_b = \frac{\partial F}{\partial y'^b} = \text{const.} \quad (8.277)$$

If F has no explicit dependence on x , the translation $x^* = x + \alpha$ has $\tau = 1$ and $\xi^a = 0$. Hence

$$F - \sum_{a=1}^m p_a y'^a = \text{const.} \quad (8.278)$$

With the Hamiltonian

$$H = \sum_{a=1}^m p_a y'^a - F, \quad (8.279)$$

this becomes $H = \text{const.}$

8.9.2 Several Independent Variables and Several Fields

We now pass to the general multiple-integral functional

$$J[\mathbf{u}] = \int_R F(\mathbf{x}, \mathbf{u}, \partial \mathbf{u}) \, d^n x, \quad (8.280)$$

where

$$\mathbf{x} = (x^1, \dots, x^n), \quad \mathbf{u} = (u^1, \dots, u^m), \quad u^a_{,i} = \partial_i u^a. \quad (8.281)$$

To keep the formulas readable, repeated coordinate indices are summed from 1 to n when no explicit summation sign is shown.

Consider the infinitesimal transformation

$$\begin{aligned} x^{*i} &= x^i + \alpha \phi^i + o(\alpha), \quad i = 1, \dots, n, \\ u^{*a} &= u^a + \alpha \psi^a + o(\alpha), \quad a = 1, \dots, m. \end{aligned} \quad (8.282)$$

The change of the field at a fixed point is

$$Q^a = \psi^a - \sum_{i=1}^n u^a_{,i} \phi^i. \quad (8.283)$$

Definition 8.12 (Invariance of a multiple-integral functional) The functional (8.280) is invariant under (8.282) if

$$\int_{R^*} F(\mathbf{x}^*, \mathbf{u}^*, \partial^* \mathbf{u}^*) \, d^n x^* = \int_R F(\mathbf{x}, \mathbf{u}, \partial \mathbf{u}) \, d^n x \quad (8.284)$$

for every sufficiently regular region R . More generally, it is a divergence symmetry if

$$\delta(F \, d^n x) = \partial_i B^i \, d^n x. \quad (8.285)$$

Introduce

$$\mathcal{E}_a(F) = \frac{\partial F}{\partial u^a} - \sum_{i=1}^n \partial_i \left(\frac{\partial F}{\partial u^a_{,i}} \right), \quad P_a^i = \frac{\partial F}{\partial u^a_{,i}}. \quad (8.286)$$

The coefficient of the first-order change of the volume element is

$$\delta(d^n x) = (\partial_i \phi^i) \, d^n x. \quad (8.287)$$

Using the chain rule and the product rule gives the exact decomposition

$$\delta(F \, d^n x) = \left[\sum_{a=1}^m \mathcal{E}_a(F) Q^a + \sum_{i=1}^n \partial_i \left(\sum_{a=1}^m P_a^i Q^a + F \phi^i \right) \right] d^n x. \quad (8.288)$$

Equation (8.288) is the multidimensional form of (8.273). The first term contains the field equations. The second term is a divergence and therefore becomes a boundary integral under the divergence theorem.

Theorem 8.6 (Noether's first theorem for multiple integrals) Suppose (8.280) is invariant up to the divergence in (8.285). Define

$$j^i = \sum_{a=1}^m P_a^i Q^a + F \phi^i - B^i. \quad (8.289)$$

Then the identity

$$\partial_i j^i = - \sum_{a=1}^m \mathcal{E}_a(F) Q^a \quad (8.290)$$

holds before the field equations are imposed. On every extremal field,

$$\partial_i j^i = 0. \quad (8.291)$$

Proof. Subtract (8.285) from (8.288). The result is (8.290). If the field satisfies the Euler Lagrange equations, every $\mathcal{E}_a(F)$ vanishes and the current is divergence free. $\square$

For an r parameter family of transformations, let $\phi_{(A)}^i$, $\psi_{(A)}^a$, and $B_{(A)}^i$ denote the infinitesimal data associated with the parameter α^A , where $A = 1, \dots, r$. Then

$$Q_{(A)}^a = \psi_{(A)}^a - \sum_{i=1}^n u_{,i}^a \phi_{(A)}^i \quad (8.292)$$

produces the current

$$j_{(A)}^i = \sum_{a=1}^m P_a^i Q_{(A)}^a + F \phi_{(A)}^i - B_{(A)}^i. \quad (8.293)$$

Hence r independent constant parameters produce r conservation laws

$$\partial_i j_{(A)}^i = 0, \quad A = 1, \dots, r, \quad (8.294)$$

on every extremal field.

8.9.3 From a Conserved Current to a Conserved Charge

In field theory, let $x^0 = t$ and let $x^1, \dots, x^d$ be spatial coordinates. The local conservation law reads

$$\partial_0 j^0 + \sum_{k=1}^d \partial_k j^k = 0. \quad (8.295)$$

Integrating over a spatial region Σ gives

$$\frac{d}{dt} \int_{\Sigma} j^0 d^d x = - \int_{\partial \Sigma} \mathbf{j} \cdot \mathbf{n} dS. \quad (8.296)$$

If the flux through the boundary vanishes, the quantity

$$\boxed{Q = \int_{\Sigma} j^0 d^d x} \quad (8.297)$$

is independent of time. Thus the divergence equation is the local statement, while the conserved charge is its integrated form.

8.9.4 Spacetime Translations and the Energy Momentum Tensor

Suppose F has no explicit dependence on the independent variables x^i . For a constant translation,

$$\phi^i = a^i, \quad \psi^a = 0, \quad Q^a = - \sum_{k=1}^n a^k u_{,k}^a. \quad (8.298)$$

Taking $B^i = 0$, the Noether current becomes

$$j^i = - \sum_{k=1}^n a^k T^i_k, \quad (8.299)$$

where

$$\boxed{T^i_k = \sum_{a=1}^m \frac{\partial F}{\partial u_{,i}^a} u_{,k}^a - \delta^i_k F} \quad (8.300)$$

is the canonical energy momentum tensor. Since the constants a^k are independent,

$$\partial_i T^i_k = 0, \quad k = 1, \dots, n. \quad (8.301)$$

For a spacetime action, the corresponding integrated quantities are the energy and momentum of the field.

8.9.5 Noether's Second Theorem and Gauge Symmetry

Noether's first theorem concerns transformations depending on finitely many constant parameters. A gauge transformation depends on arbitrary functions of the independent variables. This greater freedom produces identities among the Euler Lagrange expressions themselves.

Consider r arbitrary functions $\varepsilon^A(\mathbf{x})$ and a characteristic of the form

$$Q^a = \sum_{A=1}^r \sum_{s=0}^N R_A^{a i_1 \dots i_s} \partial_{i_1} \dots \partial_{i_s} \varepsilon^A, \quad (8.302)$$

where the term with $s = 0$ means $R_A^a \varepsilon^A$. Assume the action is invariant for every smooth compactly supported choice of the functions ε^A .

Theorem 8.7 (Noether's second theorem) *For every gauge function ε^A , the Euler Lagrange expressions satisfy the identity*

$$\boxed{\sum_{a=1}^m \sum_{s=0}^N (-1)^s \partial_{i_1} \dots \partial_{i_s} (\mathcal{E}_a(F) R_A^{a i_1 \dots i_s}) \equiv 0, \quad A = 1, \dots, r.} \quad (8.303)$$

These are differential identities and hold without solving the field equations.

Proof. For compactly supported gauge functions, the boundary term in the first variation vanishes. Gauge invariance gives

$$0 = \delta J = \int_R \sum_{a=1}^m \mathcal{E}_a(F) Q^a \, d^n x. \quad (8.304)$$

Insert (8.302) and integrate each derivative of ε^A by parts. After s integrations by parts, the corresponding term acquires the factor $(-1)^s$. Since every ε^A is arbitrary, the fundamental lemma gives (8.303). $\square$

For electromagnetism, the gauge transformation is

$$\delta A_\nu = \partial_\nu \varepsilon. \quad (8.305)$$

Its characteristic contains one derivative of the arbitrary function. Noether's second theorem gives

$$\partial_\nu \mathcal{E}^\nu \equiv 0. \quad (8.306)$$

For the free electromagnetic field,

$$\mathcal{E}^\nu = \frac{1}{\mu_0} \partial_\mu F^{\mu\nu}, \quad (8.307)$$

so the identity becomes

$$\partial_\nu \partial_\mu F^{\mu\nu} = 0, \quad (8.308)$$

which follows from the antisymmetry of $F^{\mu\nu}$. If an external current is coupled to the field, gauge invariance requires

$$\partial_\nu J^\nu = 0. \quad (8.309)$$

Thus Noether's first theorem converts finite-dimensional continuous symmetries into conserved currents, while Noether's second theorem converts gauge freedom into identities among the field equations.

8.10 Mathematical Structure of Field Theory

Classical field theory is variational calculus with several independent variables. A mechanical coordinate $q_i(t)$ becomes a field $\phi_a(x^\mu)$, and the Lagrangian becomes a density over spacetime. We use the metric signature $(-, +, +, +)$ and take $x^0 = ct$. Natural units are used when factors of c are not essential.

8.10.1 Principle of Stationary Action for Fields

For fields ϕ_a defined on a spacetime region Ω , the action is

$$S[\phi_a] = \int_{\Omega} \mathcal{L}(\phi_a, \partial_\mu \phi_a, x^\mu) d^4x. \quad (8.310)$$

Take $\phi_a \mapsto \phi_a + \alpha \eta_a$ with $\eta_a = 0$ on the boundary. Then

$$\begin{aligned} \delta S &= \int_{\Omega} \sum_a \left[\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) \right] \eta_a d^4x \\ &\quad + \int_{\partial\Omega} \sum_a n_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \eta_a d^3\Sigma. \end{aligned} \quad (8.311)$$

The boundary term vanishes for fixed boundary data. The stationary action condition $\delta S = 0$ then gives the field Euler-Lagrange equations in (8.240).

For a real scalar field,

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi). \quad (8.312)$$

Its equation of motion is

$$\square \phi - \frac{dV}{d\phi} = 0, \quad \square = \partial_\mu \partial^\mu. \quad (8.313)$$

With $V(\phi) = m^2 \phi^2/2$, this becomes the Klein-Gordon equation.

Electromagnetism fits the same structure. With $A_0 = -\phi/c$ and

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (8.314)$$

take

$$\mathcal{L} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu. \quad (8.315)$$

Varying A_ν gives

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu. \quad (8.316)$$

The gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \chi$ leaves $F_{\mu\nu}$ unchanged. This symmetry explains why the potential contains redundant information.

8.10.2 Conservation Laws for Fields

A local conservation law has the form

$$\partial_\mu j^\mu = 0. \quad (8.317)$$

Writing $j^\mu = (c\rho, \mathbf{j})$, this becomes

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0. \quad (8.318)$$

Integrating over a spatial volume V gives

$$\frac{d}{dt} \int_V \rho \, d^3x = - \int_{\partial V} \mathbf{j} \cdot \mathbf{n} \, dS. \quad (8.319)$$

If the flux through the boundary vanishes, the charge

$$Q = \int_V \rho \, d^3x \quad (8.320)$$

is constant. Noether's theorem explains where the current comes from. The field equations turn a continuous symmetry of the action into a divergence equation.

8.10.3 Conservation of Energy and Momentum

If $\mathcal{L}$ has no explicit dependence on x^ν , spacetime translation symmetry gives the canonical energy momentum tensor

$$T^\mu{}_\nu = \sum_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\nu \phi_a - \delta^\mu{}_\nu \mathcal{L}. \quad (8.321)$$

Using the field equations, one obtains

$$\partial_\mu T^\mu{}_\nu = 0. \quad (8.322)$$

The four conserved quantities are obtained by integrating the time component over space. They are the total energy and the three components of total momentum, with the exact signs fixed by the chosen index positions and metric convention.

The canonical momentum conjugate to ϕ_a is

$$\pi_a = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_a)}, \quad (8.323)$$

and the Hamiltonian density is

$$\mathcal{H} = \sum_a \pi_a \partial_0 \phi_a - \mathcal{L}. \quad (8.324)$$

The canonical tensor in (8.321) need not be symmetric. Adding the divergence of a suitable superpotential produces the Belinfante tensor, which has the same conserved total four momentum and is more convenient when angular momentum or gravity is involved.

8.10.4 Conservation of Angular Momentum

Lorentz invariance gives the angular momentum current

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu} + S^{\lambda\mu\nu}. \quad (8.325)$$

The first two terms describe orbital angular momentum. The spin current $S^{\lambda\mu\nu}$ appears when the field itself transforms nontrivially under Lorentz transformations. For a scalar field, it vanishes.

If the energy momentum tensor is symmetric and the spin contribution has been included consistently, then

$$\partial_\lambda M^{\lambda\mu\nu} = 0. \quad (8.326)$$

The conserved charges are

$$J^{\mu\nu} = \int M^{0\mu\nu} d^3x. \quad (8.327)$$

The purely spatial components generate rotations and give ordinary angular momentum. The components with one time index generate Lorentz boosts. Energy, momentum, and angular momentum are therefore different parts of one symmetry structure of spacetime.

8.11 Electromagnetic Fields

Electromagnetism is the cleanest full example of the field variational method developed in the preceding sections. We will build it in three layers. First, we recover the familiar three dimensional fields from potentials. Next, we derive Maxwell's equations from a spacetime action. Finally, we study gauge symmetry, energy momentum, wave solutions, and the canonical structure.

8.11.1 Spacetime and Sign Conventions

We use

$$x^\mu = (ct, x, y, z), \quad \eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1). \quad (8.328)$$

Greek indices run from 0 to 3, while Latin spatial indices run from 1 to 3. Repeated spacetime indices are summed. The four potential and four current are

$$A^\mu = \left(\frac{\phi}{c}, \mathbf{A} \right), \quad A_\mu = \left(-\frac{\phi}{c}, \mathbf{A} \right) \quad (8.329)$$

and

$$J^\mu = (c\rho, \mathbf{J}), \quad J_\mu = (-c\rho, \mathbf{J}). \quad (8.330)$$

In particular,

$$A_0 = -\frac{\phi}{c}, \quad (8.331)$$

as required by the metric convention. Also,

$$\partial_0 = \frac{1}{c} \frac{\partial}{\partial t}, \quad \partial^0 = -\frac{1}{c} \frac{\partial}{\partial t}. \quad (8.332)$$

The scalar and vector potentials define the electric and magnetic fields by

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (8.333)$$

These definitions immediately imply two of Maxwell's equations:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}. \quad (8.334)$$

The first follows because the divergence of a curl is zero. The second follows because the curl of a gradient is zero and spatial derivatives commute with the time derivative.

8.11.2 The Electromagnetic Field Tensor

Definition 8.13 (Electromagnetic field tensor) Define the antisymmetric tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (8.335)$$

Its time space components are

$$F_{0i} = \frac{1}{c} \frac{\partial A_i}{\partial t} - \frac{\partial}{\partial x^i} \left(-\frac{\phi}{c} \right) = -\frac{E_i}{c}, \quad (8.336)$$

while the spatial components are

$$F_{ij} = \epsilon_{ijk} B_k. \quad (8.337)$$

Thus

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & B_z & -B_y \\ E_y/c & -B_z & 0 & B_x \\ E_z/c & B_y & -B_x & 0 \end{pmatrix}. \quad (8.338)$$

Raising both indices changes the sign of the time space entries:

$$F^{0i} = \frac{E_i}{c}, \quad F^{i0} = -\frac{E_i}{c}, \quad F^{ij} = F_{ij}. \quad (8.339)$$

This is the first place where the metric signature matters. It is worth checking these signs once rather than memorizing a matrix without knowing its convention.

The simplest Lorentz scalar built from the field tensor is

$$F_{\mu\nu} F^{\mu\nu} = 2 \sum_i F_{0i} F^{0i} + \sum_{i,j} F_{ij} F^{ij} = 2 \left(\mathbf{B}^2 - \frac{\mathbf{E}^2}{c^2} \right). \quad (8.340)$$

Therefore the usual free electromagnetic Lagrangian density is

$$-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} = \frac{\epsilon_0}{2} \mathbf{E}^2 - \frac{1}{2\mu_0} \mathbf{B}^2, \quad (8.341)$$

where $\epsilon_0 \mu_0 c^2 = 1$.

8.11.3 The Homogeneous Equations as an Identity

Because $F_{\mu\nu}$ is built from one potential, it satisfies

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0. \quad (8.342)$$

This is the Bianchi identity. Its proof is simply the commutativity of partial derivatives:

$$\begin{aligned} \partial_\lambda (\partial_\mu A_\nu - \partial_\nu A_\mu) + \partial_\mu (\partial_\nu A_\lambda - \partial_\lambda A_\nu) \\ + \partial_\nu (\partial_\lambda A_\mu - \partial_\mu A_\lambda) = 0. \end{aligned} \quad (8.343)$$

Every second derivative appears twice with opposite signs.

For three spatial indices, (8.342) gives

$$\nabla \cdot \mathbf{B} = 0. \quad (8.344)$$

For one time index and two spatial indices, it gives

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (8.345)$$

Thus the homogeneous Maxwell equations are identities once a globally defined four potential exists. The inhomogeneous equations, by contrast, are dynamical equations obtained by varying the action.

8.11.4 The Electromagnetic Action

For a prescribed conserved current J^μ , take

$$S[A_\mu] = \int_\Omega \left[-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu \right] d^4x. \quad (8.346)$$

With our conventions,

$$J^\mu A_\mu = -\rho\phi + \mathbf{J} \cdot \mathbf{A}. \quad (8.347)$$

Therefore the three dimensional form of the Lagrangian density is

$$\mathcal{L}_{\text{EM}} = \frac{\epsilon_0}{2} \mathbf{E}^2 - \frac{1}{2\mu_0} \mathbf{B}^2 + \mathbf{J} \cdot \mathbf{A} - \rho\phi. \quad (8.348)$$

The sign of the source term is tied to the definition $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and to $A_0 = -\phi/c$. Changing one convention can move a minus sign to the covariant Maxwell equation, but the three dimensional equations must remain the same.

► Variation of the action.

Take $A_\nu \mapsto A_\nu + \alpha\eta_\nu$, where $\eta_\nu = 0$ on $\partial\Omega$. Then

$$\delta F_{\mu\nu} = \partial_\mu \eta_\nu - \partial_\nu \eta_\mu. \quad (8.349)$$

The first variation is

$$\delta S = -\frac{1}{2\mu_0} \int_{\Omega} F^{\mu\nu} (\partial_{\mu}\eta_{\nu} - \partial_{\nu}\eta_{\mu}) d^4x + \int_{\Omega} J^{\nu}\eta_{\nu} d^4x. \quad (8.350)$$

Antisymmetry gives

$$F^{\mu\nu} (\partial_{\mu}\eta_{\nu} - \partial_{\nu}\eta_{\mu}) = 2F^{\mu\nu} \partial_{\mu}\eta_{\nu}. \quad (8.351)$$

Hence

$$\begin{aligned} \delta S &= -\frac{1}{\mu_0} \int_{\Omega} F^{\mu\nu} \partial_{\mu}\eta_{\nu} d^4x + \int_{\Omega} J^{\nu}\eta_{\nu} d^4x \\ &= \int_{\Omega} \left[\frac{1}{\mu_0} \partial_{\mu} F^{\mu\nu} + J^{\nu} \right] \eta_{\nu} d^4x - \frac{1}{\mu_0} \int_{\partial\Omega} n_{\mu} F^{\mu\nu} \eta_{\nu} d^3\Sigma. \end{aligned} \quad (8.352)$$

The boundary term vanishes for fixed boundary potential. Since the four components of η_{ν} are arbitrary in the interior, stationary action gives

$$\partial_{\mu} F^{\mu\nu} = -\mu_0 J^{\nu}. \quad (8.353)$$

This is the inhomogeneous half of Maxwell's equations.

8.11.5 Recovering the Four Maxwell Equations

Let us check the covariant equation component by component. For $\nu = 0$, use $F^{i0} = -E_i/c$ and $J^0 = c\rho$:

$$\begin{aligned} \partial_{\mu} F^{\mu 0} &= \partial_i F^{i0} = -\frac{1}{c} \nabla \cdot \mathbf{E} \\ &= -\mu_0 c \rho. \end{aligned} \quad (8.354)$$

Since $\mu_0 c^2 = 1/\epsilon_0$,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (8.355)$$

For a spatial index $\nu = i$,

$$\partial_0 F^{0i} + \partial_j F^{ji} = \frac{1}{c^2} \frac{\partial E_i}{\partial t} - (\nabla \times \mathbf{B})_i. \quad (8.356)$$

Equation (8.353) therefore becomes

$$\nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}. \quad (8.357)$$

Combining these with the Bianchi identity gives the complete system

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & \nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \mathbf{J}, \\ \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0.\end{aligned}\tag{8.358}$$

The first row is dynamical and follows from the action. The second row is geometric and follows from the definition of the field tensor.

8.11.6 Gauge Symmetry and Charge Conservation

The transformation

$$A_\mu \mapsto A_\mu + \partial_\mu \chi \tag{8.359}$$

leaves $F_{\mu\nu}$ unchanged because

$$\partial_\mu \partial_\nu \chi - \partial_\nu \partial_\mu \chi = 0. \tag{8.360}$$

In three dimensional language,

$$\mathbf{A} \mapsto \mathbf{A} + \nabla \chi, \quad \phi \mapsto \phi - \frac{\partial \chi}{\partial t}. \tag{8.361}$$

Substitution into (8.333) confirms that $\mathbf{E}$ and $\mathbf{B}$ do not change.

The free field part of the action is exactly gauge invariant. The source part changes by

$$\begin{aligned}\Delta S_{\text{source}} &= \int_{\Omega} J^\mu \partial_\mu \chi \, d^4x \\ &= \int_{\partial\Omega} n_\mu J^\mu \chi \, d^3\Sigma - \int_{\Omega} \chi \partial_\mu J^\mu \, d^4x.\end{aligned}\tag{8.362}$$

For gauge functions that vanish at the boundary, invariance for arbitrary χ requires

$$\partial_\mu J^\mu = 0. \tag{8.363}$$

Since $J^\mu = (c\rho, \mathbf{J})$, this is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \tag{8.364}$$

The same result follows directly from the field equation. Apply ∂_ν to (8.353):

$$\partial_\nu \partial_\mu F^{\mu\nu} = -\mu_0 \partial_\nu J^\nu. \tag{8.365}$$

The left hand side is zero because $\partial_\nu \partial_\mu$ is symmetric in (μ, ν) while $F^{\mu\nu}$ is antisymmetric. Hence current conservation is a consistency condition of Maxwell's equations. In the language of Noether's second theorem, gauge redundancy produces a differential identity among the Euler Lagrange equations.

8.11.7 Lorenz Gauge and Wave Equations

The Lorenz gauge condition is

$$\partial_\mu A^\mu = 0. \quad (8.366)$$

In terms of the scalar and vector potentials,

$$\frac{1}{c^2} \frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{A} = 0. \quad (8.367)$$

Under a gauge transformation,

$$\partial_\mu A^\mu \mapsto \partial_\mu A^\mu + \square \chi, \quad \square = \partial_\mu \partial^\mu = -\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2. \quad (8.368)$$

Thus one reaches Lorenz gauge by solving

$$\square \chi = -\partial_\mu A^\mu. \quad (8.369)$$

There is still residual gauge freedom: any further χ satisfying $\square \chi = 0$ preserves the gauge condition.

Since

$$\begin{aligned} \partial_\mu F^{\mu\nu} &= \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) \\ &= \square A^\nu - \partial^\nu (\partial_\mu A^\mu), \end{aligned} \quad (8.370)$$

the field equation in Lorenz gauge is

$$\square A^\nu = -\mu_0 J^\nu. \quad (8.371)$$

The components are

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \phi = -\frac{\rho}{\epsilon_0} \quad (8.372)$$

and

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{A} = -\mu_0 \mathbf{J}. \quad (8.373)$$

In vacuum, every component of the four potential propagates with speed c , although gauge symmetry means that only two transverse combinations are physical.

8.11.8 A Charged Particle in an Electromagnetic Field

The field action tells us how charges produce electromagnetic fields. The particle Lagrangian tells us how those fields act back on a charge. For a particle of mass m and charge q ,

$$\mathcal{L}_{\text{particle}} = -mc^2 \sqrt{1 - \frac{\mathbf{v}^2}{c^2}} + q\mathbf{A} \cdot \mathbf{v} - q\phi. \quad (8.374)$$

The canonical momentum is

$$\mathbf{p} = \frac{\partial \mathcal{L}_{\text{particle}}}{\partial \mathbf{v}} = \gamma m \mathbf{v} + q \mathbf{A}, \quad \gamma = \frac{1}{\sqrt{1 - \mathbf{v}^2/c^2}}. \quad (8.375)$$

The Euler Lagrange equation gives

$$\frac{d}{dt} (\gamma m \mathbf{v} + q \mathbf{A}) = q \nabla (\mathbf{A} \cdot \mathbf{v}) - q \nabla \phi. \quad (8.376)$$

Using

$$\nabla (\mathbf{A} \cdot \mathbf{v}) - (\mathbf{v} \cdot \nabla) \mathbf{A} = \mathbf{v} \times (\nabla \times \mathbf{A}), \quad (8.377)$$

we obtain

$$\frac{d}{dt} (\gamma m \mathbf{v}) = q (\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (8.378)$$

The Lorentz force is therefore an Euler Lagrange equation.

The Hamiltonian follows from the Legendre transform. Since the mechanical momentum is $\mathbf{p} - q \mathbf{A}$,

$$\mathcal{H}_{\text{particle}} = c \sqrt{(\mathbf{p} - q \mathbf{A})^2 m^2 c^2} + q \phi. \quad (8.379)$$

Under (8.359), the particle Lagrangian changes by

$$\Delta \mathcal{L}_{\text{particle}} = q \frac{d\chi}{dt}. \quad (8.380)$$

This is a total time derivative, so the trajectory is unchanged. Gauge transformations are therefore a concrete example of the canonical principle that adding a total derivative does not change the Euler Lagrange equations.

8.11.9 Poynting's Theorem

Take the scalar product of the Ampere Maxwell equation with $\mathbf{E}/\mu_0$:

$$\frac{1}{\mu_0} \mathbf{E} \cdot \nabla \times \mathbf{B} - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} = \mathbf{J} \cdot \mathbf{E}. \quad (8.381)$$

Take the scalar product of Faraday's law with $\mathbf{B}/\mu_0$:

$$\frac{1}{\mu_0} \mathbf{B} \cdot \nabla \times \mathbf{E} + \frac{1}{\mu_0} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} = 0. \quad (8.382)$$

Subtract the first curl term using

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{E} - \mathbf{E} \cdot \nabla \times \mathbf{B}. \quad (8.383)$$

The result is

$$\frac{\partial u_{\text{EM}}}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}, \quad (8.384)$$

where

$$u_{\text{EM}} = \frac{\epsilon_0}{2} \mathbf{E}^2 + \frac{1}{2\mu_0} \mathbf{B}^2, \quad \mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (8.385)$$

Here u_{EM} is the electromagnetic energy density and $\mathbf{S}$ is the Poynting vector. Integrating over a fixed volume gives

$$\frac{d}{dt} \int_V u_{\text{EM}} d^3x = - \int_{\partial V} \mathbf{S} \cdot \mathbf{n} dS - \int_V \mathbf{J} \cdot \mathbf{E} d^3x. \quad (8.386)$$

The first term on the right is energy flowing out through the boundary. The second is the work done by the field on matter.

8.11.10 The Symmetric Energy Momentum Tensor

The gauge invariant symmetric energy momentum tensor is

$$T_{\text{EM}}^{\mu\nu} = \frac{1}{\mu_0} \left(F^{\mu\lambda} F^{\nu}{}_{\lambda} - \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right). \quad (8.387)$$

To compute its divergence, differentiate and use both the inhomogeneous equation and the Bianchi identity. The terms in which the derivative hits the second field tensor combine with the derivative of the scalar $F_{\alpha\beta} F^{\alpha\beta}$ and cancel. The remaining term is

$$\partial_\mu T_{\text{EM}}^{\mu\nu} = -F^{\nu\lambda} J_\lambda. \quad (8.388)$$

The four vector

$$f^\nu = F^{\nu\lambda} J_\lambda \quad (8.389)$$

is the Lorentz force density on matter. Its spatial part is

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B}. \quad (8.390)$$

Thus (8.388) says that the field loses exactly the energy momentum gained by matter.

The components of (8.387) are

$$T_{\text{EM}}^{00} = u_{\text{EM}}, \quad cT_{\text{EM}}^{0i} = S_i, \quad T_{\text{EM}}^{ij} = -\sigma_{ij}, \quad (8.391)$$

where the Maxwell stress tensor is

$$\sigma_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} \mathbf{E}^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} \mathbf{B}^2 \right). \quad (8.392)$$

The field momentum density is

$$\mathbf{g}_{\text{EM}} = \frac{\mathbf{S}}{c^2} = \epsilon_0 \mathbf{E} \times \mathbf{B}. \quad (8.393)$$

The spatial part of (8.388) becomes

$$\frac{\partial g_i}{\partial t} - \partial_j \sigma_{ij} = -f_i. \quad (8.394)$$

This is local momentum balance. The divergence of the Maxwell stress transmits force through the field, just as mechanical stress transmits force through a continuous medium.

Because $T_{\text{EM}}^{\mu\nu}$ is symmetric, the source free angular momentum current

$$M^{\lambda\mu\nu} = x^\mu T_{\text{EM}}^{\lambda\nu} - x^\nu T_{\text{EM}}^{\lambda\mu} \quad (8.395)$$

satisfies

$$\partial_\lambda M^{\lambda\mu\nu} = 0. \quad (8.396)$$

With sources, the field angular momentum is not conserved by itself. The conserved current is obtained only after the matter contribution is included.

8.11.11 Dual Tensor and Lorentz Invariants

Choose $\epsilon^{0123} = +1$ and define the dual field tensor by

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}. \quad (8.397)$$

The Bianchi identity is equivalent to

$$\partial_\mu \tilde{F}^{\mu\nu} = 0. \quad (8.398)$$

The two independent quadratic Lorentz invariants are

$$F_{\mu\nu} F^{\mu\nu} = 2 \left(\mathbf{B}^2 - \frac{\mathbf{E}^2}{c^2} \right) \quad (8.399)$$

and

$$F_{\mu\nu} \tilde{F}^{\mu\nu} = -\frac{4}{c} \mathbf{E} \cdot \mathbf{B}. \quad (8.400)$$

These numbers have the same values in every inertial frame. They classify the local field:

- (1) if $\mathbf{E} \cdot \mathbf{B} \neq 0$, no Lorentz frame can make either field vanish;
- (2) if $\mathbf{E} \cdot \mathbf{B} = 0$ and $\mathbf{B}^2 > \mathbf{E}^2/c^2$, there is locally a frame with a purely magnetic field;
- (3) if $\mathbf{E} \cdot \mathbf{B} = 0$ and $\mathbf{B}^2 < \mathbf{E}^2/c^2$, there is locally a frame with a purely electric field;
- (4) if both invariants vanish, the field is null, as for an ideal plane wave.

In source free vacuum, the equations are invariant under a continuous duality rotation

$$\begin{aligned} F^{\mu\nu} &\mapsto F^{\mu\nu} \cos \theta + \tilde{F}^{\mu\nu} \sin \theta, \\ \tilde{F}^{\mu\nu} &\mapsto \tilde{F}^{\mu\nu} \cos \theta - F^{\mu\nu} \sin \theta. \end{aligned} \quad (8.401)$$

Electric sources break this symmetry unless magnetic sources are introduced as a second current.

8.11.12 Plane Waves and Physical Polarizations

In source free Lorenz gauge,

$$\square A^\mu = 0. \quad (8.402)$$

Take the complex plane wave ansatz

$$A^\mu(x) = \epsilon^\mu e^{ik_\nu x^\nu}, \quad k^\mu = \left(\frac{\omega}{c}, \mathbf{k} \right). \quad (8.403)$$

Substitution into the wave equation gives

$$k_\mu k^\mu = 0, \quad (8.404)$$

or

$$\omega = c |\mathbf{k}|. \quad (8.405)$$

The Lorenz gauge condition gives

$$k_\mu \epsilon^\mu = 0. \quad (8.406)$$

Residual gauge freedom removes one more component of ϵ^μ . One may choose radiation gauge,

$$\phi = 0, \quad \nabla \cdot \mathbf{A} = 0. \quad (8.407)$$

Then $\mathbf{k} \cdot \mathbf{A} = 0$, so only two independent transverse polarizations remain.

For the real physical fields,

$$\mathbf{B} = \frac{1}{\omega} \mathbf{k} \times \mathbf{E} = \frac{1}{c} \hat{\mathbf{k}} \times \mathbf{E}. \quad (8.408)$$

Therefore

$$\mathbf{E} \perp \mathbf{B}, \quad |\mathbf{E}| = c |\mathbf{B}|, \quad (8.409)$$

and both invariants in (8.400) vanish. The Poynting vector is parallel to the propagation direction:

$$\mathbf{S} = cu_{\text{EM}} \hat{\mathbf{k}}. \quad (8.410)$$

The covariant formalism therefore reproduces the familiar picture: electromagnetic waves are transverse, carry energy and momentum, and propagate along null directions.

8.11.13 Canonical Structure and the Gauss Constraint

Return to the source free three dimensional Lagrangian density

$$\mathcal{L}_{\text{EM}} = \frac{\epsilon_0}{2} \mathbf{E}^2 - \frac{1}{2\mu_0} \mathbf{B}^2. \quad (8.411)$$

Since

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} - \nabla \phi, \quad (8.412)$$

the momentum conjugate to A_i is

$$\Pi_i = \frac{\partial \mathcal{L}_{\text{EM}}}{\partial(\partial A_i / \partial t)} = -\epsilon_0 E_i. \quad (8.413)$$

There is no time derivative of ϕ , so

$$\Pi_\phi = 0. \quad (8.414)$$

This means that the Legendre transformation is singular. It is not a failure of the theory. It is the Hamiltonian signature of gauge redundancy.

The canonical Hamiltonian density is

$$\mathcal{H}_{\text{EM}} = \mathbf{\Pi} \cdot \frac{\partial \mathbf{A}}{\partial t} - \mathcal{L}_{\text{EM}} = \frac{\mathbf{\Pi}^2}{2\epsilon_0} + \frac{\mathbf{B}^2}{2\mu_0} - \mathbf{\Pi} \cdot \nabla \phi. \quad (8.415)$$

After one spatial integration by parts,

$$-\int \mathbf{\Pi} \cdot \nabla \phi \, d^3x = \int \phi \nabla \cdot \mathbf{\Pi} \, d^3x - \int_{\partial V} \phi \mathbf{\Pi} \cdot \mathbf{n} \, dS. \quad (8.416)$$

Ignoring the fixed boundary term, variation with respect to ϕ imposes

$$\nabla \cdot \mathbf{\Pi} = 0, \quad (8.417)$$

which is the source free Gauss law because $\mathbf{\Pi} = -\epsilon_0 \mathbf{E}$. On the constraint surface, the Hamiltonian is the positive field energy

$$\mathcal{H}_{\text{EM}} = \frac{\epsilon_0}{2} \mathbf{E}^2 + \frac{1}{2\mu_0} \mathbf{B}^2 \quad (8.418)$$

up to the boundary term.

This is an important final lesson. For an ordinary regular mechanical system, the Legendre transformation replaces every velocity by a momentum. For electromagnetism, gauge symmetry prevents this map from being invertible. The missing velocity becomes a constraint, and that constraint is Gauss's law.

8.11.14 Boundary and Interface Conditions

The differential equations imply the standard interface conditions. Let $\mathbf{n}$ point from medium 1 to medium 2, and let σ and $\mathbf{K}$ be the surface charge and surface current. Integrating the divergence equations over a thin pillbox gives

$$\mathbf{n} \cdot (\mathbf{E}_2 - \mathbf{E}_1) = \frac{\sigma}{\epsilon_0}, \quad \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0. \quad (8.419)$$

Integrating the curl equations around a thin rectangular loop gives, in the limit of vanishing height,

$$\mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0, \quad \mathbf{n} \times (\mathbf{B}_2 - \mathbf{B}_1) = \mu_0 \mathbf{K}. \quad (8.420)$$

The last equation assumes there is no singular displacement current concentrated on the interface. These relations are the integral form of the same local field equations derived from the action.

Appendix A

Some Definitions in Linear Algebra

Here we list some definitions that are not mentioned in this book but are among the most important in linear algebra.

A.1 Vector Space

Definition A.1 (Vector Space) Let $\mathbb{F}$ be a field. A *vector space* V over $\mathbb{F}$ is a set equipped with two operations:

- vector addition $+: V \times V \rightarrow V$
- scalar multiplication $\cdot: \mathbb{F} \times V \rightarrow V$

such that $\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $a, b \in \mathbb{F}$, the following axioms hold:

1. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ (associativity)
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutativity)
3. There exists $\mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (additive identity)
4. For each $\mathbf{v} \in V$, $\exists \mathbf{u} \in V$ such that $\mathbf{v} + \mathbf{u} = \mathbf{0}$ (additive inverse)
5. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (compatibility)
6. $1\mathbf{v} = \mathbf{v}$ (scalar identity)
7. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ (distributivity over vector addition)
8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$ (distributivity over scalar addition)

A.2 Subspace

Definition A.2 (Subspace) Let V be a vector space over a field $\mathbb{F}$. A subset $W \subset V$ is called a *subspace* of V if W is itself a vector space under the vector addition and scalar multiplication inherited from V .

Theorem A.1 (Subspace Test) *Let V be a vector space over $\mathbb{F}$, and let $W \subset V$ be a nonempty subset. Then W is a subspace of V if and only if the following conditions hold:*

1. $\forall \mathbf{u}, \mathbf{v} \in W$, we have $\mathbf{u} + \mathbf{v} \in W$.
2. $\forall \mathbf{v} \in W$ and all $a \in \mathbb{F}$, we have $a\mathbf{v} \in W$.

Proof. Assume first that W is a subspace of V . Since W is a vector space under the operations inherited from V , it is closed under vector addition and scalar multiplication. Hence for any $\mathbf{u}, \mathbf{v} \in W$, we have $\mathbf{u} + \mathbf{v} \in W$, and for any $a \in \mathbb{F}$ and $\mathbf{v} \in W$, we have $a\mathbf{v} \in W$.

Conversely, suppose that W is nonempty and satisfies the two closure conditions. Since W is nonempty, choose some $\mathbf{w} \in W$. By closure under scalar multiplication,

$$0\mathbf{w} = \mathbf{0} \in W.$$

Thus W contains the zero vector. For any $\mathbf{v} \in W$, again by closure under scalar multiplication,

$$(-1)\mathbf{v} = -\mathbf{v} \in W.$$

Therefore every vector in W has its additive inverse in W . The closure of addition and scalar multiplication is already assumed. All remaining vector space axioms, such as associativity, commutativity, distributivity, and scalar identity, are inherited from the ambient vector space V . Hence W is a vector space under the inherited operations, and therefore W is a subspace of V . $\square$

A.3 Trace of a Matrix

Definition A.3 (Trace of a Matrix) Let $A = (a_{ij})$ be a square matrix in $\mathbb{F}^{n \times n}$. The *trace* of A , denoted by $\text{tr}(A)$, is defined as

$$\text{tr}(A) = \sum_{ij} a_{ij} \delta_{ij}$$

Proposition A.1 (Properties of the Trace) *Let $A, B \in \mathbb{F}^{n \times n}$ and $c \in \mathbb{F}$. Then:*

1. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$.
2. $\text{tr}(cA) = c \text{tr}(A)$.
3. $\text{tr}(AB) = \text{tr}(BA)$.
4. $\text{tr}(A^T) = \text{tr}(A)$.

A.4 The Sum of Sets

Definition A.4 (Sum of Sets) Let V be a vector space over a field $\mathbb{F}$, and let $S_1, S_2 \subseteq V$ be nonempty subsets. The *sum* of S_1 and S_2 is defined as

$$S_1 + S_2 = \{ \mathbf{v}_1 + \mathbf{v}_2 \mid \mathbf{v}_1 \in S_1, \mathbf{v}_2 \in S_2 \}$$

Proposition A.2 (Basic Properties) Let $S_1, S_2, S_3 \subseteq V$ be nonempty subsets. Then:

1. $S_1 + S_2 = S_2 + S_1$ (commutativity)
2. $(S_1 + S_2) + S_3 = S_1 + (S_2 + S_3)$ (associativity)
3. $S + \{0\} = S$

A.5 Direct Sum

Definition A.5 (Direct Sum) Let V be a vector space over a field $\mathbb{F}$, and let $W_1, W_2 \subseteq V$ be subspaces and satisfy $W_1 \cap W_2 = \{0\}$. The sum $W_1 + W_2 = V$ (by A.4) is called a *direct sum*, denoted by

$$W_1 \oplus W_2 = V$$

Proposition A.3 (Equivalent Conditions for Direct Sum) Let W_1, W_2 be subspaces of V . The following statements are equivalent:

1. $V = W_1 \oplus W_2$.
2. Every vector $\mathbf{v} \in V$ can be written uniquely as

$$\mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2, \quad \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2$$

3. $V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$.

A.6 Span

Definition A.6 (Span) Let V be a vector space over a field $\mathbb{F}$, and let $S \subseteq V$ be a nonempty subset. The *span* of S , denoted by $\text{span}(S)$, is defined as

$$\text{span}(S) = \left\{ \sum_{i=1}^n a_i \mathbf{v}_i \mid \mathbf{v}_i \in S, a_i \in \mathbb{F}, n \in \mathbb{N} \right\}$$

Proposition A.4 The span of a subset $S \subseteq V$ is the smallest subspace of V containing S .

Proposition A.5 If $\text{span}(S) = V$, then we say S generates/spans the vector space V .

A.7 Basis

Definition A.7 (Basis) Let V be a vector space over a field $\mathbb{F}$. A subset $\mathcal{B} \subset V$ is called a *basis* of V if

1. $\mathcal{B}$ spans (generates) V , i.e.

$$\text{span}(\mathcal{B}) = V$$

2. $\mathcal{B}$ is linearly independent.

Proposition A.6 (Equivalent Characterization of a Basis) A subset $\mathcal{B} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of V is a basis if and only if every vector $\mathbf{v} \in V$ can be written uniquely as

$$\mathbf{v} = \sum_i a^i \mathbf{e}_i, \quad a^i \in \mathbb{F}.$$

Remark A.1 Given a basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of V , the coefficients $(a^1, \dots, a^n)$ are called the *coordinates* of $\mathbf{v}$ with respect to this basis.

A.8 Dimension

Definition A.8 (Dimension) Let V be a vector space over a field $\mathbb{F}$. If V has a finite basis consisting of n vectors, then V is called *finite-dimensional*, and the integer n is called the *dimension* of V , denoted by

$$\dim(V) = n$$

Theorem A.2 All bases of a finite-dimensional vector space have the same number of elements.

Proof. Let

$$\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_m\}, \quad \mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$$

be two bases of V . Since $\mathcal{C}$ spans V and $\mathcal{B}$ is linearly independent, we claim that $m \leq n$.

To see this, use the replacement argument. Since $\mathcal{C}$ spans V , the vector $\mathbf{b}_1$ can be written as a linear combination of the vectors in $\mathcal{C}$,

$$\mathbf{b}_1 = a^1 \mathbf{c}_1 + \dots + a^n \mathbf{c}_n.$$

At least one coefficient is nonzero, otherwise $\mathbf{b}_1 = \mathbf{0}$, contradicting the linear independence of $\mathcal{B}$. After relabeling the vectors in $\mathcal{C}$ if necessary, we may assume $a^1 \neq 0$. Then $\mathbf{c}_1$ can be solved in terms of $\mathbf{b}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$. Hence

$$\{\mathbf{b}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$$

still spans V .

Repeating this process, each independent vector $\mathbf{b}_k$ can replace one of the remaining $\mathbf{c}_j$ while preserving the spanning property. This replacement process cannot continue for more than n steps. If $m > n$, then after replacing all n vectors of $\mathcal{C}$, the set $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ would span V . In that case $\mathbf{b}_{n+1}$ would be a linear combination of $\mathbf{b}_1, \dots, \mathbf{b}_n$, contradicting the linear independence of $\mathcal{B}$. Therefore $m \leq n$.

By the same argument with the roles of $\mathcal{B}$ and $\mathcal{C}$ exchanged, since $\mathcal{B}$ spans V and $\mathcal{C}$ is linearly independent, we get $n \leq m$. Hence $m = n$. Therefore all bases of a finite-dimensional vector space have the same number of elements. $\square$

Proposition A.7 *Let W be a subspace of a finite-dimensional vector space V . Then*

$$\dim W \leq \dim V.$$

Moreover, $\dim W = \dim V$ if and only if $W = V$.

A.9 Null Space

Definition A.9 (Null Space) Let V and W be vector spaces over a field $\mathbb{F}$, and let $T : V \rightarrow W$ be a linear transformation. The *null space* (or *kernel*) of T is defined as

$$\ker(T) = \{ \mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0} \}$$

Definition A.10 (Nullity) The *nullity* of a linear transformation $T : V \rightarrow W$ is defined as

$$\text{nullity}(T) = \dim(\ker T)$$

Proposition A.8 *The null space of a linear transformation is a subspace of its domain.*

A.10 Image of the Linear Transformation

Definition A.11 (Image) Let V and W be vector spaces over a field $\mathbb{F}$, and let $T : V \rightarrow W$ be a linear transformation. The *image* (or *range*) of T is defined as

$$\text{Im}(T) = \{ T(\mathbf{v}) \mid \mathbf{v} \in V \} \subseteq W$$

Proposition A.9 *The image of a linear transformation is a subspace of its codomain.*

Appendix B

A Short Introduction to Differential Forms and Hamiltonian Mechanics

Differential forms provide a coordinate-efficient language for integration, vector calculus, symplectic geometry, and Hamiltonian mechanics. Although the notation may look abstract at first, the essential idea is simple. A differential form is an object designed to be integrated over curves, surfaces, volumes, and higher-dimensional geometric objects. In Hamiltonian mechanics, differential forms give the most compact way to describe phase space, canonical coordinates, and Hamilton's equations.

B.1 Cotangent Space and One-Forms

Definition B.1 (Tangent Space) Let M be a smooth manifold and let $p \in M$. The *tangent space* at p , denoted by $T_p M$, is the vector space whose elements are tangent vectors at p . In local coordinates $(x^1, \dots, x^n)$, a tangent vector can be written as

$$X = X^i \frac{\partial}{\partial x^i}.$$

Definition B.2 (Cotangent Space) Let M be a smooth manifold and let $p \in M$. The *cotangent space* at p , denoted by $T_p^* M$, is the dual vector space of $T_p M$:

$$T_p^* M = \text{Hom}(T_p M, \mathbb{R}).$$

Elements of $T_p^* M$ are called *covectors* or *one-forms*.

Definition B.3 (Coordinate One-Forms) Let $(x^1, \dots, x^n)$ be local coordinates on M . The coordinate one-forms $dx^1, \dots, dx^n$ are defined by their action on the coordinate basis vectors:

$$dx^i \left(\frac{\partial}{\partial x^j} \right) = \delta^i_j.$$

Thus, $\{dx^1, \dots, dx^n\}$ is the basis of $T_p^* M$ dual to the basis $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ of $T_p M$.

Definition B.4 (One-Form) A *one-form* α on M is a smooth assignment of a covector $\alpha_p \in T_p^*M$ to each point $p \in M$. In local coordinates, a one-form is written as

$$\alpha = \alpha_i dx^i.$$

Remark B.1 A vector eats a function and produces a directional derivative. A one-form eats a vector and produces a number:

$$\alpha(X) = \alpha_i X^i.$$

B.2 Differential Forms

Definition B.5 (k -Form) Let M be a smooth manifold. A *k -form* ω is a smooth assignment to each point $p \in M$ of an alternating multilinear map

$$\omega_p : \underbrace{T_p M \times \cdots \times T_p M}_{k \text{ times}} \rightarrow \mathbb{R}.$$

The space of all smooth k -forms on M is denoted by $\Omega^k(M)$.

Definition B.6 (Alternating Property) A k -form ω is alternating if exchanging any two inputs changes the sign:

$$\omega(\dots, X_i, \dots, X_j, \dots) = -\omega(\dots, X_j, \dots, X_i, \dots).$$

In particular, if two inputs are equal, then

$$\omega(\dots, X, \dots, X, \dots) = 0.$$

Definition B.7 (Local Expression of a k -Form) In local coordinates $(x^1, \dots, x^n)$, a k -form can be written as

$$\omega = \frac{1}{k!} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k},$$

where the coefficients $\omega_{i_1 \dots i_k}$ are completely antisymmetric.

Remark B.2 A 0-form is simply a smooth function:

$$\Omega^0(M) = C^\infty(M).$$

A 1-form can be integrated along curves. A 2-form can be integrated over surfaces. An n -form on an n -dimensional manifold can be integrated over regions.

B.3 Wedge Product

Definition B.8 (Wedge Product) Let $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^\ell(M)$. Their *wedge product* is a $(k + \ell)$ -form

$$\alpha \wedge \beta \in \Omega^{k+\ell}(M).$$

The wedge product is bilinear and antisymmetric in the graded sense.

Proposition B.1 (Basic Properties of the Wedge Product) Let $\alpha \in \Omega^k(M)$, $\beta \in \Omega^\ell(M)$, and $\gamma \in \Omega^m(M)$. Then:

1. $\alpha \wedge (\beta + \gamma) = \alpha \wedge \beta + \alpha \wedge \gamma$.
2. $(\alpha + \beta) \wedge \gamma = \alpha \wedge \gamma + \beta \wedge \gamma$.
3. $(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma)$.
4. $\alpha \wedge \beta = (-1)^{k\ell} \beta \wedge \alpha$.

Corollary B.1 For coordinate one-forms,

$$dx^i \wedge dx^j = -dx^j \wedge dx^i.$$

In particular,

$$dx^i \wedge dx^i = 0.$$

Example B.1 In $\mathbb{R}^3$, a general two-form can be written as

$$\omega = A \, dyz + B \, dzx + C \, dxy.$$

This is the differential-form version of the vector field (A, B, C) appearing as an oriented flux density.

B.4 Exterior Derivative

Definition B.9 (Exterior Derivative) The *exterior derivative* is a linear operator

$$d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$$

which generalizes gradient, curl, and divergence.

Definition B.10 (Exterior Derivative in Local Coordinates) If

$$\omega = \frac{1}{k!} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k},$$

then

$$d\omega = \frac{1}{k!} \frac{\partial \omega_{i_1 \dots i_k}}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

Proposition B.2 (Properties of the Exterior Derivative) Let $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^\ell(M)$. Then:

1. $d(\alpha + \beta) = d\alpha + d\beta$.
2. $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$.
3. $d(d\alpha) = 0$.

The identity

$$d^2 = 0$$

is one of the most important identities in differential geometry. It is the differential-form version of

$$\nabla \times (\nabla f) = 0, \quad \nabla \cdot (\nabla \times \mathbf{A}) = 0.$$

B.5 Exact and Closed Forms

Definition B.11 (Closed Form) A differential form $\omega \in \Omega^k(M)$ is called *closed* if

$$d\omega = 0.$$

Definition B.12 (Exact Form) A differential form $\omega \in \Omega^k(M)$ is called *exact* if there exists a $(k-1)$ -form $\alpha \in \Omega^{k-1}(M)$ such that

$$\omega = d\alpha.$$

Proposition B.3 *Every exact form is closed.*

Proof. If $\omega = d\alpha$, then

$$d\omega = d(d\alpha) = d^2\alpha = 0.$$

Hence ω is closed. □

Remark B.3 The converse is not always true. A closed form need not be exact. The failure of closed forms to be exact contains topological information about the underlying space.

B.6 Pullback

Definition B.13 (Pullback of a Function) Let $F : M \rightarrow N$ be a smooth map and let $f \in C^\infty(N)$. The *pullback* of f by F is the function $F^*f \in C^\infty(M)$ defined by

$$F^*f = f \circ F.$$

Definition B.14 (Pullback of a One-Form) Let $F : M \rightarrow N$ be a smooth map and let $\alpha \in \Omega^1(N)$. The pullback $F^*\alpha \in \Omega^1(M)$ is defined by

$$(F^*\alpha)_p(X_p) = \alpha_{F(p)}(dF_p(X_p)).$$

Proposition B.4 (Pullback and Exterior Derivative) Let $F : M \rightarrow N$ be a smooth map and let $\omega \in \Omega^k(N)$. Then

$$F^*(d\omega) = d(F^*\omega).$$

Proposition B.5 (Pullback and Wedge Product) Let α, β be differential forms on N . Then

$$F^*(\alpha \wedge \beta) = F^*\alpha \wedge F^*\beta.$$

B.7 Stokes' Theorem

Theorem B.1 (Stokes' Theorem) Let M be an oriented smooth k -dimensional manifold with boundary ∂M . If $\omega \in \Omega^{k-1}(M)$ has compact support, then

$$\int_M d\omega = \int_{\partial M} \omega.$$

Remark B.4 Stokes' theorem contains several familiar theorems from vector calculus:

1. Fundamental theorem of calculus.
2. Green's theorem.
3. Classical Stokes' theorem.
4. Divergence theorem.

Differential forms are useful because they make integration coordinate-independent. The same theorem

$$\int_M d\omega = \int_{\partial M} \omega$$

simultaneously describes many integral theorems that are usually taught separately.

B.8 Differential Forms in Vector Calculus

Example B.2 (Gradient) Let $f \in \Omega^0(\mathbb{R}^3)$. Then

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz.$$

This corresponds to the gradient ∇f .

Example B.3 (Curl) Let

$$\alpha = P dx + Q dy + R dz.$$

Then

$$d\alpha = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dyz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dzx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxy.$$

This corresponds to $\nabla \times (P, Q, R)$.

Example B.4 (Divergence) Let

$$\beta = P dyz + Q dzx + R dxy.$$

Then

$$d\beta = \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dxyz.$$

This corresponds to $\nabla \cdot (P, Q, R)$.

B.9 Symplectic Form

Definition B.15 (Symplectic Form) Let M be a smooth manifold. A *symplectic form* on M is a two-form $\omega \in \Omega^2(M)$ such that:

1. ω is closed:

$$d\omega = 0.$$

2. ω is nondegenerate:

$$\omega_p(X, Y) = 0 \quad \forall Y \in T_p M \quad \implies \quad X = 0.$$

A pair (M, ω) is called a *symplectic manifold*.

Remark B.5 A symplectic form is the geometric structure behind Hamiltonian mechanics. It allows us to convert the differential of a function into a vector field.

Proposition B.6 *If (M, ω) is a symplectic manifold, then M must be even-dimensional.*

Definition B.16 (Canonical Symplectic Form) On the phase space $\mathbb{R}^{2n}$ with canonical coordinates

$$(q^1, \dots, q^n, p_1, \dots, p_n),$$

the canonical symplectic form is

$$\omega = dq^i \wedge dp_i.$$

Definition B.17 (Canonical One-Form) On the cotangent bundle T^*Q , the canonical one-form is

$$\theta = p_i dq^i.$$

The canonical symplectic form is

$$\omega = dq^i p_i = -d\theta.$$

B.10 Hamiltonian Vector Field

Definition B.18 (Hamiltonian Function) Let (M, ω) be a symplectic manifold. A smooth function

$$H : M \rightarrow \mathbb{R}$$

is called a *Hamiltonian*. In classical mechanics, H usually represents the total energy of a system.

Definition B.19 (Hamiltonian Vector Field) Let (M, ω) be a symplectic manifold and let $H \in C^\infty(M)$. The *Hamiltonian vector field* X_H is defined by

$$\iota_{X_H} \omega = dH,$$

where $\iota_{X_H} \omega$ denotes the interior product of ω with X_H .

Remark B.6 Some books use the sign convention

$$\iota_{X_H} \omega = -dH.$$

The two conventions differ by a sign in the definition of X_H . In mechanics, always check which convention is being used.

Theorem B.2 (Hamilton's Equations) Let the phase space have canonical coordinates (q^i, p_i) and symplectic form

$$\omega = dq^i p_i.$$

If

$$X_H = \dot{q}^i \frac{\partial}{\partial q^i} + \dot{p}_i \frac{\partial}{\partial p_i},$$

then the equation

$$\iota_{X_H} \omega = dH$$

is equivalent to Hamilton's equations:

$$\boxed{\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

Proof. First compute

$$dH = \frac{\partial H}{\partial q^i} dq^i + \frac{\partial H}{\partial p_i} dp_i.$$

Using

$$\omega = dq^i p_i,$$

we obtain

$$\iota_{X_H} \omega = \dot{q}^i dp_i - \dot{p}_i dq^i.$$

Comparing coefficients in

$$\iota_{X_H} \omega = dH,$$

we get

$$-\dot{p}_i = \frac{\partial H}{\partial q^i}, \quad \dot{q}^i = \frac{\partial H}{\partial p_i}.$$

Therefore,

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

□

B.11 Poisson Bracket

Definition B.20 (Poisson Bracket) Let $f, g \in C^\infty(\mathbb{R}^{2n})$. The *Poisson bracket* of f and g is defined as

$$\{f, g\} = \frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i}.$$

Proposition B.7 (Properties of the Poisson Bracket) Let $f, g, h \in C^\infty(\mathbb{R}^{2n})$ and $a, b \in \mathbb{R}$. Then:

1. $\{af + bg, h\} = a\{f, h\} + b\{g, h\}$.
2. $\{f, g\} = -\{g, f\}$.
3. $\{fg, h\} = f\{g, h\} + g\{f, h\}$.
4. $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$.

Proposition B.8 (Time Evolution) Let $f(q, p, t)$ be an observable. Along a Hamiltonian trajectory,

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \{f, H\}.$$

In particular, if f has no explicit time dependence, then

$$\frac{df}{dt} = \{f, H\}.$$

Corollary B.2 (Conserved Quantity) If f has no explicit time dependence and

$$\{f, H\} = 0,$$

then f is conserved along the Hamiltonian flow.

B.12 Hamiltonian Flow and Symplectic Preservation

Definition B.21 (Hamiltonian Flow) Let X_H be the Hamiltonian vector field associated with H . The *Hamiltonian flow* is the family of maps $\varphi_t : M \rightarrow M$ generated by

$$\frac{d}{dt}\varphi_t(x) = X_H(\varphi_t(x)).$$

Theorem B.3 (Symplectic Preservation) *The Hamiltonian flow preserves the symplectic form:*

$$\varphi_t^*\omega = \omega.$$

Remark B.7 This theorem says that Hamiltonian motion preserves the geometric structure of phase space. It is the differential-form version of the idea that Hamiltonian dynamics preserves phase-space volume and canonical structure.

The central geometric picture of Hamiltonian mechanics is:

Hamiltonian $H \Rightarrow dH \Rightarrow X_H \Rightarrow$ motion in phase space.

The symplectic form ω is the object that converts dH into the Hamiltonian vector field X_H .

Appendix C

Runge Kutta Method of Fourth Order

The Runge Kutta method of fourth order, usually called RK4, is a numerical method for solving initial value problems of the form

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0.$$

The unknown y may be a scalar, a vector, or a state vector containing positions and velocities. The main idea is to replace the exact average slope on one short time interval by a weighted average of four carefully chosen slopes.

C.1 Integral Form of the Problem

Let $t_{n+1} = t_n + h$. Integrating the differential equation from t_n to $t_n + h$ gives

$$y(t_n + h) = y(t_n) + \int_{t_n}^{t_n+h} f(t, y(t)) dt.$$

After the change of variable $t = t_n + \theta h$, where $0 \leq \theta \leq 1$, this becomes

$$y(t_n + h) = y(t_n) + h \int_0^1 f(t_n + \theta h, y(t_n + \theta h)) d\theta.$$

A one step method is therefore an approximation of the average slope over $0 \leq \theta \leq 1$. Euler's method uses only the slope at the left endpoint. RK4 uses four slopes.

C.2 General Explicit Runge Kutta Form

An explicit s stage Runge Kutta method has the form

$$\begin{aligned}
\mathbf{K}_i &= \mathbf{f}(t_n + c_i h, \mathbf{Y}_i), \quad i = 1, \dots, s, \\
\mathbf{Y}_i &= \mathbf{y}_n + h \sum_{j < i} a_{ij} \mathbf{K}_j, \\
\mathbf{y}_{n+1} &= \mathbf{y}_n + h \sum_{i=1}^s b_i \mathbf{K}_i.
\end{aligned}$$

The numbers a_{ij} determine the internal stage values, the numbers c_i determine the stage times, and the numbers b_i determine the final weighted average. A basic consistency requirement relates a_{ij} and c_i . For a constant slope field $\mathbf{f}(t, \mathbf{y}) = \mathbf{F}$, the exact solution at the stage time $t_n + c_i h$ is

$$\mathbf{y}(t_n + c_i h) = \mathbf{y}_n + c_i h \mathbf{F}.$$

The Runge Kutta stage formula gives

$$\mathbf{Y}_i = \mathbf{y}_n + h \sum_{j < i} a_{ij} \mathbf{F}.$$

For these two expressions to agree for every constant vector $\mathbf{F}$, one must have

$$c_i = \sum_{j < i} a_{ij}.$$

This is the first source of the internal coefficients. The stage time must match the stage displacement when the slope is constant.

C.3 Choosing the Classical Four Stages

The classical RK4 method uses four stages. The stage times are chosen as

$$c_1 = 0, \quad c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{2}, \quad c_4 = 1.$$

These are the left endpoint, two midpoint evaluations, and the right endpoint. The two midpoint stages are not redundant. The first midpoint slope is predicted from the initial slope, while the second midpoint slope corrects that prediction.

The simplest explicit choice uses only the most recent available slope to predict the next stage. Using $c_i = \sum_{j < i} a_{ij}$, this gives

$$a_{21} = \frac{1}{2}, \quad a_{32} = \frac{1}{2}, \quad a_{43} = 1,$$

and all other a_{ij} equal to zero. Thus

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}.$$

This explains the origin of the internal coefficients $1/2$, $1/2$, and 1 . They force the second and third slopes to be midpoint slopes and the fourth slope to be an endpoint slope.

C.4 Determining the Weights

To determine the weights b_i , first test the method on equations depending only on time:

$$\frac{dy}{dt} = \mathbf{g}(t).$$

The exact one step increment is

$$\mathbf{y}(t_n + h) - \mathbf{y}(t_n) = h \int_0^1 \mathbf{g}(t_n + \theta h) d\theta.$$

The Runge Kutta increment becomes

$$h \sum_{i=1}^4 b_i \mathbf{g}(t_n + c_i h).$$

For fourth order accuracy, this quadrature rule should integrate all polynomials in θ up to degree 3 exactly. Hence the weights must satisfy

$$\sum_{i=1}^4 b_i c_i^p = \int_0^1 \theta^p d\theta = \frac{1}{p+1}, \quad p = 0, 1, 2, 3.$$

With $c_1 = 0$, $c_2 = c_3 = 1/2$, and $c_4 = 1$, the two middle points appear at the same time. The classical symmetric choice takes $b_2 = b_3$. Let $B = b_2 + b_3$. The moment equations become

$$\begin{aligned} b_1 + B + b_4 &= 1, \\ \frac{1}{2}B + b_4 &= \frac{1}{2}, \\ \frac{1}{4}B + b_4 &= \frac{1}{3}, \\ \frac{1}{8}B + b_4 &= \frac{1}{4}. \end{aligned}$$

Subtracting the third equation from the second gives

$$\frac{1}{4}B = \frac{1}{6}, \quad B = \frac{2}{3}.$$

Then

$$b_4 = \frac{1}{2} - \frac{1}{2}B = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

The first equation gives

$$b_1 = 1 - B - b_4 = 1 - \frac{2}{3} - \frac{1}{6} = \frac{1}{6}.$$

Since $b_2 = b_3$ and $B = 2/3$, one obtains

$$b_2 = b_3 = \frac{1}{3}.$$

Therefore

$$\mathbf{b}^T = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6} \right).$$

This explains the source of the weights 1, 2, 2, 1 after the common factor 1/6 is pulled out.

C.5 The Classical RK4 Formula

Combining the coefficients gives

$$\mathbf{K}_1 = \mathbf{f}(t_n, \mathbf{y}_n),$$

$$\mathbf{K}_2 = \mathbf{f} \left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2} \mathbf{K}_1 \right),$$

$$\mathbf{K}_3 = \mathbf{f} \left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2} \mathbf{K}_2 \right),$$

$$\mathbf{K}_4 = \mathbf{f}(t_n + h, \mathbf{y}_n + h \mathbf{K}_3),$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{6} (\mathbf{K}_1 + 2\mathbf{K}_2 + 2\mathbf{K}_3 + \mathbf{K}_4).$$

The corresponding Butcher tableau is

$$\begin{array}{c|cccc}
0 & 0 & 0 & 0 & 0 \\
\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\
\frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\
1 & 0 & 0 & 1 & 0 \\
\hline
& \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6}
\end{array}$$

C.6 Taylor Expansion and Order Conditions

The previous quadrature argument determines the coefficients from time dependent test equations. To prove the method is fourth order for nonlinear systems, one must also check the dependence of $\mathbf{f}$ on $\mathbf{y}$. It is enough to compare the Taylor expansion of the exact solution with the Taylor expansion of the numerical step.

For readability, consider the autonomous vector equation

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(\mathbf{y}).$$

The nonautonomous case can be reduced to this form by adding time as one more unknown with $dt/dt = 1$. Let $d\mathbf{f}$ be the Jacobian, $D^2\mathbf{f}$ the second derivative as a bilinear map, and $D^3\mathbf{f}$ the third derivative as a trilinear map. Repeated differentiation gives

$$\begin{aligned}
\mathbf{y}' &= \mathbf{f}, \\
\mathbf{y}'' &= d\mathbf{f} \mathbf{f}, \\
\mathbf{y}^{(3)} &= D^2\mathbf{f}(\mathbf{f}, \mathbf{f}) + d\mathbf{f} d\mathbf{f} \mathbf{f}, \\
\mathbf{y}^{(4)} &= D^3\mathbf{f}(\mathbf{f}, \mathbf{f}, \mathbf{f}) + 3D^2\mathbf{f}(d\mathbf{f} \mathbf{f}, \mathbf{f}) + d\mathbf{f} D^2\mathbf{f}(\mathbf{f}, \mathbf{f}) + d\mathbf{f} d\mathbf{f} d\mathbf{f} \mathbf{f}.
\end{aligned}$$

Therefore the exact solution has the expansion

$$\begin{aligned}
\mathbf{y}(t+h) &= \mathbf{y} + h\mathbf{f} + \frac{h^2}{2}d\mathbf{f} \mathbf{f} \\
&\quad + \frac{h^3}{6} [D^2\mathbf{f}(\mathbf{f}, \mathbf{f}) + d\mathbf{f} d\mathbf{f} \mathbf{f}] \\
&\quad + \frac{h^4}{24} [D^3\mathbf{f}(\mathbf{f}, \mathbf{f}, \mathbf{f}) + 3D^2\mathbf{f}(d\mathbf{f} \mathbf{f}, \mathbf{f}) + d\mathbf{f} D^2\mathbf{f}(\mathbf{f}, \mathbf{f}) + d\mathbf{f} d\mathbf{f} d\mathbf{f} \mathbf{f}] + \mathcal{O}(h^5).
\end{aligned}$$

Expanding the general Runge Kutta step in the same way and comparing each elementary differential gives the fourth order conditions

condition	required value
$\mathbf{b}^T \mathbf{e}$	1
$\mathbf{b}^T \mathbf{c}$	$\frac{1}{2}$
$\mathbf{b}^T \mathbf{c}^2$	$\frac{1}{3}$
$\mathbf{b}^T A \mathbf{c}$	$\frac{1}{6}$
$\mathbf{b}^T \mathbf{c}^3$	$\frac{1}{4}$
$\mathbf{b}^T C A \mathbf{c}$	$\frac{1}{8}$
$\mathbf{b}^T A \mathbf{c}^2$	$\frac{1}{12}$
$\mathbf{b}^T A^2 \mathbf{c}$	$\frac{1}{24}$

Here $\mathbf{e} = (1, 1, 1, 1)^T$, $C = \text{diag}(\mathbf{c})$, and powers of $\mathbf{c}$ are taken componentwise. The table is the coefficient comparison between the exact Taylor expansion and the numerical Taylor expansion. The first four conditions come from matching terms through order three. The last four conditions come from matching the four elementary fourth-order terms. For the classical RK4 coefficients,

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{6} \end{bmatrix}.$$

Now check each condition directly. First,

$$\mathbf{b}^T \mathbf{e} = \frac{1}{6} + \frac{1}{3} + \frac{1}{3} + \frac{1}{6} = 1.$$

Next,

$$\mathbf{b}^T \mathbf{c} = \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{6} = \frac{1}{2}.$$

Also,

$$\mathbf{b}^T \mathbf{c}^2 = \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{6} = \frac{1}{3},$$

and

$$\mathbf{b}^T \mathbf{c}^3 = \frac{1}{3} \cdot \frac{1}{8} + \frac{1}{3} \cdot \frac{1}{8} + \frac{1}{6} = \frac{1}{4}.$$

For the remaining conditions, compute

$$A\mathbf{c} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{4} \\ \frac{1}{2} \end{bmatrix}.$$

Therefore

$$\mathbf{b}^T A\mathbf{c} = \frac{1}{3} \frac{1}{4} + \frac{1}{6} \frac{1}{2} = \frac{1}{6}.$$

Since

$$CA\mathbf{c} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{8} \\ \frac{1}{2} \end{bmatrix},$$

we get

$$\mathbf{b}^T CA\mathbf{c} = \frac{1}{3} \frac{1}{8} + \frac{1}{6} \frac{1}{2} = \frac{1}{8}.$$

Next,

$$A\mathbf{c}^2 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{8} \\ \frac{1}{4} \end{bmatrix},$$

so

$$\mathbf{b}^T A\mathbf{c}^2 = \frac{1}{3} \frac{1}{8} + \frac{1}{6} \frac{1}{4} = \frac{1}{12}.$$

Finally,

$$A^2\mathbf{c} = A(A\mathbf{c}) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{4} \end{bmatrix}.$$

Thus

$$\mathbf{b}^T A^2\mathbf{c} = \frac{1}{6} \frac{1}{4} = \frac{1}{24}.$$

All order conditions through order four are satisfied. Hence the local truncation error is $\mathcal{O}(h^5)$, and under the usual smoothness and Lipschitz assumptions the accumulated global error over a fixed time interval is $\mathcal{O}(h^4)$. This proves that the classical RK4 method is fourth order.

C.7 Projectile Motion With Quadratic Drag

Consider a projectile moving in two dimensions under gravity and quadratic drag. The equations are

$$m\dot{v}_x = -c\sqrt{v_x^2 + v_y^2}v_x, \quad m\dot{v}_y = -mg - c\sqrt{v_x^2 + v_y^2}v_y.$$

Let

$$\mathbf{y} = \begin{bmatrix} x \\ y \\ v_x \\ v_y \end{bmatrix}.$$

Since $\dot{x} = v_x$ and $\dot{y} = v_y$, the second order physical problem becomes the first order system

$$\frac{d}{dt} \begin{bmatrix} x \\ y \\ v_x \\ v_y \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \\ -\frac{c}{m}\sqrt{v_x^2 + v_y^2}v_x \\ -g - \frac{c}{m}\sqrt{v_x^2 + v_y^2}v_y \end{bmatrix}.$$

This is exactly the form required by RK4.

C.8 Simulation Figure

The following figure is produced by the Python code in the next section. The red curve is the RK4 trajectory with quadratic drag. The blue curve is the trajectory without drag.

C.9 Python Code

```
import numpy as np
```

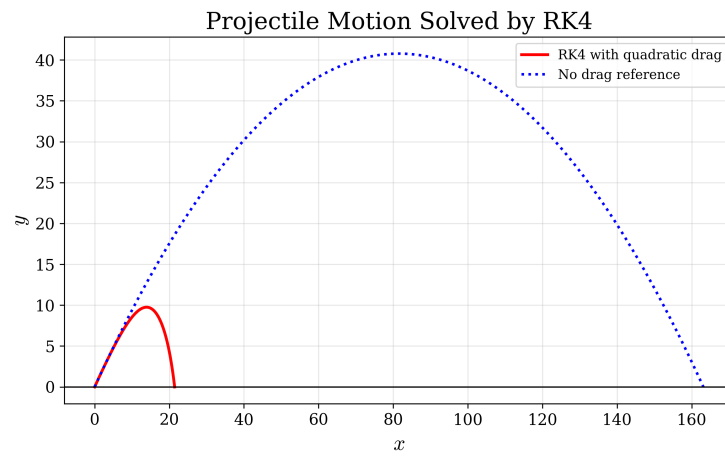


Fig. C.1 Projectile trajectory computed with RK4.

```
import matplotlib.pyplot as plt

m = 1.0
c = 0.08
g = 9.81
v0 = 40.0
theta_deg = 45.0

theta = np.deg2rad(theta_deg)
state0 = np.array([
    0.0,
    0.0,
    v0 * np.cos(theta),
    v0 * np.sin(theta),
], dtype=float)

def rhs_drag(t, state):
    x, y, vx, vy = state
    speed = np.sqrt(vx**2 + vy**2)
    return np.array([
        vx,
        vy,
        -(c / m) * speed * vx,
        -g - (c / m) * speed * vy,
    ], dtype=float)

def rhs_no_drag(t, state):
    x, y, vx, vy = state
    return np.array([
        vx,
        vy,
```

```

        0.0,
        -g,
    ], dtype=float)

def rk4_step(f, t, state, \dd t):
    k1 = f(t, state)
    k2 = f(t + 0.5 * \dd t, state + 0.5 * \dd t * k1)
    k3 = f(t + 0.5 * \dd t, state + 0.5 * \dd t * k2)
    k4 = f(t + \dd t, state + \dd t * k3)
    return state + (\dd t / 6.0) * (k1 + 2.0 * k2 +
                                   2.0 * k3 + k4)

def integrate_until_ground(f, state0, \dd t=0.001, tmax=20.0):
    t = 0.0
    state = state0.copy()
    times = [t]
    states = [state.copy()]

    while t < tmax:
        state = rk4_step(f, t, state, \dd t)
        t = t + \dd t
        times.append(t)
        states.append(state.copy())
        if t > 0.0 and state[1] < 0.0:
            break

    return np.array(times), np.array(states)

\dd t = 0.001

t_drag, states_drag = integrate_until_ground(rhs_drag, state0, \dd t=\dd t)
t_free, states_free = integrate_until_ground(rhs_no_drag, state0, \dd t=\dd t)

x_drag = states_drag[:, 0]
y_drag = states_drag[:, 1]
x_free = states_free[:, 0]
y_free = states_free[:, 1]

plt.figure(figsize=(8, 5))
plt.plot(x_drag, y_drag, color='red', linewidth=2.2,
         label='RK4 with quadratic drag')
plt.plot(x_free, y_free, color='blue', linewidth=2.0,
         linestyle='dotted', label='No drag reference')
plt.axhline(0.0, color='black', linewidth=1.0)
plt.xlabel(r'$x$')
plt.ylabel(r'$y$')
plt.title('Projectile Motion Solved by RK4')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('rk4_projectile_drag.png', dpi=300)
plt.show()

```

```
print(f"Range with drag      = {x_drag[-1]:.4f}")
print(f"Range without drag   = {x_free[-1]:.4f}")
print(f"Max height with drag = {np.max(y_drag):.4f}")
print(f"Max height no drag   = {np.max(y_free):.4f}")
```

C.10 Numerical Interpretation

The no drag case gives a much longer range because horizontal velocity remains constant. With quadratic drag, the horizontal speed decreases rapidly when the projectile is fast. The vertical motion is also strongly affected because the drag force opposes the instantaneous velocity. For the parameters in the code, the RK4 result gives a much shorter range and a lower maximum height.

The time step h should be checked by convergence. A simple test is to repeat the simulation with h , $h/2$, and $h/4$. Since RK4 has global error $\mathcal{O}(h^4)$, halving the step size should reduce the dominant numerical error by approximately a factor of 16 once the calculation is in the asymptotic regime.

Appendix D

Introduction to Special Relativity

This appendix is meant to supplement the material that could not be covered within the sixteen-week course schedule, namely special relativity. It introduces the mathematical methods mentioned in Chapter 7, together with the principles behind time dilation, length contraction, and several famous paradoxes in relativity. It may be read as an introductory popular-level supplement. These ideas appear not only in classical mechanics, but also play an important role in electrodynamics. If you are interested, you may read through it briefly.

D.1 Lorentz Transform

D.1.1 The Original Question and the Galilean Transform

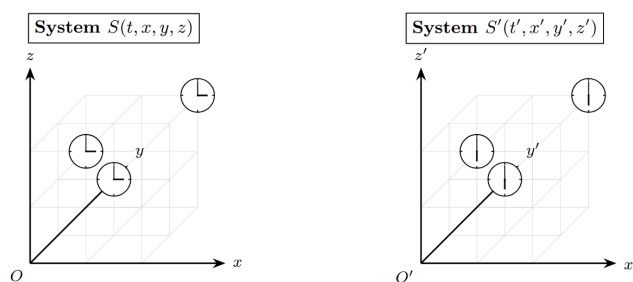


Fig. D.1 Two different reference frames with their own spacetime coordinates.

Consider two inertial reference frames

$$S(t, x, y, z)$$

and

$$S'(t', x', y', z').$$

Assume that the frame S' moves with constant velocity v along the x -direction relative to S . The Galilean transformation is defined by

$$\begin{cases} t' = t + c_0, \\ x' = x - vt + c_1, \end{cases} \quad \begin{cases} y' = y + c_2, \\ z' = z + c_3. \end{cases} \quad (\text{D.1})$$

The constants c_0, c_1, c_2, c_3 represent arbitrary choices of the origins of time and space. The term $-vt$ means that the origin of S' moves with velocity v as seen from S . Indeed, if $x' = 0$ and $c_1 = 0$, then

$$x = vt.$$

Thus the origin of S' has uniform motion in the x -direction relative to S .

The most important feature of the Galilean transformation is that it leaves acceleration invariant. Since

$$x' = x - vt + c_1$$

and

$$t' = t + c_0,$$

we have

$$\frac{dx'}{dt'} = \frac{d(x - vt + c_1)}{dt} = \frac{dx}{dt} - v. \quad (\text{D.2})$$

Therefore the velocity transforms as

$$u'_x = u_x - v. \quad (\text{D.3})$$

Taking another derivative gives

$$\frac{d^2 x'}{dt'^2} = \frac{d}{dt} \left(\frac{dx}{dt} - v \right) = \frac{d^2 x}{dt^2}. \quad (\text{D.4})$$

Hence

$$a'_x = a_x. \quad (\text{D.5})$$

The same argument applies to the y - and z -directions. Therefore Newton's second law keeps the same form in different inertial frames,

$$F = ma, \quad F' = ma'. \quad (\text{D.6})$$

This is why Newtonian mechanics is naturally compatible with the Galilean transformation.

However, equation (D.3) also shows the main limitation of the Galilean transformation. Speeds are not invariant. If a signal moves in S with speed c along the x -direction, then

$$u_x = c.$$

In the frame S' , its speed becomes

$$u'_x = c - v.$$

Thus a finite speed cannot have the same value in every inertial frame under the Galilean transformation.

Equivalently, suppose a light signal satisfies

$$x = ct$$

in the frame S . Using the Galilean transformation with $c_0 = c_1 = 0$, we obtain

$$x' = x - vt = ct - vt = (c - v)t.$$

Since $t' = t$, this becomes

$$x' = (c - v)t'.$$

Therefore the speed measured in S' is

$$c' = c - v.$$

This contradicts the relativistic requirement that the universal speed c , namely the speed of light in vacuum, should be measured to have the same value in every inertial frame.

The central question is therefore not how to make the relative velocity v universal. The relative velocity between two frames can take different values. The real question is how to modify the coordinate transformation between inertial frames so that there exists a universal speed c whose value is the same for all inertial observers.

D.1.2 Definition and Matrices in Minkowski Space

In special relativity, we add the time axis ct to the original Euclidean space. This spacetime is called **Minkowski space**. In the original coordinate system S , suppose that an object moves in the positive x -direction with speed c . Then, from the viewpoint of S , its trajectory in spacetime is

$$x = ct.$$

This trajectory is shown as the dashed line in the following figure:

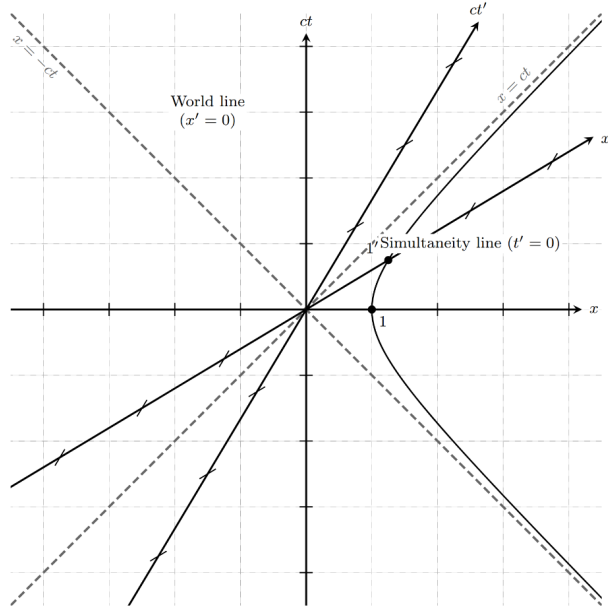


Fig. D.2 The illustration of the Lorentz Transform in Minkowski space.

Now consider another inertial coordinate system S' , which moves to the right with speed v relative to S . The new time axis ct' is tilted with respect to the old time axis ct , as shown in the figure. If we want to claim that c is a universal speed, then the trajectory $x = ct$ must represent the same light ray in both coordinate systems. In other words, the projection of this trajectory onto the time axis and the space axis should still correspond to the same speed c .

Therefore, when the inertial coordinate system changes, the new spacetime axes must be arranged symmetrically with respect to the light ray $x = ct$. This kind of transformation is called the **Lorentz transformation**, i.e.

$$(t, x, y, z) \longrightarrow (t', x', y', z'). \quad (\text{D.7})$$

If SR holds, there are five assumptions:

1. Space is in **Euclidean space**, which has **translation symmetry** and the **rotation symmetry**.
2. Time has **translation symmetry**.
3. There's an universal speed $c = \text{const}$ **in all inertia frame**.
4. The transform's tensor Λ depends on the relative $\mathbf{v}$ between S and S' .
5. If S is at $\mathbf{v}$ w.r.t. S' , then S' is at $-\mathbf{v}$ w.r.t. S .

We define **4-vector** as x^μ , which is equivalent to

$$\vec{x}^\mu \equiv (x^0, x^1, x^2, x^3) = (ct, x, y, z). \quad (\text{D.8})$$

4-vector can be used to simultaneously describe the time and the space. The coordinate (Lorentz) transform is defined to be

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu, \quad \begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \Lambda^0{}_0 & \Lambda^0{}_1 & \Lambda^0{}_2 & \Lambda^0{}_3 \\ \Lambda^1{}_0 & \Lambda^1{}_1 & \Lambda^1{}_2 & \Lambda^1{}_3 \\ \Lambda^2{}_0 & \Lambda^2{}_1 & \Lambda^2{}_2 & \Lambda^2{}_3 \\ \Lambda^3{}_0 & \Lambda^3{}_1 & \Lambda^3{}_2 & \Lambda^3{}_3 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} \quad (\text{D.9})$$

where Λ is called **Lorentz Transform matrix**. In addition, we can define the transformation of a covariant vector. To remain the inner product:

$$A_\mu V^\mu = A'_\mu V'^\mu = A'_\mu \Lambda^\mu{}_\nu V^\nu = A_\nu V^\nu$$

this yields

$$\text{Tensor form : } A'_\mu \Lambda^\mu{}_\nu = A_\nu \quad (\text{D.10})$$

$$\text{Matrix Form : } \mathbf{A}'_\mu \Lambda = \mathbf{A}_\nu \Rightarrow \mathbf{A}'_\mu = \mathbf{A}_\nu \Lambda^{-1} \quad (\text{D.11})$$

Therefore,

$$A'_\mu = A_\nu \left(\Lambda^{-1} \right)^\nu{}_\mu. \quad (\text{D.12})$$

So, the transformations of the contravariant and the covariant vectors are respectively:

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu \quad (\text{D.13})$$

$$x'_\mu = x_\nu \left(\Lambda^{-1} \right)^\nu{}_\mu \quad (\text{D.14})$$

D.1.2.1 Minkowski Metric between 2 Events

If there's two events in S :

$$\begin{cases} \text{Event 1 : } S(ct_1, x_1, y_1, z_1) \\ \text{Event 2 : } S(ct_2, x_2, y_2, z_2) \end{cases}$$

As a result, if an object in S moves from the Event 1 to Event 2 **with light speed c** , it satisfies

$$\pm c \Delta t = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \Leftrightarrow \text{speed} = c \quad (\text{D.15})$$

and (D.15) is

$$c^2 \Delta t^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \Rightarrow -(c^2 \Delta t^2) + \Delta x^2 + \Delta y^2 + \Delta z^2 = 0, \quad (\text{D.16})$$

which can be written as

$$\Delta x^\mu \eta_{\mu\nu} \Delta x^\nu = 0 \quad dx^\mu \eta_{\mu\nu} dx^\nu = 0, \quad (\text{D.17})$$

where η is called **Minkowski metric**:

$$\eta = \text{diag}(-1, 1, 1, 1) \quad (\text{D.18})$$

If the object moves with speed v different from c , we can further define the **Minkowski interval** or **line element** ds^2 :

$$ds^2 \equiv -c^2 dt^2 + dx^2 + dy^2 + dz^2 = dx^\mu \eta_{\mu\nu} dx^\nu \quad (\text{D.19})$$

with different velocity v , the Minkowski intervals are

$$\begin{cases} v = c : ds^2 = 0, \text{ since } dx^i \delta_{ij} dx^j = (dx^0)^2 = c^2 dt^2 \\ v < c : ds^2 < 0, \text{ since } dx^i \delta_{ij} dx^j < (dx^0)^2 = c^2 dt^2 \\ v > c : ds^2 > 0, \text{ since } dx^i \delta_{ij} dx^j > (dx^0)^2 = c^2 dt^2 \end{cases} \quad (\text{D.20})$$

when $v < c$, the interval of the object must be less than that of light, vice versa. In different inertial frame, the universal speed c must have:

$$\Delta x^\mu \eta_{\mu\nu} \Delta x^\nu = \Delta x'^\mu \eta_{\mu\nu} \Delta x'^\nu, \quad dx^\mu \eta_{\mu\nu} dx^\nu = dx'^\mu \eta_{\mu\nu} dx'^\nu \quad (\text{D.21})$$

by the transform (D.9), the infinitesimal displacement equals

$$dx'^\mu = \Lambda^\mu{}_\nu dx^\nu, \quad (\text{D.22})$$

hence

$$\begin{aligned} dx'^\mu \eta_{\mu\nu} dx'^\nu &= (\Lambda^\mu{}_\alpha dx^\alpha) \eta_{\mu\nu} (\Lambda^\nu{}_\beta dx^\beta) \\ &= dx^\alpha \Lambda^\mu{}_\alpha \eta_{\mu\nu} \Lambda^\nu{}_\beta dx^\beta = dx^\alpha \eta_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta dx^\beta \end{aligned}$$

By (D.21), the universal speed forces that the relation between matrix and metric is:

$$\boxed{\Lambda^\mu{}_\alpha \eta_{\mu\nu} \Lambda^\nu{}_\beta = \eta_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta = \eta_{\alpha\beta}} \quad (\text{D.23})$$

We can also rewrite (D.21) as the matrix form, we obtain

$$dx'^{\mu T} \eta dx'^\mu = \left(\Lambda dx^\mu \right)^T \eta \left(\Lambda dx^\mu \right) = dx^{\mu T} \Lambda^T \eta \Lambda dx^\mu = dx^{\mu T} \eta dx^\mu$$

Then we can obtain another representation of the result of (D.23):

$$\boxed{\Lambda^T \eta \Lambda = \eta \quad \forall \vec{x}^\mu \in S} \quad (\text{D.24})$$

In addition, when traveling with speed c , we can multiply a non-zero constant λ and the equation still remain 0, therefore

$$\begin{cases} \eta_{\mu\nu} \Lambda^\nu{}_\alpha \Lambda^\mu{}_\beta = \lambda \eta_{\alpha\beta}, & \lambda \in \mathbb{R} \setminus \{0\} \\ \Lambda^T(\mathbf{v}) \eta \Lambda(\mathbf{v}) = \lambda(\mathbf{v}) \eta \end{cases} \quad (\text{D.25})$$

By the assumption 5, S is at $-\mathbf{v}$ w.r.t. S' , therefore the Lorentz transform becomes

$$\Lambda^{-1\mu}{}_\nu dx'^\nu = dx^\mu, \quad \Lambda^{-1}(\mathbf{v}) dx'^\mu = dx^\mu \quad (\text{D.26})$$

Meanwhile, the equivalent description gives that

$$dx^\mu = \Lambda(-\mathbf{v}) dx'^\mu \quad (\text{D.27})$$

so

$$\boxed{\Lambda^{-1}(\mathbf{v}) = \Lambda(-\mathbf{v})}. \quad (\text{D.28})$$

Since $\Lambda^T \eta \Lambda = \lambda \eta$, then

$$(\Lambda^T)^{-1} \Lambda^T \eta \Lambda (\Lambda^{-1}) = (\Lambda^T)^{-1} \underbrace{(\lambda^{-1})}_{=1} \lambda \eta (\Lambda^{-1})$$

this yields

$$\lambda^{-1}(\mathbf{v}) \eta = (\Lambda^T(\mathbf{v}))^{-1} \eta \Lambda^{-1}(\mathbf{v}) = \Lambda^T(-\mathbf{v}) \eta \Lambda(-\mathbf{v}),$$

by the interval invariance under Lorentz transform:

$$dx^{\mu T} \eta dx^\mu = dx'^{\mu T} \eta dx'^\mu \Rightarrow dx'^{\mu T} \Lambda^T(-\mathbf{v}) \eta \Lambda(-\mathbf{v}) dx'^\mu = dx'^\mu \lambda(-\mathbf{v}) \eta dx'^\mu$$

the equations above give a new conclusion:

$$\boxed{\lambda^{-1}(\mathbf{v}) = \lambda(-\mathbf{v})}. \quad (\text{D.29})$$

D.1.2.2 The Components of Lorentz Transform Matrix

Suppose that S' moves with speed v relative to S along the x -direction (y, z fixed). Since $y' = y$ and $z' = z$, $\Lambda^2_2 = \Lambda^3_3 = 1$. Without the effect caused by y, z , the matrix can be written as:

$$\Lambda_{y,z \text{ Fixed}} = \left(\begin{array}{cc|cc} \Lambda^0_0 & \Lambda^0_1 & 0 & 0 \\ \Lambda^1_0 & \Lambda^1_1 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

And let $\Lambda^0_0 = A$, $\Lambda^0_1 = B$, $\Lambda^1_0 = C$ and $\Lambda^1_1 = D$, therefore the transform:

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \left(\begin{array}{cc|cc} A & B & 0 & 0 \\ C & D & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} \quad (\text{D.30})$$

which leads to

$$ct' = Act + Bx \quad (\text{D.31})$$

$$x' = Cct + Dx \quad (\text{D.32})$$

Since the reference frame S' moves along x-direction with speed v , then the trajectory of S' with respect to S is

$$x = vt = \frac{v}{c} \cdot ct \equiv \beta ct, \quad (\text{D.33})$$

where $\beta = v/c$. Substitute this into (D.31) and (D.32). At the origin of S' , $x' = 0$, so (D.32) becomes:

$$0 = Cct + D(\beta ct) \Rightarrow C = -D\beta \equiv -\gamma\beta \quad (\text{D.34})$$

In convention, we call D as γ . So (D.32) equals

$$x' = \gamma(x - \beta ct) \quad (\text{D.35})$$

Meanwhile, the relative speed of S with respect to (w.r.t.) S' is $-v$, then

$$x = \gamma(x + \beta ct) \quad (\text{D.36})$$

By interval invariance in Minkowski space, we've gotten

$$ds'^2 = ds^2 \Rightarrow -c^2 t'^2 + x'^2 = -c^2 t^2 + x^2$$

then substitute (D.31) and (D.35) into it, we have

$$\begin{aligned} & -(Act + Bx)^2 + (-\gamma\beta ct + \gamma x)^2 = -c^2 t^2 \\ \Rightarrow & -(A^2 c^2 t^2 + 2ABctx + B^2 x^2) + \gamma^2 \beta^2 c^2 t^2 - 2\gamma^2 \beta ct x + \gamma^2 x^2 = -c^2 t^2 + x^2. \end{aligned}$$

Comparing the coefficients of c^2t^2 and x^2 , we have

$$-A^2 + \gamma^2\beta^2 = -1 \Rightarrow A^2 = 1 + \gamma^2\beta^2 \quad (\text{D.37})$$

$$-2AB - 2\gamma^2\beta = 0 \Rightarrow AB = -\gamma^2\beta \quad (\text{D.38})$$

$$-B^2 + \gamma^2 = 1 \Rightarrow B^2 = \gamma^2 - 1 \quad (\text{D.39})$$

Multiply (D.37) by (D.39), the result is equal to the square of (D.38), therefore,

$$\begin{aligned} A^2B^2 &= (1 + \gamma^2\beta^2)(\gamma^2 - 1) = \gamma^4\beta^2 \\ \Rightarrow \gamma^2 - 1 + \gamma^4\beta^2 - \gamma^2\beta^2 &= \gamma^4\beta^2 \end{aligned}$$

Hence,

$$\gamma^2(1 - \beta^2) = 1 \Rightarrow \gamma^2 = \frac{1}{1 - \beta^2}, \quad (\text{D.40})$$

since if $v = 0$, the parameter $\beta = v/c = 0$ so the Lorentz transform is

$$\mathbf{x} = \Lambda \mathbf{x} \Rightarrow \Lambda = \mathbf{1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

so the parameter $\gamma > 0$, which leads to

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (\text{D.41})$$

substitute γ into (D.37), we can thus determine A :

$$A^2 = 1 + \frac{\beta^2}{1 - \beta^2} = \gamma^2 \Rightarrow A = \gamma \quad (\text{D.42})$$

as the same result of γ , $A > 0$. Then $AB = -\gamma^2\beta$ gives

$$B = -\gamma\beta \quad (\text{D.43})$$

Therefore, the Lorentz matrix is

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1-v^2/c^2}} & \frac{-v}{\sqrt{c^2-v^2}} & 0 & 0 \\ \frac{-v}{\sqrt{c^2-v^2}} & \frac{1}{\sqrt{1-v^2/c^2}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{D.44})$$

As a result, the Lorentz transform gives that

$$t' = \gamma \left(t - \frac{vx}{c^2} \right), \quad x' = \gamma(x - vt), \quad y' = y, \quad z' = z. \quad (\text{D.45})$$

This transform is observed by the relative inertial frame. That is, the time and length that the observer in the inertial frame S measures are t and x ; while the time and length that the observer in another inertial frame S' is t' and x' . The frame S' is moving to the right with the relative velocity $\hat{x}v$, so the minus signs hold. But when we change the frame from S' to S , the frame S is moving to the left with the relative velocity $-\hat{x}v$, then the time t and the length x he observe is rewritten as

$$t = \gamma \left(t' + \frac{v}{c^2} x' \right), \quad x = \gamma(x' + vt'). \quad (\text{D.46})$$

In comparison, we can conclude that

$$\text{Measurement in } S' : t' = \gamma \left(t - \frac{vx}{c^2} \right), \quad x' = \gamma(x - vt) \quad (\text{D.47})$$

$$\text{Measurement in } S : t = \gamma \left(t' + \frac{vx'}{c^2} \right), \quad x = \gamma(x' + vt'). \quad (\text{D.48})$$

Furthermore, we can also denote the Lorentz Transform by

$$\begin{cases} t' = \gamma \left(t - \frac{vx}{c^2} \right) \\ x' = \gamma(x - vt) \end{cases} \longrightarrow \begin{cases} ct' = \gamma \left(ct - \frac{v}{c}x \right) \\ x' = \gamma \left(x - \frac{v}{c}(ct) \right) \end{cases}$$

where $x \equiv x^1$ and $ct \equiv x^0$ by the above definition. Therefore:

$$\begin{cases} x'^0 = \gamma \left(x^0 - \frac{v}{c}x^1 \right) \\ x'^1 = \gamma \left(x^1 - \frac{v}{c}x^0 \right) \end{cases} \quad (\text{D.49})$$

Unlike the Galilean transformation, the Lorentz transformation mixes space and time. In the standard configuration, where the two origins coincide at $t = t' = 0$, it takes the form of (D.45). Under this transformation, if a light signal satisfies $x = ct$, then

$$x' = \gamma(ct - vt) = \gamma(c - v)t$$

and

$$t' = \gamma \left(t - \frac{vct}{c^2} \right) = \gamma \left(1 - \frac{v}{c} \right) t.$$

Therefore

$$\frac{x'}{t'} = \frac{\gamma(c - v)t}{\gamma \left(1 - \frac{v}{c} \right) t} = c.$$

Hence the speed c remains the same in the frame S' . This is the basic reason why the Lorentz transformation replaces the Galilean transformation in special relativity.

D.2 Paradoxes

In special relativity, many so-called paradoxes come from applying Newtonian intuition to relativistic spacetime. The difficulty is usually not caused by a contradiction in the theory. It comes from using ideas such as absolute simultaneity or universal time, which are no longer valid after the Lorentz transformation is introduced.

The central point is that different inertial observers may assign different time intervals, lengths, and simultaneity relations to the same physical process. However, all observers agree on invariant spacetime quantities, especially the spacetime interval and the proper time along a worldline.

D.2.1 Time Dilation

One of the most important consequences of special relativity is **time dilation**. It means that the time interval measured by an observer who sees a clock moving is longer than the time interval measured by an observer moving together with the clock. This does not mean that the clock is broken or that time is only an illusion. It means that time interval is not an absolute quantity. It depends on the inertial frame in which the two events are measured.

To make the idea precise, we consider a simple light clock on a train. The clock consists of two mirrors separated by a distance h . A pulse of light travels from the lower mirror to the upper mirror and then comes back. Suppose that the train moves with constant velocity v relative to the ground frame.

In the rest frame of the train, the light moves vertically. The distance traveled by the light during one half tick is h , so

$$\Delta\tau_{\frac{1}{2}} = \frac{h}{c}. \quad (\text{D.50})$$

Thus the full tick time measured by an observer on the train is

$$\Delta\tau = \frac{2h}{c}. \quad (\text{D.51})$$

This time interval $\Delta\tau$ is called the **proper time**, because the two events occur at the same spatial position in the train frame. More explicitly, the emission and return of the light pulse occur at the same place relative to the clock.

However, from the ground frame, the train is moving. During the same half tick, the light still travels at speed c , but its path is no longer vertical. The light follows a diagonal path, because the train moves horizontally while the light moves from one mirror to the other. If the half tick time measured from the ground frame is $\Delta t/2$, then the light travels a distance

$$c \frac{\Delta t}{2}, \quad (\text{D.52})$$

while the train moves horizontally by

$$v \frac{\Delta t}{2}. \quad (\text{D.53})$$

Therefore, by the Pythagorean theorem,

$$\left(c \frac{\Delta t}{2}\right)^2 = h^2 + \left(v \frac{\Delta t}{2}\right)^2. \quad (\text{D.54})$$

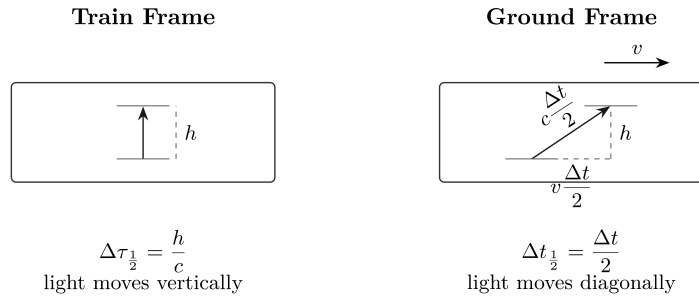


Fig. D.3 A light clock on a moving train. In the train frame, the light travels vertically between two mirrors. In the ground frame, the train moves horizontally, so the same light pulse follows a diagonal path. Since the speed of light is the same in both inertial frames, the diagonal path requires a longer coordinate time.

From (D.54), we have

$$\frac{c^2 \Delta t^2}{4} = h^2 + \frac{v^2 \Delta t^2}{4}. \quad (\text{D.55})$$

Moving the velocity term to the left hand side gives

$$\frac{\Delta t^2}{4}(c^2 - v^2) = h^2. \quad (\text{D.56})$$

Hence

$$\Delta t^2 = \frac{4h^2}{c^2 - v^2} = \frac{4h^2}{c^2} \frac{1}{1 - \frac{v^2}{c^2}}. \quad (\text{D.57})$$

Using (D.51), we know that

$$\Delta \tau^2 = \frac{4h^2}{c^2}. \quad (\text{D.58})$$

Therefore,

$$\Delta t^2 = \frac{\Delta \tau^2}{1 - \frac{v^2}{c^2}}. \quad (\text{D.59})$$

Taking the positive square root, we obtain

$$\boxed{\Delta t = \gamma \Delta \tau}, \quad \gamma := \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \quad (\text{D.60})$$

Since $0 \leq v < c$, we always have

$$\gamma \geq 1. \quad (\text{D.61})$$

Thus

$$\Delta t \geq \Delta \tau. \quad (\text{D.62})$$

This is the mathematical statement of time dilation. The moving clock is measured to tick more slowly, because a fixed proper time interval $\Delta \tau$ on the train corresponds to a longer coordinate time interval Δt in the ground frame.

The same result can also be written in terms of the spacetime interval. For a particle or clock moving with velocity v , we have

$$c^2 d\tau^2 = c^2 dt^2 - dx^2. \quad (\text{D.63})$$

Since $dx = v dt$, this becomes

$$c^2 d\tau^2 = c^2 dt^2 - v^2 dt^2 = c^2 \left(1 - \frac{v^2}{c^2}\right) dt^2. \quad (\text{D.64})$$

Thus

$$d\tau = \sqrt{1 - \frac{v^2}{c^2}} dt = \frac{dt}{\gamma}. \quad (\text{D.65})$$

Equivalently,

$$\boxed{dt = \gamma d\tau}. \quad (\text{D.66})$$

This form is often more useful in physics. The proper time τ is the time measured by the clock itself along its own worldline. The coordinate time t is the time assigned by

a chosen inertial frame. Therefore, time dilation is not a mysterious mechanical effect on the clock. It is a geometric relation between coordinate time and proper time in spacetime.

D.2.2 Twin Paradox

The twin paradox describes two observers. One observer stays on Earth, while the other travels at relativistic speed, turns around, and returns. At first glance, each observer may seem to see the other as moving, so one may incorrectly expect both twins to conclude that the other has aged less.

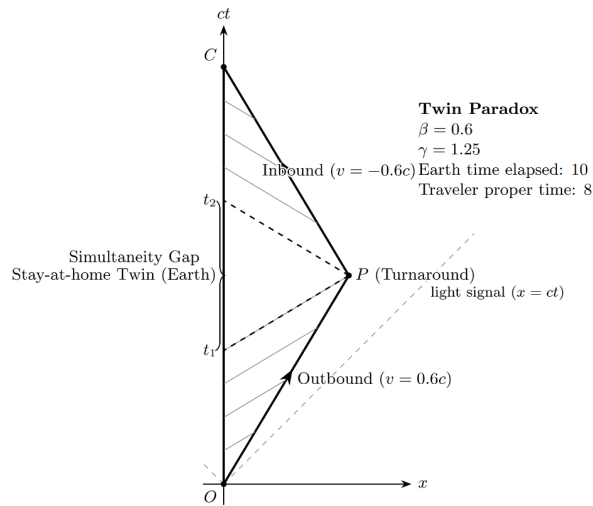


Fig. D.4 Spacetime diagram for the twin paradox. One twin remains in an inertial frame, while the other travels away and returns.

The resolution is that the two twins do not play symmetric roles. The twin who stays on Earth can be approximated as remaining in one inertial frame. The traveling twin changes inertial frames during the turnaround. This change of frame breaks the apparent symmetry.

The cleanest way to understand the result is through proper time. For an object moving with speed v , the proper time interval satisfies (See Sec ??)

$$d\tau = dt \sqrt{1 - \frac{v^2}{c^2}} = \frac{dt}{\gamma}.$$

Therefore, along a worldline,

$$\tau = \int \sqrt{1 - \frac{v^2}{c^2}} dt.$$

The stationary twin has $v = 0$ in the Earth frame, so the elapsed proper time equals the coordinate time,

$$\tau_{\text{Earth}} = \int dt.$$

The traveling twin has $v \neq 0$ for most of the trip, so

$$\tau_{\text{travel}} = \int \sqrt{1 - \frac{v^2}{c^2}} dt < \int dt = \tau_{\text{Earth}}.$$

Hence, the traveling twin experiences less proper time and returns younger.

The twin paradox is resolved by comparing proper time along two different worldlines. The traveling twin follows a path through spacetime with shorter proper time, so they age less.

D.2.3 Barn and Ladder Paradox

The barn and ladder paradox considers a ladder moving at relativistic speed toward a barn. In the rest frame of the barn, the ladder is moving, so its length is Lorentz contracted. If the ladder is sufficiently fast, it can appear short enough to fit completely inside the barn.

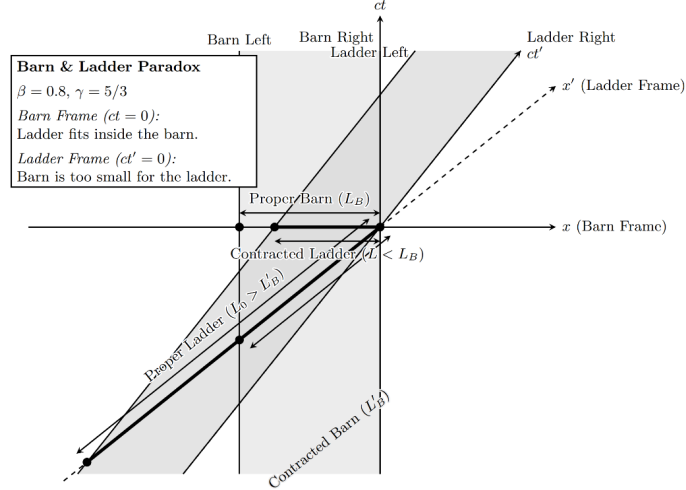


Fig. D.5 Spacetime diagram for the barn and ladder paradox. The ladder is length-contracted in the barn frame, while the barn is length-contracted in the ladder frame.

Suppose the proper length of the ladder is L_0 . In the barn frame, the moving ladder has a length

$$L = \frac{L_0}{\gamma}.$$

If L is smaller than the barn length, then there exists a moment in the barn frame when the whole ladder is inside the barn.

The paradox appears when we switch to the ladder frame. In the ladder frame, the ladder is at rest with length L_0 , while the barn is moving and length-contracted. Therefore, the barn is even shorter. It seems impossible for the ladder to fit inside the barn.

The resolution is the relativity of simultaneity. The statement "the front and back of the ladder are inside the barn at the same time" depends on the frame. In the barn frame, the front and back doors can be closed simultaneously while the ladder is inside. In the ladder frame, those two door-closing events are not simultaneous.

Mathematically, the Lorentz transformation gives

$$t' = \gamma \left(t - \frac{vx}{c^2} \right).$$

For two events that are simultaneous in the barn frame, we have $\Delta t = 0$. However, if they occur at different positions, $\Delta x \neq 0$, then

$$\Delta t' = \gamma \left(\Delta t - \frac{v\Delta x}{c^2} \right) = -\gamma \frac{v\Delta x}{c^2}.$$

Thus,

$$\Delta t' \neq 0.$$

So events that are simultaneous in the barn frame are not simultaneous in the ladder frame.

The barn and ladder paradox is resolved by the relativity of simultaneity. The ladder can fit inside the barn at one instant in the barn frame, but the corresponding front-door and back-door events are not simultaneous in the ladder frame.

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Key Terms by Chapter

This short list collects the main mathematical and physical keywords appearing in each chapter. It is meant to be used together with the final keyword index, where the corresponding page numbers are generated automatically by LaTeX.

Basic Calculus for Physics

- Basic Calculus for Physics
- Calculus
- Norm, Functions, and Continuity
- p Norm in Metric Space
- p -norm
- Metric space
- Norm
- Continuity and Differentiability
- Continuity
- Differentiability
- Types of Functions
- Injective function
- Surjective function
- Bijective function
- Function
- Differentiation
- The Chain Rule
- Chain rule
- Trigonometric Differentiation
- Trigonometric differentiation
- Derivative of Exponential Function
- Exponential function
- Implicit Differentiation
- Implicit differentiation
- Fundamental Techniques of Integration
- Fundamental Theorem of Calculus
- Fundamental theorem of calculus
- Integration by Substitution
- Substitution method
- Integration by Parts
- Integration by parts
- Exponential and Logarithmic Integral
- Exponential integral
- Logarithmic integral
- Trigonometric Substitution
- Trigonometric substitution
- Hyperbolic Functions
- Hyperbolic function
- The Method of Partial Fraction
- Partial fraction
- Taylor Expansion
- Taylor series
- Taylor approximation
- General Form of Taylor Expansion
- Taylor expansion
- Linear Approximation

- Partial Differentiation
- Partial derivative
- Functions of Several Variables
- Several variables
- Partial Derivatives
- Total Derivative
- Total derivative
- The Chain Rule of Several Variables
- Multivariable chain rule
- Constrained Partial Differentials
- Constrained differential
- Triple product rule
- Reciprocal relation
- Physical Meaning in Thermodynamics
- Multiple Integrals
- Multiple integral
- Lagrange multipliers
- Lagrange multiplier
- Jacobian Matrix and Coordinate Transformation
- Jacobian matrix
- Coordinate transformation
- Elliptic Integrals in the Pendulum Problem
- Elliptic integral
- Pendulum
- Gaussian Integral
- Gaussian integral
- Gamma Function
- Gamma function
- Beta Function
- Beta function
- Elliptic Integral and Pendulum
- Acceleration in Three Coordinates
- Polar Coordinates
- Polar coordinate
- Cylindrical Coordinates
- Cylindrical coordinate
- Spherical Coordinates
- Spherical coordinate

Introduction to Vector Analysis

- Introduction to Vector Analysis
- Vector analysis
- Vector Analysis
- Basic Vector Operations
- Vector operation
- Algebra of Vector
- Vector algebra
- Vector Products
- Dot product
- Cross product
- Scalar product
- Vector product
- Vector Identities
- Vector identity
- Definition of Vector Differentiation
- Vector Field and Curve
- Vector field
- Curve
- The Definition of Limit and Differentiation
- Directional Derivative
- Directional derivative
- Gradient
- Definition of Divergence and Curl
- Divergence
- Curl
- Identities of Vector Operators
- Vector operator identity
- Vector Integral
- Definition of Vector Integration
- Vector integration
- Line Integral and Kinetic Energy
- Line integral
- Kinetic energy
- Conservative Field and Potential Functions
- Conservative field
- Potential function
- Surface integral
- Green's Theorem
- Green's theorem
- Gauss's Divergence Theorem

- Gauss's divergence theorem
- Divergence theorem
- The Derivation of the Theorem
- Gauss's Law
- Gauss's law
- Delta Function/Step Function and Distribution
- Delta function
- Step function
- Distribution
- Derivation of Gauss's Law
- Stokes' Theorem
- Stokes' theorem

Complex Numbers in Physics

- Complex Numbers in Physics
- Complex number
- Euler's Formula and Complex Exponentials
- Euler's formula
- Complex exponential
- Euler's Formula
- De Moivre's Theorem
- De Moivre's theorem
- Trigonometric Identities
- Trigonometric identity
- Basic Complex Calculus
- Complex calculus
- Real and Imaginary Parts: Physical Interpretation
- Real part
- Imaginary part
- Real and Imaginary Parts
- Physical Interpretation
- Simple Harmonic Motion
- Simple harmonic motion
- AC Circuit
- AC circuit
- Introduction to Fourier Series and Transform
- Fourier analysis
- Fourier series
- Fourier transform
- Fourier Series and Transform
- Average Value of Sines and Cosines
- Average value
- Sine
- Cosine
- Fourier Series in the Real Domain
- Real Fourier series
- Orthogonality of Sine and Cosine
- Orthogonality
- Sine cosine orthogonality
- Kronecker delta
- Complex Form of Fourier Series
- Complex Fourier series
- Convergence and the Gibbs Phenomenon
- Convergence
- Gibbs phenomenon
- Fourier Transform of a Non-periodic Function
- Non-periodic function
- Frequency domain
- Time domain
- Delta Function Identity
- Delta function identity
- Another Perspective of Fourier Transform and Representation
- Fourier representation
- Applications: DFT and FFT
- DFT
- FFT
- Discrete Fourier transform
- Fast Fourier transform
- Spectral decomposition
- Applications
- DFT and FFT

Ordinary Differential Equation

- Ordinary Differential Equation
- Ordinary differential equation
- ODE
- First-Order ODEs
- First-order ODE
- Linearity of ODE and Differentiation
- Linearity
- Differentiation operator
- Homogeneous Problems and Separable Equations
- Homogeneous ODE
- Separable equation
- Integrating Factor
- Integrating factor
- The Bernoulli Equation
- Bernoulli equation
- Homogeneous Equation
- Second-Order ODEs
- Second-order ODE
- Homogeneous Problems
- Nonhomogeneous Problems and The Operator Descent Method
- Nonhomogeneous ODE
- Operator descent method
- Particular solution
- Generalization to Arbitrary Right-Hand Side Functions
- Other Methods for Second-Order Equation
- Differential Equations Involving Only y
- Cauchy Equation
- Cauchy equation
- A Complicated Differential Equation with One of Its Solutions
- The Examples
- Series Solution to ODE
- Series solution
- Ordinary Points and Power Series Solutions
- Ordinary point
- Power series solution
- Singular Points and the Frobenius Method
- Singular point
- Frobenius method
- Physical Meaning of Series Solutions
- Laplace Transform
- Laplace transform
- Basic Properties
- Step Functions, Impulses, and Physical Interpretation
- Step function
- Impulse
- Table of Common Laplace Transforms
- Useful Properties of the Laplace Transform
- Sturm-Liouville Theorem
- Sturm-Liouville theorem
- Sturm-Liouville Form
- Sturm-Liouville form
- Self-Adjointness and the Weighted Inner Product
- Self-adjointness
- Weighted inner product
- Eigenfunctions and Orthogonality
- Eigenfunction
- Eigenvalue
- Orthogonality
- Expansion in Normal Modes
- Normal mode expansion
- Special Function
- Special function
- Bessel Function
- Bessel function
- Example: Circular Membrane
- Example
- Circular Membrane
- Non-truncation of Bessel Series
- Legendre Polynomial
- Legendre polynomial
- Orthogonality of Legendre Polynomials
- Legendre Polynomial in Laplace's Equation
- Truncation of Legendre Series
- Spherical Harmonics
- Spherical harmonics

- Quantum Mechanics
- Comparison Between Bessel Functions and Legendre Polynomials
- Applications: Driven Oscillation and RLC Circuit
- Applications
- Driven Oscillation and RLC Circuit
- Driven Oscillation
- Driven oscillation
- RLC Circuit
- RLC circuit

Introduction to Partial Differential Equations

- Introduction to Partial Differential Equations
- Partial differential equation
- PDE
- Partial Differential Equations
- Basic Ideas of Partial Differential Equations
- Order and Linearity
- Order of PDE
- Linearity of PDE
- Three Classical Types of Second Order PDEs
- Elliptic PDE
- Parabolic PDE
- Hyperbolic PDE
- Initial and Boundary Conditions
- Initial condition
- Boundary condition
- Heat Diffusion Equation
- Heat equation
- Diffusion equation
- Derivation in One Dimension
- Derivation in Three Dimensions
- Solving the Heat Equation on a Finite Rod
- Finite rod
- Example: A Single Sine Temperature Profile
- Example
- A Single Sine Temperature Profile
- Energy Decay
- Energy decay
- Wave Equation
- Wave equation
- Derivation from a Vibrating String
- D'Alembert Solution on the Infinite Line
- D'Alembert solution
- Finite String with Fixed Ends
- Fixed boundary condition
- Example: Plucked String Released from Rest
- Plucked String Released from Rest
- Energy Conservation for the Wave Equation
- General Solution to Wave Functions
- Superposition of Waves
- Superposition
- Complex Solution in 1D
- Solution in 3D and the Wave Vector
- Wave vector
- Electromagnetic Wave Equation
- Electromagnetic wave equation
- Laplace Equation
- Laplace equation
- Harmonic function
- 3-D Problem
- Spherical Problem
- Interior Spherical Dirichlet Problem
- Dirichlet problem
- Exterior Spherical Dirichlet Problem
- Examples
- Mean Value Property and Maximum Principle
- Mean value property
- Maximum principle
- Electrostatics
- Steady Heat Conduction
- Poisson's Equation
- Poisson's equation

- Gravitational Potential of a Uniform Sphere
- Poisson's Equation and Green's Function in Electrostatics
- Green's function

Introduction to Calculus of Variations

- Introduction to Calculus of Variations
- Calculus of variations
- Calculus of Variations
- Definition of a Functional
- Functional
- Extremum of Functional
- Extremum
- Euler-Lagrange Equation and Lagrangian and Hamiltonian
- Euler-Lagrange equation
- Lagrangian
- Hamiltonian
- Euler-Lagrange Equation without Constraint Conditions
- Euler-Lagrange Equation under Constraint Condition
- Constraint condition
- The Canonical Form of the Euler Equation and Hamiltonian
- Canonical form
- Euler equation
- The Second Variation
- Second variation
- Quadratic Functional and Legendre Condition
- Quadratic functional
- Legendre condition
- Jacobi's Necessary Condition and Conjugate Points
- Jacobi's condition
- Conjugate point
- Sufficient Condition for a Weak Extremum
- Weak extremum
- Sufficient condition
- Variational Problem involving Multiple Integrals
- Multiple integral functional
- Variation of a Functional Defined on a Fixed Region
- EOM of Continuous System
- Continuous system
- Equation of motion
- Variation of a Functional Defined on a Variable Region
- Noether's Theorem in Mathematics and Field
- Noether's theorem
- Noether in a Single Independent Variable
- General Case for Noether's Theorem
- Applications of the Calculus of Variations
- No One Has Told Me Why a Straight Line is the Shortest Distance!
- Fermat's Principle
- Cycloid Problem
- Dido Problem
- Mathematical Structure of Field Theory
- Principle of Stationary Action for Fields
- Stationary action
- Field theory
- Conservation Laws for Fields
- Conservation law
- Conservation of Energy/Momentum
- Energy conservation
- Momentum conservation
- Conservation of Angular Momentum
- Angular momentum conservation

Introduction to Linear Algebra

- Introduction to Linear Algebra
- Linear algebra
- Linear Algebra
- Linear Map
- Linear map
- Mappings Between Vector Spaces
- Vector space
- Mapping
- Definition of Linearity
- Linearity
- Linear Maps and Matrices
- Matrix representation
- Physical Meaning of Linearity and Uniqueness
- Basic Rules for Matrices
- Matrix Multiplication
- Matrix multiplication
- Linear Transformation
- Linear transformation
- Cofactor of a Matrix
- Cofactor
- Definition of Determinants
- Determinant
- Permutations and Sign of a Permutation
- Permutation
- Sign of permutation
- Determinant of Order N
- Determinant Basic Formula
- Inverse Matrix
- Inverse matrix
- Meaning of the Identity Matrix
- Identity matrix
- Formula for the Inverse Matrix
- Linear Dependence and Independence
- Linear dependence
- Linear independence
- Linear Relations in a System of Equations
- Wronskian and Abel's Identity
- Wronskian
- Abel's identity
- Orthogonal Matrices
- Orthogonal matrix
- Introduction to Eigenproblems
- Eigenproblem
- Eigenproblems
- Definition of Eigenvalues and Eigenvectors
- Eigenvalue
- Eigenvector
- Diagonalization and its Meaning
- Diagonalization
- Diagonalizing Hermitian Matrices(Optional)
- Hermitian matrix
- Applications to the Eigen Problem in Physical Systems
- Coupled Spring System
- Coupled spring system
- Infinite Potential Well
- Infinite potential well
- Curvilinear Coordinates
- Curvilinear coordinate
- General Definition of Curvilinear Coordinates
- Orthogonal Form of Curvilinear Coordinates
- Orthogonal coordinate
- Gradient in Orthogonal Curvilinear Coordinates
- Gradient
- Divergence in Orthogonal Coordinate Systems
- Divergence
- Curl in Orthogonal Coordinate Systems
- Curl

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- Introduction to Tensor
- Tensor
- Some Conventions in this Chapter
- Why Tensors Appear
- Einstein Summation Convention
- Einstein summation convention
- Euclidean Space and the Kronecker Delta
- Euclidean space
- Kronecker delta
- Antisymmetrized Pseudo Tensor
- Pseudo tensor
- Levi-Civita symbol
- Tensor Transformation Laws
- Tensor transformation law
- The Inertia Tensor
- Inertia tensor
- Rotations as Matrices and Antisymmetric Generators
- Rotation matrix
- Antisymmetric generator
- From Euclidean Space to Minkowski Space
- Lorentz Transformations
- Lorentz transformation
- Four Velocity and Four Momentum
- Four-velocity
- Four-momentum
- Gauge Transformation and Field Strength
- Gauge transformation
- Field strength tensor
- Maxwell Equations in Minkowski Space
- Maxwell's equations

Some Definitions in Linear Algebra

- Some Definitions in Linear Algebra
- Vector Space
- Vector space
- Subspace
- Trace of a Matrix
- Trace
- The Sum of Sets
- Direct Sum
- Direct sum
- Span
- Basis
- Dimension
- Null Space
- Null space
- Image of the Linear Transformation
- Image

A Short Introduction to Differential Forms and Hamiltonian Mechanics

- A Short Introduction to Differential Forms and Hamiltonian Mechanics
- Cotangent Space and One-Forms
- Cotangent space
- One-form
- Differential Forms
- Differential form
- Wedge Product
- Wedge product
- Exterior Derivative
- Exterior derivative
- Exact and Closed Forms
- Exact form
- Closed form
- Pullback
- Stokes' Theorem
- Stokes' theorem
- Differential Forms in Vector Calculus

- Symplectic Form
- Symplectic form
- Hamiltonian Vector Field
- Hamiltonian vector field
- Poisson Bracket
- Poisson bracket
- Hamiltonian Flow and Symplectic Preservation
- Hamiltonian flow
- Symplectic preservation

Runge Kutta Method of Fourth Order

- Runge Kutta Method of Fourth Order
- Runge-Kutta method
- RK4
- Integral Form of the Problem
- General Explicit Runge Kutta Form
- Explicit Runge-Kutta method
- Choosing the Classical Four Stages
- Determining the Weights
- The Classical RK4 Formula
- Classical RK4 formula
- Taylor Expansion and Order Conditions
- Order condition
- Projectile Motion With Quadratic Drag
- Quadratic drag
- Projectile motion
- Simulation Figure
- Python Code
- Numerical Interpretation

Introduction to Special Relativity

- Introduction to Special Relativity
- Special relativity
- Special Relativity
- Lorentz Transform
- Lorentz transform
- The Original Question and the Galilean Transform
- Galilean transform
- Definition and Matrices in Minkowski Space
- Minkowski metric
- Minkowski Metric between 2 Events
- Spacetime interval
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- Lorentz matrix
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- Twin Paradox
- Twin paradox
- Barn and Ladder Paradox
- Barn and ladder paradox

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